

Capital Budgeting and Valuation with Leverage

NOTATION

FCF_t	free cash flows at date t
r_{wacc}	weighted average cost of capital
r_E, r_D	equity and debt costs of capital
r_D^*	equity-equivalent debt cost of capital
E	market value of equity
D	market value of debt (net of cash)
τ_c	marginal corporate tax rate
D_t	incremental debt of project on date t
V_t^L	value of a levered investment on date t
d	debt-to-value ratio
r_U	unlevered cost of capital
V^U	unlevered value of investment
T^s	value of predetermined tax shields
k	interest coverage ratio
Int_t	interest expense on date t
D^s	debt net of predetermined tax shields
ϕ	permanence of the debt level
τ_e, τ_i	tax rate on equity and interest income
τ^*	effective tax advantage of debt

IN FALL 2015, GENERAL ELECTRIC COMPANY HAD A MARKET

capitalization of approximately \$255 billion. With net debt of over \$224 billion, GE's total enterprise value was \$479 billion, making it the second most valuable business in the world (just behind Apple, and ahead of Alphabet and Exxon Mobil). GE's businesses include power generation and air transportation equipment, health care and medical equipment, consumer appliances, and consumer and commercial financing and insurance. With a debt-to-value ratio exceeding 50%, leverage is clearly part of GE's business strategy. How should a firm that uses leverage, like GE, incorporate the costs and benefits associated with leverage into its capital budgeting decisions? And how should a firm adjust for the differences in risk, and debt capacity, associated with its different business activities?

We introduced capital budgeting in Chapter 7. There, we outlined the following basic procedure: First, we estimate the incremental free cash flow generated by the project; then we discount the free cash flow based on the project's cost of capital to determine the NPV. Thus far, we have focused on all-equity financed projects. In this chapter, we integrate the lessons from Parts 4 and 5 into our capital budgeting framework and consider alternative financing arrangements. In particular, we address how the financing decision of the firm can affect both the cost of capital and the set of cash flows that we ultimately discount.

We begin by introducing the three main methods for capital budgeting with leverage and market imperfections: the weighted average cost of capital (WACC) method, the adjusted present value (APV) method, and the flow-to-equity (FTE) method. While their details differ, when appropriately applied each method produces the same estimate of an investment's (or firm's) value. As we shall see, the choice of method is thus guided by which is the simplest to use in a given setting. Ultimately, we will develop recommendations regarding the best method to use depending on the firm's financing policy.

Throughout this chapter, we focus on the intuition and implementation of the main capital budgeting methods. The appendix to this chapter provides additional details about the justification for and assumptions behind some of the results we use in the chapter. It also introduces advanced computational techniques that can be used in Excel to solve for leverage and value simultaneously.

18.1 Overview of Key Concepts

We introduce the three main methods of capital budgeting in Sections 18.2 through 18.4. Before we turn to the specifics, we will revisit some important ideas we encountered earlier in the text that underpin the valuation methods.

Chapter 15 demonstrated that because interest payments are deductible as an expense for the corporation, debt financing creates a valuable interest tax shield for the firm. We can include the value of this tax shield in the capital budgeting decision in several ways. First, we can use the *WACC method*, explained in Section 18.2, in which we discount the unlevered free cash flows using the weighted-average cost of capital, or WACC. Because we calculate the WACC using the effective *after-tax* interest rate as the cost of debt, this method incorporates the tax benefit of debt implicitly through the cost of capital.

Alternatively, we can first value a project's free cash flows without leverage by discounting them using the unlevered cost of capital. We can then separately estimate and add the present value of the interest tax shields from debt. This method, in which we explicitly add the value of the interest tax shields to the project's unlevered value, is called the *adjusted present value (APV) method*, which we explain in Section 18.3.

Our third method makes use of the observation in Chapter 9 that rather than value the firm based on its free cash flows, we can also value its equity based on the total payouts to shareholders. The *flow-to-equity (FTE) method*, introduced in Section 18.4, applies this idea to value the incremental payouts to equity associated with a project.

To illustrate these methods most clearly, we begin the chapter by applying each method to a single example in which we have made a number of simplifying assumptions:

1. *The project has average risk.* We assume initially that the market risk of the project is equivalent to the average market risk of the firm's investments. In that case, the project's cost of capital can be assessed based on the risk of the firm.
2. *The firm's debt-equity ratio is constant.* Initially, we consider a firm that adjusts its leverage to maintain a constant debt-equity ratio in terms of market values. This policy determines the amount of debt the firm will take on when it accepts a new project. It also implies that the risk of the firm's equity and debt, and therefore its weighted average cost of capital, will not fluctuate due to leverage changes.
3. *Corporate taxes are the only imperfection.* We assume initially that the main effect of leverage on valuation is due to the corporate tax shield. We ignore personal taxes and issuance costs, and we assume that other imperfections (such as financial distress or agency costs) are not significant at the level of debt chosen.

While these assumptions are restrictive, they are also a reasonable approximation for many projects and firms. The first assumption is likely to fit typical projects of firms with investments concentrated in a single industry. The second assumption, while unlikely to hold exactly, reflects the fact that firms tend to increase their levels of debt as they grow larger; some may even have an explicit target for their debt-equity ratio. Finally, for firms

without very high levels of debt, the interest tax shield is likely to be the most important market imperfection affecting the capital budgeting decision. Hence, the third assumption is a reasonable starting point to begin our analysis.

Of course, while these three assumptions are reasonable in many situations, there are certainly projects and firms for which they do not apply. The remainder of the chapter therefore relaxes these assumptions and shows how to generalize the methods to more complicated settings. In Section 18.5, we adjust these methods for projects whose risk or debt capacity is substantially different from the rest of the firm. These adjustments are especially important for multidivisional firms, such as GE. In Section 18.6, we consider alternative leverage policies for the firm (rather than maintaining a constant debt-equity ratio) and adapt the APV method to handle such cases. We consider the consequence of other market imperfections, such as issuance, distress, and agency costs, on valuation in Section 18.7. Finally, in Section 18.8, we investigate a number of advanced topics, including periodically adjusted leverage policies and the effect of investor taxes.

CONCEPT CHECK

1. What are the three methods we can use to include the value of the tax shield in the capital budgeting decision?
2. In what situation is the risk of a project likely to match that of the overall firm?

18.2 The Weighted Average Cost of Capital Method

The WACC method takes the interest tax shield into account by using the after-tax cost of capital as the discount rate. When the market risk of the project is similar to the average market risk of the firm's investments, then its cost of capital is equal to the firm's weighted average cost of capital (WACC). As we showed in Chapter 15, the WACC incorporates the benefit of the interest tax shield by using the firm's *after-tax* cost of capital for debt:

$$r_{wacc} = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - \tau_c) \quad (18.1)$$

In this formula,

$$\begin{aligned} E &= \text{market value of equity} & r_E &= \text{equity cost of capital} \\ D &= \text{market value of debt (net of cash)} & r_D &= \text{debt cost of capital} \\ \tau_c &= \text{marginal corporate tax rate} \end{aligned}$$

For now, we assume that the firm maintains a constant debt-equity ratio and that the WACC calculated in Eq. 18.1 remains constant over time. Because the WACC incorporates the tax savings from debt, we can compute the *levered value* of an investment, which is its value including the benefit of interest tax shields given the firm's leverage policy, by discounting its future free cash flow using the WACC. Specifically, if FCF_t is the expected free cash flow of an investment at the end of year t , then the investment's initial levered value, V_0^L , is¹

$$V_0^L = \frac{FCF_1}{1 + r_{wacc}} + \frac{FCF_2}{(1 + r_{wacc})^2} + \frac{FCF_3}{(1 + r_{wacc})^3} + \dots \quad (18.2)$$

¹See this chapter's appendix for a formal justification of this result.

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QUESTION: *When developing a model to analyze a new investment, how do you deal with the uncertainty around future cash flows? Does it differ with new technologies?*

ANSWER: Dealing with uncertainty is an art as well as a science. Companies must take on risk to grow, and that requires a solid investment strategy. The model should support the strategy, not drive it. You can fixate on the model and whether the WACC is 2 points too high or too low, however what's more important is having a team that understands the business proposition and key drivers, such as top-line growth and the expected business environment. By choosing the right inputs for the model, they develop appropriate scenarios to help make the decision.

In the case of an airline's fleet acquisition, you could be fine-tuning a well-defined model with more certain inputs. If you acquire another company—in any industry—you focus on the business rationale for the investment. This involves much greater uncertainty.

QUESTION: *How does the capital budgeting process differ at VMware versus United Continental?*

ANSWER: A mature, capital-intensive, lower margin company such as an airline may emphasize more traditional measures such as NPV, IRR, and WACC. It has more history and lower volatility on the inputs for assumptions—and thus less variability. Airline expenditures can be very large, for example, the decision to purchase or lease new aircraft. The WACC and the timing of cash flows often drive the analysis.

With technology, it's sometimes more difficult to quantify the expected impact of a new product or acquisition. We adapt our model to include more variability and place greater focus on revenue growth and margin assumptions than on certain more detailed factors such as components of WACC. These tend to be more important drivers of the

INTERVIEW WITH ZANE ROWE



investment's overall value than the precise cost of capital.

QUESTION: *What key financial metrics do you use to make investment decisions? How does it differ between these firms?*

ANSWER: Regardless of the type of company—public or private, technology or capital-intensive—both the fundamentals and metrics are important. Capital budgeting models are only as good as their inputs. The most important element is engaging with all business units affected by the investment to ensure the right variables are in the model.

At the airline we looked at traditional metrics such as NPV and IRR, driven by the right WACC, as well as market multiples. A tech company will use some of those variables, too, but weight them differently. Top-line assumptions such as revenue and margins may carry a larger weight on the ultimate decision. Also, every capital budgeting decision must take into account the competitive and economic environments, the company's growth rate, and geographic risk.

QUESTION: *What is the importance of considering Free Cash Flows as opposed to just the earnings implications of a financial decision?*

ANSWER: Free cash flow (FCF) is important regardless of type of company or industry. Both FCF and earnings affect a financial decision. The airline business can be more sensitive to FCF than earnings, depending on the stage of the business cycle. We went through challenging periods—9/11, oil shocks, financial crisis—where FCF was the decision driver because of our highly levered balance sheet. We may choose a project that is FCF positive but has a minimal or potentially negative short-term impact on earnings.

In technology, FCF carries a fair amount of weight, too. That said, we communicate a number of variables and financial drivers to our analyst and investor base. When earnings and FCF don't perfectly align, we convey the purpose of our decision in greater detail. In an industry or company with a higher growth profile, you have more flexibility to continue to make investments that help drive future growth, placing less emphasis on the details of the capital budgeting model itself.

Using the WACC to Value a Project

Let's apply the WACC method to value a project. Avco, Inc., is a manufacturer of custom packaging products. Avco is considering introducing a new line of packaging, the RFX series, that will include an embedded radio-frequency identification (RFID) tag, which is a miniature radio antenna and transponder that allows a package to be tracked much more efficiently and with fewer errors than standard bar codes.

Avco engineers expect the technology used in these products to become obsolete after four years. During the next four years, however, the marketing group expects annual sales of \$60 million per year for this product line. Manufacturing costs and operating expenses are expected to be \$25 million and \$9 million, respectively, per year. Developing the product will require upfront R&D and marketing expenses of \$6.67 million, together with a \$24 million investment in equipment. The equipment will be obsolete in four years and will be depreciated via the straight-line method over that period. Avco bills the majority of its customers in advance, and it expects no net working capital requirements for the project. Avco pays a corporate tax rate of 40%. Given this information, the spreadsheet in Table 18.1 forecasts the project's expected free cash flow.²

The market risk of the RFX project is expected to be similar to that for the company's other lines of business. Thus, we can use Avco's equity and debt to determine the weighted average cost of capital for the new project. Table 18.2 shows Avco's current market value balance sheet and equity and debt costs of capital. Avco has built up \$20 million in cash for investment needs, so that its *net* debt is $D = 320 - 20 = \$300$ million. Avco's enterprise value, which is the market value of its non-cash assets, is $E + D = \$600$ million. Avco intends to maintain a similar (net) debt-equity ratio for the foreseeable future, including any financing related to the RFX project.

TABLE 18.1
SPREADSHEET

Expected Free Cash Flow from Avco's RFX Project

	Year	0	1	2	3	4
Incremental Earnings Forecast (\$ million)						
1 Sales	—	60.00	60.00	60.00	60.00	60.00
2 Cost of Goods Sold	—	(25.00)	(25.00)	(25.00)	(25.00)	(25.00)
3 Gross Profit	—	35.00	35.00	35.00	35.00	35.00
4 Operating Expenses	(6.67)	(9.00)	(9.00)	(9.00)	(9.00)	(9.00)
5 Depreciation	—	(6.00)	(6.00)	(6.00)	(6.00)	(6.00)
6 EBIT	(6.67)	20.00	20.00	20.00	20.00	20.00
7 Income Tax at 40%	2.67	(8.00)	(8.00)	(8.00)	(8.00)	(8.00)
8 Unlevered Net Income	(4.00)	12.00	12.00	12.00	12.00	12.00
Free Cash Flow						
9 Plus: Depreciation	—	6.00	6.00	6.00	6.00	6.00
10 Less: Capital Expenditures	(24.00)	—	—	—	—	—
11 Less: Increases in NWC	—	—	—	—	—	—
12 Free Cash Flow	(28.00)	18.00	18.00	18.00	18.00	18.00

²The spreadsheets shown in this chapter are available for download in [MyFinanceLab](#).

TABLE 18.2**Avco's Current Market Value Balance Sheet (\$ million)
and Cost of Capital without the RFX Project**

Assets		Liabilities		Cost of Capital	
Cash	20	Debt	320	Debt	6%
Existing Assets	600	Equity	300	Equity	10%
Total Assets	620	Total Liabilities and Equity	620		

With this capital structure, Avco's weighted average cost of capital is

$$r_{wacc} = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - \tau_c) = \frac{300}{600}(10.0\%) + \frac{300}{600}(6.0\%)(1 - 0.40) = 6.8\%$$

We can determine the value of the project, including the tax shield from debt, by calculating the present value of its future free cash flows, V_0^L , using the WACC:

$$V_0^L = \frac{18}{1.068} + \frac{18}{1.068^2} + \frac{18}{1.068^3} + \frac{18}{1.068^4} = \$61.25 \text{ million}$$

Because the upfront cost of launching the product line is only \$28 million, this project is a good idea—taking the project results in an NPV of $61.25 - 28 = \$33.25$ million for the firm.

Summary of the WACC Method

To summarize, the key steps in the WACC valuation method are as follows:

1. Determine the free cash flow of the investment.
2. Compute the weighted average cost of capital using Eq. 18.1.
3. Compute the value of the investment, including the tax benefit of leverage, by discounting the free cash flow of the investment using the WACC.

In many firms, the corporate treasurer performs the second step, calculating the firm's WACC. This rate can then be used throughout the firm as the companywide cost of capital for new investments *that are of comparable risk to the rest of the firm and that will not alter the firm's debt-equity ratio*. Employing the WACC method in this way is very simple and straightforward. As a result, it is the method that is most commonly used in practice for capital budgeting purposes.

EXAMPLE 18.1

Valuing an Acquisition Using the WACC Method

Problem

Suppose Avco is considering the acquisition of another firm in its industry that specializes in custom packaging. The acquisition is expected to increase Avco's free cash flow by \$3.8 million the first year, and this contribution is expected to grow at a rate of 3% per year from then on. Avco has negotiated a purchase price of \$80 million. After the transaction, Avco will adjust its capital

structure to maintain its current debt-equity ratio. If the acquisition has similar risk to the rest of Avco, what is the value of this deal?

Solution

The free cash flows of the acquisition can be valued as a growing perpetuity. Because its risk matches the risk for the rest of Avco, and because Avco will maintain the same debt-equity ratio going forward, we can discount these cash flows using the WACC of 6.8%. Thus, the value of the acquisition is

$$V^L = \frac{3.8}{6.8\% - 3\%} = \$100 \text{ million}$$

Given the purchase price of \$80 million, the acquisition has an NPV of \$20 million.

Implementing a Constant Debt-Equity Ratio

Thus far, we have simply assumed the firm adopted a policy of keeping its debt-equity ratio constant. In fact, an important advantage of the WACC method is that you do not need to know how this leverage policy is implemented in order to make the capital budgeting decision. Nevertheless, keeping the debt-equity ratio constant has implications for how the firm's total debt will change with new investment. For example, Avco currently has a debt-equity ratio of $300/300 = 1$ or, equivalently, a debt-to-value ratio $[D/(E + D)]$ of 50%. To maintain this ratio, the firm's new investments must be financed with debt equal to 50% of their market value.

By undertaking the RFX project, Avco adds new assets to the firm with initial market value $V_0^L = \$61.25$ million. Therefore, to maintain its debt-to-value ratio, Avco must add $50\% \times 61.25 = \$30.625$ million in new net debt.³ Avco can add this net debt either by reducing cash or by borrowing and increasing debt. Suppose Avco decides to spend its \$20 million in cash and borrow an additional \$10.625 million. Because only \$28 million is required to fund the project, Avco will pay the remaining $30.625 - 28 = \$2.625$ million to shareholders through a dividend (or share repurchase). Table 18.3 shows Avco's market value balance sheet with the RFX project in this case.

TABLE 18.3

Avco's Current Market Value Balance Sheet (\$ million) with the RFX Project

Assets		Liabilities	
Cash	—	Debt	330.625
Existing Assets	600.00		
RFX Project	61.25	Equity	330.625
Total Assets	661.25	Total Liabilities and Equity	661.25

³We can also evaluate the project's debt as follows: Of the \$28 million upfront cost of the project, 50% (\$14 million) will be financed with debt. In addition, the project generates an NPV of \$33.25 million, which will increase the market value of the firm. To maintain a debt-equity ratio of 1, Avco must add debt of $50\% \times 33.25 = \$16.625$ million at the time when the NPV of the project is anticipated (which could occur before the new investment is made). Thus, the total new debt is $14 + 16.625 = \$30.625$ million.

This financing plan maintains Avco's 50% debt-to-value ratio. The market value of Avco's equity increases by $330.625 - 300 = \$30.625$ million. Adding the dividend of \$2.625 million, the shareholders' total gain is $30.625 + 2.625 = \$33.25$ million, which is exactly the NPV we calculated for the RFX project.

What happens over the life of the project? First, we define an investment's **debt capacity**, D_t , as the amount of debt at date t that is required to maintain the firm's target debt-to-value ratio, d . If V_t^L is the project's levered continuation value on date t —that is, the levered value of its free cash flow after date t —then

$$D_t = d \times V_t^L \quad (18.3)$$

We compute the debt capacity for the RFX project in the spreadsheet in Table 18.4. Starting with the project's free cash flow, we compute its levered continuation value at each date (line 2) by discounting the future free cash flow at the WACC as in Eq. 18.2. Because the continuation value at each date includes the value of all subsequent cash flows, it is even simpler to compute the value at each date by working backward from period 4, discounting next period's free cash flow and continuation value:

$$V_t^L = \frac{\overbrace{FCF_{t+1} + V_{t+1}^L}^{\text{Value of FCF in year } t+2 \text{ and beyond}}}{1 + r_{wacc}} \quad (18.4)$$

Once we have computed the project's value V_t^L at each date, we apply Eq. 18.3 to compute the project's debt capacity at each date (line 3). As the spreadsheet shows, the project's debt capacity declines each year, and falls to zero by the end of year 4.

TABLE 18.4
SPREADSHEET

Continuation Value and Debt Capacity of the RFX Project over Time

	Year	0	1	2	3	4
Project Debt Capacity (\$ million)						
1 Free Cash Flow		(28.00)	18.00	18.00	18.00	18.00
2 Levered Value, V_t^L (at $r_{wacc} = 6.8\%$)		61.25	47.41	32.63	16.85	—
3 Debt Capacity, D_t (at $d = 50\%$)		30.62	23.71	16.32	8.43	—

EXAMPLE 18.2

Debt Capacity for an Acquisition

Problem

Suppose Avco proceeds with the acquisition described in Example 18.1. How much debt must Avco use to finance the acquisition and still maintain its debt-to-value ratio? How much of the acquisition cost must be financed with equity?

Solution

From the solution to Example 18.2, the market value of the assets acquired in the acquisition, V^L , is \$100 million. Thus, to maintain a 50% debt-to-value ratio, Avco must increase its debt by \$50 million. The remaining \$30 million of the \$80 million acquisition cost will be financed with new equity. In addition to the \$30 million in new equity, the value of Avco's existing shares will increase by the \$20 million NPV of the acquisition, so in total the market value of Avco's equity will rise by \$50 million.

CONCEPT CHECK

1. Describe the key steps in the WACC valuation method.
2. How does the WACC method take into account the tax shield?

18.3 The Adjusted Present Value Method

The **adjusted present value (APV)** method is an alternative valuation method in which we determine the levered value V^L of an investment by first calculating its *unlevered value* V^U , which is its value without any leverage, and then adding the value of the interest tax shield. That is, as we showed in Chapter 15:⁴

The APV Formula

$$V^L = APV = V^U + PV(\text{Interest Tax Shield}) \quad (18.5)$$

As we did with the WACC method, we focus solely on the corporate tax benefits of debt for now and defer the discussion of other consequences of leverage to Section 18.7. As Eq. 18.5 shows, the APV method incorporates the value of the interest tax shield directly, rather than by adjusting the discount rate as in the WACC method. Let's demonstrate the APV method by returning to Avco's RFX project.

The Unlevered Value of the Project

From the free cash flow estimates in Table 18.1, the RFX project has an upfront cost of \$28 million, and it generates \$18 million per year in free cash flow for the next four years. The first step in the APV method is to calculate the value of these free cash flows using the project's cost of capital if it were financed without leverage.

What is the project's unlevered cost of capital? Because the RFX project has similar risk to Avco's other investments, its unlevered cost of capital is the same as for the firm as a whole. Thus, as we did in Chapter 12, we can calculate the unlevered cost of capital using Avco's pretax WACC, the average return the firm's investors expect to earn:

Unlevered Cost of Capital with a Target Leverage Ratio

$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D = \text{Pretax WACC} \quad (18.6)$$

To understand why the firm's unlevered cost of capital equals its pretax WACC, note that the pretax WACC represents investors' required return for holding the entire firm (equity and debt). Thus, it will depend only on the firm's overall risk. So long as the firm's leverage choice does not change the overall risk of the firm, the pretax WACC must be the same whether the firm is levered or unlevered—recall Figure 15.2 on page 559.

Of course, this argument relies on the assumption that the overall risk of the firm is independent of the choice of leverage. As we showed in Chapter 14, this assumption always holds in a perfect market. It will also hold in a world with taxes whenever the risk of the tax shield is the same as the risk of the firm (so the size of the tax shield will not change the overall riskiness of the firm). In this chapter's appendix, we show that the tax shield will have the same risk as the firm if the firm maintains a *target leverage ratio*. A **target leverage ratio**

⁴Stewart Myers developed the application of APV to capital budgeting, see "Interactions of Corporate Financing and Investment Decisions—Implications for Capital Budgeting," *Journal of Finance* 29 (1974): 1–25.

means that the firm adjusts its debt proportionally to the project's value, or its cash flows, so that a constant debt-equity ratio is a special case.

Applying Eq. 18.6 to Avco, we find its unlevered cost of capital to be

$$r_U = 0.50 \times 10.0\% + 0.50 \times 6.0\% = 8.0\%$$

Avco's unlevered cost of capital is less than its equity cost of capital of 10.0% (which includes the financial risk of leverage), but is more than its WACC of 6.8% (which incorporates the tax benefit of leverage).

Given our estimate of the unlevered cost of capital r_U and the project's free cash flows, we calculate the project's value without leverage:

$$V^U = \frac{18}{1.08} + \frac{18}{1.08^2} + \frac{18}{1.08^3} + \frac{18}{1.08^4} = \$59.62 \text{ million}$$

Valuing the Interest Tax Shield

The value of the unlevered project, V^U , calculated above does not include the value of the tax shield provided by the interest payments on debt—it is the value of the project were it purely equity financed. Given the project's debt capacity from Table 18.4, we can estimate the expected interest payments and the tax shield as shown in the spreadsheet in Table 18.5. The interest paid in year t is estimated based on the amount of debt outstanding at the end of the prior year:

$$\text{Interest paid in year } t = r_D \times D_{t-1} \quad (18.7)$$

The interest tax shield is equal to the interest paid multiplied by the corporate tax rate τ_c .

TABLE 18.5
SPREADSHEET

Expected Debt Capacity, Interest Payments, and Tax Shield for Avco's RFX Project

	Year	0	1	2	3	4
Interest Tax Shield (\$ million)						
1 Debt Capacity, D_t (at $d = 50\%$)		30.62	23.71	16.32	8.43	—
2 Interest Paid (at $r_D = 6\%$)			1.84	1.42	0.98	0.51
3 Interest Tax Shield (at $\tau_c = 40\%$)			0.73	0.57	0.39	0.20

To compute the present value of the interest tax shield, we need to determine the appropriate cost of capital. The interest tax shields shown in Table 18.5 are expected values, and the true amount of the interest tax shield each year will vary with the cash flows of the project. Because Avco maintains a fixed debt-equity ratio, if the project does well, its value will be higher, it will support more debt, and the interest tax shield will be higher. If the project goes poorly, its value will fall, Avco will reduce its debt level, and the interest tax shield will be lower. Thus, the tax shield will fluctuate with, and therefore share the risk of, the project itself.⁵

When the firm maintains a target leverage ratio, its future interest tax shields have similar risk to the project's cash flows, so they should be discounted at the project's unlevered cost of capital.

⁵In Section 18.6, we consider the case in which the debt levels are fixed in advance, and so do *not* fluctuate with the project cash flows. As we will see, in that case the tax shield has lower risk, and thus a lower cost of capital, than the project.

EXAMPLE 18.3**Risk of Tax Shields with a Constant Debt-Equity Ratio****Problem**

Suppose ABC Corporation maintains a constant debt-equity ratio of 1, the current total value of the firm is \$100 million, and its existing debt is riskless. Over the next month news will come out that will either raise or lower ABC's value by 20%. How will ABC adjust its debt level in response? What can you conclude about the risk of its interest tax shields?

Solution

Originally ABC has \$50 million in equity and \$50 million in debt. Once the news comes out, ABC's value will rise to \$120 million or fall to \$80 million. Thus, to maintain a debt equity ratio of 1, ABC will either increase its debt level to \$60 million or decrease it to \$40 million. Because the firm's interest payments and tax shield will fall proportionally, they have the same risk as the overall firm.

For Avco's RFX project, we have

$$PV(\text{interest tax shield}) = \frac{0.73}{1.08} + \frac{0.57}{1.08^2} + \frac{0.39}{1.08^3} + \frac{0.20}{1.08^4} = \$1.63 \text{ million}$$

To determine the value of the project with leverage, we add the value of the interest tax shield to the unlevered value of the project:⁶

$$V^L = V^U + PV(\text{interest tax shield}) = 59.62 + 1.63 = \$61.25 \text{ million}$$

Again, given the \$28 million initial investment required, the RFX project has an NPV with leverage of $61.25 - 28 = \$33.25$ million, which matches precisely the value we computed in Section 18.2 using the WACC approach.

Summary of the APV Method

To determine the value of a levered investment using the APV method, we proceed as follows:

1. Determine the investment's value without leverage, V^U , by discounting its free cash flows at the unlevered cost of capital, r_U . With a constant debt-equity ratio, r_U may be estimated using Eq. 18.6.
2. Determine the present value of the interest tax shield.
 - a. Determine the expected interest tax shield: Given expected debt D_t on date t , the interest tax shield on date $t + 1$ is $\tau_c r_D D_t$.⁷
 - b. Discount the interest tax shield. If a constant debt-equity ratio is maintained, using r_U is appropriate.

⁶Because we are using the same discount rate for the free cash flow and the tax shield, the cash flows of the project and the tax shield can be combined first and then discounted at the rate r_U . These combined cash flows are also referred to as the capital cash flows (CCF): $\text{CCF} = \text{FCF} + \text{Interest Tax Shield}$. This method is known as the CCF or "compressed APV" method [see S. Kaplan and R. Ruback, "The Valuation of Cash Flow Forecasts: An Empirical Analysis," *Journal of Finance* 50 (1995): 1059–1093; and R. Ruback, "Capital Cash Flows: A Simple Approach to Valuing Risky Cash Flows," *Financial Management* 31 (2002): 85–103].

⁷The return on the debt need not come solely from interest payments, so this value is an approximation. The same approximation is implicit in the definition of the WACC (for additional precision, see footnote 27 in this chapter's appendix).

3. Add the unlevered value, V^U , to the present value of the interest tax shield to determine the value of the investment with leverage, V^L .

In this case, the APV method is more complicated than the WACC method because we must compute two separate valuations: the unlevered project and the interest tax shield. Furthermore, in this example, to determine the project's debt capacity for the interest tax shield calculation, we relied on the calculation in Table 18.4, *which depended on the value of the project*. Thus, we need to know the debt level to compute the APV, but with a constant debt-equity ratio we need to know the project's value to compute the debt level. As a result, implementing the APV approach with a constant debt-equity ratio requires solving for the project's debt and value *simultaneously*. (See this chapter's appendix for an example of this calculation.)

Despite its complexity, the APV method has some advantages. As we shall see in Section 18.6, it can be easier to apply than the WACC method when the firm does not maintain a constant debt-equity ratio. It also provides managers with an explicit valuation of the tax shield itself. In the case of Avco's RFX project, the benefit of the interest tax shield is relatively small. Even if tax rates were to change, or if Avco decided for other reasons not to increase its debt, the profitability of the project would not be jeopardized. However, this need not always be the case. Consider again the acquisition in Example 18.1, where the APV method makes clear that the gain from the acquisition crucially depends on the interest tax shield.

EXAMPLE 18.4

Using the APV Method to Value an Acquisition

Problem

Consider again Avco's acquisition from Examples 18.1 and 18.2. The acquisition will contribute \$3.8 million in free cash flows the first year, which will grow by 3% per year thereafter. The acquisition cost of \$80 million will be financed with \$50 million in new debt initially. Compute the value of the acquisition using the APV method, assuming Avco will maintain a constant debt-equity ratio for the acquisition.

Solution

First, we compute the value without leverage. Given Avco's unlevered cost of capital of $r_U = 8\%$, we get

$$V^U = 3.8 / (8\% - 3\%) = \$76 \text{ million}$$

Avco will add new debt of \$50 million initially to fund the acquisition. At a 6% interest rate, the interest expense the first year is $6\% \times 50 = \$3$ million, which provides an interest tax shield of $40\% \times 3 = \$1.2$ million. Because the value of the acquisition is expected to grow by 3% per year, the amount of debt the acquisition supports—and, therefore, the interest tax shield—is expected to grow at the same rate. The present value of the interest tax shield is

$$PV(\text{interest tax shield}) = 1.2 / (8\% - 3\%) = \$24 \text{ million}$$

The value of the acquisition with leverage is given by the APV:

$$V^L = V^U + PV(\text{interest tax shield}) = 76 + 24 = \$100 \text{ million}$$

This value is identical to the value computed in Example 18.1 and implies an NPV of $100 - 80 = \$20$ million for the acquisition. Without the benefit of the interest tax shield, the NPV would be $76 - 80 = -\$4$ million.

We can easily extend the APV approach to include other market imperfections such as financial distress, agency, and issuance costs. We discuss these complexities further in Section 18.7.

CONCEPT CHECK

1. Describe the adjusted present value (APV) method.
2. At what rate should we discount the interest tax shield when a firm maintains a target leverage ratio?

18.4 The Flow-to-Equity Method

In the WACC and APV methods, we value a project based on its free cash flow, which is computed ignoring interest and debt payments. Some students find these methods confusing because, if the goal is to determine the benefit of the project to shareholders, it seems to them that we should focus on the cash flows that *shareholders* will receive.

In the **flow-to-equity (FTE)** valuation method, we explicitly calculate the free cash flow available to equity holders *after taking into account all payments to and from debt holders*. The cash flows to equity holders are then discounted using the *equity* cost of capital.⁸ Despite this difference in implementation, the FTE method produces the same assessment of the project's value as the WACC or APV methods.

Calculating the Free Cash Flow to Equity

The first step in the FTE method is to determine the project's **free cash flow to equity (FCFE)**. The FCFE is the free cash flow that remains after adjusting for interest payments, debt issuance, and debt repayment. The spreadsheet shown in Table 18.6 calculates the FCFE for Avco's RFX project.

TABLE 18.6
SPREADSHEET

Expected Free Cash Flows to Equity from Avco's RFX Project

	Year	0	1	2	3	4
Incremental Earnings Forecast (\$ million)						
1 Sales	—	60.00	60.00	60.00	60.00	60.00
2 Cost of Goods Sold	—	(25.00)	(25.00)	(25.00)	(25.00)	(25.00)
3 Gross Profit	—	35.00	35.00	35.00	35.00	35.00
4 Operating Expenses	(6.67)	(9.00)	(9.00)	(9.00)	(9.00)	(9.00)
5 Depreciation	—	(6.00)	(6.00)	(6.00)	(6.00)	(6.00)
6 EBIT	(6.67)	20.00	20.00	20.00	20.00	20.00
7 Interest Expense	—	(1.84)	(1.42)	(0.98)	(0.51)	(0.51)
8 Pretax Income	(6.67)	18.16	18.58	19.02	19.49	19.49
9 Income Tax at 40%	2.67	(7.27)	(7.43)	(7.61)	(7.80)	(7.80)
10 Net Income	(4.00)	10.90	11.15	11.41	11.70	11.70
Free Cash Flow to Equity						
11 Plus: Depreciation	—	6.00	6.00	6.00	6.00	6.00
12 Less: Capital Expenditures	(24.00)	—	—	—	—	—
13 Less: Increases in NWC	—	—	—	—	—	—
14 Plus: Net Borrowing	30.62	(6.92)	(7.39)	(7.89)	(8.43)	(8.43)
15 Free Cash Flow to Equity	2.62	9.98	9.76	9.52	9.27	9.27

⁸The FTE approach generalizes the total payout method for valuing the firm described in Chapter 9. In that method, we value the total dividends and repurchases that the firm pays to shareholders. It is also equivalent to the residual income valuation method used in accounting (see appendix).

Comparing the FCFE estimates in Table 18.6 with the free cash flow estimates in Table 18.1, we notice two changes. First, we deduct interest expenses (from Table 18.5) on line 7, before taxes. As a consequence, we compute the incremental net income of the project on line 10, rather than its *unlevered* net income as we do when computing free cash flows. The second change is line 14, where we add the proceeds from the firm's net borrowing activity. These proceeds are positive when the firm increases its net debt; they are negative when the firm reduces its net debt by repaying principal (or retaining cash). For the RFX project, Avco issues \$30.62 million in debt initially. At date 1, however, the debt capacity of the project falls to \$23.71 million (see Table 18.4), so that Avco must repay $30.62 - 23.71 = \$6.91$ million of the debt.⁹ In general, given the project's debt capacity D_t ,

$$\text{Net Borrowing at Date } t = D_t - D_{t-1} \quad (18.8)$$

As an alternative to Table 18.6, we can compute a project's FCFE directly from its free cash flow. Because interest payments are deducted before taxes in line 7, we adjust the firm's FCF by their after-tax cost. We then add net borrowing to determine FCFE:

Free Cash Flow to Equity

$$FCFE = FCF - \underbrace{(1 - \tau_c) \times (\text{Interest Payments})}_{\text{After-tax interest expense}} + (\text{Net Borrowing}) \quad (18.9)$$

We illustrate this alternative calculation for Avco's RFX project in Table 18.7. Note that the project's FCFE is lower than its FCF in years 1 through 4 due to the interest and principal payments on the debt. In year 0, however, the proceeds from the loan more than offset the negative free cash flow, so FCFE is positive (and equal to the dividend we calculated in Section 18.2).

Valuing Equity Cash Flows

The project's free cash flow to equity shows the expected amount of additional cash the firm will have available to pay dividends (or conduct share repurchases) each year. Because these cash flows represent payments to equity holders, they should be discounted at the project's equity cost of capital. Given that the risk and leverage of the RFX project are the same as for Avco overall, we can use Avco's equity cost of capital of $r_E = 10.0\%$ to discount the project's FCFE:

$$NPV(FCFE) = 2.62 + \frac{9.98}{1.10} + \frac{9.76}{1.10^2} + \frac{9.52}{1.10^3} + \frac{9.27}{1.10^4} = \$33.25 \text{ million}$$

TABLE 18.7
SPREADSHEET

Computing FCFE from FCF for Avco's RFX Project

	Year	0	1	2	3	4
Free Cash Flow to Equity (\$ million)						
1	Free Cash Flow	(28.00)	18.00	18.00	18.00	18.00
2	After-tax Interest Expense	—	(1.10)	(0.85)	(0.59)	(0.30)
3	Net Borrowing	30.62	(6.92)	(7.39)	(7.89)	(8.43)
4	Free Cash Flow to Equity	2.62	9.98	9.76	9.52	9.27

⁹The \$0.01 million difference in the spreadsheet is due to rounding.

What Counts as “Debt”?

Firms often have many types of debt as well as other liabilities, such as leases. Practitioners use different guidelines to determine which to include as debt when computing the WACC. Some use only long-term debt. Others use both long-term and short-term debt, plus lease obligations. Students are often confused by these different approaches and are left wondering: Which liabilities should be included as debt?

In fact, any choice will work if done correctly. We can view the WACC and FTE methods as special cases of a more general approach in which we *value the after-tax cash flows from a set of the firm’s assets and liabilities by discounting them at the after-tax weighted average cost of capital of the firm’s remaining assets and liabilities*. In the WACC method, the FCF does not include the interest and principal payments on

debt, so debt is included in the calculation of the weighted average cost of capital. In the FTE method, the FCFE incorporates the after-tax cash flows to and from debt holders, so debt is excluded from the weighted average cost of capital (which is simply the equity cost of capital).

Other combinations are also possible. For example, long-term debt can be included in the weighted average cost of capital, and short-term debt can be included as part of the cash flows. Similarly, other assets (such as cash) or liabilities (such as leases) can be included either in the weighted average cost of capital or as part of the cash flow. All such methods, if applied consistently, will lead to an equivalent valuation. Typically, the most convenient choice is the one for which the assumption of a constant debt-to-value ratio is a reasonable approximation.

The value of the project’s FCFE represents the gain to shareholders from the project. It is identical to the NPV we computed using the WACC and APV methods.

Why isn’t the project’s NPV lower now that we have deducted interest and debt payments from the cash flows? Recall that these costs of debt are offset by cash received when the debt is issued. Looking back at Table 18.6, the cash flows from debt in lines 7 and 14 have an NPV of zero assuming the debt is fairly priced.¹⁰ In the end, the only effect on value comes from a reduction in the tax payments, leaving the same result as with the other methods.

Summary of the Flow-to-Equity Method

The key steps in the flow-to-equity method for valuing a levered investment are as follows:

1. Determine the free cash flow to equity of the investment using Eq. 18.9.
2. Determine the equity cost of capital, r_E .
3. Compute the contribution to equity value, E , by discounting the free cash flow to equity using the equity cost of capital.

Applying the FTE method was simplified in our example because the project’s risk and leverage matched the firm’s, and the firm’s equity cost of capital was expected to remain constant. Just as with the WACC, however, this assumption is reasonable only if the firm maintains a constant debt-equity ratio. If the debt-equity ratio changes over time, the risk of equity—and, therefore, its cost of capital—will change as well.

¹⁰The interest and principal payments for the RFX project are as follows:

	Year	0	1	2	3	4
1	Net Borrowing	30.62	(6.92)	(7.39)	(7.89)	(8.43)
2	Interest Expense	—	(1.84)	(1.42)	(0.98)	(0.51)
3	Cash Flow from Debt	30.62	(8.76)	(8.81)	(8.87)	(8.93)

Because these cash flows have the same risk as the debt, we discount them at the debt cost of capital of 6% to compute their NPV:

$$30.62 + \frac{-8.76}{1.06} + \frac{-8.81}{1.06^2} + \frac{-8.87}{1.06^3} + \frac{-8.93}{1.06^4} = 0.$$

In this setting, the FTE approach has the same disadvantage associated with the APV approach: We need to compute the project's debt capacity to determine interest and net borrowing before we can make the capital budgeting decision. For this reason, in most settings the WACC is easier to apply. The FTE method can offer an advantage when calculating the value of equity for the entire firm, if the firm's capital structure is complex and the market values of other securities in the firm's capital structure are not known. In that case, the FTE method allows us to compute the value of equity directly. In contrast, the WACC and APV methods compute the firm's enterprise value, so that a separate valuation of the other components of the firm's capital structure is needed to determine the value of equity. Finally, by emphasizing a project's implication for equity, the FTE method may be viewed as a more transparent method for discussing a project's benefit to shareholders—a managerial concern.

EXAMPLE 18.5

Using the FTE Method to Value an Acquisition

Problem

Consider again Avco's acquisition from Examples 18.1, 18.2, and 18.4. The acquisition will contribute \$3.8 million in free cash flows the first year, growing by 3% per year thereafter. The acquisition cost of \$80 million will be financed with \$50 million in new debt initially. What is the value of this acquisition using the FTE method?

Solution

Because the acquisition is being financed with \$50 million in new debt, the remaining \$30 million of the acquisition cost must come from equity:

$$FCFE_0 = -80 + 50 = -\$30 \text{ million}$$

In one year, the interest on the debt will be $6\% \times 50 = \$3$ million. Because Avco maintains a constant debt-equity ratio, the debt associated with the acquisition is also expected to grow at a 3% rate: $50 \times 1.03 = \$51.5$ million. Therefore, Avco will borrow an additional $51.5 - 50 = \$1.5$ million in one year.

$$FCFE_1 = +3.8 - (1 - 0.40) \times 3 + 1.5 = \$3.5 \text{ million}$$

After year 1, FCFE will also grow at a 3% rate. Using the cost of equity $r_E = 10\%$, we compute the NPV:

$$NPV(FCFE) = -30 + 3.5/(10\% - 3\%) = \$20 \text{ million}$$

This NPV matches the result we obtained with the WACC and APV methods.

CONCEPT CHECK

1. Describe the key steps in the flow to equity method for valuing a levered investment.
2. Why does the assumption that the firm maintains a constant debt-equity ratio simplify the flow-to-equity calculation?

18.5 Project-Based Costs of Capital

Up to this point, we have assumed that both the risk and the leverage of the project under consideration matched those characteristics for the firm as a whole. This assumption allowed us, in turn, to assume that the cost of capital for a project matched the cost of capital of the firm.

In the real world, specific projects often differ from the average investment made by the firm. Consider General Electric Company, discussed in the introduction to this chapter. Projects in its health care division are likely to have different market risk than projects in air transportation equipment or at NBC Universal. Projects may also vary in the amount of leverage they will support—for example, acquisitions of real estate or capital equipment are often highly levered, whereas investments in intellectual property are not. In this section, we show how to calculate the cost of capital for the project's cash flows when a project's risk and leverage differ from those for the firm overall.

Estimating the Unlevered Cost of Capital

We begin by reviewing the method introduced in Chapter 12 to calculate the unlevered cost of capital of a project with market risk that is very different from the rest of the firm. Suppose Avco launches a new plastics manufacturing division that faces different market risks than its main packaging business. What unlevered cost of capital would be appropriate for this division?

We can estimate r_U for the plastics division by looking at other single-division plastics firms that have similar business risks. For example, suppose two firms are comparable to the plastics division and have the following characteristics:

Firm	Equity Cost of Capital	Debt Cost of Capital	Debt-to-Value Ratio, D/(E + D)
Comparable #1	12.0%	6.0%	40%
Comparable #2	10.7%	5.5%	25%

Assuming that both firms maintain a target leverage ratio, we can estimate the unlevered cost of capital for each competitor by using the pretax WACC from Eq. 18.6:

$$\text{Competitor 1: } r_U = 0.60 \times 12.0\% + 0.40 \times 6.0\% = 9.6\%$$

$$\text{Competitor 2: } r_U = 0.75 \times 10.7\% + 0.25 \times 5.5\% = 9.4\%$$

Based on these comparable firms, we estimate an unlevered cost of capital for the plastics division of about 9.5%.¹¹ With this rate in hand, we can use the APV approach to calculate the value of Avco's investment in plastic manufacturing. To use either the WACC or FTE method, however, we need to estimate the project's equity cost of capital, which will depend on the incremental debt the firm will take on as a result of the project.

Project Leverage and the Equity Cost of Capital

Suppose the firm will fund the project according to a target leverage ratio. This leverage ratio may differ from the firm's overall leverage ratio, as different divisions or types of investments may have different optimal debt capacities. We can rearrange terms in Eq. 18.6 to get the following expression for the equity cost of capital:¹²

$$r_E = r_U + \frac{D}{E}(r_U - r_D) \quad (18.10)$$

¹¹If we are using the CAPM to estimate expected returns, this procedure is equivalent to unlevering the betas of comparable firms using Eq. 12.9:

$$\beta_U = [E/(E + D)] \beta_E + [D/(D + E)] \beta_D.$$

¹²We derived this same expression with perfect capital markets in Eq. 14.5.

Equation 18.10 shows that the project's equity cost of capital depends on its unlevered cost of capital, r_U , and the debt-equity ratio of the incremental financing that will be put in place to support the project. For example, suppose that Avco plans to maintain an equal mix of debt and equity financing as it expands into plastics manufacturing, and it expects its borrowing cost to remain at 6%. Given its 9.5% unlevered cost of capital, the plastics division's equity cost of capital is

$$r_E = 9.5\% + \frac{0.50}{0.50}(9.5\% - 6\%) = 13.0\%$$

Once we have the equity cost of capital, we can use Eq. 18.1 to determine the division's WACC:

$$r_{wacc} = 0.50 \times 13.0\% + 0.50 \times 6.0\% \times (1 - 0.40) = 8.3\%$$

Based on these estimates, Avco should use a WACC of 8.3% for the plastics division, compared to the WACC of 6.8% for the packaging division that we calculated in Section 18.2.

In fact, we can combine Eq. 18.1 and Eq. 18.10 to obtain a direct formula for the WACC when the firm maintains a target leverage ratio for the project. If d is the project's debt-to-value ratio, $[D/(E + D)]$, then¹³

Project-Based WACC Formula

$$r_{wacc} = r_U - d\tau_c r_D \quad (18.11)$$

For example, in the case of Avco's plastics division:

$$r_{wacc} = 9.5\% - 0.50 \times 0.40 \times 6\% = 8.3\%$$

EXAMPLE 18.6

Computing Divisional Costs of Capital

Problem

Hasco Corporation is a multinational provider of lumber and milling equipment. Currently, Hasco's equity cost of capital is 12.7%, and its borrowing cost is 6%. Hasco has traditionally maintained a 40% debt-to-value ratio. Hasco engineers have developed a GPS-based inventory control tracking system, which the company is considering developing commercially as a separate division. Management views the risk of this investment as similar to that of other technology companies' investments, with comparable firms typically having an unlevered cost of capital of 15%. Suppose Hasco plans to finance the new division using 10% debt financing (a constant debt-to-value ratio of 10%) with a borrowing rate of 6%, and its corporate tax rate is 35%. Estimate the unlevered, equity, and weighted average costs of capital for each division.

Solution

For the lumber and milling division, we can use the firm's current equity cost of capital $r_E = 12.7\%$ and debt-to-value ratio of 40%. Then

$$r_{wacc} = 0.60 \times 12.7\% + 0.40 \times 6\% \times (1 - 0.35) = 9.2\%$$

$$r_U = 0.60 \times 12.7\% + 0.40 \times 6\% = 10.0\%$$

¹³We derive Eq. 18.11 (which is equivalent to Eq. 12.13) by comparing the WACC and pretax WACC in Eq. 18.1 and Eq. 18.6. This formula was proposed by R. Harris and J. Pringle, "Risk Adjusted Discount Rates: Transition from the Average Risk Case," *Journal of Financial Research* 8 (1985): 237–244.

For the technology division, we estimate its unlevered cost of capital using comparable firms: $r_U = 15\%$. Because Hasco's technology division will support 10% debt financing,

$$r_E = 15\% + \frac{0.10}{0.90}(15\% - 6\%) = 16\%$$

$$r_{wacc} = 15\% - 0.10 \times 0.35 \times 6\% = 14.8\%$$

Note that the cost of capital is quite different across the two divisions.

Determining the Incremental Leverage of a Project

To determine the equity or weighted average cost of capital for a project, we need to know the amount of debt to associate with the project. For capital budgeting purposes, the project's financing is the *incremental* financing that results if the firm takes on the project. That is, it is the change in the firm's total debt (net of cash) with the project versus without the project.

The incremental financing of a project need not correspond to the financing that is directly tied to the project. As an example, suppose a project involves buying a new warehouse, and the purchase of the warehouse is financed with a mortgage for 90% of its value. However, if the firm has an overall policy to maintain a 40% debt-to-value ratio, it will reduce debt elsewhere in the firm once the warehouse is purchased in an effort to maintain that ratio. In that case, the appropriate debt-to-value ratio to use when evaluating the warehouse project is 40%, not 90%.

Here are some important concepts to remember when determining the project's incremental financing.

COMMON MISTAKE

Re-Levering the WACC

When computing the WACC using its definition in Eq. 18.1, always remember that the equity and debt costs of capital, r_E and r_D , will change for different choices of the firm's leverage ratio. For example, consider a firm with a debt-to-value ratio of 25%, a debt cost of capital of 6.67%, an equity cost of capital of 12%, and a tax rate of 40%. From Eq. 18.1, its current WACC is

$$\begin{aligned} r_{wacc} &= 0.75(12\%) + 0.25(6.67\%)(1 - 0.40) \\ &= 10\% \end{aligned}$$

Suppose the firm increases its debt-to-value ratio to 50%. It is tempting to conclude that its WACC will fall to

$$0.50(12\%) + 0.50(6.67\%)(1 - 0.40) = 8\%$$

In fact, when the firm increases leverage, the risk of its equity and debt will increase, causing its equity and debt cost of capital to rise. To compute the new WACC correctly, we must first determine the firm's unlevered cost of capital from Eq. 18.6:

$$r_U = 0.75(12\%) + 0.25(6.67\%) = 10.67\%$$

If the firm's debt cost of capital rises to 7.34% with the increase in leverage, then from Eq. 18.10 its equity cost of capital will rise as well:

$$r_E = 10.67\% + \frac{0.50}{0.50}(10.67\% - 7.34\%) = 14\%$$

Using Eq. 18.1, with the new equity and debt cost of capital, we can correctly compute the new WACC:

$$\begin{aligned} r_{wacc} &= 0.50(14\%) + 0.50(7.34\%)(1 - 0.40) \\ &= 9.2\% \end{aligned}$$

We can also calculate the new WACC using Eq. 18.11:

$$r_{wacc} = 10.67\% - 0.50(0.40)(7.34\%) = 9.2\%$$

Note that if we fail to incorporate the effect of an increase in leverage on the firm's equity and debt costs of capital, we will overestimate the reduction in its WACC.

Cash Is Negative Debt. A firm's leverage should be evaluated based on its debt net of any cash. Thus, if an investment will reduce the firm's cash holdings, it is equivalent to the firm adding leverage. Similarly, if the positive free cash flow from a project will increase the firm's cash holdings, then this growth in cash is equivalent to a reduction in the firm's leverage.

A Fixed Equity Payout Policy Implies 100% Debt Financing. Consider a firm whose dividend payouts and expenditures on share repurchases are set in advance and will not be affected by a project's free cash flow. In this case, the only source of financing is *debt*—any cash requirement of the project will be funded using the firm's cash or borrowing, and any cash that the project produces will be used to repay debt or increase the firm's cash. As a result, the incremental effect of the project on the firm's financing is to change the level of debt, so this project is 100% debt financed (that is, its debt-to-value ratio $d = 1$). If the firm's payout policy is fixed for the life of a project, the appropriate WACC for the project is $r_U - \tau_c r_D$. This case can be relevant for a highly levered firm that devotes its free cash flow to paying down its debt or for a firm that is hoarding cash.

Optimal Leverage Depends on Project and Firm Characteristics. Projects with safer cash flows can support more debt before they increase the risk of financial distress for the firm. But, as we discussed in Part 5, the likelihood of financial distress that a firm can bear depends on the magnitude of the distress, agency, and asymmetric information costs that it may face. These costs are not specific to a project, but rather depend on the characteristics of the entire firm. As a consequence, the optimal leverage for a project will depend on the characteristics of both the project and the firm.

Safe Cash Flows Can Be 100% Debt Financed. When an investment has risk-free cash flows, a firm can offset these cash flows 100% with debt and leave its overall risk unchanged. If it does so, the appropriate discount rate for safe cash flows is $r_D(1 - \tau_c)$.

EXAMPLE 18.7

Debt Financing at Chipotle

Problem

In mid-2015, Chipotle Mexican Grill held nearly 880 million in cash and securities and no debt. Consider a project with an unlevered cost of capital of $r_U = 12\%$. Suppose Chipotle's payout policy is completely fixed during the life of this project, so that the free cash flow from the project will affect only Chipotle's cash balance. If Chipotle earns 4% interest on its cash holdings and pays a 35% corporate tax rate, what cost of capital should Chipotle use to evaluate the project?

Solution

Because the inflows and outflows of the project change Chipotle's cash balance, the project is financed by 100% debt; that is, $d = 1$. The appropriate cost of capital for the project is

$$r_{wacc} = r_U - \tau_c r_D = 12\% - 0.35 \times 4\% = 10.6\%$$

Note that the project is effectively 100% debt financed, because even though Chipotle itself had no debt, if the cash had not been used to finance the project, Chipotle would have had to pay taxes on the interest the cash earned.

CONCEPT CHECK

1. How do we estimate a project's unlevered cost of capital when the project's risk is different from that of a firm?
2. What is the incremental debt associated with a project?

18.6 APV with Other Leverage Policies

To this point, we have assumed that the incremental debt of a project is set to maintain a constant debt-equity (or, equivalently, debt-to-value) ratio. While a constant debt-equity ratio is a convenient assumption that simplifies the analysis, not all firms adopt this leverage policy. In this section, we consider two alternative leverage policies: constant interest coverage and predetermined debt levels.

When we relax the assumption of a constant debt-equity ratio, the equity cost of capital and WACC for a project will change over time as the debt-equity ratio changes. As a result, the WACC and FTE method are difficult to implement (see Section 18.8 for further details). The APV method, however, is relatively straightforward to use and is therefore the preferred method with alternative leverage policies.

Constant Interest Coverage Ratio

As discussed in Chapter 15, if a firm is using leverage to shield income from corporate taxes, then it will adjust its debt level so that its interest expenses grow with its earnings. In this case, it is natural to specify the firm's incremental interest payments as a target fraction, k , of the project's free cash flow:¹⁴

$$\text{Interest Paid in Year } t = k \times FCF_t \quad (18.12)$$

When the firm keeps its interest payments to a target fraction of its FCF, we say it has a **constant interest coverage ratio**.

To implement the APV approach, we must compute the present value of the tax shield under this policy. Because the tax shield is proportional to the project's free cash flow, it has the same risk as the project's cash flow and so should be discounted at the same rate—that is, the unlevered cost of capital, r_U . But the present value of the project's free cash flow at rate r_U is the unlevered value of the project. Thus,

$$\begin{aligned} PV(\text{Interest Tax Shield}) &= PV(\tau_c k \times FCF) = \tau_c k \times PV(FCF) \\ &= \tau_c k \times V^U \end{aligned} \quad (18.13)$$

That is, with a constant interest coverage policy, the value of the interest tax shield is proportional to the project's unlevered value. Using the APV method, the value of the project with leverage is given by the following formula:

Levered Value with a Constant Interest Coverage Ratio

$$\begin{aligned} V^L &= V^U + PV(\text{Interest Tax Shield}) = V^U + \tau_c k \times V^U \\ &= (1 + \tau_c k) V^U \end{aligned} \quad (18.14)$$

For example, we calculated the unlevered value of Avco's RFX project as $V^U = \$59.62$ million in Section 18.3. If Avco targets interest to be 20% of its free cash flow, the value with leverage is $V^L = [1 + 0.4 (20\%)] 59.62 = \64.39 million. (This result differs from the value of \$61.25 million for the project that we calculated in Section 18.3, where we assumed a different leverage policy of a 50% debt-to-value ratio.)

¹⁴It might be even better to specify interest as a fraction of taxable earnings. Typically, however, taxable earnings and free cash flows are roughly proportional, so the two specifications are very similar. Also, for Eq. 18.12 to hold exactly, the firm must adjust debt continuously throughout the year. We will relax this assumption in Section 18.8 to a setting in which the firm adjusts debt periodically based on its expected level of future free cash flow (see Example 18.11).

Equation 18.14 provides a simple rule to determine an investment's levered value based on a leverage policy that may be appropriate for many firms.¹⁵ Note also that if the investment's free cash flows are expected to grow at a constant rate, then the assumption of constant interest coverage and a constant debt-equity ratio are equivalent, as in the following example.

EXAMPLE 18.8

Valuing an Acquisition with Target Interest Coverage

Problem

Consider again Avco's acquisition from Examples 18.1 and 18.2. The acquisition will contribute \$3.8 million in free cash flows the first year, growing by 3% per year thereafter. The acquisition cost of \$80 million will be financed with \$50 million in new debt initially. Compute the value of the acquisition using the APV method assuming Avco will maintain a constant interest coverage ratio for the acquisition.

Solution

Given Avco's unlevered cost of capital of $r_U = 8\%$, the acquisition has an unlevered value of

$$V^U = 3.8 / (8\% - 3\%) = \$76 \text{ million}$$

With \$50 million in new debt and a 6% interest rate, the interest expense the first year is $6\% \times 50 = \$3$ million, or $k = \text{Interest} / \text{FCF} = 3 / 3.8 = 78.95\%$. Because Avco will maintain this interest coverage, we can use Eq. 18.14 to compute the levered value:

$$V^L = (1 + \tau_c k) V^U = [1 + 0.4 (78.95\%)] 76 = \$100 \text{ million}$$

This value is identical to the value computed using the WACC method in Example 18.1, where we assumed a constant debt-equity ratio.

Predetermined Debt Levels

Rather than set debt according to a target debt-equity ratio or interest coverage level, a firm may adjust its debt according to a fixed schedule that is known in advance. Suppose, for example, that Avco plans to borrow \$30.62 million and then will reduce the debt on a fixed schedule to \$20 million after one year, to \$10 million after two years, and to zero after three years. The RFX project will have no other consequences for Avco's leverage, regardless of its success. How can we value an investment like this one when its future *debt levels*, rather than the *debt-equity ratio*, are known in advance?

When the debt levels are known in advance, it is straightforward to compute the interest payments and the corresponding tax shield the firm will obtain. The question is, at what rate should we discount this tax shield to determine the present value? In Section 18.3, we used the project's unlevered cost of capital because the amount of debt—and, therefore, the tax shield—fluctuated with the value of the project itself and so had similar risk. However, with a fixed debt schedule, the amount of the debt will not fluctuate. In this case, the tax shield is less risky than the project, so it should be discounted at a lower rate.

¹⁵J. Graham and C. Harvey report that a majority of firms target a credit rating when issuing debt ["The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001)]. The interest coverage ratios are important determinants of credit ratings. Firms and rating agencies also consider the *book* debt-equity ratio, which often fluctuates more closely with a firm's cash flows, rather than with its market value. (For example, book equity increases when the firm invests in physical capital to expand, which generally results in higher cash flows.)

TABLE 18.8
SPREADSHEET**Interest Payments and Interest Tax Shield Given a Fixed Debt Schedule for Avco's RFX Project**

	Year	0	1	2	3	4
Interest Tax Shield (\$ million)						
1 Debt Capacity, D_t (fixed schedule)		30.62	20.00	10.00	—	—
2 Interest Paid (at $r_D = 6\%$)			1.84	1.20	0.60	—
3 Interest Tax Shield (at $\tau_c = 40\%$)			0.73	0.48	0.24	—

Indeed, the risk of the tax shield is similar to the risk of the debt payments. Therefore, we advise the following general rule:¹⁶

When debt levels are set according to a fixed schedule, we can discount the predetermined interest tax shields using the debt cost of capital, r_D .

Let's apply this to our Avco example, given the debt schedule and interest tax shield shown in Table 18.8. Using the debt cost of capital, $r_D = 6\%$:

$$PV(\text{Interest Tax Shield}) = \frac{0.73}{1.06} + \frac{0.48}{1.06^2} + \frac{0.24}{1.06^3} = \$1.32 \text{ million}$$

We then combine the value of the tax shield with the unlevered value of the project (which we already computed in Section 18.3) to determine the APV:

$$V^L = V^U + PV(\text{Interest Tax Shield}) = 59.62 + 1.32 = \$60.94 \text{ million}$$

The value of the interest tax shield computed here, \$1.32 million, differs from the value of \$1.63 million we computed in Section 18.3 based on constant debt-equity ratio. Comparing the firm's debt in the two cases, we see that it is paid off more rapidly in Table 18.8 than in Table 18.4. Also, because the debt-equity ratio for the project changes over time in this example, the project's WACC also changes, making it difficult to apply the WACC method to this case. We show how to do so, and verify that we get the same result, as part of the advanced topics in Section 18.8.

A particularly simple example of a predetermined debt level occurs when the firm has permanent fixed debt, maintaining the same level of debt forever. We discussed this debt policy in Section 15.2 and showed that if the firm maintains a fixed level of debt, D , the value of the tax shield is $\tau_c \times D$.¹⁷ Hence, the value of the levered project in this case is

Levered Value with Permanent Debt

$$V^L = V^U + \tau_c \times D \quad (18.15)$$

A Cautionary Note: When debt levels are predetermined, the firm will not adjust its debt based on fluctuations to its cash flows or value according to a target leverage ratio, and the risk of the interest tax shield differs from the risk of the cash flows. As a result, *the firm's pretax WACC no longer coincides with its unlevered cost of capital, so Eq. 18.6, Eq. 18.10, and Eq. 18.11 do not apply.* (For example, if we compute the WACC using Eq. 18.11 and

¹⁶The risk of the tax shield is not literally equivalent to that of the debt payments, because it is based on only the interest portion of the payments and is subject to the risk of fluctuations in the firm's marginal tax rate. Nevertheless, this assumption is a reasonable approximation absent much more detailed information.

¹⁷Because the interest tax shield is $\tau_c r_D D$ in perpetuity, using the discount rate r_D we get $PV(\text{Interest Tax Shield}) = \tau_c r_D D / r_D = \tau_c D$.

apply it in the case of permanent debt, the value we estimate will *not* be consistent with Eq. 18.15.) For the correct relationship between the firm's WACC, unlevered, and equity cost of capital, we need to use more general versions of these equations, which we provide in Eq. 18.20 and Eq. 18.21 in Section 18.8.

A Comparison of Methods

We have introduced three methods for valuing levered investments: WACC, APV, and FTE. How do we decide which method to use in which circumstances?

When used consistently, each method produces the same valuation for the investment. Thus, the choice of method is largely a matter of convenience. As a general rule, the WACC method is the easiest to use when the firm will maintain a fixed debt-to-value ratio over the life of the investment. For alternative leverage policies, the APV method is usually the most straightforward approach. The FTE method is typically used only in complicated settings for which the values of other securities in the firm's capital structure or the interest tax shield are themselves difficult to determine.

CONCEPT CHECK

1. What condition must the firm meet to have a constant interest coverage policy?
2. What is the appropriate discount rate for tax shields when the debt schedule is fixed in advance?

18.7 Other Effects of Financing

The WACC, APV, and FTE methods determine the value of an investment incorporating the tax shields associated with leverage. However, as we discussed in Chapter 16, some other potential imperfections are associated with leverage. In this section, we investigate ways to adjust our valuation to account for imperfections such as issuance costs, security mispricing, and financial distress and agency costs.

Issuance and Other Financing Costs

When a firm takes out a loan or raises capital by issuing securities, the banks that provide the loan or underwrite the sale of the securities charge fees. Table 18.9 lists the typical fees for common transactions. The fees associated with the financing of the project are a cost

TABLE 18.9

**Typical Issuance Costs for Different Securities,
as a Percentage of Proceeds¹⁸**

Financing Type	Underwriting Fees
Bank loans	< 2%
Corporate bonds	
Investment grade	1–2%
Non-investment grade	2–3%
Equity issues	
Initial public offering	8–9%
Seasoned equity offering	5–6%

¹⁸Fees vary by transaction size; estimates here are based on typical legal, underwriting, and accounting fees for a \$50 million transaction. For example, see I. Lee, S. Lochhead, J. Ritter, and Q. Zhao, "The Cost of Raising Capital," *Journal of Financial Research* 19 (1996): 59–74.

that should be included as part of the project's required investment, reducing the NPV of the project.

For example, suppose a project has a levered value of \$20 million and requires an initial investment of \$15 million. To finance the project, the firm will borrow \$10 million and fund the remaining \$5 million by reducing dividends. If the bank providing the loan charges fees (after any tax deductions) totaling \$200,000, the project NPV is

$$NPV = V^L - (\text{Investment}) - (\text{After Tax Issuance Costs}) = 20 - 15 - 0.2 = \$4.8 \text{ million}$$

This calculation presumes the cash flows generated by the project will be paid out. If instead they will be reinvested in a new project, and thereby save *future* issuance costs, the present value of these savings should also be incorporated and will offset the current issuance costs.

Security Mispricing

With perfect capital markets, all securities are fairly priced and issuing securities is a zero-NPV transaction. However, as discussed in Chapter 16, sometimes management may believe that the securities they are issuing are priced at less than (or more than) their true value. If so, the NPV of the transaction, which is the difference between the actual money raised and the true value of the securities sold, should be included when evaluating the decision. For example, if the financing of the project involves an equity issue, and if management believes that the equity will sell at a price that is less than its true value, this mispricing is a cost of the project for the *existing* shareholders.¹⁹ It can be deducted from the project NPV in addition to other issuance costs.

When a firm borrows funds, a mispricing scenario arises if the interest rate charged differs from the rate that is appropriate given the actual risk of the loan. For example, a firm may pay an interest rate that is too high if news that would improve its credit rating has not yet become public. With the WACC method, the cost of the higher interest rate will result in a higher weighted average cost of capital and a lower value for the investment. With the APV method, we must add to the value of the project the NPV of the loan cash flows when evaluated at the “correct” rate that corresponds to their actual risk.²⁰

EXAMPLE 18.9

Valuing a Loan

Problem

Gap, Inc., is considering borrowing \$100 million to fund an expansion of its stores. Given investors' uncertainty regarding its prospects, Gap will pay a 6% interest rate on this loan. The firm's management knows, however, that the actual risk of the loan is extremely low and that the appropriate rate on the loan is 5%. Suppose the loan is for five years, with all principal being repaid in the fifth year. If Gap's marginal corporate tax rate is 40%, what is the net effect of the loan on the value of the expansion?

Solution

The following table shows the cash flows (in \$ million) and interest tax shields of a fair loan, which has a 5% interest rate, and of the above-market rate loan Gap will receive, which has a 6%

¹⁹New shareholders, of course, benefit from receiving the shares at a low price.

²⁰We must also use the correct rate for r_D when leveraging or unlevering the cost of capital.

interest rate. For each loan, we compute both the NPV of the loan cash flows and the present value of the interest tax shields, using the correct rate $r_D = 5\%$.

	Year	0	1	2	3	4	5
1	Fair Loan	100.00	(5.00)	(5.00)	(5.00)	(5.00)	(105.00)
2	Interest Tax Shield		2.00	2.00	2.00	2.00	2.00
3	At $r_D = 5\%$:						
4	NPV(Loan Cash Flows)	0.00					
5	PV(Interest Tax Shield)	8.66					
6	Actual Loan	100.00	(6.00)	(6.00)	(6.00)	(6.00)	(106.00)
7	Interest Tax Shield		2.40	2.40	2.40	2.40	2.40
8	At $r_D = 5\%$:						
9	NPV(Loan Cash Flows)	(4.33)					
10	PV(Interest Tax Shield)	10.39					

For the fair loan, note that the NPV of the loan cash flows is zero. Thus, the benefit of the loan on the project's value is the present value of the interest tax shield of \$8.66 million. For the actual loan, the higher interest rate increases the value of the interest tax shield but implies a negative NPV for the loan cash flows. The combined effect of the loan on the project's value is

$$NPV(\text{Loan Cash Flows}) + PV(\text{Interest Tax Shield}) = -4.33 + 10.39 = \$6.06 \text{ million}$$

While leverage is still valuable due to the tax shields, paying the higher interest rate reduces its benefit to the firm by $8.66 - 6.06 = \$2.60$ million.

Financial Distress and Agency Costs

As discussed in Chapter 16, one consequence of debt financing is the possibility of financial distress and agency costs. Because these costs affect the future free cash flows that will be generated by the project, they can be incorporated directly into the estimates of the project's expected free cash flows. When the debt level—and, therefore, the probability of financial distress—is high, the expected free cash flow will be reduced by the expected costs associated with financial distress and agency problems. (Conversely, as we also noted in Chapter 16, in some situations the threat of default can also prompt management to improve efficiency and thereby increase the firm's free cash flow.)

Financial distress and agency costs also have consequences for the cost of capital. For example, financial distress is more likely to occur when economic times are bad. As a result, the costs of distress cause the value of the firm to fall further in a market downturn. Financial distress costs therefore tend to increase the sensitivity of the firm's value to market risk, further raising the cost of capital for highly levered firms.²¹

How do we incorporate financial distress costs into the valuation methods described in this chapter? One approach is to adjust our free cash flow estimates to account for the costs, and increased risk, resulting from financial distress. An alternative method is to first value the project ignoring these costs, and then add the present value of the incremental cash flows associated with financial distress and agency problems separately. Because these costs tend to occur only when a firm is in (or near) default, valuing them is complicated and best done using the option valuation techniques introduced in Part 7. In some special

²¹In other words, distress costs tend to have a negative beta (they are higher in bad times). Because they are a *cost*, including them in the firm's free cash flows will raise the beta of the firm.

GLOBAL FINANCIAL CRISIS**Government Loan Guarantees**

In times of crisis, firms may appeal to the federal government for financial assistance. Often, such aid comes in the form of subsidized loans or loan guarantees. For example, in the wake of the September 11, 2001 tragedy, the U.S. government made available \$10 billion in loan guarantees to enable air carriers to obtain credit. U.S. Airways received the largest loan guarantee of \$900 million, and America West Airlines received the second largest, for \$429 million. Ultimately, these loans were repaid without taxpayer expense.

Loan guarantees were also an important part of the government's response to the 2008 financial crisis. The U.S. government has insured over \$1 trillion in debt issued by financial institutions or assets held by the banks. The government also made over \$500 billion in direct loans to

distressed firms. Moreover, firms and banks viewed as “too big to fail” were thought to have implicit guarantees even if they did not have explicit ones.

These guarantees enabled firms to obtain loans at a lower interest rate than they otherwise would have received without government assistance. If these loans were fairly priced at market rates, then the loans obtained with the help of the federal guarantee had a positive NPV for the borrowers, and were equivalent to a direct cash subsidy. If, on the other hand, market rates for these loans were too high—due perhaps to asymmetric information or a lack of available lenders—then these loans and guarantees may have improved terms for borrowers at a lower cost to taxpayers than a direct cash bailout.

cases, however, we can use the values of the firm's existing securities to estimate the value of distress costs, as in the following example.

EXAMPLE 18.10**Valuing Distress Costs****Problem**

Your firm has zero coupon debt with a face value of \$100 million due in 5 years time, and no other debt outstanding. The current risk-free rate is 5%, but due to default risk the yield to maturity of the debt is 12%. You believe that in the event of default, 10% of the losses are attributable to bankruptcy and distress costs. (For example, if the debt holders lose \$60 million and recover \$40 million, \$6 million of the loss in value would not have occurred if the firm had been unlevered and thus avoided bankruptcy.) Estimate the present value of the distress costs.

Solution

With a 12% yield, the current market value of the firm's debt is $\$100/1.12^5 = \56.74 million. If the firm's debt were risk-free, its market value would be $\$100/1.05^5 = \78.35 million. The difference in these values, $\$78.35 - \$56.74 = \$21.61$ million, is the present value of the debt holders' expected losses in default. If 10% of these losses is due to bankruptcy and distress costs, then the present value of these costs is $\$21.61 \times 0.10 = \2.16 million.

CONCEPT CHECK

1. How do we deal with issuance costs and security mispricing costs in our assessment of a project's value?
2. How would financial distress and agency costs affect a firm's use of leverage?

18.8 Advanced Topics in Capital Budgeting

In the previous sections, we have highlighted the most important methods for capital budgeting with leverage and demonstrated their application in common settings. In this section, we consider several more complicated scenarios and show how our tools can be extended to these cases. First, we consider leverage policies in which firms keep debt fixed in the short run, but adjust to a target leverage ratio in the long run. Second, we look at the relationship

between a firm's equity and unlevered cost of capital for alternative leverage policies. Third, we implement the WACC and FTE methods when the firm's debt-equity ratio changes over time. We then conclude the section by incorporating the effects of personal taxes.

Periodically Adjusted Debt

To this point, we have considered leverage policies in which debt is either adjusted continuously to a target leverage ratio²² or set according to a fixed plan that will never change. More realistically, most firms allow the debt-equity ratio of the firm to stray from the target and periodically adjust leverage to bring it back into line with the target.

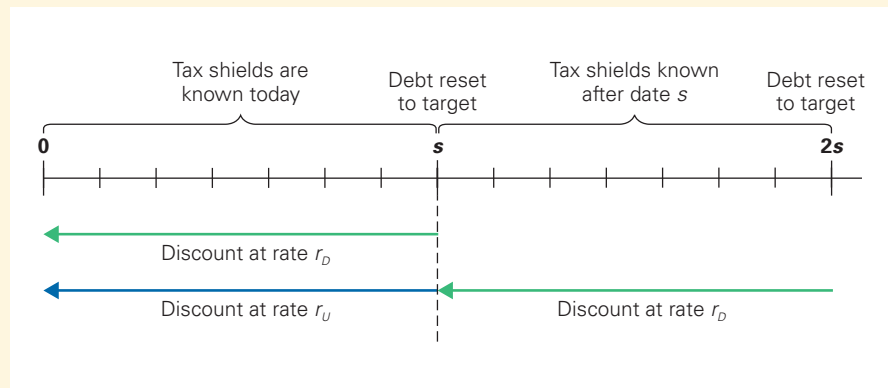
Suppose the firm adjusts its leverage every s periods, as shown in Figure 18.1. Then the firm's interest tax shields up to date s are predetermined, so they should be discounted at rate r_D . In contrast, interest tax shields that occur after date s depend on future adjustments the firm will make to its debt, so they are risky. If the firm will adjust the debt according to a target debt-equity ratio or interest coverage level, then the future interest tax shields should be discounted at rate r_D for the periods that they are known, but at rate r_U for all earlier periods when they are still risky.

An important special case is when the debt is adjusted annually. In that case, the expected interest expense on date t , Int_t , is known as of date $t - 1$. Therefore, we discount the interest tax shield at rate r_D for one period, from date t to $t - 1$ (because it will be known at that time), and then discount it from date $t - 1$ to 0 at rate r_U :

$$PV(\tau_c \times Int_t) = \frac{\tau_c \times Int_t}{(1 + r_U)^{t-1}(1 + r_D)} = \frac{\tau_c \times Int_t}{(1 + r_U)^t} \times \left(\frac{1 + r_U}{1 + r_D} \right) \quad (18.16)$$

FIGURE 18.1

Discounting the Tax Shield with Periodic Adjustments



If the debt is reset to a target leverage ratio every s periods, then interest tax shields within the first s periods are known and should be discounted at rate r_D . Interest tax shields that occur after date s are not yet known, so they should be discounted at rate r_D for the periods when they will be known and at rate r_U for earlier periods.

²²While we have simplified our exposition earlier in the chapter by calculating debt and interest payments on an annual basis, the formulas we have used in the case of a target leverage ratio or interest coverage ratio are based on the assumption that the firm maintains the target leverage ratio or interest coverage during the year.

Equation 18.16 implies that we can value the tax shield by discounting it at rate r_U as before, and then multiply the result by the factor $(1 + r_U)/(1 + r_D)$ to account for the fact that the tax shield is known one year in advance.

This same adjustment can be applied to other valuation methods as well. For example, when the debt is adjusted annually rather than continuously to a target debt-to-value ratio d , the project-based WACC formula of Eq. 18.11 becomes²³

$$r_{wacc} = r_U - d\tau_c r_D \frac{1 + r_U}{1 + r_D} \quad (18.17)$$

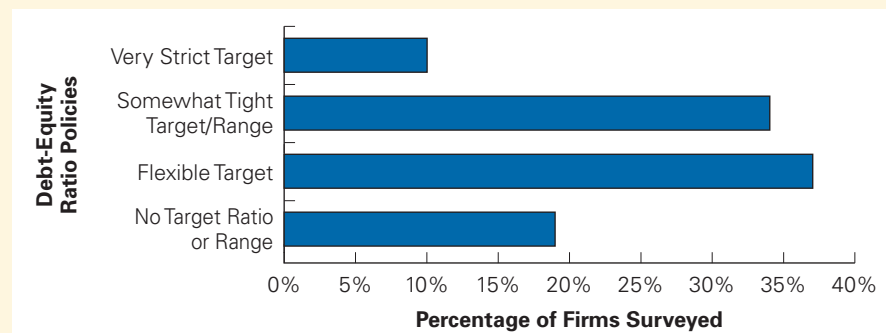
Similarly, when the firm sets debt annually based on its expected future free cash flow, the constant interest coverage model in Eq. 18.14 becomes

$$V^L = \left(1 + \tau_c k \frac{1 + r_U}{1 + r_D}\right) V^U \quad (18.18)$$

How do firms actually adjust their leverage? As Figure 18.2 shows, approximately 50% of surveyed firms attempt to keep their leverage ratio in tight range. When firms adjust their debt levels only periodically, the risk of the tax shield declines and its value increases as shown in Eqs. 18.17 and 18.18. Example 18.11 illustrates these methods in a constant growth setting.

FIGURE 18.2

Firms' Leverage Policies



Of 392 CFOs surveyed by Professors J. Graham and C. Harvey, 81% reported having a target debt-equity ratio. However, only 10% of respondents viewed the target as set in stone. Most were willing to let the debt-equity ratio of the firm stray from the target and periodically adjust leverage to bring it back into line.

Source: J. R. Graham and C. Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001): 187–243.

²³An equivalent WACC formula was proposed by J. Miles and J. Ezzell, "The Weighted Average Cost of Capital, Perfect Capital Markets and Project Life: A Clarification," *Journal of Financial and Quantitative Analysis* 15 (1980): 719–730.

EXAMPLE 18.11**Annual Debt Ratio Targeting****Problem**

Celmax Corporation expects free cash flows this year of \$7.36 million and a future growth rate of 4% per year. The firm currently has \$30 million in debt outstanding. This leverage will remain fixed during the year, but at the end of each year Celmax will increase or decrease its debt to maintain a constant debt-equity ratio. Celmax pays 5% interest on its debt, pays a corporate tax rate of 40%, and has an unlevered cost of capital of 12%. Estimate Celmax's value with this leverage policy.

Solution

Using the APV approach, the unlevered value is $V^U = 7.36/(12\% - 4\%) = \92.0 million. In the first year, Celmax will have an interest tax shield of $\tau_c r_D D = 0.40 \times 5\% \times \$30 \text{ million} = \$0.6$ million. Because Celmax will adjust its debt after one year, the tax shields are expected to grow by 4% per year with the firm. The present value of the interest tax shield is therefore

$$PV(\text{Interest Tax Shield}) = \underbrace{\frac{0.6}{(12\% - 4\%)}}_{PV \text{ at rate } r_U} \times \underbrace{\left(\frac{1.12}{1.05}\right)}_{\text{Debt is set 1 year in advance}} = \$8.0 \text{ million}$$

Therefore, $V^L = V^U + PV(\text{Interest Tax Shield}) = 92.0 + 8.0 = \100.0 million.

We can also apply the WACC method. From Eq. 18.17, Celmax's WACC is

$$\begin{aligned} r_{wacc} &= r_U - d \tau_c r_D \frac{1 + r_U}{1 + r_D} = 12\% - \frac{30}{100} (0.40)(5\%) \frac{1.12}{1.05} \\ &= 11.36\% \end{aligned}$$

Therefore, $V^L = 7.36/(11.36\% - 4\%) = \100 million.

Finally, the constant interest coverage model can be applied (in this setting with constant growth, a constant debt-equity ratio implies a constant interest coverage ratio). Given interest of $5\% \times \$30 \text{ million} = \1.50 million this year, from Eq. 18.18,

$$\begin{aligned} V^L &= \left(1 + \tau_c k \frac{1 + r_U}{1 + r_D}\right) V^U \\ &= \left(1 + 0.40 \times \frac{1.50}{7.36} \times \frac{1.12}{1.05}\right) 92.0 = \$100 \text{ million} \end{aligned}$$

Leverage and the Cost of Capital

The relationship between leverage and the project's costs of capital in Eq. 18.6, Eq. 18.10, and Eq. 18.11 relies on the assumption that the firm maintains a target leverage ratio. That relationship holds because in that case the interest tax shields have the same risk as the firm's cash flows. But when debt is set according to a fixed schedule for some period of time, the interest tax shields for the scheduled debt are known, relatively safe cash flows. These safe cash flows will reduce the effect of leverage on the risk of the firm's equity. To account for this effect, we should deduct the value of these "safe" tax shields from the debt—in the same way that we deduct cash—when evaluating a firm's leverage. That is, if T^s is the present value of the interest tax shields from predetermined debt, the risk of a firm's equity will depend on its *debt net of the predetermined tax shields*:

$$D^s = D - T^s \quad (18.19)$$

We show in this chapter's appendix that Eq. 18.6 and Eq. 18.10 continue to apply with D replaced by D^s , so that the more general relationship between the unlevered and equity costs of capital are related as follows:

Leverage and the Cost of Capital with a Fixed Debt Schedule

$$r_U = \frac{E}{E + D^s} r_E + \frac{D^s}{E + D^s} r_D \quad \text{or, equivalently, } r_E = r_U + \frac{D^s}{E} (r_U - r_D) \quad (18.20)$$

We can also combine Eq. 18.20 with the definition of the WACC in Eq. 18.1 and generalize the project-based WACC formula in Eq. 18.11:

Project WACC with a Fixed Debt Schedule

$$r_{wacc} = r_U - d \tau_c [r_D + \phi(r_U - r_D)] \quad (18.21)$$

where $d = D/(D + E)$ is the debt-to-value ratio, and $\phi = T^s/(\tau_c D)$ is a measure of the permanence of the debt level, D . Here are three cases commonly used in practice, which differ according to the frequency with which the debt is assumed to adjust to the growth of the investment:²⁴

1. Continuously adjusted debt: $T^s = 0$, $D^s = D$, and $\phi = 0$
2. Annually adjusted debt: $T^s = \frac{\tau_c r_D D}{1 + r_D}$, $D^s = D \left(1 - \tau_c \frac{r_D}{1 + r_D}\right)$, and $\phi = \frac{r_D}{1 + r_D}$
3. Permanent debt: $T^s = \tau_c D$, $D^s = D(1 - \tau_c)$, and $\phi = 1$

Finally, note that unless d and ϕ remain constant over time, the WACC and equity cost of capital must be computed period by period.

EXAMPLE 18.12

APV and WACC with Permanent Debt

Problem

International Paper Company is considering the acquisition of additional forestland in the southeastern United States. The wood harvested from the land will generate free cash flows of \$4.5 million per year, with an unlevered cost of capital of 7%. As a result of this acquisition, International Paper will permanently increase its debt by \$30 million. If International Paper's tax rate is 35%, what is the value of this acquisition using the APV method? Verify this result using the WACC method.

Solution

Using the APV method, the unlevered value of the land is $V^U = FCF/r_U = 4.5/0.07 = \64.29 million. Because the debt is permanent, the value of the tax shield is $\tau_c D = 0.35(30) = 10.50$. Therefore, $V^L = 64.29 + 10.50 = \$74.79$ million.

To use the WACC method, we apply Eq. 18.21 with $\phi = T^s/(\tau_c D) = 1$ and $d = 30/74.79 = 40.1\%$. Therefore, the WACC for the investment is

$$r_{wacc} = r_U - d \tau_c r_U = 7\% - 0.401 \times 0.35 \times 7\% = 6.017\%$$

and $V^L = 4.5/0.06017 = \$74.79$ million.

²⁴Case 1 reduces to the Harris-Pringle formula (see footnote 14), case 2 is the Miles-Ezzell formula (see footnote 23), and case 3 is equivalent to the Modigliani-Miller-Hamada formula with permanent debt. See F. Modigliani and M. Miller, "Corporate Income Taxes and the Cost of Capital: A Correction," *American Economic Review* 53 (1963): 433–443; and R. Hamada, "The Effect of a Firm's Capital Structure on the Systematic Risks of Common Stocks," *Journal of Finance* 27 (1972): 435–452.

The WACC or FTE Method with Changing Leverage

When a firm does not maintain a constant debt-equity ratio for a project, the APV method is generally the most straightforward method to apply. The WACC and FTE methods become more difficult to use because when the proportion of debt financing changes, the project's equity cost of capital and WACC will not remain constant over time. With a bit of care, however, these methods can still be used (and, of course, will lead to the same result as the APV method).

As an example, let's see how we would apply the WACC or FTE methods to the RFX project when Avco has the fixed debt schedule we analyzed earlier using the APV method. The spreadsheet in Table 18.10 computes the equity cost of capital and WACC for the RFX project each year given the fixed debt schedule shown in line 3. The project's value is computed using the APV method in line 7 as the total of the unlevered and tax shield values. With the project's equity value and net debt D^s in hand, we can use Eq. 18.20 to calculate the project's equity cost of capital each year (line 11). Note that the equity cost of capital declines over time as the project's leverage ratio D^s/E declines. By year 3, the debt is fully repaid and the equity cost of capital equals the unlevered cost of capital of 8%.

Given the project's equity cost of capital, we compute its WACC using Eq. 18.1 in line 12. For example, at the beginning of the project,

$$\begin{aligned} r_{wacc} &= \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_c) \\ &= \frac{30.32}{60.94} 9.93\% + \frac{30.62}{60.94} 6\% (1 - 0.40) = 6.75\% \end{aligned}$$

Note that as the leverage of the project falls, its WACC rises, until it eventually equals the unlevered cost of capital of 8% when the project debt is fully repaid at year 3.

Once we have computed the WACC or the equity cost of capital, we can value the project using the WACC or FTE method. Because the cost of capital changes over time,

TABLE 18.10
SPREADSHEET

Adjusted Present Value and Cost of Capital for Avco's RFX Project with a Fixed Debt Schedule

	Year	0	1	2	3	4
Unlevered Value (\$ million)						
1 Free Cash Flow		(28.00)	18.00	18.00	18.00	18.00
2 Unlevered Value, V^U (at $r_U = 8.0\%$)		59.62	46.39	32.10	16.67	—
Interest Tax Shield						
3 Debt Schedule, D_t		30.62	20.00	10.00	—	—
4 Interest Paid (at $r_d = 6\%$)		—	1.84	1.20	0.60	—
5 Interest Tax Shield (at $\tau_c = 40\%$)		—	0.73	0.48	0.24	—
6 Tax Shield Value, T^s (at $r_D = 6.0\%$)		1.32	0.67	0.23	—	—
Adjusted Present Value						
7 Levered Value, $V^L = V^U + T^s$		60.94	47.05	32.33	16.67	—
Effective Leverage and Cost of Capital						
8 Equity, $E = V^L - D$		30.32	27.05	22.33	16.67	—
9 Effective Debt, $D^s = D - T^s$		29.30	19.33	9.77	—	—
10 Effective Debt-Equity Ratio, D^s/E		0.966	0.715	0.438	0.000	—
11 Equity Cost of Capital, r_E		9.93%	9.43%	8.88%	8.00%	—
12 WACC, r_{wacc}		6.75%	6.95%	7.24%	8.00%	—

TABLE 18.11
SPREADSHEET**WACC Method for Avco's RFX Project with a Fixed Debt Schedule**

	Year	0	1	2	3	4
WACC Method (\$ million)						
1 Free Cash Flow		(28.00)	18.00	18.00	18.00	18.00
2 WACC, r_{wacc}		6.75%	6.95%	7.24%	8.00%	
3 Levered Value V^L (at r_{wacc})		60.94	47.05	32.33	16.67	—

we must use a different discount rate each year when applying these methods. For example, using the WACC method, the levered value each year is computed as

$$V_t^L = \frac{FCF_{t+1} + V_{t+1}^L}{1 + r_{wacc}(t)} \quad (18.22)$$

where $r_{wacc}(t)$ is the project's WACC in year t . This calculation is shown in Table 18.11. Note that the levered value matches the result from the APV method (line 7 in Table 18.10). The same approach can be used when applying the FTE method.²⁵

Personal Taxes

As we discussed in Chapter 15, leverage has tax consequences for both investors and corporations. For individuals, interest income from debt is generally taxed more heavily than income from equity (capital gains and dividends). So how do personal taxes affect our valuation methods?

If investors are taxed on the income they receive from holding equity or debt, it will raise the return they require to hold those securities. That is, the equity and debt cost of capital in the market *already* reflects the effects of investor taxes. As a result, *the WACC method does not change in the presence of investor taxes*; we can continue to compute the WACC according to Eq. 18.1 and compute the levered value as in Section 18.2.

The APV approach, however, requires modification in the presence of investor taxes because it requires that we compute the unlevered cost of capital. This computation *is* affected by the presence of investor taxes. Let τ_e be the tax rate investors pay on equity income (dividends) and τ_i be the tax rate investors pay on interest income. Then, given an expected return on debt r_D , define r_D^* as the expected return on equity income that would give investors the same after-tax return: $r_D^*(1 - \tau_e) = r_D(1 - \tau_i)$. So

$$r_D^* \equiv r_D \frac{(1 - \tau_i)}{(1 - \tau_e)} \quad (18.23)$$

Because the unlevered cost of capital is for a hypothetical firm that is all equity, investors' tax rates on income for such a firm are the equity rates, so we must use the rate r_D^* when computing the unlevered cost of capital. Therefore, Eq. 18.20 becomes

Unlevered Cost of Capital with Personal Taxes

$$r_U = \frac{E}{E + D^s} r_E + \frac{D^s}{E + D^s} r_D^* \quad (18.24)$$

²⁵You will notice, however, that we used the APV to compute the debt-equity ratio each period, which we needed to calculate r_E and r_{wacc} . If we had not already solved for the APV, we would need to determine the project's value and WACC simultaneously, using the approach described in this chapter's appendix.

Next, we must compute the interest tax shield using the effective tax advantage of debt, τ^* , in place of τ_e . The effective tax rate τ^* incorporates the investors' tax rate on equity income, τ_e , and on interest income, τ_i , and was defined in Chapter 15 as follows:

$$\tau^* = 1 - \frac{(1 - \tau_e)(1 - \tau_i)}{(1 - \tau_i)} \quad (18.25)$$

Then, we calculate the interest tax shield using tax rate τ^* and interest rate r_D^* :

$$\text{Interest Tax Shield in Year } t = \tau^* \times r_D^* \times D_{t-1} \quad (18.26)$$

Finally we discount the interest tax shields at rate r_U if the firm maintains a target leverage ratio or at rate r_D^* if the debt is set according to a predetermined schedule.²⁶

EXAMPLE 18.13

Using the APV Method with Personal Taxes

Problem

Apex Corporation has an equity cost of capital of 14.4% and a debt cost of capital of 6%, and the firm maintains a debt-equity ratio of 1. Apex is considering an expansion that will contribute \$4 million in free cash flows the first year, growing by 4% per year thereafter. The expansion will cost \$60 million and will be financed with \$40 million in new debt initially with a constant debt-equity ratio maintained thereafter. Apex's corporate tax rate is 40%; the tax rate on interest income is 40%; and the tax rate on equity income is 20%. Compute the value of the expansion using the APV method.

Solution

First, we compute the value without leverage. From Eq. 18.23, the debt cost of capital of 6% is equivalent to an equity rate of

$$r_D^* = r_D \frac{1 - \tau_i}{1 - \tau_e} = 6\% \times \frac{1 - 0.40}{1 - 0.20} = 4.5\%$$

Because Apex maintains a constant debt-equity ratio, $D^s = D$ and Apex's unlevered cost of capital is, using Eq. 18.24,

$$r_U = \frac{E}{E + D^s} r_E + \frac{D^s}{E + D^s} r_D^* = 0.50 \times 14.4\% + 0.50 \times 4.5\% = 9.45\%$$

Therefore, $V^U = 4/(9.45\% - 4\%) = \73.39 million.

From Eq. 18.25, the effective tax advantage of debt is

$$\tau^* = 1 - \frac{(1 - \tau_e)(1 - \tau_i)}{(1 - \tau_i)} = 1 - \frac{(1 - 0.40)(1 - 0.20)}{(1 - 0.40)} = 20\%$$

Apex will add new debt of \$40 million initially, so from Eq. 18.26 the interest tax shield is $20\% \times 4.5\% \times 40 = \0.36 million the first year (note that we use r_D^* here). With a growth rate of 4%, the present value of the interest tax shield is

$$PV(\text{Interest Tax Shield}) = 0.36/(9.45\% - 4\%) = \$6.61 \text{ million}$$

²⁶If the debt is permanent, for example, the value of the tax shield is $\tau^* r_D^* D / r_D^* = \tau^* D$, as shown in Chapter 15.

Therefore, the value of the expansion with leverage is given by the APV:

$$V^L = V^U + PV(\text{Interest Tax Shield}) = 73.39 + 6.61 = \$80 \text{ million}$$

Given the cost of \$60 million, the expansion has an NPV of \$20 million.

Let's check this result using the WACC method. Note that the expansion has the same debt-to-value ratio of $40/80 = 50\%$ as the firm overall. Thus, its WACC is equal to the firm's WACC:

$$\begin{aligned} r_{wacc} &= \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - \tau_c) \\ &= 0.50 \times 14.4\% + 0.50 \times 6\% \times (1 - 0.40) = 9\% \end{aligned}$$

Therefore, $V^L = 4/(9\% - 4\%) = \$80 \text{ million}$, as before.

As Example 18.13 illustrates, the WACC method is much simpler to apply than the APV method in the case with investor taxes. More significantly, the WACC approach does not require knowledge of investors' tax rates. This fact is important because in practice, estimating the marginal tax rate of the investor can be very difficult.

However, if the investment's leverage or risk does not match the firm's, then investor tax rates are required even with the WACC method, as we must unlever and/or re-lever the firm's cost of capital using Eq. 18.24. When the investor's tax rate on interest income exceeds that on equity income, an increase in leverage will lead to a smaller reduction in the WACC (see Problem 28).

CONCEPT CHECK

1. When a firm has pre-determined tax shields, how do we measure its net debt when calculating its unlevered cost of capital?
2. If the firm's debt-equity ratio changes over time, can the WACC method still be applied?

MyFinanceLab

Here is what you should know after reading this chapter: MyFinanceLab will help you identify what you know and where to go when you need to practice.

18.1 Overview of Key Concepts

- The three main methods of capital budgeting are weighted average cost of capital (WACC), adjusted present value (APV), and flow-to-equity (FTE).

18.2 The Weighted Average Cost of Capital

- The key steps in the WACC valuation method are as follows:
 - Determine the unlevered free cash flows of the investment.
 - Compute the weighted average cost of capital:

$$r_{wacc} = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - \tau_c) \quad (18.1)$$

- Compute the value with leverage, V^L , by discounting the free cash flows of the investment using the WACC.

18.3 The Adjusted Present Value Method

- To determine the value of a levered investment using the APV method, proceed as follows:
 - Determine the investment's value without leverage, V^U , by discounting its free cash flows at the unlevered cost of capital, r_U .

- Determine the present value of the interest tax shield.
 - a. Given debt D_t on date t , the tax shield on date $t + 1$ is $\tau_c r_D D_t$.
 - b. If the debt level varies with the investment's value or free cash flow, use discount rate r_U . (If the debt is predetermined, discount the tax shield at rate r_D . See Section 18.6.)
- Add the unlevered value V^U to the present value of the interest tax shield to determine the value of the investment with leverage, V^L .

18.4 The Flow-to-Equity Method

- The key steps in the flow-to-equity method for valuing a levered investment are as follows:
 - Determine the free cash flow to equity of the investment:

$$FCFE = FCF - (1 - \tau_c) \times (\text{Interest Payments}) + (\text{Net Borrowing}) \quad (18.9)$$

- Compute the contribution to equity value, E , by discounting the free cash flow to equity using the equity cost of capital.

18.5 Project-Based Costs of Capital

- If a project's risk is different from that of the firm as a whole, we must estimate its cost of capital separately from the firm's cost of capital. We estimate the project's unlevered cost of capital by looking at the unlevered cost of capital for other firms with similar market risk as the project.
- With a target leverage ratio, the unlevered, equity, and weighted average costs of capital are related as follows:

$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D = \text{Pretax WACC} \quad (18.6)$$

$$r_E = r_U + \frac{D}{E} (r_U - r_D) \quad (18.10)$$

$$r_{wacc} = r_U - d \tau_c r_D, \quad (18.11)$$

where $d = D/(D + E)$ is the project's debt-to-value ratio.

- When assessing the leverage associated with a project, we must consider its incremental impact on the debt, net of cash balances, of the firm overall and not just the specific financing used for that investment.

18.6 APV with Other Leverage Policies

- A firm has a constant interest coverage policy if it sets debt to maintain its interest expenses as a fraction, k , of free cash flow. The levered value of a project with such a leverage policy is $V^L = (1 + \tau_c k) V^U$.
- When debt levels are set according to a fixed schedule:
 - We can discount the predetermined interest tax shields using the debt cost of capital, r_D .
 - The unlevered cost of capital can no longer be computed as the pretax WACC (see Section 18.8).
- If a firm chooses to keep the level of debt at a constant level, D , permanently, then the levered value of a project with such a leverage policy is $V^L = V^U + \tau_c \times D$.
- In general, the WACC method is the easiest to use when a firm has a target debt-equity ratio that it plans to maintain over the life of the investment. For other leverage policies, the APV method is usually the most straightforward method.

18.7 Other Effects of Financing

- Issuance costs and any costs or gains from mispricing of issued securities should be included in the assessment of a project's value.
- Financial distress costs are likely to (1) lower the expected free cash flow of a project and (2) raise the firm's cost of capital. Taking these effects into account, together with other agency and asymmetric information costs, may limit a firm's use of leverage.

18.8 Advanced Topics in Capital Budgeting

- If a firm adjusts its debt annually to a target leverage ratio, the value of the interest tax shield is enhanced by the factor $(1 + r_U)/(1 + r_D)$.
- If the firm does not adjust leverage continuously, so that some of the tax shields are predetermined, the unlevered, equity, and weighted average costs of capital are related as follows:

$$r_U = \frac{E}{E + D^s} r_E + \frac{D^s}{E + D^s} r_D \quad \text{or, equivalently,} \quad r_E = r_U + \frac{D^s}{E} (r_U - r_D) \quad (18.20)$$

$$r_{wacc} = r_U - d \tau_c [r_D + \phi(r_U - r_D)] \quad (18.21)$$

where $d = D/(D + E)$ is the debt-to-value ratio of the project, $D^s = D - T^s$, and T^s is the value of predetermined interest tax shields, and $\phi = T^s/(\tau_c D)$ reflects the permanence of the debt level.

- The WACC method does not need to be modified to account for investor taxes. For the APV method, we use the interest rate

$$r_D^* \equiv r_D \frac{(1 - \tau_i)}{(1 - \tau_e)} \quad (18.23)$$

in place of r_D and we replace τ_c with the effective tax rate:

$$\tau^* = 1 - \frac{(1 - \tau_c)(1 - \tau_e)}{(1 - \tau_i)} \quad (18.25)$$

- If the investment's leverage or risk does not match the firm's, then investor tax rates are required even with the WACC method, as we must unlever and/or re-lever the firm's cost of capital using

$$r_U = \frac{E}{E + D^s} r_E + \frac{D^s}{E + D^s} r_D^* \quad (18.24)$$

Key Terms

adjusted present value (APV) *p. 680*
 book enterprise value *p. 722*
 constant interest coverage ratio *p. 692*
 debt capacity *p. 679*
 economic value added *p. 722*

flow-to-equity (FTE) *p. 684*
 free cash flow to equity (FCFE) *p. 684*
 residual income method *p. 721*
 target leverage ratio *p. 680*

Further Reading

For a further treatment of the valuation with leverage, see: T. Copeland, T. Koller, and J. Murrin, *Valuation: Measuring and Managing the Value of Companies* (McGraw-Hill, 2000); and S. Pratt, R. Reilly, and R. Schweih, *Valuing a Business: The Analysis and Appraisal of Closely Held Companies* (McGraw-Hill, 2000).

For a more detailed treatment of the issues discussed in this chapter, see: E. Arzac and L. Glosten, "A Reconsideration of Tax Shield Valuation," *European Financial Management* 11 (2005): 453–461; R. Harris and J. Pringle, "Risk-Adjusted Discount Rates—Extensions from the Average-Risk Case," *Journal of Financial Research* 8 (1985): 237–244; I. Inselbag and H. Kaufold, "Two DCF Approaches in Valuing Companies Under Alternative Financing Strategies (and How to Choose Between Them)," *Journal of Applied Corporate Finance* 10 (1997): 114–122; T. Luehrman, "Using APV: A Better Tool for Valuing Operations," *Harvard Business Review* 75 (1997): 145–154; J. Miles and J. Ezzell, "The Weighted Average Cost of Capital, Perfect Capital Markets, and Project Life: A Clarification," *Journal of Financial and Quantitative Analysis* 15 (1980): 719–730; J. Miles and J. Ezzell, "Reformulation Tax Shield Valuation: A Note," *Journal of Finance* 40 (1985): 1485–1492; R. Ruback, "Capital Cash Flows: A Simple Approach to Valuing Risky Cash Flows," *Financial Management* 31 (2002): 85–104; and R. Taggart, "Consistent Valuation and Cost of Capital Expressions with Corporate and Personal Taxes," *Financial Management* 20 (1991): 8–20.

Problems

All problems are available in [MyFinanceLab](#). An asterisk (*) indicates problems with a higher level of difficulty.

Overview of Key Concepts

- Explain whether each of the following projects is likely to have risk similar to the average risk of the firm.
 - The Clorox Company considers launching a new version of Armor All designed to clean and protect notebook computers.
 - Google, Inc., plans to purchase real estate to expand its headquarters.
 - Target Corporation decides to expand the number of stores it has in the southeastern United States.
 - GE decides to open a new Universal Studios theme park in China.
- Suppose Caterpillar, Inc., has 666 million shares outstanding with a share price of \$73.09 and \$24.41 billion in debt. If in three years, Caterpillar has 709 million shares outstanding trading for \$86.62 per share, how much debt will Caterpillar have if it maintains a constant debt-equity ratio?
- In 2015, Intel Corporation had a market capitalization of \$134 billion, debt of \$13.2 billion, cash of \$13.8 billion, and EBIT of nearly \$16 billion. If Intel were to increase its debt by \$1 billion and use the cash for a share repurchase, which market imperfections would be most relevant for understanding the consequence for Intel's value? Why?
- Backcountry Adventures is a Colorado-based outdoor travel agent that operates a series of winter backcountry huts. Currently, the value of the firm (debt + equity) is \$3.5 million. But profits will depend on the amount of snowfall: If it is a good year, the firm will be worth \$5 million, and if it is a bad year it will be worth \$2.5 million. Suppose managers always keep the debt to equity ratio of the firm at 25%, and the debt is riskless.
 - What is the initial amount of debt?
 - Calculate the percentage change in the value of the firm, its equity and its debt once the level of snowfall is revealed, but before the firm adjusts the debt level to achieve its target debt to equity ratio.
 - Calculate the percentage change in the value of outstanding debt once the firm adjusts to its target debt-equity ratio.
 - What does this imply about the riskiness of the firm's tax shields? Explain.

The Weighted Average Cost of Capital Method

- Suppose Goodyear Tire and Rubber Company is considering divesting one of its manufacturing plants. The plant is expected to generate free cash flows of \$1.69 million per year, growing at a rate of 2.6% per year. Goodyear has an equity cost of capital of 8.5%, a debt cost of capital of 7.1%, a marginal corporate tax rate of 33%, and a debt-equity ratio of 2.4. If the plant has average risk and Goodyear plans to maintain a constant debt-equity ratio, what after-tax amount must it receive for the plant for the divestiture to be profitable?
- Suppose Alcatel-Lucent has an equity cost of capital of 9.4%, market capitalization of \$9.49 billion, and an enterprise value of \$13 billion. Suppose Alcatel-Lucent's debt cost of capital is 7.1% and its marginal tax rate is 35%.
 - What is Alcatel-Lucent's WACC?
 - If Alcatel-Lucent maintains a constant debt-equity ratio, what is the value of a project with average risk and the following expected free cash flows?

Year	0	1	2	3
FCF (\$ million)	−100	52	100	65

- If Alcatel-Lucent maintains its debt-equity ratio, what is the debt capacity of the project in part (b)?

7. Acort Industries has 10 million shares outstanding and a current share price of \$36 per share. It also has long-term debt outstanding. This debt is risk free, is four years away from maturity, has annual coupons with a coupon rate of 10%, and has a \$115 million face value. The first of the remaining coupon payments will be due in exactly one year. The riskless interest rates for all maturities are constant at 6%. Acort has EBIT of \$101 million, which is expected to remain constant each year. New capital expenditures are expected to equal depreciation and equal \$18 million per year, while no changes to net working capital are expected in the future. The corporate tax rate is 42%, and Acort is expected to keep its debt-equity ratio constant in the future (by either issuing additional new debt or buying back some debt as time goes on).
 - a. Based on this information, estimate Acort's WACC.
 - b. What is Acort's equity cost of capital?

The Adjusted Present Value Method

8. Suppose Goodyear Tire and Rubber Company has an equity cost of capital of 8.5%, a debt cost of capital of 7.1%, a marginal corporate tax rate of 33%, and a debt-equity ratio of 2.4. Assume that Goodyear maintains a constant debt-equity ratio.
 - a. What is Goodyear's WACC?
 - b. What is Goodyear's unlevered cost of capital?
 - c. Explain, intuitively, why Goodyear's unlevered cost of capital is less than its equity cost of capital and higher than its WACC.
9. You are a consultant who has been hired to evaluate a new product line for Markum Enterprises. The upfront investment required to launch the product is \$6 million. The product will generate free cash flow of \$700,000 the first year, and this free cash flow is expected to grow at a rate of 6% per year. Markum has an equity cost of capital of 11.3%, a debt cost of capital of 6.28%, and a tax rate of 32%. Markum maintains a debt-equity ratio of 0.70.
 - a. What is the NPV of the new product line (including any tax shields from leverage)?
 - b. How much debt will Markum initially take on as a result of launching this product line?
 - c. How much of the product line's value is attributable to the present value of interest tax shields?
10. Consider Alcatel-Lucent's project in Problem 6.
 - a. What is Alcatel-Lucent's unlevered cost of capital?
 - b. What is the unlevered value of the project?
 - c. What are the interest tax shields from the project? What is their present value?
 - d. Show that the APV of Alcatel-Lucent's project matches the value computed using the WACC method.

The Flow-to-Equity Method

11. Consider Alcatel-Lucent's project in Problem 6.
 - a. What is the free cash flow to equity for this project?
 - b. What is its NPV computed using the FTE method? How does it compare with the NPV based on the WACC method?
12. In year 1, AMC will earn \$2100 before interest and taxes. The market expects these earnings to grow at a rate of 2.7% per year. The firm will make no net investments (i.e., capital expenditures will equal depreciation) or changes to net working capital. Assume that the corporate tax rate equals 42%. Right now, the firm has \$5250 in risk-free debt. It plans to keep a constant ratio of debt to equity every year, so that on average the debt will also grow by 2.7% per year. Suppose the risk-free rate equals 4.5%, and the expected return on the market equals 9.9%. The asset beta for this industry is 1.27.
 - a. If AMC were an all-equity (unlevered) firm, what would its market value be?
 - b. Assuming the debt is fairly priced, what is the amount of interest AMC will pay next year? If AMC's debt is expected to grow by 2.7% per year, at what rate are its interest payments expected to grow?

- c. Even though AMC's debt is *riskless* (the firm will not default), the future growth of AMC's debt is uncertain, so the exact amount of the future interest payments is risky. Assuming the future interest payments have the same beta as AMC's assets, what is the present value of AMC's interest tax shield?
- d. Using the APV method, what is AMC's total market value, V^L ? What is the market value of AMC's equity?
- e. What is AMC's WACC? (*Hint:* Work backward from the FCF and V^L .)
- f. Using the WACC, what is the expected return for AMC equity?
- g. Show that the following holds for AMC: $\beta_A = \frac{E}{D+E}\beta_E + \frac{D}{D+E}\beta_D$.
- h. Assuming that the proceeds from any increases in debt are paid out to equity holders, what cash flows do the equity holders expect to receive in one year? At what rate are those cash flows expected to grow? Use that information plus your answer to part (f) to derive the market value of equity using the FTE method. How does that compare to your answer in part (d)?

Project-Based Costs of Capital

13. Procter and Gamble (PKGR) has historically maintained a debt-equity ratio of approximately 0.16. Its current stock price is \$48 per share, with 2.6 billion shares outstanding. The firm enjoys very stable demand for its products, and consequently it has a low equity beta of 0.4 and can borrow at 4.7%, just 20 basis points over the risk-free rate of 4.5%. The expected return of the market is 10.2%, and PKGR's tax rate is 32%.
 - a. This year, PKGR is expected to have free cash flows of \$5.8 billion. What constant expected growth rate of free cash flow is consistent with its current stock price?
 - b. PKGR believes it can increase debt without any serious risk of distress or other costs. With a higher debt-equity ratio of 0.4, it believes its borrowing costs will rise only slightly to 5%. If PKGR announces that it will raise its debt-equity ratio to 0.4 through a leveraged recap, determine the increase in the stock price that would result from the anticipated tax savings.
14. Amarindo, Inc. (AMR), is a newly public firm with 10.5 million shares outstanding. You are doing a valuation analysis of AMR. You estimate its free cash flow in the coming year to be \$15.37 million, and you expect the firm's free cash flows to grow by 4.4% per year in subsequent years. Because the firm has only been listed on the stock exchange for a short time, you do not have an accurate assessment of AMR's equity beta. However, you do have beta data for UAL, another firm in the same industry:

	Equity Beta	Debt Beta	Debt-Equity Ratio
UAL	1.80	0.36	1.2

AMR has a much lower debt-equity ratio of 0.36, which is expected to remain stable, and its debt is risk free. AMR's corporate tax rate is 25%, the risk-free rate is 5.5%, and the expected return on the market portfolio is 11.3%.

- a. Estimate AMR's equity cost of capital.
- b. Estimate AMR's share price.



15. Remex (RMX) currently has no debt in its capital structure. The beta of its equity is 1.32. For each year into the indefinite future, Remex's free cash flow is expected to equal \$21 million. Remex is considering changing its capital structure by issuing debt and using the proceeds to buy back stock. It will do so in such a way that it will have a 28% debt-equity ratio after the change, and it will maintain this debt-equity ratio forever. Assume that Remex's debt cost of capital will be 6.76%. Remex faces a corporate tax rate of 32%. Except for the corporate tax rate of 32%, there are no market imperfections. Assume that the CAPM holds, the risk-free rate of interest is 5.2%, and the expected return on the market is 11.44%.

- a. Using the information provided, fill in the table below.

	Debt-Equity Ratio	Debt Cost of Capital	Equity Cost of Capital	Weighted Average Cost of Capital
Before change in capital structure	0	N/A		
After change in capital structure	0.28	6.76%		

- b. Using the information provided and your calculations in part (a), determine the value of the tax shield acquired by Remex if it changes its capital structure in the way it is considering.

APV with Other Leverage Policies



16. You are evaluating a project that requires an investment of \$108 today and provides a single cash flow of \$118 for sure one year from now. You decide to use 100% debt financing, that is, you will borrow \$108. The risk-free rate is 4% and the tax rate is 33%. Assume that the investment is fully depreciated at the end of the year, so without leverage you would owe taxes on the difference between the project cash flow and the investment, that is, \$10.
- Calculate the NPV of this investment opportunity using the APV method.
 - Using your answer to part (a), calculate the WACC of the project.
 - Verify that you get the same answer using the WACC method to calculate NPV.
 - Finally, show that flow-to-equity also correctly gives the NPV of this investment opportunity.
17. Tybo Corporation adjusts its debt so that its interest expenses are 21% of its free cash flow. Tybo is considering an expansion that will generate free cash flows of \$2.16 million this year and is expected to grow at a rate of 3.8% per year from then on. Suppose Tybo's marginal corporate tax rate is 40%.
- If the unlevered cost of capital for this expansion is 10.6%, what is its unlevered value?
 - What is the levered value of the expansion?
 - If Tybo pays 5.5% interest on its debt, what amount of debt will it take on initially for the expansion?
 - What is the debt-to-value ratio for this expansion? What is its WACC?
 - What is the levered value of the expansion using the WACC method?



18. You are on your way to an important budget meeting. In the elevator, you review the project valuation analysis you had your summer associate prepare for one of the projects to be discussed:

(\$ million)	0	1	2	3	4
EBIT		10.1	10.4	10.3	9.7
Interest (5%)		-4.0	-4.0	-3.0	-2.0
Earnings Before Taxes		6.1	6.4	7.3	7.7
Taxes (41%)		-2.5	-2.6	-3.0	-3.2
Depreciation		25.0	25.0	25.0	25.0
Cap Ex	-100.0				
Additions to NWC	-20.0				20.0
Net New Debt	80.0	0.0	-20.0	-20.0	-40.0
FCFE	-40.0	28.6	8.8	9.3	9.5
NPV at 11% Equity Cost of Capital	6.0				

Looking over the spreadsheet, you realize that while all of the cash flow estimates are correct, your associate used the flow-to-equity valuation method and discounted the cash flows using the *company's* equity cost of capital of 11%. While the project's risk is similar to the firm's, the project's incremental leverage is very different from the company's historical debt-equity ratio of 0.20: For this project, the company will instead borrow \$80 million upfront and repay \$20 million in year 2, \$20 million in year 3, and \$40 million in year 4. Thus, the *project's* equity cost of capital is likely to be higher than the firm's, not constant over time—invalidating your associate's calculation.

Clearly, the FTE approach is not the best way to analyze this project. Fortunately, you have your calculator with you, and with any luck you can use a better method before the meeting starts.

- a. What is the present value of the interest tax shield associated with this project?
 - b. What are the free cash flows of the project?
 - c. What is the best estimate of the project's value from the information given?
19. Your firm is considering building a \$597 million plant to manufacture HDTV circuitry. You expect operating profits (EBITDA) of \$141 million per year for the next 10 years. The plant will be depreciated on a straight-line basis over 10 years (assuming no salvage value for tax purposes). After 10 years, the plant will have a salvage value of \$297 million (which, since it will be fully depreciated, is then taxable). The project requires \$50 million in working capital at the start, which will be recovered in year 10 when the project shuts down. The corporate tax rate is 35%. All cash flows occur at the end of the year.
- a. If the risk-free rate is 4.7%, the expected return of the market is 10.9%, and the asset beta for the consumer electronics industry is 1.72, what is the NPV of the project?
 - b. Suppose that you can finance \$478 million of the cost of the plant using 10-year, 9.3% coupon bonds sold at par. This amount is incremental new debt associated specifically with this project and will not alter other aspects of the firm's capital structure. What is the value of the project, including the tax shield of the debt?

Other Effects of Financing

20. Parnassus Corporation plans to invest \$150 million in a new generator that will produce free cash flows of \$20 million per year in perpetuity. The firm is all equity financed, with an equity cost of capital of 10%.
- a. What is the NPV of the project ignoring any costs of raising funds?
 - b. Suppose the firm will issue new equity to raise the \$150 million, and has after-tax issuance costs equal to 8% of the proceeds. What is the NPV of the project including these issuance costs, assuming all future free cash flows generated by it will be paid out?
 - c. Suppose that instead of paying out the project's future free cash flows, a substantial portion of these free cash flows will be retained and invested in other projects, reducing Parnassus' required fundraising in the future. Specifically, suppose the firm will reinvest all free cash flows for the next 10 years, and then pay out the cash flows after that. If its issuance costs remain constant at 8%, what is the NPV of the project including issuance costs in this case?



21. DFS Corporation is currently an all-equity firm, with assets with a market value of \$155 million and 4 million shares outstanding. DFS is considering a leveraged recapitalization to boost its share price. The firm plans to raise a fixed amount of permanent debt (i.e., the outstanding principal will remain constant) and use the proceeds to repurchase shares. DFS pays a 25% corporate tax rate, so one motivation for taking on the debt is to reduce the firm's tax liability. However, the upfront investment banking fees associated with the recapitalization will be 1% of the amount of debt raised. Adding leverage will also create the possibility of future financial distress or agency costs; shown below are DFS's estimates for different levels of debt:

Debt amount (\$ million)	0	10	20	30	40	50
Present value of expected distress and agency costs (\$ million)	0.0	−0.16	−1.81	−3.52	−7.41	−11.46

- a. Based on this information, which level of debt shown above is the best choice for DFS?
 - b. Estimate the stock price once this transaction is announced.
22. Your firm is considering a \$120 million investment to launch a new product line. The project is expected to generate a free cash flow of \$20 million per year, and its unlevered cost of capital is 8%. To fund the investment, your firm will take on \$72 million in permanent debt.
- a. Suppose the marginal corporate tax rate is 35%. Ignoring issuance costs, what is the NPV of the investment?
 - b. Suppose your firm will pay a 4% underwriting fee when issuing the debt. It will raise the remaining \$48 million by issuing equity. In addition to the 7% underwriting fee for the equity issue, you believe that your firm's current share price of \$39 is \$4 per share less than its true value. What is the NPV of the investment including any tax benefits of leverage? (Assume all fees are on an after-tax basis.)
23. Consider Avco's RFX project from Section 18.3. Suppose that Avco is receiving government loan guarantees that allow it to borrow at the 6% rate. Without these guarantees, Avco would pay 6.5% on its debt.
- a. What is Avco's unlevered cost of capital given its true debt cost of capital of 6.5%?
 - b. What is the unlevered value of the RFX project in this case? What is the present value of the interest tax shield?
 - c. What is the NPV of the loan guarantees? (*Hint:* Because the actual loan amounts will fluctuate with the value of the project, discount the expected interest savings at the unlevered cost of capital.)
 - d. What is the levered value of the RFX project, including the interest tax shield and the NPV of the loan guarantees?

Advanced Topics in Capital Budgeting

24. Arden Corporation is considering an investment in a new project with an unlevered cost of capital of 8.5%. Arden's marginal corporate tax rate is 37%, and its debt cost of capital is 4.8%.
- a. Suppose Arden adjusts its debt continuously to maintain a constant debt-equity ratio of 0.5. What is the appropriate WACC for the new project?
 - b. Suppose Arden adjusts its debt once per year to maintain a constant debt-equity ratio of 0.5. What is the appropriate WACC for the new project now?
 - c. Suppose the project has free cash flows of \$10.2 million per year, which are expected to decline by 2.2% per year. What is the value of the project in parts (a) and (b) now?
25. XL Sports is expected to generate free cash flows of \$11.2 million per year. XL has permanent debt of \$35 million, a tax rate of 37%, and an unlevered cost of capital of 10.4%.
- a. What is the value of XL's equity using the APV method?
 - b. What is XL's WACC? What is XL's equity value using the WACC method?
 - c. If XL's debt cost of capital is 4.6%, what is XL's equity cost of capital?
 - d. What is XL's equity value using the FTE method?



- *26. Propel Corporation plans to make a \$50.2 million investment, initially funded completely with debt. The free cash flows of the investment and Propel's incremental debt from the project follow:

Year	0	1	2	3
Free cash flows (\$ million)	−50.2	38.3	18.5	24.5
Debt (\$ million)	50.2	29.3	15.1	0.0

Propel's incremental debt for the project will be paid off according to the predetermined schedule shown. Propel's debt cost of capital is 7.7%, and its tax rate is 35%. Propel also estimates an unlevered cost of capital for the project of 12.2%.

- a. Use the APV method to determine the levered value of the project at each date and its initial NPV.
 - b. Calculate the WACC for this project at each date. How does the WACC change over time? Why?
 - c. Compute the project's NPV using the WACC method.
 - d. Compute the equity cost of capital for this project at each date. How does the equity cost of capital change over time? Why?
 - e. Compute the project's equity value using the FTE method. How does the initial equity value compare with the NPV calculated in parts (a) and (c)?
- *27. Gartner Systems has no debt and an equity cost of capital of 10.8%. Gartner's current market capitalization is \$97 million, and its free cash flows are expected to grow at 2.9% per year. Gartner's corporate tax rate is 36%. Investors pay tax rates of 38% on interest income and 19% on equity income.
- a. Suppose Gartner adds \$49 million in permanent debt and uses the proceeds to repurchase shares. What will Gartner's levered value be in this case?
 - b. Suppose instead Gartner decides to maintain a 50% debt-to-value ratio going forward. If Gartner's debt cost of capital is 7.24%, what will Gartner's levered value be in this case?
-  *28. Revtek, Inc., has an equity cost of capital of 12.6% and a debt cost of capital of 6.3%. Revtek maintains a constant debt-equity ratio of 0.5, and its tax rate is 34%.
- a. What is Revtek's WACC given its current debt-equity ratio?
 - b. Assuming no personal taxes, how will Revtek's WACC change if it increases its debt-equity ratio to 2 and its debt cost of capital remains at 6.3%?
 - c. Now suppose an investors pays tax rates of 32% on interest income and 13% on income from equity. How will Revtek's WACC change if it increases its debt-equity ratio to 2 in this case?
 - d. Provide an intuitive explanation for the difference in your answers to parts (b) and (c).

Data Case

Toyota Motor Company is expanding the production of their gas-electric hybrid drive systems and plans to shift production in the United States. To enable the expansion, they are contemplating investing \$1.5 billion in a new plant with an expected 10-year life. The anticipated free cash flows from the new plant would be \$220 million the first year of operation and grow by 10% for each of the next two years and then 5% per year for the remaining seven years. As a newly hired MBA in the capital budgeting division you have been asked to evaluate the new project using the WACC, Adjusted Present Value, and Flow-to-Equity methods. You will compute the appropriate costs of capital and the net present values with each method. Because this is your first major assignment with the firm, they want you to demonstrate that you are capable of handling the different valuation methods. You must seek out the information necessary to value the free cash flows but will be provided some directions to follow. (This is an involved assignment, but at least you don't have to come up with the actual cash flows for the project!)

1. Go to Yahoo! Finance (finance.yahoo.com) and get the quote for Toyota (symbol: TM).
 - a. Under "Financials," click on the income statement. The income statements for the last three fiscal years will appear. Copy and paste the data into Excel.
 - b. Go back to the Web page and select "Balance Sheets" from the top of the page. Repeat the download procedure for the balance sheets, then copy and paste them into the same worksheet as the income statements.
 - c. Click "Historical prices" in the left column, and find Toyota's stock price for the last day of the month at the end of each of the past three fiscal years. Record the stock price on each date in your spreadsheet.

2. Create a timeline in Excel with the free cash flows for the 10 years of the project.
3. Determine the WACC using Eq. 18.1.
 - a. For the cost of debt, r_D :
 - i. Go to finra-markets.morningstar.com/BondCenter/Default.jsp and click to search. Enter Toyota's symbol, select the Corporate toggle, and press "Enter."
 - ii. Look at the average credit rating for Toyota long-term bonds. If you find that they have a rating of A or above, then you can make the approximation that the cost of debt is the risk-free rate. If Toyota's credit rating has slipped, use Table 12.3 to estimate the beta of debt from the credit rating.
 - b. For the cost of equity, r_E :
 - i. Get the yield on the 10-year U.S. Treasury Bond from Yahoo! Finance (finance.yahoo.com). Click on "Market Data." Scroll down to the Bonds Summary. Enter that yield as the risk-free rate.
 - ii. Find the beta for Toyota from Yahoo! Finance. Enter the symbol for Toyota and click "Key Statistics." The beta for Toyota will be listed there.
 - iii. Use a market risk premium of 4.50% to compute r_E using the CAPM. If you need to, repeat the exercise to compute r_D .
 - c. Determine the values for E and D for Eq. 18.1 for Toyota and the debt-to-value and equity-to-value ratios.
 - i. To compute the net debt for Toyota, add the long-term debt and the short-term debt and subtract cash and cash equivalents for each year on the balance sheet.
 - ii. Obtain the historical number of shares outstanding from Google Finance (www.google.com/finance). Enter Toyota's ticker in the search box, click "Financials." Look on the income statement for "Diluted Weighted Average Shares." Multiply the historical stock prices by the number of shares outstanding you collected to compute Toyota's market capitalization at the end of each fiscal year.
 - iii. Compute Toyota's enterprise value at the end of each fiscal year by combining the values obtained for its equity market capitalization and its net debt.
 - iv. Compute Toyota's debt-to-value ratio at the end of each year by dividing its net debt by its enterprise value. Use the average ratio from the last four years as an estimate for Toyota's target debt-to-value ratio.
 - d. Determine Toyota's tax rate by dividing the income tax by earnings before tax for each year. Take the average of the four rates as Toyota's marginal corporate tax rate.
 - e. Compute the WACC for Toyota using Eq. 18.1.
4. Compute the NPV of the hybrid engine expansion given the free cash flows you calculated using the WACC method of valuation.
5. Determine the NPV using the Adjusted Present Value Method, and also using the Flow-to-Equity method. In both cases, assume Toyota maintains the target leverage ratio you computed in Question 3c.
6. Compare the results under the three methods and explain how the resulting NPVs are achieved under each of the three different methods.

Note: Updates to this data case may be found at www.berkdemarzo.com.

CHAPTER 18
APPENDIX

Foundations and Further Details

In this appendix, we look at the foundations for the WACC method, and for the relationship between a firm's levered and unlevered costs of capital. We also address how we can solve for a firm's leverage policy and value simultaneously.

Deriving the WACC Method

The WACC can be used to value a levered investment, as in Eq. 18.2. Consider an investment that is financed by both debt and equity. Because equity holders require an expected return of r_E on their investment and debt holders require a return of r_D , the firm will have to pay investors a total of

$$E(1 + r_E) + D(1 + r_D) \quad (18A.1)$$

next year. What is the value of the investment next year? The project generates free cash flows of FCF_1 at the end of the year. In addition, the interest tax shield of the debt provides a tax savings of $\tau_c \times (\text{interest on debt}) \approx \tau_c r_D D$.²⁷ Finally, if the investment will continue beyond next year, it will have a continuation value of V_1^L . Thus, to satisfy investors, the project cash flows must be such that

$$E(1 + r_E) + D(1 + r_D) = FCF_1 + \tau_c r_D D + V_1^L \quad (18A.2)$$

Because $V_0^L = E + D$, we can write the WACC definition in Eq. 18.1 as

$$r_{wacc} = \frac{E}{V_0^L} r_E + \frac{D}{V_0^L} r_D (1 - \tau_c) \quad (18A.3)$$

If we move the interest tax shield to the left side of Eq. 18A.2, we can use the definition of the WACC to rewrite Eq. 18A.2 as follows:

$$\underbrace{E(1 + r_E) + D[1 + r_D(1 - \tau_c)]}_{V_0^L(1 + r_{wacc})} = FCF_1 + V_1^L \quad (18A.4)$$

²⁷The return on the debt r_D need not come solely from interest payments. If C_t is the coupon paid and D_t is the market value of the debt in period t , then in period t , r_D is defined as

$$r_D = \frac{E[\text{Coupon Payment} + \text{Capital Gain}]}{\text{Current Price}} = \frac{E[C_{t+1} + D_{t+1} - D_t]}{D_t}$$

The return that determines the firm's interest expense is

$$\bar{r}_D = \frac{E[C_{t+1} + \bar{D}_{t+1} - \bar{D}_t]}{D_t}$$

where D_t is the value of the debt on date t according to a fixed schedule set by the tax code based on the difference between the bond's initial price and its face value, which is called the bond's *original issue discount* (OID). (If the bond is issued at par and the firm will not default on the next coupon, then $\bar{D}_t = \bar{D}_{t+1}$ and $\bar{r}_D = C_{t+1}/D_t$, which is the bond's *current yield*.) Thus, the true after-tax cost of debt is $(r_D - \tau_c \bar{r}_D)$. In practice, the distinction between r_D and $\bar{r}_D$ is often ignored, and the after-tax cost of debt is computed as $r_D(1 - \tau_c)$. Also, the debt's yield to maturity is often used in place of r_D . Because the yield ignores default risk, it overstates r_D and thus the WACC. See Chapter 12 for alternative methods.

Dividing by $(1 + r_{wacc})$, we can express the value of the investment today as the present value of next period's free cash flows and continuation value:

$$V_0^L = \frac{FCF_1 + V_1^L}{1 + r_{wacc}} \quad (18.A5)$$

In the same way, we can write the value in one year, V_1^L , as the discounted value of the free cash flows and continuation value of the project in year 2. If the WACC is the same next year, then

$$V_0^L = \frac{FCF_1 + V_1^L}{1 + r_{wacc}} = \frac{FCF_1 + \frac{FCF_2 + V_2^L}{1 + r_{wacc}}}{1 + r_{wacc}} = \frac{FCF_1}{1 + r_{wacc}} + \frac{FCF_2 + V_2^L}{(1 + r_{wacc})^2} \quad (18.A6)$$

By repeatedly replacing each continuation value, and *assuming the WACC remains constant*, we can derive Eq. 18.2:²⁸

$$V_0^L = \frac{FCF_1}{1 + r_{wacc}} + \frac{FCF_2}{(1 + r_{wacc})^2} + \frac{FCF_3}{(1 + r_{wacc})^3} + \dots \quad (18.A7)$$

That is, *the value of a levered investment is the present value of its future free cash flows using the weighted average cost of capital.*

The Levered and Unlevered Cost of Capital

In this appendix, we derive the relationship between the levered and unlevered cost of capital for the firm. Suppose an investor holds a portfolio of all of the equity and debt of the firm. Then the investor will receive the free cash flows of the firm plus the tax savings from the interest tax shield. These are the same cash flows an investor would receive from a portfolio of the unlevered firm (which generates the free cash flows) and a separate “tax shield” security that paid the investor the amount of the tax shield each period. Because these two portfolios generate the same cash flows, by the Law of One Price they have the same market values:

$$V^L = E + D = V^U + T \quad (18.A8)$$

where T is the present value of the interest tax shield. Equation 18.A8 is the basis of the APV method. Because these portfolios have equal cash flows, they must also have identical expected returns, which implies

$$E r_E + D r_D = V^U r_U + T r_T \quad (18.A9)$$

where r_T is the expected return associated with the interest tax shields. The relationship between r_E , r_D , and r_U will depend on the expected return r_T , which is determined by the risk of the interest tax shield. Let's consider the two cases discussed in the text.

Target Leverage Ratio

Suppose the firm adjusts its debt continuously to maintain a target debt-to-value ratio, or a target ratio of interest to free cash flow. Because the firm's debt and interest payments will vary with the firm's value and cash flows, it is reasonable to expect the risk of the interest

²⁸This expansion is the same approach we took in Chapter 9 to derive the discounted dividend formula for the stock price.

tax shield will equal that of the firm's free cash flow, so $r_T = r_U$. Making this assumption, which we return to below, Eq. 18A.9 becomes

$$\begin{aligned} E r_E + D r_D &= V^U r_U + T r_U = (V^U + T) r_U \\ &= (E + D) r_U \end{aligned} \quad (18A10)$$

Dividing by $(E + D)$ leads to Eq. 18.6.

Predetermined Debt Schedule

Suppose some of the firm's debt is set according to a predetermined schedule that is independent of the growth of the firm. Suppose the value of the tax shield from the scheduled debt is T^s , and the remaining value of the tax shield $T - T^s$ is from debt that will be adjusted according to a target leverage ratio. Because the risk of the interest tax shield from the scheduled debt is similar to the risk of the debt itself, Eq. 18A.9 becomes

$$E r_E + D r_D = V^U r_U + T r_T = V^U r_U + (T - T^s) r_U + T^s r_D \quad (18A11)$$

Subtracting $T^s r_D$ from both sides, and using $D^s = D - T^s$,

$$\begin{aligned} E r_E + D^s r_D &= (V^U + T - T^s) r_U = (V^L - T^s) r_U \\ &= (E + D^s) r_U \end{aligned} \quad (18A12)$$

Dividing by $(E + D^s)$ leads to Eq. 18.20.

Risk of the Tax Shield with a Target Leverage Ratio

Above, we assumed that with a target leverage ratio, it is reasonable to assume that $r_T = r_U$. Under what circumstances should this be the case?

We define a target leverage ratio as a setting in which the firm adjusts its debt at date t to be a proportion $d(t)$ of the investment's value, or a proportion $k(t)$ of its free cash flow. (The target ratio for either policy need not be constant over time, but can vary according to a predetermined schedule.)

With either policy, the value at date t of the incremental tax shield from the project's free cash flow at date s , FCF_s , is proportional to the value of the cash flow $V_t^L(FCF_s)$, so it should be discounted at the same rate as FCF_s . Therefore, the assumption $r_T = r_U$ follows as long as at each date the cost of capital associated with the value of each future free cash flow is the same (a standard assumption in capital budgeting).²⁹

Solving for Leverage and Value Simultaneously

When we use the APV method, we need to know the debt level to compute the interest tax shield and determine the project's value. But if a firm maintains a constant debt-to-value ratio, we need to know the project's value to determine the debt level. How can we apply the APV method in this case?

When a firm maintains a constant leverage ratio, to use the APV method we must solve for the debt level and the project value simultaneously. While complicated to do by hand, it is (fortunately) easy to do in Excel. We begin with the spreadsheet shown in Table 18A.1, which illustrates the standard APV calculation outlined in Section 18.3 of the text. For now, we have just inserted arbitrary values for the project's debt capacity in line 3.

²⁹If the risk of the individual cash flows differs, then r_T will be a weighted average of the unlevered costs of capital of the individual cash flows, with the weights depending on the schedule d or k . See P. DeMarzo, "Discounting Tax Shields and the Unlevered Cost of Capital," 2005, ssrn.com/abstract=1488437.

TABLE 18A.1
SPREADSHEET**Adjusted Present Value for Avco's RFX Project
with Arbitrary Debt Levels**

	Year	0	1	2	3	4
Unlevered Value (\$ million)						
1 Free Cash Flow		(28.00)	18.00	18.00	18.00	18.00
2 Unlevered Value, V^U (at $r_u = 8.0\%$)		59.62	46.39	32.10	16.67	—
Interest Tax Shield						
3 Debt Capacity (arbitrary)		30.00	20.00	10.00	5.00	—
4 Interest Paid (at $r_d = 6\%$)		—	1.80	1.20	0.60	0.30
5 Interest Tax Shield (at $\tau_c = 40\%$)		—	0.72	0.48	0.24	0.12
6 Tax Shield Value, T (at $r_u = 8.0\%$)		1.36	0.75	0.33	0.11	—
Adjusted Present Value						
7 Levered Value, $V^L = V^U + T$		60.98	47.13	32.42	16.78	—

Note that the debt capacity specified in line 3 is not consistent with a 50% debt-to-value ratio for the project. For example, given the value of \$60.98 million in year 0, the initial debt capacity should be $50\% \times \$60.98 \text{ million} = \30.49 million in year 0. But if we change each debt capacity in line 3 to a *numerical* value that is 50% of the value in line 7, the interest tax shield and the project's value will change, and we will still not have a 50% debt-to-value ratio.

The solution is to enter in line 3 a *formula* that sets the debt capacity to be 50% of the project's value in line 7 in the same year. Now line 7 depends on line 3, and line 3 depends on line 7, creating a circular reference in the spreadsheet (and you will most likely receive an error message). By changing the calculation option in Excel to calculate the spreadsheet iteratively (File > Options > Formulas and check the Enable iterative calculation box), Excel will keep calculating until the values in line 3 and line 7 of the spreadsheet are consistent, as shown in Table 18A.2.

TABLE 18A.2
SPREADSHEET**Adjusted Present Value for Avco's RFX Project with Debt
Levels Solved Iteratively**

	Year	0	1	2	3	4
Unlevered Value (\$ million)						
1 Free Cash Flow		(28.00)	18.00	18.00	18.00	18.00
2 Unlevered Value, V^U (at $r_u = 8.0\%$)		59.62	46.39	32.10	16.67	—
Interest Tax Shield						
3 Debt Capacity (at $d = 50\%$)		30.62	23.71	16.32	8.43	—
4 Interest Paid (at $r_d = 6\%$)		—	1.84	1.42	0.98	0.51
5 Interest Tax Shield (at $\tau_c = 40\%$)		—	0.73	0.57	0.39	0.20
6 Tax Shield Value, T (at $r_u = 8.0\%$)		1.63	1.02	0.54	0.19	—
Adjusted Present Value						
7 Levered Value, $V^L = V^U + T$		61.25	47.41	32.63	16.85	—

Thus we calculate the same $NPV = V_0^L - FCF_0 = 61.25 - 28 = \33.25 million as in Section 18.1.

The same method can be applied when using the WACC method with known debt levels. In that case, we need to know the project's value to determine the debt-to-value ratio and compute the WACC, and we need to know the WACC to compute the project's value. Again, we can use iteration within Excel to determine simultaneously the project's value and debt-to-value ratio.

The Residual Income and Economic Value Added Valuation Methods

An alternative valuation method developed primarily in the accounting literature is the **residual income method**.³⁰ A company's residual income in year t is defined as its net income less a charge for the required return on stockholders' equity (using the equity cost of capital r_E):

$$\begin{aligned}\text{Residual Income}_t &= \text{Net Income}_t - \underbrace{r_E \times \text{Book Value of Equity}_{t-1}}_{\text{Equity charge}} \\ &= (ROE_t - r_E) \times \text{Book Value of Equity}_{t-1}\end{aligned}\quad (18.A13)$$

We can think of residual income as the firm's profit in excess of the required return on its equity, and can be thought of as a measure of the equity value added to the firm. The residual income is positive if and only if the firm's accounting return on equity, or ROE, exceeds its equity cost of capital.

The residual income valuation method states that the market value of the firm's equity should equal its book value plus the present value of its future residual income. That is, if we write E_0 for the market value and BE_0 for the book value of equity at the *end* of the period, and RI_t for the residual income in year t :

Residual Income Valuation Method

$$E_0 = BE_0 + PV(\text{Residual Income}) = BE_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1 + r_E)^t} \quad (18.A14)$$

The residual income method is equivalent to the Flow-to-Equity method described in Section 18.4. Recall that in the FTE method, we value equity as the present value of FCFE, the total cash flows paid out to shareholders. Because the book value of the firm's shares increases each year by net income (NI) less any payouts to shareholders (FCFE), we have

$$BE_t = BE_{t-1} + NI_t - FCFE_t \quad (18.A15)$$

Then we can rewrite FCFE, using Eq. 18.A15 and Eq. 18.A13, as

$$FCFE_t = NI_t + BE_{t-1} - BE_t = RI_t + (1 + r_E) BE_{t-1} - BE_t$$

Taking present values, we see that all of the BE terms with $t \geq 1$ cancel, so that

$$E_0 = PV(FCFE) = PV(RI) + BE_0$$

Note that Eq. 18.A14 presumes the equity cost of capital is constant over time, which only holds if the firm has a target leverage ratio. Otherwise, we must compute r_E period by period as in Table 18.10.

Note that we can also apply the Residual Income method to value a project. In that case, if we assume the initial and ending book value of equity *for the project* is zero, then

$$NPV(\text{Project}) = PV(\text{Incremental Residual Income}) \quad (18.A16)$$

Table 18A.3 applies this approach to value Avco's RFX project assuming a target leverage ratio of $d = 50\%$. Note that to compute the residual income from the project, we consider only its incremental contribution to the book value of equity from the balance sheet. In this example, because the incremental debt from the project exceeds the book value of the equipment, the incremental equity (and thus the equity charge) is negative.

³⁰See e.g. J. A. Ohlson, "Earnings, Book Values, and Dividends in Equity Valuation," *Contemporary Accounting Research* (1995): 661–687.

TABLE 18A.3
SPREADSHEET**Evaluating Avco's RFX Project Using the Residual Income Method (assuming a target leverage ratio from Table 18.4)**

	Year	0	1	2	3	4
Incremental Earnings Forecast (\$ million)						
1 Sales	—	60.00	60.00	60.00	60.00	60.00
2 Cost of Goods Sold	—	(25.00)	(25.00)	(25.00)	(25.00)	(25.00)
3 Gross Profit	—	35.00	35.00	35.00	35.00	35.00
4 Operating Expenses	(6.67)	(9.00)	(9.00)	(9.00)	(9.00)	(9.00)
5 Depreciation	—	(6.00)	(6.00)	(6.00)	(6.00)	(6.00)
6 EBIT	(6.67)	20.00	20.00	20.00	20.00	20.00
7 Interest Expense	—	(1.84)	(1.42)	(0.98)	(0.51)	(0.51)
8 Pretax Income	(6.67)	18.16	18.58	19.02	19.49	19.49
9 Income Tax at 40%	2.67	(7.27)	(7.43)	(7.61)	(7.80)	(7.80)
10 Net Income	(4.00)	10.90	11.15	11.41	11.70	11.70
Project Balance Sheet Data						
11 Property, Plant & Equipment	24.00	18.00	12.00	6.00	—	—
12 Debt	30.62	23.71	16.32	8.43	—	—
13 Incremental Equity	(6.62)	(5.71)	(4.32)	(2.43)	—	—
Residual Income						
14 Capital Charge ($r_e = 10\%$)	—	0.66	0.57	0.43	0.24	0.24
15 Net Income	(4.00)	10.90	11.15	11.41	11.70	11.70
16 Residual Income	(4.00)	11.56	11.72	11.84	11.94	11.94
17 PV at $r_e = 10\%$	33.25					

We can use a similar logic to develop an alternative equation for the WACC method. Here we begin with a measure of **economic value added** equal to the firm's *unlevered* net income less a charge for the required return on the firm's *total* invested capital (both equity and debt). We measure total invested capital as the firm's **book enterprise value**:

$$\text{Invested Capital} = \text{Book Value of Equity} + \text{Net Debt} = \text{Book Enterprise Value} \quad (18.A17)$$

Then economic value added in year t is defined as

$$\begin{aligned} \text{Economic Value Added}_t &= \underbrace{\text{EBIT}_t \times (1 - \tau_c)}_{\text{Unlevered Net Income}} - \underbrace{r_{wacc} \times \text{Book Enterprise Value}_{t-1}}_{\text{Capital Charge}} \\ &= (ROIC_t - r_{wacc}) \times \text{Book Enterprise Value}_{t-1} \end{aligned} \quad (18.A18)$$

We can see that economic value added is only positive if the firm's return on invested capital, or ROIC, exceeds its weighted-average cost of capital. Because

$$BEV_t = BEV_{t-1} + \text{EBIT}_t(1 - \tau_c) - FCF_t, \quad (18.A19)$$

we can then show that the WACC method is equivalent to the firm's current (market) enterprise value (V_0) being equal to its book enterprise value (BEV_0) plus the present value of future economic value added (EVA):

Economic Value Added Valuation Method

$$V_0 = BEV_0 + PV(\text{Economic Value Added}) = BEV_0 + \sum_{t=1}^{\infty} \frac{EVA_t}{(1 + r_{wacc})^t} \quad (18.A20)$$

Again, we can apply this method to compute the NPV of a project. Assuming the starting and ending incremental book enterprise value from the project is zero, we have

$$NPV(\text{Project}) = PV(\text{Incremental EVA}) \quad (18.A21)$$

Valuation and Financial Modeling: A Case Study

CHAPTER

19

THE GOAL OF THIS CHAPTER IS TO APPLY THE FINANCIAL TOOLS

we have developed thus far to demonstrate how they are used in practice to build a valuation model of a firm. In this chapter, we will value a hypothetical firm, Ideko Corporation. Ideko is a privately held designer and manufacturer of specialty sports eyewear based in Chicago. In mid-2005, its owner and founder, June Wong, has decided to sell the business, after having relinquished management control about four years ago. As a partner in KKP Investments, you are investigating purchasing the company. If a deal can be reached, the acquisition will take place at the end of the current fiscal year. In that event, KKP plans to implement operational and financial improvements at Ideko over the next five years, after which it intends to sell the business.

Ideko has total assets of \$87 million and annual sales of \$75 million. The firm is also quite profitable, with earnings this year of almost \$7 million, for a net profit margin of 9.3%. You believe a deal could be struck to purchase Ideko's equity at the end of this fiscal year for an acquisition price of \$150 million, which is almost double Ideko's current book value of equity. Is this price reasonable?

We begin the chapter by estimating Ideko's value using data for comparable firms. We then review KKP's operating strategies for running the business after the acquisition, to identify potential areas for improvements. We build a financial model to project cash flows that reflect these operating improvements. These cash flow forecasts enable us to value Ideko using the APV model introduced in Chapter 18 and estimate the return on KKP's investment. Finally, we explore the sensitivity of the valuation estimates to our main assumptions.

NOTATION

R_s	return on security s
r_f	risk-free rate
α_s	the alpha of security s
β_s	the beta of security s
R_{mkt}	return of the market portfolio
$E[R_{mkt}]$	expected return of the market portfolio
ε_s	the regression error term
β_U	the beta of an unlevered firm
β_E	the beta of equity
β_D	the beta of debt
r_U	unlevered cost of capital
r_{wacc}	weighted average cost of capital
r_D	debt cost of capital
V_T^L	continuing value of a project at date T
V^U	unlevered value
FCF_t	free cash flow at date t
g	growth rate
T^s	predetermined tax shield value

19.1 Valuation Using Comparables

As a result of preliminary conversations with Ideko's founder, you have estimates of Ideko's income and balance sheet information for the current fiscal year shown in Table 19.1. Ideko currently has debt outstanding of \$4.5 million, but it also has a substantial cash balance. To obtain your first estimate of Ideko's value, you decide to value Ideko by examining comparable firms.

A quick way to gauge the reasonableness of the proposed price for Ideko is to compare it to that of other publicly traded firms using the method of comparable firms introduced in Chapter 9. For example, at a price of \$150 million, Ideko's price-earnings (P/E) ratio is $150,000/6939 = 21.6$, roughly equal to the market average P/E ratio in mid-2005.

It is even more informative to compare Ideko to firms in a similar line of business. Although no firm is exactly comparable to Ideko in terms of its overall product line, three firms with which it has similarities are Oakley, Inc.; Luxottica Group; and Nike, Inc. The closest competitor is Oakley, which also designs and manufactures sports eyewear. Luxottica Group is an Italian eyewear maker, but much of its business is prescription eyewear; it also owns and operates a number of retail eyewear chains. Nike is a manufacturer of specialty sportswear products, but its primary focus is footwear. You also decide to compare Ideko to a portfolio of firms in the sporting goods industry.

A comparison of Ideko's proposed valuation to this peer set, as well as to the average firm in the sporting goods industry, appears in Table 19.2. The table not only lists P/E ratios, but also shows each firm's enterprise value (EV) as a multiple of sales and EBITDA (earnings before interest, taxes, depreciation, and amortization). Recall that enterprise value is the total value of equity plus net debt, where net debt is debt less cash and investments in marketable securities that are not required as part of normal operations. Ideko has \$4.5 million in debt, and you estimate that it holds \$6.5 million of cash in excess of its working capital needs. Thus, Ideko's enterprise value at the proposed acquisition price is $150 + 4.5 - 6.5 = \$148$ million.

At the proposed price, Ideko's P/E ratio is low relative to those of Oakley and Luxottica, although it is somewhat above the P/E ratios of Nike and the industry overall. The same

TABLE 19.1
SPREADSHEET

Estimated 2005 Income Statement and Balance Sheet
Data for Ideko Corporation

Year 2005		Year 2005		
Income Statement (\$ 000)		Balance Sheet (\$ 000)		
1	Sales	75,000	Assets	
2	Cost of Goods Sold		1 Cash and Equivalents	12,664
3	Raw Materials	(16,000)	2 Accounts Receivable	18,493
4	Direct Labor Costs	(18,000)	3 Inventories	6,165
5	Gross Profit	41,000	4 Total Current Assets	37,322
6	Sales and Marketing	(11,250)	5 Property, Plant, and Equipment	49,500
7	Administrative	(13,500)	6 Goodwill	—
8	EBITDA	16,250	7 Total Assets	86,822
9	Depreciation	(5,500)	Liabilities and Stockholders' Equity	
10	EBIT	10,750	8 Accounts Payable	4,654
11	Interest Expense (net)	(75)	9 Debt	4,500
12	Pretax Income	10,675	10 Total Liabilities	9,154
13	Income Tax	(3,736)	11 Stockholders' Equity	77,668
14	Net Income	6,939	12 Total Liabilities and Equity	86,822

TABLE 19.2**Ideko Financial Ratios Comparison, Mid-2005**

Ratio	Ideko (Proposed)	Oakley, Inc.	Luxottica Group	Nike, Inc.	Sporting Goods Industry
P/E	21.6×	24.8×	28.0×	18.2×	20.3×
EV/Sales	2.0×	2.0×	2.7×	1.5×	1.4×
EV/EBITDA	9.1×	11.6×	14.4×	9.3×	11.4×
EBITDA/Sales	21.7%	17.0%	18.5%	15.9%	12.1%

can be said for Ideko's valuation as a multiple of sales. Thus, based on these two measures, Ideko looks "cheap" relative to Oakley and Luxottica, but is priced at a premium relative to Nike and the average sporting goods firm. The deal stands out, however, when you compare Ideko's enterprise value relative to EBITDA. The acquisition price of just over nine times EBITDA is below that of all of the comparable firms as well as the industry average. Note also Ideko's high profit margins: At $16,250/75,000 = 21.7\%$, its EBITDA margin exceeds that of all of the comparables.

While Table 19.2 provides some reassurance that the acquisition price is reasonable compared to other firms in the industry, it by no means establishes that the acquisition is a good investment opportunity. As with any such comparison, the multiples in Table 19.2 vary substantially. Furthermore, they ignore important differences such as the operating efficiency and growth prospects of the firms, and they do not reflect KKP's plans to improve Ideko's operations. To assess whether this investment is attractive requires a careful analysis both of the operational aspects of the firm and of the ultimate cash flows the deal is expected to generate and the return that should be required.

EXAMPLE 19.1**Valuation by Comparables****Problem**

What range of acquisition prices for Ideko is implied by the range of multiples for P/E, EV/Sales, and EV/EBITDA in Table 19.2?

Solution

For each multiple, we can find the highest and lowest values across all three firms and the industry portfolio. Applying each multiple to the data for Ideko in Table 19.1 yields the following results:

Multiple	Range		Price (in \$ million)	
	Low	High	Low	High
P/E	18.2×	28.0×	126.3	194.3
EV/Sales	1.4×	2.7×	107.0	204.5
EV/EBITDA	9.3×	14.4×	153.1	236.0

For example, Nike has the lowest P/E multiple of 18.2. Multiplying this P/E by Ideko's earnings of \$6.94 million gives a value of $18.2 \times 6.94 = \$126.3$ million. The highest multiple of enterprise value to sales is 2.7 (Luxottica); at this multiple, Ideko's enterprise value is $2.7 \times 75 = \$202.5$ million. Adding Ideko's excess cash and subtracting its debt implies a purchase price of $202.5 + 6.5 - 4.5 = \$204.5$ million. The table above demonstrates that while comparables provide a useful benchmark, they cannot be relied upon for a precise estimate of value.

CONCEPT CHECK

1. What is the purpose of the valuation using comparables?
2. If the valuation using comparables indicates the acquisition price is reasonable compared to other firms in the industry, does it establish that the acquisition is a good investment opportunity?

19.2 The Business Plan

While comparables provide a useful starting point, whether this acquisition is a successful investment for KKP depends on Ideko's post-acquisition performance. Thus, it is necessary to look in detail at Ideko's operations, investments, and capital structure, and to assess its potential for improvements and future growth.

Operational Improvements

On the operational side, you are quite optimistic regarding the company's prospects. The market is expected to grow by 5% per year, and Ideko produces a superior product. Ideko's market share has not grown in recent years because current management has devoted insufficient resources to product development, sales, and marketing. Conversely, Ideko has overspent on administrative costs. Indeed, Table 19.1 reveals that Ideko's current administrative expenses are $13,500/75,000 = 18\%$ of sales, a rate that exceeds its expenditures on sales and marketing (15% of sales). This is in stark contrast to its rivals, which spend less on administrative overhead than they do on sales and marketing.

KKP plans to cut administrative costs immediately and redirect resources to new product development, sales, and marketing. By doing so, you believe Ideko can increase its market share from 10% to 15% over the next five years. The increased sales demand can be met in the short run using the existing production lines by increasing overtime and running some weekend shifts. However, once the growth in volume exceeds 50%, Ideko will definitely need to undertake a major expansion to increase its manufacturing capacity.

The spreadsheet in Table 19.3 shows sales and operating cost assumptions for the next five years based on this plan. In the spreadsheet, numbers in blue represent data that has been entered, whereas numbers in black are calculated based on the data provided. For example, given the current market size of 10 million units and an expected growth rate of 5% per year, the spreadsheet calculates the expected market size in years 1 through 5. Also shown is the expected growth in Ideko's market share.

TABLE 19.3
SPREADSHEET

Ideko Sales and Operating Cost Assumptions

		Year	2005	2006	2007	2008	2009	2010
Sales Data		Growth/Year						
1	Market Size (000 units)	5.0%	10,000	10,500	11,025	11,576	12,155	12,763
2	Market Share	1.0%	10.0%	11.0%	12.0%	13.0%	14.0%	15.0%
3	Average Sales Price (\$/unit)	2.0%	75.00	76.50	78.03	79.59	81.18	82.81
Cost of Goods Data								
4	Raw Materials (\$/unit)	1.0%	16.00	16.16	16.32	16.48	16.65	16.82
5	Direct Labor Costs (\$/unit)	4.0%	18.00	18.72	19.47	20.25	21.06	21.90
Operating Expense and Tax Data								
6	Sales and Marketing (% sales)		15.0%	16.5%	18.0%	19.5%	20.0%	20.0%
7	Administrative (% sales)		18.0%	15.0%	15.0%	14.0%	13.0%	13.0%
8	Tax Rate		35.0%	35.0%	35.0%	35.0%	35.0%	35.0%

Note that Ideko's average selling price is expected to increase because of a 2% inflation rate each year. Likewise, manufacturing costs are expected to rise. Raw materials are forecast to increase at a 1% rate and, although you expect some productivity gains, labor costs will rise at a 4% rate due to additional overtime. The table also shows the reallocation of resources from administration to sales and marketing over the five-year period.

EXAMPLE 19.2

Production Capacity Requirements

Problem

Based on the data in Table 19.3, what production capacity will Ideko require each year? When will an expansion be necessary?

Solution

Production volume each year can be estimated by multiplying the total market size and Ideko's market share in Table 19.3:

	Year	2005	2006	2007	2008	2009	2010
Production Volume (000 units)							
1	Market Size	10,000	10,500	11,025	11,576	12,155	12,763
2	Market Share	10.0%	11.0%	12.0%	13.0%	14.0%	15.0%
3	Production Volume (1 × 2)	1,000	1,155	1,323	1,505	1,702	1,914

Based on this forecast, production volume will exceed its current level by 50% by 2008, necessitating an expansion then.

Capital Expenditures: A Needed Expansion

The spreadsheet in Table 19.4 shows the forecast for Ideko's capital expenditures over the next five years. Based on the estimates for capital expenditures and depreciation, this spreadsheet tracks the book value of Ideko's plant, property, and equipment starting from its level at the beginning of 2005. Note that investment is expected to remain at its current level over the next two years, which is roughly equal to the level of depreciation. Ideko will expand its production during this period by using its existing plant more efficiently. In 2008, however, a major expansion of the plant will be necessary, leading to a large increase in capital expenditures in 2008 and 2009.

The depreciation entries in Table 19.4 are based on the appropriate depreciation schedule for each type of property. Those calculations are quite specific to the nature of the property and are not detailed here. The depreciation shown will be used for tax purposes.¹

TABLE 19.4
SPREADSHEET

Ideko Capital Expenditure Assumptions

	Year	2005	2006	2007	2008	2009	2010
Fixed Assets and Capital Investment (\$ 000)							
1	Opening Book Value	50,000	49,500	49,050	48,645	61,781	69,102
2	Capital Investment	5,000	5,000	5,000	20,000	15,000	8,000
3	Depreciation	(5,500)	(5,450)	(5,405)	(6,865)	(7,678)	(7,710)
4	Closing Book Value	49,500	49,050	48,645	61,781	69,102	69,392

¹Firms often maintain separate books for accounting and tax purposes, and they may use different depreciation assumptions for each. Remember that because depreciation affects cash flows through its tax consequences, tax depreciation is more relevant for valuation.

Working Capital Management

To compensate for its weak sales and marketing efforts, Ideko has sought to retain the loyalty of its retailers in part by maintaining a very lax credit policy. This policy affects Ideko's working capital requirements: For every extra day that customers take to pay, another day's sales revenue is added to accounts receivable (rather than received in cash). From Ideko's current income statement and balance sheet (Table 19.1), we can estimate the number of days of receivables:

$$\begin{aligned}\text{Accounts Receivable Days} &= \frac{\text{Accounts Receivable (\$)}}{\text{Sales Revenue (\$/yr)}} \times 365 \text{ days/yr} \\ &= \frac{18,493}{75,000} \times 365 \text{ days} = 90 \text{ days}\end{aligned}\quad (19.1)$$

The standard for the industry is 60 days, and you believe that Ideko can tighten its credit policy to achieve this goal without sacrificing many sales.

You also hope to improve Ideko's inventory management. Ideko's balance sheet in Table 19.1 lists inventory of \$6.165 million. Of this amount, approximately \$2 million corresponds to raw materials, while the rest is finished goods. Given raw material expenditures of \$16 million for the year, Ideko currently holds $(2/16) \times 365 = 45.6$ days' worth of raw material inventory. While maintaining a certain amount of inventory is necessary to avoid production stoppages, you believe that, with tighter controls of the production process, 30 days' worth of inventory will be adequate.

Capital Structure Changes: Levering Up

With little debt, excess cash, and substantial earnings, Ideko appears to be significantly underleveraged. You plan to greatly increase the firm's debt, and have obtained bank commitments for loans of \$100 million should an agreement be reached. These term loans will have an interest rate of 6.8%, and Ideko will pay interest only during the next five years. The firm will seek additional financing in 2008 and 2009 associated with the expansion of its manufacturing plant, as shown in the spreadsheet in Table 19.5. While Ideko's credit quality should improve over time, the steep slope of the yield curve suggests interest rates may increase; therefore, on balance, you expect Ideko's borrowing rate to remain at 6.8%.

Given Ideko's outstanding debt, its interest expense each year is computed as²

$$\text{Interest in Year } t = \text{Interest Rate} \times \text{Ending Balance in Year } (t - 1) \quad (19.2)$$

The interest on the debt will provide a valuable tax shield to offset Ideko's taxable income.

TABLE 19.5
SPREADSHEET

Ideko's Planned Debt and Interest Payments

	Year	2005	2006	2007	2008	2009	2010
Debt and Interest Table (\$ 000)							
1	Outstanding Debt	100,000	100,000	100,000	115,000	120,000	120,000
2	Interest on Term Loan	6.80%	(6,800)	(6,800)	(6,800)	(7,820)	(8,160)

²Equation 19.2 assumes that changes in debt occur at the end of the year. If debt changes throughout the year, it is more accurate to compute interest expenses based on the average level of debt during the year.

TABLE 19.6
SPREADSHEET**Sources and Uses of Funds for the Ideko Acquisition**

Acquisition Financing (\$ 000)			
Sources		Uses	
1 New Term Loan	100,000	Purchase Ideko Equity	150,000
2 Excess Ideko Cash	6,500	Repay Existing Ideko Debt	4,500
3 KKP Equity Investment	53,000	Advisory and Other Fees	5,000
4 Total Sources of Funds	159,500	Total Uses of Funds	159,500

In addition to the tax benefit, the loan will allow KKP to limit its investment in Ideko and preserve its capital for other investments and acquisitions. The sources and uses of funds for the acquisition are shown in Table 19.6. In addition to the \$150 million purchase price for Ideko's equity, \$4.5 million will be used to repay Ideko's existing debt. With \$5 million in advisory and other fees associated with the transaction, the acquisition will require \$159.5 million in total funds. KKP's sources of funds include the new loan of \$100 million as well as Ideko's own excess cash (which KKP will have access to). Thus, KKP's required equity contribution to the transaction is $159.5 - 100 - 6.5 = \$53$ million.

CONCEPT CHECK

1. What are the different operational improvements KKP plans to make?
2. Why is it necessary to consider these improvements to assess whether the acquisition is attractive?

19.3 Building the Financial Model

The value of any investment opportunity arises from the future cash flows it will generate. To estimate the cash flows resulting from the investment in Ideko, we begin by projecting Ideko's future earnings. We then consider Ideko's working capital and investment needs and estimate its free cash flow. With these data in hand, we can forecast Ideko's balance sheet and statement of cash flows.

Forecasting Earnings

We can forecast Ideko's income statement for the five years following the acquisition based on the operational and capital structure changes proposed. This income statement is often referred to as a **pro forma** income statement, because it is not based on actual data but rather depicts the firm's financials under a given set of hypothetical assumptions. The pro forma income statement translates our expectations regarding the operational improvements KKP can achieve at Ideko into consequences for the firm's earnings.

To build the pro forma income statement, we begin with Ideko's sales. Each year, sales can be calculated from the estimates in Table 19.3 as follows:

$$\text{Sales} = \text{Market Size} \times \text{Market Share} \times \text{Average Sales Price} \quad (19.3)$$

For example, in 2006, Ideko has projected sales of $10.5 \text{ million} \times 11\% \times 76.5 = \88.358 million. The spreadsheet in Table 19.7 shows Ideko's current (2005) sales as well as projections for five years after the acquisition (2006–2010).

Joseph L. Rice, III is a founding partner and the former chairman of Clayton, Dubilier & Rice (CD&R). The firm is among the most respected private equity firms in the world. Its investments span a number of industry segments with enterprise values ranging from \$1 billion to \$15 billion.

QUESTION: *How has private equity business changed since you began in the industry?*

ANSWER: The term “private equity” is very broad and today can cover virtually every kind of investing, short of investing in the stock or bond markets. The buyout business represents a significant component of the private equity market. Since I started in 1966, I’ve seen many changes as the asset class has matured. In the 1960s and 1970s, the buyout business had relatively little following. Limited capital availability kept transactions small, and we relied on unconventional funding sources. The total purchase price of my first transaction was approximately \$3 million, financed through a secured bank line and from individuals contributing amounts ranging from \$25,000 to \$50,000. In contrast, in 2005, we bought Hertz from Ford for approximately \$15 billion.

As the industry has evolved, the attractive returns generated from buyout investments has attracted broader interest from both institutions and high net worth individuals. Buyout firms apply a variety of value creation models, including financial engineering, multiple arbitrage, and industry sector bets, such as technology or healthcare. Today there is more focus on generating returns from improving business performance—which has always been CD&R’s underlying investment approach. The character of the businesses that we buy has also changed. Traditionally, this was an asset-heavy business, with much of the financing coming from banks that lent against percentages of inventory and receivables and the liquidation value of hard assets. Now it’s become more of a cash flow business.

QUESTION: *What makes a company a good buyout candidate?*

ANSWER: We look to acquire good businesses at fair prices. Acquiring non-core, underperforming divisions of large companies and making them more effective

INTERVIEW WITH JOSEPH L. RICE, III



has been a fertile investment area for CD&R. These divestiture buyouts tend to be complex and require experience and patience to execute. For example, we were in discussions with Ford management for three years prior to leading the Hertz division acquisition.

After running a series of projections based on information from management, we develop a capital structure designed to insure the viability of the acquisition candidate. We are relatively unconcerned with EPS but are very return conscious, focusing on cash and creating long-term shareholder value. We must also believe that we can generate a return on equity that meets our standards and justifies our investors’ commitments to us.

We also acquire businesses confronting strategic issues where our operating expertise can bring value, such as Kinko’s, a great brand franchise that we reorganized and expanded. We prefer service and distribution businesses to large manufacturers because of the wage differential between Asia and the United States and Europe. We also prefer businesses with a diversity of suppliers and customers and where there are multiple levers under our control to improve operating performance.

QUESTION: *Post acquisition, what is the role of the private equity firm?*

ANSWER: CD&R brings both a hands-on ownership style and capital. After closing a transaction, we assess current management’s capability to do the job our investment case calls for. If necessary, we build and strengthen the management team. Then we work with them to determine the appropriate strategy to produce outstanding results. Finally, we aggressively pursue productivity, cost reduction, and growth initiatives to enhance operating and financial performance. At Kinko’s, we restructured 129 separate S-corporations into one centralized corporation and installed a new management team. Our key strategic decision was transforming Kinko’s from a loose confederation of consumer and small business-oriented copy shops into a highly networked company serving major corporations. In the end, that is what made the company an attractive acquisition for FedEx in 2004.

TABLE 19.7
SPREADSHEET**Pro Forma Income Statement for Ideko, 2005–2010**

	Year	2005	2006	2007	2008	2009	2010
Income Statement (\$ 000)							
1 Sales		75,000	88,358	103,234	119,777	138,149	158,526
2 Cost of Goods Sold							
3 Raw Materials		(16,000)	(18,665)	(21,593)	(24,808)	(28,333)	(32,193)
4 Direct Labor Costs		(18,000)	(21,622)	(25,757)	(30,471)	(35,834)	(41,925)
5 Gross Profit		41,000	48,071	55,883	64,498	73,982	84,407
6 Sales and Marketing		(11,250)	(14,579)	(18,582)	(23,356)	(27,630)	(31,705)
7 Administrative		(13,500)	(13,254)	(15,485)	(16,769)	(17,959)	(20,608)
8 EBITDA		16,250	20,238	21,816	24,373	28,393	32,094
9 Depreciation		(5,500)	(5,450)	(5,405)	(6,865)	(7,678)	(7,710)
10 EBIT		10,750	14,788	16,411	17,508	20,715	24,383
11 Interest Expense (net)		(75)	(6,800)	(6,800)	(6,800)	(7,820)	(8,160)
12 Pretax Income		10,675	7,988	9,611	10,708	12,895	16,223
13 Income Tax		(3,736)	(2,796)	(3,364)	(3,748)	(4,513)	(5,678)
14 Net Income		6,939	5,193	6,247	6,960	8,382	10,545

The next items in the income statement detail the cost of goods sold. The raw materials cost can be calculated from sales as

$$\text{Raw Materials} = \text{Market Size} \times \text{Market Share} \times \text{Raw Materials per Unit} \quad (19.4)$$

In 2006, the cost of raw materials is $10.5 \text{ million} \times 11\% \times 16.16 = \18.665 million . The same method can be applied to determine the direct labor costs. Sales, marketing, and administrative costs can be computed directly as a percentage of sales. For example:

$$\text{Sales and Marketing} = \text{Sales} \times (\text{Sales and Marketing \% of Sales}) \quad (19.5)$$

Therefore, sales and marketing costs are forecast to be $\$88.358 \text{ million} \times 16.5\% = \14.579 million in 2006.

Deducting these operating expenses from Ideko's sales, we can project EBITDA over the next five years as shown in line 8 of Table 19.7. Subtracting the depreciation expenses we estimated in Table 19.4, we arrive at Ideko's earnings before interest and taxes. Next, we deduct interest expenses according to the schedule given in Table 19.5.³ The final expense is the corporate income tax, which we computed using the tax rate in Table 19.3 as

$$\text{Income Tax} = \text{Pretax Income} \times \text{Tax Rate} \quad (19.6)$$

After income taxes, we are left with Ideko's projected pro forma net income as the bottom line in Table 19.7. Based on our projections, net income will rise by 52% from \$6.939 million to \$10.545 million at the end of five years, although it will drop in the near term due to the large increase in interest expense from the new debt.

³This interest expense should be offset by any interest earned on investments. As we discuss later in this chapter, we assume that Ideko does not invest its excess cash balances, but instead pays them out to its owner, KKP. Thus, net interest expenses are solely due to Ideko's outstanding debt.

EXAMPLE 19.3**Forecasting Income****Problem**

By what percentage is Ideko's EBITDA expected to grow over the five-year period? By how much would it grow if Ideko's market share remained at 10%?

Solution

EBITDA will increase from \$16.25 million to \$32.09 million, or $(32.09/16.25) - 1 = 97\%$, over the five years. With a 10% market share rather than a 15% market share, sales will be only $(10\%/15\%) = 66.7\%$ of the forecast in Table 19.7. Because Ideko's operating expenses are proportional to its unit sales, its expenses and EBITDA will also be 66.7% of the current estimates. Thus, EBITDA will grow to $66.7\% \times 32.09 = \$21.40$ million, which is an increase of only $(21.40/16.25) - 1 = 32\%$.

Working Capital Requirements

The spreadsheet in Table 19.8 lists Ideko's current working capital requirements and forecasts the firm's future working capital needs. (See Chapter 26 for a further discussion of working capital requirements and their determinants.) This forecast includes the plans to tighten Ideko's credit policy, speed up customer payments, and reduce Ideko's inventory of raw materials.

Based on these working capital requirements, the spreadsheet in Table 19.9 forecasts Ideko's net working capital (NWC) over the next five years. Each line item in the spreadsheet is found by computing the appropriate number of days' worth of the corresponding revenue or expense from the income statement (Table 19.7). For example, accounts receivable in 2006 is calculated as⁴

$$\begin{aligned} \text{Accounts Receivable} &= \text{Days Required} \times \frac{\text{Annual Sales}}{365 \text{ days/yr}} \\ &= 60 \text{ days} \times \frac{\$88.358 \text{ million/yr}}{365 \text{ days/yr}} = \$14.525 \text{ million} \quad (19.7) \end{aligned}$$

TABLE 19.8
SPREADSHEET**Ideko's Working Capital Requirements**

		Year	2005	>2005
Working Capital Days				
Assets		Based on:	Days	Days
1	Accounts Receivable	Sales Revenue	90	60
2	Raw Materials	Raw Materials Costs	45	30
3	Finished Goods	Raw Materials + Labor Costs	45	45
4	Minimum Cash Balance	Sales Revenue	30	30
Liabilities				
5	Wages Payable	Direct Labor + Admin Costs	15	15
6	Other Accounts Payable	Raw Materials + Sales and Marketing	45	45

⁴If products are highly seasonal, large fluctuations in working capital may occur over the course of the year. When these effects are important, it is best to develop forecasts on a quarterly or monthly basis so that the seasonal effects can be tracked.

TABLE 19.9
SPREADSHEET**Ideko's Net Working Capital Forecast**

	Year	2005	2006	2007	2008	2009	2010
Working Capital (\$ 000)							
Assets							
1	Accounts Receivable	18,493	14,525	16,970	19,689	22,709	26,059
2	Raw Materials	1,973	1,534	1,775	2,039	2,329	2,646
3	Finished Goods	4,192	4,967	5,838	6,815	7,911	9,138
4	Minimum Cash Balance	6,164	7,262	8,485	9,845	11,355	13,030
5	Total Current Assets	30,822	28,288	33,067	38,388	44,304	50,872
Liabilities							
6	Wages Payable	1,294	1,433	1,695	1,941	2,211	2,570
7	Other Accounts Payable	3,360	4,099	4,953	5,938	6,900	7,878
8	Total Current Liabilities	4,654	5,532	6,648	7,879	9,110	10,448
Net Working Capital							
9	Net Working Capital (5 – 8)	26,168	22,756	26,419	30,509	35,194	40,425
10	Increase in Net Working Capital		(3,412)	3,663	4,089	4,685	5,231

Similarly, Ideko's inventory of finished goods will be $45 \times (18.665 + 21.622)/365 = \4.967 million.

Table 19.9 also lists Ideko's minimum cash balance each year. This balance represents the minimum level of cash needed to keep the business running smoothly, allowing for the daily variations in the timing of income and expenses. Firms generally earn little or no interest on these balances, which are held in cash or in a checking or short-term savings accounts. As a consequence, we account for this opportunity cost by including the minimal cash balance as part of the firm's working capital.

We assume that Ideko will earn no interest on this minimal balance. (If it did, this interest would reduce the firm's net interest expense in the income statement.) We also assume that Ideko will pay out as dividends all cash not needed as part of working capital. Therefore, Ideko will hold no excess cash balances or short-term investments above the minimal level reported in Table 19.9. If Ideko were to retain excess funds, these balances would be considered part of its financing strategy (reducing its net debt), and not as part of its working capital.⁵

Ideko's net working capital for each year is computed in Table 19.9 as the difference between the forecasted current assets and current liabilities. Increases in net working capital represent a cost to the firm. Note that as a result of the improvements in accounts receivable and inventory management, Ideko will reduce its net working capital by more than \$3.4 million in 2006. After this initial savings, working capital needs will increase in conjunction with the growth of the firm.

Forecasting Free Cash Flow

We now have the data needed to forecast Ideko's free cash flows over the next five years. Ideko's earnings are available from the income statement (Table 19.7), as are its depreciation and interest expenses. Capital expenditures are available from Table 19.4, and changes in net working capital can be found in Table 19.9. We combine these items to estimate the free cash flows in the spreadsheet in Table 19.10.

⁵Firms often hold excess cash in anticipation of future investment needs or possible cash shortfalls. Because Ideko can rely on KKP to provide needed capital, excess cash reserves are unnecessary.

TABLE 19.10
SPREADSHEET**Ideko's Free Cash Flow Forecast**

	Year	2005	2006	2007	2008	2009	2010
Free Cash Flow (\$ 000)							
1 Net Income			5,193	6,247	6,960	8,382	10,545
2 Plus: After-Tax Interest Expense			4,420	4,420	4,420	5,083	5,304
3 Unlevered Net Income			9,613	10,667	11,380	13,465	15,849
4 Plus: Depreciation			5,450	5,405	6,865	7,678	7,710
5 Less: Increases in NWC			3,412	(3,663)	(4,089)	(4,685)	(5,231)
6 Less: Capital Expenditures			(5,000)	(5,000)	(20,000)	(15,000)	(8,000)
7 Free Cash Flow of Firm			13,475	7,409	(5,845)	1,458	10,328
8 Plus: Net Borrowing			—	—	15,000	5,000	—
9 Less: After-Tax Interest Expense			(4,420)	(4,420)	(4,420)	(5,083)	(5,304)
10 Free Cash Flow to Equity			9,055	2,989	4,735	1,375	5,024

To compute Ideko's free cash flow, which excludes cash flows associated with leverage, we first adjust net income by adding back the after-tax interest payments associated with the net debt in its capital structure:⁶

$$\text{After-Tax Interest Expense} = (1 - \text{Tax Rate}) \times (\text{Interest on Debt} - \text{Interest on Excess Cash}) \quad (19.8)$$

Because Ideko has no excess cash, its after-tax interest expense in 2006 is $(1 - 35\%) \times 6.8 = \4.42 million, providing unlevered net income of $5.193 + 4.42 = \$9.613$ million. We could also compute the unlevered net income in Table 19.10 by starting with EBIT and deducting taxes. In 2006, for example, EBIT is forecasted as \$14.788 million, which amounts to $14.788 \times (1 - 35\%) = \9.613 million after taxes.

To compute Ideko's free cash flow from its unlevered net income, we add back depreciation (which is not a cash expense), and deduct Ideko's increases in net working capital and capital expenditures. The free cash flow on line 7 of Table 19.10 shows the cash the firm will generate for its investors, both debt and equity holders. While Ideko will generate substantial free cash flow over the next five years, the level of free cash flow varies substantially from year to year. It is highest in 2006 (due mostly to the large reduction in working capital) and is forecasted to be negative in 2008 (when the plant expansion will begin).

To determine the free cash flow to equity, we first add Ideko's net borrowing (that is, increases to net debt):

$$\text{Net Borrowing in Year } t = \text{Net Debt in Year } t - \text{Net Debt in Year } (t - 1) \quad (19.9)$$

Ideko will borrow in 2008 and 2009 as part of its expansion. We then deduct the after-tax interest payments that were added in line 2.

As shown in the last line of Table 19.10, during the next five years Ideko is expected to generate a positive free cash flow to equity, which will be used to pay dividends to KKP. The free cash flow to equity will be highest in 2006; by 2010, KKP will recoup a significant fraction of its initial investment.

⁶If Ideko had some interest income or expenses from working capital, we would *not* include that interest here. We adjust only for interest that is related to the firm's *financing*—that is, interest associated with debt and *excess* cash (cash not included as part of working capital).

EXAMPLE 19.4**Leverage and Free Cash Flow****Problem**

Suppose Ideko does not add leverage in 2008 and 2009, but instead keeps its debt fixed at \$100 million until 2010. How would this change in its leverage policy affect its expected free cash flow? How would it affect the free cash flow to equity?

Solution

Because free cash flow is based on unlevered net income, it will not be affected by Ideko's leverage policy. Free cash flow to equity will be affected, however. Net borrowing will be zero each year, and the firm's after-tax interest expense will remain at the 2006 level of \$4.42 million:

	Year	2005	2006	2007	2008	2009	2010
Free Cash Flow (\$ 000)							
1	Free Cash Flow of Firm		13,475	7,409	(5,845)	1,458	10,328
2	Plus: Net Borrowing		—	—	—	—	—
3	Less: After-Tax Interest Expense		(4,420)	(4,420)	(4,420)	(4,420)	(4,420)
4	Free Cash Flow to Equity		9,055	2,989	(10,265)	(2,962)	5,908

In this case, Ideko will have a negative free cash flow to equity in 2008 and 2009. That is, without additional borrowing, KKP will have to invest additional capital in the firm to fund the expansion.

USING EXCEL**Summarizing Model Outputs**

After completing our earnings and cash flow forecasts, it is helpful to create a summary table with key outputs from the model. The spreadsheet below highlights Ideko's revenue, earnings, and cash flow forecasts. In addition to the raw numbers, we provide further context by calculating annual growth rates, margins, cumulative cash flows, and leverage multiples:

	AB	C	D	E	F	G	H	I	J
1									2005-2010
2	Ideko Key Financials	2005	2006	2007	2008	2009	2010		CAGR
3	Revenue	75,000	88,358	103,234	119,777	138,149	158,526		16.1%
4	% Growth		17.8%	16.8%	16.0%	15.3%	14.8%		
5	EBITDA	16,250	20,238	21,816	24,373	28,393	32,094		14.6%
6	% Margin	21.7%	22.9%	21.1%	20.3%	20.6%	20.2%		
7	EBIT	10,750	14,788	16,411	17,508	20,715	24,383		17.8%
8	% Margin	14.3%	16.7%	15.9%	14.6%	15.0%	15.4%		
9	Net Income	6,939	5,193	6,247	6,980	8,382	10,545		8.7%
10	% Margin	9.3%	5.9%	6.1%	5.8%	6.1%	6.7%		
11	Cash Flows								Cumulative
12	FCF		13,475	7,409	(5,845)	1,458	10,328		26,825
13	Dividends (FCFE)		9,055	2,989	4,735	1,375	5,024		23,178
14	Leverage								
15	Debt / EBITDA	6.2x	4.9x	4.6x	4.7x	4.2x	3.7x		
16	EBITDA / Interest		3.0x	3.2x	3.6x	3.6x	3.9x		

The summary above quickly reveals insights not immediately apparent from the raw numbers. First, note that while revenue is growing by 16.1% per year on average (as indicated by the compound annual growth rate), we see that year-over-year growth is declining over the forecast period. Second, EBITDA is growing more slowly than revenue (14.6% versus 16.1%) due to falling margins, indicating that Ideko's operating expenses are growing faster than revenues. EBIT growth is higher, however, at 17.8% per year, because the firm's improved capital efficiency means depreciation grows more slowly than revenues. Finally, net income growth is only 8.7% because of Ideko's increased interest payments, though net profits margins begin to improve after 2008.

Note the higher variability of Ideko's cash flows compared to earnings due to its fluctuating investment needs. The cumulative cash flows show that equity holders hope to recoup over \$23 million, or nearly half their investment, from the first five years of dividends alone. Finally, the leverage multiples show that while initial leverage is high, Ideko will quickly reduce leverage and improve its interest coverage ratio.

The Balance Sheet and Statement of Cash Flows (Optional)

The information we have calculated so far can be used to project Ideko's balance sheet and statement of cash flows through 2010. While these statements are not critical for our valuation, they often prove helpful in providing a more complete picture of how a firm will grow during the forecast period. These statements for Ideko are shown in the spreadsheets in Table 19.11 and Table 19.12.

The statement of cash flows in Table 19.11 starts with net income. Cash from operating activities includes depreciation as well as changes to working capital items (other than cash) from Table 19.9. Note that increases in accounts receivable or inventory are a *use* of cash, whereas an increase in accounts payable is a *source* of cash. Cash from investing activities includes the capital expenditures in Table 19.4. Cash from financing activities includes changes in outstanding debt from Table 19.5, and dividends or stock issuance determined by the free cash flow to equity in Table 19.10. (If free cash flow to equity were negative in any year, it would appear as stock issuance in line 13 in Table 19.11.) As a final check on the calculations, note that the change in cash and cash equivalents on line 15 in Table 19.11 equals the change in the minimum cash balance shown in Table 19.9.

For the balance sheet, we must begin by adjusting Ideko's closing 2005 balance sheet from Table 19.1 to account for the transaction. The transaction will affect the firm's goodwill, stockholders' equity, and its cash and debt balances. We show these changes by constructing a new "pro forma" 2005 balance sheet that represents the firm just after the transaction closes (see Table 19.12). Let's consider how each change is determined.

First, we calculate goodwill as the difference between the acquisition price and the value of the net assets acquired:

$$\text{New Goodwill} = \text{Acquisition Price} - \text{Value of Net Assets Acquired} \quad (19.10)$$

In this case, the acquisition price is the \$150 million purchase price for Ideko's existing equity. For the net assets acquired, we use Ideko's existing book value of equity, \$77.668

TABLE 19.11
SPREADSHEET

Pro Forma Statement of Cash Flows for Ideko, 2005–2010

	Year	2005	2006	2007	2008	2009	2010
Statement of Cash Flows (\$ 000)							
1	Net Income		5,193	6,247	6,960	8,382	10,545
2	Depreciation		5,450	5,405	6,865	7,678	7,710
3	Changes in Working Capital						
4	Accounts Receivable		3,968	(2,445)	(2,719)	(3,020)	(3,350)
5	Inventory		(336)	(1,112)	(1,242)	(1,385)	(1,544)
6	Accounts Payable		878	1,116	1,231	1,231	1,338
7	Cash from Operating Activities		15,153	9,211	11,095	12,885	14,699
8	Capital Expenditures		(5,000)	(5,000)	(20,000)	(15,000)	(8,000)
9	Other Investment		—	—	—	—	—
10	Cash from Investing Activities		(5,000)	(5,000)	(20,000)	(15,000)	(8,000)
11	Debt Issuance (or Repayment)		—	—	15,000	5,000	—
12	Dividends		(9,055)	(2,989)	(4,735)	(1,375)	(5,024)
13	Sale (or Purchase) of Stock		—	—	—	—	—
14	Cash from Financing Activities		(9,055)	(2,989)	10,265	3,625	(5,024)
15	Change in Cash (7 + 10 + 14)		1,098	1,223	1,360	1,510	1,675

TABLE 19.12
SPREADSHEET**Pro Forma Balance Sheet for Ideko, 2005–2010**

	Year	2005PF	2006	2007	2008	2009	2010
Balance Sheet (\$ 000)							
Assets							
1 Cash and Cash Equivalents		6,164	7,262	8,485	9,845	11,355	13,030
2 Accounts Receivable		18,493	14,525	16,970	19,689	22,709	26,059
3 Inventories		6,165	6,501	7,613	8,854	10,240	11,784
4 Total Current Assets		30,822	28,288	33,067	38,388	44,304	50,872
5 Property, Plant, and Equipment		49,500	49,050	48,645	61,781	69,102	69,392
6 Goodwill		72,332	72,332	72,332	72,332	72,332	72,332
7 Total Assets		152,654	149,670	154,044	172,501	185,738	192,597
Liabilities							
8 Accounts Payable		4,654	5,532	6,648	7,879	9,110	10,448
9 Debt		100,000	100,000	100,000	115,000	120,000	120,000
10 Total Liabilities		104,654	105,532	106,648	122,879	129,110	130,448
Stockholders' Equity							
11 Starting Stockholders' Equity			48,000	44,138	47,396	49,621	56,628
12 Net Income		(5,000)	5,193	6,247	6,960	8,382	10,545
13 Dividends			(9,055)	(2,989)	(4,735)	(1,375)	(5,024)
14 Sale (or Purchase) of Stock		53,000	—	—	—	—	—
15 Stockholders' Equity		48,000	44,138	47,396	49,621	56,628	62,149
16 Total Liabilities and Equity		152,654	149,670	154,044	172,501	185,738	192,597

million, *excluding* any pre-existing goodwill (\$0). Thus, new goodwill is $150 - (77.668 - 0) = \$72.332$ million.⁷

Next, consider stockholders' equity, which we calculate as

$$\text{New Stockholders' Equity} = \text{Equity Contributions} - \text{Expensed Transaction Fees} \quad (19.11)$$

In this case, we deduct the \$5 million in advisory fee expenses from KKP's initial equity contribution of \$53 million to calculate new shareholders' equity of \$48 million.⁸

Finally, we adjust Ideko's cash and debt balances by deducting the excess cash of \$6.5 million used to help fund the transaction, eliminating Ideko's existing debt (which will be repaid), and adding the new debt incurred of \$100 million. These steps complete the pro forma 2005 balance sheet as of the completion of the transaction and are shown in Table 19.12.

To construct the balance sheet going forward, adjust the cash balance to reflect the change in cash from line 15 of the statement of cash flows in Table 19.11. All other current assets and liabilities come from the net working capital spreadsheet (Table 19.9). The inventory entry on the balance sheet includes both raw materials and finished goods. Property, plant, and equipment information comes from the capital expenditure spreadsheet

⁷We have simplified somewhat the balance sheet adjustments in the goodwill calculation. In particular, the book values of Ideko's assets and liabilities may be "stepped up" to reflect current fair values, and a portion of the excess purchase price may be allocated to intangible assets as opposed to goodwill.

⁸Under SFAS 141R, as of December 2008, acquisition-related advisory and legal fees should be expensed as incurred. To illustrate this method, we have applied it here. Financing fees related to debt issuance, on the other hand, are not deducted from shareholders' equity, but are instead capitalized as a new asset and amortized over the loan's term. For simplicity, we have ignored these fees and any related tax consequences.

USING EXCEL

Auditing Your Financial Model

Building a three-statement financial model is complex, and even experienced financial modelers make mistakes. Audit your model to check for errors using the following best practices.

Do Not Use Stockholders' Equity as a Plug

Because the balance sheet must balance, it is tempting to calculate stockholders' equity in the spreadsheet as the difference between total assets and total liabilities. Doing so, however, nullifies a key benefit of the balance sheet as an audit tool on the rest of the financial model. As a best practice, always calculate stockholders' equity directly as shown here. Then, in a line below the balance sheet, include a "check" line that tests whether total assets equal total liabilities and equity each year—and if the difference is non-zero, display it in a bold red font so the error is apparent.

If the model does have an error, the amount of the difference will provide a clue to its source: If the difference is a constant each year, it is likely to result from one of the adjustments in the initial pro forma balance sheet; if it varies over time, check each line of the balance sheet to be sure the trends make sense (e.g., that inventories grow with sales), and check the sign of each line item in the statement of cash flows to make sure all sources of cash are positive and all uses of cash are negative (especially working capital).

Check the Cash and Payout Policy

Check that the firm's cash and payouts match its intended policy. In this case, we forecast that Ideko would pay out all excess cash going forward. Thus, its cash balances in the balance sheet should match the minimum balances specified in Table 19.9. If not, be sure to audit the free cash flow to equity calculation in Table 19.10. Note also that in this example, free cash flow to equity (i.e., dividends) was an output of the model. We could have alternatively assumed Ideko would retain its excess cash. In that case, free cash flow to equity would be zero (no dividends would be paid), and changes in cash (and thus the firm's net borrowing) would be outputs.

(Table 19.4), and the debt comes from Table 19.5. From Eq. 2.7, stockholders' equity increases each year through retained earnings (net income less dividends) and new capital contributions (stock issuances net of repurchases). Dividends after 2005 are equal to positive free cash flows to equity from Table 19.10, whereas negative free cash flows to equity represent a stock issuance. As a check on the calculations, note that the balance sheet does, indeed, balance: Total assets equal total liabilities and equity.⁹

Ideko's book value of equity will decline in 2006, as Ideko reduces its working capital and pays out the savings as part of a large dividend. The firm's book value will then rise as it expands. Ideko's book debt-equity ratio will decline from $100,000/48,000 = 2.1$ to $120,000/62,149 = 1.9$ during the five-year period.

CONCEPT CHECK

1. What is a pro forma income statement?
2. How do we calculate the firm's free cash flow, and the free cash flow to equity?

⁹In Table 19.12, goodwill is assumed to remain constant. If the transaction were structured as an acquisition of assets (as opposed to stock), the goodwill would be amortizable over 15 years for tax reporting as per section 197 of the Internal Revenue Code. For financial accounting purposes, goodwill is not amortized but is subject to an impairment test at least once a year as specified in FASB 142 (though changes in goodwill due to impairment have no tax accounting consequence).

19.4 Estimating the Cost of Capital

To value KKP's investment in Ideko, we need to assess the risk associated with Ideko and estimate an appropriate cost of capital. Because Ideko is a private firm, we cannot use its own past returns to evaluate its risk, but must instead rely on comparable publicly traded firms. In this section, we use data from the comparable firms identified earlier to estimate a cost of capital for Ideko.

Our approach is as follows. First, we use the techniques developed in Part 4 to estimate the equity cost of capital for Oakley, Luxottica Group, and Nike. Then, we estimate the unlevered cost of capital for each firm based on its capital structure. We use the unlevered cost of capital of the comparable firms to estimate Ideko's unlevered cost of capital. Once we have this estimate, we can use Ideko's capital structure to determine its equity cost of capital or WACC, depending on the valuation method employed.

CAPM-Based Estimation

To determine an appropriate cost of capital, we must first determine the appropriate measure of risk. KKP's investment in Ideko will represent a large fraction of its portfolio. As a consequence, KKP itself is not well diversified. But KKP's investors are primarily pension funds and large institutional investors, which are themselves well diversified and which evaluate their performance relative to the market as a benchmark. Thus, you decide that estimating market risk using the CAPM approach is justified.

Using the CAPM, we can estimate the equity cost of capital for each comparable firm based on the beta of its equity. As outlined in Chapter 12, the standard approach to estimating an equity beta is to determine the historical sensitivity of the stock's returns to the market's returns by using linear regression to estimate the slope coefficient in the equation:

$$\underbrace{R_s - r_f}_{\text{Excess return of stock } s} = \alpha_s + \beta_s \underbrace{(R_{mkt} - r_f)}_{\text{Excess return of market portfolio}} + \varepsilon_s \quad (19.12)$$

As a proxy for the market portfolio, we will use a value-weighted portfolio of all NYSE, AMEX, and NASDAQ stocks. With data from 2000 to 2004, we calculate the excess return—the realized return minus the yield on a one-month Treasury security—for each firm and for the market portfolio. We then estimate the equity beta for each firm by regressing its excess return onto the excess return of the market portfolio. We perform the regression for both monthly returns and 10-day returns. The estimated equity betas, together with their 95% confidence intervals, are shown in Table 19.13.

TABLE 19.13

Equity Betas with Confidence Intervals for Comparable Firms

Firm	Monthly Returns		10-Day Returns	
	Beta	95% C.I.	Beta	95% C.I.
Oakley	1.99	1.2 to 2.8	1.37	0.9 to 1.9
Luxottica	0.56	0.0 to 1.1	0.86	0.5 to 1.2
Nike	0.48	−0.1 to 1.0	0.69	0.4 to 1.0

While we would like to assess risk and, therefore, estimate beta based on longer horizon returns (consistent with our investors' investment horizon), the confidence intervals we obtain using monthly data are extremely wide. These confidence intervals narrow somewhat when we use 10-day returns. In any case, the results make clear that a fair amount of uncertainty persists when we estimate the beta for an individual firm.

Unlevering Beta

Given an estimate of each firm's equity beta, we next "unlever" the beta based on the firm's capital structure. Here we use Eq. 12.9 (which is equivalent, in terms of returns, to calculating the pretax WACC as in Eq. 18.6):

$$\beta_U = \left(\frac{\text{Equity Value}}{\text{Enterprise Value}} \right) \beta_E + \left(\frac{\text{Net Debt Value}}{\text{Enterprise Value}} \right) \beta_D \quad (19.13)$$

Recall that we must use the *net* debt of the firm—that is, we must subtract any cash from the level of debt—so we use the enterprise value of the firm as the sum of debt and equity in the formula.¹⁰ Table 19.14 shows the capital structure for each comparable firm. Oakley has no debt, while Luxottica has about 17% debt in its capital structure. Nike holds cash that exceeds its debt, leading to a negative net debt in its capital structure.

Table 19.14 also estimates the unlevered beta of each firm. Here, we have used an equity beta for each firm within the range of the results from Table 19.13. Given the low or negative debt levels for each firm, assuming a beta for debt of zero is a reasonable approximation. Then, we compute an unlevered beta for each firm according to Eq. 19.13.

The range of the unlevered betas for these three firms is large. Both Luxottica and Nike have relatively low betas, presumably reflecting the relative noncyclicality of their core businesses (prescription eyewear for Luxottica and athletic shoes for Nike). Oakley has a much higher unlevered beta, perhaps because the high-end specialty sports eyewear it produces is a discretionary expense for most consumers.

Ideko's Unlevered Cost of Capital

The data from the comparable firms provides guidance to us for estimating Ideko's unlevered cost of capital. Ideko's products are not as high end as Oakley's eyewear, so their sales are unlikely to vary as much with the business cycle as Oakley's sales do. However, Ideko does not have a prescription eyewear division, as Luxottica does. Ideko's products are also fashion items rather than exercise items, so we expect Ideko's cost of capital to be closer to

TABLE 19.14 Capital Structure and Unlevered Beta Estimates for Comparable Firms

Firm	$\frac{E}{E+D}$	$\frac{D}{E+D}$	β_E	β_D	β_U
Oakley	1.00	0.00	1.50	—	1.50
Luxottica	0.83	0.17	0.75	0	0.62
Nike	1.05	−0.05	0.60	0	0.63

¹⁰Recall from Chapter 18 that Eq. 19.13 assumes that the firm will maintain a target leverage ratio. If the debt is expected to remain fixed for some period, we should also deduct the value of the predetermined tax shields from the firm's net debt.

Oakley's than to Nike's or Luxottica's. Therefore, we use 1.20 as our preliminary estimate for Ideko's unlevered beta, which is somewhat above the average of the comparables in Table 19.14.

We use the security market line of the CAPM to translate this beta into a cost of capital for Ideko. In mid-2005, one-year Treasury rates were approximately 4%; we use this rate for the risk-free interest rate. We also need an estimate of the market risk premium. Since 1960, the average annual return of the value-weighted market portfolio of U.S. stocks has exceeded that of one-year Treasuries by approximately 5%. However, this estimate is a backward-looking number. As we mentioned in Chapter 12, some researchers believe that future stock market excess returns are likely to be lower than this historical average. To be conservative in our valuation of Ideko, we will use 5% as the expected market risk premium.

Based on these choices, our estimate of Ideko's unlevered cost of capital is

$$\begin{aligned} r_U &= r_f + \beta_U(E[R_{mkt}] - r_f) = 4\% + 1.20(5\%) \\ &= 10\% \end{aligned}$$

Of course, as our discussion has made clear, this estimate contains a large amount of uncertainty. Thus, we will include sensitivity analysis with regard to the unlevered cost of capital in our analysis.

EXAMPLE 19.5

Estimating the Unlevered Cost of Capital

Problem

Using the monthly equity beta estimates for each firm in Table 19.13, what range of unlevered cost of capital estimates is possible?

Solution

Oakley has the highest equity beta of 1.99, which is also its unlevered beta (it has no debt). With this beta, the unlevered cost of capital would be $r_U = 4\% + 1.99(5\%) = 13.95\%$. At the other extreme, given its capital structure, Luxottica's equity beta of 0.56 implies an unlevered beta of $(0.56)(0.83) = 0.46$. With this beta, the unlevered cost of capital would be $r_U = 4\% + 0.46(5\%) = 6.3\%$.

As with any analysis based on comparables, experience and judgment are necessary to come up with a reasonable estimate of the unlevered cost of capital. In this case, our choice would be guided by industry norms, an assessment of which comparable is closest in terms of market risk, and possibly knowledge of how cyclical Ideko's revenues have been historically.

CONCEPT CHECK

1. What is a standard approach to estimate an equity beta?
2. How do we estimate a firm's unlevered cost of capital using data from comparable publicly traded firms?

19.5 Valuing the Investment

Thus far, we have forecasted the first five years of cash flows from KKP's investment in Ideko, and we have estimated the investment's unlevered cost of capital. In this section, we combine these inputs to estimate the value of the opportunity. The first step is to develop an estimate of Ideko's value at the end of our five-year forecast horizon. To do so, we

consider both a multiples approach and a discounted cash flow (DCF) valuation using the WACC method. Given Ideko's free cash flow and continuation value, we then estimate its total enterprise value in 2005 using the APV method. Deducting the value of debt and KKP's initial investment from our estimate of Ideko's enterprise value gives the NPV of the investment opportunity. In addition to NPV, we look at some other common metrics, including IRR and cash multiples.

The Multiples Approach to Continuation Value

Practitioners generally estimate a firm's continuation value (also called the terminal value) at the end of the forecast horizon using a valuation multiple. While forecasting cash flows explicitly is useful in capturing those specific aspects of a company that distinguish the firm from its competitors in the short run, in the long run firms in the same industry typically have similar expected growth rates, profitability, and risk. As a consequence, multiples are likely to be relatively homogeneous across firms. Thus, applying a multiple is potentially as reliable as estimating the value based on an explicit forecast of distant cash flows.

Of the different valuation multiples available, the EBITDA multiple is most often used in practice. In most settings, the EBITDA multiple is more reliable than sales or earnings multiples because it accounts for the firm's operating efficiency and is not affected by leverage differences between firms. We estimate the continuation value using an EBITDA multiple as follows:

$$\text{Continuation Enterprise Value at Forecast Horizon} = \text{EBITDA at Horizon} \times \text{EBITDA Multiple at Horizon} \quad (19.14)$$

From the income statement in Table 19.7, Ideko's EBITDA in 2010 is forecast to be \$32.09 million. If we assume its EBITDA multiple in 2010 is unchanged from the value of 9.1 that we calculated at the time of the original purchase, then Ideko's continuation value in 2010 is $32.09 \times 9.1 = \$292.05$ million. This calculation is shown in the spreadsheet in Table 19.15. Given Ideko's outstanding debt of \$120 million in 2010, this estimate corresponds to an equity value of \$172.05 million.

Table 19.15 also shows Ideko's sales and P/E multiples based on this continuation value. The continuation value is 1.8 times Ideko's 2010 sales, and the equity value is 16.3 times Ideko's 2010 earnings. Because the P/E multiple is affected by leverage, we also report Ideko's **unlevered P/E ratio**, which is calculated as its continuing enterprise value divided by its unlevered net income in 2010 (listed in Table 19.10). Ideko would have this P/E ratio if it had no debt in 2010, so this information is useful when comparing Ideko to unlevered firms in the industry.

TABLE 19.15
SPREADSHEET

Continuation Value Estimate for Ideko

Continuation Value: Multiples Approach (\$ 000)			Common Multiples	
1	EBITDA in 2010	32,094	EV/Sales	1.8×
2	EBITDA multiple	9.1×	P/E (levered)	16.3×
3	Continuation Enterprise Value	292,052	P/E (unlevered)	18.4×
4	Debt	(120,000)		
5	Continuation Equity Value	172,052		

We can use the various multiples to assess the reasonableness of our estimated continuation value. While the value-to-sales ratio is high compared to the overall sporting goods industry, these multiples are otherwise low relative to the comparables in Table 19.2. If we expect these industry multiples to remain stable, this estimate of Ideko's continuation value seems reasonable (if not relatively conservative).

The Discounted Cash Flow Approach to Continuation Value

One difficulty with relying solely on comparables when forecasting a continuation value is that we are comparing *future* multiples of the firm with *current* multiples of its competitors. In 2010, the multiples of Ideko and the comparables we have chosen may all be very different, especially if the industry is currently experiencing abnormal growth. To guard against such a bias, it is wise to check our estimate of the continuation value based on fundamentals using a discounted cash flow approach.

To estimate a continuation value in year T using discounted cash flows, we assume a constant expected growth rate, g , and a constant debt-equity ratio. As explained in Chapter 18, when the debt-equity ratio is constant, the WACC valuation method is the simplest to apply:

$$\text{Enterprise Value in Year } T = V_T^L = \frac{FCF_{T+1}}{r_{wacc} - g} \quad (19.15)$$

To estimate free cash flow in year $T + 1$, recall that free cash flow is equal to unlevered net income plus depreciation, less capital expenditures and increases in net working capital (see Table 19.10):

$$\begin{aligned} FCF_{T+1} = & \text{Unlevered Net Income}_{T+1} + \text{Depreciation}_{T+1} \\ & - \text{Increases in NWC}_{T+1} - \text{Capital Expenditures}_{T+1} \end{aligned} \quad (19.16)$$

Suppose the firm's sales are expected to grow at a nominal rate g . If the firm's operating expenses remain a fixed percentage of sales, then its unlevered net income will also grow at rate g . Similarly, the firm's receivables, payables, and other elements of net working capital will grow at rate g .

What about capital expenditures? The firm will need new capital to offset depreciation; it will also need to add capacity as its production volume grows. Given a sales growth rate g , we may expect that the firm will need to expand its investment in fixed assets at about the same rate. In that case,¹¹

$$\text{Capital Expenditures}_{T+1} = \text{Depreciation}_{T+1} + g \times \text{Fixed Assets}_T$$

Thus, given a growth rate of g for the firm, we can estimate its free cash flow as

$$\begin{aligned} FCF_{T+1} = & (1 + g) \times \text{Unlevered Net Income}_T \\ & - g \times \text{Net Working Capital}_T - g \times \text{Fixed Assets}_T \end{aligned} \quad (19.17)$$

Together, Eq. 19.15 and Eq. 19.17 allow us to estimate a firm's continuation value based on its long-run growth rate.

¹¹Here, fixed assets are measured according to their book value net of accumulated depreciation. This level of capital expenditures is required to maintain the firm's ratio of sales to fixed assets (also called its fixed asset turnover ratio). An alternative approach—which is preferable if we anticipate a change in the turnover ratio—is to estimate increases in NWC and net investment (capital expenditures in excess of depreciation) in Eq. 19.16 as a target percentage of the *change* in sales, as we did in Chapter 9 (see Example 9.7), where the target percentage is the expected long-run ratio of NWC and PP&E to sales.

EXAMPLE 19.6**A DCF Estimate of the Continuation Value****Problem**

Estimate Ideko's continuation value in 2010 assuming a future expected growth rate of 5%, a future debt-to-value ratio of 40%, and a debt cost of capital of 6.8%.

Solution

In 2010, Ideko's unlevered net income is forecasted to be \$15.849 million (Table 19.10), with working capital of \$40.425 million (Table 19.9). It has fixed assets of \$69.392 million (Table 19.4). From Eq. 19.17, we can estimate Ideko's free cash flow in 2011:

$$FCF_{2011} = (1.05)(15.849) - (5\%)(40.425) - (5\%)(69.392) = \$11.151 \text{ million}$$

This estimate represents nearly an 8% increase over Ideko's 2010 free cash flow of \$10.328 million. It exceeds the 5% growth rate of sales due to the decline in the required additions to Ideko's net working capital as its growth rate slows.

With a debt-to-value ratio of 40%, Ideko's WACC can be calculated from Eq. 18.11 or Eq. 12.13:

$$r_{wacc} = r_U - d \tau_c r_D = 10\% - 0.40(0.35) 6.8\% = 9.05\%$$

Given the estimate of Ideko's free cash flow and WACC, we can estimate Ideko's continuation value in 2010:

$$V_{2010}^L = \frac{11.151}{9.05\% - 5\%} = \$275.33 \text{ million}$$

This continuation value represents a terminal EBITDA multiple of $275.33/32.09 = 8.6$.

Both the multiples approach and the discounted cash flow approach are useful in deriving a realistic continuation value estimate. Our recommendation is to combine both approaches, as we do in Table 19.16. As shown in the spreadsheet, our projected EBITDA multiple of 9.1 can be justified by the discounted cash flow method with a nominal long-term growth rate of about 5.3%.¹² Given an expected future inflation rate of 2.5%, this

TABLE 19.16
SPREADSHEET**Discounted Cash Flow Estimate of Continuation Value, with Implied EBITDA Multiple****Continuation Value: DCF and EBITDA Multiple (\$ 000)**

1	Long-Term Growth Rate	5.3%		
2	Target $D/(E + D)$	40.0%		
3	Projected WACC	9.05%		
Free Cash Flow in 2011				
4	Unlevered Net Income	16,695	Continuation Enterprise Value	292,052
5	Less: Increase in NWC	(2,158)		
6	Less: Net Investment*	(3,705)	Implied EBITDA Multiple	9.1×
7	Free Cash Flow	10,832		

*Net investment equals the difference between capital expenditures and depreciation, so subtracting this amount is equivalent to adding back depreciation and subtracting capital expenditures (see Eq. 9.20).

¹²The exact nominal growth rate needed to match an EBITDA multiple of 9.1 is 5.33897%, which can be found using Solver in Excel.

COMMON MISTAKE**Continuation Values and Long-Run Growth**

The continuation value is one of the most important estimates when valuing a firm. A common mistake is to use an overly optimistic continuation value, which will lead to an upward bias in the estimated current value of the firm. Be aware of the following pitfalls:

Using multiples based on current high growth rates. Continuation value estimates are often based on current valuation multiples of existing firms. But if these firms are currently experiencing high growth that will eventually slow down, their multiples can be expected to decline over time. In this scenario, if we estimate a continuation value based on today's multiples without accounting for this decline as growth slows, the estimate will be biased upward.

Ignoring investment necessary for growth. When using the discounted cash flow method, we cannot assume

that $FCF_{T+1} = FCF_T(1 + g)$ if the firm's growth rate has changed between T and $T + 1$. Whenever the growth rate changes, expenditures on working and fixed capital will be affected, and we must take this effect into account as we do in Eq. 19.17.

Using unsustainable long-term growth rates. When using the discounted cash flow method, we must choose a long-term growth rate for the firm. By choosing a high rate, we can make the continuation value estimate extremely high. In the long run, however, firms cannot continue to grow faster than the overall economy. Thus, we should be suspicious of long-term growth rates that exceed the expected rate of GDP growth, which has averaged between 2.5% and 3.5% in *real* terms (that is, not including inflation) in the United States over the past several decades.

nominal rate represents a real growth rate of about 2.8%. This implied growth rate is another important reality check for our continuation value estimate. If it is much higher than our expectations of long-run growth for the industry as a whole, we should be more skeptical of the estimate being used.

APV Valuation of Ideko's Equity

Our estimate of Ideko's continuation value summarizes the value of the firm's free cash flow beyond the forecast horizon. We can combine it with our forecast for free cash flow through 2010 (Table 19.10, line 7) to estimate Ideko's value today. Recall from Chapter 18 that because the debt is paid on a fixed schedule during the forecast period, the APV method is the easiest valuation method to apply.

The steps to estimate Ideko's value using the APV method are shown in the spreadsheet in Table 19.17. First, we compute Ideko's unlevered value V^U , which is the firm's value if we were to operate the company without leverage during the forecast period and sell it for its continuation value at the end of the forecast horizon. Thus, the final value in 2010 would be the continuation value we estimated in Table 19.15. The value in earlier periods

TABLE 19.17
SPREADSHEET

APV Estimate of Ideko's Initial Equity Value

	Year	2005	2006	2007	2008	2009	2010
APV Method (\$ 000)							
1 Free Cash Flow			13,475	7,409	(5,845)	1,458	10,328
2 Unlevered Value V^U (at 10%)		202,732	209,530	223,075	251,227	274,891	292,052
3 Interest Tax Shield			2,380	2,380	2,380	2,737	2,856
4 Tax Shield Value T^S (at 6.8%)		10,428	8,757	6,972	5,067	2,674	—
5 APV: $V^L = V^U + T^S$		213,160	218,287	230,047	256,294	277,566	292,052
6 Debt		(100,000)	(100,000)	(100,000)	(115,000)	(120,000)	(120,000)
7 Equity Value		113,160	118,287	130,047	141,294	157,566	172,052

includes the free cash flows paid by the firm (from Table 19.10) discounted at the unlevered cost of capital r_U that we estimated in Section 19.4¹³:

$$V_{t-1}^U = \frac{FCF_t + V_t^U}{1 + r_U} \quad (19.18)$$

Next, we incorporate Ideko's interest tax shield during the forecast horizon. The interest tax shield equals the tax rate of 35% (Table 19.3) multiplied by Ideko's scheduled interest payments (see Table 19.5). Because the debt levels are predetermined, we compute the value T^s of the tax shield by discounting the tax savings at the debt interest rate, $r_D = 6.80\%$:

$$T_{t-1}^s = \frac{\text{Interest Tax Shield}_t + T_t^s}{1 + r_D} \quad (19.19)$$

Combining the unlevered value and the tax shield value gives the APV, which is Ideko's enterprise value given the planned leverage policy. By deducting debt, we obtain our estimate for the value of Ideko's equity during the forecast period.

Thus, our estimate for Ideko's initial enterprise value is \$213 million, with an equity value of \$113 million. As KKP's initial cost to acquire Ideko's equity is \$53 million (see Table 19.6), based on these estimates the deal looks attractive, with an NPV of \$113 million – \$53 million = \$60 million.

A Reality Check

At this point, it is wise to step back and assess whether our valuation results make sense. Does an initial enterprise value of \$213 million for Ideko seem reasonable compared to the values of other firms in the industry?

Here again, multiples are helpful. Let's compute the initial valuation multiples that would be implied by our estimated enterprise value of \$213 million and compare them to Ideko's closest competitors as we did in Table 19.2. Table 19.18 provides our results.

Naturally, the valuation multiples based on the estimated enterprise value of \$213 million, which would correspond to a purchase price of \$215 million given Ideko's existing debt and excess cash, are higher than those based on a purchase price of \$150 million. They are now at the top end or somewhat above the range of the values of the other firms

COMMON MISTAKE

Missing Assets or Liabilities

When computing the enterprise value of a firm from its free cash flows, remember that we are valuing only those assets and liabilities whose cash flow consequences are included in our projections. Any “missing” assets or liabilities must be added to the APV estimate to determine the value of equity. In this case, we deduct the firm's debt and add any excess cash or other marketable securities that have not been included (for Ideko, excess cash has already been paid out and will remain at zero, so

no adjustment is needed). We also adjust for any other assets or liabilities that have not been explicitly considered. For example, if a firm owns vacant land, or if it has patents or other rights whose potential cash flows were not included in the projections, the value of these assets must be accounted for separately. The same is true for liabilities such as stock option grants, potential legal liabilities, leases (if the lease payments were not included in earnings), or underfunded pension liabilities.

¹³Note that in 2010, we use the continuation value of \$292 million as the final “unlevered” value, even though this value might include future tax shields (as in Example 19.6). This approach is correct as we are using the APV calculation only to value the additional tax shields earned during the forecast horizon. Combining discounting methodologies as we have done here is very useful in situations like this one in which the firm is likely to be recapitalized at the exit horizon.

TABLE 19.18**Ideko Financial Ratios Comparison, Mid-2005, Based on Discounted Cash Flow Estimate Versus Proposed Purchase Price**

Ratio	Ideko (Estimated Value)	Ideko (Purchase Price)	Oakley, Inc.	Luxottica Group	Nike, Inc.	Sporting Goods
P/E	31.0×	21.6×	24.8×	28.0×	18.2×	20.3×
EV/Sales	2.8×	2.0×	2.0×	2.7×	1.5×	1.4×
EV/EBITDA	13.1×	9.1×	11.6×	14.4×	9.3×	11.4×

that we used for comparison. While these multiples are not unreasonable given the operational improvements that KKP plans to implement, they indicate that our projections may be somewhat optimistic and depend critically on KKP's ability to achieve the operational improvements it plans.

Our estimated initial EBITDA multiple of 13.1 also exceeds the multiple of 9.1 that we assumed for the continuation value. Thus, our estimate forecasts a decline in the EBITDA multiple, which is appropriate given our expectation that growth will be higher in the short run. If the multiple did not decline, we should question whether our continuation value is too optimistic.

IRR and Cash Multiples

While the NPV method is the most reliable method when evaluating a transaction like KKP's acquisition of Ideko, real-world practitioners often use IRR and the *cash multiple* (or multiple of money) as alternative valuation metrics. We discuss both of these methods in this section.

To compute the IRR, we must compute KKP's cash flows over the life of the transaction. KKP's initial investment in Ideko, from Table 19.6, is \$53 million. KKP will then receive cash dividends from Ideko based on the free cash flow to equity reported in Table 19.10. Finally, we assume that KKP will sell its equity share in Ideko at the end of five years, receiving the continuation equity value. We combine these data to determine KKP's cash flows in the spreadsheet in Table 19.19. Given the cash flows, we compute the IRR of the transaction, which is 33.3%.

While an IRR of 33.3% might sound attractive, it is not straightforward to evaluate in this context. To do so, we must compare it to the appropriate cost of capital for KKP's investment. Because KKP holds an equity position in Ideko, we should use Ideko's equity

TABLE 19.19
SPREADSHEET**IRR and Cash Multiple for KKP's Investment in Ideko**

	Year	2005	2006	2007	2008	2009	2010
IRR and Cash Multiple							
1	Initial Investment	(53,000)					
2	Free Cash Flow to Equity		9,055	2,989	4,735	1,375	5,024
3	Continuation Equity Value						172,052
4	KKP Cash Flows	(53,000)	9,055	2,989	4,735	1,375	177,077
5	IRR	33.3%					
6	Cash Multiple	3.7×					

cost of capital. Of course, Ideko's leverage ratio changes over the five-year period, which will change the risk of its equity. Thus, there is no single cost of capital to compare to the IRR.¹⁴

The spreadsheet in Table 19.19 also computes the cash multiple for the transaction. The **cash multiple** (also called the **multiple of money** or **absolute return**) is the ratio of the total cash received to the total cash invested. The cash multiple for KKP's investment in Ideko is

$$\begin{aligned}\text{Cash Multiple} &= \frac{\text{Total Cash Received}}{\text{Total Cash Invested}} \\ &= \frac{9055 + 2989 + 4735 + 1375 + 177,077}{53,000} = 3.7 \quad (19.20)\end{aligned}$$

That is, KKP expects to receive a return that is 3.7 times its investment in Ideko. The cash multiple is a common metric used by investors in transactions such as this one. It has an obvious weakness: The cash multiple does not depend on the amount of time it takes to receive the cash, nor does it account for the risk of the investment. It is therefore useful only for comparing deals with similar time horizons and risk.

CONCEPT CHECK

1. What are the main methods of estimating the continuation value of the firm at the end of the forecast horizon?
2. What are the potential pitfalls of analyzing a transaction like this one based on its IRR or cash multiple?

19.6 Sensitivity Analysis

Any financial valuation is only as accurate as the estimates on which it is based. Before concluding our analysis, it is important to assess the uncertainty of our estimates and to determine their potential impact on the value of the deal.

Once we have developed the spreadsheet model for KKP's investment in Ideko, it is straightforward to perform a sensitivity analysis to determine the impact of changes in different parameters on the deal's value. For example, the spreadsheet in Table 19.20 shows the sensitivity of our estimates of the value of KKP's investment to changes in our assumptions regarding the exit EBITDA multiple that KKP obtains when Ideko is sold, as well as Ideko's unlevered cost of capital.

In our initial analysis, we assumed an exit EBITDA multiple of 9.1. Table 19.20 shows that each 1.0 increase in the multiple represents about \$20 million in initial value.¹⁵ KKP will break even on its \$53 million investment in Ideko with an exit multiple of slightly more than 6.0. The table also shows, however, that an exit multiple of 6.0 is consistent with a future growth rate for Ideko of less than 2%, which is even less than the expected rate of inflation and perhaps unrealistically low.

Table 19.20 also illustrates the effect of a change to our assumption about Ideko's unlevered cost of capital. A higher unlevered cost of capital reduces the value of KKP's

¹⁴See the appendix to this chapter for a calculation of Ideko's annual equity cost of capital.

¹⁵In fact, we can calculate this directly as the present value of Ideko's projected EBITDA in 2010: (\$32.094 million)/(1.10⁵) = \$19.928 million.

TABLE 19.20
SPREADSHEET**Sensitivity Analysis for KKP's Investment in Ideko**

Exit EBITDA Multiple	6.0	7.0	8.0	9.1	10.0	11.0
Implied Long-Run Growth Rate	1.60%	3.43%	4.53%	5.34%	5.81%	6.21%
Ideko Enterprise Value (in \$ million)	151.4	171.3	191.2	213.2	231.1	251.0
KKP Equity Value (in \$ million)	51.4	71.3	91.2	113.2	131.1	151.0
KKP IRR	14.8%	22.1%	28.0%	33.3%	37.1%	40.8%

Unlevered Cost of Capital	9.0%	10.0%	11.0%	12.0%	13.0%	14.0%
Implied Long-Run Growth Rate	3.86%	5.34%	6.81%	8.29%	9.76%	11.24%
Ideko Enterprise Value (in \$ million)	222.1	213.2	204.7	196.7	189.1	181.9
KKP Equity Value (in \$ million)	122.1	113.2	104.7	96.7	89.1	81.9

investment; yet, even with a rate as high as 14%, the equity value exceeds KKP's initial investment. However, if the unlevered cost of capital exceeds 12%, the implied long-term growth rate that justifies the assumed exit EBITDA multiple of 9.1 is probably unrealistically high. Thus, if we believe the unlevered cost of capital falls within this range, we should lower our forecast for the exit EBITDA multiple, which will further reduce the value of KKP's equity. Conversely, if we are confident in our estimate of the exit multiple, this analysis lends further support to our choice for the unlevered cost of capital.

The exercises at the end of this chapter continue the sensitivity analysis by considering different levels of market share growth and changes to working capital management.

CONCEPT CHECK

1. What is the purpose of the sensitivity analysis?
2. Table 19.20 shows the sensitivity analysis for KKP's investment in Ideko. Given this information, do you recommend the acquisition of Ideko?

MyFinanceLab

Here is what you should know after reading this chapter. MyFinanceLab will help you identify what you know and where to go when you need to practice.

19.1 Valuation Using Comparables

- Valuation using comparables may be used as a preliminary way to estimate the value of a firm. Common multiples include the price-earnings ratio, and the ratio of enterprise value to sales or EBITDA.

19.2 The Business Plan

- When evaluating an acquisition it is necessary to look in detail at the company's operations, investments, and capital structure, and to assess its potential for improvements and future growth.

19.3 Building the Financial Model

- The value of an investment ultimately depends on the firm's future cash flows. To estimate cash flows, one must first develop forecasts for the target firm's operations, investments, and capital structure.
- A financial model may be used to project the future cash flows from an investment.
 - A pro forma income statement projects the firm's earnings under a given set of hypothetical assumptions.
 - The financial model should also consider future working capital needs and capital expenditures to estimate future free cash flows.
 - Based on these estimates, we can forecast the balance sheet and statement of cash flows.

- When forecasting the balance sheet, we must first adjust the firm's initial balance sheet to reflect the transaction, including:

$$\text{New Goodwill} = \text{Acquisition Price} - \text{Value of Net Assets Acquired} \quad (19.10)$$

$$\text{New Stockholders' Equity} = \text{Equity Contributions} - \text{Expensed Transaction Fees} \quad (19.11)$$

- Checking whether the balance sheet balances is an important audit tool for the consistency of the financial model.

19.4 Estimating the Cost of Capital

- To value an investment, we need to assess its risk and estimate an appropriate cost of capital. One method for doing so is to use the CAPM.
 - Use the CAPM to estimate the equity cost of capital for comparable firms, based on their equity betas.
 - Given an estimate of each comparable firm's equity beta, unlever the beta based on the firm's capital structure.
 - Use the CAPM and the estimates of unlevered betas for comparable firms to estimate the unlevered cost of capital for the investment.

19.5 Valuing the Investment

- In addition to forecasting cash flows for a few years, we need to estimate the firm's continuation value at the end of the forecast horizon.
 - One method is to use a valuation multiple based on comparable firms.
 - To estimate a continuation value in year T using discounted cash flows, it is common practice to assume a constant expected growth rate g and a constant debt-equity ratio:

$$\text{Enterprise Value in Year } T = V_T^L = \frac{FCF_{T+1}}{r_{wacc} - g} \quad (19.15)$$

- Given the forecasted cash flows and an estimate of the cost of capital, the final step is to combine these inputs to estimate the value of the opportunity. We may use the valuation methods described in Chapter 18 to calculate firm value.
- While the NPV method is the most reliable approach for evaluating an investment, practitioners often use the IRR and cash multiple as alternative valuation metrics.
 - We use the cash flows over the lifetime of the investment to calculate the IRR.
 - The cash multiple for an investment is the ratio of the total cash received to the total cash invested:

$$\text{Cash Multiple} = \frac{\text{Total Cash Received}}{\text{Total Cash Invested}} \quad (19.20)$$

19.6 Sensitivity Analysis

- Sensitivity analysis is useful for evaluating the uncertainty of estimates used for valuation, and the impact of this uncertainty on the value of the deal.

Key Terms

cash multiple (multiple of money, absolute return) *p.* 748

pro forma *p.* 729

unlevered P/E ratio *p.* 742

Further Reading

For more detail on the issues involved in the valuation and financial modeling of companies and projects, see: T. Koller, M. Goedhart, and D. Wessels, *Valuation: Measuring and Managing the Value of Companies* (John Wiley & Sons, 2010); S. Benninga and O. Sarig, *Corporate Finance: A Valuation Approach* (McGraw-Hill/Irwin, 1996); E. Arzac, *Valuation for Mergers, Buyouts and Restructuring*

(John Wiley & Sons, 2007); S. Pratt, R. Reilly, and R. Schweihs, *Valuing a Business: The Analysis and Appraisal of Closely Held Companies* (McGraw-Hill, 2007); and J. Rosenbaum and J. Pearl, *Investment Banking* (John Wiley & Sons, 2009).

Problems

All problems are available in [MyFinanceLab](#). An asterisk (*) indicates problems with a higher level of difficulty.

Valuation Using Comparables

1. You would like to compare Ideko's profitability to its competitors' profitability using the EBITDA/sales multiple. Given Ideko's current sales of \$75.6 million, use the information in Table 19.2 to compute a range of EBITDA for Ideko assuming it is run as profitably as its competitors.

The Business Plan

2. Assume that Ideko's market share will increase by 0.4% per year rather than the 1% used in the chapter. What production capacity will Ideko require each year? When will an expansion become necessary (when production volume will exceed the current level by 50%)?
3. Under the assumption that Ideko market share will increase by 0.5% per year, you determine that the plant will require an expansion in 2010. The cost of this expansion will be \$15 million. Assuming the financing of the expansion will be delayed accordingly, calculate the projected interest payments and the amount of the projected interest tax shields (assuming that the interest rates on the term loans remain the same as in the chapter) through 2010.

Building the Financial Model



4. Under the assumption that Ideko's market share will increase by 0.5% per year (and the investment and financing will be adjusted as described in Problem 3), you project the following depreciation:

	Year	2005	2006	2007	2008	2009	2010
Fixed Assets and Capital Investment (\$ 000)							
2	New Investment	5,000	5,000	5,000	5,000	5,000	20,000
3	Depreciation	(5,500)	(5,450)	(5,405)	(5,365)	(5,328)	(6,795)

Using this information, project net income through 2010 (that is, reproduce Table 19.7 under the new assumptions).



5. Under the assumptions that Ideko's market share will increase by 0.5% per year (implying that the investment, financing, and depreciation will be adjusted as described in Problems 3 and 4) and that the forecasts in Table 19.8 remain the same, calculate Ideko's working capital requirements through 2010 (that is, reproduce Table 19.9 under the new assumptions).



6. Under the assumptions that Ideko's market share will increase by 0.5% per year (implying that the investment, financing, and depreciation will be adjusted as described in Problems 3 and 4) but that the projected improvements in net working capital do not transpire (so the numbers in Table 19.8 remain at their 2005 levels through 2010), calculate Ideko's working capital requirements through 2010 (that is, reproduce Table 19.9 under these assumptions).



7. Forecast Ideko's free cash flow (reproduce Table 19.10), assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (that is, under the assumptions in Problem 5).



8. Forecast Ideko's free cash flow (reproduce Table 19.10), assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly;

and the projected improvements in working capital do *not* occur (that is, under the assumptions in Problem 6).



- *9. Reproduce Ideko's balance sheet and statement of cash flows, assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (that is, under the assumptions in Problem 5).



- *10. Reproduce Ideko's balance sheet and statement of cash flows, assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital do *not* occur (that is, under the assumptions in Problem 6).

Estimating the Cost of Capital

11. Calculate Ideko's unlevered cost of capital when Ideko's unlevered beta is 1.1 rather than 1.2, and all other required estimates are the same as in the chapter.
12. Calculate Ideko's unlevered cost of capital when the market risk premium is 6% rather than 5%, the risk-free rate is 5% rather than 4%, and all other required estimates are the same as in the chapter.

Valuing the Investment

13. Using the information produced in the income statement in Problem 4, use EBITDA as a multiple to estimate the continuation value in 2010, assuming the current value remains unchanged (reproduce Table 19.15). Infer the EV/sales and the unlevered and levered P/E ratios implied by the continuation value you calculated.
14. How does the assumption on future improvements in working capital affect your answer to Problem 13?
15. Approximately what expected future long-run growth rate would provide the same EBITDA multiple in 2010 as Ideko has today (i.e., 9.1)? Assume that the future debt-to-value ratio is held constant at 40%; the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year until 2010; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (i.e., the assumptions in Problem 5).
16. Approximately what expected future long-run growth rate would provide the same EBITDA multiple in 2010 as Ideko has today (i.e., 9.1)? Assume that the future debt-to-value ratio is held constant at 40%; the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital do *not* occur (i.e., the assumptions in Problem 6).
17. Using the APV method, estimate the value of Ideko and the NPV of the deal using the continuation value you calculated in Problem 13 and the unlevered cost of capital estimate in Section 19.4. Assume that the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year until 2010; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (i.e., the assumptions in Problem 5).
18. Using the APV method, estimate the value of Ideko and the NPV of the deal using the continuation value you calculated in Problem 13 and the unlevered cost of capital estimate in Section 19.4. Assume that the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital do *not* occur (i.e., the assumptions in Problem 6).
19. Use your answers from Problems 17 and 18 to infer the value today of the projected improvements in working capital under the assumptions that Ideko's market share will increase by 0.5% per year and that investment, financing, and depreciation will be adjusted accordingly.

CHAPTER 19 APPENDIX

NOTATION

r_E equity
cost of
capital

Compensating Management

The success of KKP's investment critically depends on its ability to execute the operational improvements laid out in its business plan. KKP has learned from experience that it is much more likely to achieve its goals if the management team responsible for implementing the changes is given a strong incentive to succeed. KKP therefore considers allocating 10% of Ideko's equity to a management incentive plan. This equity stake would be vested over the next five years, and it would provide Ideko's senior executives with a strong financial interest in the success of the venture. What is the cost to KKP of providing this equity stake to the management team? How will this incentive plan affect the NPV of the acquisition?

To determine the value of the acquisition to KKP, we must include the cost of the 10% equity stake granted to management. Because the grant vests after five years, management will not receive any of the dividends paid by Ideko during that time. Instead, management will receive the equity in five years time, at which point we have estimated the value of Ideko's equity to be \$172 million (see Table 19.15). Thus, the cost of management's stake in 2010 is equal to $10\% \times \$172 \text{ million} = \17.2 million according to our estimate. We must determine the present value of this amount today.

Because the payment to the managers is an equity claim, to compute its present value we must use an equity cost of capital. We take an FTE valuation approach to estimate the cost of management's share in Ideko, shown in the spreadsheet in Table 19A.1.

To compute Ideko's equity cost of capital r_E , we use Eq. 18.20, which applies when the debt levels of the firm follow a known schedule:

$$r_E = r_U + \frac{D - T^s}{E} (r_U - r_D)$$

Using the debt, equity, and tax shield values from the spreadsheet in Table 19.17 to compute the effective leverage ratio $(D - T^s)/E$, we compute r_E each year as shown in the spreadsheet. We then compute the cost of management's equity share by discounting at this rate:

$$\text{Cost of Management's Share}_t = \frac{\text{Cost of Management's Share}_{t+1}}{1 + r_E(t)} \quad (19A.1)$$

Once we have determined the cost of management's equity share, we deduct it from the total value of Ideko's equity (from Table 19.17) to determine the value of KKP's share of Ideko's equity, shown as the last line of the spreadsheet. Given the initial cost of the acquisition to KKP of \$53 million, KKP's NPV from the investment, including the cost of management's compensation, is $\$103.58 \text{ million} - \$53 \text{ million} = \$50.58 \text{ million}$.

TABLE 19A.1
SPREADSHEET

**FTE Estimate of the Cost of Management's Share
and KKP's Equity Value**

	Year	2005	2006	2007	2008	2009	2010
Management/KKP Share (\$ 000)							
1 Management Payoff (10% share)							17,205
2 Effective Leverage $(D - T^s)/E$		0.792	0.771	0.715	0.778	0.745	
3 Equity Cost of Capital r_E		12.53%	12.47%	12.29%	12.49%	12.38%	
4 Cost of Management's Share		(9,576)	(10,777)	(12,120)	(13,610)	(15,309)	(17,205)
5 Ideko Equity Value		113,160	118,287	130,047	141,294	157,566	172,052
6 KKP Equity		103,583	107,511	117,927	127,684	142,256	154,847

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Working Capital Management

NOTATION

CCC cash conversion cycle

NPV net present value

EAR effective annual rate

IN CHAPTER 2, WE DEFINED A FIRM'S NET WORKING CAPITAL

as its current assets minus its current liabilities. Net working capital is the capital required in the short term to run the business. Thus, working capital management involves short-term asset accounts such as cash, inventory, and accounts receivable, as well as short-term liability accounts such as accounts payable.

The level of investment in each of these accounts differs from firm to firm and from industry to industry. It also depends on factors such as the type of business and industry standards. Some firms, for example, require heavy inventory investments because of the nature of their business. Consider The Kroger Company, a retail grocery chain, and Carnival Corporation, a cruise ship operator. Inventory amounted to nearly 20% of Kroger's total assets at the start of 2015, whereas Carnival's investment in inventory was less than 1%. A grocery store requires a large investment in inventory, while a cruise line's profitability is generated primarily from its investment in plant, property, and equipment—that is, its 100 cruise ships.

There are opportunity costs associated with investing in inventories and accounts receivable, and from holding cash. Excess funds invested in these accounts could instead be used to pay down debt or returned to shareholders in the form of a dividend or share repurchase. This chapter focuses on the tools firms use to manage their working capital efficiently and thereby minimize these opportunity costs. We begin by discussing why firms have working capital and how it affects firm value. In a perfectly competitive market, many of the working capital accounts would be irrelevant. Not surprisingly, the existence of these accounts for real firms can be traced to market frictions. We discuss the costs and benefits of trade credit and evaluate the trade-offs firms make in managing various working capital accounts. Finally, we discuss the cash balance of a firm and provide an overview of the short-term investments in which a firm may choose to invest its cash.

26.1 Overview of Working Capital

Most projects require the firm to invest in net working capital. The main components of net working capital are cash, inventory, receivables, and payables. Working capital includes the cash that is needed to run the firm on a day-to-day basis. It does not include excess cash, which is cash that is not required to run the business and can be invested at a market rate. As we discussed in Chapter 14, excess cash may be viewed as part of the firm's capital structure, offsetting firm debt. In Chapter 8, we discussed how any increases in net working capital represent an investment that reduces the cash that is available to the firm. Therefore, working capital alters a firm's value by affecting its free cash flow. In this section, we examine the components of net working capital and their effects on the firm's value.

The Cash Cycle

The level of working capital reflects the length of time between when cash goes out of a firm at the beginning of the production process and when it comes back in. A company first buys inventory from its suppliers, in the form of either raw materials or finished goods. A firm typically buys its inventory on credit, which means that the firm does not have to pay cash immediately at the time of purchase. After receiving the inventory, even if the inventory is in the form of finished goods, it may sit on the shelf for some time. Finally, when the inventory is ultimately sold, the firm may extend credit to its customers, delaying when it will receive the cash. A firm's **cash cycle** is the length of time between when the firm pays cash to purchase its initial inventory and when it receives cash from the sale of the output produced from that inventory. Figure 26.1 illustrates the cash cycle.

Some practitioners measure the cash cycle by calculating the cash conversion cycle. The **cash conversion cycle (CCC)** is defined as

$$\text{CCC} = \text{Accounts Receivable Days} + \text{Inventory Days} - \text{Accounts Payable Days} \quad (26.1)$$

where

$$\text{Accounts Receivable Days} = \frac{\text{Accounts Receivable}}{\text{Average Daily Sales}}$$

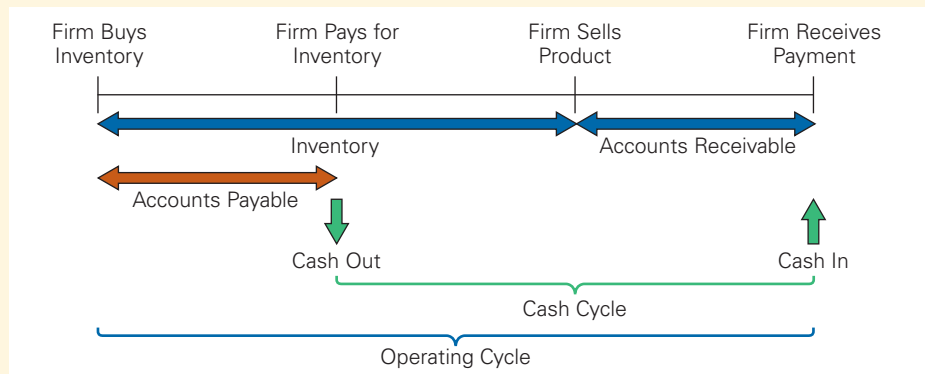
$$\text{Inventory Days} = \frac{\text{Inventory}}{\text{Average Daily Cost of Goods Sold}}$$

$$\text{Accounts Payable Days} = \frac{\text{Accounts Payable}}{\text{Average Daily Cost of Goods Sold}}$$

FIGURE 26.1

The Cash and Operating Cycles for a Firm

The cash cycle is the average time between when a firm pays for its inventory and when it receives cash from the sale of its product.



The firm's **operating cycle** is the average length of time between when a firm originally purchases its inventory and when it receives the cash back from selling its product. If the firm pays cash for its inventory, this period is identical to the firm's cash cycle. However, most firms buy their inventory on credit, which reduces the amount of time between the cash investment and the receipt of cash from that investment.

The longer a firm's cash cycle, the more working capital it has, and the more cash it needs to carry to conduct its daily operations. Table 26.1 provides data on the working capital needs for selected firms in a variety of industries.

Because of the characteristics of different industries, working capital levels vary significantly. For example, grocery stores (Kroger) and fast-food restaurants (Chipotle Mexican Grill) typically sell on a cash-only basis, and thus have very short accounts receivable days.¹ Similar results hold for Southwest Airlines, because many of its customers pay in advance for airline tickets with cash or credit cards. Inventory represents the largest percentage of

TABLE 26.1

Working Capital in Various Industries (2015)

Company	Ticker	Industry	Accounts Receivable Days	Inventory Days	Accounts Payable Days	CCC
Molson Coors Brewing	TAP	Brewing	48	37	184	-99
Pepsico	PEP	Beverages	40	40	164	-84
Verizon Communications	VZ	Telecommunications	37	8	117	-72
Apple	AAPL	Computer Hardware	20	7	76	-49
Bristol-Myers Squibb	BMJ	Pharmaceuticals	72	126	223	-25
Southwest Airlines	LUV	Airlines	9	9	32	-14
Amazon.com	AMZN	Internet Retail	19	45	73	-9
The New York Times	NYT	Publishing	41	0	49	-8
Chipotle Mexican Grill	CMG	Restaurants	2	2	10	-6
The Kroger Co.	KR	Grocery Stores	4	24	23	5
Wal-Mart Stores	WMT	Superstores	4	47	39	12
Microsoft	MSFT	Software	62	33	77	18
Macy's	SHLD	Department Stores	4	147	133	18
FedEx	FDX	Air Freight	43	5	22	26
Starbucks	SBUX	Restaurants	12	30	16	26
Nordstrom	JWN	Department Stores	29	79	62	46
Nike	NKE	Footwear	36	91	41	86
Brown-Forman	BF.B	Distillers and Vintners	72	440	90	422
Lennar	LEN	Homebuilding	3	463	23	443
Sotheby's	BID	Auction Services	393	169	117	445
Tiffany & Co.	TIF	Luxury Goods	17	546	66	497
Major U.S. Firms (value-weighted Average)			48	71	76	43

Source: www.capitaliq.com

¹When you use your Visa or MasterCard to pay for your groceries, it is a cash sale for the store. The credit card company pays the store cash upon receipt of the credit slip, even if you do not pay your credit card bill on time.

sales for firms such as Tiffany, Brown-Forman, and Lennar, whose products have long production and sales cycles, whereas Chipotle and The New York Times hold minimal inventory. Note also the wide variation in the firms' cash conversion cycles; Amazon.com and Southwest's cash conversion cycle is negative, reflecting the fact that they receive cash from their customers before having to pay suppliers, whereas firms such as Tiffany must spend cash to produce their products more than one year before they receive revenues from them.

Firm Value and Working Capital

Any reduction in working capital requirements generates a positive free cash flow that the firm can distribute immediately to shareholders. For example, if a firm is able to reduce its required net working capital by \$50,000 permanently, it will be able to distribute this \$50,000 as a dividend to its shareholders immediately and thus increase firm value by the same amount. Similarly, when evaluating a project, reducing the project's net working capital needs over the project's life reduces the opportunity cost associated with this use of capital.

EXAMPLE 26.1

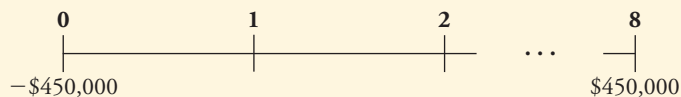
Costly Working Capital for a Project

Problem

Emerald City Paints would like to construct a new facility that will manufacture paint. In addition to the capital expenditure on the plant, management estimates that the project will require an investment today of \$450,000 for net working capital. The firm will recover the investment in net working capital eight years from today, when management anticipates closing the plant. The discount rate for this type of cash flow is 6% per year. What is the present value of the cost of working capital for the paint facility? What is the value of an inventory policy that would halve the plant's net working capital requirements?

Solution

The cash flows for the investment in net working capital are $-\$450,000$ today and $+\$450,000$ eight years from today. Putting this on a timeline:



Given a discount rate of 6% per year, the NPV of these cash flows is

$$NPV = -\$450,000 + \frac{\$450,000}{(1 + 0.06)^8} = -\$167,664$$

Although Emerald City Paints receives back all of its investment in working capital, it loses the time value of money on this cash. If a new inventory policy could halve this requirement, it would be worth $\$167,664/2 = \$83,832$ to the firm.

Managing working capital efficiently will maximize firm value. We now turn our attention to some specific working capital accounts.

CONCEPT CHECK

1. What is the firm's cash cycle? How does it differ from the operating cycle?
2. How does working capital impact a firm's value?

26.2 Trade Credit

When a firm allows a customer to pay for goods at some date later than the date of purchase, it creates an account receivable for the firm and an account payable for the customer. Accounts receivable represent the credit sales for which a firm has yet to receive payment. The accounts payable balance represents the amount that a firm owes its suppliers for goods that it has received but for which it has not yet paid. The credit that the firm is extending to its customer is known as **trade credit**. A firm would, of course, prefer to be paid in cash at the time of purchase, but a “cash-only” policy may cause it to lose its customers to competition. In this section, we demonstrate how managers can compare the costs and benefits of trade credit to determine optimal credit policies.

Trade Credit Terms

To see how the terms of trade credit are quoted, let's consider some examples. If a supplier offers its customers terms of “Net 30,” payment is not due until 30 days from the date of the invoice. Essentially, the supplier is letting the customer use its money for an extra 30 days. (Note that “30” is not a magic number; the invoice could specify “Net 40,” “Net 15,” or any other number of days as the payment due date.)

Sometimes the selling firm will offer the buying firm a discount if payment is made early. The terms “2/10, Net 30” mean that the buying firm will receive a 2% discount if it pays for the goods within 10 days; otherwise, the full amount is due in 30 days. Firms offer discounts to encourage customers to pay early so that the selling firm gets cash from the sale sooner. However, the amount of the discount also represents a cost to the selling firm because it does not receive the full selling price for the product.

Trade Credit and Market Frictions

In a perfectly competitive market, trade credit is just another form of financing. Under the Modigliani-Miller assumptions of perfect capital markets, the amounts of payables and receivables are therefore irrelevant. In reality, product markets are rarely perfectly competitive, so firms can maximize their value by using their trade credit options effectively.

Cost of Trade Credit. Trade credit is, in essence, a loan from the selling firm to its customer. The price discount represents an interest rate. Often, firms offer favorable interest rates on trade credit as a price discount to their customers. Therefore, financial managers should evaluate the terms of trade credit to decide whether to use it.

How do we compute the interest rate on trade credit? Suppose a firm sells a product for \$100 but offers its customer terms of 2/10, Net 30. The customer doesn't have to pay anything for the first 10 days, so it effectively has a zero-interest loan for this period. If the customer takes advantage of the discount and pays within the 10-day discount period, the customer pays only \$98 for the product. The cost of the discount to the selling firm is equal to the discount percentage times the selling price. In this case, it is $0.02 \times \$100$, or \$2.

Rather than pay within 10 days, the customer has the option to use the \$98 for an additional 20 days ($30 - 10 = 20$). The interest rate for the 20-day term of the loan is $\$2/\$98 = 2.04\%$. With a 365-day year, this rate over 20 days corresponds to an effective annual rate of²

$$EAR = (1.0204)^{365/20} - 1 = 44.6\%$$

²See Eq. 5.1 in Chapter 5.

Thus, by not taking the discount, the firm is effectively paying 2.04% to borrow the money for 20 days, which translates to an effective annual rate of 44.6%! If the firm can obtain a bank loan at a lower interest rate, it would be better off borrowing at the lower rate and using the cash proceeds of the loan to take advantage of the discount offered by the supplier.

EXAMPLE 26.2

Estimating the Effective Cost of Trade Credit

Problem

Your firm purchases goods from its supplier on terms of 1/15, Net 40. What is the effective annual cost to your firm if it chooses not to take advantage of the trade discount offered?

Solution

Because the discount is 1%, for a \$100 purchase your firm must pay either \$99 in 15 days or \$100 in 40 days. Given the difference of 25 days ($40 - 15$), these terms correspond to an effective annual rate of $(100/99)^{365/25} - 1 = 15.8\%$.

Benefits of Trade Credit. For a number of reasons, trade credit can be an attractive source of funds. First, trade credit is simple and convenient to use, and it therefore has lower transaction costs than alternative sources of funds. For example, no paperwork must be completed, as would be the case for a loan from a bank. Second, it is a flexible source of funds, and can be used as needed. Finally, it is sometimes the only source of funding available to a firm.

Trade Credit Versus Standard Loans. You might wonder why companies would ever provide trade credit. After all, most companies are not banks, so why are they in the business of making loans? Several reasons explain their willingness to offer trade credit.³ First, providing financing at below-market rates is an indirect way to lower prices for only certain customers. Consider, for example, an automobile manufacturer. Rather than lower prices on all cars, the financing division may offer specific credit terms that are attractive to customers with bad credit, but unattractive to customers with good credit. In this way, the car manufacturer is able to discount the price only for those customers with bad credit who otherwise might not be able to afford the car.

Second, because a supplier may have an ongoing business relationship with its customer, it may have more information about the credit quality of the customer than a traditional outside lender such as a bank. The supplier may also be able to increase the likelihood of payment by threatening to cut off future supplies if payment is not made. Finally, if the buyer defaults, the supplier may be able to seize the inventory as collateral. This inventory is likely to be more valuable to a company within the industry such as the supplier (which presumably has other customers) than to an outsider.

Managing Float

One factor that contributes to the length of a firm's receivables and payables is the delay between the time a bill is paid and the cash is actually received. This delay, or processing float, will impact a firm's working capital requirements.

³For a detailed discussion of these issues, see B. Biais and C. Gollier, "Trade Credit and Credit Rationing," *Review of Financial Studies* 10 (1997): 903–937; and M. Petersen and R. Rajan, "Trade Credit: Theories and Evidence," *Review of Financial Studies* 10 (1997): 661–691.

Collection Float. **Collection float** is the amount of time it takes for a firm to be able to use funds after a customer has paid for its goods. Firms can reduce their working capital needs by reducing their collection float. Collection float is determined by three factors:

- **Mail float:** How long it takes the firm to receive the check after the customer has mailed it
- **Processing float:** How long it takes the firm to process the check and deposit it in the bank
- **Availability float:** How long it takes before the bank gives the firm credit for the funds

Disbursement Float. **Disbursement float** is the amount of time it takes before payments to suppliers actually result in a cash outflow for the firm. Like collection float, it is a function of mail time, processing time, and check-clearing time. Although a firm may try to extend its disbursement float in order to lengthen its payables and reduce its working capital needs, it risks making late payments to suppliers. In such a case, the firm may be charged an additional fee for paying late or may be required to pay cash before delivery (CBD) or on delivery (COD) for future purchases. In some cases, the supplier may refuse to do business in the future with the delinquent firm.

Electronic Check Processing. Firms can employ several methods to reduce their collection and disbursement floats. The **Check Clearing for the 21st Century Act (Check 21)**, which became effective on October 28, 2004, eliminated the disbursement float due to the check-clearing process. Under the Act, banks can process check information electronically, and the funds are deducted from a firm's checking account on the same day that the firm's supplier deposits the check in its bank in most cases. Unfortunately, even though under Check 21 the funds are taken out of the check writer's account almost immediately, the check recipient's account is not credited as quickly. As a result, the act does not serve to reduce collection float.

There are, however, several ways that a firm *can* reduce its collection float. For example, the firm may streamline its in-house check-processing procedures. In addition, with electronic collection, funds are automatically transferred from the customer's bank account to the firm's bank account on the payment date, reducing the collection float to zero. The methods a firm employs to reduce its collection float are not without costs, of course. Therefore, to decide which, if any, to employ, the firm must compare the costs and benefits of systems that allow it to use its cash for a longer period.

CONCEPT CHECK

1. What does the term "2/10, Net 30" mean?
2. Why do companies provide trade credit?

26.3 Receivables Management

So far, we have discussed the costs and benefits of trade credit in general. Next, we look at some issues that arise specifically from the management of a firm's accounts receivable. In particular, we focus on how a firm adopts a policy for offering credit to its customers and how it monitors its accounts receivable on an ongoing basis.

Determining the Credit Policy

Establishing a credit policy involves three steps that we will discuss in turn:

1. Establishing credit standards
2. Establishing credit terms
3. Establishing a collection policy

Establishing Credit Standards. Management must first decide on its credit standards. Will it extend credit to anyone who applies for it? Or will it be selective and extend credit only to those customers who have very low credit risk? Unless the firm adopts the former policy, it will need to assess the credit risk of each customer before deciding whether to grant credit. Large firms often perform this analysis in-house with their own credit departments. Many small firms purchase credit reports from credit rating agencies such as Dun & Bradstreet.

The decision of how much credit risk to assume plays a large role in determining how much money a firm ties up in its receivables. While a restrictive policy can result in a lower sales volume, the firm will have a smaller investment in receivables. Conversely, a less selective policy will produce higher sales, but the level of receivables will also rise.

Establishing Credit Terms. After a firm decides on its credit standards, it must next establish its credit terms. The firm decides on the length of the period before payment must be made (the “net” period) and chooses whether to offer a discount to encourage early payments. If it offers a discount, it must also determine the discount percentage and the discount period. If the firm is relatively small, it will probably follow the lead of other firms in the industry in establishing these terms.

Establishing a Collection Policy. The last step in the development of a credit policy is to decide on a collection policy. The content of this policy can range from doing nothing if a customer is paying late (generally not a good choice), to sending a polite letter of inquiry, to charging interest on payments extending beyond a specified period, to threatening legal action at the first late payment.

Monitoring Accounts Receivable

After establishing a credit policy, a firm must monitor its accounts receivable to analyze whether its credit policy is working effectively. Two tools that firms use to monitor the accounts receivable are the accounts receivable days (or average collection period) and the aging schedule.

Accounts Receivable Days. The accounts receivable days is the average number of days that it takes a firm to collect on its sales. A firm can compare this number to the payment policy specified in its credit terms to judge the effectiveness of its credit policy. If the credit terms specify “Net 30” and the accounts receivable days outstanding is 50 days, the firm can conclude that its customers are paying 20 days late, on average.

The firm should also look at the trend in the accounts receivable days over time. If the accounts receivable days ratio of a firm has been approximately 35 days for the past few years and it is 43 days this year, the firm may want to reexamine its credit policy. Of course, if the economy is sluggish, the entire industry may be affected. Under these circumstances, the increase might have little to do with the firm itself.

Because accounts receivable days can be calculated from the firm’s financial statements, outside investors commonly use this measure to evaluate a firm’s credit management policy. A major weakness of the accounts receivable days is that it is merely one number and conceals much useful information. Seasonal sales patterns may cause the number calculated for the accounts receivable days to change, depending on when the calculation takes place. The number can also look reasonable even when a substantial percentage of the firm’s customers are paying late.

Aging Schedule. An **aging schedule** categorizes accounts by the number of days they have been on the firm's books. It can be prepared using either the number of accounts or the dollar amount of the accounts receivable outstanding. For example, assume that a firm selling on terms of 2/15, Net 30, has \$530,000 in accounts receivable that has been on the books for 15 or fewer days in 220 accounts. Another \$450,000 has been on the books for 16 to 30 days and is made up of 190 accounts, and \$350,000 has been on the books for 31 to 45 days and represents 80 accounts. The firm has \$200,000 that has been on the books for 46 to 60 days in 60 accounts. Yet another \$70,000 has been on the books for more than 60 days and is made up of 20 accounts. Table 26.2 includes aging schedules based on the number of accounts and dollar amounts outstanding.

In this case, if the firm's average daily sales is \$65,000, its accounts receivable days is $\$1,600,000/\$65,000 = 25$ days. But on closer examination, using the aging schedules in Table 26.2, we can see that 28% of the firm's credit customers (and 39% by dollar amounts) are paying late.

TABLE 26.2**Aging Schedules**

Days Outstanding	Number of Accounts	Percentage of Accounts (%)	Amount Outstanding (\$)	Percentage Outstanding (%)
1–15	220	38.6	530,000	33.1
16–30	190	33.3	450,000	28.1
31–45	80	14.0	350,000	21.9
46–60	60	10.5	200,000	12.5
60+	20	3.5	70,000	4.4
	570	100.0	1,600,000	100.0

EXAMPLE 26.3**Aging Schedules****Problem**

Financial Training Systems (FTS) bills its accounts on terms of 3/10, Net 30. The firm's accounts receivable include \$100,000 that has been outstanding for 10 or fewer days, \$300,000 outstanding for 11 to 30 days, \$100,000 outstanding for 31 to 40 days, \$20,000 outstanding for 41 to 50 days, \$10,000 outstanding for 51 to 60 days, and \$2,000 outstanding for more than 60 days. Prepare an aging schedule for FTS.

Solution

With the available information, we can calculate the aging schedule based on dollar amounts outstanding.

Days Outstanding	Amount Outstanding (\$)	Percentage Outstanding (%)
1–10	100,000	18.8
11–30	300,000	56.4
31–40	100,000	18.8
41–50	20,000	3.8
51–60	10,000	1.9
60+	2,000	0.3
	532,000	100.0

If the aging schedule gets “bottom-heavy”—that is, if the percentages in the lower half of the schedule begin to increase—the firm will likely need to revisit its credit policy. The aging schedule is also sometimes augmented by analysis of the **payments pattern**, which provides information on the percentage of monthly sales that the firm collects in each month after the sale. By examining past data, a firm may observe that 10% of its sales are usually collected in the month of the sale, 40% in the month following the sale, 25% two months after the sale, 20% three months after the sale, and 5% four months after the sale. Management can compare this normal payments pattern to the current payments pattern. Knowledge of the payments pattern is also useful for forecasting the firm’s working capital requirements.

CONCEPT CHECK

1. Describe three steps in establishing a credit policy.
2. What is the difference between accounts receivable days and an aging schedule?

26.4 Payables Management

A firm should choose to borrow using accounts payable only if trade credit is the cheapest source of funding. The cost of the trade credit depends on the credit terms. The higher the discount percentage offered, the greater the cost of forgoing the discount. The cost of forgoing the discount is also higher with a shorter loan period. When a company has a choice between trade credit from two different suppliers, it should take the less expensive alternative.

In addition, a firm should always pay on the latest day allowed. For example, if the discount period is 10 days and the firm is taking the discount, payment should be made on day 10, not on day 2. If the discount is not taken and the terms are 2/10, Net 30, the full payment should be made on day 30, not on day 16. A firm should strive to keep its money working for it as long as possible without developing a bad relationship with its suppliers or engaging in unethical practices. In this section, we examine two techniques that firms use to monitor their accounts payable.

Determining Accounts Payable Days Outstanding

Similar to the situation with its accounts receivable, a firm should monitor its accounts payable to ensure that it is making its payments at an optimal time. One method is to calculate the accounts payable days outstanding and compare it to the credit terms. The accounts payable days outstanding is the accounts payable balance expressed in terms of the number of days of cost of goods sold. If the accounts payable outstanding is 40 days and the terms are 2/10, Net 30, the firm can conclude that it generally pays late and may be risking supplier difficulties. Conversely, if the accounts payable days outstanding is 25 days and the firm has not been taking the discount, the firm is paying too early. It could be earning another five days’ interest on its money.

EXAMPLE 26.4

Accounts Payable Management

Problem

The Rowd Company has an average accounts payable balance of \$250,000. Its average daily cost of goods sold is \$14,000, and it receives terms of 2/15, Net 40, from its suppliers. Rowd chooses to forgo the discount. Is the firm managing its accounts payable well?

Solution

The firm is not managing its accounts payable well. Rowd’s accounts payable days outstanding is $\$250,000 / \$14,000 = 17.9$ days. If Rowd made payment three days earlier, it could take advantage of the 2% discount. If for some reason it chooses to forgo the discount, it should not be paying the full amount until the fortieth day.

Stretching Accounts Payable

Some firms ignore the payment due period and pay later, in a practice referred to as **stretching the accounts payable**. Given terms of 2/10, Net 30, for example, a firm may choose to not pay until 45 days have passed. Doing so reduces the direct cost of trade credit because it lengthens the time that a firm has use of the funds. While the interest rate per period remains the same— $\$2/\$98 = 2.04\%$ —the firm is now using the \$98 for 35 days beyond the discount period, rather than 20 days as provided by the trade credit terms.

EXAMPLE 26.5

Cost of Trade Credit with Stretched Accounts Payable

Problem

What is the effective annual cost of credit terms of 1/15, Net 40, if the firm stretches the accounts payable to 60 days?

Solution

The interest rate per period is $\$1/\$99 = 1.01\%$. If the firm delays payment until the sixtieth day, it has use of the funds for 45 days beyond the discount period. There are $365/45 = 8.11$ 45-day periods in one year. Thus, the effective annual cost is $(1.0101)^{8.11} - 1 = 8.49\%$.

Firms may also make a payment on the thirtieth day but pay only the discounted price. Some may pay only the discounted price and pay even later than the thirtieth day. While all of these actions will reduce the effective annual rate associated with the trade credit, the firm may incur costs as a result of these actions. Suppliers may react to a firm whose payments are always late by imposing terms of cash on delivery (COD) or cash before delivery (CBD). The delinquent firm then bears the additional costs associated with these terms and may have to negotiate a bank loan to have the cash available to pay. The supplier may also discontinue business with the delinquent customer, leaving the customer to find another source, which may be more expensive or of lower quality. A poor credit rating might also result, making it difficult for the firm to obtain good terms with any other supplier. Moreover, when a firm explicitly agrees to the terms of the sale, violating these terms constitutes unethical business behavior in most people's minds.

CONCEPT CHECK

1. What is accounts payable days outstanding?
2. What are the costs of stretching accounts payable?

26.5 Inventory Management

As we discussed earlier, in a perfect markets setting, firms would not need payables or receivables. Interest rates on trade credit would be competitive, and firms could use alternative sources of financing. However, unlike trade credit, inventory represents one of the required factors of production. Therefore, even in a perfect markets setting in which the Modigliani-Miller propositions hold, firms still need inventory.

Inventory management receives extensive coverage in courses on operations management. Nevertheless, it is the firm's financial manager who must arrange for the financing necessary to support the firm's inventory policy and who is responsible for ensuring the firm's overall profitability. Therefore, the role of the inventory manager is to balance the costs and benefits associated with inventory. Because excessive inventory uses cash, efficient management of inventory increases firm value.

Benefits of Holding Inventory

A firm needs its inventory to operate for several reasons. First, inventory helps minimize the risk that the firm will not be able to obtain an input it needs for production. If a firm holds too little inventory, **stock-outs**, the situation when a firm runs out of goods, may occur, leading to lost sales. Disappointed customers may switch to one of the firm's competitors.

Second, firms may hold inventory because factors such as seasonality in demand mean that customer purchases do not perfectly match the most efficient production cycle. Consider the case of the Sandpoint Toy Company. As is typical for many toy manufacturers, 80% of Sandpoint's annual sales occur between September and December, during the holiday gift season. It is more efficient for Sandpoint to manufacture toys at relatively constant levels throughout the year. If Sandpoint produces its toys at a constant rate, its inventory levels will increase to very high levels by August, in anticipation of the increase in sales beginning in September. In contrast, Sandpoint may consider a seasonal manufacturing strategy, producing more toys between September and December when sales are high. Under this strategy, inventory would not accumulate, freeing up cash flow from working capital and reducing the costs of inventory. However, seasonal manufacturing incurs additional costs, such as increased wear and tear on the manufacturing equipment during peak demand and the need to hire and train seasonal workers. Sandpoint must weigh the costs of the inventory buildup under constant production against the benefits of more efficient production. The optimal choice is likely to involve a compromise between the two extremes, so that Sandpoint will carry some inventory throughout the year.

Costs of Holding Inventory

As suggested by the Sandpoint Toy example, tying up capital in inventory is costly for a firm. We can classify the direct costs associated with inventory into three categories:

- *Acquisition costs* are the costs of the inventory itself over the period being analyzed (usually one year).
- *Order costs* are the total costs of placing an order over the period being analyzed.
- *Carrying costs* include storage costs, insurance, taxes, spoilage, obsolescence, and the opportunity cost of the funds tied up in the inventory.

Minimizing these total costs involves some trade-offs. For example, if we assume no quantity discounts are available, the lower the level of inventory a firm carries, the lower its carrying cost, but the higher its annual order costs, because it needs to place more orders during the year.

That said, the benefits from reducing inventory requirements can be substantial. In 2003, the apparel chain GAP reduced its investment in inventory significantly by reducing its inventory days outstanding by 24%. This change freed up \$344 million for other purposes. GAP invested some of this cash in short-term securities—primarily in U.S.

government and agency securities and in bank certificates of deposits with maturities between three months and one year. The firm reported an increase of \$1.2 million in interest income in fiscal year 2003 compared with fiscal year 2002. It attributed the increase to increases in the average cash balances available for investment.⁴

Some firms seek to reduce their carrying costs as much as possible. With “**just-in-time**” (**JIT**) **inventory management**, a firm acquires inventory precisely when needed so that its inventory balance is always zero, or very close to it. This technique requires exceptional coordination with suppliers as well as a predictable demand for the firm’s products. In addition, there may be a trickle-down effect when one firm in an industry adopts JIT. For example, in 1999, Toys ‘R Us instituted JIT, which caused one of its suppliers, toy manufacturer Hasbro, to make changes in its production schedule.⁵

CONCEPT CHECK

1. What are the benefits and costs of holding inventory?
2. Describe “just-in-time” inventory management.

26.6 Cash Management

In the Modigliani-Miller setting, the level of cash is irrelevant. With perfect capital markets, a firm is able to raise new money instantly at a fair rate, so it can never be short of cash. Similarly, the firm can invest excess cash at a fair rate to earn an NPV of zero.

In the real world, of course, markets are not perfect. Liquidity has a cost; for example, holding liquid assets may earn a below-market return, and a firm may face transaction costs if it needs to raise cash quickly. Similarly, recall from Chapters 15 and 17 that due to the double taxation of corporate interest income, holding excess cash has a tax disadvantage. In these cases, the optimal strategy for a firm is to hold cash in anticipation of seasonalities in its operating or investment cash flows, as well as to buffer random shocks that affect the business. Risky firms and firms with high-growth opportunities tend to hold a relatively high percentage of assets as cash. Firms with easy access to capital markets (for which the transaction costs of raising cash are lower) tend to hold less cash.⁶ In this section, we examine the firm’s motivation for holding cash, tools for managing cash, and the short-term securities in which firms invest.

Motivation for Holding Cash

There are three reasons why a firm holds cash:

- To meet its day-to-day needs
- To compensate for the uncertainty associated with its cash flows
- To satisfy bank requirements

Let’s discuss each of these motivations in more detail.

Transactions Balance. Just like you, a firm must hold enough cash to pay its bills. The amount of cash a firm needs to be able to pay its bills is sometimes referred to as a **transactions balance**. The amount of cash a firm needs to satisfy the transactions balance

⁴GAP 2003 annual report.

⁵Hasbro 1999 annual report.

⁶See T. Opler, L. Pinkowitz, R. Stulz, and R. Williamson, “The Determinants and Implications of Corporate Cash Holdings,” *Journal of Financial Economics* 52(1) (1999): 3–46.

requirement depends on both the average size of the transactions made by the firm and the firm's cash cycle, discussed earlier in the chapter. Firms set the transactions balance so that the firm's cash and other liquid securities are adequate to pay its near-term liabilities. One common measure used to assess whether the firm has adequate liquidity to meet short-term needs is its quick ratio, which is the ratio of current assets other than inventory to current liabilities. By increasing its cash balance, the firm can raise its quick ratio to its desired level.

Precautionary Balance. The amount of cash a firm holds to counter the uncertainty surrounding its future cash needs is known as a **precautionary balance**. The size of this balance depends on the degree of uncertainty surrounding a firm's cash flows. The more uncertain future cash flows are, the harder it is for a firm to predict its transactions need, so the larger the precautionary balance must be. A useful measure to assess the firm's desired precautionary balance is the volatility of its operating cash flows over different horizons. Firms will typically choose to maintain liquid assets equal to some multiple of this volatility, in order to avoid the risk of experiencing a cash shortfall.

Compensating Balance. A firm's bank may require it to hold a **compensating balance** in an account at the bank as compensation for services that the bank performs. Compensating balances are typically deposited in accounts that either earn no interest or pay a very low interest rate. This arrangement is similar to a bank offering individuals free checking so long as their balances do not fall below a certain level—say, \$1000. Essentially, the customer has \$1000 cash that he cannot use unless he is willing to pay a service charge. Similarly, the cash that a firm has tied up to meet a compensating balance requirement is unavailable for other uses.

Alternative Investments

In our discussion of collection and disbursement floats, we assumed that the firm will invest any cash in short-term securities. In fact, the firm may choose from a variety of short-term securities that differ somewhat with regard to their default risk and liquidity risk. The greater

Hoarding Cash

Corporate liquidity is measured as corporate investments in short-term, marketable securities. According to IRS statistics, the cash holdings of U.S. corporations more than tripled between 1999 and 2009, rising from \$1.6 trillion to more than \$4.8 trillion. Indeed, in 2015 more than 39% of U.S. publicly traded firms were net investors, having more cash and short-term investments than debt outstanding.

Why have companies been accumulating more cash? Factors include (i) a shift away from industries such as manufacturing that spend heavily on plant and equipment, (ii) growth in service sectors that have low capital expenditures and high cash flows, (iii) avoidance of repatriation taxes by multi-national firms (see box on page 557), and (iv) a desire by companies to preserve liquidity and financial flexibility. As a result, corporate savings has reached an all-time high.

How are companies investing their cash? A 2007 survey by Treasury Strategies indicated that 20% is invested in money market funds (which in turn invest in a diversified portfolio of the short-term securities described in Table 26.3), 18% is invested in bonds and notes, and the remainder is invested directly in commercial paper, CDs, repurchase agreements, and other investments.

During the 2008 financial crisis, short-term credit markets froze and many businesses that relied on short-term credit found themselves unable to conduct business. Even firms with sufficient cash balances had to worry about how secure their cash was: many banks were in or on the verge of default, and even traditionally safe money market funds risked losses. As a result, the breakdown in credit markets in the early days of the financial crisis was potentially as disruptive for firms that relied on cash balances to conduct business as it was for firms that relied on credit.

the risk, the higher the expected return on the investment. The financial manager must decide how much risk she is willing to accept in return for a higher yield. If her firm expects to need the funds within the next 30 days, the manager will probably avoid the less liquid options. Table 26.3 briefly describes the most frequently used short-term investments; these short-term debt securities are collectively referred to as **money market** securities. Firms may invest in these securities directly, or through money market mutual funds.

TABLE 26.3 Money Market Investment Options

Investment	Description	Maturity	Risk	Liquidity
Treasury Bills	Short-term debt of the U.S. government.	Four weeks, three months (91 days), six months (182 days), or one year when newly issued	Default risk free.	Very liquid and marketable.
Certificates of Deposit (CDs)	Short-term debt issued by banks, minimum denomination of \$100,000.	Varying maturities up to one year	If the issuing bank is insured by the FDIC, any amount up to \$250,000 is free of default risk because it is covered by the insurance. Any amount in excess of \$250,000 is not insured and is subject to default risk.	Unlike CDs purchased by individuals, these CDs sell on the secondary market, but are less liquid than Treasury bills.
Repurchase Agreements	Essentially a loan arrangement wherein a securities dealer is the “borrower” and the investor is the “lender”; the investor buys securities, such as U.S. Treasury bills, from the securities dealer, with an agreement to sell the securities back to the dealer at a later date for a specified higher price.	Very short term, ranging from overnight to approximately three months in duration	The security serves as collateral for the loan, and therefore the investor is exposed to very little risk. However, the investor needs to consider the creditworthiness of the securities dealer when assessing the risk.	No secondary market for repurchase agreements.
Banker's Acceptances	Drafts written by the borrower and guaranteed by the bank on which the draft is drawn, typically used in international trade transactions; the borrower is an importer who writes the draft in payment for goods.	Typically one to six months	Because both the borrower and a bank have guaranteed the draft, there is usually very little risk.	When the exporter receives the draft, he may hold it until maturity and receive its full value or he may sell the draft at a discount prior to maturity.
Commercial Paper	Short-term, unsecured debt issued by large corporations. The minimum denomination is \$25,000, but most commercial paper has a face value of \$100,000 or more.	Typically one to six months	Default risk depends on the creditworthiness of the issuing corporation.	No active secondary market, but issuer may repurchase commercial paper.
Short-Term Tax Exempts	Short-term debt of state and local governments; these instruments pay interest that is exempt from federal taxation, so their pre-tax yield is lower than that of a similar-risk, fully taxable investment.	Typically one to six months	Default risk depends on the creditworthiness of the issuing government.	Moderate secondary market.

Thus, a financial manager who wants to invest the firm's funds in the least risky security will choose to invest in Treasury bills. However, if the financial manager wishes to earn a higher return on the firm's short-term investments, she may opt to invest some or all of the firm's excess cash in a riskier alternative, such as commercial paper.

CONCEPT CHECK

1. List three reasons why a firm holds cash.
2. What trade-off does a firm face when choosing how to invest its cash?

MyFinanceLab

Here is what you should know after reading this chapter. MyFinanceLab will help you identify what you know and where to go when you need to practice.

26.1 Overview of Working Capital

- Working capital management involves managing the firm's short-term assets and short-term liabilities.
- A firm's cash cycle is the length of time between when the firm pays cash to purchase its initial inventory and when it receives cash from the sale of the output produced from that inventory. The operating cycle is the average length of time between when a firm originally purchases its inventory and when it receives the cash back from selling its product.

26.2 Trade Credit

- Trade credit is effectively a loan from the selling firm to its customer. The cost of trade credit depends on the credit terms. The cost of not taking a discount that is offered by a supplier implies an interest rate for the loan.
- Companies provide trade credit to their customers for two reasons: (1) as an indirect way to lower prices, and (2) because they may have advantages in making loans to their customers relative to other potential sources of credit.
- A firm should compare the cost of trade credit with the cost of alternative sources of financing in deciding whether to use the trade credit offered.
- Establishing a credit policy involves three steps: establishing credit standards, establishing credit terms, and establishing a collection policy.

26.3 Receivables Management

- Accounts receivable days and the aging schedule are two methods used to monitor the effectiveness of a firm's credit policy.

26.4 Payables Management

- Firms should monitor accounts payable to ensure that they are making payments at an optimal time.

26.5 Inventory Management

- Firms hold inventory to avoid lost sales due to stock-outs and because of factors such as seasonal demand.
 - Because excessive inventory uses cash, efficient inventory management increases the firm's free cash flow and thus increases firm value.
 - The costs of inventory include acquisition costs, order costs, and carrying costs.

26.6 Cash Management

- If a firm's need to hold cash is reduced, the funds can be invested in a number of different short-term securities, including Treasury bills, certificates of deposit, commercial paper, repurchase agreements, banker's acceptances, and short-term tax exempts.

Key Terms

aging schedule <i>p.</i> 958	mail float <i>p.</i> 956
availability float <i>p.</i> 956	money market <i>p.</i> 964
cash conversion cycle (CCC) <i>p.</i> 951	operating cycle <i>p.</i> 952
cash cycle <i>p.</i> 951	payments pattern <i>p.</i> 959
Check Clearing for the 21st Century Act (Check 21) <i>p.</i> 956	precautionary balance <i>p.</i> 963
collection float <i>p.</i> 956	processing float <i>p.</i> 956
compensating balance <i>p.</i> 963	stock-outs <i>p.</i> 961
disbursement float <i>p.</i> 956	stretching the accounts payable <i>p.</i> 960
"just-in-time" (JIT) inventory management <i>p.</i> 962	trade credit <i>p.</i> 954
	transactions balance <i>p.</i> 962

Further Reading

For more advanced analysis of working capital management, consult the following textbooks: R. Cole and L. Mishler, *Consumer and Business Credit Management* (McGraw-Hill, 1998); F. Fabozzi, S. Mann, and M. Choudhry, *The Global Money Markets* (John Wiley, 2002); and T. Maness and J. Zietlow, *Short-Term Financial Management* (South-Western, 2004).

The following articles address some of the research questions in working capital management.

Cash Management Issues. H. Almeida, M. Campello, and M. Weisbach, "The Cash Flow Sensitivity of Cash," *Journal of Finance* 59(4), (2004): 1777–1804; W. Baumol, "The Transactions Demand for Cash: An Inventory Theoretic Approach," *Quarterly Journal of Economics* 66(4) (1952): 545–556; J. Gentry, "State of the Art of Short-Run Financial Management," *Financial Management* 17(2) (1988): 41–57; M. Miller and D. Orr, "A Model of the Demand for Money by Firms," *Quarterly Journal of Economics* 80(3) (1966): 413–435; T. Opler, L. Pinkowitz, R. Stulz, and R. Williamson, "The Determinants and Implications of Corporate Cash Holdings," *Journal of Financial Economics* 52(1) (1999): 3–46; C. Payne, "The ABCs of Cash Management," *Journal of Corporate Accounting and Finance* 16(1) (2004): 3–8; L. Pinkowitz and R. Williamson, "What Is the Market Value of a Dollar of Corporate Cash?" *Journal of Applied Corporate Finance* 19 (2007): 74–81; and L. Pinkowitz, R. Stulz, and R. Williamson, "Does the Contribution of Corporate Cash Holdings and Dividends to Firm Value Depend on Governance? A Cross-Country Analysis," *Journal of Finance* 61 (2006): 2725–2751.

Trade Credit. Y. Lee and J. Stowe, "Product Risk, Asymmetric Information and Trade Credit," *Journal of Financial and Quantitative Analysis* 28(2) (1993): 285–300; M. Long, I. Malitz, and S. A. Ravid, "Trade Credit, Quality Guarantees, and Product Marketability," *Financial Management* 22(4) (1993): 117–127; S. Mian and C. Smith, "Extending Trade Credit and Financing Receivables," *Journal of Applied Corporate Finance* 7(1) (1994): 75–84; S. Mian and C. Smith, "Accounts Receivable Management Policy: Theory and Evidence," *Journal of Finance* 47(1) (1992): 169–200; O. Ng, J. Smith, and R. Smith, "Evidence on the Determinants of Credit Terms Used in Interfirm Trade," *Journal of Finance* 54(3) (1999): 1109–1129; F. Scherr, "Optimal Trade Credit Limits," *Financial Management* 25(1) (Spring 1996): 71–85; and J. Smith, "Trade Credit and Information Asymmetry," *Journal of Finance* 42(4) (1987): 863–872.

Problems

All problems are available in [MyFinanceLab](#). An asterisk (*) indicates problems with a higher level of difficulty.

Overview of Working Capital

1. Answer the following questions:
 - a. What is the difference between a firm's cash cycle and its operating cycle?
 - b. How will a firm's cash cycle be affected if a firm increases its inventory, all else being equal?
 - c. How will a firm's cash cycle be affected if a firm begins to take the discounts offered by its suppliers, all else being equal?
2. Does an increase in a firm's cash cycle necessarily mean that a firm is managing its cash poorly?
3. Aberdeen Outboard Motors is contemplating building a new plant. The company anticipates that the plant will require an initial investment of \$2.11 million in net working capital today. The plant will last 11 years, at which point the full investment in net working capital will be recovered. Given an annual discount rate of 6.4%, what is the net present value of this working capital investment?
4. The Greek Connection had sales of \$32 million and a cost of goods sold of \$12.8 million in 2015. A simplified balance sheet for the firm appears below:



THE GREEK CONNECTION
Balance Sheet
As of December 31, 2015
(in \$ thousand)

Assets		Liabilities and Equity	
Cash	\$ 1,954	Accounts payable	\$ 1,300
Accounts receivable	4,414	Notes payable	1,000
Inventory	1,414	Accruals	1,220
Total current assets	\$ 7,782	Total current liabilities	\$ 3,520
		Long-term debt	\$ 3,000
Net plant, property, and equipment	\$ 8,500	Total liabilities	\$ 6,520
Total assets	\$ 16,282	Common equity	\$ 9,762
		Total liabilities and equity	\$ 16,282

- a. Calculate The Greek Connection's net working capital in 2015.
- b. Calculate the cash conversion cycle of The Greek Connection in 2015.
- c. The industry average accounts receivable days is 30 days. What would the cash conversion cycle for The Greek Connection have been in 2015 had it matched the industry average for accounts receivable days?

Trade Credit

5. Assume the credit terms offered to your firm by your suppliers are 2.5/5, Net 30. Calculate the cost of the trade credit if your firm does not take the discount and pays on day 30.
6. Your supplier offers terms of 1.4/12, Net 45. What is the effective annual cost of trade credit if you choose to forgo the discount and pay on day 45?
7. The Fast Reader Company supplies bulletin board services to numerous hotel chains nationwide. The owner of the firm is investigating the benefit of employing a billing firm to do her billing and collections. Because the billing firm specializes in these services, collection float will be reduced by 17 days. Average daily collections are \$1300, and the owner can earn 7% annually (expressed as an APR with monthly compounding) on her investments. If the billing firm charges \$325 per month, should the owner employ the billing firm?

8. The Saban Corporation is trying to decide whether to switch to a bank that will accommodate electronic funds transfers from Saban's customers. Saban's financial manager believes the new system would decrease its collection float by as much as seven days. The new bank would require a compensating balance of \$29,000, whereas its present bank has no compensating balance requirement. Saban's average daily collections are \$12,000, and it can earn 9% on its short-term investments. Should Saban make the switch? (Assume the compensating balance at the new bank will be deposited in a non-interest earning account.)

Receivables Management

9. What are the three steps involved in establishing a credit policy?
10. The Manana Corporation had sales of \$83 million this year. Its accounts receivable balance averaged \$2 million. How long, on average, does it take the firm to collect on its sales?
11. The Mighty Power Tool Company has the following accounts on its books:

Customer	Amount Owed (\$)	Age (days)
ABC	47,000	35
DEF	35,000	3
GHI	11,000	11
KLM	71,000	23
NOP	46,000	42
QRS	13,000	10
TUV	84,000	52
WXY	38,000	90

The firm extends credit on terms of 1/15, Net 30. Develop an aging schedule using 15-day increments through 60 days, and then indicate any accounts that have been outstanding for more than 60 days.

Payables Management

12. What is meant by "stretching the accounts payable"?
-  13. Simple Simon's Bakery purchases supplies on terms of 1.6/10, Net 30. If Simple Simon's chooses to take the discount offered, it must obtain a bank loan to meet its short-term financing needs. A local bank has quoted Simple Simon's owner an interest rate of 10.1% on borrowed funds. Should Simple Simon's enter the loan agreement with the bank and begin taking the discount?
14. Your firm purchases goods from its supplier on terms of 1.7/15, Net 35.
- What is the effective annual cost to your firm if it chooses not to take the discount and makes its payment on day 35?
 - What is the effective annual cost to your firm if it chooses not to take the discount and makes its payment on day 45?
-  *15. Use the financial statements supplied on the next page for International Motor Corporation (IMC) to answer the following questions.
- Calculate the cash conversion cycle for IMC for both 2015 and 2016. What change has occurred, if any? All else being equal, how does this change affect IMC's need for cash?
 - IMC's suppliers offer terms of Net 30. Does it appear that IMC is doing a good job of managing its accounts payable?

INTERNATIONAL MOTOR CORPORATION

Income Statement (in \$ million)
for the Years Ending December 31

	2015	2016
Sales	\$ 60,000	\$ 75,000
Cost of goods sold	52,000	61,000
Gross profit	\$ 8,000	\$ 14,000
Selling and general and administrative expenses	6,000	8,000
Operating profit	\$ 2,000	\$ 6,000
Interest expense	1,400	1,300
Earnings before tax	\$ 600	\$ 4,700
Taxes	300	2,350
Earnings after tax	\$ 300	\$ 2,350

INTERNATIONAL MOTOR CORPORATION

Balance Sheet (in \$ million)
as of December 31

	2015	2016		2015	2016
Assets			Liabilities		
Cash	\$ 3,080	\$ 6,100	Accounts payable	\$ 3,600	\$ 4,600
Accounts receivable	2,800	6,900	Notes payable	1,180	1,250
Inventory	6,200	6,600	Accruals	5,600	6,211
Total current assets	\$ 12,080	\$ 19,600	Total current liabilities	\$ 10,380	\$ 12,061
Net plant, property, and equipment	\$ 23,087	\$ 20,098	Long-term debt	\$ 6,500	\$ 7,000
Total assets	\$ 35,167	\$ 39,698	Total liabilities	\$ 16,880	\$ 19,061
			Equity		
			Common stock	\$ 2,735	\$ 2,735
			Retained earnings	\$ 15,552	\$ 17,902
			Total equity	\$ 18,287	\$ 20,637
			Total liabilities and equity	\$ 35,167	\$ 39,698

Inventory Management

16. Ohio Valley Homecare Suppliers, Inc. (OVHS) had \$11 million in sales in 2015. Its cost of goods sold was \$4.4 million, and its average inventory balance was \$1.9 million.
- Calculate the number of inventory days outstanding for OVHS.
 - The average days of inventory days in the industry is 73 days. By how much would OVHS reduce its investment in inventory if it could improve its inventory days to meet the industry average?

Cash Management

17. Which of the following short-term securities would you expect to offer the highest before-tax return: Treasury bills, certificates of deposit, short-term tax exempts, or commercial paper? Why?

Data Case

You are the Chief Financial Officer (CFO) of BP. This afternoon you played golf with a member of the company's board of directors. Somewhere during the back nine, the board member enthusiastically described a recent article she had read in a leading management journal. This article noted several companies that had improved their stock price performance through effective working capital management, and the board member was intrigued. She wondered whether BP was managing its working capital effectively and, if not, whether BP could accomplish something similar. How was BP managing its working capital, and how does it compare to its competitors?

Upon returning home, you decide to do a quick preliminary investigation using information freely available on the Internet.

1. Obtain BP's financial statements for the past three years from Yahoo! Finance (finance.yahoo.com).
 - a. Enter the stock symbol (BP) in the box and click "Get Quotes."
 - b. Under "Financials," click "Income Statement." Copy and paste the statement into Excel (if using Internet Explorer, place the cursor in the statement and right-click the mouse, then choose "Export to Microsoft Excel" from the menu).
 - c. Go back to the Web page and under "Financials," click "Balance Sheet"; repeat the download procedure for the balance sheet.
 - d. Copy and paste the balance sheet so that it is on the same worksheet as the income statement.
2. Obtain the competitors' ratios for comparison from Yahoo! Finance (finance.yahoo.com).
 - a. Enter ExxonMobil Corporation's stock symbol (XOM) in the box at the top and click "Get Quotes."
 - b. Follow the steps in Part 1 to obtain "net receivables" and "inventory" from the most recent annual balance sheet, and "total revenue" and "cost of revenue" from the most recent annual income statement.
 - c. Repeat the two steps above for Chevron Corporation (CVX).
3. Compute the cash conversion cycle for BP for each of the last three years.
 - a. Compute the inventory days using "cost of revenue" as cost of goods sold and a 365-day year.
 - b. Compute accounts receivable days using a 365-day year.
 - c. Compute accounts payable days.
 - d. Compute the cash conversion cycle for each year.
4. How has BP's CCC changed over the last few years?
5. Compare BP's inventory and receivables turnover ratios for the most recent year to those of its competitors.
 - a. Compute BP's inventory turnover ratio as cost of revenue/inventory.
 - b. Compute BP's receivable turnover ratio as total revenue/net receivables.
 - c. Compute the average inventory turnover ratio and average receivable turnover ratio of Chevron and ExxonMobil. How do BP's numbers compare to the average ratios of its competitors? Do they confirm or refute your answer to Question 4?
6. Determine how BP's free cash flow would change if BP's inventory and accounts receivable balances were adjusted to meet the industry averages.
7. Determine the amount of additional free cash flow that would be available if BP adjusted its accounts payable days to 75 days.
8. Determine the net amount of additional free cash flow and BP's cash conversion cycle if its inventory and receivables turnover ratios were at the industry average and its payable days were 75 days.

9. What are your impressions regarding BP's working capital management based on this preliminary analysis? Discuss any advantages and disadvantages of bringing the cash conversion cycle more in line with the industry averages.
10. You are somewhat concerned about the reliability of financial data from the Internet, and decide to check the data versus BP's SEC filings. To obtain these filings, go to the SEC Web site and use the EDGAR system to search for BP's financial statements: www.sec.gov/edgar/searchedgar/companysearch.html. Enter BP in the ticker symbol field and then click "Search." Enter 20-F in the "Filing Type" field, and then click "Search." Click on the latest 20-F, "Annual and transition report of foreign private issuers," and then click on the 20-F document link. Once you are in the document, scroll down to the table of contents and select "Financial Statements," and then find either the balance sheet or income statement to find the numbers you need. Is there any discrepancy between these numbers and the data you downloaded originally?

Note: Updates to this data case may be found at www.berkdemarzo.com.

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Short-Term Financial Planning

MATTEL, INC. IS A COMPANY IN THE STANDARD AND POOR'S 500 INDEX, with year-end 2014 assets of over \$6.7 billion. Mattel designs and manufactures toys throughout the world; its major product lines include the Barbie, Fisher-Price, and American Girl brands. The demand for toys is typically highly seasonal, peaking during the fall in anticipation of December's holiday retailing season. As a result, Mattel's revenues vary dramatically throughout the calendar year. For example, revenues during the fourth quarter of the calendar year are typically more than twice as high as revenues in the first quarter.

Mattel's varying business revenues cause its cash flows to be highly cyclical. The firm generates surplus cash during some months; it has a great demand for capital during other months. These seasonal financing requirements are quite different from its ongoing, long-term demand for permanent capital. How does a company such as Mattel manage its short-term cash needs within each calendar year?

In this chapter, we analyze short-term financial planning. We begin by showing how companies forecast their cash flows to determine their short-term financing needs, and we explore reasons why firms use short-term financing. We next discuss financing policies that guide these financing decisions. Finally, we compare alternative ways a company can finance a shortfall during periods when it is not generating enough cash, including short-term financing with bank loans, commercial paper, and secured financing.

NOTATION

EAR effective annual rate

APR annual percentage rate

27.1 Forecasting Short-Term Financing Needs

The first step in short-term financial planning is to forecast the company's future cash flows. This exercise has two distinct objectives. First, a company forecasts its cash flows to determine whether it will have surplus cash or a cash deficit for each period. Second, management needs to decide whether that surplus or deficit is temporary or permanent. If it is permanent, it may affect the firm's long-term financial decisions. For example, if a company anticipates an ongoing surplus of cash, it may choose to increase its dividend payout. Deficits resulting from investments in long-term projects are often financed using long-term sources of capital, such as equity or long-term bonds.

In this chapter, we focus specifically on short-term financial planning. With this perspective, we are interested in analyzing the types of cash surpluses or deficits that are temporary and, therefore, short term in nature. When a company analyzes its short-term financing needs, it typically examines cash flows at quarterly intervals. To illustrate, let's assume that it is currently December 2018 and consider the case of Springfield Snowboards, Inc. Springfield manufactures snowboarding equipment, which it sells primarily to sports retailers. Springfield anticipates that in 2019 its sales will grow by 10% to \$20 million and its total net income will be \$1,950,000. Assuming that both sales and production will occur uniformly throughout the year, management's forecast of its quarterly net income and statement of cash flows for 2019 is presented in the spreadsheet in Table 27.1 (also shown, in gray, is the income statement for the fourth quarter of 2018).¹

From this forecast, we see that Springfield is a profitable company. Its quarterly net income is almost \$500,000. Springfield's capital expenditures are equal to depreciation, and while Springfield's working capital requirements increase in the first quarter due to the increase in sales, they remain constant thereafter and so have no further cash flow consequence. Based on these projections, Springfield will be able to fund projected sales growth from its operating profit and, in fact, will accumulate excess cash on an ongoing basis. Given similar growth forecasts for next year and beyond, this surplus is likely to be long term. Springfield could reduce the surplus by paying some of it out as a dividend or by repurchasing shares.

Let's now turn to Springfield's potential short-term financing needs. Firms typically require short-term financing for three reasons: seasonalities, negative cash flow shocks, and positive cash flow shocks.

Seasonalities

In many industries, sales are seasonal. Figure 27.1 shows the seasonal pattern of sales for department stores, sporting goods, and building materials. Department store and sporting good sales are concentrated during the Christmas holiday season, while for building materials, sales peak in the spring ahead of the summer building season. When sales are concentrated during a few months, sources and uses of cash are also likely to be seasonal. Firms in this position may find themselves with a surplus of cash during some months that is sufficient to compensate for a shortfall during other months. However, because of timing differences, such firms often have short-term financing needs.

¹Given the extensive coverage we have provided in Chapters 2 and 19 on how to construct pro forma financial statements, we do not rehash those details here. The full Excel model for all spreadsheets in this chapter is available from [MyFinanceLab](https://myfinancelab.com) or www.pearsonglobaleditions.com/berk_demarzo. Note that, for simplicity, we have assumed Springfield has no debt, and earns no interest on retained cash.

TABLE 27.1
SPREADSHEET**Projected Financial Statements for Springfield Snowboards, 2019, Assuming Level Sales**

	Quarter	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4
Income Statement (\$000)						
1 Sales		4,545	5,000	5,000	5,000	5,000
2 Cost of Goods Sold		(2,955)	(3,250)	(3,250)	(3,250)	(3,250)
3 Selling, General and Administrative		(455)	(500)	(500)	(500)	(500)
4 EBITDA		1,136	1,250	1,250	1,250	1,250
5 Depreciation		(455)	(500)	(500)	(500)	(500)
6 EBIT		682	750	750	750	750
7 Taxes		(239)	(263)	(263)	(263)	(263)
8 Net Income		443	488	488	488	488
Statement of Cash Flows						
9 Net Income			488	488	488	488
10 Depreciation			500	500	500	500
11 Changes in Working Capital						
12 Accounts Receivable			(136)	—	—	—
13 Inventory			—	—	—	—
14 Accounts Payable			48	—	—	—
15 Cash from Operating Activities			899	988	988	988
16 Capital Expenditures			(500)	(500)	(500)	(500)
17 Other Investment			—	—	—	—
18 Cash from Investing Activities			(500)	(500)	(500)	(500)
19 Net Borrowing			—	—	—	—
20 Dividends			—	—	—	—
21 Capital Contributions			—	—	—	—
22 Cash from Financing Activities			—	—	—	—
23 Change in Cash and Equivalents (15 + 18 + 22)			399	488	488	488

To illustrate, let's return to the example of Springfield Snowboards. In Table 27.1, management assumed that Springfield's sales occur uniformly throughout the year. In reality, for a snowboard manufacturer, sales are likely to be highly seasonal. Assume that 20% of sales occur during the first quarter, 10% during each of the second and third quarters (largely Southern Hemisphere sales), and 60% occur in the fourth quarter, in anticipation

FIGURE 27.1**Sales Seasonality (2010–2015)**

Monthly sales specified as a multiple of average monthly sales for each industry.

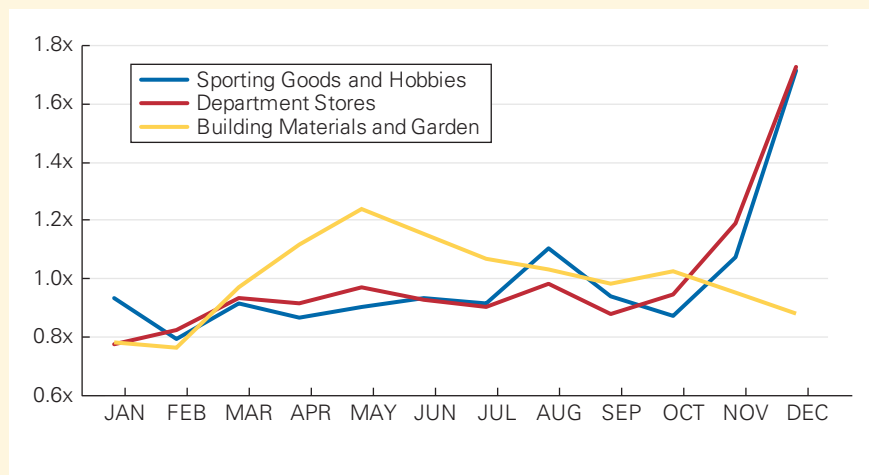


TABLE 27.2
SPREADSHEET**Projected Financial Statements for Springfield Snowboards, 2019, Assuming Seasonal Sales**

	Quarter	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4
Income Statement (\$000)						
1 Sales		10,909	4,000	2,000	2,000	12,000
2 Cost of Goods Sold		(7,091)	(2,600)	(1,300)	(1,300)	(7,800)
3 Selling, General and Administrative		(773)	(450)	(350)	(350)	(850)
4 EBITDA		3,045	950	350	350	3,350
5 Depreciation		(455)	(500)	(500)	(500)	(500)
6 EBIT		2,591	450	(150)	(150)	2,850
7 Taxes		(907)	(158)	53	53	(998)
8 Net Income		1,684	293	(98)	(98)	1,853
Statement of Cash Flows						
9 Net Income			293	(98)	(98)	1,853
10 Depreciation			500	500	500	500
11 Changes in Working Capital						
12 Accounts Receivable			2,073	600	—	(3,000)
13 Inventory			(650)	(1,950)	(1,950)	4,550
14 Accounts Payable			48	—	—	—
15 Cash from Operating Activities			2,263	(948)	(1,548)	3,903
16 Capital Expenditures			(500)	(500)	(500)	(500)
17 Other Investment			—	—	—	—
18 Cash from Investing Activities			(500)	(500)	(500)	(500)
19 Net Borrowing			—	—	—	—
20 Dividends			—	—	—	—
21 Capital Contributions			—	—	—	—
22 Cash from Financing Activities			—	—	—	—
23 Change in Cash and Equivalents (15 + 18 + 22)			1,763	(1,448)	(2,048)	3,403

of the (Northern Hemisphere) winter snowboarding season. The spreadsheet in Table 27.2 presents the resulting statement of cash flows. These forecasts continue to assume production occurs uniformly throughout the year.

From Table 27.2, we see that Springfield is still a profitable company, and its annual net income still totals \$1,950,000. However, the introduction of seasonal sales creates some dramatic swings in Springfield's short-term cash flows. There are two effects of seasonality on cash flows. First, while cost of goods sold fluctuates proportionally with sales, other costs (such as administrative overhead and depreciation) do not, leading to large changes in the firm's net income by quarter. Second, net working capital changes are more pronounced. In the first quarter, Springfield receives cash by collecting the receivables from last year's high fourth quarter sales. During the second and third quarters, the company's inventory balance increases. Given capacity constraints in its manufacturing equipment, Springfield produces snowboards throughout the year, even though sales during the summer are low. Because production occurs uniformly, accounts payable do not vary over the year. Inventory, however, builds up in anticipation of fourth quarter sales—and increases in inventory use cash. As a consequence, Springfield has negative net cash flows during the second and third quarters, primarily to fund its inventory. By the fourth quarter, high sales recover cash for the company.

Seasonal sales create large short-term cash flow deficits and surpluses. During the second and third quarters, the company will need to find additional short-term sources of cash to fund inventory. During the fourth quarter, Springfield will have a large short-term surplus. Given that its seasonal cash flow needs are likely to recur next year, Springfield may choose to invest this cash in one of the short-term investment options discussed in Chapter 26.

TABLE 27.3
SPREADSHEET**Projected Financial Statements for Springfield Snowboards, 2019, Assuming Level Sales and a Negative Cash Flow Shock**

	Quarter	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4
Income Statement (\$000)						
1 Sales		4,545	5,000	5,000	5,000	5,000
2 Cost of Goods Sold		(2,955)	(3,250)	(3,250)	(3,250)	(3,250)
3 Selling, General and Administrative		(455)	(500)	(500)	(500)	(500)
4 EBITDA		1,136	1,250	1,250	1,250	1,250
5 Depreciation		(455)	(500)	(500)	(525)	(525)
6 EBIT		682	750	750	725	725
7 Taxes		(239)	(263)	(263)	(254)	(254)
8 Net Income		443	488	488	471	471
Statement of Cash Flows						
9 Net Income			488	488	471	471
10 Depreciation			500	500	525	525
11 Changes in Working Capital						
12 Accounts Receivable			(136)	—	—	—
13 Inventory			—	—	—	—
14 Accounts Payable			48	—	—	—
15 Cash from Operating Activities			899	988	996	996
16 Capital Expenditures			(500)	(1,500)	(525)	(525)
17 Other Investment			—	—	—	—
18 Cash from Investing Activities			(500)	(1,500)	(525)	(525)
19 Net Borrowing			—	—	—	—
20 Dividends			—	—	—	—
21 Capital Contributions			—	—	—	—
22 Cash from Financing Activities			—	—	—	—
23 Change in Cash and Equivalents (15 + 18 + 22)			399	(513)	471	471

Management can then use this cash to fund some of its short-term working capital needs during the following year.

Negative Cash Flow Shocks

Occasionally, a company will encounter circumstances in which cash flows are temporarily negative for an unexpected reason. We refer to such a situation as a negative cash flow shock. Like seasonalities, negative cash flow shocks can create short-term financing needs.

Returning to the Springfield Snowboards example, assume that during April 2019, management learns that some manufacturing equipment has broken unexpectedly. It will cost an additional \$1,000,000 to replace the equipment.² To illustrate the effect of this negative cash flow shock, we return to the base case in which Springfield's sales are level rather than seasonal. (The marginal impact of this negative shock given seasonal sales would be similar.) The spreadsheet in Table 27.3 presents cash flows with level sales and the broken equipment.

In this case, the one-time expenditure of \$1 million to replace equipment results in a negative net cash flow of \$513,000 during the second quarter of 2019. If its cash reserves

²For simplicity, assume the book value of the replaced equipment is zero, so the equipment change does not have any immediate tax implications. Also, assume that Springfield obtains the replacement equipment quickly, so that any interruption in production is negligible. Finally, assume that Springfield increases investment by an amount equal to any increase in depreciation. The general results discussed here still hold absent these assumptions, although the calculations become more complicated.

are insufficient, Springfield will have to borrow (or arrange for another financing source) to cover the \$513,000 shortfall. However, the company continues to generate positive cash flow in subsequent quarters, and by the fourth quarter it will have generated enough in cumulative cash flow to repay any loan. Therefore, this negative cash flow shock has created the need for short-term financing.

Positive Cash Flow Shocks

We next analyze a case in which a positive cash flow shock affects short-term financing needs. Although this surprise is good news, it still creates demand for short-term financing.

During the first quarter of 2019, the director of marketing at Springfield Snowboards announces a deal with a chain of outdoor sporting goods stores located in the Midwest. Springfield will be the exclusive supplier to this customer, leading to an overall sales increase of 20% for the firm, with other operating expenses expected to increase accordingly. The increased sales will begin in the second quarter. As part of the deal, Springfield has agreed to a one-time expense of \$500,000 for marketing in areas where the stores are located. An extra \$1 million in capital expenditures is also required during the first quarter to increase production capacity. Likewise, sales growth will affect required working capital.

Managers at Springfield prepared the projected financial statement in Table 27.4 to reflect this new business. Notice that net income is lower during the first quarter, reflecting the \$500,000 increase in marketing expenses. By contrast, net income in subsequent quarters is higher, reflecting the higher sales. Sales increases in each of the first two quarters result in increases in accounts receivable and accounts payable.

TABLE 27.4
SPREADSHEET

Projected Financial Statements for Springfield Snowboards, 2019, Assuming Level Sales and a Growth Opportunity

	Quarter	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4
Income Statement (\$000)						
1 Sales		4,545	5,000	6,000	6,000	6,000
2 Cost of Goods Sold		(2,955)	(3,250)	(3,900)	(3,900)	(3,900)
3 Selling, General and Administrative		(455)	(1,000)	(600)	(600)	(600)
4 EBITDA		1,136	750	1,500	1,500	1,500
5 Depreciation		(455)	(500)	(525)	(525)	(525)
6 EBIT		682	250	975	975	975
7 Taxes		(239)	(88)	(341)	(341)	(341)
8 Net Income		443	163	634	634	634
Statement of Cash Flows						
9 Net Income			163	634	634	634
10 Depreciation			500	525	525	525
11 Changes in Working Capital						
12 Accounts Receivable			(136)	(300)	—	—
13 Inventory			—	—	—	—
14 Accounts Payable			48	105	—	—
15 Cash from Operating Activities			574	964	1,159	1,159
16 Capital Expenditures			(1,500)	(525)	(525)	(525)
17 Other Investment			—	—	—	—
18 Cash from Investing Activities			(1,500)	(525)	(525)	(525)
19 Net Borrowing			—	—	—	—
20 Dividends			—	—	—	—
21 Capital Contributions			—	—	—	—
22 Cash from Financing Activities			—	—	—	—
23 Change in Cash and Equivalents (15 + 18 + 22)			(926)	439	634	634

Even though the unexpected event in this case—the opportunity to grow more rapidly—is positive, it results in a negative net cash flow during the first quarter, due primarily to the new marketing expenses and capital expenditures. However, because the company will be even more profitable in subsequent quarters, this financing need is temporary.

Now that we have explained how a company determines its short-term needs, let's explore how these needs are financed.

CONCEPT CHECK

1. How do we forecast the firm's future cash requirements?
2. What is the effect of seasonalities on short-term cash flows?

27.2 The Matching Principle

In a perfect capital market, the choice of financing is irrelevant; thus, how the firm chooses to finance its short-term cash needs cannot affect value. In reality, important market frictions exist, including transaction costs. For example, one transaction cost is the opportunity cost of holding cash in accounts that pay little or no interest. Firms also face high transaction costs if they need to negotiate a loan on short notice to cover a cash shortfall. Firms can increase their value by adopting a policy that minimizes these kinds of costs. One such policy is known as the matching principle. The **matching principle** states that short-term needs should be financed with short-term debt and long-term needs should be financed with long-term sources of funds.³

Permanent Working Capital

Permanent working capital is the amount that a firm must keep invested in its short-term assets to support its continuing operations. Because this investment in working capital is required so long as the firm remains in business, it constitutes a long-term investment. The matching principle indicates that the firm should finance this permanent investment in working capital with long-term sources of funds. Such sources have lower transaction costs than short-term sources of funds, which would have to be replaced more often.

Temporary Working Capital

Another portion of a firm's investment in its accounts receivable and inventory is temporary and results from seasonal fluctuations in the firm's business or unanticipated shocks. This **temporary working capital** is the difference between the actual level of investment in short-term assets and the permanent working capital investment. Because temporary working capital represents a short-term need, the firm should finance this portion of its investment with short-term financing.

To illustrate the distinction between permanent and temporary working capital, we return to the Springfield Snowboards example. Table 27.2 presented cash flow forecasts assuming seasonal sales. In the spreadsheet in Table 27.5, we report the underlying levels of working capital that correspond to these forecasts.

³Some evidence indicates that most firms appear to follow the matching principle. See W. Beranek, C. Cornwell, and S. Choi, "External Financing, Liquidity, and Capital Expenditures," *Journal of Financial Research* (Summer 1995): 207–222; and M. Stohs and D. Mauer, "The Determinants of Corporate Debt Maturity Structure," *Journal of Business* 69(3) (1996): 279–312.

TABLE 27.5
SPREADSHEET**Projected Levels of Working Capital for Springfield Snowboards, 2019, Assuming Seasonal Sales**

	Quarter	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4
Net Working Capital Requirements (\$000)						
1	Minimum Cash Balance	500	500	500	500	500
2	Accounts Receivable	3,273	1,200	600	600	3,600
3	Inventory	300	950	2,900	4,850	300
4	Accounts Payable	(477)	(525)	(525)	(525)	(525)
5	Net Working Capital	3,595	2,125	3,475	5,425	3,875

In Table 27.5, we see that working capital for Springfield varies from a minimum of \$2,125,000 in the first quarter of 2019 to \$5,425,000 in the third quarter. The minimum level of working capital, or \$2,125,000, can be thought of as the firm's permanent working capital. The difference between this minimum level and the higher levels in subsequent quarters (for example, $\$5,425,000 - \$2,125,000 = \$3,300,000$ in the third quarter) reflects Springfield's temporary working capital requirements.

Financing Policy Choices

Following the matching principle should, in the long run, help minimize a firm's transaction costs. But what if, instead of using the matching principle, a firm financed its permanent working capital needs with short-term debt? When the short-term debt comes due, the firm will have to negotiate a new loan. This new loan will involve additional transaction costs, and it will carry whatever market interest rate exists at the time. As a result, the firm is also exposed to interest rate risk. Financing part or all of the permanent working capital with short-term debt is known as an **aggressive financing policy**. An ultra-aggressive policy would involve financing even some of the plant, property, and equipment with short-term sources of funds.

When the yield curve is upward sloping, the interest rate on short-term debt is lower than the rate on long-term debt. In that case, short-term debt may appear cheaper than long-term debt. However, we know that with perfect capital markets, Modigliani and Miller's results from Chapter 14 apply: The benefit of the lower rate from short-term debt is offset by the risk that the firm will have to refinance the debt in the future at a higher rate. This risk is borne by the equity holders, and so the firm's equity cost of capital will rise to offset any benefit from the lower borrowing rate.

Why, then, might a firm choose an aggressive financing policy? Such a policy might be beneficial if the market imperfections mentioned in Chapter 16, such as agency costs and asymmetric information, are important. The value of short-term debt is less sensitive to the firm's credit quality than long-term debt; therefore, its value will be less affected by management's actions or information. As a result, short-term debt can have lower agency and "lemons" costs than long-term debt, and an aggressive financing policy can benefit shareholders. On the other hand, by relying on short-term debt, the firm exposes itself to **funding risk**, which is the risk of incurring financial distress costs, should the firm not be able to refinance its debt in a timely manner or at a reasonable rate.

Alternatively, a firm could finance its short-term needs with long-term debt, a practice known as a **conservative financing policy**. For example, when following such a policy, a firm would use long-term sources of funds to finance its fixed assets, permanent working capital, and some of its seasonal needs. The firm would use short-term debt very sparingly to meet its peak seasonal needs. To implement such a policy effectively, there will necessarily be periods when excess cash is available—those periods when the firm requires little or no investment in temporary working capital. While a conservative financing policy reduces funding risk, it entails other costs: First, excess cash may earn a below-market interest rate,

thereby reducing the firm's value. Second, even if the cash is invested at a competitive rate, interest income on the cash will be subject to double taxation, an additional cost. Finally, holding excess cash within the firm also increases the possibility that managers of the firm will use it nonproductively—for example, on perquisites for themselves.

Once a firm determines its short-term financing needs, it must choose which instruments it will use for this purpose. In the rest of this chapter, we survey the specific financing options available: bank loans, commercial paper, and secured financing.

CONCEPT CHECK

1. What is the matching principle?
2. What is the difference between temporary and permanent working capital?

27.3 Short-Term Financing with Bank Loans

One of the primary sources of short-term financing, especially for small businesses, is the commercial bank. Bank loans are typically initiated with a **promissory note**, which is a written statement that indicates the amount of the loan, the date payment is due, and the interest rate. In this section, we examine three types of bank loans: single, end-of-period payment loans; lines of credit; and bridge loans. In addition, we compare the interest rates, common stipulations, and fees associated with these bank loans.

Single, End-of-Period Payment Loan

The most straightforward type of bank loan is a single, end-of-period-payment loan. Such a loan agreement requires that the firm pay interest on the loan and pay back the principal in one lump sum at the end of the loan. The interest rate may be fixed or variable. With a fixed interest rate, the specific rate that the commercial bank will charge is stipulated at the time the loan is made. With a variable interest rate, the terms of the loan may indicate that the rate will vary with some spread relative to a benchmark rate, such as the yield on one-year Treasury securities or the prime rate. The **prime rate** is the rate banks charge their most creditworthy customers. However, large corporations can sometimes negotiate bank loans at an interest rate that is *below* the prime rate. For example, in its 2007 annual report, Mattel indicated that the weighted average interest rate it paid on average short-term borrowings from domestic institutions was 5.5% in 2007. By comparison, the average prime rate in 2007 was 8.05%.⁴ Another common benchmark rate is the **London Inter-Bank Offered Rate**, or **LIBOR**, which is the rate of interest at which banks borrow funds from each other in the London interbank market. It is quoted for maturities of one day to one year for 10 major currencies. As it is a rate paid by banks with the highest credit quality, most firms will borrow at a rate that exceeds LIBOR.

Line of Credit

Another common type of bank loan arrangement is a **line of credit**, in which a bank agrees to lend a firm any amount up to a stated maximum. This flexible agreement allows the firm to draw upon the line of credit whenever it chooses.

Firms frequently use lines of credit to finance seasonal needs.⁵ The line of credit may be an **uncommitted line of credit**, meaning it is an informal agreement that does not legally bind

⁴Mattel 2007 annual report and *Federal Reserve Statistical Release* Web site. On the other hand, in 2015 Mattel was paying over 10% while the prime rate was only 3.25%.

⁵Lines of credit may be used for other purposes as well. For example, Gartner, Inc., which provides research and analysis on information technology, announced that it would use both cash on hand and an existing bank line of credit to finance its 2005 acquisition of competitor Meta Group (Craig Schneider, "Dealwatch," www2.CFO.com, January 5, 2005).

the bank to provide the funds. As long as the borrower's financial condition remains good, the bank is happy to advance additional funds. A **committed line of credit** consists of a written, legally binding agreement that obligates the bank to provide the funds regardless of the financial condition of the firm (unless the firm is bankrupt), as long as the firm satisfies any restrictions in the agreement. These arrangements are typically accompanied by a compensating balance requirement (that is, a requirement that the firm maintain a minimum level of deposits with the bank) and restrictions regarding the level of the firm's working capital. The firm pays a commitment fee based on a percentage of the unused portion of the line of credit plus interest on the amount that the firm borrowed. The line of credit agreement may also stipulate that at some point in time the outstanding balance must be zero. This policy ensures that the firm does not use the short-term financing to finance its long-term obligations.

Banks usually renegotiate the terms of a line of credit on an annual basis. A **revolving line of credit** is a committed line of credit that involves a solid commitment from the bank for a longer period of time, typically two to three years. A revolving line of credit with no fixed maturity is called **evergreen credit**. In its 2014 annual report, Mattel reported that it relied on a \$1.6 billion revolving credit facility as a source of financing for its seasonal working capital requirements.

Bridge Loan

A **bridge loan** is another type of short-term bank loan that is often used to “bridge the gap” until a firm can arrange for long-term financing. For example, a real estate developer may use a bridge loan to finance the construction of a shopping mall. After the mall is completed, the developer will obtain long-term financing. Other firms use bridge loans to finance their plant and equipment costs until they receive the proceeds from the sale of a long-term debt or an equity issue. After a natural disaster, lenders may provide businesses with short-term loans to serve as bridges until they receive insurance payments or long-term disaster relief.

Bridge loans are often quoted as discount loans with fixed interest rates. With a **discount loan**, the borrower is required to pay the interest at the *beginning* of the loan period. The lender deducts interest from the loan proceeds when the loan is made.

Common Loan Stipulations and Fees

We now turn to common loan stipulations and fees that affect the effective interest rate on a loan. Specifically, we look at loan commitment fees, loan origination fees, and compensating balance requirements.

Commitment Fees. Various loan fees charged by banks affect the effective interest rate that the borrower pays. For example, the commitment fee associated with a committed line of credit increases the effective cost of the loan to the firm. The “fee” can really be considered an interest charge under another name. Suppose that a firm has negotiated a committed line of credit with a stated maximum of \$1 million and an interest rate of 10% (EAR) with a bank. The commitment fee is 0.5% (EAR). At the beginning of the year, the firm borrows \$800,000. It then repays this loan at the end of the year, leaving \$200,000 unused for the rest of the year. The total cost of the loan is

Interest on borrowed funds = $0.10(\$800,000)$	=	\$80,000
Commitment fee paid on unused portion = $0.005(\$200,000)$	=	\$ 1,000
Total cost		<u>\$81,000</u>

Loan Origination Fee. Another common type of fee is a **loan origination fee**, which a bank charges to cover credit checks and legal fees. The firm pays the fee when the loan is initiated; like a discount loan, it reduces the amount of usable proceeds that the firm receives. And like the commitment fee, it is effectively an additional interest charge.

To illustrate, assume that Timmons Towel and Diaper Service is offered a \$500,000 loan for three months at an APR of 12%. This loan has a loan origination fee of 1%. The loan origination fee is charged on the principal of the loan. In this case, the fee amounts to $0.01 \times \$500,000 = \$5,000$, so the actual amount borrowed is \$495,000. The interest payment for three months is $\$500,000(0.12/4) = \$15,000$. Putting these cash flows on a timeline:



Thus, the actual three-month interest rate paid is

$$\frac{515,000}{495,000} - 1 = 4.04\%$$

Expressing this rate as an EAR gives $1.0404^4 - 1 = 17.17\%$.

Compensating Balance Requirements. Regardless of the loan structure, the bank may include a compensating balance requirement in the loan agreement that reduces the usable loan proceeds. Recall from Chapter 26 that a compensating balance requirement means that the firm must hold a certain percentage of the principal of the loan in an account at the bank. Assume that, rather than charging a loan origination fee, Timmons Towel and Diaper Service's bank requires that the firm keep an amount equal to 10% of the loan principal in a non-interest-bearing account with the bank as long as the loan remains outstanding. The loan was for \$500,000, so this requirement means that Timmons must hold $0.10 \times 500,000 = \$50,000$ in an account at the bank. Thus, the firm has only \$450,000 of the loan proceeds actually available for use, although it must pay interest on the full loan amount. At the end of the loan period, the firm owes $\$500,000 \times (1 + 0.12/4) = \$515,000$, and so must pay $\$515,000 - 50,000 = \$465,000$ after using its compensating balance. Putting these cash flows on a timeline:



The actual three-month interest rate paid is

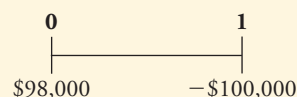
$$\frac{465,000}{450,000} - 1 = 3.33\%$$

Expressing this as an EAR gives $1.0333^4 - 1 = 14.01\%$.

We assumed that Timmons' compensating balance is held in a non-interest-earning account. Sometimes a bank will allow the compensating balance to be held in an account that pays a small amount of interest to offset part of the interest expense of the loan.

Solution

Let's put the firm's cash flows on a timeline:



The actual three-month interest rate paid is

$$\frac{100,000}{98,000} - 1 = 2.04\%$$

Expressing this as an EAR gives $1.0204^4 - 1 = 8.42\%$.

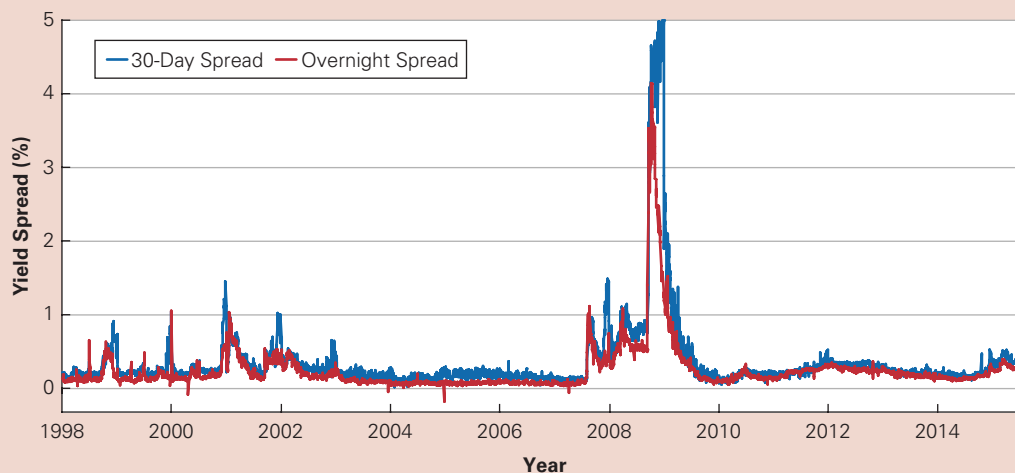
GLOBAL FINANCIAL CRISIS**Short-Term Financing in Fall 2008**

One of the biggest problems firms faced during the financial crisis was short-term financing. In the weeks following the bankruptcy of Lehman Brothers, the short-term credit markets froze. Many investors lost confidence in money market mutual funds and withdrew their capital. In response, fund managers liquidated their short-term investments, causing the availability of short-term credit to contract dramatically and short-term yields to skyrocket. Nowhere was this more evident than in the commercial paper market. The figure below shows the spread (difference) between the yields on the highest-rated commercial paper (P1) and the second-highest-rated commercial paper (P2) for two maturities: overnight (red) and 30 day (blue).

Before the collapse of housing prices and the subprime crisis in 2007, spreads were tiny—less than 0.2%. Then, in the latter half of 2007, spreads increased dramatically, at times exceeding 1% and consistently remaining above

0.5%. The onset of the financial crisis in the fall of 2008 precipitated an explosion in the spread. By October, spreads were over 4%. Government intervention helped reduce the overnight spread, but the more risky 30-day spread remained elevated and actually topped 6% on the last day of the year. The new year brought more calm to the short-term debt markets and by late 2009 spreads were back to their fall 2007 levels. Since that time, spreads have remained low (although have trended mildly upward through 2015).

Credit spreads at the level witnessed during the financial crisis effectively shut firms like Mattel out of the commercial paper market and severely hampered their ability to conduct business. In addition, the increased uncertainty in financial markets made firms less likely to invest. Both effects were significant contributors to the global recession that accompanied the financial crisis.



Source: Federal Reserve Board (www.federalreserve.gov/DataDownload/Choose.aspx?rel=CP)

CONCEPT CHECK

1. What is commercial paper?
2. How is interest paid on commercial paper?

27.5 Short-Term Financing with Secured Financing

Businesses can also obtain short-term financing by using **secured loans**, which are loans collateralized with short-term assets—most typically the firm's accounts receivables or inventory. Commercial banks, finance companies, and **factors**, which are firms that purchase the receivables of other companies, are the most common sources for secured short-term loans.

Accounts Receivable as Collateral

Firms can use accounts receivable as security for a loan by pledging or factoring.

Pledging of Accounts Receivable. In a **pledging of accounts receivable** agreement, the lender reviews the invoices that represent the credit sales of the borrowing firm and decides which credit accounts it will accept as collateral for the loan, based on its own credit standards. The lender then typically lends the borrower some percentage of the value of the accepted invoices—say, 75%. If the borrowing firm's customers default on their bills, the firm is still responsible to the lender for the money.

Factoring of Accounts Receivable. In a **factoring of accounts receivable** arrangement, the firm sells receivables to the lender (i.e., the factor), and the lender agrees to pay the firm the amount due from its customers at the end of the firm's payment period. For example, if a firm sells its goods on terms of Net 30, then the factor will pay the firm the face value of its receivables, less a factor's fee, at the end of 30 days. The firm's customers are usually instructed to make payments directly to the lender. In many cases, the firm

A Seventeenth-Century Financing Solution

In recent years, it has become more difficult for small businesses to obtain funding in order to purchase inventory. Several factors have contributed to this trend. First, bigger banks have acquired many small, regional banks that were traditionally important sources of loans to small businesses. Second, large banks have tightened lending requirements for small borrowers. Third, many small businesses rely increasingly on foreign suppliers that demand payment upfront, increasing the immediate demand for capital by small businesses.

Some small businesses have started to rely on a 400-year-old solution: venture merchant financing. This type of financing arrangement began in the seventeenth century, when groups of investors would provide capital for the voyages of Dutch sea captains. The captains would travel the seas, using the capital to purchase exotic merchandise. On their return, the merchant bankers would take about one-third of the captain's profits when the goods were sold as compensation for the financing.

Now consider the Kosher Depot, which sells exotic kosher foods to restaurants and supermarkets in Westbury, New York. It wanted to grow but lacked access to capital to purchase more specialty-foods inventory. Kosher Depot arranged a two-year, \$3.3 million venture merchant financing arrangement with Capstone Business Credit. Kosher Depot would prearrange sales and notify Capstone, which would use its capital to buy the goods for Kosher Depot. Capstone would purchase and import the goods, storing them in its own warehouses. The warehouses then filled the orders received by Kosher Depot. For its services, Capstone received about 30% of the profits.

The cost of this arrangement—the 30% margin charged by the venture merchant—may be expensive relative to some of the alternative financing arrangements discussed in this chapter. However, the price may be worthwhile for a small business with no other short-term alternatives.

Source: Marie Leone, "Capital Ideas: A Little Cash'll Do Ya," [ww2.CFO.com](#), March 3, 2005

can borrow as much as 80% of the face value of its receivables from the factor, thereby receiving its funds in advance. In such a case, the lender will charge interest on the loan in addition to the factor's fee. The lender charges the factor's fee, which may range from 1% to 2% of the face value of the accounts receivable, whether or not the firm borrows any of the available funds. Both the interest rate and the factor's fee vary depending on such issues as the size of the borrowing firm, the dollar volume of its receivables, and the creditworthiness of its customers. The dollar amounts involved in factoring agreements may be substantial. In 2014, for example, Mattel had sold nearly \$23 million of its accounts receivable under factoring arrangements.

A factoring arrangement may be **with recourse**, meaning that the lender can seek payment from the borrower should the borrower's customers default on their bills. Alternatively, the financing arrangement may be **without recourse**, in which case the lender bears the risk of bad-debt losses. In this latter case, the factor will pay the firm the amount due regardless of whether the factor receives payment from the firm's customers. If the arrangement is with recourse, the lender may not require that it approve the customers' accounts before sales are made. If the factoring agreement is without recourse, the borrowing firm must receive credit approval for a customer from the factor prior to shipping the goods. If the factor gives its approval, the firm ships the goods and the customer is directed to make payment directly to the lender.

Inventory as Collateral

Inventory can be used as collateral for a loan in one of three ways: as a floating lien, as a trust receipt, or in a warehouse arrangement.

Floating Lien. In a **floating lien**, **general lien**, or **blanket lien** arrangement, all of the inventory is used to secure the loan. This arrangement is the riskiest setup from the standpoint of the lender because the value of the collateral used to secure the loan dwindles as inventory is sold. When a firm becomes financially distressed, management may be tempted to sell the inventory without making payments on the loan. In such a case, the firm may not have enough funds to replenish its inventory. As a result, the loan may become under-collateralized. To counter this risk, this type of loan bears a higher interest rate than the next two arrangements that we discuss. In addition, lenders will loan a low percentage of the value of the inventory.

Trust Receipt. With a **trust receipts loan** or **floor planning**, distinguishable inventory items are held in a trust as security for the loan. As these items are sold, the firm remits the proceeds from their sale to the lender in repayment of the loan. The lender will periodically send someone to ensure that the borrower has not sold some of the specified inventory and failed to make a repayment on the loan. Car dealerships often use this type of secured financing arrangement to obtain the funds needed to purchase vehicles from the manufacturer.

Warehouse Arrangement. In a **warehouse arrangement**, the inventory that serves as collateral for the loan is stored in a warehouse. A warehouse arrangement is the least risky collateral arrangement from the standpoint of the lender. This type of arrangement can be set up in one of two ways.

The first method is to use a **public warehouse**, which is a business that exists for the sole purpose of storing and tracking the inflow and outflow of the inventory. The lender extends a loan to the borrowing firm, based on the value of the inventory stored. When the

Loan Guarantees: The Ex-Im Bank Controversy

Exporters looking to finance working capital needs often face difficulty obtaining loans from domestic lenders. Lenders may feel unable to evaluate adequately the credit risk of foreign customers, for example, making them unwilling to lend against the exporter's accounts receivables, or to finance its purchases of inventory.

The Export-Import Bank of the United States, or Ex-Im Bank, was established in 1934 by President Franklin D. Roosevelt as an independent export credit agency that serves to “assist in financing the export of U.S. goods and services to international markets.” It does so by providing loan guarantees which insure exporters' receivables against the risk of default by foreign customers, as well as risks of non-payment due to war or inconvertibility of the foreign currency. After obtaining such a loan guarantee, it is then usually possible for the exporter to arrange for short-term

financing of its working capital needs from traditional lenders.

Critics of the Ex-Im Bank argue that because the bank is government sponsored, these loan guarantees implicitly pass the risk of the loans to U.S. taxpayers. If defaults were to exceed the bank's reserves, the federal government would need to bail out the bank. Supporters of the bank emphasize that the bank creates U.S. jobs by supporting the growth of U.S. exporters, and that without these guarantees the firms it serves would be unable to obtain competitive financing terms.

After facing controversial reauthorization votes in 2012 and 2014, the U.S. Congress let the Ex-Im Bank's authorization expire on June 30, 2015, closing the bank. After months of strong lobbying by U.S. businesses, Congress voted in a rare bipartisan effort to reauthorize and reopen the bank in December 2015.

borrowing firm needs the inventory to sell, it returns to the warehouse and retrieves it upon receiving permission from the lender. This arrangement provides the lender with the tightest control over the inventory. Public warehouses work well for some types of inventory, such as wine and tobacco products, which must age before they are ready to be sold. It is not practical for items that are subject to spoilage or are bulky and, therefore, difficult to transport to and from the warehouse.

The second option, a **field warehouse**, is operated by a third party, but is set up on the borrower's premises in a separate area so that the inventory collateralizing the loan is kept apart from the borrower's main plant. This type of arrangement is convenient for the borrower but gives the lender the added security of having the inventory that serves as collateral controlled by a third party.

Warehouse arrangements are expensive. The business operating the warehouse charges a fee on top of the interest that the borrower must pay the lender for the loan. However, the borrower may also save on the costs of storing the inventory herself. Because the warehouse is a professional at inventory control, there is likely to be little loss due to damaged goods or theft, which in turn lowers insurance costs. Because the control of the inventory remains in the hands of a third party, lenders may be willing to lend a greater percentage of the market value of the inventory than they would under other inventory arrangements.

EXAMPLE 27.3

Calculating the Effective Annual Cost of Warehouse Financing

Problem

The Row Cannery wants to borrow \$2 million for one month. Using its inventory as collateral, it can obtain a 12% (APR) loan. The lender requires that a warehouse arrangement be used. The warehouse fee is \$10,000, payable at the end of the month. Calculate the effective annual rate of this loan for Row Cannery.

- A bridge loan is a short-term bank loan that is used to bridge the gap until the firm can arrange for long-term financing.
- The number of compounding periods and other loan stipulations, such as commitment fees, loan origination fees, and compensating balance requirements, affect the effective annual rate of a bank loan.

27.4 Short-Term Financing with Commercial Paper

- Commercial paper is a method of short-term financing that is usually available only to large, well-known firms. It is a low-cost alternative to a short-term bank loan for those firms with access to the commercial paper market.

27.5 Short-Term Financing with Secured Financing

- Short-term loans may also be structured as secured loans. The accounts receivable and inventory of a firm typically serve as collateral in short-term secured financing arrangements.
- Accounts receivable may be either pledged as security for a loan or factored. In a factoring arrangement, the accounts receivable are sold to the lender (or factor), and the firm's customers are usually instructed to make payments directly to the factor.
- Inventory can be used as collateral for a loan in several ways: a floating lien (also called a general or blanket lien), a trust receipts loan (or floor planning), or a warehouse arrangement. These arrangements vary in the extent to which specific items of inventory are identified as collateral; consequently, they vary in the amount of risk the lender faces.

Key Terms

aggressive financing policy <i>p.</i> 980	line of credit <i>p.</i> 981
blanket lien <i>p.</i> 987	loan origination fee <i>p.</i> 983
bridge loan <i>p.</i> 982	London Inter-Bank Offered Rate (LIBOR) <i>p.</i> 981
commercial paper <i>p.</i> 984	matching principle <i>p.</i> 979
committed line of credit <i>p.</i> 982	permanent working capital <i>p.</i> 979
conservative financing policy <i>p.</i> 980	pledging of accounts receivable <i>p.</i> 986
dealer paper <i>p.</i> 984	prime rate <i>p.</i> 981
direct paper <i>p.</i> 984	promissory note <i>p.</i> 981
discount loan <i>p.</i> 982	public warehouse <i>p.</i> 987
evergreen credit <i>p.</i> 982	revolving line of credit <i>p.</i> 982
factors <i>p.</i> 986	secured loans <i>p.</i> 986
factoring of accounts receivable <i>p.</i> 986	temporary working capital <i>p.</i> 979
field warehouse <i>p.</i> 988	trust receipts loan <i>p.</i> 987
floating lien <i>p.</i> 987	uncommitted line of credit <i>p.</i> 981
floor planning <i>p.</i> 987	warehouse arrangement <i>p.</i> 987
funding risk <i>p.</i> 980	with recourse <i>p.</i> 987
general lien <i>p.</i> 987	without recourse <i>p.</i> 987

Further Reading

For an in-depth discussion of short-term financing, see: G. Gallinger and B. Healey, *Liquidity Analysis and Management* (Addison-Wesley, 1991); N. Hill and W. Sartoris, *Short-Term Financial Management: Text and Cases* (Prentice-Hall, 1994); J. Kallberg and K. Parkinson, *Corporate Liquidity: Management and Measurement* (Irwin/McGraw Hill, 1996); F. Scherr, *Modern Working Capital Management: Text and Cases* (Prentice-Hall, 1989); and K. Smith and G. Gallinger, *Readings on Short-Term Financial Management* (West, 1988).

Problems

All problems are available in [MyFinanceLab](#).

Forecasting Short-Term Financing Needs

- Which of the following companies are likely to have high short-term financing needs? Why?
 - A clothing retailer
 - A professional sports team
 - An electric utility
 - A company that operates toll roads
 - A restaurant chain
- Sailboats Etc. is a retail company specializing in sailboats and other sailing-related equipment. The following table contains financial forecasts as well as current (month 0) working capital levels. During which months are the firm's seasonal working capital needs the greatest? When does it have surplus cash?

	Month						
(\$000)	0	1	2	3	4	5	6
Net Income		\$10	\$12	\$15	\$25	\$30	\$18
Depreciation		2	3	3	4	5	4
Capital Expenditures		1	0	0	1	0	0
Levels of Working Capital							
Accounts Receivable	\$2	3	4	5	7	10	6
Inventory	3	2	4	5	5	4	2
Accounts Payable	2	2	2	2	2	2	2

The Matching Principle



- What is the difference between permanent working capital and temporary working capital?
- Quarterly working capital levels for your firm for the next year are included in the following table. What are the permanent working capital needs of your company? What are the temporary needs?

	Quarter			
(\$000)	1	2	3	4
Cash	\$100	\$100	\$100	\$100
Accounts Receivable	199	96	99	604
Inventory	201	505	905	52
Accounts Payable	104	95	95	97

- Why might a company choose to finance permanent working capital with short-term debt?

Short-Term Financing with Bank Loans



- Hand-to-Mouth (H2M) is currently cash-constrained, and must make a decision about whether to delay paying one of its suppliers, or take out a loan. They owe the supplier \$10,000 with terms of 2/10 Net 40, so the supplier will give them a 2% discount if they pay today (when the

discount period expires). Alternatively, they can pay the full \$10,000 in one month when the invoice is due. H2M is considering three options:

Alternative A: Forgo the discount on its trade credit agreement, wait and pay the full \$10,000 in one month.

Alternative B: Borrow the money needed to pay its supplier today from Bank A, which has offered a one-month loan at an APR of 12%. The bank will require a (no-interest) compensating balance of 5% of the face value of the loan and will charge a \$100 loan origination fee. Because H2M has no cash, it will need to borrow the funds to cover these additional amounts as well.

Alternative C: Borrow the money needed to pay its supplier today from Bank B, which has offered a one-month loan at an APR of 15%. The loan has a 1% loan origination fee, which again H2M will need to borrow to cover.

Which alternative is the cheapest source of financing for Hand-to-Mouth?

7. Consider two loans with a 1-year maturity and identical face values: a 7.6% loan with a 0.97% loan origination fee and a 7.6% loan with a 4.7% (no-interest) compensating balance requirement. Which loan would have the higher effective annual rate? Why?
8. What is the difference between evergreen credit and a revolving line of credit?
9. Which of the following one-year \$1000 bank loans offers the lowest effective annual rate?
 - a. A loan with an APR of 5.9%, compounded monthly
 - b. A loan with an APR of 5.9%, compounded annually, that also has a compensating balance requirement of 9.7% (on which no interest is paid)
 - c. A loan with an APR of 5.9%, compounded annually, that has a 1.2% loan origination fee
10. The Needy Corporation borrowed \$14,000 from Bank Ease. According to the terms of the loan, Needy must pay the bank \$488 in interest every three months for the three-year life of the loan, with the principal to be repaid at the maturity of the loan. What effective annual rate is Needy paying?

Short-Term Financing with Commercial Paper

11. The Treadwater Bank wants to raise \$1.1 million using three-month commercial paper. The net proceeds to the bank will be \$1,084,600. What is the effective annual rate of this financing for Treadwater?
12. Magna Corporation has an issue of commercial paper with a face value of \$1,000,000 and a maturity of six months. Magna received net proceeds of \$973,710 when it sold the paper. What is the effective annual rate of the paper to Magna?
13. What is the difference between direct paper and dealer paper?
14. The Signet Corporation has issued four-month commercial paper with a \$6.3 million face value. The firm netted \$6,113,898 on the sale. What effective annual rate is Signet paying for these funds?

Short-Term Financing with Secured Financing

15. What is the difference between pledging accounts receivable to secure a loan and factoring accounts receivable?
16. The Ohio Valley Steel Corporation has borrowed \$5.1 million for one month at a stated annual rate of 8.7%, using inventory stored in a field warehouse as collateral. The warehouse charges a \$5100 fee, payable at the end of the month. What is the effective annual rate of this loan?
17. Discuss the three different arrangements under which a firm may use inventory to secure a loan.
18. The Rasputin Brewery is considering using a public warehouse loan as part of its short-term financing. The firm will require a loan of \$540,000. Interest on the loan will be 9.6% (APR, annual compounding) to be paid at the end of the year. The warehouse charges 1% of the face value of the loan, payable at the beginning of the year. What is the effective annual rate of this warehousing arrangement?

CHAPTER 17

FINANCING

INTERNATIONAL TRADE

LEARNING OBJECTIVES

The specific objectives of this chapter are to:

- describe methods of payment for international trade,
- explain common trade finance methods, and
- describe the major agencies that facilitate international trade with export insurance and/or loan programmes.

The international trade activities of MNCs have grown in importance over time. This trend is attributable to the increased globalization of the world economies and the availability of trade finance from the international banking community. Although banks also finance domestic trade, their role in financing international trade is more critical due to the additional complications involved. First, the exporter might question the importer's ability to make payment. Second, even if the importer is credit-worthy, the government might impose exchange controls that prevent payment to the exporter. Third, the importer might not trust the exporter to ship the goods ordered. Fourth, even if the exporter does ship the goods, trade barriers or time lags in international transportation might delay arrival time. Financial managers must recognize methods that they can use to finance international trade so that they can conduct exporting or importing in a manner that maximizes the value of an MNC.

PAYMENT METHODS FOR INTERNATIONAL TRADE

In any international trade transaction, credit is provided by the supplier (exporter), the buyer (importer), one or more financial institutions, or any combination of these. The supplier may have sufficient cash flow to finance the entire trade cycle, beginning with the production of the product by the exporter until payment is eventually made by the importer. This form of credit is known as **supplier credit**. In some cases, the exporter may require bank financing to augment its cash flow. On the other hand, the exporter may not desire to provide financing, in which case the buyer will have to finance the transaction itself, either internally or externally, through its bank. Banks on both sides of the transaction can thus play an integral role in trade financing.

In general, five basic methods of payment are used to settle international transactions, each with a different degree of risk to the exporter and importer (Exhibit 17.1):

- Prepayment
- Letters of credit
- Drafts (sight/time)
- Consignment
- Open account

Prepayment

Under the **prepayment** method, the exporter will not ship the goods until the buyer has remitted payment to the exporter. This method affords the supplier the greatest degree of protection, and it is normally requested of first-time buyers whose creditworthiness is unknown or whose countries are in financial difficulty. Most buyers, however, are not willing to bear all the risk by prepaying an order.

Letters of credit (L/C)

A **letter of credit (L/C)** is an instrument issued by a bank on behalf of the importer (buyer) promising to pay the exporter (beneficiary) upon presentation of shipping documents in compliance with the terms stipulated therein. In effect, the bank is substituting its credit for that of the buyer. The exporter is assured of receiving payment from the issuing bank as long as it presents documents in accordance with the L/C. The importer must still rely upon the exporter to ship the goods as described in the documents, since the L/C does not guarantee that the goods purchased will be those invoiced and shipped. Letters of credit will be described in greater detail later in this chapter.

Drafts

A **draft** (or **bill of exchange**) is an unconditional promise drawn by one party, usually the exporter, instructing the buyer to pay the face amount of the draft upon presentation. It can be thought of as a you-owe-me note. The person owed the money usually draws up the bill simply to ensure that it is in a legally enforceable format. The draft represents the exporter's formal demand for payment from the buyer. A draft affords the exporter less protection than an L/C, because the banks are not obliged to honour payments on the buyer's behalf.

Most trade transactions handled on a draft basis are processed through banking channels. In banking terminology, these transactions are known as **documentary collections**. If shipment is made under a sight draft, the exporter is paid once shipment has been made and the draft is presented to the buyer for payment. The buyer's bank will not release the shipping documents to the buyer until the buyer has paid the draft. This is known as **documents against payment**.

If a shipment is made under a time draft, the exporter instructs the buyer's bank to release the shipping documents against acceptance (signing) of the draft. This method of payment is sometimes referred to as **documents against acceptance**. By accepting the draft, the buyer is promising to pay the exporter at the specified future date. This accepted draft is also known as a **trade acceptance**. This method affords some credit to the buyer. The added risk is that if the buyer fails to pay the draft at maturity, the bank is not obliged to honour payment. The exporter is assuming all the risk.

Consignment

Under a **consignment** arrangement, the exporter ships the goods to the importer while still retaining actual title to the merchandise. The importer has access to the items but does not have to pay for the goods until they have been sold to a third party. The exporter is trusting the importer to remit payment for the goods sold at that time. If the importer fails to pay, the exporter has limited recourse because no draft is involved and the goods have already been sold. As a result of the high risk, consignments are seldom used except by affiliated and subsidiary companies trading with the parent company.

Open account

The opposite of prepayment is the **open account transaction** in which the exporter ships the merchandise and expects the buyer to remit payment according to the agreed upon terms. The exporter is relying fully upon the financial creditworthiness, integrity and reputation of the buyer. Open account transactions are widely utilized, particularly among the industrialized countries in North America and Europe. Although there is no legal recourse for non payment, where transactions are relatively small and frequent there is an implied security in the sense that failure to pay will result in no more business and possible difficulty in obtaining supplies from elsewhere.

TRADE FINANCE METHODS

As mentioned in the previous section, banks on both sides of the transaction play a critical role in financing international trade. The following are some of the more popular methods of financing international trade:

- Accounts receivable financing
- Factoring
- Letters of credit (L/Cs)
- Banker's acceptance
- Working capital financing
- Medium-term capital goods financing (forfaiting)
- Countertrade

Each of these methods is described in turn.

Accounts receivable financing

In some cases, the exporter of goods may be willing to ship goods to the importer without an assurance of payment from a bank. This could take the form of an open account shipment or a time draft. If the exporter is willing to wait for payment, it will extend credit to the buyer.

EXHIBIT 17.1 Comparison of payment methods

Method	Usual Time of Payment	Goods Available to Buyers	Risk to Exporter	Risk to Importer
Prepayment	Before shipment	After payment	None	Relies completely on exporter to ship goods as ordered
Letter of credit	When shipment is made	After payment	Very little or none, depending on credit terms	Assured shipment made, but relies on exporter to ship goods described in documents
Sight draft; documents against payments	On presentation of draft to buyer	After payment	If draft unpaid must dispose of goods	Same as above unless importer can inspect goods before payment
Time draft; documents against acceptance	On maturity of drafts	Before payment	Relies on buyer to pay drafts	Same as above
Consignment	At time of sale by buyer	Before payment	Allows importer to sell inventory before paying exporter	None; improves cash flow of buyer
Open account	As agreed	Before payment	Relies completely on buyer to pay account as agreed	None

If the exporter needs funds immediately, it may require financing from a bank. In what is referred to as **accounts receivable financing**, the bank will provide a loan to the exporter secured by an assignment of the account receivable (to be paid by the importer) as security. The length of a financing term is usually one to six months. To mitigate the additional risk of a foreign receivable, exporters and banks often require export credit insurance before financing foreign receivables.

Factoring

When an exporter ships goods before receiving payment, the accounts receivable (or debtors) balance increases. Unless the exporter has received a loan from a bank, it is initially financing the transaction and must monitor the collections of receivables. Since there is a danger that the buyer will never pay at all, the exporting firm may consider selling the accounts receivable to a third party, known as a **factor**. The factor then assumes all administrative responsibilities involved in collecting from the buyer and the associated credit exposure. The factor usually purchases the receivable at a discount and also receives a flat processing fee.

Since it is the importer who must be creditworthy from a factor's point of view, **cross-border factoring** is often used. This involves a network of factors in various countries who assess credit risk. The exporter's factor contacts a correspondent factor in the buyer's country to assess the importer's creditworthiness and handle the collection of the receivable.

Letters of credit (L/C)

Introduced earlier, the letter of credit (L/C) is one of the oldest forms of trade finance still in existence. Because of the protection and benefits it accords to both exporter and importer, it is a critical component of many international trade transactions. The L/C is an undertaking by a bank to make payments on behalf of a specified party to a beneficiary under specified conditions. The beneficiary (exporter) is paid upon

presentation of the required documents in compliance with the terms of the L/C. The L/C process normally involves two banks, the exporter's bank and the importer's bank. The issuing bank is substituting its credit for that of the importer. It has essentially guaranteed payment to the exporter, provided the exporter complies with the terms and conditions of the L/C.

EXAMPLE

Nike can attribute part of its international business growth in the 1970s to the use of L/Cs. In 1971, Nike (which was then called BSR) was not well known to businesses in Japan or anywhere else. Nevertheless, by using L/Cs, it was still able to subcontract the production of athletic shoes in Japan. The L/Cs assured the Japanese shoe producer that it would receive payment for the shoes it would send to the United States

and thus facilitated the flow of trade without concern about credit risk. Banks served as the guarantors in the event that the Japanese shoe company was not paid in full after transporting shoes to the United States. Without such agreements, Nike (and many other firms) would not be able to order shipments of goods.

Types of letters of credit. Trade-related letters of credit are known as **commercial letters of credit** or **import/export letters of credit** (For an example of an **irrevocable letter of credit**, see Exhibit 17.2). The documentary credit procedure is depicted in the flowchart in Exhibit 17.3. In what is commonly referred to as a *re-financing of a sight L/C*, the bank arranges to fund a loan to pay out the L/C instead of charging the importer's account immediately. The importer is responsible for repaying the bank both the principal and interest at maturity. This is just another method of providing extended payment terms to a buyer when the exporter insists upon payment at sight.

The typical documentation required under an L/C before payment can be made includes a draft (sight or time), a commercial invoice and a bill of lading. Depending upon the agreement, product, or country, other documents (such as a certificate of origin, inspection certificate, packing list, or insurance certificate) might be required. The three most common documents to accompany L/Cs are drafts, bills of lading and commercial invoices.

EXHIBIT 17.2 Example of an irrevocable letter of credit

Name of issuing bank

Address of issuing bank

Name of exporter

Address of exporter

We establish our irrevocable letter of credit: for the account of (*importer name*), in the amount of (*value of exports*), expiring (*date*), available by your draft at (*time period*) days sight and accompanied by: (any invoices, packing lists, bills of lading, etc., that need to be presented with the letter of credit) Insurance provided by (*exporter or importer*) covering shipment of (*merchandise description*)

From: (*port of shipment*)

To: (*port of arrival*)

(Authorized Signature)

Draft. Also known as a bill of exchange, a draft (introduced earlier). A **banker's acceptance** is a time draft drawn on and accepted by a bank. When presented under an L/C, the draft represents the exporter's formal demand for payment. The time period, or **tenor**, of most time drafts is usually anywhere from 30 to 180 days.

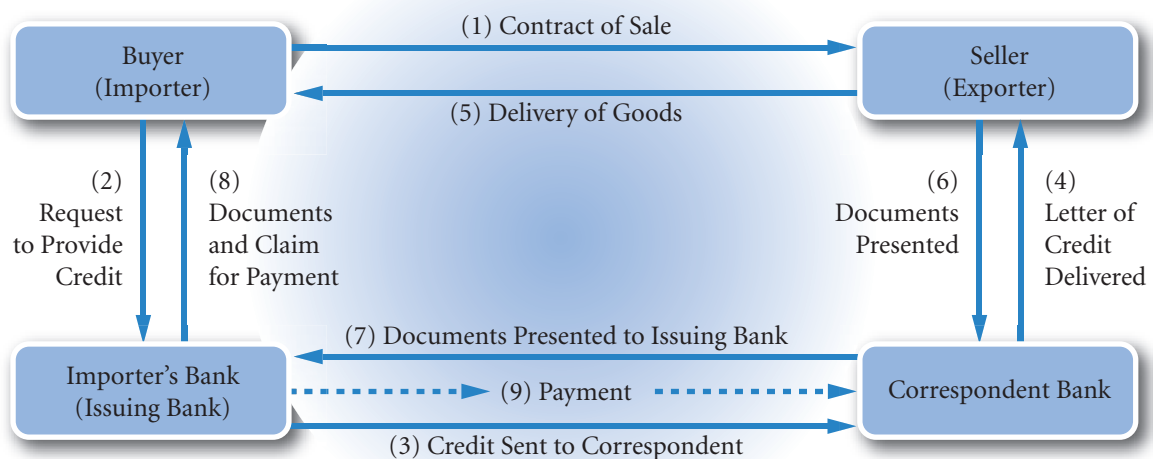
Bill of lading. The key document in an international shipment under an L/C is the **bill of lading (B/L)**. It serves as a *receipt for shipment* and a summary of freight charges; most importantly, it conveys title to the merchandise. If the merchandise is to be shipped by boat, the carrier will issue what is known as an **ocean bill of lading**. When the merchandise is shipped by air, the carrier will issue an **airway bill**. The carrier presents the bill to the exporter (shipper), who in turn presents it to the bank along with the other required documents for payment.

A significant feature of a B/L is its negotiability. When a B/L is made out to order, it is said to be in negotiable form. The exporter normally endorses the B/L to the bank once payment is received from the bank, the bank may now sell the bill and recoup most of the payment.

The bank holding the bill when due will not endorse the B/L over to the importer until payment has been made. The importer needs the original B/L to pick up the merchandise. With a **negotiable B/L**, title passes to the holder of the endorsed B/L. Because a negotiable B/L grants title to the holder, banks can take the merchandise as collateral. A B/L usually includes the following provisions:

- A description of the merchandise
- Identification marks on the merchandise
- Evidence of loading (receiving) ports
- Name of the exporter (shipper)
- Name of the importer
- Status of freight charges (prepaid or collect)
- Date of shipment.

EXHIBIT 17.3 Documentary credit procedure



Commercial invoice. The exporter's (seller's) description of the merchandise being sold to the buyer is the **commercial invoice**, which normally contains the following information:

- Name and address of seller
- Name and address of buyer
- Date
- Terms of payment
- Price, including freight, handling and insurance if applicable
- Quantity, weight, packaging, etc.
- Shipping information.

Under an L/C shipment, the description of the merchandise outlined in the invoice must correspond exactly to that contained in the L/C.

Variations of the L/C. There are several variations of the L/C that are useful in financing trade. A **standby letter of credit** can be used to guarantee invoice payments to a supplier. It promises to pay the beneficiary if the buyer fails to pay as agreed. In an international or domestic trade transaction, the seller will agree to ship to the buyer on standard open account terms as long as the buyer provides a standby L/C for a specified amount and term. As long as the buyer pays the seller as agreed, the standby L/C is never funded. The buyer's bank is essentially guaranteeing that the buyer will make payment to the seller.

A **transferable letter of credit** is a variation of the standard commercial L/C that allows the first beneficiary to transfer all or a part of the original L/C to a third party. The new beneficiary has the same rights and protection as the original beneficiary. This type of L/C is used extensively by brokers, who are not the actual suppliers.

EXAMPLE

A broker asks a foreign buyer to issue an L/C for £100 000 in his favour. The L/C must contain a clause stating that the L/C is transferable. The broker has located a supplier who will provide the product for £80 000, but requests payment in advance from the broker. With a transferable L/C, the broker can transfer £80 000 of the original L/C to the supplier under the same terms and conditions, except for the amount,

the latest shipment date, the invoice, and the period of validity. When the supplier ships the product, it presents its documents to the bank. When the bank pays the L/C, £80 000 is paid to the supplier and £20 000 goes to the broker. In effect, the broker has utilized the credit of the buyer to finance the entire transaction.

<http://>

Many banks have a website that explains the variety of trade financing that they can provide for firms. Try, for example, searching for 'NatWest bank letter of credit' in an internet search engine.

EXHIBIT 17.4 Banker's acceptance

January 15 2004

\$ 1,000,000

Ninety (90) Days after sight

Pay to the

Order of *Colombian Coffee Traders Ltd.*

Value received and to be paid to the order of *Colombian Coffee Traders Ltd.*

To *International Bank, N.A.*

of Coffee from Colombia to U.S.A.

ACCEPTED

International Bank N.A.
One Grand Central Station
New York, N.Y. 10017-4499
U.S.A.

Under International Bank L/C #155

Colombian Coffee Traders Ltd/
Bogota, Colombia

Dollars

DRAFT

the transaction which gives rise to this instrument is the importation of Coffee from Colombia to U.S.A.

Banker's acceptance

Introduced earlier, a banker's acceptance (shown in Exhibit 17.4) is a bill of exchange, or time draft, drawn on and accepted by a bank. It is the accepting bank's obligation to pay the holder of the draft at maturity.

For the first step in creating a banker's acceptance, the importer orders goods from the exporter. The importer then requests its local bank to issue an L/C on its behalf. The L/C will allow the exporter to draw a time draft on the bank in payment for the exported goods. The exporter presents the time draft along with shipping documents to its local bank, and the exporter's bank sends the time draft along with shipping documents to the importer's bank. The importer's bank accepts the draft, thereby creating the banker's acceptance. If the exporter does not want to wait until the specified date to receive payment, it can request that the banker's acceptance be sold in the money market. By doing so, the exporter will receive less funds from the sale of the banker's acceptance than if it had waited to receive payment. This discount reflects the time value of money.

If the exporter holds the acceptance until maturity, it provides the financing for the importer as it does with accounts receivable financing. In this case, the key difference between a banker's acceptance and accounts receivable financing is that a banker's acceptance guarantees payment to the exporter by a bank.

A banker's acceptance can be beneficial to the exporter, importer and issuing bank. The exporter does not need to worry about the credit risk of the importer and can therefore penetrate new foreign markets without concern about the credit risk of potential customers. In addition, the exporter faces little exposure to political risk or to exchange controls imposed by a government because banks normally are allowed to meet their payment commitments even if controls are imposed. In contrast, controls could prevent an importer from paying, so without a banker's acceptance, an exporter might not receive payment even though the importer is willing to pay. Finally, the exporter can sell the banker's acceptance at a discount before payment is due and thus obtain funds up front from the issuing bank.

The importer benefits from a banker's acceptance by obtaining greater access to foreign markets when purchasing supplies and other products. Without banker's acceptances, exporters may be unwilling to accept the credit risk of importers. In addition, due to the documents presented along with the acceptance, the importer is assured that goods have been shipped. Even though the importer has not paid in advance, this assurance is valuable because it lets the importer know if and when supplies and other products will arrive. Finally, because the banker's acceptance allows the importer to pay at a later date, the importer's payment is financed until the maturity date of the banker's acceptance. Without an acceptance, the importer is likely to be forced to pay in advance, thereby tying up funds.

The bank accepting the drafts benefits in that it earns a commission for creating an acceptance. The commission that the bank charges the customer reflects the customer's perceived creditworthiness. The interest rate charged to the customer, commonly referred to as the **all-in-rate**, consists of the discount rate plus the

acceptance commission. In general, the all-in-rate for acceptance financing is lower than prime-based borrowings, as shown in the following comparison:

	Loan	Acceptance
Amount:	£1 000 000	£1 000 000
Term:	180 days	180 days
Rate:	Prime + 1.5%	Banker's Acceptance rate + 1.5%
	10.0% + 1.5% = 11.5%	7.60% + 1.5% = 9.10%
Interest cost:	£57 500	£45 500

In this case, the savings in interest for a six-month period is £12 000 (i.e. £57 500 – £45 500). Since the banker's acceptance is a marketable instrument with an active secondary market, the rates on acceptances usually fall between the rates on short-term Treasury bills and the rates on commercial paper. Investors are usually willing to purchase acceptances as an investment because of their yield, safety and liquidity. When a bank creates, accepts and sells the acceptance, it is actually using the investor's money to finance the bank's customer. As a result, the bank has created an asset at one price, sold it at another and retained a commission (spread) as its fee.

EXAMPLE

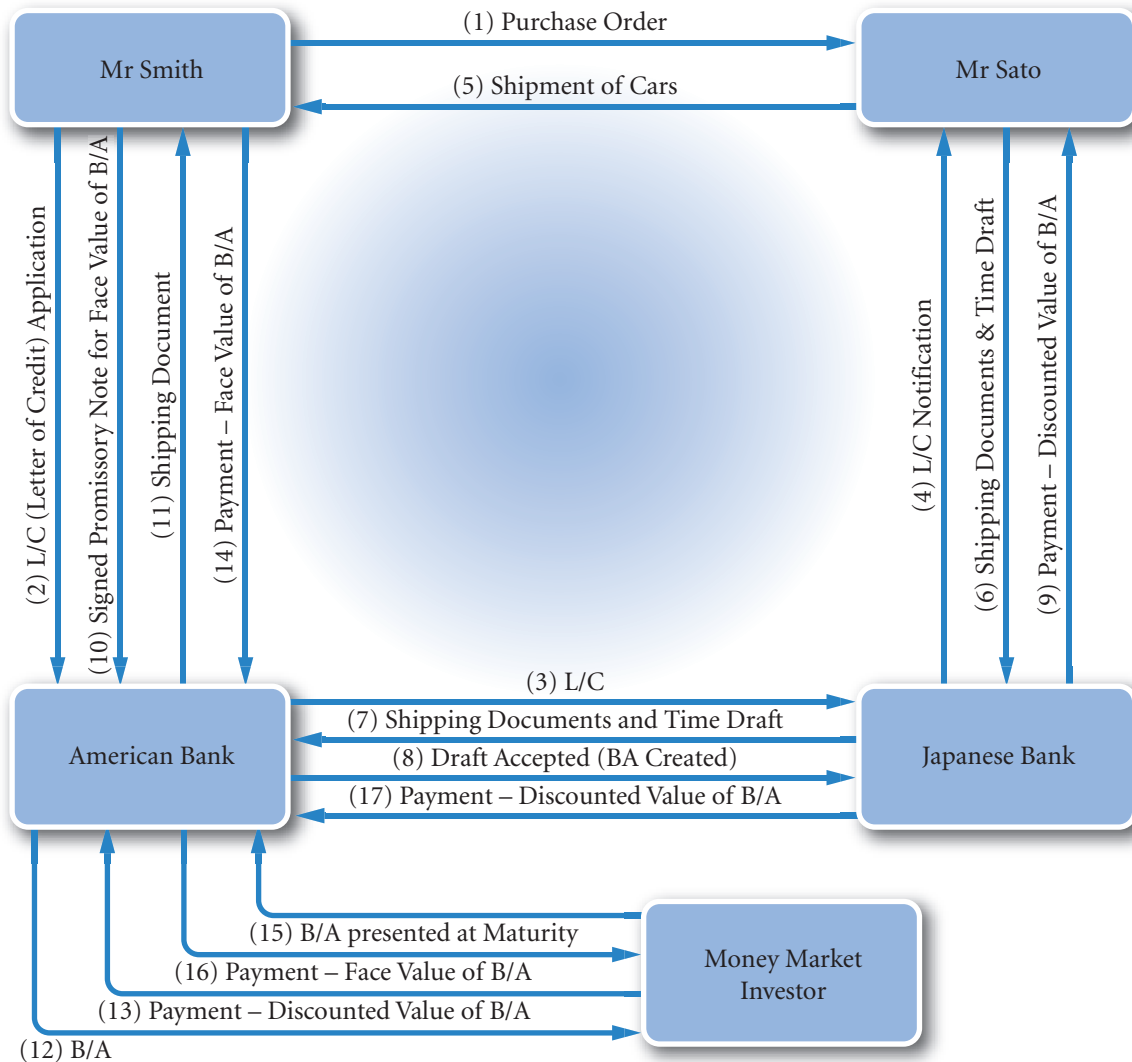
Slow Bank has accepted a time draft from WellKnown plc for £1 500 000 to be paid to WellKnown in six months' time. WellKnown wants the money now and has asked its own bank, MyBank, to sell the bill. MyBank charges a 10% discount to WellKnown. So WellKnown receives $£1\,500\,000 \times (1 - 10\%) = £1\,350\,000$. MyBank then sells the bill to BigInsurance plc for £1 400 000. In effect, MyBank has operated as a go-between and has created an asset, the bill to be paid by Slow Bank, for a cost of £1 350 000 (buying it

from WellKnown) and sold it for £1 400 000. BigInsurance plc has purchased this asset and in effect has used its spare cash ultimately to help provide finance to WellKnown. Big Insurance will collect £1 500 000 from Slow bank in six months' time a return of $(£1\,500\,000 - £1\,400\,000) / £1\,500\,000 = 0.0666$ or 6.66%. This is a return of $1.066^2 - 1 = 0.136$ or 13.6% per year and presumably it is a good return given the low risk of non-payment by Slow Bank.

Banker's acceptance financing can also be arranged through the refinancing of a sight letter of credit. In this case, the beneficiary of the L/C (the exporter) may insist on payment at sight. The bank arranges to finance the payment of the sight L/C under a separate acceptance-financing agreement. The importer (borrower) simply draws drafts upon the bank (a 'you, the bank, owe me' note), the bank in turn accepts and discounts the drafts (i.e. sells the bills on the money markets) on condition that the importer will pay with say a promissory note. The proceeds are used to pay the exporter. At maturity of the discounted acceptance, the money market investor presents the bill to the bank, who pay, and the importer then pays the money to the bank honouring the promissory note. Why is it done in this way? Well, it is important that to the money markets the bank is the debtor – the discount will be less. So the bank creates this debt to the importer with the backing that the importer will pay.

Acceptance financing can also be arranged without the use of an L/C under a separate acceptance agreement. Similar to a regular loan agreement, it stipulates the terms and conditions under which the bank is

EXHIBIT 17.5 Life cycle of a typical banker's acceptance (B/A)



prepared to finance the borrower using acceptances instead of promissory notes. As long as the acceptances meet one of the underlying transaction requirements, the bank and borrower can utilize banker's acceptances as an alternative financing mechanism. The life cycle of a banker's acceptance is illustrated in Exhibit 17.5.

Working capital financing

As just explained, a banker's acceptance can allow an exporter to receive funds immediately, yet allow an importer to delay its payment until a future date. The bank may even provide short-term loans beyond the banker's acceptance period. In the case of an importer, the purchase from overseas usually represents the acquisition of inventory. The loan finances the working capital cycle that begins with the purchase of inventory and continues with the sale of the goods, creation of an account receivable, and finally conversion to cash. With an exporter, the short-term loan might finance the manufacture of the merchandise destined for export (pre-export financing) or the time period from when the sale is made until payment is received from

the buyer. For example, the firm may have imported foreign beer, which it plans to distribute to grocery and liquor stores. The bank can provide a letter of credit for trade finance, and also finance the importer's cost from the time of distribution and collection of payment.

Medium-term capital goods financing (forfaiting)

Because capital goods are often quite expensive, an importer may not be able to make payment on the goods within a short time period. Thus, longer-term financing may be required here. The exporter might be able to provide financing for the importer but may not desire to do so, since the financing may extend over several years. In this case, a type of trade finance known as **forfaiting** could be used. The forfaiter buys the bills or promissory notes from the exporter. These financial obligations are usually guaranteed by the importer (payer's) bank. The importer thus finds that payment has to be made to a financial institution rather than the exporter. Thus, for a fee, the forfaiter enables the exporter to receive the money immediately and the importer at the same time has a period of credit before payment has to be made.

In some respects, forfaiting is similar to factoring, in that the forfaiter (or factor) assumes responsibility for the collection of payment from the buyer, the underlying credit risk and the risk pertaining to the countries involved. Since the forfaiter rather than the exporter now bears the risk of non-payment, it should assess the creditworthiness of the importer as if it were extending a medium-term loan. Forfait transactions normally are supported (collateralized) by a bank guarantee or letter of credit issued by the importer's bank for the term of the transaction. Since obtaining financial information about the importer is usually difficult, the forfaiter (a bank or financial institution) places a great deal of reliance on the bank guarantee as the collateral in the event the buyer fails to pay as agreed. It is this guarantee backing the transaction that has fostered the growth of the forfait market, particularly in Europe, as a practical means of trade finance.

Forfaiting transactions are usually in excess of £250 000 and can be denominated in most currencies. For some larger transactions, more than one bank may be involved as the forfaiter. In this case, a syndicate is formed wherein each participant assumes a proportionate share of the underlying risk and profit. A forfaiting firm may decide to sell the promissory notes of the importer to other financial institutions willing to purchase them. However, the forfaiting firm is still responsible for payment on the notes in the event the importer is unable to pay.

Countertrade

The term **countertrade** denotes all types of foreign trade transactions in which the sale of goods to one country is linked to the purchase or exchange of goods from that same country. Some types of countertrade, such as barter, have been in existence for thousands of years. Only recently, however, has countertrade gained popularity and importance. The growth in various types of countertrade has been fuelled by large balance-of-payment disequilibriums, foreign currency shortages, the debt problems of less developed countries and stagnant worldwide demand. As a result, many MNCs have encountered countertrade opportunities, particularly in Asia, Latin America, and Eastern Europe. The most common types of countertrade include barter, compensation, and counterpurchase.

Barter is the exchange of goods between two parties without the use of any currency as a medium of exchange. Most barter arrangements are one-time transactions governed by one contract. An example would be the exchange of 100 tons of wheat from Canada for 20 tons of shrimp from Ecuador.

In a **compensation** or clearing-account arrangement, the delivery of goods to one party is compensated for by the seller's buying back a certain amount of the product from that same party. The transaction is governed by one contract, and the value of the goods is expressed in monetary terms. The buy-back arrangement could be for a fraction of the original sale (**partial compensation**) or more than 100% of the original sale (**full compensation**). An example of compensation would be the sale of phosphate from Morocco to France in exchange for purchasing a certain percentage of fertilizer. In some countries, this is also referred to as an industrial cooperation arrangement. Such arrangements often involve the construction

of large projects, such as power plants, in exchange for the purchase of the project's output over an extended period of time. For example, Brazil sold a hydroelectric plant to Argentina and in exchange purchased a percentage of the plant's output under a long-term contract.

The term **counterpurchase** denotes the exchange of goods between two parties under two distinct contracts expressed in monetary terms. Delivery and payment of both goods are technically separate transactions.

Despite the economic inefficiencies of countertrade, it has become much more important in recent years. The primary participants are governments and MNCs, with assistance provided by specialists in the field, such as attorneys, financial institutions and trading companies. The transactions are usually large and very complex. Many variations of countertrade exist, and the terminology used by the various market participants is still forming as the countertrade market continues to develop.

GOVERNMENT AGENCIES FOR INTERNATIONAL TRADE

Due to the inherent risks of international trade, government institutions and the private sector offer various forms of export credit, export finance and guarantee programmes to reduce risk and stimulate foreign trade. The private sector is generally reluctant to provide financial support for investments with high risk. Usually such investments are in countries with poorly developed or unreliable financial systems. To provide cover for such transactions most developed countries have a government backed agency whose principal goal is to provide credit and finance for exports.

In the UK export services are provided by the Export Credit Guarantee Department (ECGD). For a fee, it provides a guarantee of payment to UK exporters on transactions ranging from £25 000 to over £100 million. As with a forfaiter, it will also offer loans to importers thus enabling exporters to receive payment immediately and at the same time give time for foreign importers to pay in accordance with the conditions of the loan. Government backing of the ECGD and like organizations elsewhere ensures that the guarantees are well respected in the marketplace and may of themselves be sufficient for a company to then get private finance to support an export transaction. The ECGD also offers insurance against political risk for UK investors of up to 15 years.

As with all government agencies a moral as well as a practical rationale is required to justify their role. On the virtuous side, Third World countries termed Heavily Indebted Poor Countries (HIPC – this is an official IMF term) often have poor credit records. Credit guarantees and loans to importers in those countries encourage trade with HIPCs helping their economies to grow. Thus the ECGD, by offering generous credit terms to HIPCs is providing a form of foreign aid. It is arguable that such help is much better than straight gift-based aid in that trade is encouraged, and trade is generally seen as the principal means of becoming a developed economy.

Some trade is, of course, with governments. The ECGD holds most of the sovereign (government) debt owed to the UK. The agency therefore has a crucial role in any debt relief programme. HIPCs are supported by IMF and World Bank programmes. Satisfactory compliance is measured as achieving a 'completion point' that triggers a debt relief programme. Developed countries in the form of the Paris Club will reduce payments to a sustainable level for countries that achieve their completion point, and in the case of the UK, will write off old debts completely. Thus the ECGD is heavily involved in aid packages.

There is also, surprisingly, a less virtuous argument. First, IMF debt relief programmes are seen by some as intrusive, unaccountable and ineffective. Visit the IMF website – <http://www.imf.org> – and search under the term 'ineffective IMF' – a good example of openness. The progress towards developed nation status is generally recognized as having been slow to non-existent for most HIPCs over the last 50 years. This leads to the second counter argument that the true purpose of the ECGD is to finance and thereby subsidize domestic industry with little real concern for the recipient country. The aid given to Third World countries to purchase exports from developed countries is in fact a way of helping MNCs sell goods to customers of doubtful credit. Although such aid purports to be to help the customer, the suspicion is that the real purpose is to help the MNC. Arms sales, in particular, are difficult to justify as necessary for a developing

country. In tacit acknowledgement of this critique the ECGD sets out a broad mission statement that includes a commitment that:

ECGD will when considering support look not only at the payment risks but also at the underlying quality of the project including its environmental, social and human rights impact.

(Statement of ECGD Business Principles, 2000)

There is also a special procedure over financial help for arms exports (some 24% of ECGD business). In addition to the special licences, for less developed countries it also applies a 'productive expenditure test'. The relevant form asks:

*Please explain how it (the project) will contribute to economic and social development:
You may care to give details of how the project will:*

- *Meet priority human development needs;*
- *Promote the inclusion of the poor in economic activity;*
- *Build or rehabilitate essential infrastructure;*
- *Promote economic growth; or*
- *Promote development of the indigenous private sector.*

For some, these stated intentions are not borne out by practice and there is a campaign for European Credit Agencies (ECAs) to improve their practices (see <http://www.fern.org/pages/eca/eucamp.html>). Undoubtedly, financing international trade has a political dimension. Similar issues can be found in examining 'soft loans' policies, for more information see <http://www.bankwatch.org> and search for 'EIB'. MNCs need to be aware that requesting government help requires an ethical as well as a financial justification.

SUMMARY

- The common methods of payment for international trade are: (1) prepayment (before goods are sent), (2) letters of credit, (3) drafts, (4) consignment, and (5) open account.
- The most popular methods of financing international trade are: (1) accounts receivable financing, (2) factoring, (3) letters of credit, (4) banker's acceptances, (5) working capital financing, (6) medium-term capital goods financing (forfaiting), and (7) countertrade.
- The major agencies that facilitate international trade with export insurance and/or loan programmes are: (1) Export-Import Bank, (2) Private Export Funding Corporation, and (3) Overseas Private Investment Corporation.

CRITICAL DEBATE

Do government agencies promoting exports benefit the recipient country?

Proposition. Yes. They enable countries to import vital supplies needed to develop their economies. The commercial world would not necessarily export to these countries because of the risks that they face. They are in a catch 22 position, they cannot get credit unless they are

developed and they cannot develop unless they have credit to purchase goods. Government agencies are the ideal bodies to break this vicious circle. In addition, the agencies can and do promote a social and economic agenda working with the World Bank and the IMF.

Opposing view. No. The commercial world will lend to these countries and has done so to a greater extent in

colonial days before World War I. Developing countries must realize that a stable currency and a stable economy is a top priority and that instability is more than just a technical problem. Corruption, over ambitious economic policies and social unrest caused by oppression all result in unstable economies. Giving credit and guarantees to these countries has unfortunately served to support poor government practice rather than cure it. Developed

countries are concerned with helping their own firms in competition with other developed countries and any desire to promote standards is lost in the need to protect home industry against competition.

With whom do you agree? Reread the section on the ECGD and follow the website links.

SELF TEST

Answers are provided in Appendix A at the back of the text.

- 1 Explain why so many international transactions require international trade credit facilitated by commercial banks.
- 2 Explain the difference in the risk to the exporter between accounts receivable financing and factoring.
- 3 Explain how the Export-Import Bank can encourage UK firms to export to less developed countries where there is political risk.

QUESTIONS AND APPLICATIONS

- 1 **Banker's acceptances.**
 - a Describe how foreign trade would be affected if banks did not provide trade-related services.
 - b How can a banker's acceptance be beneficial to an exporter, an importer, and a bank?
- 2 **Export financing.**
 - a Why would an exporter provide financing for an importer?
 - b Is there much risk in this activity? Explain.
- 3 **Role of factors.** What is the role of a factor in international trade transactions?
- 4 **ECGD**
 - a What is the role today of the ECGD?
 - b Describe the basis of the political opposition to the ECGD.
- 5 **Bills of lading.** What are bills of lading, and how do they facilitate international trade transactions?
- 6 **Forfaiting.** What is forfaiting? Specify the type of traded goods for which forfaiting is applied.
- 7 Should the **World Bank** be in charge of providing finance for exports?
- 8 **Government programmes.** What motivates a government to establish/intervene in the credit markets?
- 9 **Countertrade.** What is countertrade?
- 10 **Impact of a financial crisis.** What steps should Brunch take if a crisis occurs in country X which is a major exporting destination for Brunch Ltd?

ADVANCED QUESTIONS

- 11 **Letters of credit.** Channel Traders (UK) is a firm based in Grimsby, that specializes in seafood exports and commonly uses letters of credit (L/Cs) to ensure payment. It recently experienced a problem, however. Channel Traders had an irrevocable L/C issued by a Russian bank to ensure that it would receive payment upon shipment of 16 000 tons of fish to a Russian firm. This bank backed out of its obligation, however, stating that it was not authorized to guarantee commercial transactions.
 - a Explain how an irrevocable L/C would normally facilitate the business transaction between the Russian importer and Channel Traders (the UK exporter).
 - b Explain how the cancellation of the L/C could create a trade crisis between the UK and Russian firms.

- c Why do you think situations like this (the cancellation of the L/C) are rare in industrialized countries?
- d Can you think of any alternative strategy that the UK exporter could have used to protect itself better when dealing with a Russian importer?

PROJECT WORKSHOP

- 12** Compare and contrast the web provision for exporters from the UK and the US. For the US provision

visit <http://www.exim.gov> and for the UK provision visit <http://www.ecgd.gov>.

DISCUSSION IN THE BOARDROOM

This exercise can be found on the companion website at www.cengagelearning.co.uk/madura_fox2e.

RUNNING YOUR OWN MNC

This exercise can be found on the companion website at www.cengagelearning.co.uk/madura_fox2e.

Essays/discussion and articles can be found at the end of Part 5

BLADES PLC CASE STUDY

Assessment of international trade financing in Thailand

Blades plc has recently decided to establish a subsidiary in Thailand to produce 'Speedos', Blades' primary roller blade product. In establishing the subsidiary in Thailand, Blades was motivated by the high growth potential of the Thai roller blade market. Furthermore, Blades has decided to establish a subsidiary, as opposed to acquiring an existing Thai roller blade manufacturer for sale, in order to maintain its flexibility and control over the operations in Thailand. Moreover, Blades has decided to issue yen-denominated notes to partially finance the cost of establishing the subsidiary. Blades has decided to issue notes denominated in yen instead of baht to avoid the high effective interest rates associated with the baht-denominated notes.

Currently, Blades plans to sell all roller blades manufactured in Thailand to retailers in Thailand. Furthermore, Blades plans to purchase all components for roller blades manufactured in Thailand from Thai suppliers. Similarly, all of Blades' roller blades manufactured in the United Kingdom will be sold to retailers in the United Kingdom and all components needed for Blades' UK production will be purchased from suppliers in the UK. Consequently, Blades will have no exports and imports once the plant in Thailand is operational, which is expected to occur early next year.

Construction of the plant in Thailand has already begun, and Blades is currently in the process of purchasing the machinery necessary to produce Speedos. Besides these activities, Ben Holt, Blades' financial director, has been actively lining up suppliers of the needed rubber and plastic components in Thailand and identifying Thai customers, which will consist of various sports product retailers in Thailand.

Although Holt has been successful in locating both interested suppliers and interested customers, he is discovering that he has neglected certain precautions for operating a subsidiary in Thailand. First, although Blades is relatively well known in the United States, it is not recognized internationally. Consequently, the suppliers Blades would like to use in Thailand are not familiar with the firm and have no information about its reputation. Moreover, Blades' previous activities in Thailand were restricted to the export of a fixed number of Speedos annually to one customer, a Thai retailer called Entertainment Products. Holt has little information

about the potential Thai customers that would buy the roller blades produced by the new plant. He is aware, however, that although letters of credit (L/Cs) and drafts are usually employed for exporting purposes, these instruments are also used for trade within a country between relatively unknown parties.

Of the various potential customers Blades has identified in Thailand, four retailers of sports products appear particularly interested. Because Blades is not familiar with these firms and their reputations, it would like to receive payment from them as soon as possible. Ideally, Blades would like its customers to prepay for their purchases, as this would involve the least risk for Blades. Unfortunately, none of the four potential customers have agreed to a prepayment arrangement. In fact, one potential customer, Cool Runnings Inc., insists on an open account transaction. Payment terms in Thailand for purchases of this type are typically 'net 60', indicating that payment for the roller blades would be due approximately two months after a purchase was made. Two of the remaining three retailers, Sports Equipment Inc., and Major Leagues Inc., have indicated that they would also prefer an open account transaction; however, both of these retailers have indicated that their banks would act as intermediaries for a time draft. The fourth retailer, Sports Gear Inc., is indifferent as to the specific payment method but has indicated to Blades that it finds a pre-payment arrangement unacceptable.

Blades also needs a suitable arrangement with its various potential suppliers of rubber and plastic components in Thailand. Because Blades' financing of the Thai subsidiary involved a UK bank, it has virtually no contacts in the Thai banking system. Because Blades is relatively unknown in Thailand, Thai suppliers have indicated that they would prefer pre-payment or at least a guarantee from a Thai bank that Blades will be able to make payment within 30 days of purchase. Blades does not currently have accounts receivable in Thailand. It does, however, have accounts receivable in the United Kingdom resulting from its UK sales.

Ben Holt would like to please Blades' Thai customers and suppliers in order to establish strong business relationships in Thailand. However, he is worried that Blades may be at a disadvantage if it accepts all of the

Thai firms' demands. Consequently, he has asked you, a financial analyst for Blades plc to provide him with some guidance regarding international trade financing. Specifically, Holt has asked you to answer the following questions for him:

- 1 Assuming that banks in Thailand issue a time draft on behalf of Sports Equipment Inc., and Major Leagues Inc., would Blades receive payment for its roller blades before it delivers them? Do the banks issuing the time drafts guarantee payment on behalf of the Thai retailers if they default on the payment?
- 2 What payment method should Blades suggest to Sports Gear Inc.? Substantiate your answer.
- 3 What organization could Blades contact in order to insure its sales to the Thai retailers? What type of insurance does this organization provide?
- 4 How could Blades use accounts receivable financing or factoring, considering that it does not currently have accounts receivable in Thailand? If Blades uses a Thai bank to obtain this financing, how do you think the fact that Blades does not have receivables in Thailand would affect the terms of the financing?
- 5 Assuming that Blades is unable to locate a Thai bank that is willing to issue an L/C on Blades' behalf, can you think of a way Blades could utilize its bank in the United Kingdom to effectively obtain an L/C from a Thai bank?
- 6 What organizations could Blades contact to obtain working capital financing? If Blades is unable to obtain working capital financing from these organizations, what are its other options to finance its working capital needs in Thailand?

SMALL BUSINESS DILEMMA

Ensuring payment for products exported by the Sports Exports Company

The Sports Exports Company (Ireland) produces basketballs and exports them to a distributor in the United Kingdom. It typically sends basketballs in bulk and then receives payment after the distributor receives the shipment. The business relationship with the distributor is based on trust. Although the relationship has worked thus far, Jim Logan (owner of the Sports Exports Company) is concerned about the possibility that the distributor will not make its payment.

- 1 How could Jim use a letter of credit to ensure that he will be paid for the products he exports?
- 2 Jim has discussed the possibility of expanding his export business through a second sporting goods distributor in the United States; this second distributor would cover a different territory than the first distributor. The second distributor is only willing to engage in a consignment arrangement when selling basketballs to retail stores. Explain the risk to Jim beyond the typical types of risk he incurs when dealing with the first distributor. Should Jim pursue this type of business?

CHAPTER 18

SHORT-TERM FINANCING

LEARNING OBJECTIVES

The specific objectives of this chapter are to:

- explain why MNCs consider foreign financing,
- explain how MNCs determine whether to use foreign financing, and
- illustrate the possible benefits of financing with a portfolio of currencies.

All firms make short-term financing decisions periodically. Beyond the trade financing discussed in the previous chapter, MNCs obtain short-term financing to support other operations as well. Because MNCs have access to additional sources of funds, their short-term financing decisions are more complex than those of other companies. Financial managers must understand the possible advantages and disadvantages of short-term financing with foreign currencies so that they can make short-term financing decisions that maximize the value of the MNC.

SOURCES OF SHORT-TERM FINANCING

MNC parents and their subsidiaries typically use various methods of obtaining short-term funds to satisfy their liquidity needs. When financing in a foreign currency, it is increasingly the case that the bank or financial market providing the finance will not be in the country of the currency. When this is the case the term ‘euro’ is used. Thus a eurodollar bank account is an account that will probably be situated in London and will be denominated in dollars. A eurobank is a bank that makes loans and accepts deposits in foreign currencies.

Euronotes

One method increasingly used in recent years is the issuing of **euronotes**, or unsecured debt securities. These are bonds issued by MNCs with an interest payment and a fixed term after which the MNC will repay the nominal (agreed) amount. The interest rates on these notes are based on LIBOR (the interest rate eurobanks charge on interbank loans). Euronotes typically have maturities of one, three, or six months. Some MNCs continually roll them over as a form of intermediate-term financing. That is to say, they issue new euronotes to repay ones that are maturing need to be repaid by the MNC. The MNCs in this way extend the term of the loan to themselves. Commercial banks underwrite the notes for MNCs, and some commercial banks purchase them for their own investment portfolios.

Euro-commercial paper

In addition to euronotes, MNCs also issue **euro-commercial paper** to obtain short-term financing. Dealers issue this paper for MNCs without the backing of an underwriting syndicate, so a selling price is not guaranteed to the issuers. Maturities can be tailored to the issuer’s preferences. Dealers make a secondary market by offering to repurchase euro-commercial paper before maturity.

Eurobank loans

Direct loans from eurobanks, which are typically utilized to maintain a relationship with eurobanks, are another popular source of short-term funds for MNCs. If other sources of short-term funds become unavailable, MNCs rely more heavily on direct loans from eurobanks. Most MNCs maintain credit arrangements with various banks around the world. Some MNCs have credit arrangements with more than 100 foreign and domestic banks.

INTERNAL FINANCING BY MNCs

Before an MNC’s parent or subsidiary in need of funds searches for outside funding, it should check other subsidiaries’ cash flow positions to determine whether any internal funds are available.

EXAMPLE

The French subsidiary of Shrimport Ltd (UK) has experienced strong earnings and invested a portion of the earnings locally in money market securities. Meanwhile, Shrimport’s Mexican subsidiary has generated

lower earnings recently but needs funding to support expansion. Shrimport can instruct the French subsidiary to loan some of its excess funds to the Mexican subsidiary.

This process is especially useful during periods when the cost of obtaining funds in the parent's home country is relatively high.

Parents of MNCs can also attempt to obtain financing from their subsidiaries by increasing the markups on supplies they send to the subsidiaries. In this case, the funds the subsidiary gives to the parent will never be returned. This method of supporting the parent can sometimes be more feasible than obtaining loans from the subsidiary because it may circumvent restrictions or taxes imposed by national governments. In some cases, though, this method itself may be restricted or limited by host governments where subsidiaries are located. The practice may also transgress double taxation agreements where intra (within) company trade should be conducted 'at arm's-length', i.e. as if they were independent companies.

WHY MNCs CONSIDER FOREIGN FINANCING

Regardless of whether an MNC parent or subsidiary decides to obtain financing from subsidiaries or from some other source, it must also decide which currency to borrow. Even if it needs its home currency, it may prefer to borrow a foreign currency. Reasons for this preference follow.

Foreign financing to offset foreign currency inflows

A large firm may finance in a foreign currency to offset a net receivables position in that foreign currency.

EXAMPLE

Plenty Ltd (UK) has net receivables denominated in dollars and needs pounds now for liquidity purposes. It can borrow dollars and convert them to British pounds to obtain the needed funds. Then, the net receivables in dollars will be used to pay off the loan. In

this example, financing in a foreign currency reduces the firm's exposure to fluctuating exchange rates. This strategy is especially appealing if the interest rate of the foreign currency is low and the currency is not appreciating.

How Avon used foreign financing during the Asian crisis. During the Asian crisis in 1997 and 1998, many MNCs with Asian subsidiaries were adversely affected by the weakening of Asian currencies against the dollar. Avon Products Inc. used various methods to reduce its economic exposure to the weak Asian currencies. Given that Avon had more cash inflows than cash outflows in Asian currencies, it used strategies that reduced the excess of cash inflows denominated in those currencies. First, it purchased more materials locally. Second, it borrowed funds locally to finance its operations so that it could use some of its cash inflows in Asian currencies to repay the debt. Third, it hired more local salespeople (rather than relying on marketing from the United States) to help sell its products locally. Fourth, it began to remit its earnings more frequently so that excess cash flows denominated in Asian currencies would not accumulate.

Foreign financing to reduce costs

Even when an MNC parent or subsidiary is not attempting to cover foreign net receivables, it may still consider borrowing foreign currencies if the interest rates on those currencies are relatively low and the currency stable. Since interest rates vary among currencies, the cost of borrowing can vary substantially among countries. MNCs that conduct business in countries with high interest rates and a currency that is

not depreciating to compensate, incur a high cost of short-term financing if they finance in the local currency. Thus, they may consider financing with another currency that has a lower interest rate. By shaving 1 percentage point off its financing rate, an MNC can save £1 million annually on debt of £100 million. Thus, MNCs are motivated to consider in detail various currencies when financing their operations.

Exhibit 18.1 compares interest rates among countries for a given point in time. In most periods, the interest rate in Japan is relatively low, while the interest rates in many developing countries are relatively high. Countries with a high rate of inflation tend to have high interest rates.

EXHIBIT 18.1 Short-term interest rates for various countries August 2010

World Interest Rates Table	
Australia	4.25%
Brazil	8.75%
Canada	0.25%
China	5.31%
Czech Republic	1.00%
Egypt	8.25%
European Monetary Union	1.00%
Hong Kong SAR	0.50%
Hungary	5.50%
Iceland	9.00%
India	5.00%
Japan	0.10%
Korea, Republic of	2%
New Zealand	2.50%
Norway	1.75%
Poland	3.50%
Slovakia	1.75%
South Africa	6.50%
Sweden	0.25%
Switzerland	0.25%
Taiwan	1.25%
Turkey	6.50%
United Kingdom	0.50%
United States	0.25%

Source: <http://www.fxstreet.com/fundamental/interest-rates-table/>

EXAMPLE

Salon SA is a French firm that needs euros to expand its EU operations. Assume the euro financing rate is 9%, while the Japanese yen financing rate is 4%. Salon can borrow Japanese yen and immediately convert those yen to euros for use. When the loan repayment is due, Salon will need to obtain Japanese yen to pay off the loan. If the value of the Japanese yen in terms of euros has not changed since the time Salon obtained the loan, it will pay 4% on that loan. If, however, the Japanese yen has increased in value by 2% over the period then the total cost of the loan will be the 4% plus the extra 2% of the cost of buying the

yen to repay the loan (using the approximate method). The total cost will therefore be $4\% + 2\% = 6\%$, still much cheaper than the 9% cost of borrowing in euros. So when choosing a currency for the loan, Salon has to decide whether the lower interest rate in a foreign currency is going to be taken up by the increasing value of that currency. The international Fisher effect (IFE) suggests that the currency will increase in value such that any saving in interest rates will be taken up entirely by the increase in value of the currency and hence the cost of repayment. Salon is hoping that the IFE will not hold if it decides to borrow in yen.

USING THE WEB

Forecasts of interest rates

When an MNC borrows funds in a specific currency, its choice of a maturity is partially based on expectations of future interest rates in that country. Forecasts of interest rates in the near future for each country are provided by a number of websites that can be accessed by search engines; try <http://www.investica.co.uk/rateforecasts.htm>.

DETERMINING THE EFFECTIVE FINANCING RATE

As has been pointed out, the actual cost of financing by the debtor firm will depend on: (1) the interest rate charged by the bank that provided the loan, and (2) the movement in the borrowed currency's value over the life of the loan. Thus, the actual or 'effective' financing rate may differ from the quoted interest rate. This point is further illustrated in the following example.

EXAMPLE

Deckborn Ltd (UK) obtains a two-year loan in New Zealand dollars (NZ\$) worth £1 000 000 at the quoted interest rate of 8% – a full 2% less than an equivalent loan offered to Deckborn in the UK. When Deckborn receives the New Zealand loan, it converts the New Zealand dollars to British pounds to pay a supplier for materials. The

exchange rate at that time is £0.40 per New Zealand dollar, so the NZ\$ 2 500 000 is converted to £1 000 000 (computed as $\text{NZ\$ } 2\,500\,000 \times \text{£}0.40 \text{ per NZ\$} = \text{£}1\,000\,000$). One year later, Deckborn pays interest of NZ\$200 000 (computed as $8\% \times \text{NZ\$ } 2\,500\,000$). Two years later, Deckborn pays back the loan of

(Continued)

NZ\$2 500 000 plus a second interest payment instalment of NZ\$200 000 a total of NZ\$2 700 000. Assume that the New Zealand dollar unexpectedly appreciates from £0.40 to £0.50 shortly after receiving the loan and before any repayments are made. The interest payment

at the end of the first year will therefore cost Deckborn $\text{NZ\$}200\,000 \times 0.50$ or £100 000. At the end of the second year Deckborn will have a bill for NZ\$2 700 000, this will cost £1 350 000 (computed as $\text{NZ\$}2\,700\,000 \times \text{£}0.50$ per NZ\$). The cash flows are as follows:

	Start of period 1 or t_0	End of period 1 or t_1	End of period 2 or t_2
Cash flows in NZ\$s	NZ\$2 500 000	NZ\$200 000	NZ\$2 700 000
Exchange rate £s per NZ\$	£0.40	£0.40	£0.40
Cash flows in £s	+£1 000 000	-£80 000	-£1 080 000
	Start of period 1 or t_0	End of period 1 or t_1	End of period 2 or t_2
Cash flows in NZ\$s	NZ\$2 500 000	NZ\$200 000	NZ\$2 700 000
Exchange rate £s per NZ\$	£0.40	£0.50	£0.50
Cash flows in £s	+£1 000 000	-£100 000	-£1 350 000

The effective cost is the interest rate that 'equates' the repayments with the initial loan. The interest rate is a

cost so the basic formula is repayment less cost equals the original loan. The formula is:

$$\text{Initial loan} + \text{£1 000 000} = \text{repayment} \frac{-\text{£100 000}}{(1+r)^1} + \text{repayment} \frac{-\text{£1 350 000}}{(1+r)^2}$$

Cost of borrowing

A two-period model can be solved by formula but the simplest approach is to solve such problems by trial and error using a spreadsheet (where the cost of borrowing is placed in cell C9 (10% = 0.10 etc.) the formula for the present value in another cell is $=1\,000\,000 - 100\,000 / (1+C9) - 1\,350\,000 / (1+C9)^2$). A sufficiently close result is achieved with an interest rate of 21.3% thus:

$$+£1\,000\,000 = \frac{-£100\,000}{(1+0.213)^1} + \frac{-1\,350\,000}{(1+0.213)^2}$$

Originally, Deckborn was seeking to obtain a loan for 8% some 2% cheaper than the UK equivalent. But because of the increase in the value of the NZ dollar, Deckborn is now having to pay 21.3%.

MANAGING FOR VALUE

Outsourcing short-term asset and liability management

SkyePharma is a UK-based drug delivery company, focused on the development of prescription pharmaceuticals utilizing proprietary controlled release technologies. In May 1996 the company listed on the main

board of the London Stock Exchange and in 1998 achieved a US NASDAQ listing.

In June 2000, SkyePharma issued a five year 6% convertible bond raising £60 million to fund key

(Continued)

research and development projects. The company's strategy is to earn a greater share of product revenues and profits through: 1) providing customers with a wider range of services, and 2) taking selected key products to later stages in the development process independently.

Given this background, SkyePharma's finance function was focused on value added support activities such as strategic risk management rather than day-to-day operational treasury activities.

SkyePharma has specific requirements with regard to the management of the convertible bond proceeds:

- Credit risk has to be diversified across a number of institutions to a minimum credit rating of A+.
- Cash has to be managed in line with stated cash burn obtaining a consistently competitive rate of return.

In addition it was recognized that, to help support the stated business strategy, SkyePharma would require access to an active foreign exchange and interest rate deal execution and hedging resource going forward.

Driven by the costs and management time involved in providing these services in house, a strategic decision was taken to outsource this function. The Royal Bank of Scotland's Agency Treasury Services team was chosen as the preferred outsourcing supplier. The bank was mandated to act as SkyePharma's agent and deal with all aspects of treasury risk.

The services provided fall into three categories:

- Liquidity management.
- Foreign exchange and interest rate risk management.
- Treasury administration and support.

Liquidity management

As a first step, an investment model was created for management of the convertible bond proceeds, which reflected SkyePharma's cash flow requirements and risk/reward parameters. This model covered:

- Credit limits – maximum maturity to be authorized.
- Counterparty list – list of authorized banks/financial institutions and credit ratings.
- Instruments permitted – range of authorized financial instruments and approved amount per instrument.
- Time limits – tenor per counterparty and financial instrument.

- Authorized signatories – list of designated personnel who could approve changes to the agreed investment policy.

This model provides the basis for SkyePharma's investment portfolio. Liquidity is managed out to a pre-agreed panel of counterparties using a competitive tender approach. Working to an ongoing cashflow forecast, cash is repatriated in line with requirements and the portfolio is constantly updated to reflect both ongoing investment needs and alterations in the shape of the sterling yield curve.

Foreign exchange and interest rate risk management

With overseas operations in Europe and the United States, SkyePharma has forex exposures in the Swiss franc, euro and US dollar. To ensure effective control over management of foreign exchange risk, a hedging policy has been pre-agreed. Similar in nature to the liquidity management policy, it covers products, tenors, credit limits and authorized signatories.

In practice, as soon as a foreign exchange exposure has been identified, it is notified to the bank. Under the terms of the outsourcing mandate, it is analyzed and a range of solutions to hedge the risk arising are created for approval. SkyePharma select the solution most appropriate to business needs, and this is then executed on the company's behalf in the marketplace.

In terms of interest rate risk management, early on an opportunity was identified to lower the servicing costs on the convertible bond through interest rate risk management. The structuring, execution and settlement aspects of this issue were also outsourced.

Treasury administration and support

To reduce costs within the finance function, SkyePharma has outsourced all confirmations, settlements, payments and transfers activity. A performance benchmarking service forms part of the outsourcing mandate, with performance reported on a monthly basis against a pre-agreed benchmark. This benchmark is appropriate to the service being utilized, i.e. a LIBID-related benchmark for liquidity management and appropriate foreign exchange budget rates (LIBID is the London Interbank bid rate, the rate banks offer to buy eurocurrency deposits hence determining the interest rate).

(Continued)

Conclusion

Outsourcing is a growing trend in finance, following proven successes in IT, HR and other operational areas. In today's uncertain economic climate, corporations are looking for innovative ways to generate further efficiencies and protect competitive positions. For Skye-

Pharma, outsourcing this part of our business has allowed us to concentrate on strategies for growing the company and has helped us to optimize the value our treasury exposures represent.

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Interest rates can be broken down into differing elements. The relationship is multiplicative. Thus if the overall rate is called the effective financing rate (denoted as r_{eff}) then breaking the rate down into two elements can be represented as:

$$r_{eff} = (1 + r_1) (1 + r_2) - 1$$

or three elements as:

$$r_{eff} = (1 + r_1) (1 + r_2) (1 + r_3) - 1$$

When managing international loans a natural division to examine is how much of the effective interest rate is due to foreign interest rates (i_f) and how much due to the percentage change in the exchange rate (e_f). Thus:

$$r_{eff} = (1 + i_f) (1 + e_f)$$

Taking the case of Deckborn, $i_f = 0.08$ or 8% and the effective rate is 21.3%. So the portion of the effective rate due to changes in the exchange rate (e_f) is:

$$0.213 = (1 + 0.08) (1 + e_f) - 1$$

thus:

$$\begin{aligned} e_f &= 1.213/1.08 - 1 \\ &= 0.123 \text{ or } 12.3\% \end{aligned}$$

Deckborn was charged 8% interest but incurred a further 12.3% due to the fact that the value of the NZ\$ went up from £0.40 to £0.50. As a result interest payments in NZ\$ were more expensive and the loan was also more expensive to repay. But the value of the NZ\$ could have fallen in which case the loan could have been even cheaper.

Assume the same details as in the previous example, only the value of the NZ\$ fell from £0.40 per NZ\$ to £0.35 per NZ\$.

	Start of period 1 or t_0	End of period 1 or t_1	End of period 2 or t_2
Cash flows in NZ\$s	NZ\$2 500 000	NZ\$200 000	NZ\$2 700 000
Exchange rate £s per NZ\$	£0.40	£0.35	£0.35
Cash flows in £s	+£1 000 000	-£70 000	-£945 000

We are now seeking a rate of return that will solve the following equation:

$$+£1\,000\,000 = \frac{-£70\,000}{(1+r)^1} + \frac{-£945\,000}{(1+r)^2}$$

Again using trial and error and a spreadsheet the solution is 0.77% so:

$$+£1\,000\,000 = \frac{-£70\,000}{(1 + 0.0077)^1} + \frac{-945\,000}{(1 + 0.0077)^2}$$

If the NZ dollar were to fall in value in this way, the loan would virtually be free. Thus Deckborn is taking a risk by borrowing in a foreign currency: the cost could be much cheaper than a domestic loan, but if the value of the foreign currency increases, the cost of the loan could be greater. An MNC should therefore consider the expected rate of appreciation or depreciation as well as the quoted interest rates of foreign currencies.

USING THE WEB

Short-term foreign interest rates

Short-term interest rates for major currencies such as the Canadian dollar, Japanese yen and British pound for various maturities are provided at <http://www.bloomberg.com>. The short-term interest rates provided at this site reflect the government cost of borrowing; an MNC would have to pay a slightly higher interest rate than the rate shown. A review of the data illustrates how short-term interest rates can vary among currencies at a given point in time.

CRITERIA CONSIDERED FOR FOREIGN FINANCING

A more formal consideration of the factors relevant to short-term financing should include:

- Interest rate parity
- Exchange rate forecasts

These criteria can influence the MNC's decision regarding which currency or currencies to borrow. Each is discussed in turn.

Interest rate parity

Recall that covered interest arbitrage was described as a short-term foreign investment with a simultaneous forward sale of the foreign currency denominating the foreign investment. From a financing perspective, covered interest arbitrage can be conducted as follows. First, borrow a foreign currency and convert that currency to the home currency for use. Also, simultaneously purchase the foreign currency forward to lock in the exchange rate of the currency needed to pay off the loan. If the foreign currency's interest rate is low, this may appear to be a feasible strategy. However, such a currency will, in an efficient market, exhibit a forward premium that offsets the differential between its interest rate and the home interest rate. Therefore borrowing in the foreign currency will incur the same cost as borrowing domestically. So if a foreign interest rate is 2% cheaper, the currency on the forward market will exhibit a 2% premium (i.e. be more expensive by 2%) and this premium will offset the lower interest rate. Thus there is no benefit through borrowing from abroad.

More formally interest rate parity can be expressed as:

$$r_{eff} = (1 + i_f)(1 + p) - 1$$

where:

- r_{eff} = the overall cost of financing in a foreign currency
- i_f = foreign currency interest rate
- p = the forward rate premium – currency more expensive (or discount – currency cheaper).

If interest rate parity exists then

$$r_{eff} = i_b$$

where:

- i_b = home interest rate

and therefore:

$$i_b = (1 + i_f)(1 + p) - 1$$

and:

$$p = \frac{(1 + i_b)}{(1 + i_f)} - 1 \text{ and approximately for small differences } p = i_b - i_f$$

which is a more formal way of saying that the premium will reflect the difference in interest rates.

Covered interest rate arbitrage is one of the relationships that hold well in the marketplace. If there are differences with domestic returns than administration costs or restrictions will ensure that no riskless (arbitrage) profit can be made.

An MNC has a choice when borrowing internationally, it can cover its position by taking out a forward or futures contract thus ensuring a return similar to domestic returns. Alternatively it can leave its position uncovered. In such a case:

$$r_{eff} = (1 + i_f)(1 + e_f) - 1$$

where:

- e_f = the change in value of the foreign currency.

The forward rate premium (p) had to reflect the difference in interest rates to avoid arbitrage profits; but in this case interest rate parity may or may not exist as there is no implied forward or futures contract in e_f . So a firm may have experiences as in the Deckborn example.

Finally, an MNC may choose to half cover its position by taking out a derivative such as a forward or futures or option contract that protects against some of the risk. In the case of forwards and futures, taking out contracts that do not fully cover the position would mean that some risk remains. In the Deckborn example the New Zealand loan was 2% cheaper therefore the forward rate for the first year can be expected to be 2% more expensive to offset the difference. The current spot was £0.40 per NZ\$ so the forward rate should be £0.408 per NZ\$ for the first year (i.e. $(0.408 - 0.40)/0.40 = 0.02$ or 2%). If Deckborn only covers (by taking out a forward or futures contract) for NZ\$100 000 of the NZ\$200 000 due after the first year, then there is no cover for the cost of repaying the remaining NZ\$100 000, there may be gains or losses compared to domestic lending. Alternatively, an option may be taken out. In this case, Deckborn may take out a call option to buy NZ\$200 000 at £0.42. So that if the NZ\$ increased by more than 5% (i.e. $(0.42 - 0.40) / 0.40 = 0.05$ or 5%) Deckborn could exercise the option and lock in a maximum effective interest rate of 3% more than the domestic (UK) equivalent (5% extra at the £0.42 rate less the 2% lower NZ charge compared to the domestic charge). Similar calculations can be made for the year 2 payments.

Exhibit 18.2 summarizes the implications of a variety of scenarios relating to interest rate parity.

Finally, an issue dealt with in Chapter 16 and earlier, is the effect of the borrowing on the overall exchange rate exposure of the MNC. Short-term financing in a foreign currency implies cash outflows in

EXHIBIT 18.2 Financing alternatives and immediate implications

Scenario	Implications
All repayments fully covered by forward or futures contracts.	The effective interest rate will be about the same as the domestic rate (there may be differences due to transaction costs and market imperfections).
All repayments made at the prevailing exchange rate.	If interest parity holds, the effective rate will be similar to the domestic rate. If interest rate parity does not hold, the effective cost could be higher or lower than the domestic rate depending on the change in value of the foreign currency.
Some repayments covered by a futures or forwards contract.	As in scenario 1 for the elements covered by the contract and as in scenario 2 for the uncovered element.
Repayments covered by options.	Assuming that the option is out of the money (the exercise price is above the currency spot), if the value of the foreign currency goes above the level determined by the strike or exercise price, the option will be exercised and the increased cost of payment restricted to that of the exercise price.

that currency. Where there are earnings in that currency as well, the borrowing will give a degree of protection from exchange rate exposure. So, if the value of the currency falls, the adverse effect on translation of earnings into the home currency will be offset partially (or even fully) by the saving made by the cheaper cost home currency of payments of interest and capital in the foreign currency. The matching is not an exact science but setting any cash outflow in a currency against an inflow will help protect against the effects of changes in the value of that currency.

Exchange rate forecasts

While the forecasting capabilities of firms are somewhat limited, some firms may make decisions based on their estimation of the future value of the currency. The estimate may be based on market forecasts through a stockbroker subscription service or the quoted futures rate may be taken as a prediction of the future rate (effectively assuming that interest rate parity holds). Or a firm may use its own in-house economists to make an estimate. Methods of prediction used by staff may range from sophisticated econometric techniques to purely subjective judgements.

A point estimate or prediction of a single rate is rarely sufficient. Most MNCs will want to have an idea of the potential variation in the rate that could occur in the future. Only by knowing these possibilities can some impression of risk be gained. In looking at future possibilities two approaches are common. The first is to develop a probability distribution, looking at the varying rates and their probabilities. This is the dominant approach in financial theory leading to measures of spread such as the standard deviation and also to measures of association as in covariance or correlation. A second approach is to assess scenarios. Probabilities may or may not be used, for instance, if a firm wants to adopt a policy of protecting against the three worst scenarios, no probability calculations would be required. Scenarios are not confined to predicting the value of the exchange rate. There may, for instance, be a series of scenarios whereby the value of the foreign currency falls by 5%, but the differences in the scenarios is what is happening to other exchange rates or to international regulations or government policies. So this approach is more than merely approximating a distribution of exchange rates.

Starting with a prediction of a single future rate, once the firm develops a forecast or series of forecasts for the exchange rate's percentage change over the financing period (e_f), it can use this forecast along with the foreign interest rate to forecast the effective financing rate of a foreign currency. The forecasted rate can then be compared to the domestic financing rate.

EXAMPLE

Danya Ltd (Ireland) needs funds for one year and is aware that the one-year interest rate in euros is 12% while the interest rate from borrowing Swiss francs is 8%. Danya forecasts that the Swiss franc will appreciate from its current rate of 0.64 euros to 0.6528 euros or by 2% over the next year. The expected value for e_f [written as $E(e_f)$] will therefore be 2%. Thus, the expected effective financing rate $E(r_f)$ will be:

$$\begin{aligned} E(r_{\text{eff}}) &= (1 + i_f)(1 + E(e_f)) - 1 \\ &= (1 + 0.08)(1 + 0.02) - 1 \\ &= 0.1016 \text{ or } 10.16\% \end{aligned}$$

In this example, financing in Swiss francs is expected to be less expensive than financing in euros. However, the value for e_f is forecasted and therefore is not known with certainty. Thus, there is no guarantee that foreign financing will truly be less costly.

Deriving a value for e_f that equates domestic and foreign rates. Continuing from the previous example, Danya Ltd may attempt at least to determine what value of e_f would make the effective rate from foreign financing the same as domestic financing. To determine this value, note that e_f plays the same role as ' p ' above so:

$$e_f = \frac{(1 + i_b)}{(1 + i_f)} - 1 \text{ and approximately for small differences } e_f = i_b - i_f$$

$i_b = 12\%$ and $i_f = 8\%$ so the change in value of foreign currency (e_f) that equates domestic and foreign rates is $12\% - 8\% = 4\%$. The more accurate calculation is:

$$e_f = \frac{(1 + 0.12)}{(1 + 0.08)} - 1 = 0.037037 \text{ or } 3.704\%$$

This suggests that the Swiss franc would have to appreciate by about 3.7% over the loan period to make the Swiss franc loan as costly as a loan in euros. Any smaller degree of appreciation would make the Swiss franc loan less costly. Danya can use this information when determining whether to borrow euros or Swiss francs. If it expects the Swiss franc to appreciate by more than 3.7% over the loan life, it should prefer borrowing in euros. If it expects the Swiss franc to appreciate by less than 3.7% or to depreciate, its decision is more complex. If the potential savings from financing with the foreign currency outweigh the risk involved, then the firm should choose that route. The final decision here will be influenced by Danya's degree of risk aversion.

Use of probability distributions. To gain more insight about the financing decision, a firm may wish to develop a probability distribution for the percentage change in value for a particular foreign currency over the financing horizon. Since forecasts are not always accurate, it is sometimes useful to develop a probability distribution instead of relying on a single point estimate. Using the probability distribution of possible percentage changes in the currency's value, along with the currency's interest rate, the firm can determine the probability distribution of the possible effective financing rates for the currency. Then, it can compare this distribution to the known financing rate of the home currency in order to make its financing decision.

EXAMPLE

Caroline Ltd (UK) is deciding whether to borrow Swiss francs for one year. It finds that the bank quote for Caroline to borrow in Swiss francs is 8% and the quoted rate for the pound is 15%. It then develops a probability distribution for the Swiss franc's possible percentage change in value over the life of the loan.

The probability distribution is displayed in Exhibit 18.3. The first row in Exhibit 18.3 shows that there is a 5% probability of a 6% depreciation in the Swiss franc over the loan life. If the Swiss franc does depreciate by 6%, the effective financing rate would be 1.52%. Thus, there is a 5% probability that Caroline will incur a 1.52% effective financing rate on its loan. The second row shows that there is a 10% probability of a 4% depreciation in the Swiss franc over the loan life. If the

Swiss franc does depreciate by 4%, the effective financing rate would be 3.68%. Thus, there is a 10% probability that Caroline will incur a 3.68% effective financing rate on its loan.

For each possible percentage change in the Swiss franc's value, there is a corresponding effective financing rate. We can associate each possible effective financing rate (third column) with its probability of occurring (second column). By multiplying each possible effective financing rate by its associated probability, we can compute an expected value for the effective financing rate of the Swiss franc. Based on the information in Exhibit 18.3, the expected value of the effective financing rate, referred to as $E(r_f)$, is computed as:

$$\begin{aligned} E(r_f) &= 5\%(1.52\%) + 10\%(3.68\%) + 15\%(6.92\%) + 20\%(9.08\%) + 20\%(12.32\%) + 15\%(14.48\%) \\ &\quad + 10\%(16.64\%) + 5\%(18.80\%) \\ &= 0.076\% + 0.368\% + 1.038\% + 1.816\% + 2.464\% + 2.172\% + 1.664\% + 0.94\% \\ &= 10.538\% \end{aligned}$$

Thus, the decision for Caroline is whether to borrow pounds (at 15% interest) or Swiss francs (with an expected value of 10.538% for the effective financing rate). Using Exhibit 18.3, the risk reflects the 5% chance (probability) that the effective financing rate on

Swiss francs will be 18.8% and the 10% chance that the effective financing rate on Swiss francs will be 16.64%. Either of these possibilities represents a greater expense to Caroline than it would incur if it borrowed British pounds.

To further assess the decision regarding which currency to borrow, the information in the second and third columns of Exhibit 18.3 is used to develop the probability distribution in Exhibit 18.4. This exhibit illustrates the probability of each possible effective financing rate that may occur if Caroline borrows Swiss francs. Notice that the UK interest rate (15%) is included in Exhibit 18.4 for comparison purposes. There is no distribution of possible outcomes for the UK rate since the rate of 15% is known with certainty (no exchange rate risk exists). There is a 15% probability that the UK rate will be lower than the effective rate on Swiss francs and an 85% chance that the UK rate will be higher than the effective rate on Swiss francs. This information can assist the firm in its financing decision. Given the potential savings relative to the small degree of risk, Caroline decides to borrow Swiss francs.

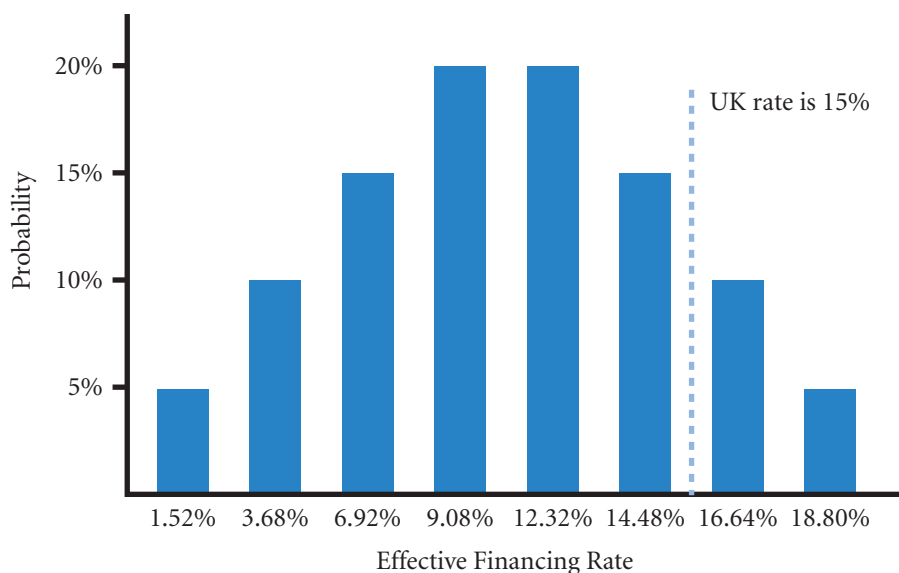
ACTUAL RESULTS FROM FOREIGN FINANCING

The fact that some firms utilize foreign financing suggests that they believe reduced financing costs can be achieved. To assess this issue, the effective financing rates of a Swiss Franc loan to a UK company compared to a UK loan are compared in Exhibit 18.5. The somewhat artificial assumption is that for both loans the variable monthly deposit rate is charged for the month and then the loan is repaid at the end of the month – the rates have been annualized. We look at the cost as a percentage of the original loan for each month

EXHIBIT 18.3 Analysis of financing with a foreign currency

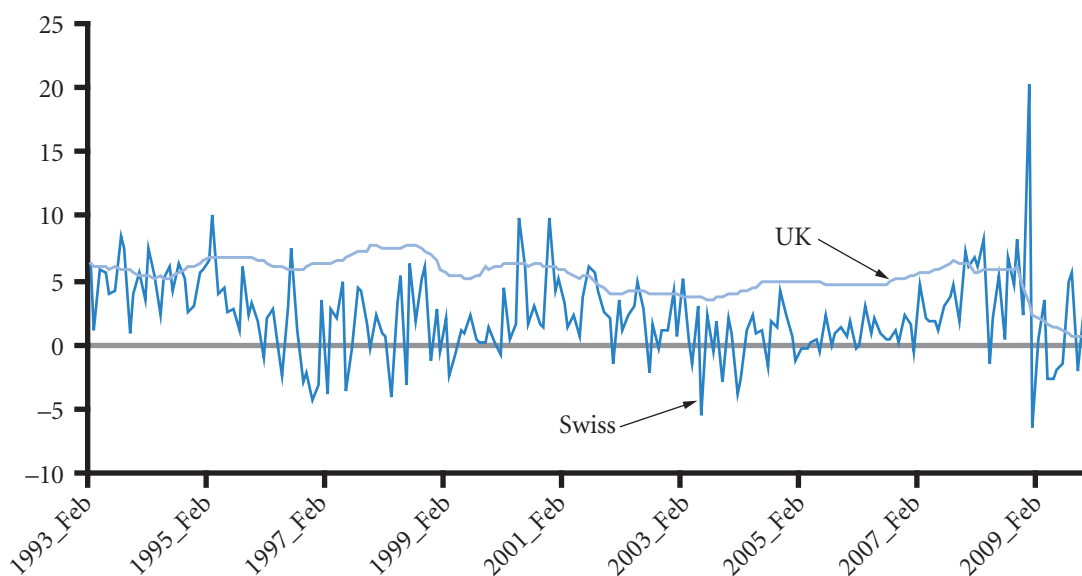
Possible rate of change in the Swiss franc over the life of the loan	Probability of occurrence	Effective financing rate if this rate of change in the (e_t) Swiss franc does occur (r_{eff})
-6%	5%	$(1.08)[1 + (-6\%)] - 1 = 1.52\%$
-4	10	$(1.08)[1 + (-4\%)] - 1 = 3.68$
-1	15	$(1.08)[1 + (-1\%)] - 1 = 6.92$
+1	20	$(1.08)[1 + (1\%)] - 1 = 9.08$
+4	20	$(1.08)[1 + (4\%)] - 1 = 12.32$
+6	15	$(1.08)[1 + (6\%)] - 1 = 14.48$
+8	10	$(1.08)[1 + (8\%)] - 1 = 16.64$
+10	5	$(1.08)[1 + (10\%)] - 1 = 18.80$
	100%	

EXHIBIT 18.4 Probability distribution of effective financing rates



from January 1995 to November 2009. For the Swiss loan, the payment is in Swiss francs; so if the Swiss franc has fallen by more than the interest rate, the repayment could be less than the original loan! This indeed happened on just over half the occasions. The average interest rate of the Swiss loan was 3.2% compared to 3.5% for a domestic loan. However, the standard deviation of the Swiss loan was 36% compared to 1% for the UK loan! The Swiss loan is very much a gamble on the exchange rate and unless it is offsetting earnings does not seem to be a sensible policy for an MNC.

EXHIBIT 18.5 Comparison of the monthly British pound cost of financing a Swiss Franc loan compared with a UK loan



FINANCING WITH A PORTFOLIO OF CURRENCIES

MNCs may be able to achieve lower financing costs without excessive risk by financing with a portfolio of foreign currencies, as demonstrated here.

EXAMPLE

Niels BV (Netherlands) needs to borrow 100,000 euros for one year and obtains the following interest rate quotes:

- Interest rate for a one-year loan in euros = 15%
- Interest rate for a one-year loan in Swiss francs = 8%
- Interest rate for a one-year loan in Japanese yen = 9%

Since the quotes for a loan in Swiss francs or Japanese yen are relatively low, Niels may desire to borrow in a foreign currency. If Niels decides to use foreign financing, it has three choices based on the information given: (1) borrow only Swiss francs, (2) borrow only Japanese yen, or (3) borrow a portfolio of Swiss francs and Japanese yen. Assume that Niels BV has established possi-

ble percentage changes in the spot rate for both the Swiss franc and the Japanese yen from the time the loan would begin until loan repayment, as shown in the second column of Exhibit 18.6. The third column shows the probability that each possible percentage change might occur.

Based on the assumed interest rate of 8% for the Swiss franc, the effective financing rate is computed for each possible percentage change in the Swiss franc's spot rate over the loan life. There is a 30% chance that the Swiss franc will appreciate by 1% over the loan life. In that case, the effective financing rate will be 9.08%. Thus, there is a 30% chance that the effective financing rate will be 9.08%. Furthermore, there is a 50% chance that the effective financing rate will be 11.24% and a 20% chance that it will be

(Continued)

EXHIBIT 18.6 Derivation of possible effective financing rates

Currency	Possible change in the spot rate over the loan life	Probability of that percentage change in the spot rate occurring	Computation of effective financing rate based on the percentage change in the spot rate
Swiss franc	1%	30%	$(1.08)[1 + (0.01)] - 1 = 0.0908$, or 9.08%
Swiss franc	3	50	$(1.08)[1 + (0.03)] - 1 = 0.1124$, or 11.24%
Swiss franc	9	20	$(1.08)[1 + (0.09)] - 1 = 0.1772$, or 17.72%
		100%	
Japanese yen	-1%	35%	$(1.09)[1 + (-0.01)] - 1 = 0.0791$, or 7.91%
Japanese yen	3	40	$(1.09)[1 + (0.03)] - 1 = 0.1227$, or 12.27%
Japanese yen	7	25	$(1.09)[1 + (0.07)] - 1 = 0.1663$, or 16.63%
		100%	

17.72%. Given that the euro loan rate is 15%, there is only a 20% chance that financing in Swiss francs will be more expensive than domestic financing.

The lower section of Exhibit 18.6 provides information on the Japanese yen. For example, the yen has a 35% chance of depreciating by 1% over the loan life, and so on. Based on the assumed 9% interest rate and the exchange rate fluctuation forecasts, there is a 35% chance that the effective financing rate will be 7.91%, a 40% chance that it will be 12.27%, and a 25% chance that it will be 16.63%. Given the 15% rate on euro financing, there is a 25% chance that financing in Japanese yen will be more costly than domestic financing. Before examining the third possible foreign financing strategy (the portfolio approach), determine the expected value of the effective financing rate for each foreign currency by itself. This is accomplished by totaling the products of each possible effective financing rate and its associated probability as follows:

Currency	Computation of expected value of effective financing rate
Swiss francs	$30\% (9.08\%) + 50\% (11.24\%) + 20\% (17.72\%) = 11.888\%$
Japanese yen	$35\% (7.91\%) + 40\% (12.27\%) + 25\% (16.63\%) = 11.834\%$

The expected financing costs of the two currencies are almost the same. The individual degree of risk (that the costs of financing will turn out to be higher than domestic financing) is about the same for each currency. If Niels chooses to finance with only one of these foreign currencies, it is difficult to pinpoint (based on our analysis) which currency is more appropriate. Now, consider the third and final foreign financing strategy: the portfolio approach.

Based on the information in Exhibit 18.6, there are three possibilities for the Swiss franc's effective financing rate. The same holds true for the Japanese yen. If Niels BV borrows half of its needed funds in each of the foreign currencies, then there will be nine possibilities for this portfolio's effective financing rate, as shown in Exhibit 18.7. Columns 1 and 2 list all possible joint effective financing rates. Column 3 computes the joint probability of that occurrence assuming that exchange rate movements of the Swiss franc and Japanese yen are independent. Column 4 shows the computation of the portfolio's effective financing rate based on the possible rates shown for the individual currencies.

An examination of the top row will help to clarify the table. This row indicates that one possible outcome of borrowing both Swiss francs and Japanese yen is that they will exhibit effective financing rates of 9.08% and 7.91%, respectively. The probability of the Swiss franc's effective financing rate occurring is 30%, while

the probability of the Japanese yen rate occurring is 35%. Recall that these percentages were given in Exhibit 18.6. The joint probability that both of these rates will occur simultaneously is $(30\%)(35\%) = 10.5\%$ (given that there is no correlation between the two rates). Assuming that half (50%) of the funds needed are to be borrowed from each currency, the portfolio's effective financing rate will be $0.5(9.08\%) + 0.5(7.91\%) = 8.495\%$ (if those individual effective financing rates occur for each currency).

A similar procedure was used to develop the remaining eight rows in Exhibit 18.7. From this table, there is a 10.5% chance that the portfolio's effective financing rate will be 8.495%, a 12% chance that it will be 10.675%, and so on.

Exhibit 18.8 displays the probability distribution for the portfolio's effective financing rate that was derived in Exhibit 18.7. This exhibit shows that financing with a portfolio (50% financed in Swiss francs with the remaining 50% financed in Japanese yen) has only a 5% chance of being more costly than domestic financing. These results are more favourable than those of either individual foreign currency. Therefore, Niels decides to borrow the portfolio of currencies.

Portfolio diversification effects

When both foreign currencies are borrowed, the only way the portfolio will exhibit a higher effective financing rate than the domestic rate is if *both* currencies experience their maximum possible level of appreciation (which is 9% for the Swiss franc and 7% for the Japanese yen). If only one does, the severity of its appreciation will be somewhat offset by the other currency's not appreciating to such a large extent. The probability of maximum appreciation is 20% for the Swiss franc and 25% for the Japanese yen. The joint probability of both of these events occurring simultaneously is $(20\%)(25\%) = 5\%$. This is an advantage of financing in a portfolio of foreign currencies. Niels has a 95% chance of attaining lower costs with the foreign portfolio than with domestic financing.

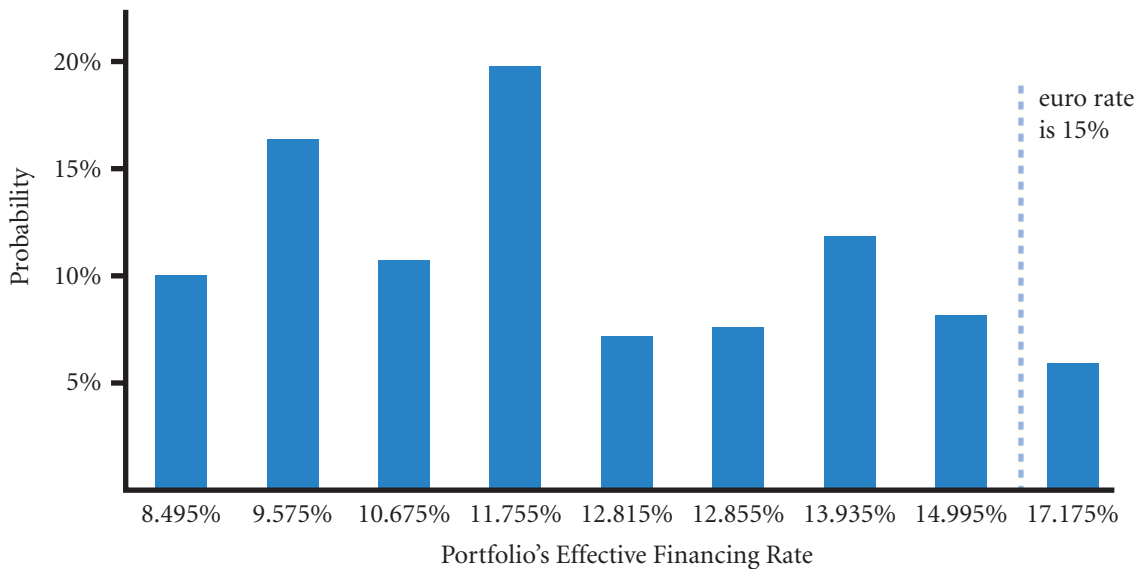
The expected value of the effective financing rate for the portfolio can be determined by multiplying the percentage financed in each currency by the expected value of that currency's individual effective financing rate. Recall that the expected value was 11.888% for the Swiss franc and 11.834% for the Japanese yen. Thus, for a portfolio representing 50% of funds

EXHIBIT 18.7 Analysis of financing with two foreign currencies based on no correlation between yen and Swiss franc changes in value

(1)	(2)	(3)	(4)
Swiss franc	Japanese yen	Computation of joint probability assuming no correlation between yen and Swiss franc changes in value	Computation of effective financing rate of portfolio (50% of total funds borrowed in each currency)
9.08%	7.91%	$(30\%)(35\%) = 10.5\%$	$0.5(9.08\%) + 0.5(7.91\%) = 8.495\%$
9.08	12.27	$(30\%)(40\%) = 12.0$	$0.5(9.08\%) + 0.5(12.27\%) = 10.675$
9.08	16.63	$(30\%)(25\%) = 7.5$	$0.5(9.08\%) + 0.5(16.63\%) = 12.855$
11.24	7.91	$(50\%)(35\%) = 17.5$	$0.5(11.24\%) + 0.5(7.91\%) = 9.575$
11.24	12.27	$(50\%)(40\%) = 20.0$	$0.5(11.24\%) + 0.5(12.27\%) = 11.755$
11.24	16.63	$(50\%)(25\%) = 12.5$	$0.5(11.24\%) + 0.5(16.63\%) = 13.935$
17.72	7.91	$(20\%)(35\%) = 7.0$	$0.5(17.72\%) + 0.5(7.91\%) = 12.815$
17.72	12.27	$(20\%)(40\%) = 8.0$	$0.5(17.72\%) + 0.5(12.27\%) = 14.995$
17.72	16.63	$(20\%)(25\%) = 5.0$	$0.5(17.72\%) + 0.5(16.63\%) = 17.175$
		100.0%	

(Continued)

EXHIBIT 18.8 Probability distribution of the portfolio's effective financing rate



borrowed in each currency, the expected value of the effective financing rate is $0.5 (11.888\%) + 0.5 (11.834\%) = 11.861\%$. Based on an overall comparison, the expected value of the portfolio's effective

financing rate is very similar to that from financing solely in either foreign currency. However, the risk (of incurring a higher effective financing rate than the domestic rate) is substantially less when financing with the portfolio.

In the example, the computation of joint probabilities requires the assumption that the two currencies move independently. If movements of the two currencies are actually highly positively correlated, then financing with a portfolio of currencies will not be as beneficial as demonstrated because there is a strong likelihood of both currencies experiencing a high level of appreciation simultaneously. If the two currencies are not highly correlated, they are less likely to simultaneously appreciate to such a degree. Thus, the chances that the portfolio's effective financing rate will exceed the UK rate are reduced when the currencies included in the portfolio are not highly positively correlated.

The example included only two currencies in the portfolio. Financing with a more diversified portfolio of additional currencies that exhibit low interest rates might increase the probability that foreign financing will be less costly than domestic financing; several currencies are unlikely to move in tandem and therefore unlikely to simultaneously appreciate enough to offset the advantage of their low interest rates. Again, the degree to which these currencies are correlated with each other is important. If all currencies are highly positively correlated with each other, financing with such a portfolio would not be very different from financing with a single foreign currency.

Repeated financing with a currency portfolio

A firm that repeatedly finances with a currency portfolio would normally prefer to compose a financing package that exhibits a somewhat predictable effective financing rate on a periodic basis. The more volatile a portfolio's effective financing rate over time; the more uncertainty (risk) there is about the effective financing rate that will exist in any period. The degree of volatility depends on the standard deviations and paired correlations of effective financing rates of the individual currencies within the portfolio.

We can use the portfolio variance as a measure of the degree of volatility. The variance of a two-currency portfolio's effective financing rate [$VAR(r_p)$] over time is computed as:

$$VAR(r_p) = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B CORR_{AB}$$

where w_A and w_B represent the percentage of total funds financed from Currencies A and B, respectively; σ_A^2 and σ_B^2 represent the individual variances of each currency's effective financing rate over time; and $CORR_{AB}$ reflects the correlation coefficient of the two currencies' effective financing rates. Since the percentage change in the exchange rate plays an important role in influencing the effective financing rate, it should not be surprising that $CORR_{AB}$ is strongly affected by the correlation between the exchange rate fluctuations of the two currencies. A low correlation between movements of the two currencies may force $CORR_{AB}$ to be low.

EXAMPLE

Valparaiso SA (Spain) considers borrowing a portfolio of Japanese yen and Swiss francs to finance its euro operations. Half of the needed funding would come from each currency. To determine how the variance in this portfolio's effective financing rate is related to characteristics of the component currencies, assume the following information based on historical information for several three-month periods:

- Mean effective financing rate of Swiss franc for three months = 3% or 0.03
- Mean effective financing rate of Japanese yen for three months = 2% or 0.02
- Standard deviation of Swiss franc's effective financing rate = 0.04
- Standard deviation of Japanese yen's effective financing rate = 0.09
- Correlation coefficient of effective financing rates of these two currencies = 0.10

Given this information, the mean effective rate on a portfolio (r_p) of funds financed 50% by Swiss francs and 50% by Japanese yen is determined by totalling the weighted individual effective financing rates:

$$\begin{aligned} r_p &= W_{SFr} f_{SFr} + W_{Yen} f_{Yen} \\ &= 0.50(0.03) + 0.50(0.02) \\ &= 0.015 + 0.01 \\ &= 0.025 \text{ or } 2.5\% \end{aligned}$$

The variance of this portfolio's effective financing rate over time is:

$$\begin{aligned} VAR(r_p) &= 0.5^2(0.04)^2 + 0.5^2(0.09)^2 \\ &\quad + 2(0.5)(0.5)(0.04)(0.09)(0.10) \\ &= 0.25(0.0016) + 0.25(0.0081) + 0.00018 \\ &= 0.002605 \end{aligned}$$

The standard deviation of the portfolio is:

$$\sqrt{0.002605} = 0.051 \text{ or } 5.1\%$$

Valparaiso can use this same process to compare various financing packages to see which package would be most appropriate. It may be more interested in estimating the mean return and variability for repeated financing in a particular portfolio in the future. There is no guarantee that past data will be indicative of the future. Yet, if the individual variability and paired correlations are somewhat stable over time, the historical variability of the portfolio's effective financing rate should provide a reasonable forecast.

To recognize the benefits from financing with two currencies that are not highly correlated, reconsider how the variance of the portfolio's effective financing rate would have been affected if the correlation

between the two currencies was 0.90 (very high correlation) instead of 0.10. The variance would be 0.004045, which is more than 50% higher than the variance when the correlation was assumed to be 0.10.

The assessment of a currency portfolio's effective financing rate and variance is not restricted to just two currencies. The mean effective financing rate for a currency portfolio of any size will be determined by totaling the respective individual effective financing rates weighted by the percentage of funds financed with each currency. Solving the variance of a portfolio's effective financing rate becomes more complex as more currencies are added to the portfolio; the underlying model, however, is the variance/covariance matrix first encountered in Chapter 3.

SUMMARY

- MNCs may use foreign financing to offset anticipated cash inflows in foreign currencies so that exposure to exchange rate risk will be minimized. Alternatively, some MNCs may use foreign financing in an attempt to reduce their financing costs. Foreign financing costs may be lower if the foreign interest rate is relatively low or if the foreign currency borrowed depreciates over the financing period.
- MNCs can determine whether to use foreign financing by estimating the effective financing rate for any foreign currency over the period in which financing will be needed. The expected effective financing rate is dependent on the quoted interest rate of the foreign currency and the forecasted percentage change in the currency's value over the financing period.
- When MNCs borrow a portfolio of currencies that have low interest rates, they can increase the probability of achieving relatively low financing costs if the currencies' values are not highly correlated.

CRITICAL DEBATE

Do MNCs increase their risk when borrowing foreign currencies?

Proposition. Yes. MNCs should borrow the currency that matches their cash inflows. If they borrow a foreign currency to finance business in a different currency, they are essentially speculating on the future exchange rate movements. The results of the strategy are uncertain, which represents risk to the MNC and its shareholders.

Opposing view. No. If MNCs expect that they can reduce the effective financing rate by borrowing a foreign

currency, they should consider borrowing that currency. This enables them to achieve lower costs and improves their ability to compete. If they take the most conservative approach by borrowing whatever currency matches their inflows, they may incur higher costs and have a greater chance of failure.

With whom do you agree? Which argument do you support? Offer your own opinion on this issue.

SELF TEST

Answers are provided in Appendix A at the back of the text.

- 1 Assume that the interest rate in New Zealand is 9%. A UK firm plans to borrow New Zealand dollars, convert them to UK pounds, and repay the loan in one year. What will be the effective financing rate if the New Zealand dollar depreciates by 6%? If the New Zealand dollar appreciates by 3%?
- 2 Using the information in question 1 and assuming a 50% chance of either scenario occurring, determine the expected value of the effective financing rate.

- 3 Assume that the Japanese one-year interest rate is 5%, while the UK one-year interest rate is 8%. What percentage change in the Japanese yen would cause a UK firm borrowing yen to incur the same effective financing rate as it would if it borrowed pounds?
- 4 The spot rate of the Australian dollar is £0.40. The one-year forward rate of the Australian dollar is £0.38. The Australian one-year interest rate is 9%. Assume that the forward rate is used to forecast the future spot rate. Determine the expected effective

financing rate for a UK firm that borrows Australian dollars to finance its UK business.

- 5 Omaha SA (France) plans to finance its UK operations by repeatedly borrowing two currencies with low interest rates whose exchange rate movements are highly correlated. Will the variance of the two-currency portfolio's effective financing rate be much lower than the variance of either individual currency's effective financing rate? Explain.

QUESTIONS AND APPLICATIONS

- 1 **Financing from subsidiaries.** Explain why an MNC parent would consider financing from its subsidiaries.
- 2 **Foreign financing.**
 - a Explain how a firm's degree of risk aversion enters into its decision of whether to finance in a foreign currency or a local currency.
 - b Discuss the use of specifying a break-even point when financing in a foreign currency.
- 3 **Probability distribution.**
 - a Discuss the development of a probability distribution of effective financing rates when financing in a foreign currency. How is this distribution developed?
 - b Once the probability distribution of effective financing rates from financing in a foreign currency is developed, how can this distribution be used in deciding whether to finance in the foreign currency or the home currency?
- 4 **Financing and exchange rate risk.** How can a UK firm finance in euros and not necessarily be exposed to exchange rate risk?
- 5 **Short-term financing analysis.** Assume that Tilly Ltd needs £3 million for a one-year period. Within one year, it will generate enough British pounds to pay off the loan. It is considering three options: (1) borrowing pounds at an interest rate of 6%, (2) borrowing Japanese yen at an interest rate of 3%, or (3) borrowing Canadian dollars at an interest rate of 4%. Tilly expects that the Japanese yen will appreciate by 1% over the next year and that the Canadian dollar will appreciate by 3%. What is the expected 'effective' financing rate for each of the three options? Which option appears to be most feasible? Why might Tilly not necessarily choose the option reflecting the lowest effective financing rate?
- 6 **Effective financing rate.** How is it possible for a firm to incur a negative effective financing rate?
- 7 **IRP application to short-term financing.**
 - a If interest rate parity does not hold, what strategy should Connect Ltd consider when it needs short-term financing?
 - b Assume that Connect Ltd needs pounds. It borrows euros at a lower interest rate than that for pounds. If interest rate parity exists and if the forward rate of the euro is a reliable predictor of the future spot rate, what does this suggest about the feasibility of such a strategy?
 - c If Connect Ltd expects the current spot rate to be a more reliable predictor of the future spot rate, what does this suggest about the feasibility of such a strategy?
- 8 **Break-even financing.** Akron Ltd needs dollars. Assume that the local one-year loan rate is 15%, while a one-year loan rate on euros is 7%. By how much must the euro appreciate to cause the loan in euros to be more costly than a UK loan?
- 9 **IRP application to short-term financing.** Assume that interest rate parity exists. If a firm believes that the forward rate is an unbiased predictor of the future spot rate, will it expect to achieve lower financing costs by consistently borrowing a foreign currency with a low interest rate?
- 10 **Effective financing rate.** Boca, SA (Spain) needs 4 million euros for one year. It currently has no business in Japan but plans to borrow Japanese yen from a Japanese bank because the Japanese interest rate is 3 percentage points lower than the euro rate. Assume that interest rate parity exists; also assume that Boca believes that the one-year forward rate of the Japanese yen will exceed the future spot rate one

year from now. Will the expected effective financing rate be higher, lower, or the same as financing with euros? Explain.

- 11 IRP application to short-term financing.** Assume that the UK interest rate is 7% and the euro's interest rate is 4%. Assume that the euro's forward rate has a premium of 4% (is 4% more expensive in terms of £s?). Determine whether the following statement is true: 'Interest rate parity does not hold; therefore, UK firms could lock in a lower financing cost by borrowing euros and purchasing euros forward for one year'. Explain your answer.
- 12 Break-even financing.** Orlando plc is a UK-based MNC with a subsidiary in Mexico. Its Mexican subsidiary needs a one-year loan of 10 million pesos for operating expenses. Since the Mexican interest rate

is 70%, Orlando is considering borrowing pounds, which it would convert to pesos to cover the operating expenses. By how much would the pound have to appreciate against the peso to cause such a strategy to backfire? (The one-year UK interest rate is 9%.)

- 13 Financing in a crisis.** What actions can a firm take with regard to its short-term financing when there are concerns that there might be a currency crisis in the country of one of its subsidiaries.
- 14 Effects of currency change.** Homewood Ltd commonly finances some of its expansion in Europe by borrowing in euros. The interest rate for the euro then increases with no prospect of a fall in value of the euro against the pound. Should Homewood re-finance in pounds? Explain.

ADVANCED QUESTIONS

- 15 Probability distribution of financing costs.** Missoula Ltd decides to borrow Japanese yen for one year. The interest rate on the borrowed yen is 8%. Missoula has developed the following probability distribution for the yen's degree of fluctuation against the pound:

Possible degree of fluctuation of yen against the pound	Percentage probability
−4%	20%
−1	30
0	10
3	40

Given this information, what is the expected value of the effective financing rate of the Japanese yen from Missoula's perspective?

- 16 Analysis of short-term financing.** Jackson Ltd is a UK-based firm that needs £600 000. It has no business in Japan but is considering one-year financing with Japanese yen because the annual interest rate would be 5% versus 9% in the United Kingdom. Assume that interest rate parity exists.
- a** Can Jackson benefit from borrowing Japanese yen and simultaneously purchasing yen one year forward to avoid exchange rate risk? Explain.

- b** Assume that Jackson does not cover its exposure and uses the forward rate to forecast the future spot rate. Determine the expected effective financing rate. Should Jackson finance with Japanese yen? Explain.
- c** Assume that Jackson does not cover its exposure and expects that the Japanese yen will appreciate by either 5%, 3%, or 2%, and with equal probability of each occurrence. Use this information to determine the probability distribution of the effective financing rate. Should Jackson finance with Japanese yen? Explain.

- 17 Financing with a portfolio.** Pfeffer GmbH (Germany) considers obtaining 40% of its one-year financing in Canadian dollars and 60% in Japanese yen. The forecasts of appreciation in the Canadian dollar and Japanese yen for the next year are as follows:

Currency	Possible percentage change in the spot rate over the loan life	Probability of that percentage change in the spot rate occurring
Canadian dollar	4%	70%
Canadian dollar	7	30
Japanese yen	6	50
Japanese yen	9	50

The interest rate on the Canadian dollar is 9%, and the interest rate on the Japanese yen is 7%. Develop the possible effective financing rates of the overall portfolio and the probability of each possibility based on the use of joint probabilities (assume independence).

18 Financing with a portfolio.

- a Does borrowing a portfolio of currencies offer any possible advantages over the borrowing of a single foreign currency?
- b If a firm borrows a portfolio of currencies, what characteristics of the currencies will affect the potential variability of the portfolio's effective financing rate? What characteristics would be desirable from a borrowing firm's perspective?

- 19 Financing with a portfolio.** Raleigh Ltd needs to borrow funds for one year to finance an expenditure in the United States. The following interest rates are available:

	Borrowing Rate
US	10%
Canada	6
Japan	5

The percentage changes in the spot rates of the Canadian dollar and Japanese yen over the next year are as follows:

Canadian dollar		Japanese yen	
Probability	Percentage change in spot rate	Probability	Percentage change in spot rate
10%	5%	20%	6%
90	2	80	1

If Raleigh Ltd borrows a portfolio, 50% of funds from Canadian dollars and 50% of funds from yen, determine the probability distribution of the effective financing rate of the portfolio. What is the probability that Raleigh will incur a higher effective financing rate from borrowing this portfolio than from borrowing US dollars?

PROJECT WORKSHOP

- 20 Yields for foreign currencies.** The Bloomberg website provides interest rate data for many different foreign currencies over various maturities. Its address is: <http://www.bloomberg.com>.

- a Go to the section that shows yields for different foreign currencies. Review the three-month yields of currencies. Assume that you could borrow at a

rate 1 percentage point above the quoted yield for each currency. Which currency would offer you the lowest quoted yield?

- b As a cash manager of a UK-based MNC that needs dollars to support UK operations, where would you borrow funds for the next three months? Explain.

DISCUSSION IN THE BOARDROOM

This exercise can be found on the companion website at www.cengagelearning.co.uk/madura_fox2e.

RUNNING YOUR OWN MNC

This exercise can be found on the companion website at www.thomsonlearning.co.uk/madura_fox.

Essays/discussion and articles can be found at the end of Part 5

BLADES PLC CASE STUDY

Use of foreign short-term financing

Blades plc just received a special order for 120 000 pairs of 'Speedos', its primary roller blade product. Ben Holt, Blades' financial director, needs short-term financing to finance this large order from the time Blades orders its supplies until the time it will receive payment. Blades will charge a price of 5000 baht per pair of Speedos. The materials needed to manufacture these 120 000 pairs will be purchased from Thai suppliers. Blades expects the cost of the components for one pair of Speedos to be approximately 3500 baht in its first year of operating the Thai subsidiary.

Because Blades is relatively unknown in Thailand, its suppliers have indicated that they would like to receive payment as early as possible. The customer that placed this order insists on open account transactions, which means that Blades will receive payment for the roller blades approximately three months subsequent to the sale. Furthermore, the production cycle necessary to produce Speedos, from purchase of the materials to the eventual sale of the product, is approximately three months. Because of these considerations, Blades expects to collect its revenues approximately six months after it has paid for the materials, such as rubber and plastic components, needed to manufacture Speedos.

Ben Holt has identified at least two alternatives for satisfying Blades' financing needs. First, Blades could borrow Japanese yen for six months, convert the yen to Thai baht, and use the baht to pay the Thai suppliers. When the accounts receivable in Thailand are collected, Blades would convert the baht received to yen and repay the Japanese yen loan. Second, Blades could borrow Thai baht for six months in order to pay its Thai suppliers. When Blades collects its accounts receivable, it would use these receipts to repay the baht loan. Thus, Blades will use revenue generated in Thailand to repay the loan, whether it borrows the money in yen or in baht.

Holt's initial research indicates that the 180-day interest rates available to Blades in Japan and in Thailand are 4% and 6%, respectively. Consequently, Holt favours borrowing the Japanese yen, as he believes this loan will be cheaper than the baht-denominated loan. He is aware that he should somehow incorporate the future movements of the yen–baht exchange rate in his analysis, but he is unsure how to accomplish this. However, he has identified the following probabil-

ity distribution of the change in the value of the Japanese yen with respect to the Thai baht and of the change in the value of the Thai baht with respect to the dollar over the six-month period of the loan:

Possible rate of change in the Japanese yen relative to the Thai baht over the life of the loan	Possible rate of change in the Thai baht relative to the pound over the life of the loan	Probability of occurrence
2%	–3%	30%
1	–2	30
0	–1	20
1	0	15
2	1	5

Holt has also informed you that the current spot rate of the yen (in baht) is THB0.347826, while the current spot rate of the baht (in pounds) is £0.0015.

As a financial analyst for Blades, you have been asked to answer the following questions for Ben Holt:

- 1 What is the amount, in baht, that Blades needs to borrow to cover the payments due to the Thai suppliers? What is the amount, in yen, that Blades needs to borrow to cover the payments due to the Thai suppliers?
- 2 Given that Blades will use the receipts from the receivables in Thailand to repay the loan and that Blades plans to remit all baht-denominated cash flows to the UK parent whether it borrows in baht or yen, does the future value of the yen with respect to the baht affect the cost of the loan if Blades borrows in yen?
- 3 Using a spreadsheet, compute the expected amount (in pounds) that will be remitted to the United Kingdom in six months if Blades finances its working capital requirements by borrowing baht versus borrowing yen. Based on your analysis, should Blades obtain a yen- or baht-denominated loan?

SMALL BUSINESS DILEMMA

Short-term financing by the Sports Exports Company

At the current time, the Sports Exports Company (Ireland) focuses on producing basketballs and exporting them to a distributor in the United Kingdom. The exports are denominated in British pounds. Jim Logan, the owner, plans to develop other sporting goods products besides the basketballs that he produces. His entire expansion will be focused on the United Kingdom, where he is trying to make a name for his firm. He remains concerned about his firm's exposure to exchange rate risk but does not plan to let that get in the way of his expansion plans because he believes that his firm can continue to penetrate the British sporting goods market. He has just negotiated a joint venture with a British firm that will produce other sporting goods products that are popular in the United States but will be sold in the United Kingdom. Jim will pay the British manufacturer in British pounds. These products will be delivered directly to the British distributor rather than to Jim, and the distributor will pay Jim with British pounds.

Jim's expansion plans will result in the need for additional funding. Jim would prefer to borrow on a short-term basis now. Jim has an excellent credit rating and collateral and therefore should be able to obtain short-term financing. The British interest rate is one-fourth of a percentage point above the US interest rate.

- 1 Should Jim borrow dollars or pounds to finance his joint venture business? Why?
- 2 Jim could also borrow euros at an interest rate that is lower than the US or British rate. The values of the euro and pound tend to move in the same direction against the dollar but not always by the same degree. Would borrowing euros to support the British joint venture result in more exposure to exchange rate risk than borrowing pounds? Would it result in more exposure to exchange rate risk than borrowing dollars?

CHAPTER 19

INTERNATIONAL CASH MANAGEMENT

LEARNING OBJECTIVES

The specific objectives of this chapter are to:

- explain the difference in analyzing cash flows from a subsidiary perspective and from a parent perspective,
- explain the various techniques used to optimize cash flows,
- explain common complications in optimizing cash flows, and
- explain the potential benefits and risks from foreign investing.

The term cash management can be broadly defined to mean optimization of cash flows and investment of excess cash. From an international perspective, cash management is very complex because laws pertaining to cross-border cash transfers differ among countries. In addition, exchange rate fluctuations can affect the value of cross-border cash transfers. Financial managers need to understand the advantages and disadvantages of investing cash in foreign markets so that they can make international cash management decisions that maximize the value of the MNC.

CASH FLOW ANALYSIS: SUBSIDIARY PERSPECTIVE

The management of working capital (such as inventory, accounts receivable and payable – debtors and creditors – and cash) has a direct influence on the amount and timing of cash flows. Working capital management and the management of cash flow are integrated. First, we discuss these elements before focusing on cash management.

Subsidiary expenses

Begin with outflow payments by the subsidiary to purchase raw materials or supplies. The subsidiary will normally have a more difficult time forecasting future outflow payments if its purchases are international rather than domestic because of exchange rate fluctuations. In addition, there is a possibility that payments will be substantially higher due to appreciation of the invoice currency. Consequently, the firm may wish to maintain a large inventory of supplies and raw materials so that it can draw from its inventory and cut down on purchases if the invoice currency appreciates. Still another possibility is that imported goods from another country could be restricted by the host government (through quotas, etc.). In this event, a larger inventory would give a firm more time to search for alternative sources of supplies or raw materials. A subsidiary with domestic supply sources would not experience such a problem and therefore would not need such a large inventory.

Outflow payments for supplies will be influenced by future sales. If the sales volume is substantially influenced by exchange rate fluctuations, its future level becomes more uncertain, which makes its need for supplies more uncertain. Such uncertainty may force the subsidiary to maintain larger cash balances to cover any unexpected increase in supply requirements.

Subsidiary revenue

If subsidiaries export their products, their sales volume may be more volatile than if the goods were only sold domestically. This volatility could be due to the fluctuating exchange rate of the invoice currency. Importers' demand for these finished goods will most likely decrease if the invoice currency appreciates. The sales volume of exports is also susceptible to business cycles of the importing countries. If the goods were sold domestically, the exchange rate fluctuations would not have a direct impact on sales, although they would still have an indirect impact since the fluctuations would influence prices paid by local customers for imports from foreign competitors.

Sales can often be increased when credit standards are relaxed. However, it is important to focus on cash inflows due to sales rather than on sales themselves. Looser credit standards may cause a slowdown in cash inflows from sales, which could offset the benefits of increased sales. Accounts receivable management is an important part of the subsidiary's working capital management because of its potential impact on cash inflows.

Subsidiary dividend payments

The subsidiary may be expected to periodically send dividend payments and other fees to the parent. These fees could represent royalties or charges for overhead costs incurred by the parent that benefit the subsidiary. An example is research and development costs incurred by the parent, which improve the quality of goods produced by the subsidiary. Whatever the reason, payments by the subsidiary to the parent are often necessary. When dividend payments and fees are known in advance and denominated in the subsidiary's currency, forecasting cash flows is easier for the subsidiary. The level of dividends paid by subsidiaries to the parent is dependent on the liquidity needs of each subsidiary, potential uses of funds at various subsidiary locations, expected movements in the currencies of the subsidiaries, and regulations of the host country government.

Subsidiary liquidity management

After accounting for all outflow and inflow payments, the subsidiary will find itself with either excess or deficient cash. It uses liquidity management to either invest its excess cash or borrow to cover its cash deficiencies. If it anticipates a cash deficiency, short-term financing is necessary, as described in the previous chapter. If it anticipates excess cash, it must determine how the excess cash should be used. Investing in foreign currencies can sometimes be attractive, but exchange rate risk makes the effective yield uncertain. This issue is discussed later in this chapter.

Liquidity management is a crucial component of a subsidiary's working capital management. Subsidiaries commonly have access to numerous lines of credit and overdraft facilities in various currencies. Therefore, they may maintain adequate liquidity without substantial cash balances. While liquidity is important for the overall MNC, it cannot be properly measured by liquidity ratios. Potential access to funds is more relevant than cash on hand.

CENTRALIZED CASH MANAGEMENT

Each subsidiary should manage its working capital by simultaneously considering all of the points discussed thus far. Often, though, each subsidiary is more concerned with its own operations than with the overall operations of the MNC. Thus, a **centralized cash management** group may need to monitor, and possibly manage, the parent–subsidiary and intersubsidiary cash flows. This role is critical since it can often benefit individual subsidiaries in need of funds or overly exposed to exchange rate risk.

EXAMPLE

The treasury department of Kraft Foods is centralized to manage liquidity, funding and foreign exchange requirements of its global operations. And Monsanto

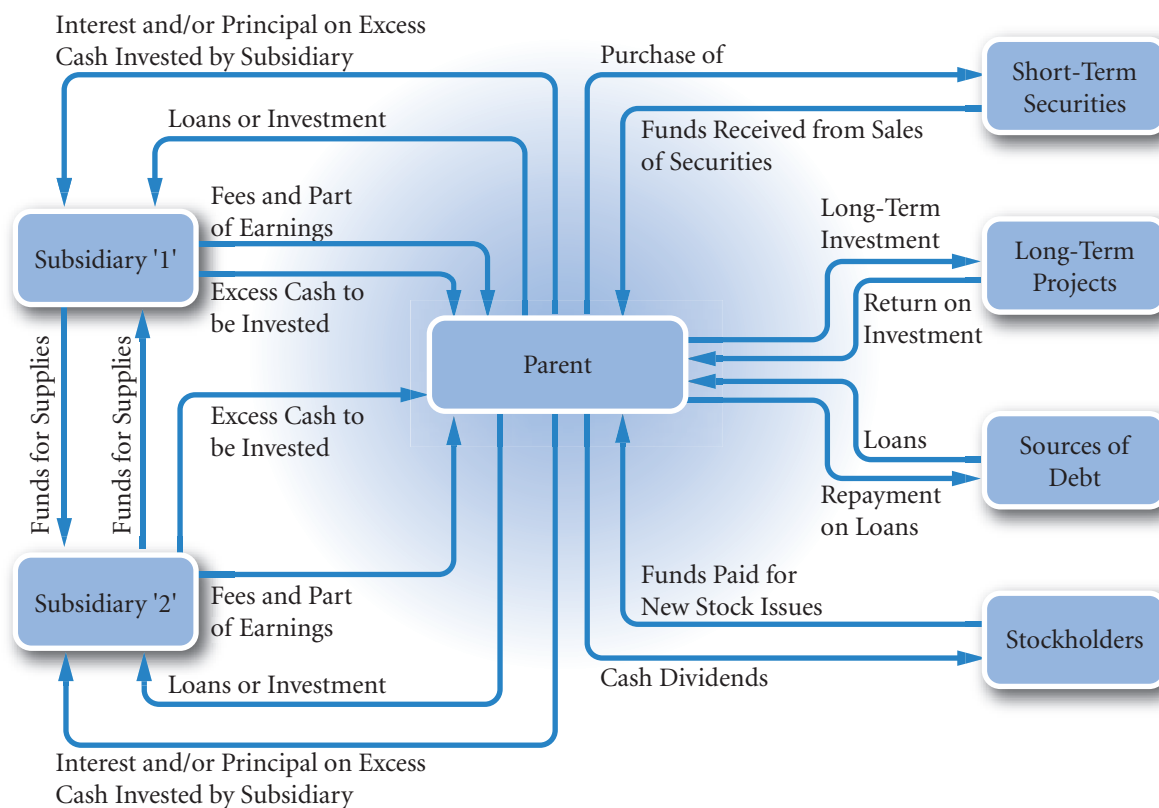
has a centralized system for pooling different currency balances from various subsidiaries in Asia that saves hundreds of thousands of dollars per year.

Exhibit 19.1 is a complement to the following discussion of cash flow management. It is a simplified cash flow diagram for an MNC with two subsidiaries in different countries. Although each MNC may handle its payments in a different manner, Exhibit 19.1 is based on simplified assumptions that will help illustrate some key concepts of international cash management. The exhibit reflects the assumption that the two subsidiaries periodically send loan repayments and dividends to the parent or send excess cash to the parent (where the centralized cash management process is assumed to take place). These cash flows represent the incoming cash to the parent from the subsidiaries. The parent's cash outflows to the subsidiaries can include loans and the return of cash previously invested by the subsidiaries. The subsidiaries also have cash flows between themselves because they purchase supplies from each other.

While each subsidiary is managing its working capital, there is a need to monitor and manage the cash flows between the parent and the subsidiaries, as well as between the individual subsidiaries. This task of international cash management should be delegated to a centralized cash management group. International cash management can be segmented into two functions: (1) optimizing cash flow movements, and (2) investing excess cash. These two functions are discussed in turn.

The centralized cash management division of an MNC cannot always accurately forecast events that affect parent–subsidiary or intersubsidiary cash flows. It should, however, be ready to react to any event by considering: (1) any potential adverse impact on cash flows, and (2) how to avoid such an adverse impact. If the cash flow situation between the parent and subsidiaries results in a cash squeeze on the parent, it

EXHIBIT 19.1 Cash flow of the overall MNC



should have sources of funds (credit lines) available. On the other hand, if it has excess cash after considering all outflow payments, it must consider where to invest funds. This decision is thoroughly examined shortly.

MANAGING FOR VALUE

Flexsys' decision to use a multibank payments system

Flexsys is a large chemical company with more than 20 subsidiaries in Europe, the United States, and Asia. To improve its liquidity, it wanted to create a system that would enable its treasury department to continuously monitor all payments at each subsidiary. Since the inception of the euro, Flexsys has had euro balances at several subsidiaries and consolidates these balances so that it can maximize the interest earned on the balances. It uses technology that allows all payments due to

be recorded and transmitted to the treasury department. The treasury nets all payments due between subsidiaries so that only net payments need to be made to cover the payments due between subsidiaries. Flexsys' system is especially effective because it is 'transparent', meaning that the payments due at each subsidiary can be easily monitored by the subsidiaries.

Flexsys uses multiple banks rather than branches of a single bank so that each subsidiary can use the

bank it prefers. Having a multibank system complicates the reporting of payments, but allows each subsidiary to choose the bank that provides it with the best service. Thus, Flexsys' decision to use a multi-

bank system can ensure optimal service at its subsidiaries. At the same time, however, its centralized payments network ensures that cash is utilized properly so that it can maximize its value.

TECHNIQUES TO OPTIMIZE CASH FLOWS

Cash inflows can be optimized by the following techniques:

- Accelerating cash inflows
- Minimizing currency conversion costs
- Managing blocked funds
- Managing intersubsidiary cash transfers

Each of these techniques is discussed in turn.

Accelerating cash inflows

The first goal in international cash management is to accelerate cash inflows, since the more quickly the inflows are received, the more quickly they can be invested or used for other purposes. Several managerial practices are advocated for this endeavour, some of which may be implemented by the individual subsidiaries. First, a corporation may establish **lockboxes** around the world, which are post office boxes to which customers are instructed to send payment. When set up in appropriate locations, lockboxes can help reduce mailing time (**mail float**). A bank usually processes incoming checks at a lockbox on a daily basis. Second, cash inflows can be accelerated by using **preauthorized payments**, which allow a corporation to charge a customer's bank account up to some limit. Both preauthorized payments and lockboxes are also used in a domestic setting. Because international transactions may have a relatively long mailing time, these methods of accelerating cash inflows can be quite valuable for an MNC.

Minimizing currency conversion costs

Another technique for optimizing cash flow movements, **netting**, can be implemented with the joint effort of subsidiaries or by the centralized cash management group. This technique optimizes cash flows by reducing the administrative and transaction costs that result from currency conversion.

EXAMPLE

Mount plc has subsidiaries located in France and in Hungary. Whenever the French subsidiary needs to purchase supplies from the Hungarian subsidiary, it

needs to convert euros into Hungary's currency (the forint) to make payment. Hungary's subsidiary must convert its forint into euros when purchasing supplies

(Continued)

from the French subsidiary. Mount plc has instructed both subsidiaries to net their transactions on a monthly basis so that only one net payment is made at the end

of each month. By using this approach, both subsidiaries avoid (or at least reduce) the transaction costs of currency conversion.

Over time, netting has become increasingly popular because it offers several key benefits. First, it reduces the number of cross-border transactions between subsidiaries, thereby reducing the overall administrative cost of such cash transfers. Second, it reduces the need for foreign exchange conversion since transactions occur less frequently, thereby reducing the transaction costs associated with foreign exchange conversion. Third, the netting process imposes tight control over information on transactions between subsidiaries. Thus, all subsidiaries engage in a more coordinated effort to accurately report and settle their various accounts. Finally, cash flow forecasting is easier since only net cash transfers are made at the end of each period, rather than individual cash transfers throughout the period. Improved cash flow forecasting can enhance financing and investment decisions.

A **bilateral netting system** involves transactions between two units: between the parent and a subsidiary, or between two subsidiaries. A **multilateral netting system** usually involves a more complex interchange among the parent and several subsidiaries. For most large MNCs, a multilateral netting system would be necessary to effectively reduce administrative and currency conversion costs. Such a system is normally centralized so that all necessary information is consolidated. From the consolidated cash flow information, net cash flow positions for each pair of units (subsidiaries, or whatever) are determined, and the actual reconciliation at the end of each period can be dictated. The centralized group may even maintain inventories of various currencies so that currency conversions for the end-of-period net payments can be completed without significant transaction costs.

MNCs commonly monitor the cash flows between their subsidiaries with the use of an intersubsidiary payments matrix.

EXAMPLE

Exhibit 19.2(a) is an example of an intersubsidiary payments matrix that totals each subsidiary's individual payments to each of the other subsidiaries. The first row indicates that the Canadian subsidiary owes the equivalent of £40 000 to the French subsidiary, the equivalent of £90 000 to the Japanese subsidiary, and so on. During this same period, these subsidiaries have also received goods from the Canadian subsidiary, for which payment is due. The second column (under Canada) shows that the Canadian subsidiary is owed the equivalent of £60 000 by the French subsidiary, the equivalent of £100 000 by the Japanese subsidiary, and so on.

Since subsidiaries owe each other, the first step is to charge only the net amounts between each other as illustrated in 19.2 (b) and (c). Since the Canadian

subsidiary owes the French subsidiary the equivalent of £40 000 but is owed the equivalent of £60 000 by the French subsidiary, the net payment required is the equivalent of £20 000 from the French subsidiary to the Canadian subsidiary. This is known as bilateral netting.

Further economies can be made through multilateral netting (Exhibit 19.3). Canada owes £40 000 in total and is owed £30 000 in total and therefore can clear net indebtedness by paying £10 000. The Canadian subsidiary would not be especially concerned as to which other subsidiary it paid the net indebtedness. Similarly, France owes a net amount of £20 000.

The only country owed money is the UK, so if Canada pays £10 000, France £20 000 and Switzerland £30 000 all to the UK, all debts will have been

(Continued)

EXHIBIT 19.2(a) Intersubsidiary payments matrix

Payments OWED BY subsidiary located in:	British pound value (in thousands) OWED TO subsidiary located in:				
	Canada	France	Japan	Switzerland	UK
Canada	—	40	90	20	40
France	60	—	30	60	50
Japan	100	30	—	20	30
Switzerland	10	50	10	—	50
UK	10	60	20	20	—

EXHIBIT 19.2(b) Bilateral netting (arrows show the example of Canada)

Payments OWED BY subsidiary located in:	British pound value (in thousands) OWED TO subsidiary located in:				
	Canada	France	Japan	Switzerland	UK
Canada	—	40	90	20	40
France	60	—	30	60	50
Japan	100	30	—	20	30
Switzerland	10	50	10	—	50
UK	10	60	20	20	—

EXHIBIT 19.2(c) Bilateral netting – the net position (arrows show the example of Canada)

Payments OWED BY subsidiary located in:	British pound value (in thousands) OWED TO subsidiary located in:				
	Canada	France	Japan	Switzerland	UK
Canada	—	0	0	10	30
France	20	—	0	10	0
Japan	10	0	—	10	10
Switzerland	0	0	0	—	30
UK	0	10	0	0	—

(Continued)

EXHIBIT 19.3 Multilateral netting schedule

Payments OWED BY subsidiary located in:	British pound value (in Thousands) OWED TO subsidiary located in:						
	Canada	France	Japan	Switzerland	UK	Total owed by:	Net payments by:
Canada	—	0	0	10	30	40	10
France	20	—	0	10	0	30	20
Japan	10	0	—	10	10	30	30
Switzerland	0	0	0	—	30	30	0
UK	0	10	0	0	—	10	0
Total owed to:	30	10	0	30	70		
Net owed to:	0	0	0	0	60		60

paid. Thus three payments will have accounted for a payments schedule that at the outset amounted to 20 payments! Note that the bilateral stage could be

skipped by simply totalling the columns and rows in the original matrix and making the net payers pay the net receivers.

Managing blocked funds

Cash flows can also be affected by a host government's blockage of funds, which might occur if the government requires all funds to remain within the country in order to create jobs and reduce unemployment. To deal with funds blockage, the MNC may implement the same strategies used when a host country government imposes high taxes. To make efficient use of these funds, the MNC may instruct the subsidiary to set up a research and development division, which incurs costs and possibly generates revenues for other subsidiaries.

Another strategy is to use transfer pricing in a manner that will increase the expenses incurred by the subsidiary. A host country government is likely to be more lenient on funds sent to cover expenses than on earnings remitted to the parent. Most tax treaties require that transfer pricing be conducted at market rates. Therefore a policy of artificially inflated transfer prices resulting in subsidiaries in developing countries paying very high prices for 'services' or 'goods' from Head Office is of dubious legality and ethics.

When subsidiaries are restricted from transferring funds to the parent, the parent may instruct the subsidiary to obtain financing from a local bank rather than from the parent. By borrowing through a local intermediary, the subsidiary is assured that its earnings can be distributed to pay off previous financing. Overall, most methods of managing blocked funds are intended to make efficient use of the funds by using them to cover expenses that are transferred to that country.

EXAMPLE

Wittenberg GmbH, a German-based MNC, has a subsidiary in the Philippines. During a turbulent period, the subsidiary was prevented from exchanging its Philippine pesos into euros to be sent home. Wittenberg held its corporate meeting in Manila so that it could use the pesos to pay the expenses of the meeting (hotel, food, etc.) in

pesos. In this way, it was able to use local funds to cover an expense that it would have incurred anyway. Ordinarily, the corporate meeting would have been held in the parent's country, and the parent would have paid the expenses.

Managing intersubsidiary cash transfers

Proper management of cash flows can also be beneficial to a subsidiary in need of funds.

EXAMPLE

Essex Ltd has two foreign subsidiaries called Short Sub and Long Sub. Short Sub needs funds, while Long Sub has excess funds. If Long Sub purchases supplies from Short Sub, it can provide financing by paying for its supplies earlier than necessary. This

technique is often called leading. Alternatively, if Long Sub sells supplies to Short Sub, it can provide financing by allowing Short Sub to lag its payments. This technique is called lagging.

The leading or lagging strategy can make efficient use of cash and thereby reduce debt. Some host governments prohibit the practice by requiring that a payment between subsidiaries occur at the time the goods are transferred. Thus, an MNC needs to be aware of any laws that restrict the use of this strategy.

COMPLICATIONS IN OPTIMIZING CASH FLOW

Most complications encountered in optimizing cash flow can be classified into three categories:

- Company-related characteristics
- Government restrictions
- Characteristics of banking systems

Each complication is discussed in turn.

Company-related characteristics

In some cases, optimizing cash flow can become complicated due to characteristics of the MNC. If one of the subsidiaries delays payments to other subsidiaries for supplies received, the other subsidiaries may be

forced to borrow until the payments arrive. A centralized approach that monitors all intersubsidiary payments should be able to minimize such problems.

Government restrictions

The existence of government restrictions can disrupt a cash flow optimization policy. Some governments prohibit the use of a netting system, as noted earlier. In addition, some countries periodically prevent cash from leaving the country, thereby preventing net payments from being made. These problems can arise even for MNCs that do not experience any company-related problems. Countries in Latin America commonly impose restrictions that affect an MNC's cash flows.

Characteristics of banking systems

The abilities of banks to facilitate cash transfers for MNCs vary among countries. Banks in the developed world are advanced in this field, but banks in some other countries do not offer services. MNCs prefer some form of zero-balance account, where excess funds can be used to make payments but earn interest until they are used. In addition, some MNCs benefit from the use of lockboxes. Such services are not available in some countries.

In addition, a bank may not update the MNC's bank account information sufficiently or provide a detailed breakdown of fees for banking services. Without full use of banking resources and information, the effectiveness of international cash management is limited. In addition, an MNC with subsidiaries in, say, eight different countries will typically be dealing with eight different banking systems. Much progress has been made in foreign banking systems in recent years. As time passes and a more uniform global banking system emerges, such problems may be alleviated.

INVESTING EXCESS CASH

Many MNCs have at least £100 million in cash balances across banks in various countries. If they can find a way to earn an extra 1% on those funds, they will generate an extra £1 million each year on cash balances of £100 million. Thus, their short-term investment decision affects the amount of their cash inflows. Their excess funds can be invested in domestic or foreign short-term securities. In some periods, foreign short-term securities will have higher interest rates than domestic interest rates. The differential can be substantial, as illustrated in Exhibit 19.4. However, firms must account for the possible exchange rate movements when assessing the potential yield on foreign investments.

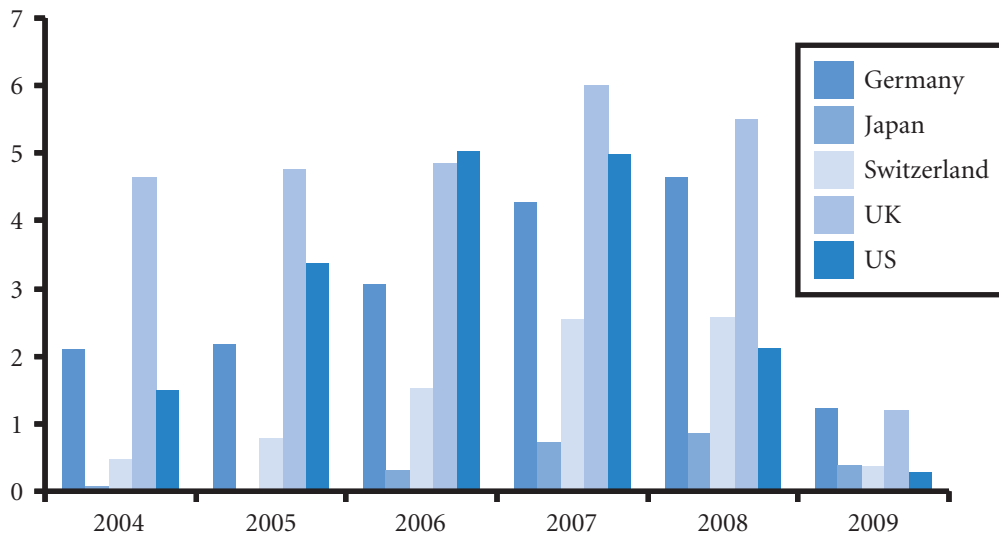
How to invest excess cash

International money markets have grown to accommodate corporate investments of excess cash. MNCs may use international money markets in an attempt to earn higher returns than they can achieve domestically.

Eurocurrency deposits are one of the most commonly used international money market instruments. Many MNCs establish large deposits in various currencies in the eurocurrency market, with eurodollar deposits being the most popular. The dollar volume of eurodollar deposits has more than doubled since 1980. eurodollar deposits commonly offer MNCs a slightly higher yield than bank deposits in the United States. Though eurodollar deposits still dominate the market, the relative importance of nondollar currencies has increased over time.

In addition to using the eurocurrency market, MNCs can also purchase foreign Treasury bills and commercial paper. Improved telecommunications systems have increased access to these securities in foreign markets and allow for a greater degree of integration among money markets in various countries.

EXHIBIT 19.4 Short-term annualized interest rates



Centralized cash management

An MNC's short-term investing policy can either maintain separate investments for all subsidiaries or employ a centralized approach. Recall that the function of optimizing cash flows can be improved by a centralized approach, since all subsidiary cash positions can be monitored simultaneously. With regard to the investing function, centralization allows for more efficient usage of funds and possibly higher returns. Here the term *centralized* means that excess cash from each subsidiary is pooled until it is needed by a particular subsidiary.

Centralization when subsidiaries use the same currency. To understand the advantages of a centralized system, consider that the rates paid on short-term investments such as bank deposits are often higher for larger amounts. Thus, if two subsidiaries have excess cash of £50 000 each for one month, the rates on their individual bank deposits may be lower than the rate they could obtain if they pooled their funds into a single £100 000 bank deposit. In this manner, the centralized (pooling) approach generates a higher rate of return on excess cash.

The centralized approach can also facilitate the transfer of funds from subsidiaries with excess funds to those that need funds.

EXAMPLE

Subsidiary A of Moorhead Ltd has excess cash of £50 000 during the next month, while Subsidiary B of Moorhead Ltd needs to borrow £50 000 for one

month. If cash management is not centralized, Subsidiary A may use the £50 000 to purchase a one-month bank certificate earning, say, 10% (on an annualized

(Continued)

basis). At the same time, Subsidiary B may borrow from a bank for one month at a rate of, say, 12%. The bank must charge a higher rate on loans than it offers

on deposits. With a centralized approach, Subsidiary B can borrow Subsidiary A's excess funds, thereby reducing its financing costs.

Centralized cash management of multiple currencies. Centralized cash management is more complicated when the MNC uses multiple currencies. All excess funds could be pooled and converted to a single currency for investment purposes. However, the advantage of pooling may be offset by the transaction costs incurred when converting to a single currency.

Centralized cash management can still be valuable, though. The short-term cash available among subsidiaries can be pooled together so that there is a separate pool for each currency. Then excess cash in a particular currency can still be used to satisfy other subsidiary deficiencies in that currency. In this way, funds can be transferred from one subsidiary to another without incurring transaction costs that banks charge for exchanging currencies. This strategy is especially feasible when all subsidiary funds are deposited in branches of a single bank so that the funds can easily be transferred among subsidiaries.

Another possible function of centralized cash management is to invest funds in securities denominated in the foreign currencies that will be needed by the subsidiaries in the future. MNCs can use excess cash to invest cash in international money market instruments so that they can cover any payables positions in specific foreign currencies. If they have payables in foreign currencies that are expected to appreciate, they can cover such positions by creating short-term deposits in those currencies. The maturity of a deposit would ideally coincide with the date at which the funds are needed.

Impact of technology on centralized cash management. International cash management requires timely information across subsidiaries regarding each subsidiary's cash positions in each currency, along with interest rate information about each currency. A centralized cash management system needs a continual flow of information about currency positions so that it can determine whether one subsidiary's shortage of cash can be covered by another subsidiary's excess cash in that currency. Given the major improvements in online technology in recent years, all MNCs can easily and efficiently create a multinational communications network among their subsidiaries to ensure that information about cash positions is continually updated.

EXAMPLE

To understand how such a communications network works, consider Jax plc which creates a cash balances website that specifies the cash balance of every currency for each subsidiary. Near the end of each day, each subsidiary revises the website to provide the latest update of its cash balance for each currency. Each subsidiary also specifies the period of time in which the excess or deficiency will persist. The parent's treasury department monitors the updated data and determines whether any cash needs identified by a subsidiary in a particular currency can be accommodated by another subsidiary that has excess cash in that same currency. The treasury department then e-mails instructions to the subsidiaries about

fund transfers. If it notices that the Canadian subsidiary has an excess of Canadian dollars for the next 26 days, and the Belgian subsidiary needs Canadian dollars tomorrow (but will have inflows of Canadian dollars in 17 days), it provides the following instructions: 'The Canadian subsidiary should transfer C\$60 000 to the Belgian subsidiary and will be repaid by the Belgian subsidiary in 17 days.' The fund transfers are essentially short-term loans, so a subsidiary that borrows funds will repay them with interest. The interest charged on a loan creates an incentive for subsidiaries to make their excess cash available and an incentive for subsidiaries with cash deficiencies to return the funds as soon as possible.

The electronic communications network may be more sophisticated than the one described here, but this description illustrates how easy it is for an MNC's parent to continuously monitor the cash balances of each subsidiary and communicate instructions among subsidiaries. The process of transferring funds among subsidiaries may be especially easy when all the MNC's subsidiaries use branches of the same bank. A communications network allows the MNC to make the best use of each subsidiary's cash, which can reduce the amount of external financing needed and reduce the MNC's exchange rate risk.

Determining the effective yield

Firms commonly consider investing in a deposit denominated in a currency with a high interest rate and then converting the funds back to dollars when the deposit matures. This strategy will not necessarily be feasible, since the currency denominating the deposit may depreciate over the life of the deposit. If it does, the advantage of a higher interest rate may be more than offset by the depreciation in the currency representing the deposit.

Consequently, it is the deposit's **effective yield**, not its interest rate, that is most important to the cash manager. The effective yield of a bank deposit considers both the interest rate and the rate of appreciation (or depreciation) of the currency denominating the deposit and can therefore be very different from the quoted interest rate on a deposit denominated in a foreign currency. An example follows to illustrate this point.

EXAMPLE

Quant plc a large UK corporation with £1 000 000 in excess cash, could invest in a one-year deposit at 6% but is attracted to higher interest rates in Australia. It creates a one-year deposit denominated in Australian dollars (A\$) at 9%. The exchange rate of the Australian dollar at the time of the deposit is £0.45. The British pounds are first converted to A\$2 222 222 (since £1 000 000 / £0.45 = \$2 222 222) and then deposited in a bank.

One year later, Quant receives A\$2 422 222, which is equal to the initial deposit plus 9% interest on the deposit. At this time, Quant plc has no use for Australian dollars and converts them into pounds. Assume that the exchange rate at this time is £0.48. The funds will convert to £1 162 667 (computed as A\$2 422 222 × £0.48 per A\$). Thus, the yield on this investment is:

$$\frac{£1\,162\,667 - £1\,000\,000}{£1\,000\,000} = 0.1627 \text{ or } 16.27\%$$

The high yield is attributed to the relatively high interest rate earned on the deposit, plus the appreciation in the currency denominating the deposit over the investment period.

If the currency had depreciated over the investment period, however, the effective yield to Quant plc would have been less than the interest rate on the deposit and could even have been lower than the interest rate available on UK investments. For example, if the Australian dollar had *depreciated* from £0.45 at the beginning of the investment period to £0.43 by the end of the investment period, Quant plc would have received £1 041 555 (computed as A\$2 422 222 × £0.43 per A\$). In this case, the yield on the investment to the UK company would have been:

$$\frac{£1\,041\,555 - £1\,000\,000}{£1\,000\,000} = 0.0416 \text{ or } 4.16\%$$

The preceding example illustrates how appreciation of the currency denominating a foreign deposit over the deposit period will force the effective yield to be above the quoted interest rate. Conversely, depreciation will create the opposite effect.

The previous computation of the effective yield on foreign deposits was conducted in a real world setting. A quicker method is shown here:

$$r = (1 + i_f)(1 + e_f) - 1$$

The effective yield on the foreign deposit is represented by r (for simpler notation the r_{eff} from the previous chapter has been dropped here), i_f is the quoted interest rate, and e_f is the percentage change (from the day of deposit to the day of withdrawal) in the value of the currency representing the foreign deposit. The term i_f was used in Chapter 20 to represent the interest rate when borrowing a foreign currency. In this chapter, the interest rate of concern is the deposit rate on the foreign currency.

EXAMPLE

Given the information for Quant plc, the effective yield on the Australian deposit can be estimated. The term e_f represents the percentage change in the Australian dollar (against the pound) from the date Australian dollars are purchased (and deposited) until the day they are withdrawn (and converted back to pounds). The Australian dollar appreciated from £0.45 to £0.48, or by 6.66% over the life of the deposit. Using this information as well as the quoted deposit rate of 9%, the

effective yield to the UK firm on this deposit denominated in Australian dollars is:

$$\begin{aligned} r &= (1 + i_f)(1 + e_f) - 1 \\ &= (1 + 0.09)(1 + 0.0666) \\ &= 0.1627 \text{ or } 16.27\% \end{aligned}$$

This estimate of the effective yield corresponds with the return on investment determined earlier for Quant plc.

If the currency had depreciated, Quant plc would have earned an effective yield that was less than the interest rate.

EXAMPLE

In the revised example for Quant plc the Australian dollar depreciated from £0.45 to £0.43, or by 4.44%. Based on the quoted interest rate of 9% and the depreciation of 4.44%, the effective yield is:

$$\begin{aligned} r &= (1 + i_f)(1 + e_f) - 1 \\ &= (1 + 0.09)(1 + 0.0444) \\ &= 0.0416 \text{ or } 4.16\% \end{aligned}$$

which is the same rate computed earlier for this revised example.

The effective yield can be negative if the currency denominating the deposit depreciates to an extent that more than offsets the interest accrued from the deposit.

EXAMPLE

Northampton Ltd invests in a bank deposit denominated in euros that provides a yield of 9%. The euro depreciates against the dollar by 12% over the one-year period. The effective yield is:

$$\begin{aligned} r &= (1 + i_f)(1 + e_f) - 1 \\ &= (1 + 0.09)(1 + -0.12) \\ &= 0.0408 \text{ or } 4.08\% \end{aligned}$$

This result indicates that Northampton will end up with 4.08% less in funds than it initially deposited.

As with bank deposits, the effective yield on all other securities denominated in a foreign currency is influenced by the fluctuation of that currency's exchange rate. Our discussion will continue to focus on bank deposits for short-term foreign investment, but the implications of the discussion can be applied to other short-term securities as well.

Implications of interest rate parity

Recall that covered interest arbitrage is described as a short-term foreign investment with a simultaneous forward sale of the foreign currency denominating the foreign investment. One might think that a foreign currency with a high interest rate would be an ideal candidate for covered interest arbitrage. However, such a currency will normally exhibit a forward discount that reflects the differential between its interest rate and the investor's home interest rate. This relationship is based on the theory of interest rate parity. Investors cannot lock in a higher return when attempting covered interest arbitrage if interest rate parity exists.

Even if interest rate parity does exist, short-term foreign investing may still be feasible but would have to be conducted on an uncovered basis (without use of the forward market). That is, short-term foreign investing may result in a higher effective yield than domestic investing, but it cannot be guaranteed.

Use of the forward rate as a forecast

If interest rate parity exists, the forward rate serves as a break-even point to assess the short-term investment decision. When investing in the foreign currency (and not covering the foreign currency position), the effective yield will be more than the domestic yield if the spot rate of the foreign currency after one year is more than the forward rate at the time the investment is undertaken. Conversely, the yield of a foreign investment will be lower than the domestic yield if the spot rate of the foreign currency after one year turns out to be less than the forward rate at the time the investment is undertaken.

Relationship with the international Fisher effect. When interest rate parity exists, MNCs in the UK that use the forward rate as a predictor of the future spot rate expect the yield on foreign deposits to equal that on UK deposits. Though the forward rate is not necessarily an accurate predictor, it may provide unbiased forecasts of the future spot rate. If the forward rate is unbiased, it does not consistently underestimate or overestimate the future spot rate with equal frequency. Thus, the effective yield on foreign deposits is equal to the domestic yield, on average. MNCs that consistently invest in foreign short-term securities would earn a yield similar on average to what they could earn on domestic securities.

Our discussion here is closely related to the international Fisher effect (IFE). Recall that the IFE suggests that the exchange rate of a foreign currency is expected to change by an amount reflecting the differential between its interest rate and the UK interest rate. The rationale behind this theory is that a high nominal interest rate reflects an expectation of high inflation, which could weaken the currency (according to purchasing power parity).

If interest rate parity exists, the forward premium or discount reflects that interest rate differential and represents the expected percentage change in the currency's value when the forward rate is used as a predictor of the future spot rate. The IFE suggests that firms cannot consistently earn short-term yields on foreign securities that are higher than those on domestic securities because the exchange rate is expected to adjust to the interest rate differential on average. If interest rate parity holds and the forward rate is an unbiased predictor of the future spot rate, we can expect the IFE to hold.

A look back in time reveals that the IFE is supported for some currencies in some periods. Moreover, it may be difficult for an MNC to anticipate when the IFE will hold and when it will not. For virtually any currency, it is possible to identify previous periods when the forward rate substantially underestimated the future spot rate, and an MNC would have earned very high returns from investing short-term funds in a foreign money market security. However, it is also possible to identify other periods when the forward rate substantially overestimated the future spot rate, and the MNC would have earned low or even negative returns from investing in that same foreign money market security.

Conclusions about the forward rate. The key implications of interest rate parity and the forward rate as a predictor of future spot rates for foreign investing are summarized in Exhibit 19.5. This exhibit explains the conditions in which investment in foreign short-term securities is feasible.

EXHIBIT 19.5 Considerations when investing excess cash

Scenario	Implications for Investing in Foreign Money Markets
1. Interest rate parity exists.	Covered interest arbitrage is not worthwhile.
2. Interest rate parity exists, and the forward rate is an accurate forecast of the future spot rate.	An uncovered investment in a foreign security is not worthwhile.
3. Interest rate parity exists, and the forward rate is an unbiased forecast of the future spot rate.	An uncovered investment in a foreign security will on average earn an effective yield similar to an investment in a domestic security.
4. Interest rate parity exists, and the forward rate is expected to overestimate the future spot rate.	An uncovered investment in a foreign security is expected to earn a lower effective yield than an investment in a domestic security.
5. Interest rate parity exists, and the forward rate is expected to underestimate the future spot rate.	An uncovered investment in a foreign security is expected to earn a higher effective yield than an investment in a domestic security.
6. Interest rate parity does not exist; the forward premium (discount) exceeds (is less than) the interest rate differential.	Covered interest arbitrage is feasible for investors residing in the home country.
7. Interest rate parity does not exist; the forward premium (discount) is less than (exceeds) the interest rate differential.	Covered interest arbitrage is feasible for foreign investors but not for investors residing in the home country.

Use of exchange rate forecasts

Although MNCs do not know how a currency's value will change over the investment horizon, they can use the formula for the effective yield provided earlier in this chapter and plug in their forecast for the percentage change in the foreign currency's exchange rate (e_f). Since the interest rate of the foreign currency deposit (i_f) is known, the effective yield can be forecasted given a forecast of e_f . This projected effective yield on a foreign deposit can then be compared with the yield when investing in the firm's local currency.

EXAMPLE

Latrobe SA is a French firm with funds available to invest for one year. It is aware that the one-year interest rate on a euro deposit is 11% and the interest rate on an Australian deposit is 14%. Assume that the French firm forecasts that the Australian deposit will depreciate from its current rate of 0.60 euros to 0.594 euros, or a 1% decrease. The expected value for e_f [$E(e_f)$] will therefore be -1% . Thus, the expected effective yield [$E(r)$] on an Australian dollar-denominated deposit is:

$$\begin{aligned} E(r) &= (1 + i_f)(1 + E(e_f)) - 1 \\ &= (1 + 0.14)(1 + -0.01) \\ &= 0.1286 \text{ or } 12.86\% \end{aligned}$$

Thus, this example, investing in an Australian dollar deposit is expected to be more rewarding than investing in a euro denominated deposit.

Keep in mind that the value for e_f is forecasted and therefore is not known with certainty. Thus, there is no guarantee that foreign investing will truly be more lucrative.

Deriving the value of e_f that equates foreign and domestic yields. From the preceding example, Latrobe may attempt to at least determine what value of e_f would make the effective yield from foreign investing the same as that from investing in a euro deposit. To determine this value, begin with the effective yield formula and solve for e_f as follows:

$$\begin{aligned} r &= (1 + i_f)(1 + e_f) - 1 \\ (1 + r) &= (1 + i_f)(1 + e_f) \\ (1 + e_f) &= \frac{(1 + r)}{(1 + i_f)} \\ e_f &= \frac{(1 + r)}{(1 + i_f)} - 1 \end{aligned}$$

Since the euro deposit rate was 11% in our previous example, that is the rate to be plugged in for r . We can also plug in 14% for i_f , so the break-even value of e_f would be:

$$\begin{aligned} e_f &= \frac{(1 + r)}{(1 + i_f)} - 1 \\ &= \frac{(1 + 0.11)}{(1 + 0.14)} - 1 \\ &= -2.63\% \end{aligned}$$

This suggests that the Australian dollar must depreciate by about 2.63% to make the Australian dollar deposit generate the same effective yield as a deposit in euros. With any smaller degree of depreciation, the Australian dollar deposit would be more rewarding. Latrobe SA can use this information when determining whether to invest in a euro or Australian dollar deposit. If it expects the Australian dollar to depreciate by more than 2.63% over the deposit period, it will prefer investing in euros. If it expects the Australian dollar to depreciate by less than 2.63%, or to appreciate, its decision is more complex. If the potential reward from investing in the foreign currency outweighs the risk involved, then the firm should choose that route. The final decision here will be influenced by the firm's degree of risk aversion.

Use of probability distributions. Since even expert forecasts are not always accurate, it is sometimes useful to develop a probability distribution instead of relying on a single prediction. An example of how a probability distribution is applied follows.

EXAMPLE

Opera Ltd (UK) is deciding whether to invest in Australian dollars for one year. It finds that the quoted interest rate for the Australian dollar is 14%, and the quoted interest rate for a pound deposit is 11%. It then develops a probability distribution for the Australian dollar's possible percentage change in value over the life of the deposit.

The probability distribution is displayed in Exhibit 19.6. From the first row in the exhibit, we see that there is a 5% probability of a 10% depreciation in the Australian dollar over the deposit's life.

If the Australian dollar does depreciate by 10%, the effective yield will be 2.60%. This indicates that there is a 5% probability that Opera Ltd will earn a 2.60% effective yield on its funds. From the second row in the exhibit, there is a 10% probability of an 8% depreciation in the Australian dollar over the deposit period. If the Australian dollar does depreciate by 8%, the effective yield will be 4.88%, which means there is a 10% probability that Opera will generate a 4.88% effective yield on this deposit.

EXHIBIT 19.6 Analysis of investing in a foreign currency

Possible rate of change in the Australian dollar over the life of the investment (e_t)	Probability of occurrence	Effective yield if this rate of change in the Australian dollar does occur (combining the 14% interest rate with the change in the value of the currency)
-10%	5%	$(1.14) [1 + (-0.10)] - 1 = 0.0260$, or 2.60%
-8	10	$(1.14) [1 + (-0.08)] - 1 = 0.0488$, or 4.88%
-4	15	$(1.14) [1 + (-0.04)] - 1 = 0.0944$, or 9.44%
-2	20	$(1.14) [1 + (-0.02)] - 1 = 0.1172$, or 11.72%
+1	20	$(1.14) [1 + (0.01)] - 1 = 0.1514$, or 15.14%
+2	15	$(1.14) [1 + (0.02)] - 1 = 0.1628$, or 16.28%
+3	10	$(1.14) [1 + (0.03)] - 1 = 0.1742$, or 17.42%
+4	5	$(1.14) [1 + (0.04)] - 1 = 0.1856$, or 18.56%
	100%	

(Continued)

For each possible percentage change in the Australian dollar's value, there is a corresponding effective yield. Each possible effective yield (third column) is associated with a probability of that yield occurring (second column). An *expected value* of the effective

yield of the Australian dollar is derived by multiplying each possible effective yield by its corresponding probability. Based on the information in Exhibit 19.6, the expected value of the effective yield, referred to as $E(r)$, is computed this way:

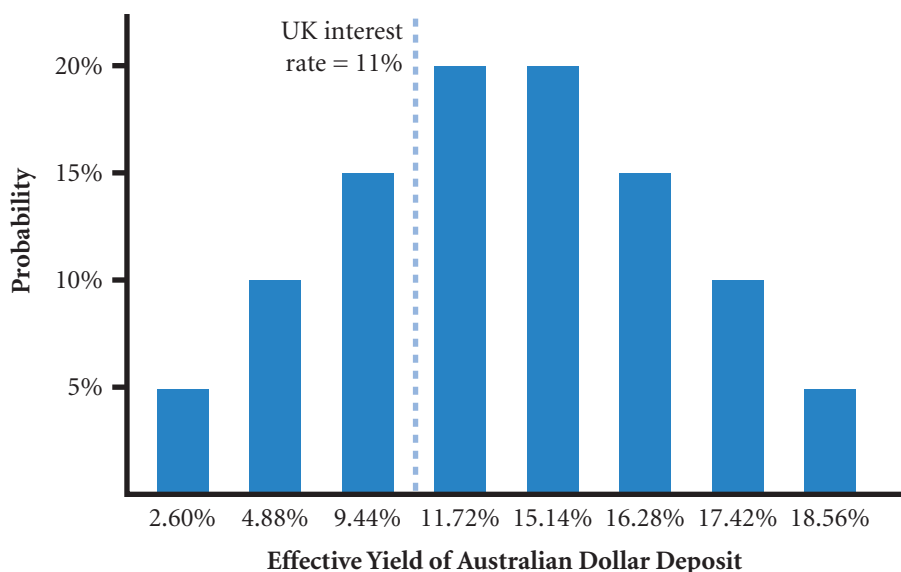
$$\begin{aligned}
 E(r) &= 5\%(2.60\%) + 10\%(4.88\%) + 15\%(9.44\%) + 20\%(11.72\%) + 20\%(15.14\%) + 15\%(16.28\%) \\
 &\quad + 10\%(17.42\%) + 5\%(18.56\%) \\
 &= 0.13\% + 0.488\% + 1.416\% + 2.344\% + 3.028\% + 2.422\% + 1.742\% + 0.928\% \\
 &= 12.58\%
 \end{aligned}$$

Thus, the expected value of the effective yield when investing in Australian dollars is approximately 12.5%.

To further assess the question of which currency to invest in, the information in the second and third columns from Exhibit 19.6 is used to develop a probability distribution in Exhibit 19.7, which illustrates the probability of each possible effective yield that may occur if Opera Ltd invests in Australian dollars. Notice that the UK interest rate (11%) is known with certainty and is included in Exhibit 19.7 for comparison purposes. A comparison of the Australian dollar's probability distribution against the UK interest rate suggests that there is a 30% probability that the UK rate will be more than the effective yield from investing in Australian dollars and a 70% chance that it will be less.

If Opera Ltd invests in a British pound deposit, it knows with certainty the yield it will earn from its investment. If it invests in Australian dollars, its risk is the 5% chance (probability) that the effective yield on the Australian dollar deposit will be 2.60%, or the 10% chance that the effective yield on the Australian dollar deposit will be 4.88%, or the 15% chance that the effective yield on Australian dollars will be 9.44%. Each of these possibilities represents a lower return to Opera Ltd than what it would have earned had it invested in a UK deposit. Opera Ltd concludes that the potential return on the Australian deposit is not high enough to compensate for the risk and decides to invest in the UK deposit.

EXHIBIT 19.7 Probability distribution of effective yields



Diversifying cash across currencies

Because an MNC is not sure how exchange rates will change over time, it may prefer to diversify cash among securities denominated in different currencies. Limiting the percentage of excess cash invested in each currency will reduce the MNC's exposure to exchange rate risk.

The degree to which a portfolio of investments denominated in various currencies will reduce risk depends on the currency correlations. Ideally, the currencies represented within the portfolio will exhibit low or negative correlations with each other. When currencies are likely to be affected by the same underlying event, their movements tend to be more highly correlated, and diversification among these types of currencies does not substantially reduce exposure to exchange rate risk.

EXAMPLE

In 1997, the interest rates in most Asian countries were higher than the interest rate in the United Kingdom. However, Asian currencies, such as the Indonesian rupiah, the Malaysian ringgit, the South Korean won and the Thailand baht, depreciated by more than 50% against the pound in less than one year. Consequently, subsidiaries based outside Asia that attempted to ben-

efit from the high Asian interest rates earned negative effective yields on their investments, so they received less than what they initially invested. Diversification of cash among these currencies was not beneficial in this case because all of the currencies weakened in response to the Asian crisis.

Dynamic hedging

Some MNCs continually adjust their short-term positions in currencies in response to revised expectations of each currency's future movement. They may engage in **dynamic hedging**, which is a strategy of applying a hedge when the currencies held are expected to depreciate and removing the hedge when the currencies held are expected to appreciate. In essence, the objective is to protect against downside risk while benefiting from the favourable movement of exchange rates.

For example, consider a treasurer of a UK firm who plans to invest in US money market securities. If the British pound begins to decline and is expected to depreciate further, the treasurer may sell dollars forward in the foreign exchange market for a future date at which the dollar's value is expected to turn upward. The treasurer is likely to take out a futures contract as these contracts are easier to buy and sell than forward contracts. If the treasurer is very confident that the dollar will depreciate in the short run, most or all of the position will be hedged.

Now assume that the pound begins to appreciate before the futures contract date. Since the contract will preclude the potential benefits from the pound's appreciation, the treasurer may buy pounds forward to offset the existing futures sale contracts. In this way, the treasurer has removed the existing hedge (or closed out the position). Of course, if the forward rate at the time of the futures purchase exceeds the forward rate that existed at the time of the futures sale, a cost is incurred to offset the hedge.

The treasurer may decide to remove only part of the hedge, offsetting only some of the existing futures sales with futures purchases. With this approach, the position is still partially protected if the pound depreciates further. Overall, the performance from using dynamic hedging is dependent on the treasurer's ability to forecast the direction of exchange rate movements.

SUMMARY

- Each subsidiary of an MNC can assess its cash flows by estimating expected cash inflows and outflows to forecast its balance in each currency. This will indicate whether it will have excess cash to invest or a cash deficiency. The MNC's parent may prefer to use a centralized perspective, which consolidates the cash flow positions of all subsidiaries. In this way, funds can be transferred among subsidiaries to accommodate cash deficiencies at particular subsidiaries.
- The common techniques to optimize cash flows are: (1) accelerating cash inflows, (2) minimizing currency conversion costs, (3) managing blocked funds, and (4) implementing intersubsidiary cash transfers.
- The efforts by MNCs to optimize cash flows are complicated by: (1) company-related characteristics, (2) government restrictions, and (3) characteristics of banking systems.
- MNCs can possibly achieve higher returns when investing excess cash in foreign currencies that either have relatively high interest rates or may appreciate over the investment period. If the foreign currency depreciates over the investment period, however, this may offset any interest rate advantage of that currency.

CRITICAL DEBATE

Should an MNC's subsidiaries operate their own cash management policies?

Proposition. Yes. Ultimately cash management means that everything is controlled from the centre. Different countries and indeed different products have very different working capital requirements. Centralization could easily lead to poor working capital management and cash flow difficulties.

Opposing view. No. Trade between subsidiaries accounts for a large percentage of world exports, so

there has to be coordination between subsidiaries. Also customers can be MNC's so coordination is also required in debt collection. Also, there are considerable exchange rate savings to be made.

With whom do you agree? Think carefully about the arguments for and against allowing subsidiaries to manage their own cash. What are the problems with each of the arguments? Is there a solution that avoids the main drawbacks?

SELF TEST

Answers are provided in Appendix A at the back of the text.

- Country X typically has a high interest rate, and its currency is expected to strengthen against the pound over time. Country Y typically has a low interest rate, and its currency is expected to weaken against the pound over time. Both countries have imposed a 'blocked funds' restriction over the next four years on the two subsidiaries owned by a UK firm. Which subsidiary will be more adversely affected by the blocked funds, assuming that there are limited opportunities for corporate expansion in both countries?
- Assume that the Australian one-year interest rate is 14%. Also assume that the Australian dollar is expected to appreciate by 8% over the next year against the British pound. What is the expected effective yield on a one-year deposit in Australia by a UK firm?
- Assume that the one-year forward rate is used as the forecast of the future spot rate. The Malaysian ringgit's spot rate is £0.12, while its one-year forward rate is £0.11. The Malaysian one-year interest rate is 11%. What is the expected effective yield on a one-year deposit in Malaysia by a UK firm?

- 4 Assume that the Venezuelan one-year interest rate is 90%, while the UK one-year interest rate is 6%. Determine the break-even value for the percentage change in Venezuela's currency (the bolivar) that would cause the effective yield to be the same for a

one-year deposit in Venezuela as for a one-year deposit in the United Kingdom.

- 5 Assume interest rate parity exists. Would European firms possibly consider placing deposits in countries with high interest rates? Explain.

QUESTIONS AND APPLICATIONS

- 1 **International cash management.** Discuss the general functions involved in international cash management. Explain how the MNC's optimization of cash flow can distort the profits of each subsidiary.
- 2 **Netting.** Explain the benefits of netting. How can a centralized cash management system be beneficial to the MNC?
- 3 **Leading and lagging.** How can an MNC implement leading and lagging techniques to help subsidiaries in need of funds?
- 4 **International Fisher effect.** If a US firm believes that the international Fisher effect holds, what are the implications regarding a strategy of continually attempting to generate high returns from investing in currencies with high interest rates?
- 5 **Investing strategy.** Trumpington Ltd has £2 million in excess cash that it has invested in Mexico at an annual interest rate of 60%. The UK interest rate is 9%. By how much would the Mexican peso have to depreciate to cause such a strategy to backfire?
- 6 **Investing strategy.** Why would a UK firm consider investing short-term funds in euros even when it does not have any future cash outflows in euros?
- 7 **Covered interest arbitrage.** Granville SA has 2 million euro in cash available for 90 days. It is considering the use of covered interest arbitrage, since the euro's 90-day interest rate is higher than the UK interest rate. What will determine whether this strategy is feasible?
- 8 **Effective yield.** Corlins plc has £1 million in cash available for 30 days. It can earn 1% on a 30-day investment in the United Kingdom. Alternatively, if it converts the pounds to South African rand, it can earn 1½% on a rand deposit. The spot rate of the rand is £0.09. The spot rate 30 days from now is expected to be £0.08. Should Corlins invest its cash in the United Kingdom or in South Africa? Explain your answer.
- 9 **Effective yield.** Rollins plc has £3 million in cash available for 180 days. It can earn 7% on a UK Treasury bill or 9% on a US Treasury bill. The US investment does require conversion of pounds to dollars. Assume that interest rate parity holds and that Rollins believes the 180-day forward rate is a reliable predictor of the spot rate to be realized 180 days from now. Would the British investment provide an effective yield that is below, above, or equal to the yield on the US investment? Explain your answer.
- 10 **Effective yield.** Repeat question 9, but this time assume that Rollins plc expects the 180-day forward rate of the dollar to substantially overestimate the spot rate to be realized in 180 days.
- 11 **Effective yield.** Repeat question 9, but this time assume that the Rollins plc expects the 180-day forward rate of the dollar to substantially underestimate the spot rate to be realized in 180 days.
- 12 **Effective yield.** Assume that the one-year UK interest rate is 10% and the one-year US interest rate is 13%. If a UK firm invests its funds in the US, by what percentage will the dollar have to depreciate to make its effective yield the same as the UK interest rate from the UK firm's perspective?
- 13 **Investing in a currency portfolio.** Why would a firm consider investing in a portfolio of foreign currencies instead of just a single foreign currency?
- 14 **Interest rate parity.** Trellis Ltd has determined that the interest rate on euros is 16% while the UK interest rate is 11% for one-year Treasury bills. The one-year forward rate of the euro has a discount of 7%. Does interest rate parity exist? Can Trellis achieve a higher effective yield by using covered interest arbitrage than by investing in UK Treasury bills? Explain.
- 15 **Diversified investments.** Hofstra Ltd has no business outside the UK but has cash invested in six European countries, each of which uses the euro as its local currency. Are Hofstra's short-term investments

well diversified and subject to a low degree of exchange rate risk? Explain.

- 16 Investing strategy.** Should McNeese Ltd consider investing funds in Latin American countries where it may expand facilities? The interest rates are high, and the proceeds from the investments could be used to help support the expansion. When would this strategy backfire?

ADVANCED QUESTIONS

- 18 Investing in a portfolio.** Poppleton Ltd plans to invest its excess cash in South African rand for one year. The one-year South African interest rate is 19%. The probability of the rand's percentage change in value during the next year is shown below:

Possible rate of change in the South African rand over the life of the investment	Probability of occurrence
−15%	20%
−4	50%
0	30%

What is the expected value of the effective yield based on this information? Given that the UK interest rate for one year is 7%, what is the probability that a one-year investment in pesos will generate a lower effective yield than could be generated if Poppleton Ltd simply invested domestically?

- 19 Effective yield of portfolio.** Ithaca (Greece) considers placing 30% of its excess funds in a one-year Singapore dollar deposit and the remaining 70% of its funds in a one-year US dollar deposit. The Singapore one-year interest rate is 15%, while the US one-year interest rate is 10%. The possible percentage changes

- 17 Impact of a crisis.** Palos SA (Spain) commonly invests some of its excess euros in foreign government short-term securities in order to earn a higher short-term interest rate on its cash. Describe how the potential return and risk of this strategy may be affected by financial crisis.

in the two currencies for the next year are forecasted as follows:

Currency	Possible percentage change in the spot rate over the investment horizon	Probability of that change in the spot rate occurring
Singapore dollar	−2%	20%
Singapore dollar	1	60
Singapore dollar	3	20
US dollar	1	50
US dollar	4	40
US dollar	6	10

Given this information, determine the possible effective yields of the portfolio and the probability associated with each possible portfolio yield. Given a one-year euro interest rate of 8%, what is the probability that the portfolio's effective yield will be lower than the yield achieved from investing in the United States? (Assume that the movements on the two currencies are not correlated.)

PROJECT WORKSHOP

- 20 Foreign currency yields.** The Bloomberg website provides interest rate data for many different foreign currencies over various maturities. Its address is <http://www.bloomberg.com>.

a Go to the section that shows yields for different foreign currencies. Review the one-year yields of

currencies. Assume that you could borrow at a rate 1 percentage point above the quoted yield for each currency. Which currency would offer you the highest quoted yield?

- b As a cash manager of an MNC based in the United Kingdom that has extra pounds that can be invested for one year, where would you invest funds for the next year? Explain.
- c If you were working for a foreign subsidiary based in Japan and could invest Japanese yen for one year until the yen are needed to support local operations, where would you invest the yen? Explain.

DISCUSSION IN THE BOARDROOM

This exercise can be found on the companion website at www.cengagelearning.co.uk/madura_fox2e.

RUNNING YOUR OWN MNC

This exercise can be found on the companion website at www.cengagelearning.co.uk/madura_fox2e.

Essays/discussion and articles can be found at the end of Part 5

BLADES PLC CASE STUDY

International cash management

Recall from Chapter 18 that the new Thailand subsidiary of Blades plc received a one-time order from a customer for 120 000 pairs of 'Speedos', Blades' primary product. There is a six-month lag between the time when Blades needs funds to purchase material for the production of the Speedos and the time when it will be paid by the customer. Ben Holt, Blades' chief financial officer (CFO), has decided to finance the cost by borrowing Thai baht at an interest rate of 6% over a six-month period. Since the average cost per pair of Speedos is approximately 3500 baht, Blades will borrow 420 million baht. The payment for the order will be used to repay the loan's principal and interest.

Ben Holt is currently planning to instruct the Thai subsidiary to remit any remaining baht-denominated cash flows back to the United Kingdom. Just before Blades receives payment for the large order, however, Holt notices that interest rates in Thailand have increased substantially. Blades would be able to invest funds in Thailand at a relatively high interest rate compared to the UK rate. Specifically, Blades could invest the remaining baht-denominated funds for one year in Thailand at an interest rate of 15%.

If the funds are remitted back to the UK parent, the excess pound volume resulting from the conversion of baht will either be used to support the UK production of Speedos, if needed, or be invested in the United Kingdom. Specifically, the funds will be used to cover the cost of goods sold in the UK manufacturing plant. Since Blades used a significant amount of cash to finance the initial investment to build the plant in Thailand and purchase the necessary equipment, its UK operations are strapped for cash. Consequently, if the subsidiary's earnings are not remitted back to the United Kingdom, Blades will have to borrow funds at an interest rate of 10% to support its UK operations. Any funds remitted by the subsidiary that are not used to support UK operations will be invested in the United States at an interest rate of 8%. Holt estimates that approximately 60% of the remitted funds will be

needed to support UK operations and that the remaining 40% will be invested in the United Kingdom. Consequently, Holt must choose between two alternative plans. First, he could instruct the Thai subsidiary to repay the baht loan (with interest) and invest any remaining funds in Thailand at an interest rate of 15%. Second, he could instruct the Thai subsidiary to repay the baht loan and remit any remaining funds back to the United Kingdom, where 60% of the funds would be used to support UK operations and 40% would be invested at an interest rate of 8%. Assume no income or withholding taxes on the earnings generated in Thailand.

Ben Holt has contacted you, a financial analyst at Blades plc to help him analyze these two options. Holt has informed you that the current spot rate of the Thai baht is £0.015 and that the baht is expected to depreciate by 5% over the coming year. He has provided you with the following list of questions he would like you to answer.

- 1 There is a tradeoff between the higher interest rates in Thailand and the delayed conversion of baht into pounds. Explain what this means.
- 2 If the net baht received from the Thailand subsidiary are invested in Thailand, how will UK operations be affected?
- 3 Construct a spreadsheet that compares the cash flows resulting from the two plans. Under the first plan, net baht-denominated cash flows (received today) will be invested in Thailand at 15% for a one-year period, after which the baht will be converted to pounds. Under the second plan, net baht-denominated cash flows are converted to pounds immediately and 60% of the funds will be used to support UK operations, while 40% are invested in the United Kingdom for one year at 8%. Which plan is superior given the expectation of the baht's value in one year?

SMALL BUSINESS DILEMMA

Cash management at the Sports Exports Company

Ever since Jim Logan began his Sports Exports Company (Ireland), he has been concerned about his exposure to exchange rate risk. The firm produces basketballs and exports them to a distributor in the United Kingdom, with the exports being denominated in British pounds. Jim has just entered into a joint venture in the United Kingdom in which a British firm produces sporting goods for Jim's firm and sells the goods to the British distributor. The distributor pays pounds to Jim's firm for these products. Jim recently borrowed pounds to finance this venture, which created some cash outflows (interest payments) that partially offset his cash inflows in pounds. The interest paid on this loan is equal to the British Treasury bill rate plus 3 percentage points. His original business of exporting has been very successful recently, which

has caused him to have revenue (in pounds) that will be retained as excess cash. Jim must decide whether to pay off part of the existing British loan, invest the cash in the euro denominated Treasury bills, or invest the cash in British Treasury bills.

- 1 If Jim invests the excess cash in UK Treasury bills, would this reduce the firm's exposure to exchange rate risk?
- 2 Jim decided to use the excess cash to pay off the British loan. However, a friend advised him to invest the cash in British Treasury bills, stating that 'the loan provides an offset to the pound receivables, so you would be better off investing in British Treasury bills than paying off the loan'. Is Jim's friend correct? What should Jim do?

PART 5 Integrative problem

SHORT-TERM ASSET AND LIABILITY MANAGEMENT

Kent plc is a large UK firm with no international business. It has two branches within the UK, a southern branch and a northern branch. Each branch currently makes investing or financing decisions independently, as if it were a separate entity. The northern branch has excess cash of £15 million to invest for the next year. It can invest its funds in Treasury bills denominated in pounds or in any of four foreign currencies. The only restriction enforced by the parent is that a maximum of £5 million can be invested or financed in any foreign currency.

The southern branch needs to borrow £15 million over one year to support its UK operations. It can borrow funds in any of these same currencies (although any foreign funds borrowed would need to be converted to pounds to finance the UK operations). The only restriction enforced by the parent is that a maximum equivalent of £5 million can be borrowed in any single currency. A large bank serving the international money market has offered Kent plc the following terms:

Currency	Annual interest rate on deposits	Annual interest rate charged on loans
British pound	6%	9%
Australian dollar	11	14
Canadian dollar	7	10
New Zealand dollar	9	12
Japanese yen	8	11

The parent of Kent plc has created one-year forecasts of each currency for the branches to use in making their investing or financing decisions:

Currency	Today's spot exchange rate	Forecasted annual percentage change in exchange rate
Australian dollar	11	14
Canadian dollar	7	10
New Zealand dollar	9	12
Japanese yen	8	11

Questions

- 1 Determine the investment portfolio composition for Kent's northern branch that would maximize the expected effective yield while satisfying the restriction imposed by the parent.
- 2 What is the expected effective yield of the investment portfolio?
- 3 Based on the expected effective yield for the portfolio and the initial investment amount of £15 million, determine the annual interest to be earned on the portfolio.
- 4 Determine the financing portfolio composition for Kent's southern branch that would minimize the expected effective financing rate while satisfying the restriction imposed by the parent.
- 5 What is the expected effective financing rate of the total amount borrowed?
- 6 Based on the expected effective financing rate for the portfolio and the total amount of £15 million borrowed, determine the expected loan repayment amount beyond the principal borrowed.
- 7 When the expected interest received by the northern branch and paid by the southern branch of Kent plc are consolidated, what is the net amount of interest received?
- 8 If the northern branch and the southern branch worked together, the northern branch could loan its £15 million to the southern branch. Nevertheless, one could argue that the branches could not take advantage of interest rate differentials or expected exchange rate effects among currencies. Given the data provided in this example, would you recommend that the two branches make their short-term investment or financing decisions independently, or should the northern branch lend its excess cash to the southern branch? Explain.

PART 5 Essays/discussion and academic articles

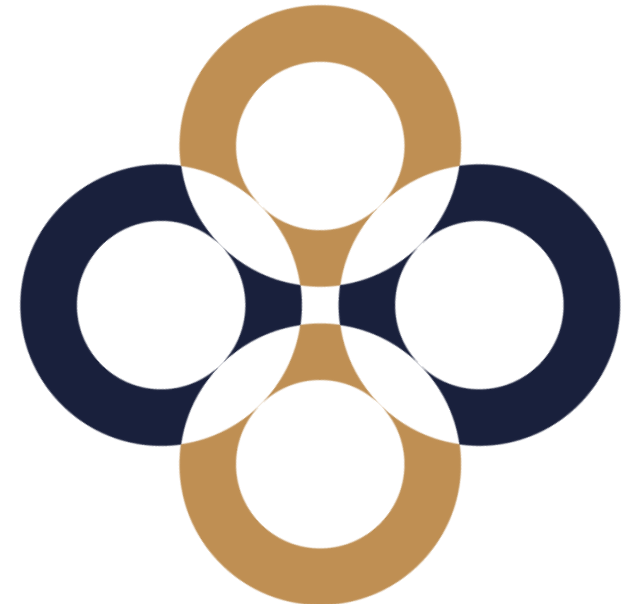
- 1 Shenk, C.R. (1997) 'The Origins of the Eurodollar Market in London 1955–1963', *Explorations in Economic History*, 35, 221–38. A readable account of financial innovation being a mixture of market competition, margins and governments.
Q *To what extent is the control of national financial economics out of the hands of any one government? Discuss in relation to international financial arrangements (the euro-dollar market being one such example).*
- 2 Easterly, W. (2002) 'How Did Heavily Indebted Poor Countries Become Heavily Indebted? Reviewing two decades of debt relief', *World Development*, 30 (10), 1677–96. Required reading for any discussion of debt relief.
Q *Is it sensible for governments of rich countries to offer better trade terms for poorer countries? Discuss.*
- 3 Menzler-Hokkanen, I. (1989) 'Countertrade Arrangements in International Trade: A Tool for Creating Competitive Advantage', *Scandinavian Journal of Management*, 5 (2), 105–22. Never mind the date; this paper looks at the relationship between business and finance – a subject that deserves more attention.
Q *Use Menzler-Hokkanen's article to assess the advantages and disadvantages of using financial schemes to promote trade. In particular, what factors need to be considered?*
- 4 Durbin, E. and D. Ng (2005) 'The Sovereign Ceiling and Emerging Market Corporate Bond Spreads', *Journal of International Money and Finance*, 24, 631–49. A look at bond spreads.
Q *Based on the study by Durbin and Ng, assess the extent to which companies' credit is limited by their home countries' credit worthiness.*
- 5 De Boyrie, M.E., S.J. Pak and J.S. Zdaniwicz (2005) 'Estimating the Magnitude of Capital Flight Due to Abnormal Pricing in International Trade, the Russia–USA Case', *Accounting Forum*, 29, 249–70. Over and under invoicing.
Q *Describe how the authors establish mispricing of exports and imports. Discuss the effects of such mispricing – is it all bad?*

Short term Financing

2023.10.10.

International Corporate Finance Lecture

Dr. Judit Burucs



Topics

- Long term Finance
- Short term Finance
- Methods of payment for international trade.
- Common trade finance methods.
- Major agencies that facilitate international trade with export insurance and/or loan programs.

Part IV Long-Term Asset and Liability Management

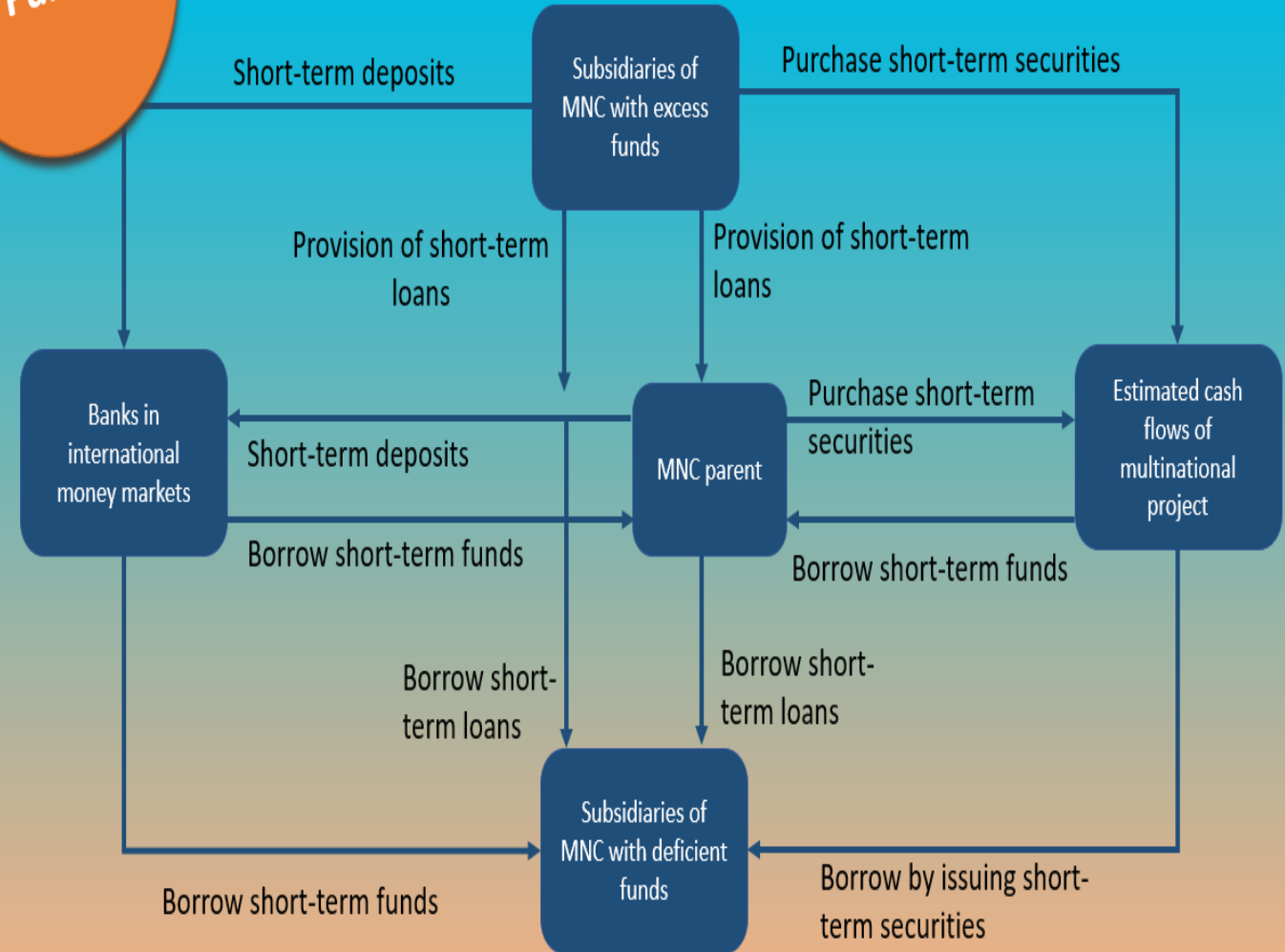
Macro-level distortion



Firm specific

Part 5

Short-Term Asset and Liability Management

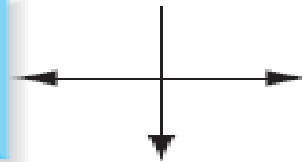


Local Market Access

Global Market Access

Firm-Specific Characteristics

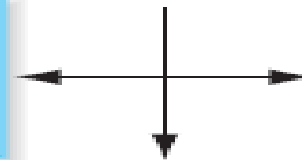
Firm's securities appeal only to domestic investors



Firm's securities appeal to international portfolio investors

Market Liquidity for Firm's Securities

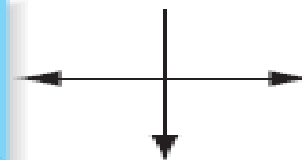
Illiquid domestic securities market and limited international liquidity



Highly liquid domestic market and broad international participation

Effect of Market Segmentation on Firm's Securities and Cost of Capital

Segmented domestic securities market that prices shares according to domestic standards



Access to global securities market that prices shares according to international standards

Source: Eitmann

Long-Term Financing Decision

Since MNCs commonly invest in **long-term projects**, they rely **heavily on long-term financing**.

The MNC must consider :

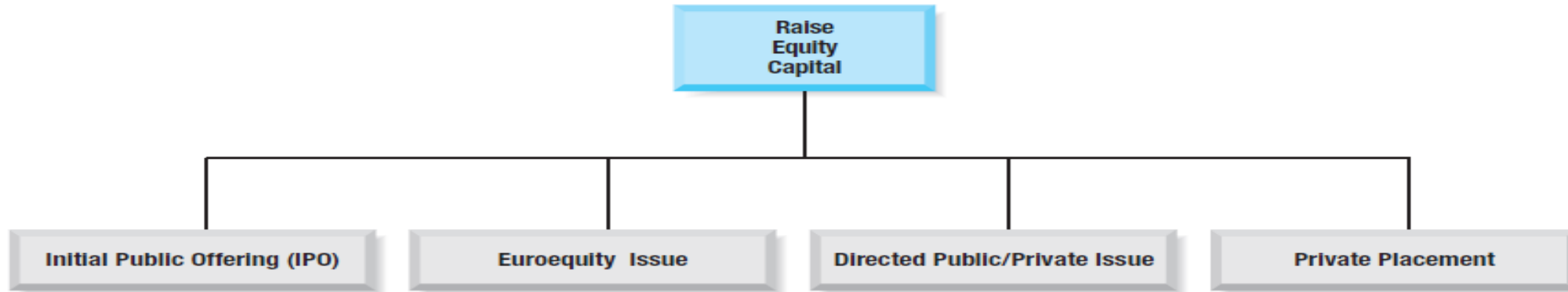
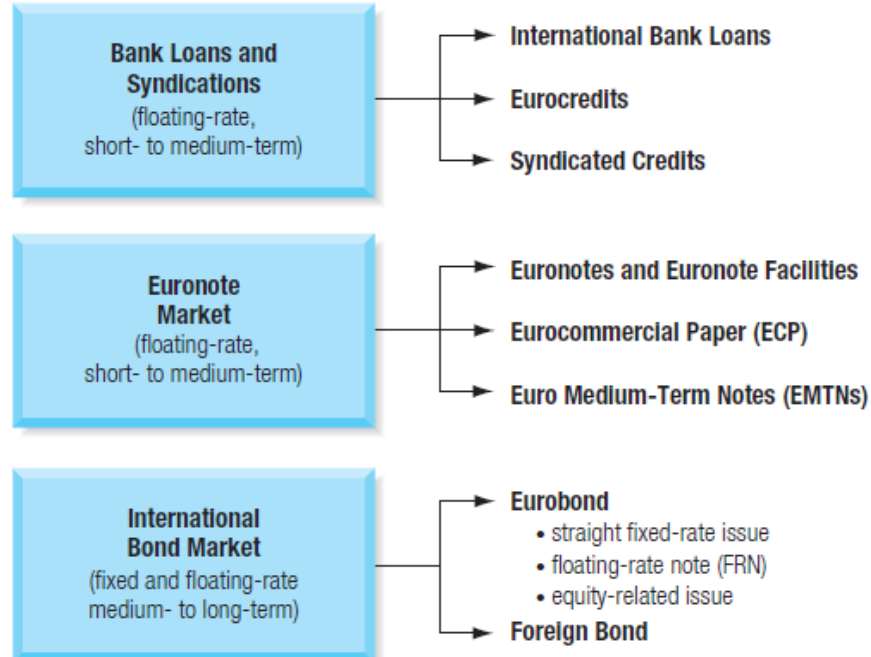
the possible sources of equity or debt,
and the costs and risks associated with each source.

- Sources of equity and debt include:
 - domestic offering in local currency
 - global offering in multiple currencies
 - private placement to home country or financial institutions to host country financial institutions.

To estimate the cost, an MNC needs to:

- determine the amount of funds needed
- forecast
 - interest rate ,
 - issue price of the bond
 - the exchange rates for the times when it must pay the bondholders

Many MNCs obtain equity funding in their home country and engage in debt financing in foreign countries.

EXHIBIT 14.4**Equity Alternatives in the Global Market****International Debt Markets and Instruments****Debt financing decision:**

- 1) Maturity
- 2) Currency
- 3) Repricing: fix or floating
- 4) Hedging

International short term financing

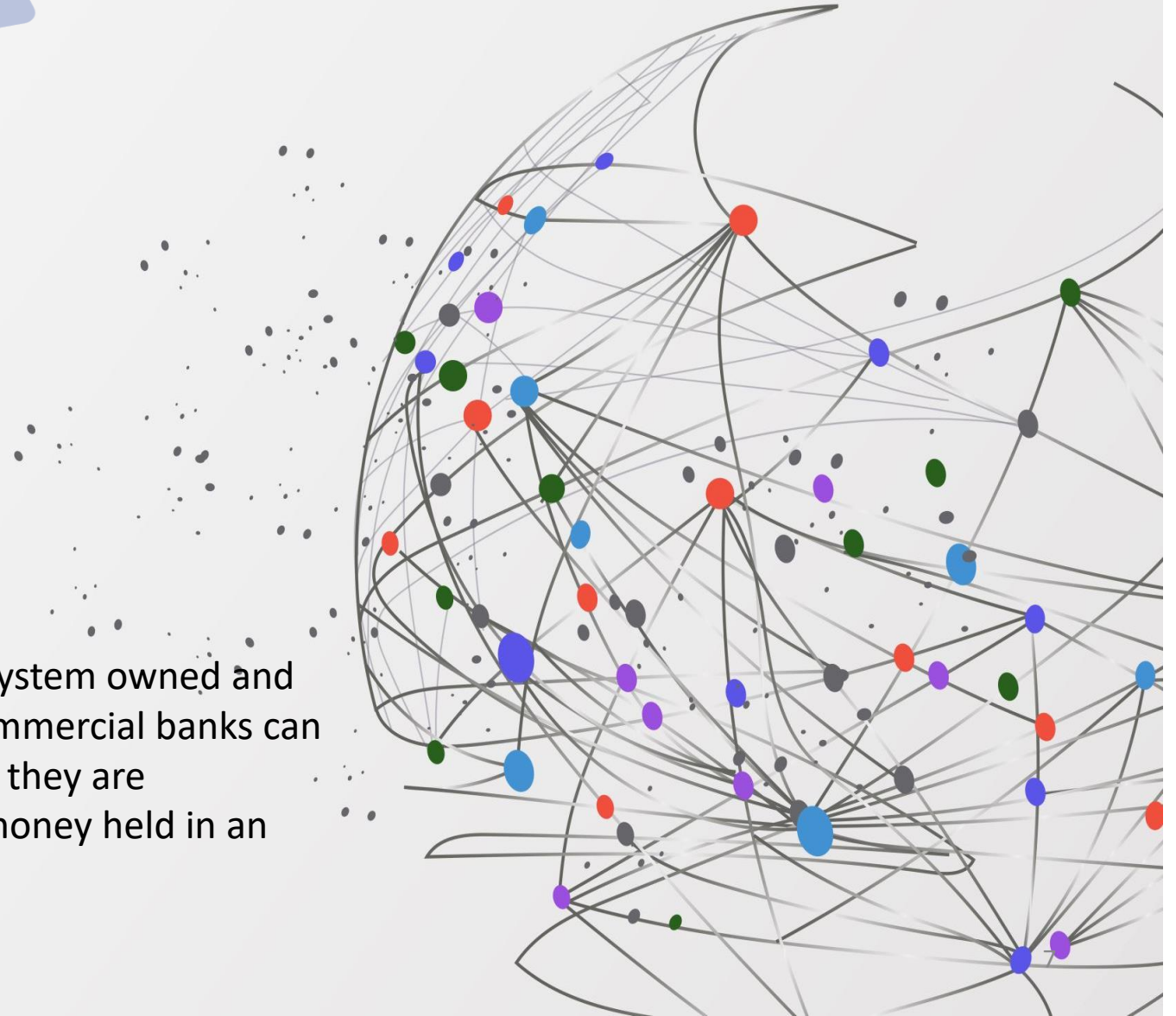
- The inter-company finance,
- The local currency loan
- Eurobank loans
- Factoring
- Euro notes
- Eurocommercial Paper
- Trade finance



PAYMENT SYSTEM

- TARGET 2
- <https://www.youtube.com/watch?v=G6x8NIYofPI>
- <https://www.federalreserve.gov/aboutthefed/payment-systems-video.htm>

TARGET2 is the real-time gross settlement (RTGS) system owned and operated by the Eurosystem. Central banks and commercial banks can submit payment orders in euro to TARGET2, where they are processed and settled in central bank money, i.e. money held in an account with a central bank.



Payment systems

Time: How long does it take?

The **payment system** is an operational network that links bank accounts and provides the **functionality for monetary exchange using bank deposits**.

- Large-value Payment Systems

Real Time Gross Settlement Systems (RTGS are funds transfer systems where transfer of money or securities takes place from one bank to another on a "real-time" and on "gross" basis.

Others: Cheques, etc

- Retail Payment Systems

- Settlement Of Foreign Exchange Transactions

- Payment Systems Worldwide

- Securities Settlement Systems

- **Clearing: is the process** where the banks use the payment information to transfer money between themselves on behalf of the payor (buyer) and the beneficiary.
- **Settlement** is the final step in the basic process and occurs when the beneficiary's bank account is credited and the payor's bank account is debited. Final settlement occurs when the banks pass value among themselves.
- **Remittance:** is a transfer of money by a foreign worker to an individual in his or her home country

http://siteresources.worldbank.org/FINANCIALSECTOR/Resources/282044-1323805522895/121534_text_corrections_3-15.pdf

RETAIL PAYMENT SYSTEMS

RETAIL PAYMENT SYSTEMS

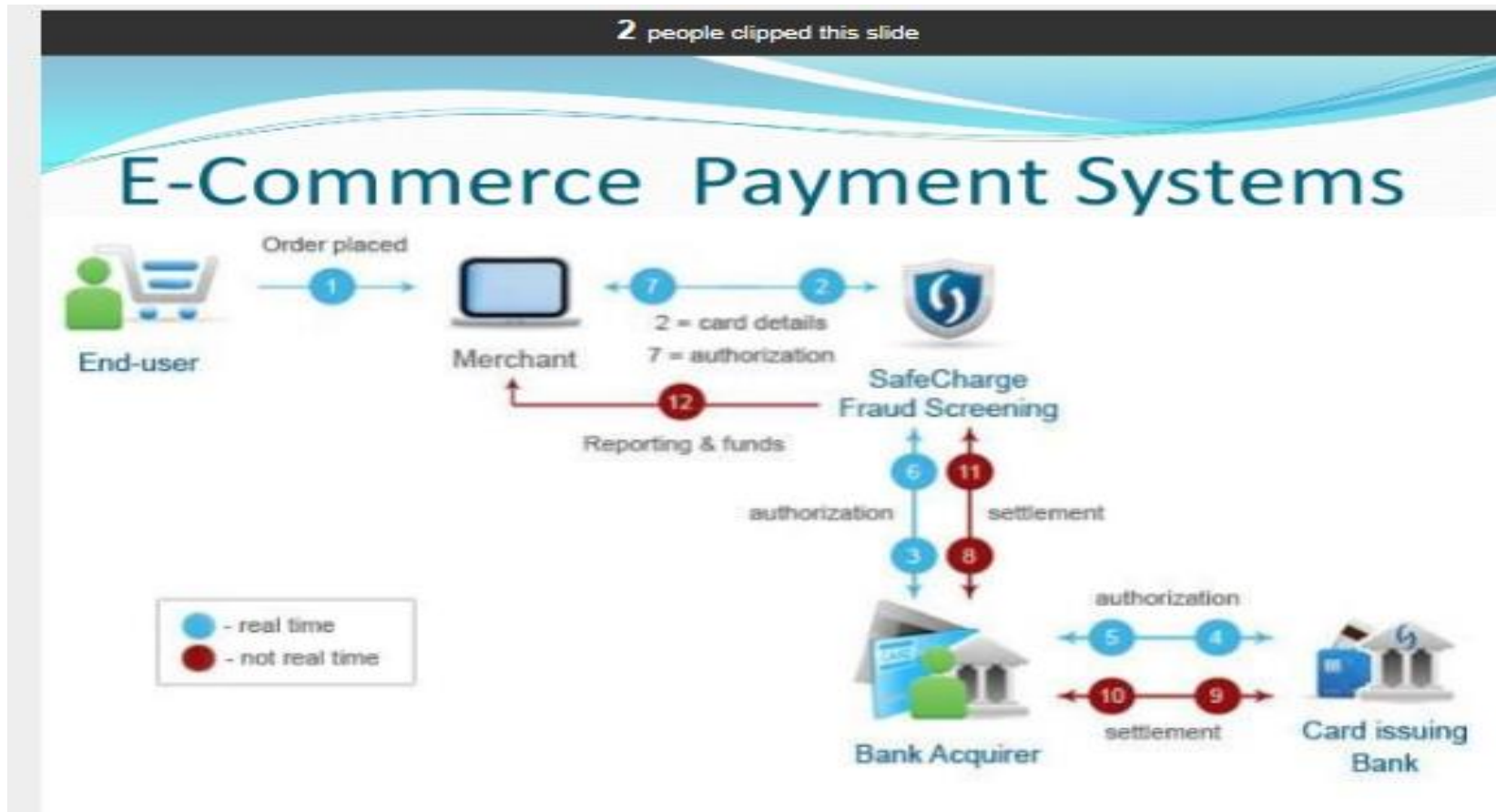
- Cheques and Cheque Clearinghouses,
- Credit Transfers, Direct Debits and Automated Clearinghouses,
- Payment Cards and Related Systems,
- Government payments



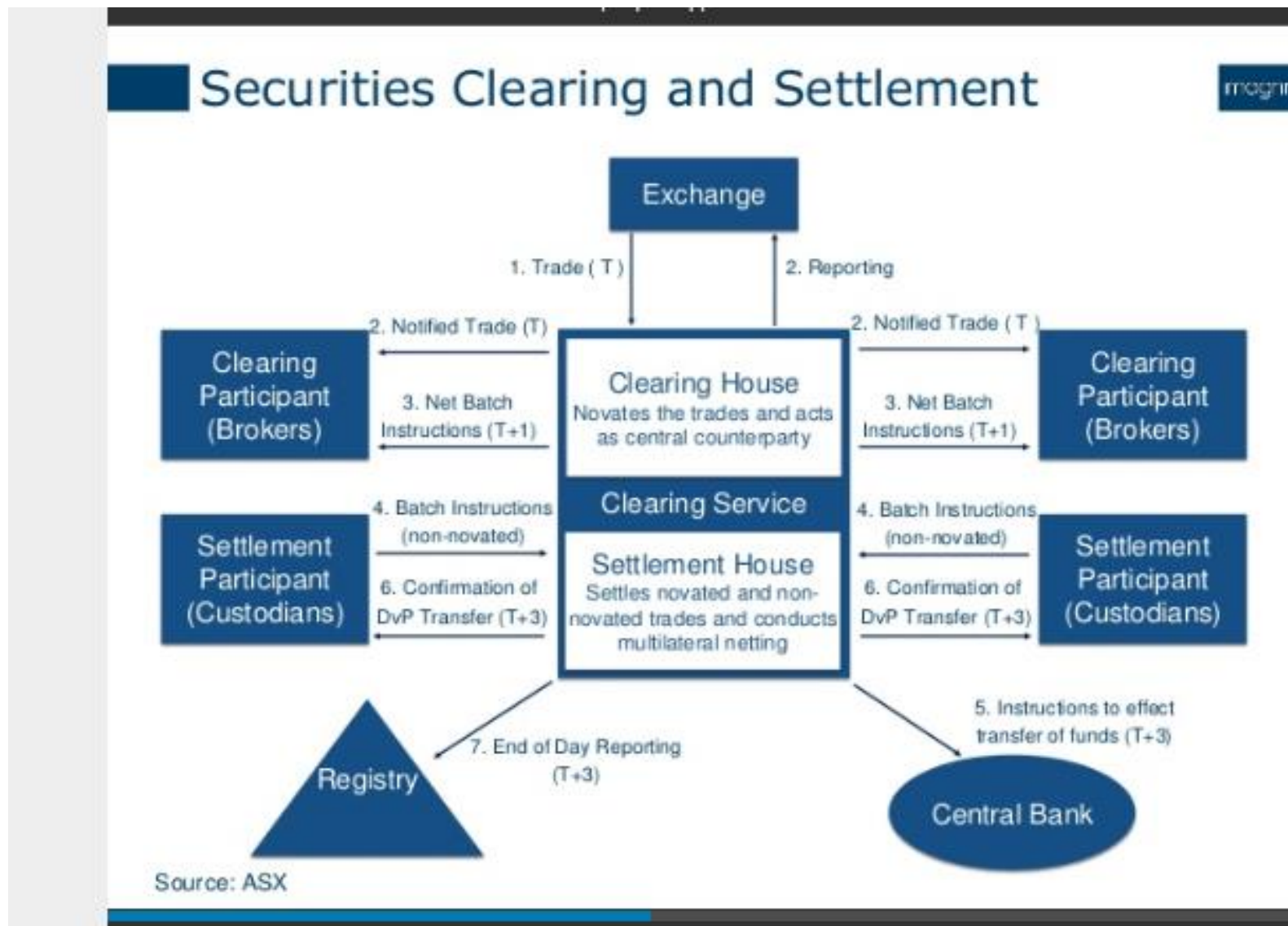
- **The slow development of access channels** to initiate and deliver cashless payments – e.g. point of sale (POS) terminals **in many developing countries, coupled with limited interoperability of the infrastructure already deployed;**
- **The limited access** by individuals to modern payment instruments in most developing countries, **especially outside urban areas;**
- **The limited competition among** banking institutions and, where available, between banks and other payment services providers, **typically resulting in higher costs and more limited coverage of these services;**

E-commerce Payment system

Listen:How Electronic Payments Work: Visa



SECURITY Clearing And Settlement



International Payment system

The **Society for Worldwide Interbank Financial Telecommunication (SWIFT)** provides a network that enables [financial institutions](#) worldwide to send and receive information about financial transactions in a secure, standardized and reliable environment. SWIFT payments are a type of international transfer sent via the SWIFT international payment network.

The SWIFT international payment network is one of the largest financial messaging systems in the world. TransferWise can send or receive certain currencies via SWIFT payment.

The Swift code consists of 8 or 11 characters. When 8-digits code is given, it refers to the primary office.

First 4 characters - bank code (only letters)

Next 2 characters - ISO 3166-1 alpha-2

country code (only letters)

Next 2 characters - location code (letters and digits) (passive participant will have "1" in the second character)

Last 3 characters - branch code, optional ('XXX' for primary office) (letters and digits)

SEPA (Single Euro Payment Area), is a European-based payment network. Fully implemented in 2014, SEPA is currently used by 38 European countries, including several not covered by the eurozone or in the European Union, such as Iceland, Switzerland and the UK.

The **Cross-Border Interbank Payment System (CIPS)** is a Chinese payment system that offers, clearing and settlement services for its participants' in cross-border RMB payments and trade.

An Example of How International Payments Work

if a company are located in the United States and want to send a funds transfer to India, the steps are the following:

1. The company must first setup the beneficiary and then transfer funds from company's account to the beneficiary.
2. The next step is when company's bank in the United States debits company's account and credits its account with the funds.
3. After this, it transfers the money to its partner bank in India or if it does not have any dealings with Indian banks, it contacts a bank in the United States that has such dealings and in both these cases;
4. The funds are then transferred from the banks in the United States to the bank in India.
5. Once the banks in India receive the funds, they must then send it to the ultimate beneficiary wherein the funds are debited from its account and credited to the recipient. This step might be a single or two step processes depending on the recipient holding an account with the concerned bank that receives the funds.

Example In Hungary)

Table 2 Typical payments performed by the systems overseen by the MNB		
Overseen systems		Typical transactions
VIBER		– bank-to-bank items: financial market transactions (i.e. HUF leg of HUF FX transactions, HUF cash leg of securities transactions) – settlement of the ICS intraday clearing cycles – central bank operations (e.g. cash and monetary policy operations) – customer transactions: large value urgent corporate or household payments
ICS	overnight clearing	– paper based credit transfers (i.e. salary or pension payments) – direct debits (i.e. bill payments) – transactions of the State Treasury
	intraday clearing	– retail payments: electronic credit transfers submitted by clients
KELER		– clearing and settlement of (1) regulated market transactions not guaranteed by KELER CCP; (2) OTC transactions; and (3) international transactions. – settlement of regulated market transactions guaranteed by KELER CCP Settlement includes the delivery and payment of financial instrument related to the transactions
KELER CCP		– clearing of regulated market (guaranteed) transactions including derivative transactions Calculation of margining and collateral requirements

http://english.mnb.hu/Root/Dokumentumtar/ENMNB/Kiadvanyok/report-on-payment-systems/MNB_JFR_2014_junius_ENG_digitalis.pdf

Payment Methods for International Trade

- In any international trade transaction, credit is provided by either
 - the supplier (exporter)
 - the buyer (importer)
 - one or more financial institutions, or
 - any combination of the above.
- The form of credit whereby the supplier funds the entire trade cycle is known as **supplier credit**.



Payment Methods for International Trade

Method 1: Prepayments

The goods will not be shipped until the buyer has paid the seller.

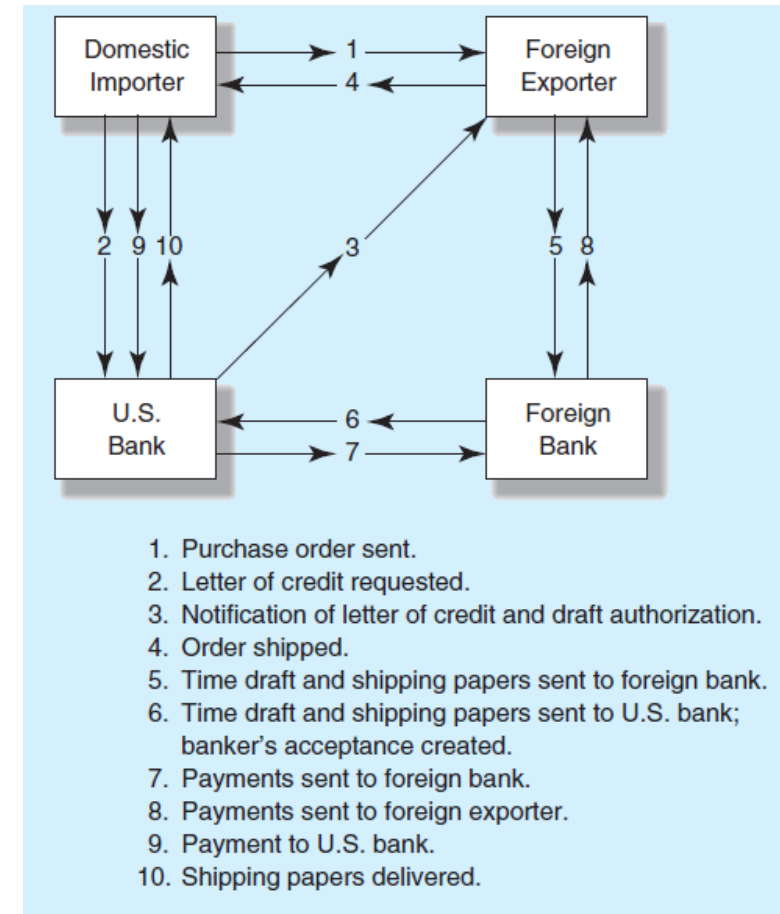
- *Time of payment*: before shipment
- *Goods available to buyers*: after payment
- *Risk to exporter*: none
- *Risk to importer*: relies completely on exporter to ship goods as ordered.

Payment Methods for International Trade

Method 2: Letters of credit (L/C)

These are issued by a bank on behalf of the importer promising to pay the exporter upon presentation of the shipping documents.

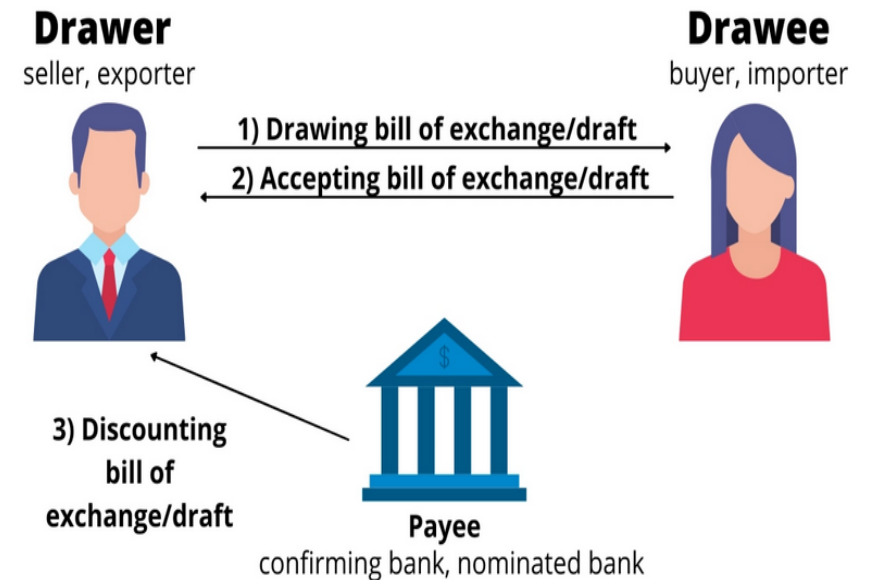
- *Time of payment*: when shipment is made
- *Goods available to buyers*: after payment
- *Risk to exporter*: very little or none
- *Risk to importer*: relies on exporter to ship goods as described in documents.



Payment Methods for International Trade

Method 3: Drafts (Bills of Exchange)

- These are unconditional promises drawn by the exporter instructing the buyer to pay the face amount of the drafts.
- Banks on both ends usually act as intermediaries in the processing of shipping documents and the collection of payment. In banking terminology, the transactions are known as documentary collections
- *Sight drafts (documents against payment)*: when the shipment has been made, the draft is presented to the buyer for payment.
- *Time of payment*: on presentation of draft
- *Goods available to buyers*: after payment
- *Risk to exporter*: disposal of unpaid goods
- *Risk to importer*: relies on exporter to ship goods as described in documents.



Payment Methods for International Trade

Method 4: **Consignments**

The exporter retains actual title to the goods that are shipped to the importer.

- *Time of payment*: at time of sale by buyer to third party.
- *Goods available to buyers*: before payment.
- *Risk to exporter*: allows importer to sell inventory before paying exporter.
- *Risk to importer*: none.

Payment Methods for International Trade (8)

Method 5: Open Accounts

The exporter ships the merchandise and expects the buyer to remit payment according to the agreed-upon terms.

- *Time of payment*: as agreed upon.
- *Goods available to buyers*: before payment.
- *Risk to exporter*: relies completely on buyer to pay account as agreed upon.
- *Risk to importer*: none.

Open Account Market

As global economies become more integrated, it is easier for exporters and importers themselves to access dependable information about foreign-trade partners, and they are less willing to pay for the risk protection afforded by traditional methods.

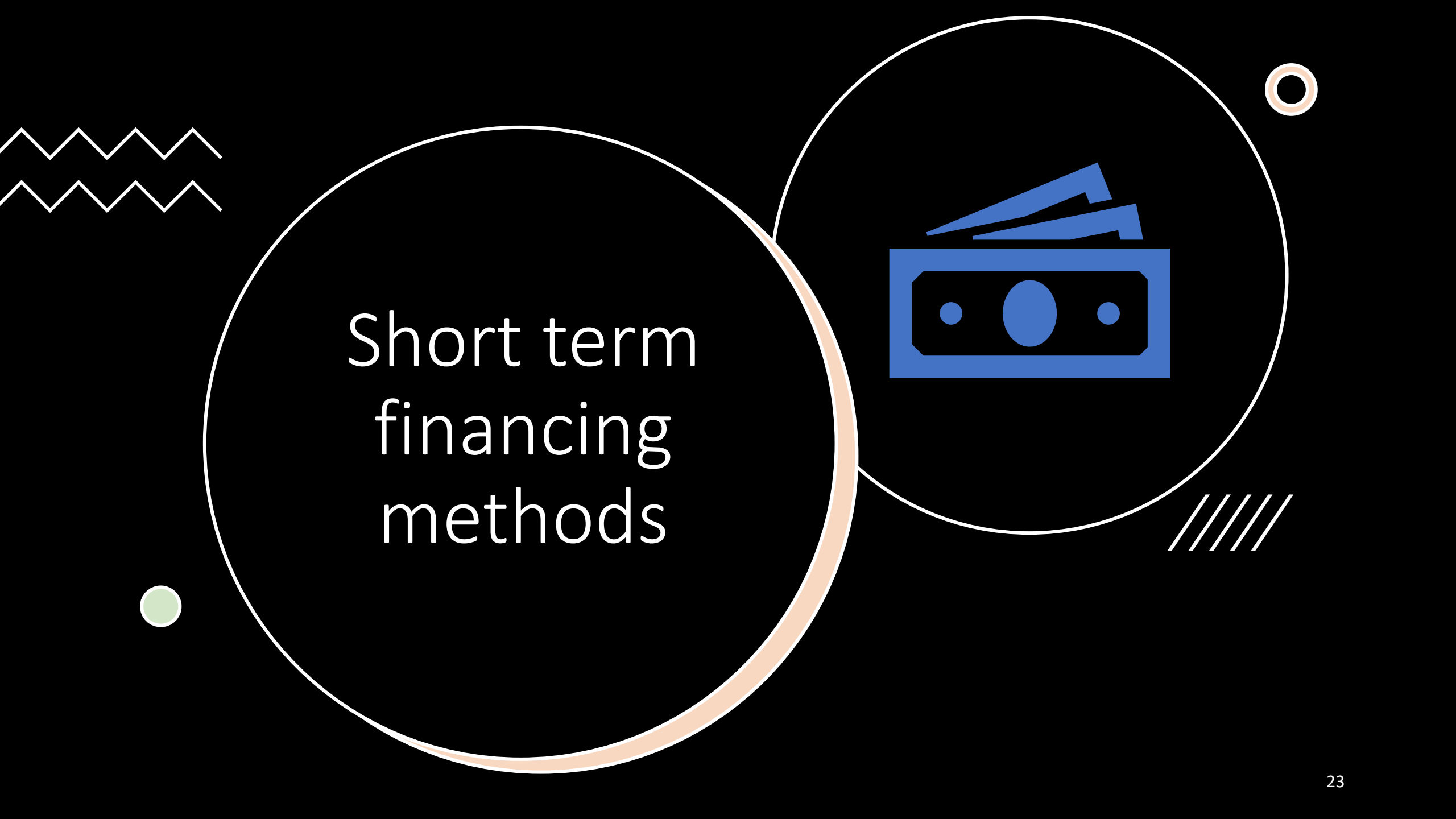
This has led to a preference for open accounts, and banks can provide value through [supply chain finance \(SCF\)](#), taking care of invoices and funding suppliers (and buyers) against invoices. There is a huge potential market in open account transactions, and banks can take advantage of these volumes by promoting their **foreign exchange** and SCF services more consistently.

- [Reverse factoring](#) allows sellers to sell their receivables and/or drafts relating to a particular buyer to a bank at a discount as soon as they are approved by the buyer.

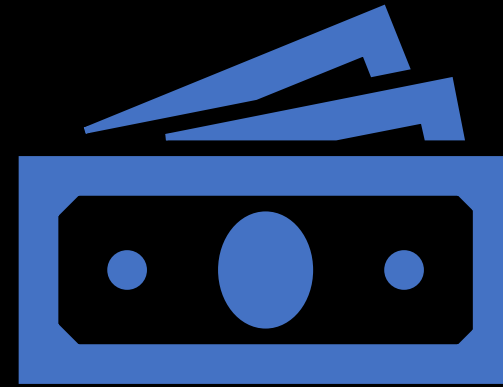
<https://tfig.unece.org/contents/supply-chain-finance.htm>

Comparison of Payment Methods

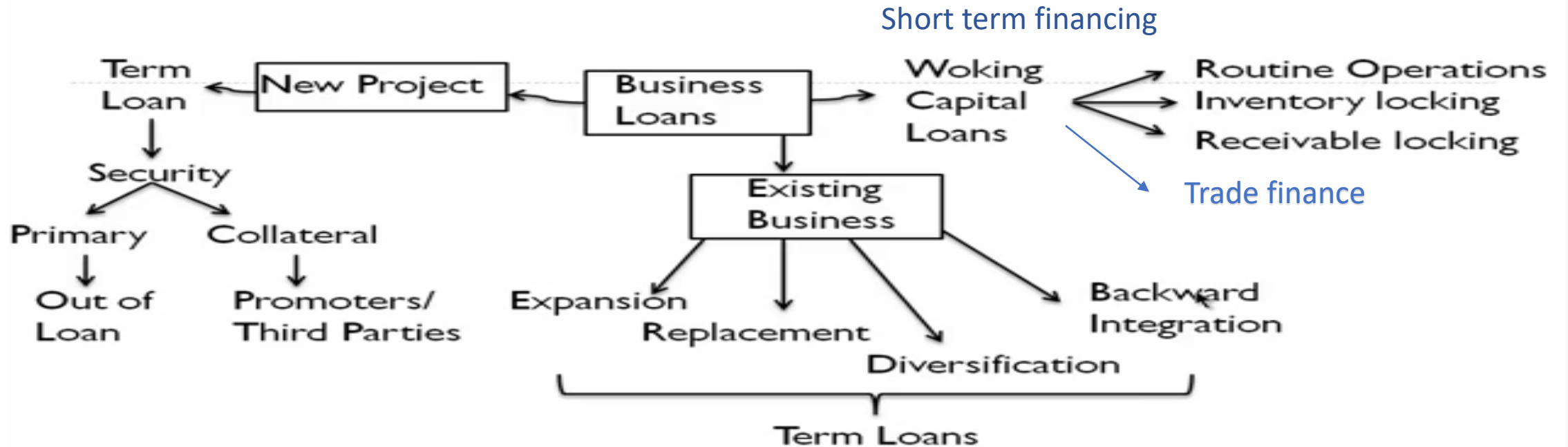
Method	Usual Time of Payment	Goods Available to Buyers	Risk to Exporter	Risk to Importer
Prepayment	Before shipment	After payment	None	Relies completely on exporter to ship goods as ordered
Letter of credit	When shipment is made	After payment	Very little or none, depending on credit terms	Assured shipment made, but relies on exporter to ship goods described in documents
Sight draft; documents against payment	On presentation of draft to buyer	After payment	If draft unpaid, must dispose of goods	Same as above unless importer can inspect goods before payment
Time draft; documents against acceptance	On maturity of drafts	Before payment	Relies on buyer to pay drafts	Same as above
Consignment	At time of sale by buyer	Before payment	Allows importer to sell inventory before paying exporter	None; improves cash flow of buyer
Open account	As agreed	Before payment	Relies completely on buyer to pay account as agreed	None



Short term financing methods



Types of Business loans



Types of Bank Loans for Business



CA N Raja Natarajan

Video:

<https://www.investopedia.com/terms/s/syndicatedloan.asp>

Syndicated lending

- Banks engage in syndicated lending **to diversify their portfolio**, both as a matter of commercial prudence and to comply with capital-adequacy requirements.
- Syndicated lending benefits the borrowers in several ways.
 - It offers a wide range of maturities—from 364 days of revolving credit to 10-year project finance loans.
 - it provides necessary flexibility in loan drawdown during project construction.
 - Bank loans can usually be repaid without penalty, creating flexibility for later refinancing



Commercial Bank loans

- **End-of-period-payment loan.**
- Such a loan agreement requires that the firm pay interest on the loan and pay back the principal in one lump sum at the end of the loan.
- The interest rate may be fixed or variable.
- The **prime rate** is the rate banks charge their most creditworthy customers.

Line of credit, in which a bank agrees to lend a firm any amount up to a stated maximum. This flexible agreement allows the firm to draw upon the line of credit whenever it chooses. Firms frequently use lines of credit to finance seasonal needs.

- The line of credit may be an **uncommitted line of credit**, meaning it is an informal agreement that does not legally bind the bank to provide the funds. As long as the borrower's financial condition remains good, the bank is happy to advance additional funds.
- A **committed line of credit** consists of a written, legally binding agreement that obligates the bank to provide the funds regardless of the financial condition of the firm (unless the firm is bankrupt), as long as the firm satisfies any restrictions in the agreement.

A **bridge loan** is often used to “bridge the gap” until a firm can arrange for long-term financing. For example, a real estate developer may use a bridge loan to finance the construction of a shopping mall. After the mall is completed, the developer will obtain long-term financing.

FEEs

Commitment Fees. Various loan fees charged by banks affect the effective interest rate that the borrower pays.

For example, the **commitment fee associated with a committed line of credit** increases the effective cost of the loan to the firm. The “fee” can really be considered an interest charge under another name.

- **Loan Origination Fee.** Another common type of fee is a **loan origination fee**, which a bank charges to cover credit checks and legal fees.

Suppose that a firm has negotiated a committed line of credit with a stated maximum of \$1 million and an interest rate of 10% (EAR) with a bank. The commitment fee is 0.5% (EAR). At the beginning of the year, the firm borrows \$800,000. It then repays this loan at the end of the year, leaving \$200,000 unused for the rest of the year.

$$\text{Interest on borrowed funds} = 0.10(\$800,000) = \$80,000$$

$$\text{Commitment fee paid on unused portion} = 0.005(\$200,000) = \underline{\$1,000}$$

$$\text{Total cost} = \$81,000$$

Commercial paper

Commercial paper is short-term, unsecured debt used by large corporations that is usually a cheaper source of funds than a short-term bank loan.

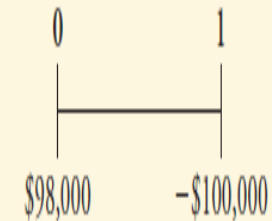
The minimum face value is \$25,000, and most commercial paper has a face value of at least \$100,000. The interest on commercial paper is typically paid by selling it at an initial discount.

Extending the maturity beyond 270 days triggers a registration requirement with the Securities and Exchange Commission (SEC), which increases issue costs and creates a time delay in the sale of the issue. Commercial paper is referred to as either direct paper or dealer paper.

- With **direct paper**, the firm sells the security directly to investors.
- With **dealer paper**, dealers sell the commercial paper to investors in exchange for a spread (or fee) for their services. The spread decreases the proceeds that the issuing firm receives, thereby increasing the effective cost of the paper. Like long-term debt, commercial paper is rated by credit rating agencies.

Solution

Let's put the firm's cash flows on a timeline:



The actual three-month interest rate paid is

$$\frac{100,000}{98,000} - 1 = 2.04\%$$

Expressing this as an EAR gives $1.0204^4 - 1 = 8.42\%$.

Accounts Receivable as Collateral

Firms can use accounts receivable as security for a loan by pledging or factoring.

Pledging of Accounts Receivable.

In a **pledging of accounts receivable** agreement, the lender reviews the invoices that represent the credit sales of the borrowing firm and decides **which credit accounts it will accept as collateral** for the loan, based on its own credit standards. The lender then typically lends the borrower some percentage of the value of the accepted invoices—say, 75%.

If the borrowing firm's customers default on their bills, the firm is still responsible to the lender for the money.

Factoring of Accounts Receivable.

In a **factoring of accounts receivable** arrangement, the firm sells receivables to the lender (i.e., the factor), and the lender agrees to pay the firm the amount due from its customers at the end of the firm's payment period.

Inventory as a collateral

- In a **floating lien**, **general lien**, or **blanket lien (zálogjog)** arrangement, all of the inventory is used to secure the loan. This arrangement is the riskiest setup from the standpoint of the lender because the value of the collateral used to secure the loan dwindles as inventory is sold.
- With a **trust receipts loan** or **floor planning**, distinguishable inventory items are held in a trust as security for the loan.

As these items are sold, the firm remits the proceeds from their sale to the lender in repayment of the loan. The lender will periodically send someone to ensure that the borrower has not sold some of the specified inventory and failed to make a repayment on the loan.

Car dealerships often use this type of secured financing arrangement to obtain the funds needed to purchase vehicles from the manufacturer.

In a **warehouse arrangement**, the inventory that serves as collateral for the loan is stored in a warehouse. A warehouse arrangement is the least risky collateral arrangement from the standpoint of the lender. This type of arrangement can be set up in one of two ways.

- The first method is to use a **public warehouse (közraktár)**.
- The second:, a **field warehouse**, is operated by a third party, but is set up on the borrower's premises in a separate area so that the inventory collateralizing the loan is kept apart from the borrower's main plant.

This type of arrangement is convenient for the borrower but gives the lender the added security of having the inventory that serves as collateral controlled by a third party.

Trade Finance Methods

- Accounts Receivable Financing
 - An exporter that needs funds immediately may obtain a bank loan that is secured by an **assignment of the account receivable**.
- Banker's Acceptance (BA)
 - This is a time draft that is drawn on and accepted by a bank (the importer's bank). The accepting bank is obliged to pay the holder of the draft at maturity.
- Factoring (Cross-Border Factoring)
 - **The accounts receivable are sold to a third party** (the factor), that then assumes all the responsibilities and exposure associated with collecting from the buyer.
- Working Capital Financing
 - Banks may provide short-term loans that finance the working capital cycle, from the purchase of inventory until the eventual conversion to cash. Generally, working capital includes the following: **Current assets** (Cash Accounts receivable within one year, Inventory) **Current liabilities** (Accounts payable due within a year, Short-term debt payments due within one year)

The key players profiled in the trade finance market

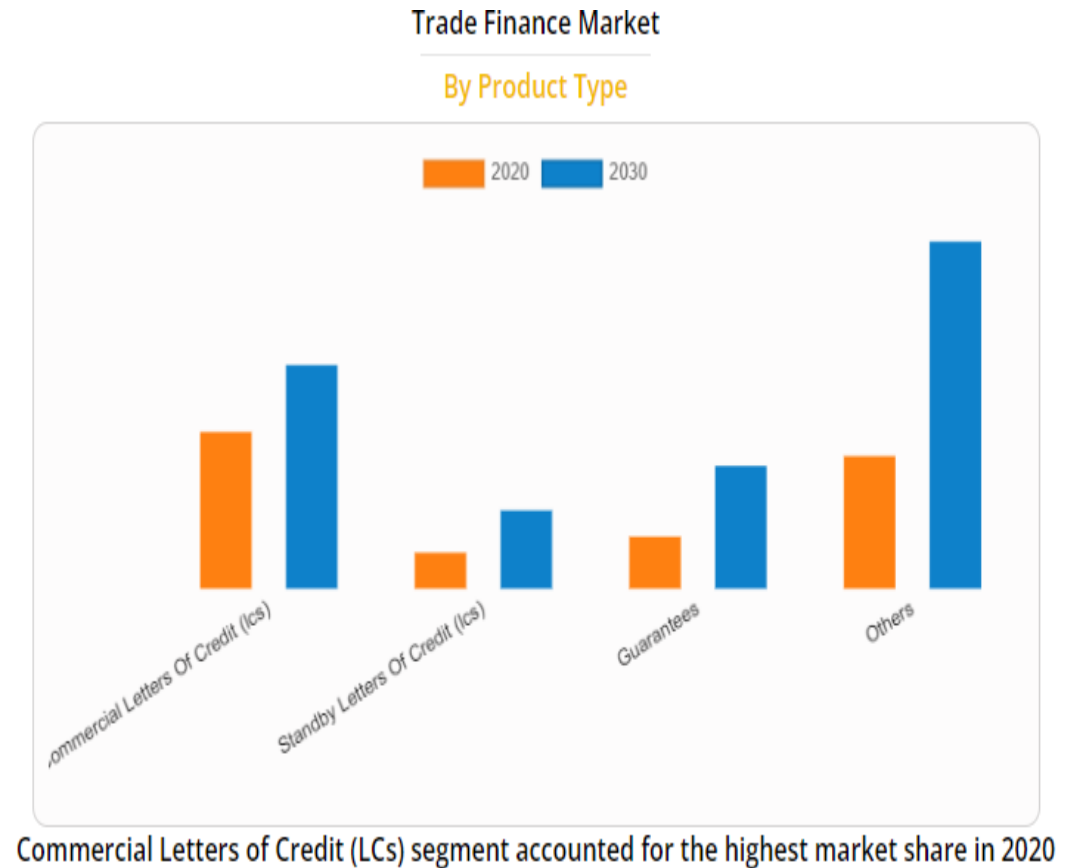
The trade finance market size is segmented into

- banks,
 - trade finance houses, and
 - others.
- The banks segment generated the highest revenue in 2020 and is anticipated to continue the same during the forecast period
 - The **banks act as intermediaries in trade finance ecosystem to provide inter-firm trade credits to buyers, sellers, and other parties involved in the trade.**
 - Support from banks to firm's ability to mitigate payment risk by purchasing trade credit insurance boosts the market growth.
 - Furthermore, banks are accelerating trade finance processes by **transforming their paper-based methods to more efficient and transparent digitized models**, thus becoming the highest service providers in the trade finance market.

- **The key player:**
 - Asian Development Bank,
 - Bank of America Corporation,
 - BNP Paribas S.A.,
 - Citigroup Inc.,
 - Euler Hermes Group,
 - HSBC Holdings PLC,
 - JPMorgan Chase & Co,
 - Mitsubishi UFJ Financial Inc.,
 - The Royal Bank of Scotland Group plc, and
 - Standard Chartered PLC.

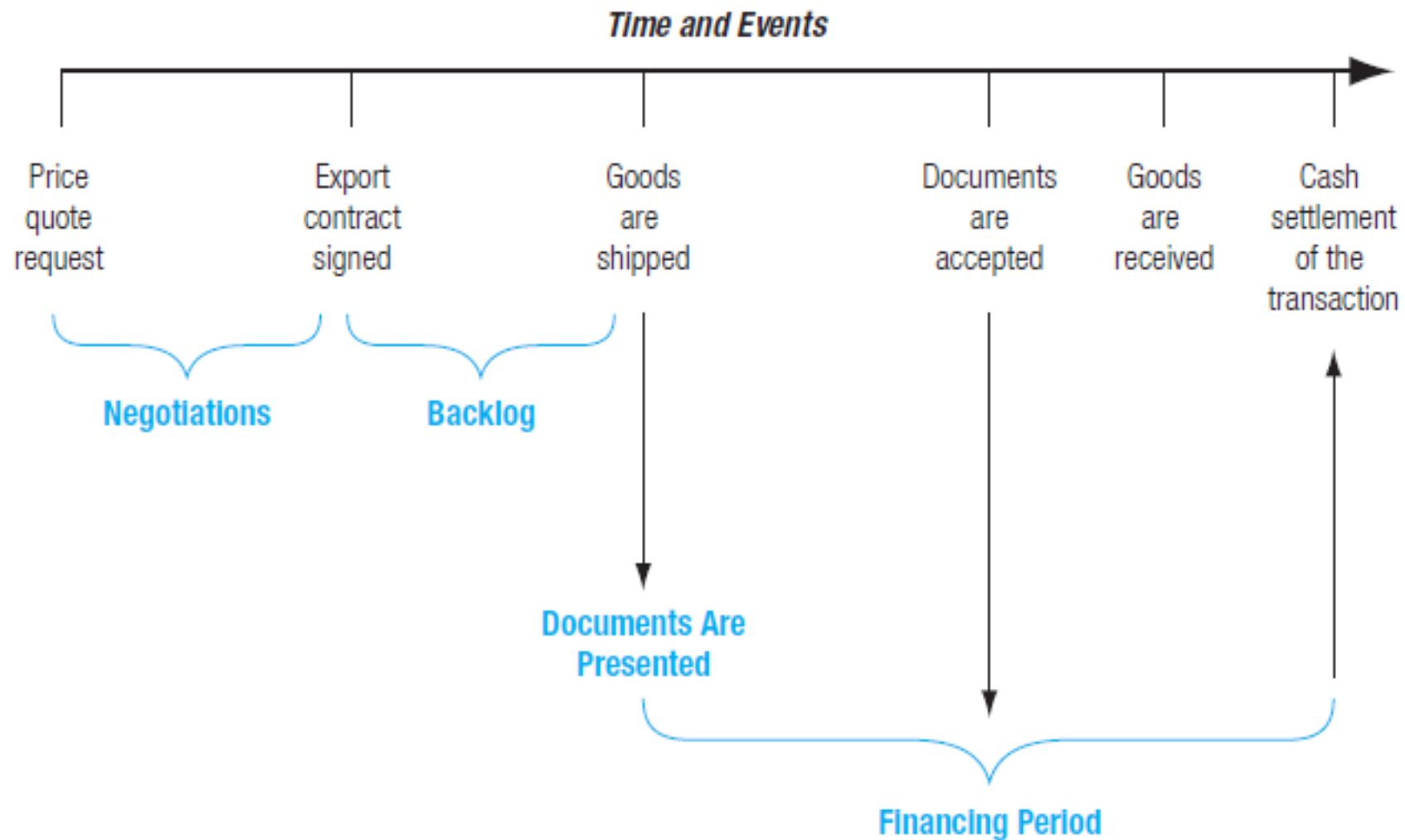
Trade Finance Market

- The global trade finance market size was valued at **\$44.09 billion** in 2020,
- and is projected to **reach \$90.21 billion** by 2030,
- registering a CAGR of **7.4%** from 2021 to 2030.
- The traditional trade is changing the **historic favorable letter of credit** and promissory note and bill of exchange has become less common and **replaced by an increase in open account trade and finance**.
- **The development of technologies** such as Optical Character Recognition (**OCR**) to read container numbers, Radio Frequency Identification (**RFID**) and Quick Response (QR) codes to identify and trace shipments, blockchain, and enhancing digitization of trade documents drive the trade finance market growth.



Source: report published by Allied Market Research, titled, “Trade Finance Market by Product Type, Provider, Application, and End User: Global Opportunity Analysis and Industry Forecast, 2021–2030,”

IBIT 20.5 The Trade Transaction Timeline and Structure



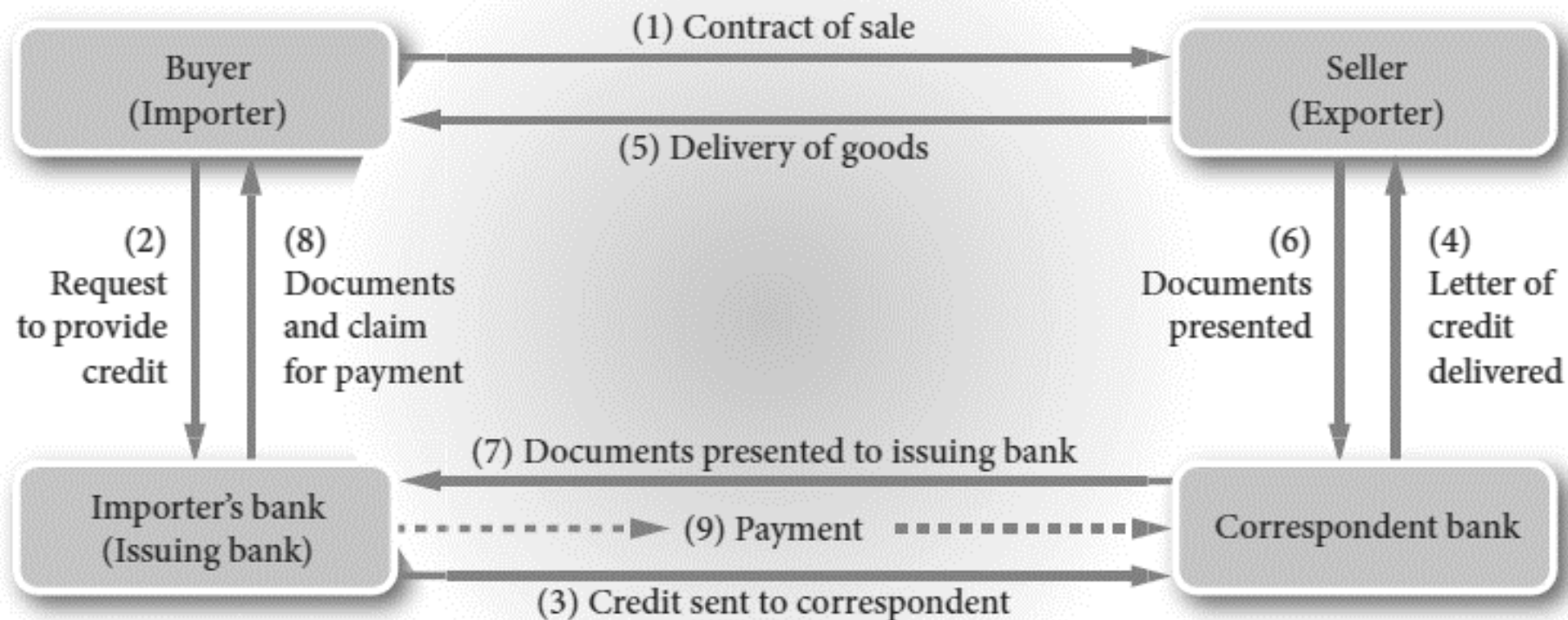
Letters of Credit

- Letters of Credit (L/C)
 - These are issued by a bank on behalf of the importer promising to pay the exporter upon presentation of the shipping documents.
 - The importer pays the issuing bank the amount of the L/C plus associated fees.
 - Commercial or import/export L/Cs are usually irrevocable.
 - Sometimes, the exporter may request that a local bank confirm (guarantee) the L/C.
 - The required documents typically include a draft (sight or time), a commercial invoice, and a bill of lading (receipt for shipment).

Example of an Irrevocable Letter of Credit

	Name of issuing bank
	Address of issuing bank
Name of exporter	
Address of exporter	
 We establish our irrevocable letter of credit: for the account of (<i>importer name</i>), in the amount of (<i>value of exports</i>), expiring (<i>date</i>), available by your draft at (<i>time period</i>) days sight and accompanied by: (any invoices, packing lists, bills of lading, etc., that need to be presented with the letter of credit) Insurance provided by (<i>exporter or importer</i>) covering shipment of (<i>merchandise description</i>) From: (<i>port of shipment</i>) To: (<i>port of arrival</i>)	
	(<i>Authorized Signature</i>)

Documentary Credit Procedure



Letters of Credit

Variations include

- *standby L/Cs*: funded only if the buyer does not pay the seller as agreed upon
- *transferable L/Cs*: the first beneficiary can transfer all or part of the original L/C to a third party
- *assignments of proceeds under an L/C*: the original beneficiary assigns the proceeds to the end supplier.

Real-Life Example of a Letter of Credit

Citibank offers letters of credit for buyers in Latin America, Africa, Eastern Europe, Asia, and the Middle East who may have difficulty obtaining international credit on their own. Citibank's letters of credit help exporters minimize the importer's country risk and the issuing bank's commercial credit risk.

Letters of credit are typically provided within two business days, guaranteeing payment by the confirming Citibank branch.

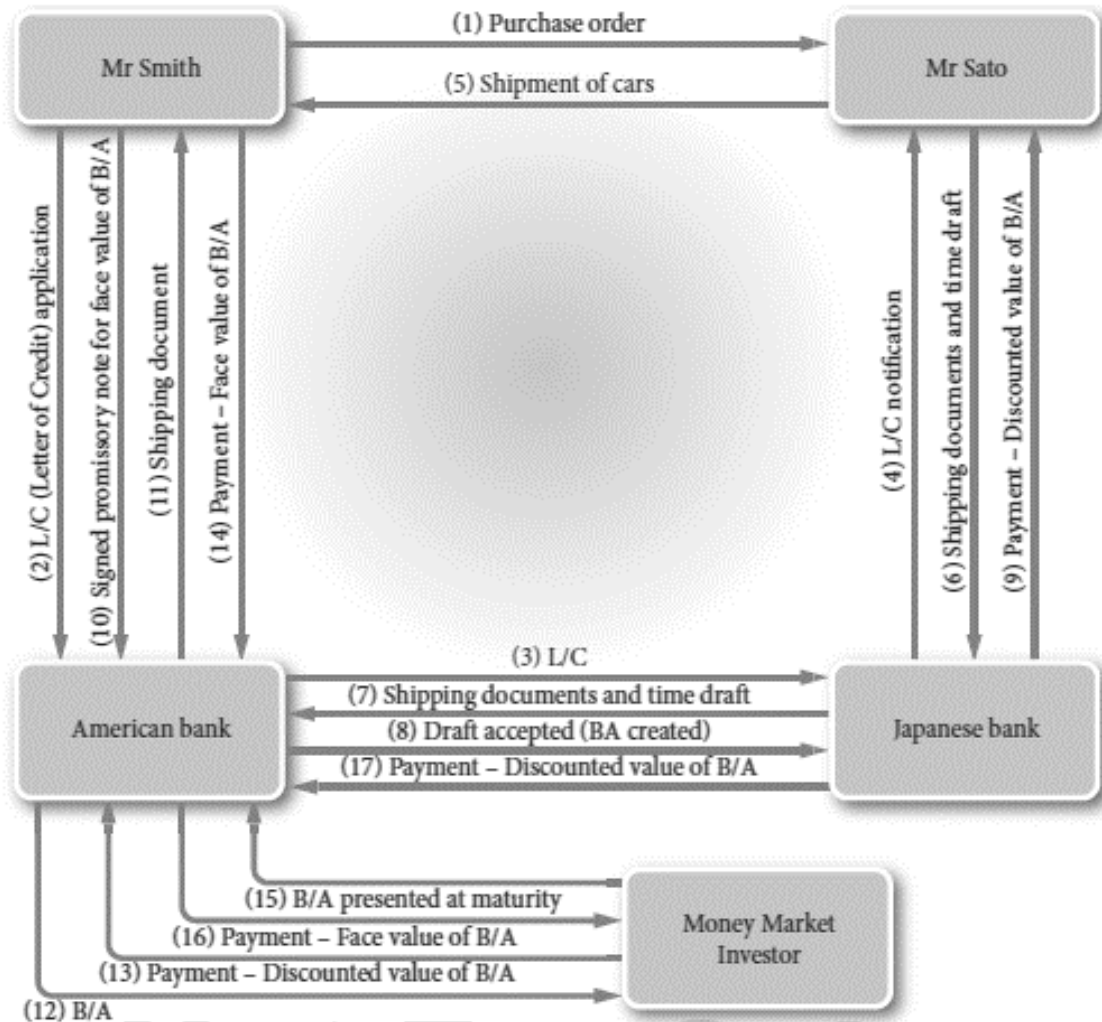
This benefit is especially valuable when a client is located in a potentially unstable economic environment.

Banker's Acceptance (BA)

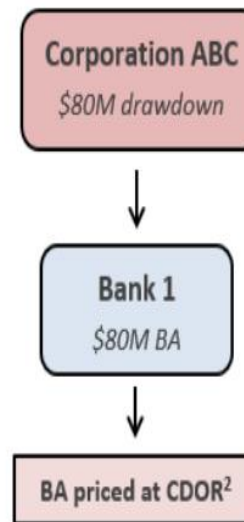
- This is a time draft that is drawn on and accepted by a bank (the importer's bank). The accepting bank is obliged to pay the holder of the draft at maturity.
- If the exporter does not want to wait for payment, it can request that the BA be sold in the money market. **Trade financing is provided by the holder of the BA.**
- **In general, all-in-rates are lower than bank loan rates.** They usually fall between the rates of short-term Treasury bills and commercial papers.
- The bank accepting the drafts charges an *all-in-rate* (interest rate) **that consists of the discount rate plus the acceptance commission.**



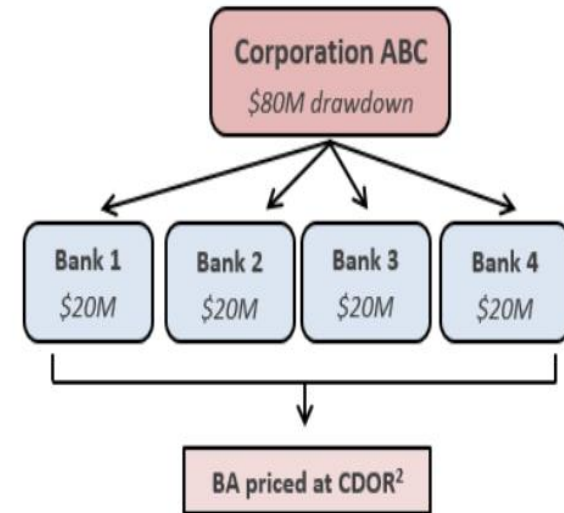
Life Cycle of a Typical Banker's Acceptance



Non-syndicated BA



Syndicated BA



¹ Allocation amounts for syndicated banker's acceptances (BA) issuance are determined by the underlying loan agreement.

² Not all BA facilities reference the Canadian Dollar Offered Rate (CDOR) in their loan agreement.

<https://www.bankofcanada.ca/wp-content/uploads/2018/06/SDP-2018-6.pdf>

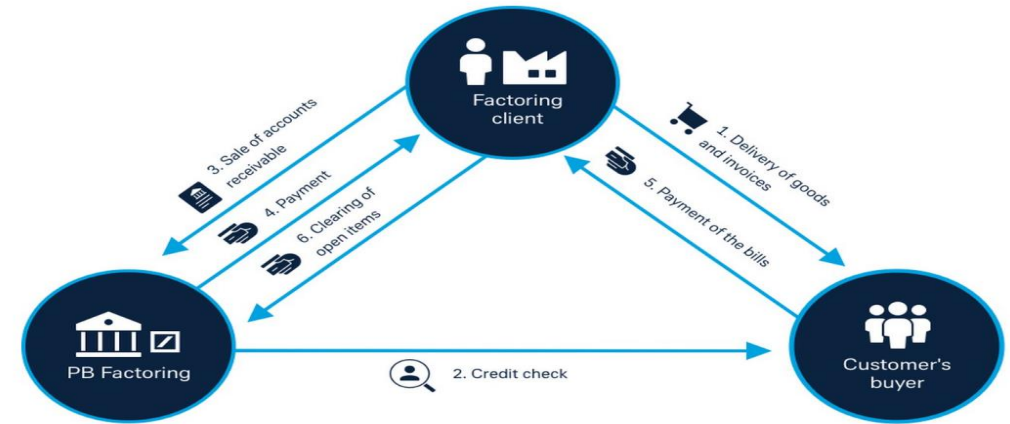
Factoring

A finanszírozási faktorügylet folyamata



You are here: Home > Solutions > Finance factoring

Finance factoring



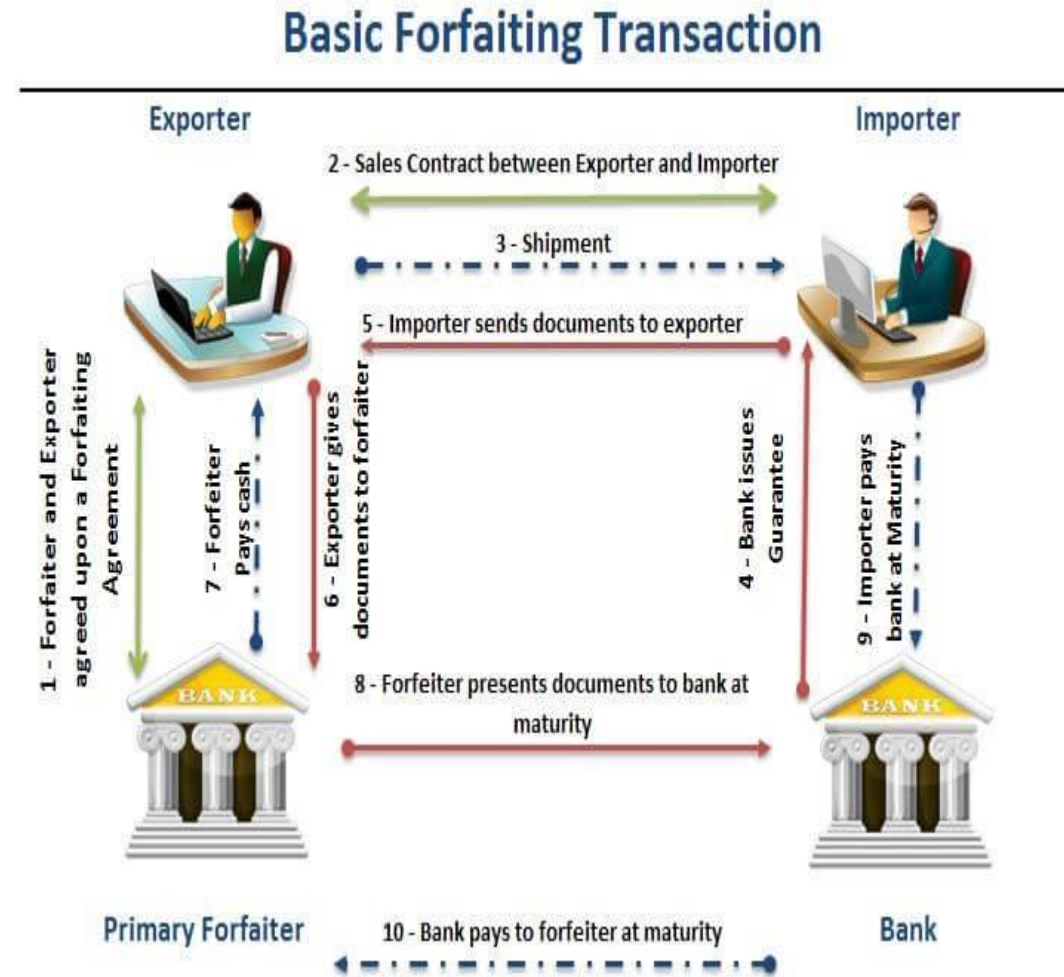
Step 1

The factoring fee in the market right now is basically consists of two factors:

1. for the service and risk-taking of the factor from the factoring fee payable: to the gross invoice amount projected 0.5 to 2.5 percent of that bill will be accounted for once upon submission.
2. and the price of money, interest: interest mainly the development of interest rates and the customer's rating and solvency

Trade Finance Methods (8)

- Medium-Term Capital Goods Financing (**Forfaiting**)
 - The importer issues a **promissory note** to the exporter to pay for its imported capital goods over a period that generally ranges from 3 to 7 years.
 - The exporter then sells the note, without recourse, to a bank (the forfaiting bank).



Difference between Factoring and Forfaiting

Sl. No.	Characteristic	Factoring	Forfaiting
1.	Suitability	For transactions with short-term maturity	For transactions with medium-term maturity period
2.	Recourse	Can be either with or without recourse	Can be without recourse only
3.	Risk	Risk can be transferred to seller	All risks are assumed by the forfaiter
4.	Cost	Cost of factoring is usually borne by the seller	Cost of forfeiting is borne by the overseas buyer (importer)
5.	Coverage	Covers a whole set of jobs at a predetermined price	Structuring and costing is done on a case-to-case basis
6.	Extent of Financing	Only a certain percent of receivables factors is advanced	Hundred percent finance is available
7.	Basis of financing	Financing depends on the credit standing of the exporter	Financing depends on the financial standing of the availing bank
8.	Services	Besides financing a Factor also provides other services such as ledger administration etc.	It is a pure financing arrangement
9.	Exchange fluctuations	No security against exchange rate fluctuations	A forfeiter guards against exchange rate fluctuations for a premium charge
10.	Contract	Between seller and Factor	Between exporter and Forfaiter

Countertrade

- Countertrade
 - These are foreign trade transactions in which the sale of goods to one country is linked to the purchase or exchange of goods from that same country.
 - Common countertrade types include barter, compensation (product buy-back), and counter purchase.
 - The primary participants are governments and MNCs.

Agencies that Motivate International Trade (1)

- Due to the inherent risks of international trade, government institutions and the private sector offer various forms of export credit, export finance, and guarantee programs to reduce risk and stimulate foreign trade.

Export-Import Bank of the US (Ex-Imbank)

- This US government agency aims to create jobs by financing and facilitating the export of US goods and services and maintaining the competitiveness of US companies in overseas markets.
- It offers guarantees of commercial loans, direct loans, and export credit insurance.

Private Export Funding Corporation (PEFCO)

- PEFCO is a private corporation that is owned by a consortium of commercial banks and industrial companies.
- In cooperation with Ex-Imbank, PEFCO provides medium- and long-term fixed-rate financing for foreign buyers through the issuance of long-term bonds.

Agencies that Motivate International Trade (4)

- In the UK export services are provided by **the Export Credit Guarantee Department (ECGD)**. For a fee, it provides a guarantee of payment to UK exporters on transactions ranging from £25,000 to over £100 million.
- As with all government agencies a moral as well as a practical rationale is required to justify their role. On the virtuous side, Third World countries termed Heavily Indebted Poor Countries (HIPC – this is an official IMF term) often have poor credit records.
- There is also, surprisingly, a less virtuous argument. The true purpose of the ECGD is to finance and thereby subsidize domestic industry with little real concern for the recipient country. Arms sales (some 24 per cent of business) are particularly difficult to justify as necessary for a developing country.



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EXPORT EXPERT

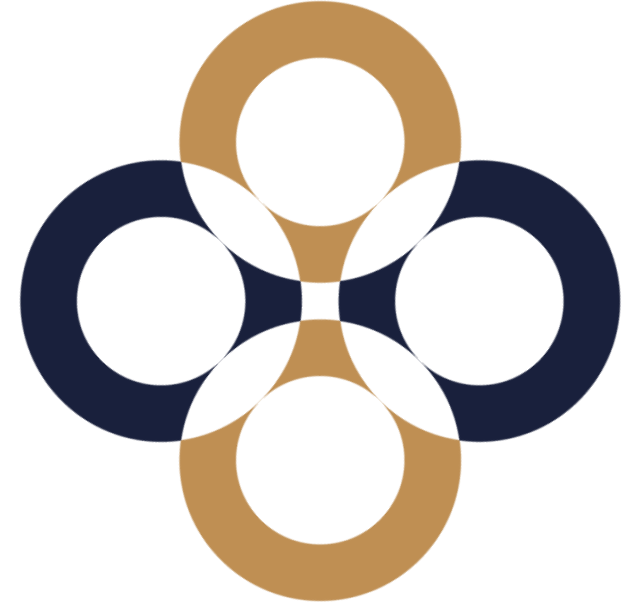
<https://exim.hu/en/index.php#products>

Thank you for your
attention !

International Cash Management

International Corporate Finance
2023.10.12

Dr. Judit Burucs



Topics

- Cash cycle
- Trade credit
- Cash Management
- Cash Pooling
- Multinational company cash management

The Cash Cycle

The level of working capital reflects the length of time between when cash goes out of a firm at the beginning of the production process and when it comes back in.

1. A company first buys inventory from its suppliers, in the form of either raw materials or finished goods. **A firm typically buys its inventory on credit**, which means that the firm does not have to pay cash immediately at the time of purchase.
2. The firm's **cash cycle** is the length of time between when the firm pays cash to purchase its initial inventory and when it receives cash from the sale of the output produced from that inventory.

Some practitioners measure the cash cycle by calculating **the cash conversion cycle (CCC)** that is defined as
CCC = Accounts Receivable Days + Inventory Days - Accounts Payable Days

$$CCC = \text{Accounts Receivable Days} + \text{Inventory Days} - \text{Accounts Payable Days} \quad (26.1)$$

where

$$\text{Accounts Receivable Days} = \frac{\text{Accounts Receivable}}{\text{Average Daily Sales}}$$

$$\text{Inventory Days} = \frac{\text{Inventory}}{\text{Average Daily Cost of Goods Sold}}$$

$$\text{Accounts Payable Days} = \frac{\text{Accounts Payable}}{\text{Average Daily Cost of Goods Sold}}$$

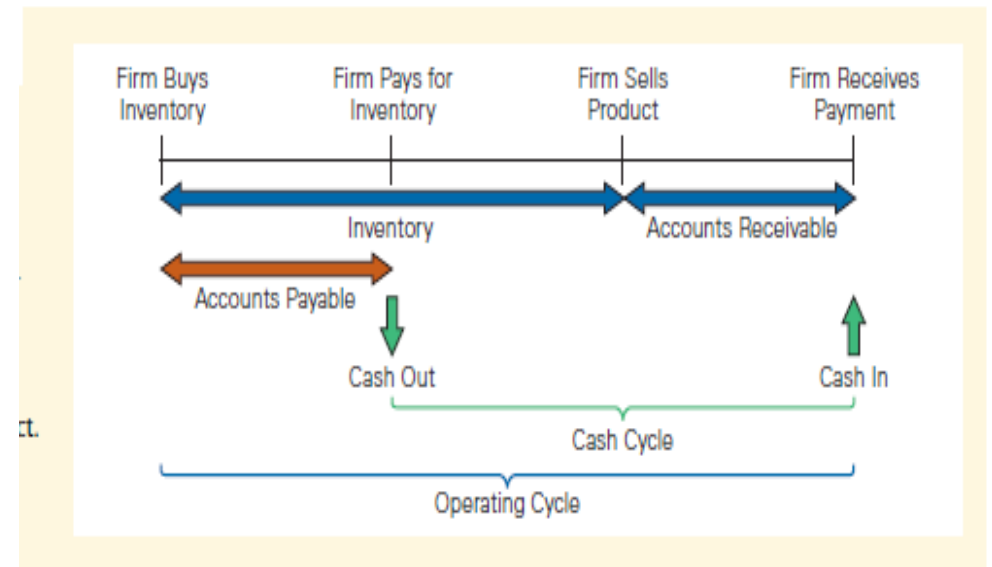


TABLE 26.1**Working Capital in Various Industries (2015)**

Company	Ticker	Industry	Accounts Receivable Days	Inventory Days	Accounts Payable Days	CCC
Molson Coors Brewing	TAP	Brewing	48	37	184	−99
Pepsico	PEP	Beverages	40	40	164	−84
Verizon Communications	VZ	Telecommunications	37	8	117	−72
Apple	AAPL	Computer Hardware	20	7	76	−49
Bristol-Myers Squibb	BMJ	Pharmaceuticals	72	126	223	−25
Southwest Airlines	LUV	Airlines	9	9	32	−14
Amazon.com	AMZN	Internet Retail	19	45	73	−9
The New York Times	NYT	Publishing	41	0	49	−8
Chipotle Mexican Grill	CMG	Restaurants	2	2	10	−6
The Kroger Co.	KR	Grocery Stores	4	24	23	5
Wal-Mart Stores	WMT	Superstores	4	47	39	12
Microsoft	MSFT	Software	62	33	77	18
Macy's	SHLD	Department Stores	4	147	133	18
FedEx	FDX	Air Freight	43	5	22	26
Starbucks	SBUX	Restaurants	12	30	16	26
Nordstrom	JWN	Department Stores	29	79	62	46
Nike	NKE	Footwear	36	91	41	86
Brown-Forman	BFB	Distillers and Vintners	72	440	90	422
Lennar	LEN	Homebuilding	3	463	23	443
Sotheby's	BID	Auction Services	393	169	117	445
Tiffany & Co.	TIF	Luxury Goods	17	546	66	497
Major U.S. Firms (value-weighted Average)			48	71	76	43

Source: www.capitaliq.com

The main sources of short-term financing

1. **Working Capital Financing** (commercial bank loans- secured and unsecured)

Banks may provide short-term loans that finance the working capital cycle, from the purchase of inventory until the eventual conversion to cash.

Generally, working capital includes the following:. **Current assets** (Cash Accounts receivable within one year, Inventory) **Current liabilities** (Accounts payable due within a year, Short-term debt payments due within one year)

2. Line of Credit

3. Commercial paper, a specific type of promissory note,

4. **Trade credit, (sale discount)**

A firm customarily buys its supplies and materials on credit from other firms, recording the debt as an account payable. This trade credit, is the largest single category of short-term credit.

Trade Finance Methods

- Accounts Receivable Financing
 - An exporter that needs funds immediately may obtain a bank loan that is secured by an **assignment of the account receivable**.
- Banker's Acceptance (BA)
 - This is a time draft that is drawn on and accepted by a bank (the importer's bank).
 - The accepting bank is obliged to pay the holder of the draft at maturity.
 - A banker's acceptance is a short-term issuance by a bank that guarantees payment at a later time.
 - A banker's acceptance is often used in importing and exporting, with the importer's bank guaranteeing payment to the exporter.
 - Applying for a banker's acceptance is similar to applying for a short-term, fixed-rate loan; the borrower goes through a credit check and sometimes extra underwriting processes.
 - Similar to buying a Treasury bill, an investor on the secondary market might buy the acceptance at a discounted price, but still get the full value at the time of maturity.
- Factoring (Cross-Border Factoring)
 - **The accounts receivable are sold to a third party** (the factor), that then assumes all the responsibilities and exposure associated with collecting from the buyer.

Forfaiting

Forfaiting

- Exporters sell their trade receivables from the importers to a third party for cash. Now third party receives payments from importer this third party known as Forfeiter. This process is called as Forfaiting

Key points

- Excellent source of funds
- For both long & short term
- Value of forfaiting <\$2,50,000
- Guarantee from importers bank in form of "Letter of credit"
- Exporters receivable in form of negotiable instrument

How it Works?

Exporter approaches forfeiter(Bank)

Exporter incorporates amount of discount rate in deal price

Importer agrees & signs the deal

Exporter delivers goods to importer

Exporter issues documents & bills

After verification bank pays to importer

On maturity bank collects payment from bank

Trade credit

The Trade credit is, a loan from the selling firm to its customer.

When a firm allows a customer to pay for goods at some date later than the date of the purchase, it creates an account receivable for the firm and an account payable for customer.

Accounts receivable represent the credit sales for which a firm has yet to receive payment.

If a supplier offers its customers **terms of “Net 30,”** payment is not due until 30 days from the date.

The terms “2/10, Net 30” mean that the buying firm will receive a 2% discount if it pays for the goods within 10 days;

Cost of Trade Credit.

- The price discount represents an interest rate.
- Often, firms offer favorable interest rates on trade credit as a price discount to their customers.

How do we compute the interest rate on trade credit?

Suppose a firm sells a product for \$100 but offers its customer **terms of 2/10, Net 30**.

- The customer doesn't have to pay anything for the first 10 days, so it effectively has a zero-interest loan for this period.
- If the customer takes advantage of the discount and pays within the 10-day discount period, the customer pays only \$98 for the product.

The cost of the discount to the selling firm is equal to the discount percentage times the selling price.

In this case, it is $0.02 * \$100$, or \$2.

Rather than pay within 10 days, the customer has the option to use the \$98 for an additional 20 days ($30 - 10 = 20$).

The interest rate for the 20-day term of the loan is $\$2/\$98 = 2.04\%$.

With a 365-day year, this rate over 20 days corresponds to an **effective annual rate of** 2

$$EAR = (1.0204)^{365/20} - 1 = 44.6\%$$

Problem

Your firm purchases goods from its supplier on terms of 1/15, Net 40.

What is the **effective annual cost to your firm** if it chooses not to take advantage of the trade discount offered?

Problem

Your firm purchases goods from its supplier on terms of 1/15, Net 40. What is the effective annual cost to your firm if it chooses not to take advantage of the trade discount offered?

Solution

Because the discount is 1%, for a \$100 purchase your firm must pay either \$99 in 15 days or \$100 in 40 days. Given the difference of 25 days ($40 - 15$), these terms correspond to an effective annual rate of $(100/99)^{365/25} - 1 = 15.8\%$.

Problems

1. Assume the credit terms offered to your firm by your suppliers are 2.5/5, Net 30. Calculate the cost of the trade credit if your firm does not take the discount and pays on day 30.
2. Your supplier offers terms of 1.4/12, Net 45. What is the effective annual cost of trade credit if you choose to forgo the discount and pay on day 45?

Trade Credit Versus Standard Loans

Benefits of Trade Credit.

- 1) Trade credit is simple and convenient to use, and it therefore has lower transaction costs than alternative sources of funds.
For example, no paperwork must be completed, as would be the case for a loan from a bank.
- 2) It is a flexible source of funds, and can be used as needed.
- 3) It is sometimes the only source of funding available to a firm.
Providing financing at below-market rates is an indirect way to lower prices for only certain customers. Consider, for example, an automobile manufacturer. **Rather than lower prices on all cars**, the financing division may offer specific credit terms that are attractive.
Second, because a supplier may have an ongoing business relationship with its customer, **it may have more information about the credit quality of the customer** than a traditional outside lender such as a bank.

Collection Float is the amount of time it takes for a firm to be able to use funds after a customer has paid for its goods.

Firms can reduce their working capital needs by reducing their collection float.

Collection float is determined by three factors:

- **Mail float:** How long it takes the firm to receive the check after the customer has mailed it
- **Processing float:** How long it takes the firm to process the check and deposit it in the bank
- **Availability float:** How long it takes before the bank gives the firm credit for the funds

Receivable and payable management

Receivable management

1) Establishing a credit policy involves three steps that we will discuss in turn:

1. Establishing credit standards
2. Establishing credit terms
3. Establishing a collection policy

2) After establishing a credit policy, **a firm must monitor its accounts receivable to analyze whether its credit policy is working effectively.** Two tools that firms use to monitor the accounts receivable are

- the accounts receivable days (or average collection period)
- the aging schedule.

TABLE 26.2

Aging Schedules

Days Outstanding	Number of Accounts	Percentage of Accounts (%)	Amount Outstanding (\$)	Percentage Outstanding (%)
1-15	220	38.6	530,000	33.1
16-30	190	33.3	450,000	28.1
31-45	80	14.0	350,000	21.9
46-60	60	10.5	200,000	12.5
60+	20	3.5	70,000	4.4
	570	100.0	1,600,000	100.0

Payable management

- A firm should choose to borrow using accounts payable only if trade credit is the cheapest source of funding. The cost of the trade credit depends on the credit terms.
- The cost of forgoing the discount is also higher with a shorter loan period.
- When a company has a choice between trade credit from two different suppliers, it should take the less expensive alternative.
- In addition, a firm should always pay on the latest day allowed.

Cost of Trade Credit with Stretched Accounts Payable

Problem

What is the effective annual cost of credit terms of 1/15, Net 40, if the firm stretches the accounts payable to 60 days?

Solution

The interest rate per period is $\$1/\$99 = 1.01\%$. If the firm delays payment until the sixtieth day, it has use of the funds for 45 days beyond the discount period. There are $365/45 = 8.11$ 45-day periods in one year. Thus, the effective annual cost is $(1.0101)^{8.11} - 1 = 8.49\%$.

Aging Schedules

Problem

Financial Training Systems (FTS) bills its accounts on terms of 3/10, Net 30. The firm's accounts receivable include \$100,000 that has been outstanding for 10 or fewer days, \$300,000 outstanding for 11 to 30 days, \$100,000 outstanding for 31 to 40 days, \$20,000 outstanding for 41 to 50 days, \$10,000 outstanding for 51 to 60 days, and \$2000 outstanding for more than 60 days. Prepare an aging schedule for FTS.

Problem

Financial Training Systems (FTS) bills its accounts on terms of 3/10, Net 30. The firm's accounts receivable include \$100,000 that has been outstanding for 10 or fewer days, \$300,000 outstanding for 11 to 30 days, \$100,000 outstanding for 31 to 40 days, \$20,000 outstanding for 41 to 50 days, \$10,000 outstanding for 51 to 60 days, and \$2000 outstanding for more than 60 days. Prepare an aging schedule for FTS.

Solution

With the available information, we can calculate the aging schedule based on dollar amounts outstanding.

Days Outstanding	Amount Outstanding (\$)	Percentage Outstanding (%)
1–10	100,000	18.8
11–30	300,000	56.4
31–40	100,000	18.8
41–50	20,000	3.8
51–60	10,000	1.9
60+	2,000	0.3
	532,000	100.0

Inventory and Cash management

Costs of Holding Inventory

the direct costs associated with inventory into three categories are :

1. *Acquisition costs* are the costs of the inventory itself over the period being analyzed (usually one year).
2. *Order costs* are the total costs of placing an order over the period being analyzed.
3. *Carrying costs* include storage costs, insurance, taxes, spoilage, obsolescence, and the opportunity cost of the funds tied up in the inventory. Minimizing these total costs involves some trade-offs.

Motivation for Holding Cash

There are three reasons why a firm holds cash:

1. to meet its day-to-day needs
2. To compensate for the uncertainty associated with its cash flows
3. To satisfy bank requirements

Short-term financial planning

The first step in short-term financial planning is to forecast the company's future cash flows. This exercise has two distinct objectives.

- First, a company forecasts its cash flows to determine whether it will have surplus cash or a cash deficit for each period.
- Second, management needs to decide whether that surplus or deficit is temporary or permanent.

Firms typically require short-term financing for three reasons:

- seasonalities,
- negative cash flow shocks, and (not expected events)
- positive cash flow shocks (more sale than expected, more material)

TABLE 27.2
SPREADSHEET

Projected Financial Statements for Springfield Snowboards, 2019, Assuming Seasonal Sales

	Quarter	2018Q4	2019Q1	2019Q2	2019Q3	2019Q4
Income Statement (\$000)						
1 Sales		10,909	4,000	2,000	2,000	12,000
2 Cost of Goods Sold		(7,091)	(2,600)	(1,300)	(1,300)	(7,800)
3 Selling, General and Administrative		(773)	(450)	(350)	(350)	(850)
4 EBITDA		3,045	950	350	350	3,350
5 Depreciation		(455)	(500)	(500)	(500)	(500)
6 EBIT		2,591	450	(150)	(150)	2,850
7 Taxes		(907)	(158)	53	53	(998)
8 Net Income		1,684	293	(98)	(98)	1,853
Statement of Cash Flows						
9 Net Income			293	(98)	(98)	1,853
10 Depreciation			500	500	500	500
11 Changes in Working Capital						
12 Accounts Receivable			2,073	600	—	(3,000)
13 Inventory			(650)	(1,950)	(1,950)	4,550
14 Accounts Payable			48	—	—	—
15 Cash from Operating Activities			2,263	(948)	(1,548)	3,903
16 Capital Expenditures			(500)	(500)	(500)	(500)
17 Other Investment			—	—	—	—
18 Cash from Investing Activities			(500)	(500)	(500)	(500)
19 Net Borrowing			—	—	—	—
20 Dividends			—	—	—	—
21 Capital Contributions			—	—	—	—
22 Cash from Financing Activities			—	—	—	—
23 Change in Cash and Equivalents (15 + 18 + 22)			1,763	(1,448)	(2,048)	3,403

Financing policy and Matching principle

The **matching principle** states that short-term needs should be financed with short-term debt and long-term needs should be financed with long-term sources of funds.³

- **Permanent working capital** is the amount that a firm must keep invested. The matching principle indicates that the firm should finance this permanent investment in working capital with long-term sources of funds. Such sources have lower transaction costs than short-term sources of funds, which would have to be replaced more often.
- This **temporary working capital** is the difference between the actual level of investment in short-term assets and the permanent working capital investment. It represents a short-term need, the firm should finance this portion of its investment with short-term financing.

Aggressive financing policy : financing part or all of the **permanent working capital with short-term debt**.

An ultra-aggressive policy would involve financing even some of the plant, property, and equipment with short-term sources of funds. **When the yield curve is upward sloping**, the interest rate on short-term debt is lower than the rate on long-term debt.

Conservative financing policy : a firm could finance its short-term needs with long-term debt.

For example, when following such a policy, a firm would use long-term sources of funds to finance its fixed assets, permanent working capital, and some of its seasonal needs. The firm would use short-term debt very sparingly to meet its peak seasonal needs.

Cash Management

In the Modigliani-Miller setting, the level of cash is irrelevant.

With perfect capital markets, a firm is able to raise new money instantly at a fair rate, so it can never be short of cash. Similarly, the firm can invest excess cash at a fair rate to earn an NPV of zero.

In the real world, of course, markets are not perfect

Motivation for Holding Cash

- To meet its day-to-day needs: **transactions balance**

The amount of cash a firm needs to be able to pay its bills is sometimes referred to as a **transactions balance**.

- To compensate for the uncertainty associated with its cash flows (**Precautionary Balance**).

The size of this balance depends on the degree of uncertainty surrounding a firm's cash flows. The more uncertain future cash flows are, the harder it is for a firm to predict its transactions need, so the larger the precautionary balance must be

- To satisfy bank requirements: **compensating balance**

A firm's bank may require it to hold a **compensating balance** in an account at the bank as compensation for services that the bank performs. Compensating balances are typically deposited in accounts that either earn no interest or pay a very low interest rate.

TABLE 26.3**Money Market Investment Options**

Investment	Description	Maturity	Risk	Liquidity
Treasury Bills	Short-term debt of the U.S. government.	Four weeks, three months (91 days), six months (182 days), or one year when newly issued	Default risk free.	Very liquid and marketable.
Certificates of Deposit (CDs)	Short-term debt issued by banks, minimum denomination of \$100,000.	Varying maturities up to one year	If the issuing bank is insured by the FDIC, any amount up to \$250,000 is free of default risk because it is covered by the insurance. Any amount in excess of \$250,000 is not insured and is subject to default risk.	Unlike CDs purchased by individuals, these CDs sell on the secondary market, but are less liquid than Treasury bills.
Repurchase Agreements	Essentially a loan arrangement wherein a securities dealer is the "borrower" and the investor is the "lender"; the investor buys securities, such as U.S. Treasury bills, from the securities dealer, with an agreement to sell the securities back to the dealer at a later date for a specified higher price.	Very short term, ranging from overnight to approximately three months in duration	The security serves as collateral for the loan, and therefore the investor is exposed to very little risk. However, the investor needs to consider the creditworthiness of the securities dealer when assessing the risk.	No secondary market for repurchase agreements.
Banker's Acceptances	Drafts written by the borrower and guaranteed by the bank on which the draft is drawn, typically used in international trade transactions; the borrower is an importer who writes the draft in payment for goods.	Typically one to six months	Because both the borrower and a bank have guaranteed the draft, there is usually very little risk.	When the exporter receives the draft, he may hold it until maturity and receive its full value or he may sell the draft at a discount prior to maturity.
Commercial Paper	Short-term, unsecured debt issued by large corporations. The minimum denomination is \$25,000, but most commercial paper has a face value of \$100,000 or more.	Typically one to six months	Default risk depends on the creditworthiness of the issuing corporation.	No active secondary market, but issuer may repurchase commercial paper.
Short-Term Tax Exempts	Short-term debt of state and local governments; these instruments pay interest that is exempt from federal taxation, so their pre-tax yield is lower than that of a similar-risk, fully taxable investment.	Typically one to six months	Default risk depends on the creditworthiness of the issuing government.	Moderate secondary market.

problem

4. The Greek Connection had sales of \$32 million and a cost of goods sold of \$12.8 million in 2015. A simplified balance sheet for the firm appears below:

THE GREEK CONNECTION

Balance Sheet

As of December 31, 2015

(in \$ thousand)

Assets		Liabilities and Equity	
Cash	\$ 1,954	Accounts payable	\$ 1,300
Accounts receivable	4,414	Notes payable	1,000
Inventory	1,414	Accruals	1,220
Total current assets	\$ 7,782	Total current liabilities	\$ 3,520
		Long-term debt	\$ 3,000
Net plant, property, and equipment	\$ 8,500	Total liabilities	\$ 6,520
Total assets	\$ 16,282	Common equity	\$ 9,762
		Total liabilities and equity	\$ 16,282

- Calculate The Greek Connection's net working capital in 2015.
- Calculate the cash conversion cycle of The Greek Connection in 2015.
- The industry average accounts receivable days is 30 days. What would the cash conversion cycle for The Greek Connection have been in 2015 had it matched the industry average for accounts receivable days?

Cash conversion cycle

CCC = Accounts Receivable Days + Inventory Days - Accounts Payable Days = 30days+

cash conversion cycle (CCC) that is defined as

CCC = Accounts Receivable Days + Inventory Days - Accounts Payable Days

Some practitioners measure the cash cycle by calculating **the cash conversion cycle (CCC) that is defined as**

CCC = Accounts Receivable Days + Inventory Days - Accounts Payable Days

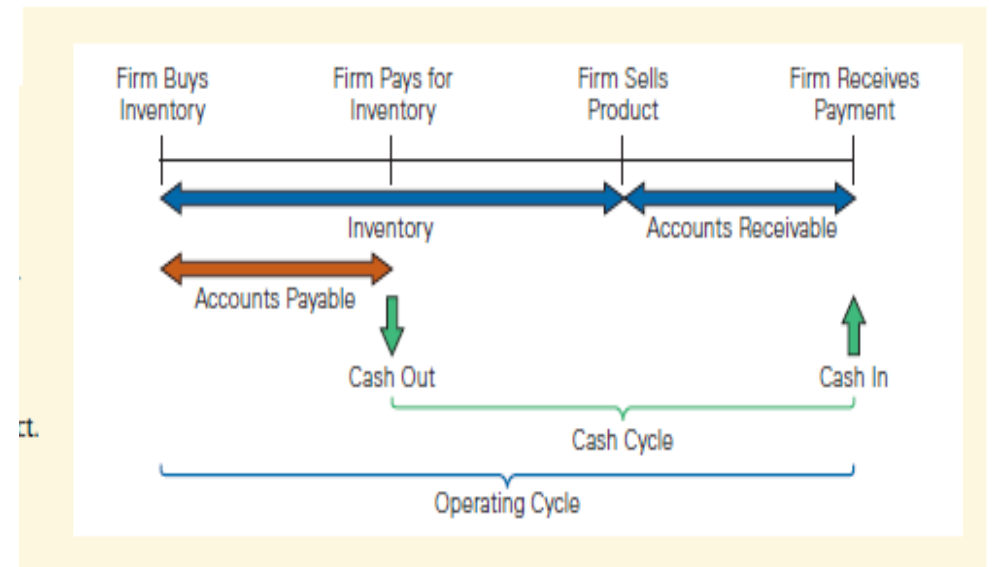
$$CCC = \text{Accounts Receivable Days} + \text{Inventory Days} - \text{Accounts Payable Days} \quad (26.1)$$

where

$$\text{Accounts Receivable Days} = \frac{\text{Accounts Receivable}}{\text{Average Daily Sales}}$$

$$\text{Inventory Days} = \frac{\text{Inventory}}{\text{Average Daily Cost of Goods Sold}}$$

$$\text{Accounts Payable Days} = \frac{\text{Accounts Payable}}{\text{Average Daily Cost of Goods Sold}}$$



What is cash pooling?

The **principle of cash pooling** is to centralise cash flow management with the holding company and to balance the bank accounts of all the subsidiaries.

Indeed, some entities of a group may be experiencing **cash deficits** and have to resort to rather expensive **short-term cash flow financing** options (high-interest loans, overdrafts that are subject to fees or **bank commissions** and so on), even though the group is financially healthy on the whole. At the same time, other entities within the group may have cash surpluses that just sit in their accounts and yield little.

Cash pooling solves this problem:

- If there is a **cash deficit**, cash pooling lets the group help those subsidiaries in difficulty and limit banking charges by having just one bank account in deficit
- If there is a **cash surplus**, cash pooling lets the group capitalise on all of the profits from each subsidiary by making better investments and increasing interest

Assume the credit terms offered to your firm by your suppliers are 2.5/5, Net 30. Calculate the cost of the trade credit if your firm does not take the discount and pays on day 30.

6. Your supplier offers terms of 1.4/12, Net 45. What is the effective annual cost of trade credit if you choose to forgo the discount and pay on day 45?

There are two main methods of cash pooling:

- Physical cash pooling:** this is when a real cash amount is swept to the master account; it is often automated by the bank
- Notional cash pooling:** this is when cash is centralised virtually by merging interest statements, which allows each entity to operate its own credit lines without funds being moved from the master account; it is generally only used by very large corporate groups

Some groups use a combination of the two methods, which is referred to as hybrid cash pooling. This is often the approach taken by large international groups.

International cash pooling

When a company has **subsidiaries** around the world, the cash pooling methods that are possible in the United Kingdom are not always an option.

Rules surrounding cash pooling differ across countries, particularly with regard to remuneration and interest rates.

Some foreign laws do not allow intra-group treasuries to be centralised, while others enforce different rules from those applicable in the UK. In Poland, for example, the law prohibits cash pooling.

International cash pooling (in a defined currency) requires pooling at two levels.

A clearing bank in each country where accounts are held completes the first level of pooling.

An international, cross-border bank then manages international pooling.

1. The first level of pooling is completed at a country level,
2. then the resulting balances are pooled at an international level.

Multi-currency cash pooling

Multi-currency cash pooling

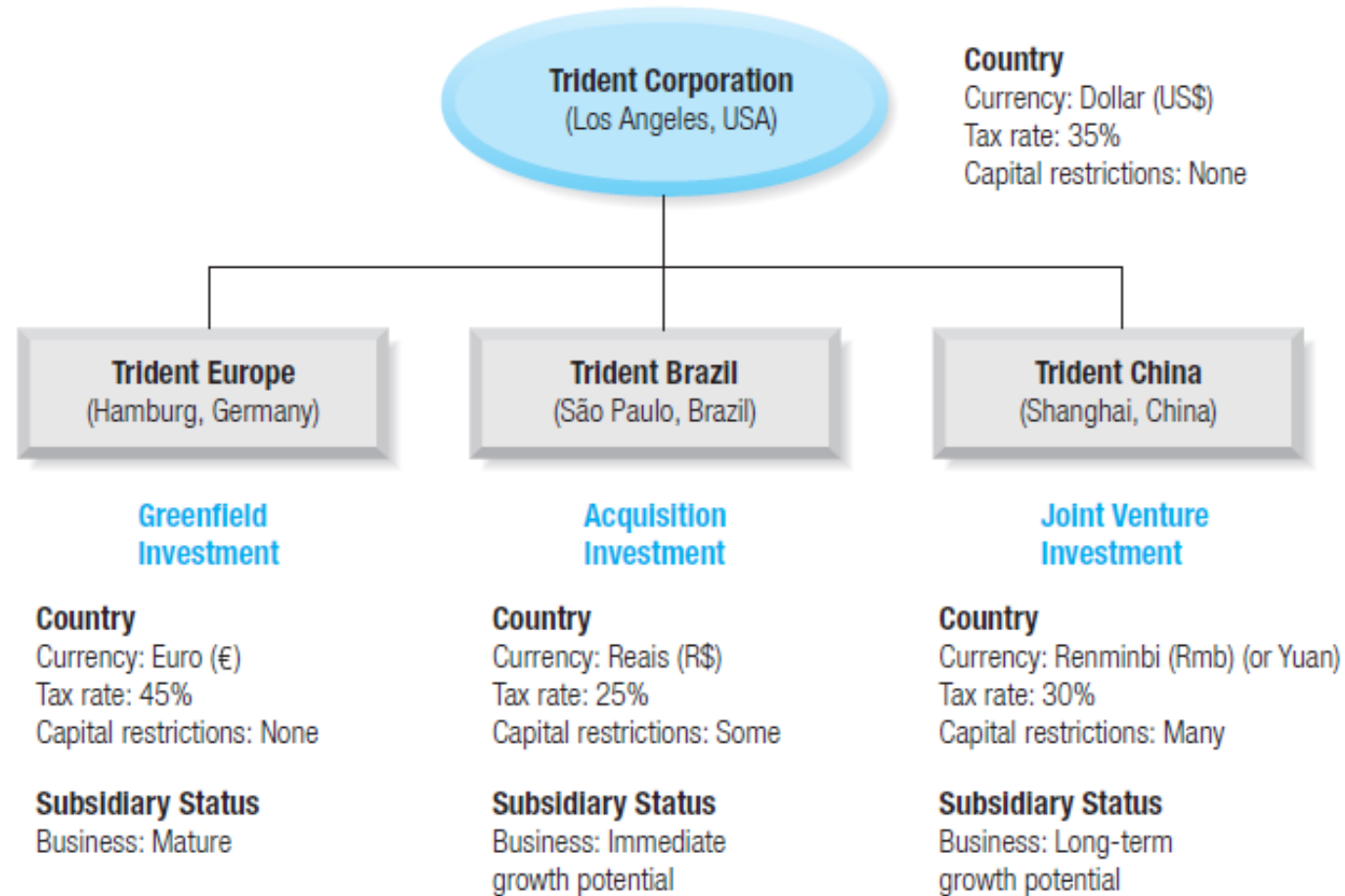
While physical cash pooling is typically used with just **one currency**, it can also be applied to multiple currencies.

If the cash pooling is managed by the bank, the scope is **restricted because bank-managed cash pooling** is generally limited to single-bank and single-currency processing.

Therefore, it is necessary to **reproduce the cash pooling systems** for each currency handled by the group. Multi-currency cash pooling is only possible using notional cash pooling with the help of a dedicated tool.

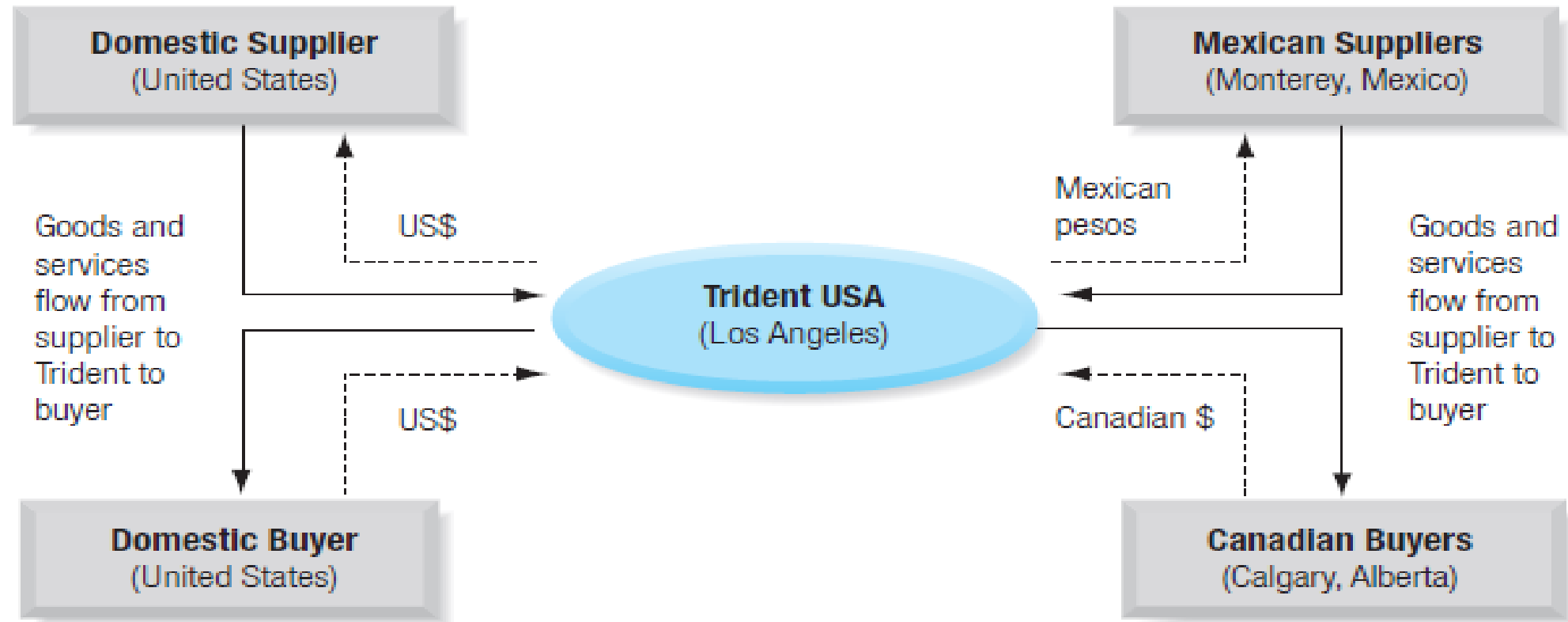
Multinacional companies

EXHIBIT 19.2 Trident's Foreign Subsidiaries



1

Financing Trade: The Flow of Goods and Funds



Determining the effective financing rate

The actual cost of financing by the debtor firm will depend on:

- (1) the interest rate charged by the bank that provided the loan, and
- (2) the movement in the borrowed currency's value over the life of the loan.

Deckborn Ltd (UK) obtains a **two-year loan in New Zealand dollars (NZ\$) worth £1 000 000** at the quoted **interest rate of 8%— a full 2% less** than an equivalent loan offered to Deckborn in the UK.

- When **Deckborn receives the New Zealand loan**, it **converts** the New Zealand dollars to **British pounds** to pay a supplier for materials. The exchange rate at that time is **£0.40** per New Zealand dollar, so the NZ\$ 2 500 000 is converted **to £1 000 000**.
- **One year later**, Deckborn pays interest of NZ\$200 000 (computed as 8% NZ\$2 500 000).
- Two years later, Deckborn pays back the loan of NZ\$2 500 000 plus a second interest payment instalment of NZ\$200 000 a **total of NZ\$2 700 000**.

Assume that the **New Zealand dollar unexpectedly appreciates from £0.40 to £0.50 shortly** after receiving the loan and before any repayments are made.

- The interest payment at the end of the first year will therefore cost Deckborn NZ\$200 000 **0.50** or £100 000.
- At the end of the second year Deckborn will have a bill for NZ\$2 700 000, this will cost £1 350 000 (computed as NZ\$2 700 000 £0.50 per NZ\$).

	Start of period 1 or t_0	End of period 1 or t_1	End of period 2 or t_2
Cash flows in NZ\$s	NZ\$2 500 000	NZ\$200 000	NZ\$2 700 000
Exchange rate £s per NZ\$	£0.40	£0.40	£0.40
Cash flows in £s	+£1 000 000	-£80 000	-£1 080 000
	Start of period 1 or t_0	End of period 1 or t_1	End of period 2 or t_2
Cash flows in NZ\$s	NZ\$2 500 000	NZ\$200 000	NZ\$2 700 000
Exchange rate £s per NZ\$	£0.40	£0.50	£0.50
Cash flows in £s	+£1 000 000	-£100 000	-£1 350 000

The effective cost is the interest rate that 'equates' the repayments with the initial loan. The interest rate is a

cost so the basic formula is repayment less cost equals the original loan. The formula is:

$$\text{Initial loan} + £1\,000\,000 = \text{repayment} \frac{-£100\,000}{(1+r)^1} + \text{repayment} \frac{-£1\,350\,000}{(1+r)^2}$$

↙ ↗
Cost of borrowing

A two-period model can be solved by formula but the simplest approach is to solve such problems by trial and error using a spreadsheet (where the cost of borrowing is placed in cell C9 (10% = 0.10 etc.) the formula for the present value in another cell is $=1\,000\,000 - 100\,000 / (1+C9) - 1\,350\,000 / (1+C9)^2$). A sufficiently close result is achieved with an interest rate of 21.3% thus:

$$+£1\,000\,000 = \frac{-£100\,000}{(1+0.213)^1} + \frac{-1\,350\,000}{(1+0.213)^2}$$

Originally, Deckborn was seeking to obtain a loan for 8% some 2% cheaper than the UK equivalent. But because of the increase in the value of the NZ dollar, Deckborn is now having to pay 21.3%.

$$r_{eff} = (1 + i_f) (1 + e_f)$$

Taking the case of Deckborn, $i_f = 0.08$ or 8% and the effective rate is 21.3%. So the portion of the effective rate due to changes in the exchange rate (e_f) is:

$$0.213 = (1 + 0.08) (1 + e_f) - 1$$

thus:

$$\begin{aligned} e_f &= 1.213/1.08 - 1 \\ &= 0.123 \text{ or } 12.3\% \end{aligned}$$

Deckborn was charged 8% interest but incurred a further 12.3% due to the fact that the value of the NZ\$ went up from £0.40 to £0.50. As a result interest payments in NZ\$ were more expensive and the loan was also more expensive to repay. But the value of the NZ\$ could have fallen in which case the loan could have been even cheaper.

Assume the same details as in the previous example, only the value of the NZ\$ fell from £0.40 per NZ\$ to £0.35 per NZ\$.

	Start of period 1 or t_0	End of period 1 or t_1	End of period 2 or t_2
Cash flows in NZ\$s	NZ\$2 500 000	NZ\$200 000	NZ\$2 700 000
Exchange rate £s per NZ\$	£0.40	£0.35	£0.35
Cash flows in £s	+£1 000 000	-£70 000	-£945 000

We are now seeking a rate of return that will solve the following equation:

$$+£1\,000\,000 = \frac{-£70\,000}{(1+r)^1} + \frac{-945\,000}{(1+r)^2}$$

Again using trial and error and a spreadsheet the solution is 0.77% so:

$$+£1\,000\,000 = \frac{-£70\,000}{(1+0.0077)^1} + \frac{-945\,000}{(1+0.0077)^2}$$

If the NZ dollar were to fall in value in this way, the loan would virtually be free. Thus Deckborn is taking a risk by borrowing in a foreign currency: the cost could be much cheaper than a domestic loan, but if the value of the foreign currency increases, the cost of the loan could be greater. An MNC should therefore consider the expected rate of appreciation or depreciation as well as the quoted interest rates of foreign currencies.

CRITERIA CONSIDERED FOR FOREIGN FINANCING

A more formal consideration of the factors relevant to short-term financing should include:

- | Interest rate parity
- | Exchange rate forecasts

where:

r_{eff} = the overall cost of financing in a foreign currency

i_f = foreign currency interest rate

p = the forward rate premium – currency more expensive (or discount – currency cheaper).

If interest rate parity exists then

$$r_{eff} = i_b$$

where:

i_b = home interest rate

and therefore:

$$i_b = (1 + i_f)(1 + p) - 1$$

and:

$$p = \frac{(1 + i_b)}{(1 + i_f)} - 1 \text{ and approximately for small differences } p = i_b - i_f$$

which is a more formal way of saying that the premium will reflect the difference in interest rates.

Covered interest rate arbitrage is one of the relationships that hold well in the marketplace. If there are differences with domestic returns than administration costs or restrictions will ensure that no riskless (arbitrage) profit can be made.

An MNC has a choice when borrowing internationally, it can cover its position by taking out a forward or futures contract thus ensuring a return similar to domestic returns. Alternatively it can leave its position uncovered. In such a case:

$$r_{eff} = (1 + i_f)(1 + e_f) - 1$$

where:

e_f = the change in value of the foreign currency.

EXHIBIT 18.6 Derivation of possible effective financing rates

Currency	Possible change in the spot rate over the loan life	Probability of that percentage change in the spot rate occurring	Computation of effective financing rate based on the percentage change in the spot rate
Swiss franc	1%	30%	$(1.08)[1 + (0.01)] - 1 = 0.0908$, or 9.08%
Swiss franc	3	50	$(1.08)[1 + (0.03)] - 1 = 0.1124$, or 11.24%
Swiss franc	9	<u>20</u>	$(1.08)[1 + (0.09)] - 1 = 0.1772$, or 17.72%
		100%	
Japanese yen	-1%	35%	$(1.09)[1 + (-0.01)] - 1 = 0.0791$, or 7.91%
Japanese yen	3	40	$(1.09)[1 + (0.03)] - 1 = 0.1227$, or 12.27%
Japanese yen	7	<u>25</u>	$(1.09)[1 + (0.07)] - 1 = 0.1663$, or 16.63%
		100%	

EXAMPLE

Valparaiso SA (Spain) considers borrowing a portfolio of Japanese yen and Swiss francs to finance its euro operations. Half of the needed funding would come from each currency. To determine how the variance in this portfolio's effective financing rate is related to characteristics of the component currencies, assume the following information based on historical information for several three-month periods:

- Mean effective financing rate of Swiss franc for three months = 3% or 0.03
- Mean effective financing rate of Japanese yen for three months = 2% or 0.02
- Standard deviation of Swiss franc's effective financing rate = 0.04
- Standard deviation of Japanese yen's effective financing rate = 0.09
- Correlation coefficient of effective financing rates of these two currencies = 0.10

Given this information, the mean effective rate on a portfolio (r_P) of funds financed 50% by Swiss francs and 50% by Japanese yen is determined by totalling the weighted individual effective financing rates:

$$\begin{aligned}r_P &= W_{SFr}r_{SFr} + W_{Yen}r_{Yen} \\&= 0.50(0.03) + 0.50(0.02) \\&= 0.015 + 0.01 \\&= 0.025 \text{ or } 2.5\%\end{aligned}$$

The variance of this portfolio's effective financing rate over time is:

$$\begin{aligned}VAR(r_P) &= 0.5^2(0.04)^2 + 0.5^2(0.09)^2 \\&\quad + 2(0.5)(0.5)(0.04)(0.09)(0.10) \\&= 0.25(0.0016) + 0.25(0.0081) + 0.00018 \\&= 0.002605\end{aligned}$$

TECHNIQUES TO OPTIMIZE CASH FLOWS

Cash inflows can be optimized by the following techniques:

- ┆ Accelerating cash inflows
- ┆ Minimizing currency conversion costs
- ┆ Managing blocked funds
- ┆ Managing intersubsidiary cash transfers

CASH FLOW ANALYSIS: SUBSIDIARY PERSPECTIVE

Subsidiary expenses

Subsidiary revenue

Subsidiary dividend payments

Subsidiary liquidity management

EXHIBIT 19.1 Cash flow of the overall MNC

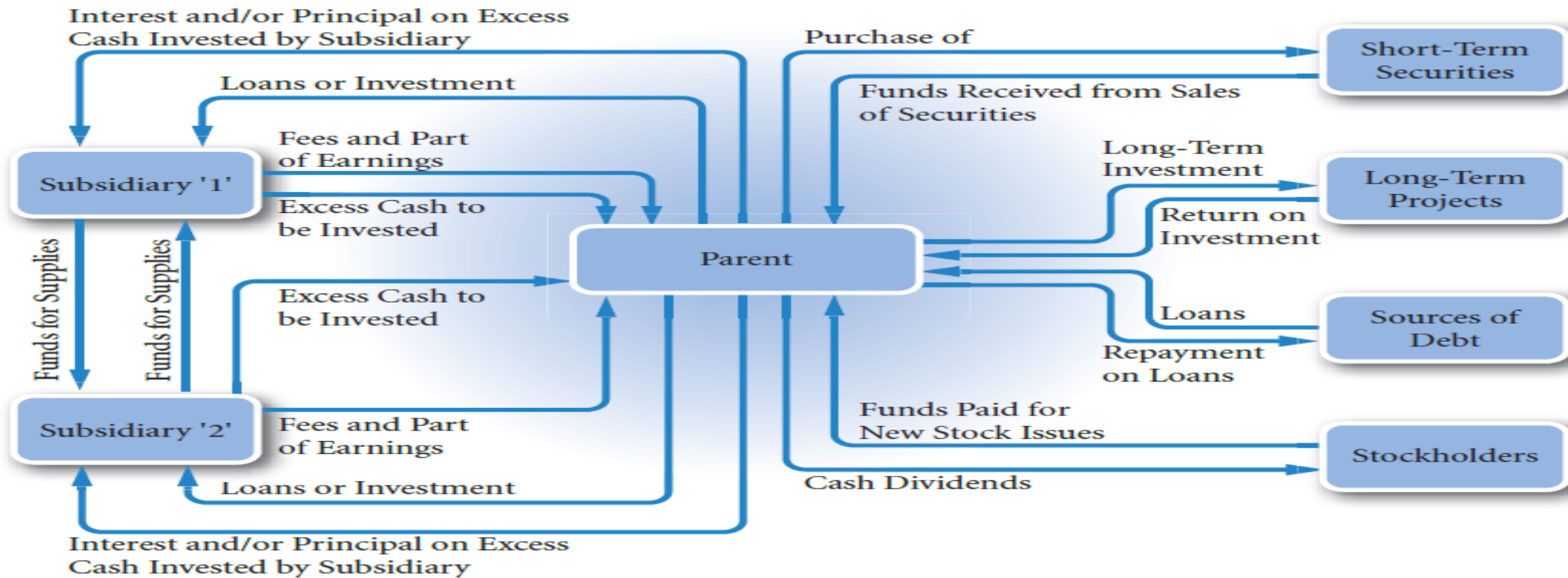


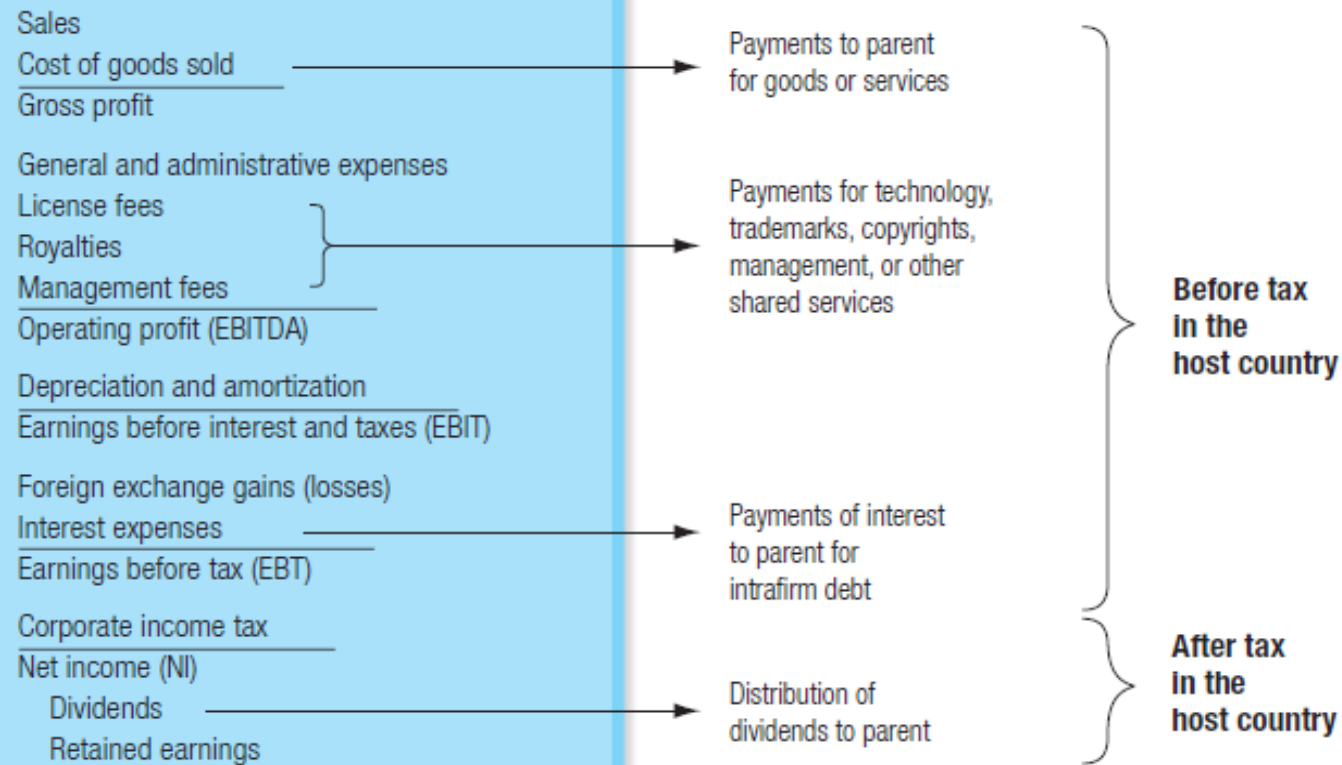
EXHIBIT 19.3 Moving Funds from Subsidiary to Parent**Foreign Subsidiary's Income Statement****Payment to Parent Company**

EXHIBIT 19.4**Trident Brazil's Net Working Capital Requirements**

Net working capital (NWC) is the net investment required of the firm to support ongoing sales. NWC components typically grow as the firm buys inputs, produces products, and sells finished goods.

Trident's Balance Sheet

Assets	Liabilities and Net Worth
Cash	Accounts payable (A/P)
Accounts receivable (A/R) Inventory	Short-term debt
Current assets	Current liabilities

$NWC = (A/R + \text{Inventory}) - A/P$

Note that NWC is *not* the same as current assets and current liabilities.

HIBIT 19.6

Trident's Multinational Working Capital Sequence

Cash inflows to Trident Brazil arise from local market sales. These cash flows are used to repay both intrafirm payables (to Trident USA) and local suppliers.



Brazilian Business Practices

Payment terms in Brazil are longer than those typical of North America. Trident Brazil must offer 60-day terms to local customers to be competitive with other firms in the local market.

U.S. Business Practices

Payment terms used by Trident USA are typical of North America, 30 days. Trident USA's local customers will expect to be paid in 30 days. Trident USA may consider extending longer terms to Brazil to reduce the squeeze.

Result: Trident Brazil is squeezed in terms of cash flow. It receives inflows in 60 days but must pay Trident USA in 30 days.

EXHIBIT 19.7 Decentralized Versus Centralized Cash Depositories**Decentralized Cash Depository**

Subsidiary	Expected Cash Need (A)	One Standard Deviation (B)	Cash Balance Budgeted for Adequate Protection ^a (A + 3B)
Trident Europe	\$10,000,000	\$1,000,000	\$13,000,000
Trident Brazil	6,000,000	2,000,000	12,000,000
Trident China	12,000,000	3,000,000	21,000,000
Total	\$ 28,000,000	\$ 6,000,000	\$46,000,000

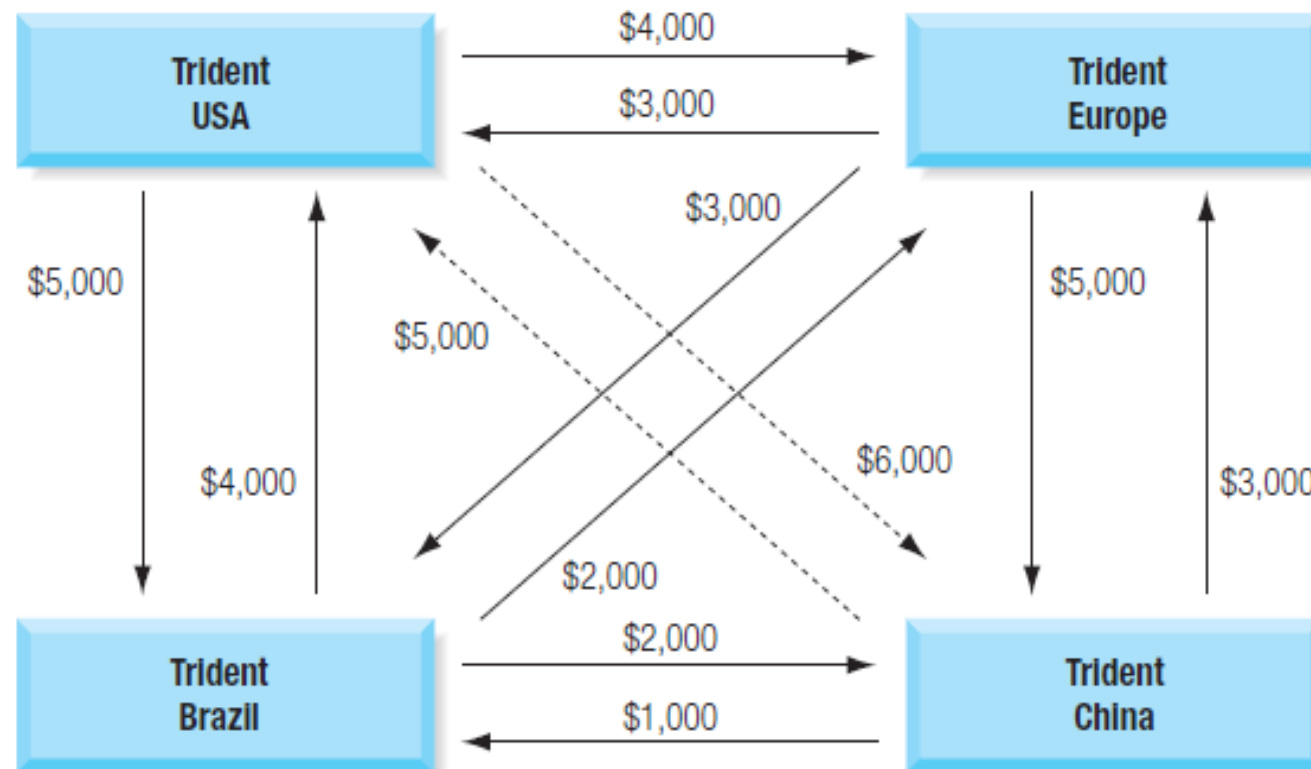
Centralized Cash Depository

Subsidiary	Expected Cash Need (A)	One Standard Deviation (B)	Cash Balance Budgeted for Adequate Protection ^a (A + 3B)
Trident Europe	\$10,000,000		
Trident Brazil	6,000,000		
Trident China	12,000,000		
Total	\$28,000,000	\$ 3,741,657 ^b	\$39,224,972

^a Adequate protection is defined as the expected cash balance plus three standard deviations, assuming that the cash flows of all three individual affiliates are normally distributed.

^b The standard deviation of the expected cash balance of the centralized depository is calculated as follows:

$$\text{Standard deviation} = \sqrt{(1,000,000)^2 + (2,000,000)^2 + (3,000,000)^2} = \$3,741,657.$$

EXHIBIT 19.8**Multilateral Matrix Before Netting (thousands of U.S. dollars)**

Prior to netting, the four sister Trident companies have numerous intrafirm payments between them. Each payment results in transfer charges.

EXHIBIT 19.2(a) Intersubsidiary payments matrix

Payments OWED BY subsidiary located in:	British pound value (in thousands) OWED TO subsidiary located in:				
	Canada	France	Japan	Switzerland	UK
Canada	—	40	90	20	40
France	60	—	30	60	50
Japan	100	30	—	20	30
Switzerland	10	50	10	—	50
UK	10	60	20	20	—

EXHIBIT 19.2(b) Bilateral netting (arrows show the example of Canada)

Payments OWED BY subsidiary located in:	British pound value (in thousands) OWED TO subsidiary located in:				
	Canada	France	Japan	Switzerland	UK
Canada	—	40	90	20	40
France	60	—	30	60	50
Japan	100	30	—	20	30
Switzerland	10	50	10	—	50
UK	10	60	20	20	—

EXHIBIT 19.2(c) Bilateral netting – the net position (arrows show the example of Canada)

Payments OWED BY subsidiary located in:	British pound value (in thousands) OWED TO subsidiary located in:				
	Canada	France	Japan	Switzerland	UK
Canada	—	0	0	10	30
France	20	—	0	10	0
Japan	10	0	—	10	10
Switzerland	0	0	0	—	30
UK	0	10	0	0	—

(Continued)

Short-term Financing Analysis. Assume that Davenport Inc. needs \$3 million for a one-year period. Within one year, it will generate enough U.S. dollars to pay off the loan. It is considering three options: (1) borrowing U.S. dollars at an interest rate of 6%, (2) borrowing Japanese yen at an interest rate of 3%, or (3) borrowing Canadian dollars at an interest rate of 4%. Davenport Inc. expects that the Japanese yen will appreciate by 1% over the next year and that the Canadian dollar will appreciate by 3%. What is the expected “effective” financing rate for each of the three options? Which option appears to be most feasible? Why might Davenport Inc. not necessarily choose the option reflecting the lowest effective financing rate?

ANSWER:

		Expected	
Interest		Expected Percentage	Effective
<u>Currency</u>	<u>Rate</u>	<u>Change in Currency</u>	<u>Financing Rate</u>
Dollars	6%	—	6.00%
Japanese yen	3%	+1%	4.03%
Canadian dollars	4%	+3%	7.12%

Short-term Financing Analysis. Assume that Davenport Inc. needs \$3 million for a one-year period. Within one year, it will generate enough U.S. dollars to pay off the loan. It is considering three options: (1) borrowing U.S. dollars at an interest rate of 6%, (2) borrowing Japanese yen at an interest rate of 3%, or (3) borrowing Canadian dollars at an interest rate of 4%. Davenport Inc. expects that the Japanese yen will appreciate by 1% over the next year and that the Canadian dollar will appreciate by 3%. What is the expected “effective” financing rate for each of the three options? Which option appears to be most feasible? Why might Davenport Inc. not necessarily choose the option reflecting the lowest effective financing rate?

ANSWER:

Expected			
InterestExpected PercentageEffective			
<u>Currency</u>	<u>Rate</u>	<u>Change in Currency</u>	<u>Financing Rate</u>
Dollars	6%	—	6.00%
Japanese yen	3%	+1%	4.03%
Canadian dollars	4%	+3%	7.12%

ANSWER: The Japanese yen option appears to be the most feasible option. Yet, the exchange rate percentage change is uncertain, which makes the effective financing rate uncertain. Thus, Tulsa Inc. will not necessarily choose this option.

Problem

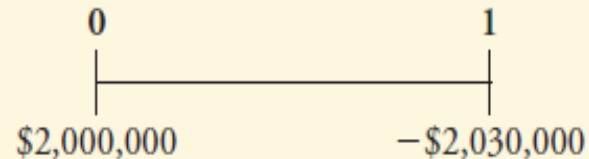
The Row Cannery wants to borrow \$2 million for one month. Using its inventory as collateral, it can obtain a 12% (APR) loan. The lender requires that a warehouse arrangement be used. The warehouse fee is \$10,000, payable at the end of the month. Calculate the effective annual rate of this loan for Row Cannery.

Problem

The Row Cannery wants to borrow \$2 million for one month. Using its inventory as collateral, it can obtain a 12% (APR) loan. The lender requires that a warehouse arrangement be used. The warehouse fee is \$10,000, payable at the end of the month. Calculate the effective annual rate of this loan for Row Cannery.

Solution

The monthly interest rate is $12\%/12 = 1\%$. At the end of the month, Row will owe $\$2,000,000 \times 1.01 = \$2,020,000$ plus the warehouse fee of \$10,000. Putting the cash flows on a timeline gives:



The actual one-month interest rate paid is

$$\frac{2,030,000}{2,000,000} - 1 = 1.5\%$$

Expressing this as an EAR gives $1.015^{12} - 1 = 19.6\%$.

receives. And like the commitment fee, it is effectively an additional interest charge.

To illustrate, assume that Timmons Towel and Diaper Service is offered a \$500,000 loan for three months at an APR of 12%. This loan has a loan origination fee of 1%. The loan origination fee is charged on the principal of the loan. In this case, the fee amounts to $0.01 \times \$500,000 = \$5,000$, so the actual amount borrowed is \$495,000. The interest payment for three months is $\$500,000(0.12/4) = \$15,000$. Putting these cash flows on a timeline:

receives. And like the commitment fee, it is effectively an additional interest charge.

To illustrate, assume that Timmons Towel and Diaper Service is offered a \$500,000 loan for three months at an APR of 12%. This loan has a loan origination fee of 1%. The loan origination fee is charged on the principal of the loan. In this case, the fee amounts to $0.01 \times \$500,000 = \$5,000$, so the actual amount borrowed is \$495,000. The interest payment for three months is $\$500,000(0.12/4) = \$15,000$. Putting these cash flows on a timeline:



Thus, the actual three-month interest rate paid is

$$\frac{515,000}{495,000} - 1 = 4.04\%$$

Expressing this rate as an EAR gives $1.0404^4 - 1 = 17.17\%$.

Investing Strategy. Tallahassee Co. has \$2 million in excess cash that it has invested in Mexico at an annual interest rate of 60 percent. The U.S. interest rate is 9 percent. By how much would the Mexican peso have to depreciate to cause such a strategy to backfire?

Investing Strategy. Tallahassee Co. has \$2 million in excess cash that it has invested in Mexico at an annual interest rate of 60 percent. The U.S. interest rate is 9 percent. By how much would the Mexican peso have to depreciate to cause such a strategy to backfire?

$$\frac{1 + 9\%}{1 + 60\%} - 1 = -31.875\%$$

ANSWER: If the peso depreciates by more than 31.875 percent, the effective yield on the Mexican deposit will be less than the domestic yield.

Problems

The Treadwater Bank wants to raise \$1.1 million using three-month commercial paper. The net proceeds to the bank will be \$1,084,600. What is the effective annual rate of this financing for Treadwater?

12. Magna Corporation has an issue of commercial paper with a face value of \$1,000,000 and a maturity of six months. Magna received net proceeds of \$973,710 when it sold the paper. What is the effective annual rate of the paper to Magna?

13. What is the difference between direct paper and dealer paper?

14. The Signet Corporation has issued four-month commercial paper with a \$6.3 million face value. The firm netted \$6,113,898 on the sale. What effective annual rate is Signet paying for these funds?

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0,058016

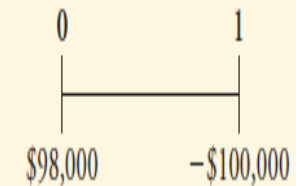
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13. What is the difference between direct paper and dealer paper?

14. The Signet Corporation has issued four-month commercial paper with a \$6.3 million face value. The firm netted \$6,113,898 on the sale. What effective annual rate is Signet paying for these funds?

Solution

Let's put the firm's cash flows on a timeline:



The actual three-month interest rate paid is

$$\frac{100,000}{98,000} - 1 = 2.04\%$$

Expressing this as an EAR gives $1.0204^4 - 1 = 8.42\%$.

Suppose that a firm has negotiated a committed line of credit with a stated maximum of \$1 million and an interest rate of 10% (EAR) with a bank. The commitment fee is 0.5% (EAR). At the beginning of the year, the firm borrows \$800,000. It then repays this loan at the end of the year, leaving \$200,000 unused for the rest of the year.

Suppose that a firm has negotiated a committed line of credit with a stated maximum of \$1 million and an interest rate of 10% (EAR) with a bank. The commitment fee is 0.5% (EAR). At the beginning of the year, the firm borrows \$800,000. It then repays this loan at the end of the year, leaving \$200,000 unused for the rest of the year.

$$\text{Interest on borrowed funds} = 0.10(\$800,000) = \$80,000$$

$$\text{Commitment fee paid on unused portion} = 0.005(\$200,000) = \underline{\$1,000}$$

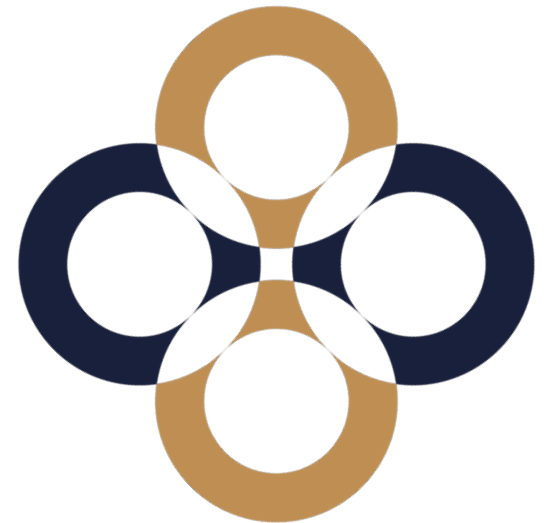
$$\text{Total cost} = \$81,000$$

The Ohio Valley Steel Corporation has borrowed \$5.1 million for one month at a stated annual rate of 8.7%, using inventory stored in a field warehouse as collateral. The warehouser charges a \$5100 fee, payable at the end of the month. What is the effective annual rate of this loan?

Thank you for your attention!

Simple groups

Group accounting under IFRS

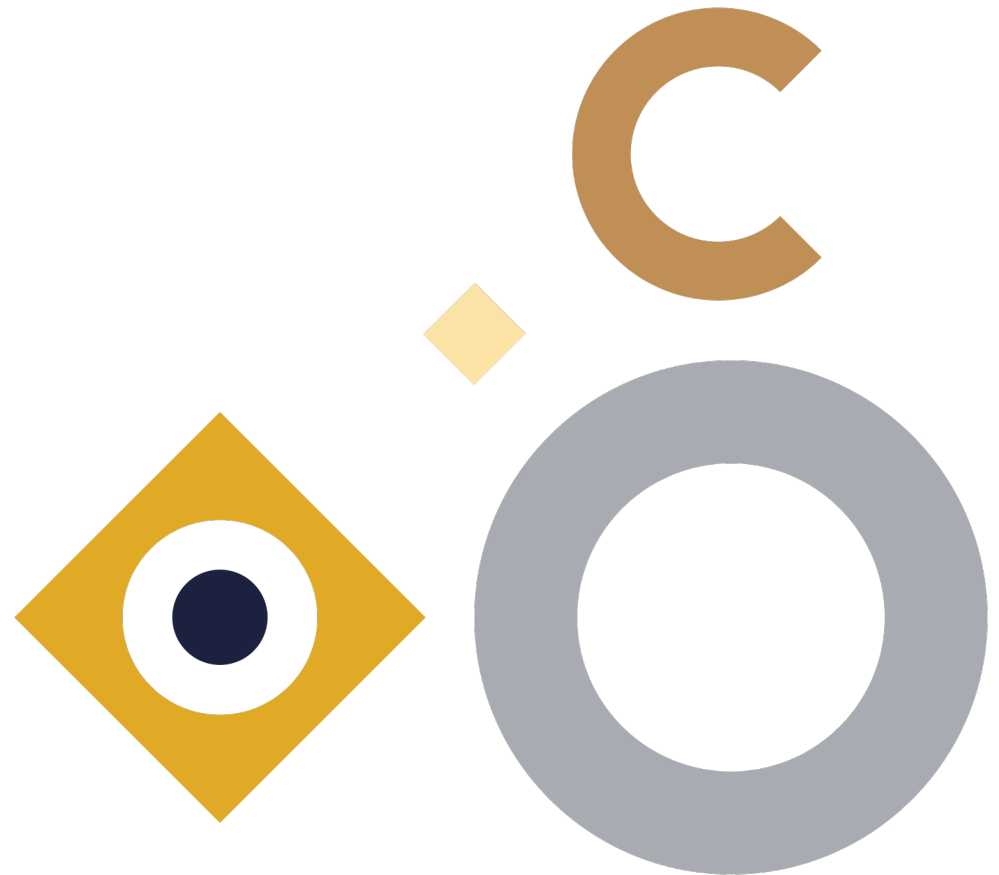


Contact information

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Why bother...?

- ▶ The owners of the parent or future investors should see the performance of the group
- ▶ There should be no puffer items
- ▶ This will tell you something...
- ▶ The technology in IFRS is not governed it is the same as everywhere (I assume you know the basics)

Legal entity vs. Business entity

Taxation, legal proceedings,
responsibility (?), etc...

Performance, Net assets,
etc. from the point of view
of the SHAREHOLDER

The relevant standards and “almost” standards

Sets the main rules of
group accounting.

IFRS 10

Explains what to do with
associates and deals with
equity accounting

IAS 28R

IFRS 11

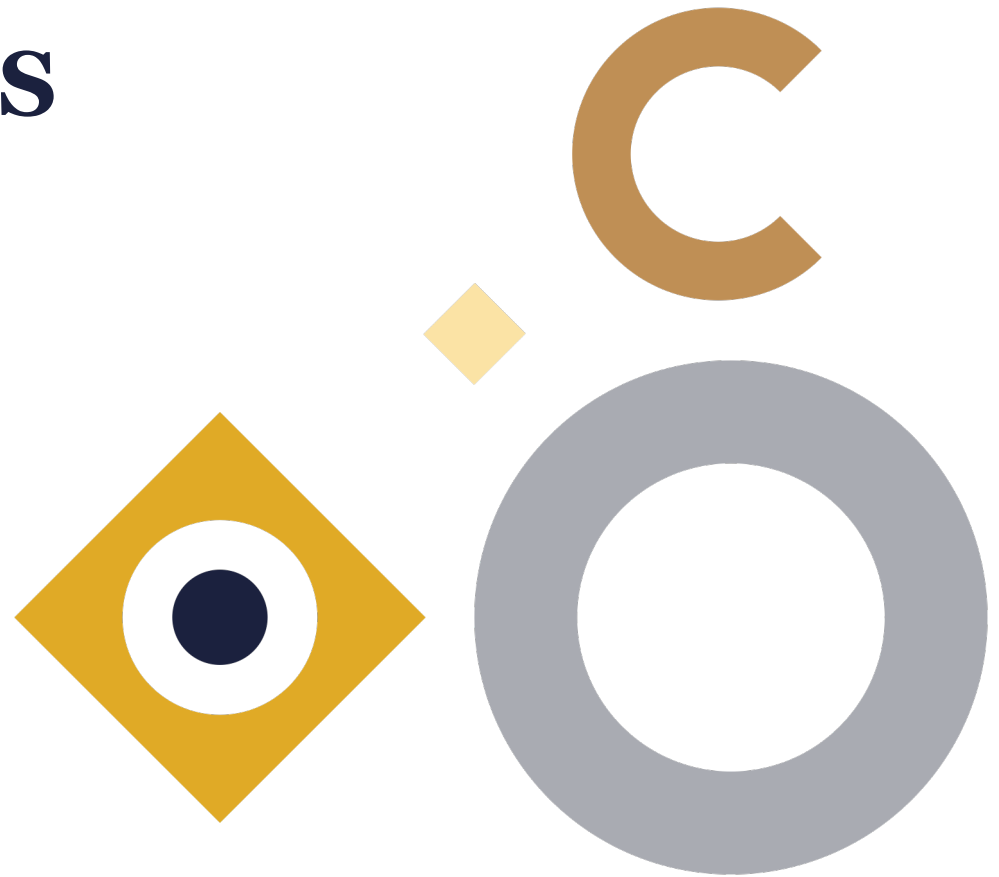
Joint arrangements!

IFRS 3

The heart of everything 😊 Deals
with business combinations
(the beginning...)

IFRS 10

CONCEPTUAL RULES



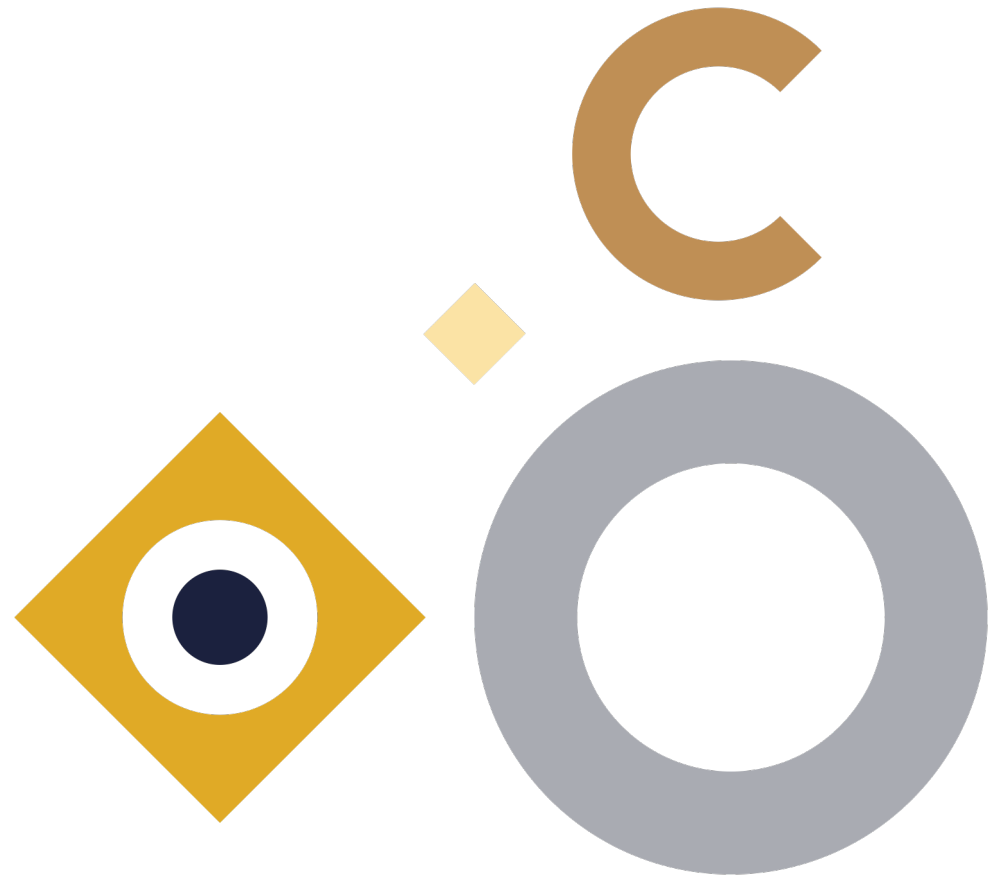
IFRS 10 Objective

- ▶ The standard applies to:
 - ▶ the preparation and presentation of **consolidated financial statements** for a **group** of entities under the **control** of a **parent**
 - ▶ accounting for investments in **subsidiaries**, joint ventures and associates when an entity presents separate financial statements
- ▶ **Consolidated financial statements** are the financial statements of a **group** presented as those of a single entity

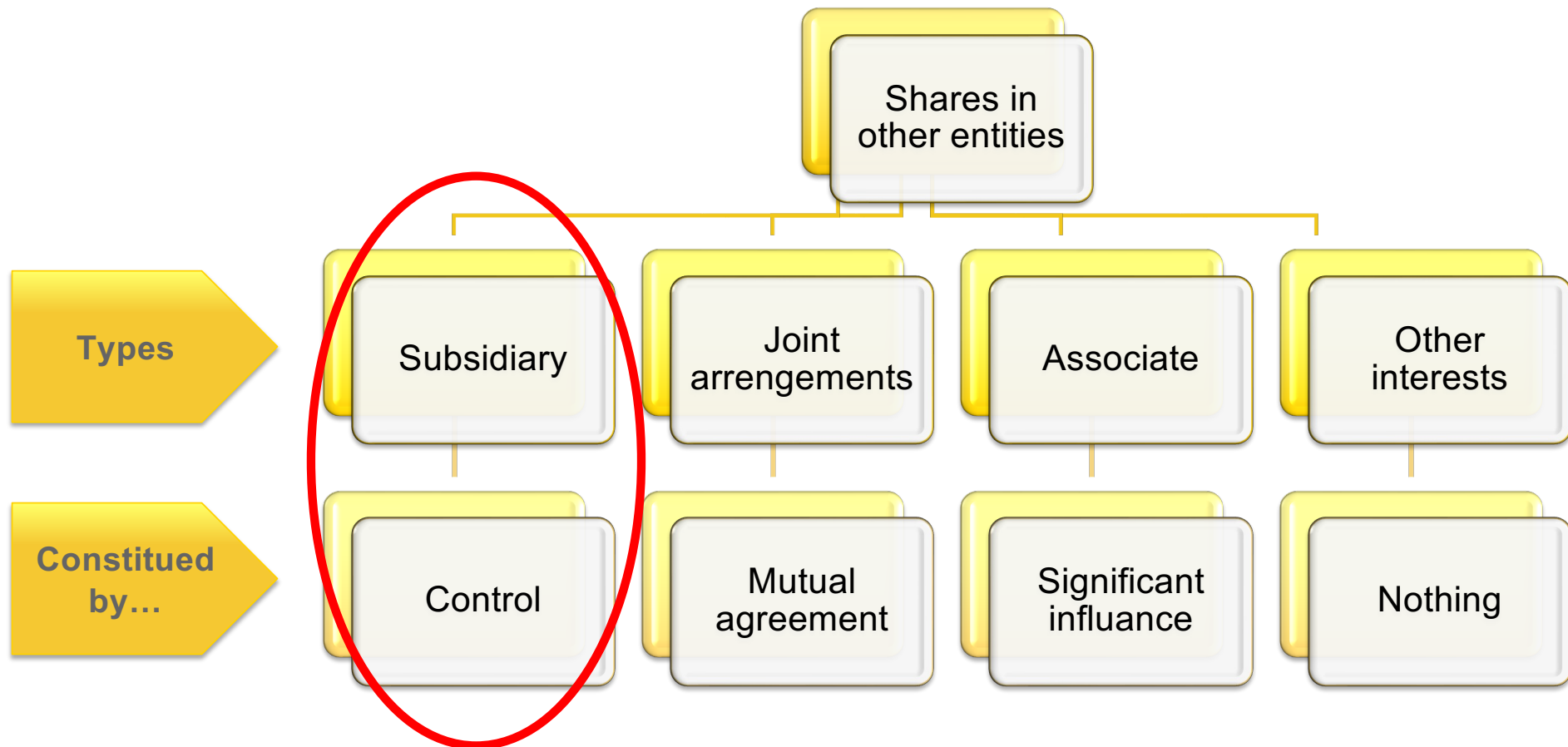
IFRS 10 Key Definitions

- ▶ A **group** is a parent and all its subsidiaries
- ▶ A **parent** is an entity that has one or more subsidiaries
- ▶ A **subsidiary** is an entity that is controlled by another entity (the parent)
- ▶ **Control** ... this is the key! See next slides!
- ▶ **Minority / Non-controlling interest** is that portion of the profit or loss and net assets attributable to equity interests that are not owned, directly or indirectly through subsidiaries by the parent

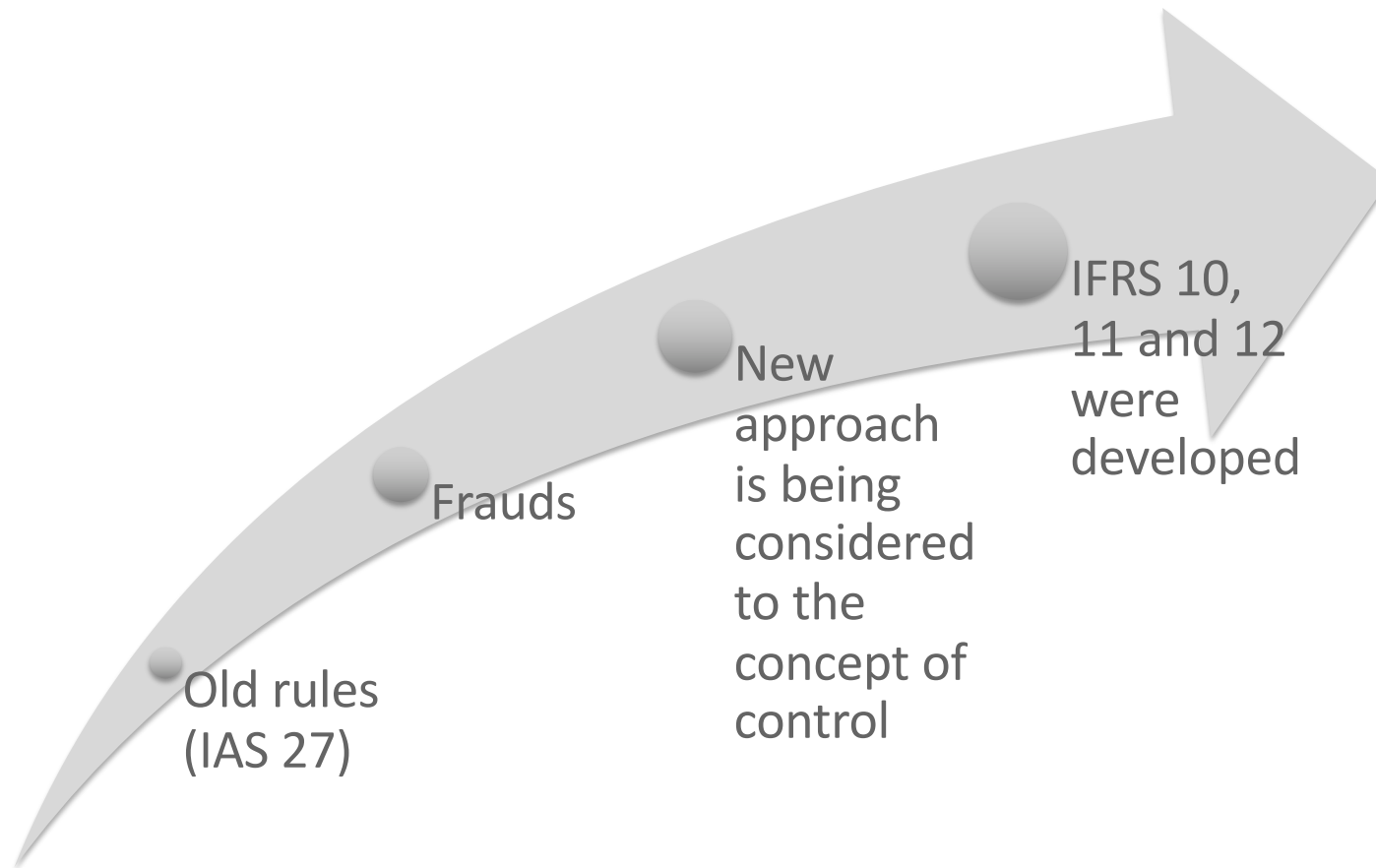
BACKGROUND, OBJECTIVES AND CORE PRINCIPLE



Classification entities



The timeline



Background, objectives and core principle

Background

- ▶ Inconsistent application of the concept of control due to perceived conflict between IAS 27 and SIC-12 (the “old rules”)
- ▶ Global financial crisis highlighted the lack of transparency about risk of exposure from off-balance-sheet entities/interests

Objectives

- ▶ Develop a single control model that is applicable to all types of entities to eliminate the perceived inconsistency between IAS 27 and SIC-12
- ▶ Improve disclosures

Core principle – remained unchanged

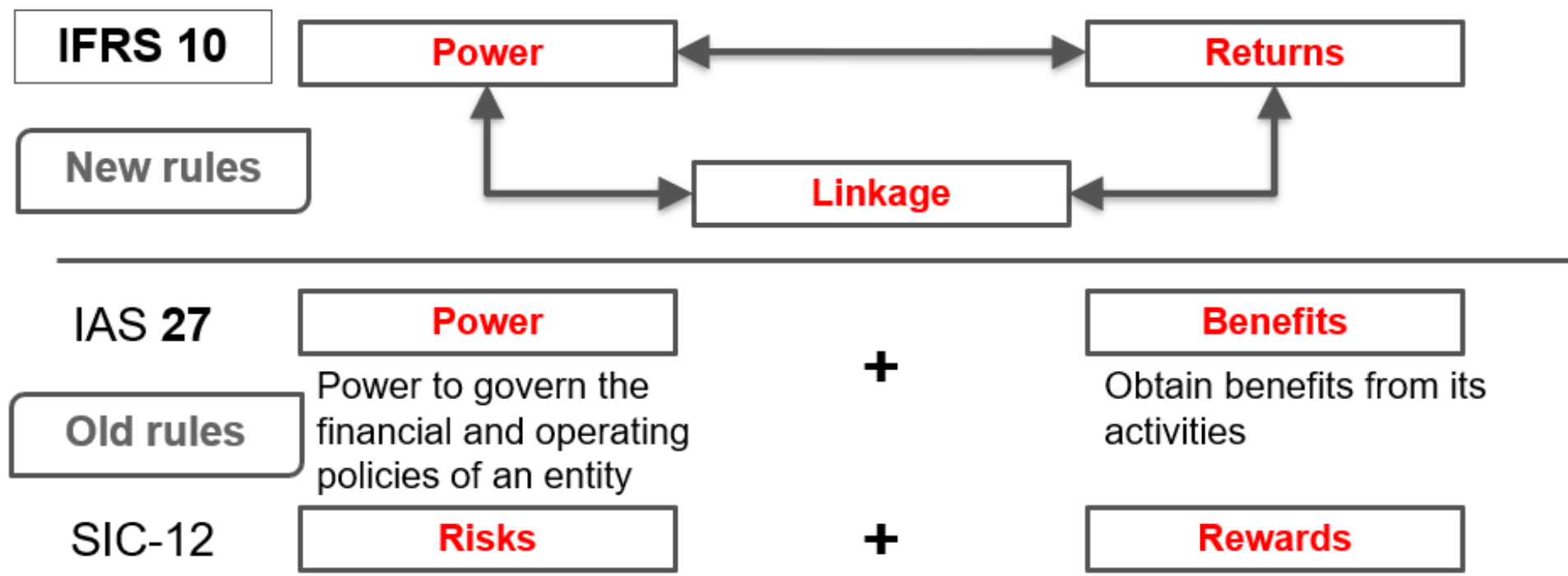
- ▶ A group presents consolidated financial statements as those of a single economic entity
- ▶ Consolidated financial statements include all controlled entities

NEW DEFINITION OF CONTROL

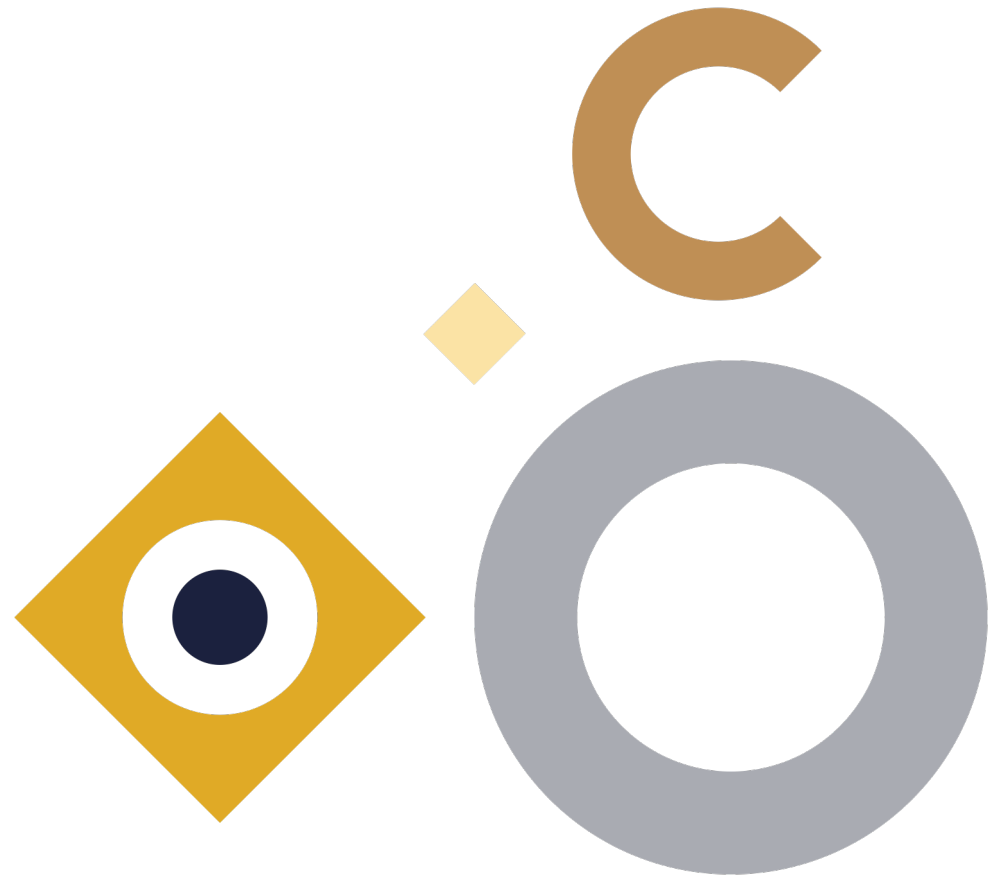


New definition of control

“An investor controls an investee when it is exposed, or has rights, to variable **returns** from its involvement with the investee and has the ability to affect those returns through its **power** over the investee.”



ASSESSING CONTROL



Assessing control: overview

Activities

Identify which activities of the investee are considered to be the 'relevant activities'; i.e., those that affect the investee's returns.

Power

Determine which party, if any, has power; that is, having existing rights that give it the current ability to direct the relevant activities.

Returns

Assess whether the investor is exposed, or has rights, to variable returns from its involvement with the investee.

Purpose and design

Continuous assessment

Identifying relevant activities: overview

Activities

Identify which activities of the investee are considered to be the 'relevant activities'; i.e., those that affect the investee's returns.

- ▶ **Examples** of activities that may affect investee's returns:
 - ▶ Determining or changing operating and financing policies
 - ▶ Making capital decision
 - ▶ Appointing, remunerating and terminating employment of service providers or key management personnel
- ▶ If two investors both direct different relevant activities, one has to identify which investor can **most** significantly affect returns
- ▶ Purpose and design

Activity 1 debrief: identifying relevant activities – example 1

- ▶ Two investors, C and D, form an investee E to acquire, dispose and manage an investment property portfolio (fund).
- ▶ Investor C is responsible for acquisitions and disposals of the investment properties of the portfolio whilst investor D is responsible for the management of the existing investment properties of the portfolio.
- ▶ The investment agreement states that the fund will hold no more than five investment properties at any one time, and indicates a planned Internal Rate of Return (IRR) on each property.
- ▶ Suppose an investment property achieves an IRR less than planned for three consecutive years. In that case, investor C is required to evaluate whether the investment property needs to be replaced, considering the forecast of the respective investment property, the overall return of the portfolio of investment properties and available properties in the market.

Activity 1 debrief: identifying relevant activities – example 1 (cont.)

Key points

- ▶ The level of activity associated with the acquisition and disposal of investment properties is based on the expected IRR of both the individual investment properties and the fund.
- ▶ If IRR levels are expected to be achievable and realistic on the properties initially acquired, then the level of acquisition and disposal activities would be low and the most relevant activity will therefore be the management of the investment properties.
- ▶ If the level of activities relating to acquisitions and disposals is high and is expected to remain high, this may indicate that the acquisitions and disposals of investment properties are the most relevant activity.

Activity 1 debrief: identifying relevant activities – example 2

- ▶ Investor A transfers receivables to investee B, an entity created solely for the purpose of purchasing and servicing those receivables.
- ▶ Investee B fully funds the acquisition of the receivables by issuing two different tranches of debt: a senior tranche (90% of debt) to the market and a junior tranche (10% of debt) to investor A.
- ▶ There are few, if any, activities to perform once investee B is set up unless the counterparties to the receivables default on payment.
- ▶ Investor A retains the customer relationships and is responsible for managing those receivables in the event of default. A third-party servicer collects the cash flows from the receivables and passes them to the investors.

Activity 1 debrief: identifying relevant activities – example 2 (cont.)

Key points

- ▶ The significant relevant activity is the management of the receivables upon default, as this activity significantly affects investee B's returns.
- ▶ A has the ability to make decisions about the relevant activity.
- ▶ A has exposure to variable returns from investee B.

Therefore, A will consolidate B.

Assessing power: overview

- ▶ Main aspects of power:
 - ▶ Power arises from rights.
 - ▶ Power is the current ability to direct and need not be exercised.
 - ▶ Power relates to direction of the relevant activities.
- ▶ Other considerations:
 - ▶ Evidence that the investor directed activities in the past is an indicator of power, but is not conclusive.
 - ▶ An investor that holds only protective rights cannot have power over an investee.

Power

Determine which party, if any, has power; that is, having existing rights that give it the current ability to direct the relevant activities.

Returns

Assess whether the investor is exposed, or has rights, to variable returns from its involvement with the investee.

Substantive rights

“Power is the existing rights that give an investor the current ability to direct the relevant activities.”

- ▶ Rights can either be substantive or protective.
- ▶ Factors to consider in assessing whether a right is substantive:
 - ▶ Are there **barriers** that would prevent the holder from exercising their rights?
 - ▶ Do the holders have the **practical ability** to exercise their rights when exercise requires agreement by more than one investor?
 - ▶ Would the investor that holds the rights **benefit** from their exercise of conversion?
 - ▶ Are the rights **currently exercisable or convertible**?

Example 1: Potential voting rights – Management intention

Cuddles, Flippy and Giggels set up Happy Tree Friends Co. Cuddles owns 60% of the voting shares, Flippy and Giggels have the rest of the voting shares (20-20%). Giggels has an option, which grants that upon request Cuddles and Flippy shall sell the half of their voting shares to her. Giggels does not want to call the option.

Who controls Happy Tree Friends Co.?

Example 2: Potential voting rights – Financial ability

Entities A and B own 55% and 45% respectively of the ordinary shares that carry voting rights of entity C. Entity B also holds debt instruments that are convertible into ordinary shares of entity C at a substantial price and would require entity B to borrow additional funds to make the payment. If the debt were to be converted, entity B would hold 70% (A's interest would reduce to 30%).

Required

Who should consolidated entity C?

Assessing returns

- ▶ Can be positive, negative or both
- ▶ Examples:
 - ▶ Dividends, distributions of economic benefits, changes in value
 - ▶ Remuneration, fees, residual interests, tax benefits
 - ▶ Synergistic returns, saving costs, economies of scale
- ▶ Returns that appear 'fixed' may actually be 'variable'
 - ▶ Bond with 'fixed' interest payments – exposed to credit risk
 - ▶ Fixed fee for managing investment – exposed to performance risk
- ▶ Exposure to returns is an indicator of control:
 - ▶ Greater the exposure greater incentive to obtain power

Returns

Assess whether the investor is exposed, or has rights, to variable returns from its involvement with the investee.

What is different/more explicit?

- ▶ Power with less than majority voting rights
- ▶ Potential voting rights
- ▶ Delegated rights
- ▶ Power over specified assets ('silos')

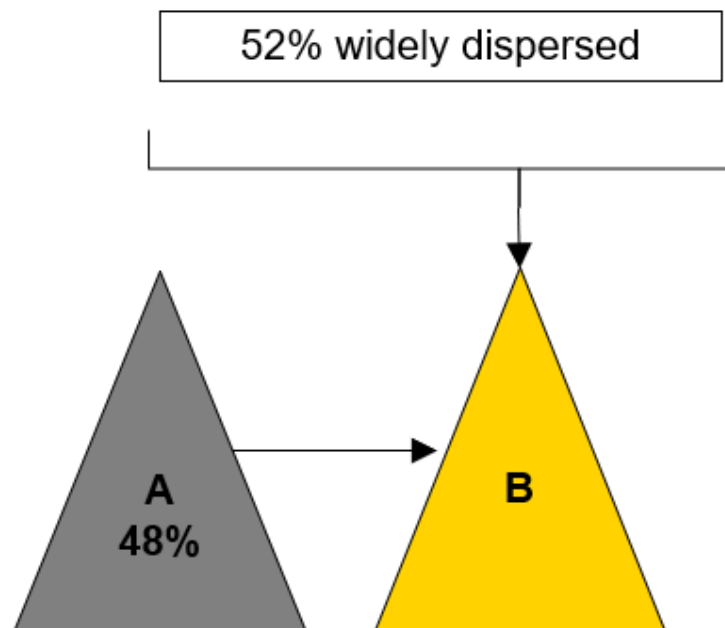
Power with less than majority of voting rights

- ▶ An investor may have the power to direct the relevant activities with less than half of the voting rights.
- ▶ All facts and circumstances must be considered:
 - ▶ Contractual arrangements with other shareholders
 - ▶ Contractual rights arising from other arrangements
 - ▶ The investor's voting rights: size of the investor's holding of voting rights relative to the size and dispersion of other vote holders (**de facto control**)
 - ▶ Potential voting rights held by the investor and other parties

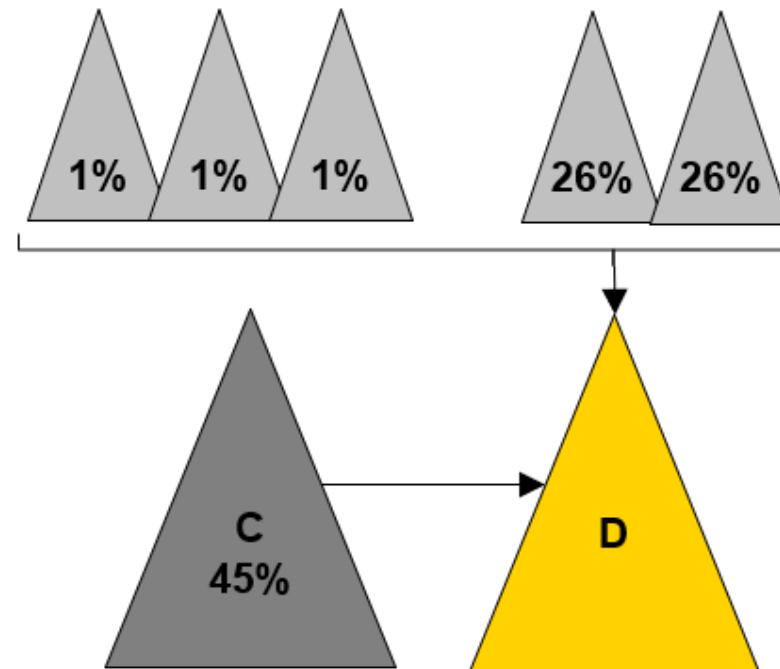
If it is not clear from considering these factors that the investor has power, the investor does not control the investee.

Assessing power – power with less than majority of voting rights

Example 1 – A has power

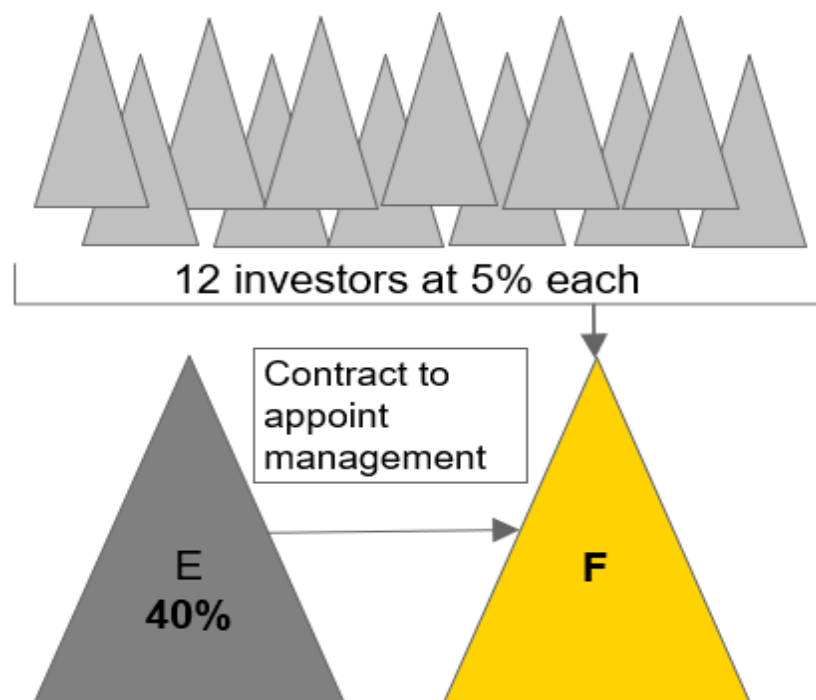


Example 2 – C does not have power

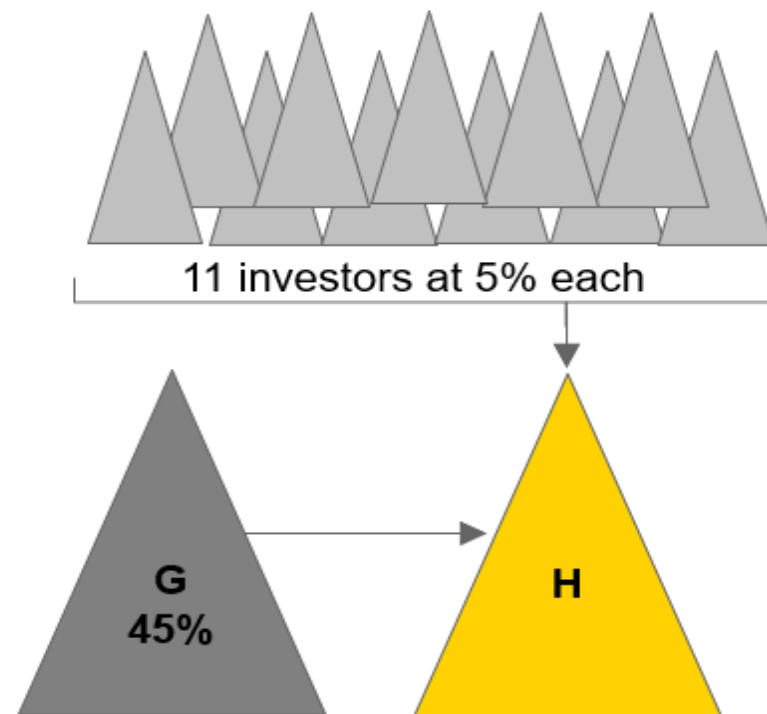


Assessing power – power with less than majority of voting rights (cont.)

Example 3 – E has power

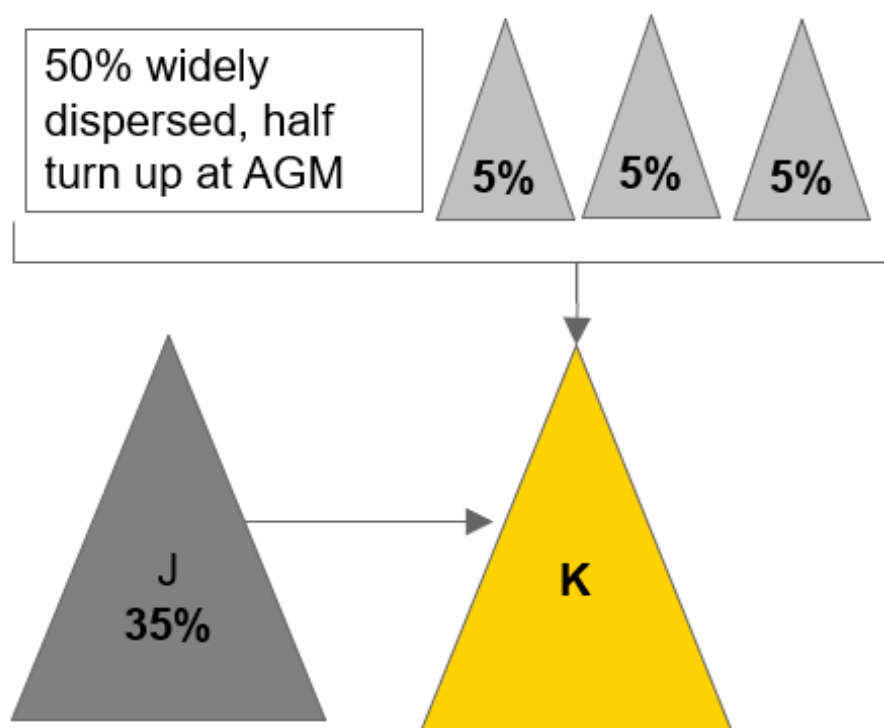


Example 4 – not conclusive

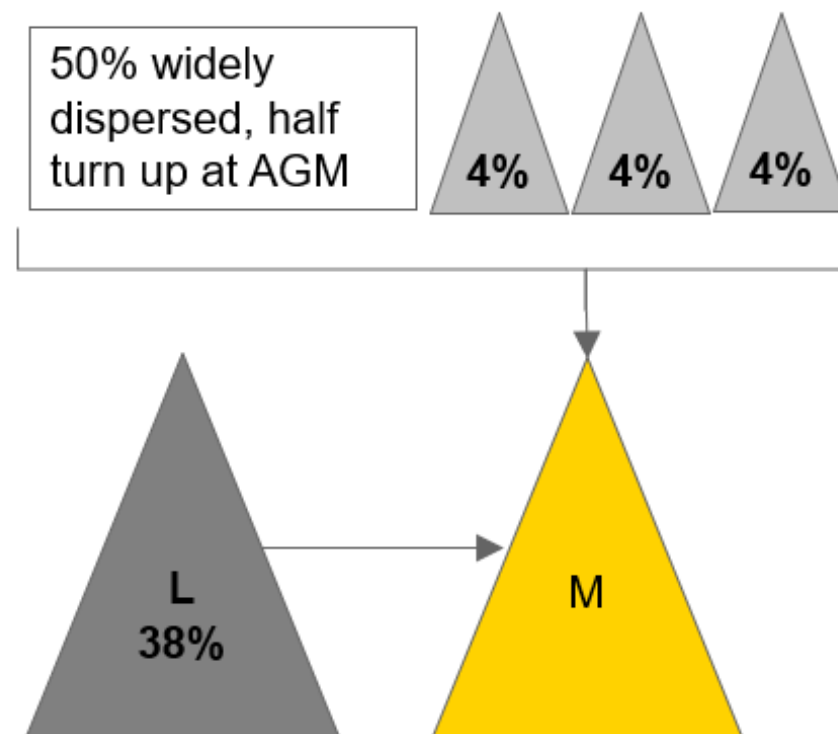


Assessing power – power with less than majority of voting rights (cont.)

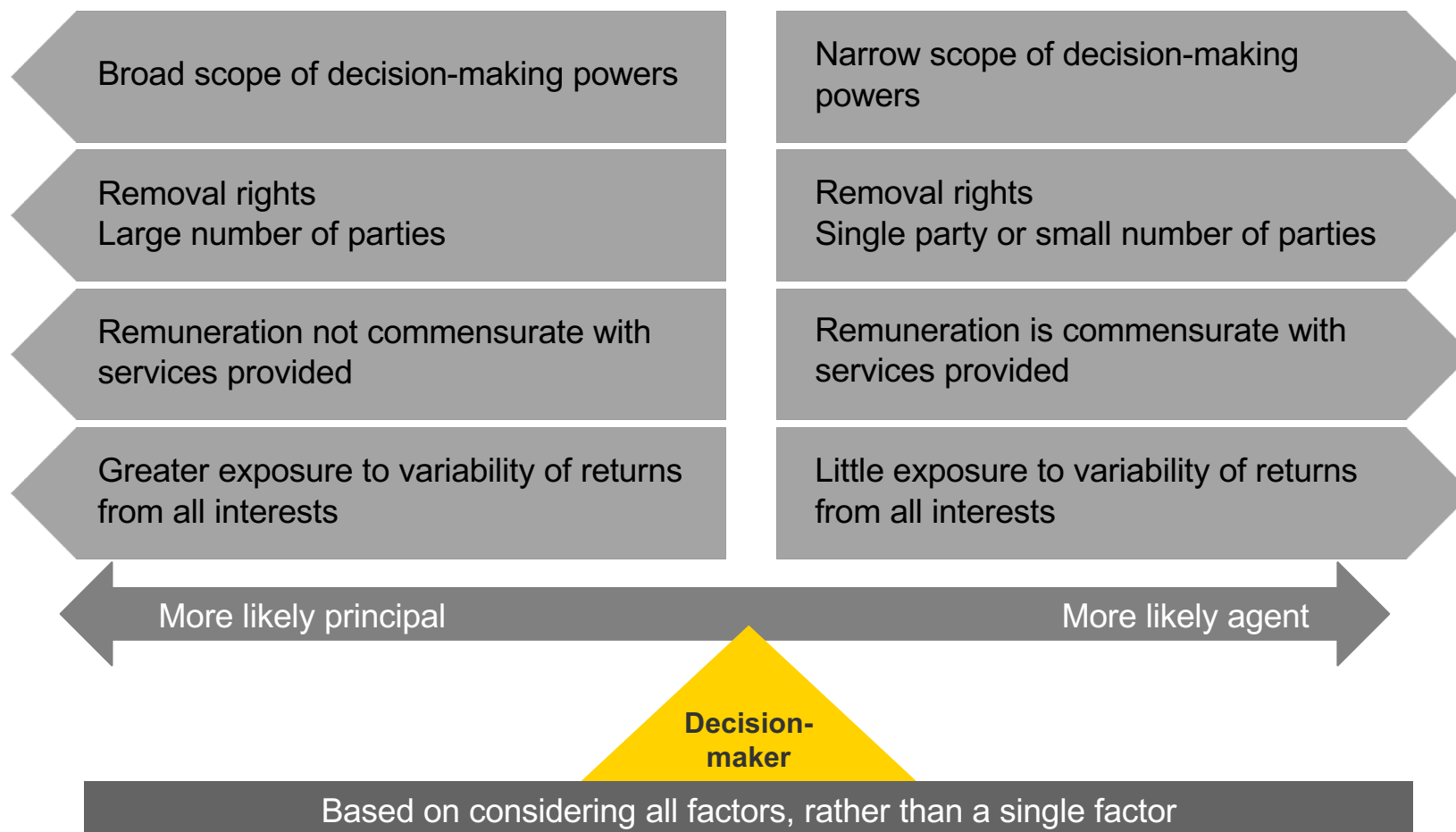
Example 5 – J does not have power



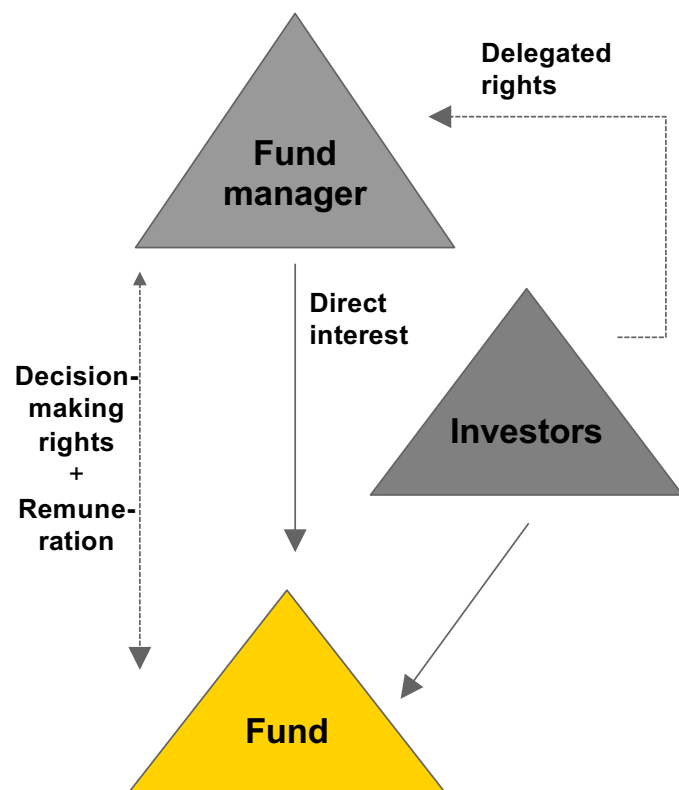
Example 6 – L has power



Delegated rights – link between power and returns



Delegated rights: examples



	Example 1	Example 2	Example 3
Decision-making scope	Broad discretion	Broad discretion	Broad discretion
Rights held by other investors	Simple majority, for cause	Simple majority, for cause	Independent board, for any reason
Remuneration	1% of NAV 20% of profits if hurdle rate achieved	1% of NAV 20% of profits if hurdle rate achieved	1% of NAV 20% of profits if hurdle rate achieved
Other interests	2% direct interest	20% direct interest	20% direct interest
Conclusion	Agent	Principal	Agent

Presentation of Consolidated Financial Statements

- ▶ **A parent SHALL present consolidated financial statements in which it consolidates its investments in subsidiaries**
- ▶ **A Parent NEED NOT present consolidated financial statements if and only if:**
 - a) the parent is a wholly-owned subsidiary, or is a partially-owned subsidiary and the minority shareholders do not object
 - b) the parent's debt or equity instruments are not traded in a public market, and
 - c) the ultimate or any intermediate parent produces consolidated financial statements available for public use that comply with IFRSs

END OF THE CONTROL CHAPTER



Non controlling interest (NCI)

- ▶ This represents the part of the group that is CONTROLLED but not owned by the shareholders of the Group!
- ▶ NCI shall be presented within equity, SEPARATELY from other items.
- ▶ NCI will be presented as a SINGLE line
- ▶ Previous name: minority interest

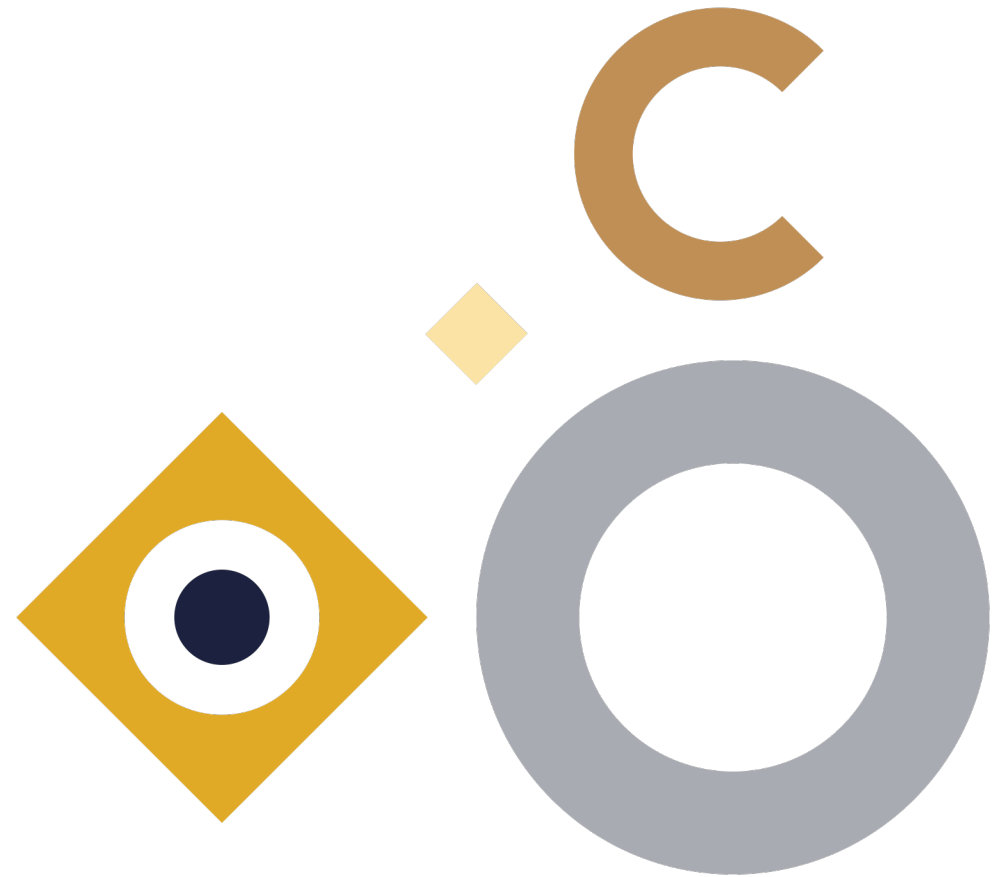
Scope of Consolidated Financial Statements

- ▶ The consolidated accounts must include ALL of the parent's subsidiaries/both domestic and foreign. There is NO exemption for:
 - ▶ a subsidiary whose business is of a different nature from the parent's. In this case, the subsidiary must be consolidated and information about the different business activities can be disclosed in accordance with *IFRS 8: Operating Segments*.
 - ▶ a subsidiary that operates under severe long-term restrictions impairing the subsidiary's ability to transfer funds to the parent.

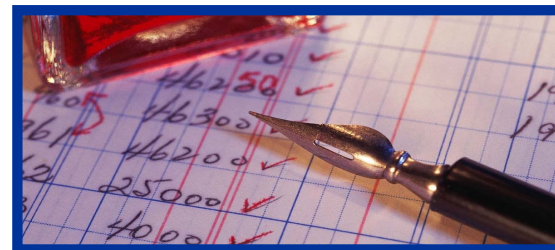
Scope of Consolidated Financial Statements

- ▶ Subsidiaries are NOT consolidated if the parent loses control
 - ▶ e.g. subsidiary becomes subject to the control of a government court, administrator or regulator or as a result of a contractual agreement
- ▶ If the parent decides to sell a subsidiary:
 - ▶ **The parent must continue to consolidate such a subsidiary until disposal date.**
 - ▶ If IFRS 5 criteria are met, the results will be shown as a discontinued operation in the consolidated income statement and assets and liabilities will be shown as "Held for Sale" in the consolidated balance sheet.

CONCEPTUAL RULES – WHAT TO DO?



Consolidation procedures



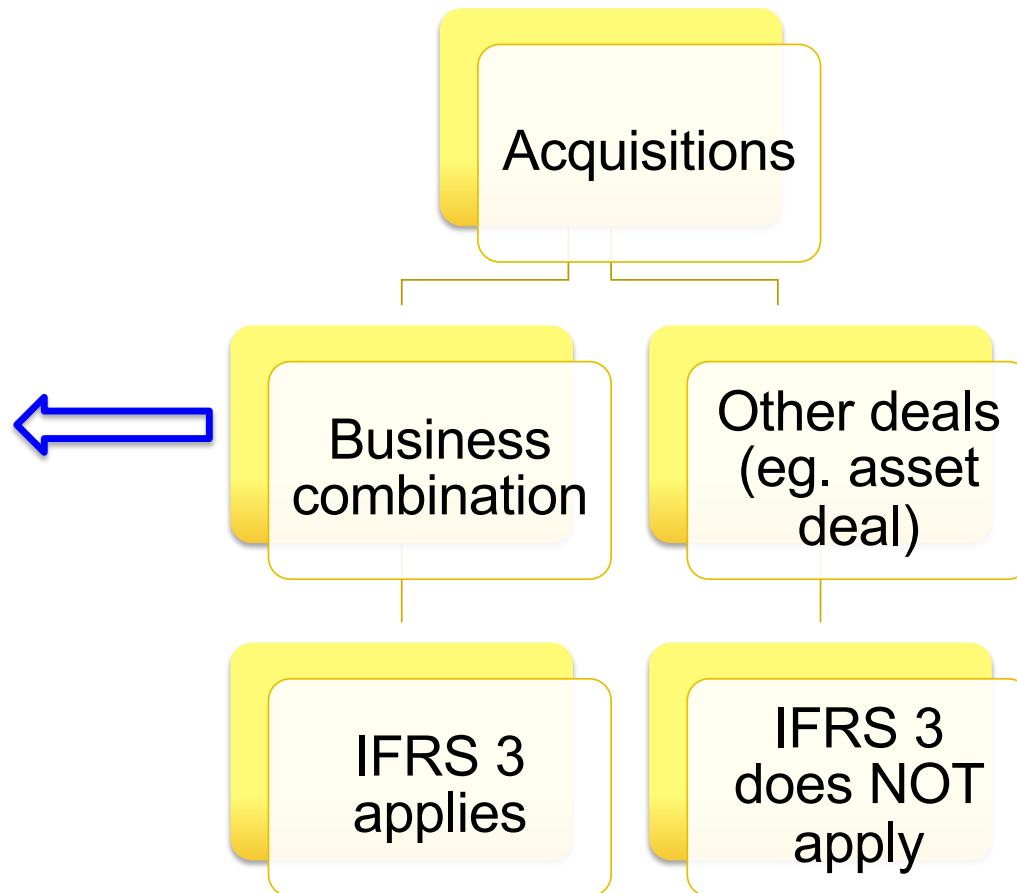
- ▶ Financial statements are combined on a line by line basis
- ▶ Consolidated financial statements should present financial information about the group as that of a single economic entity:
 - ▶ Eliminate the parent's investment in the subsidiary against their share of equity in the subsidiary
 - ▶ Non-controlling Interests are identified
 - ▶ Intra group balances, transactions, income and expenses shall be eliminated in full

Well, this is not too much... IFRS 10 is not a user friendly standard. All you know that you must eliminate in FULL!

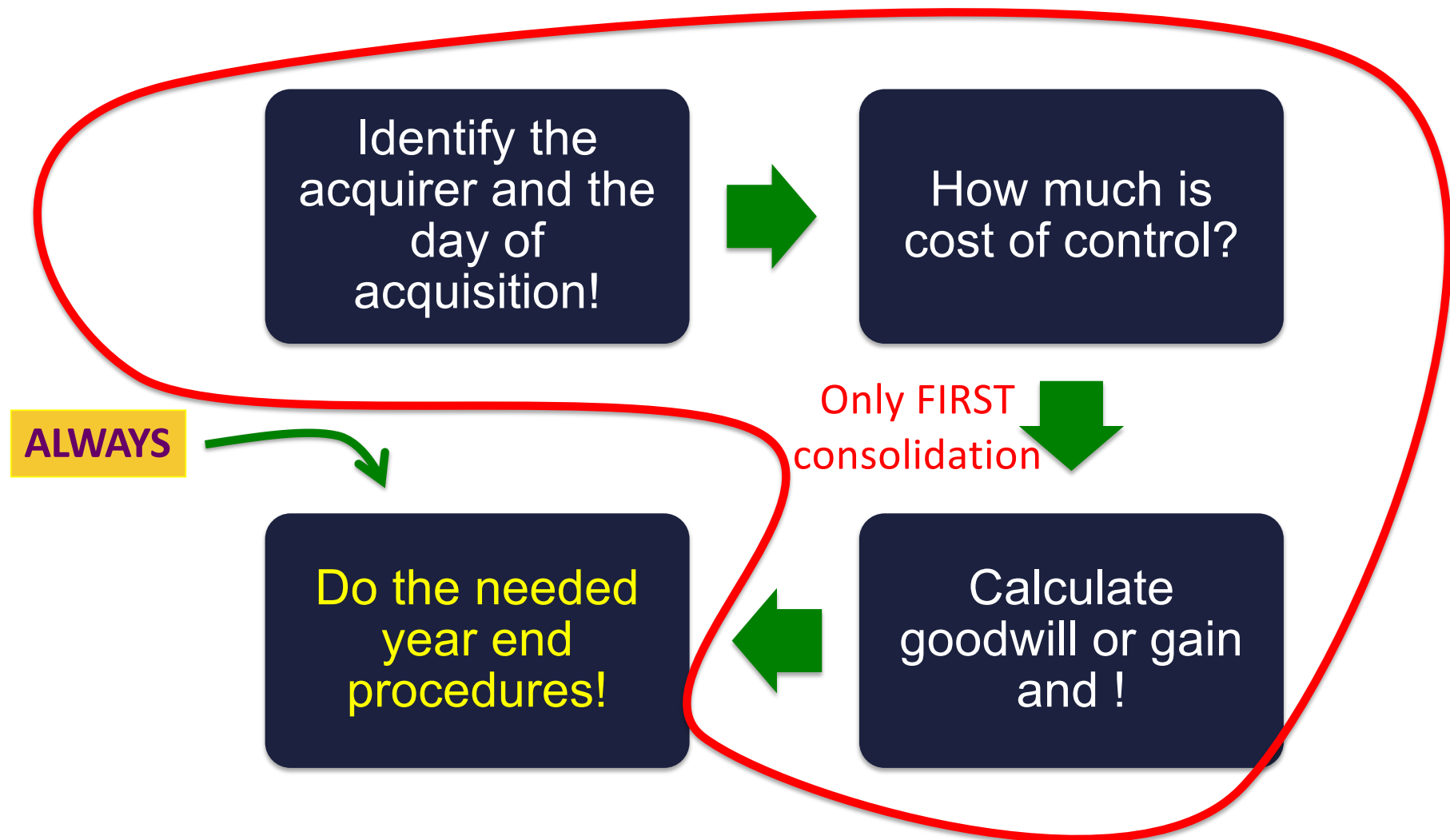
What is a business combination?

A transaction or other event in which an **acquirer obtains** control of one or more **businesses**.

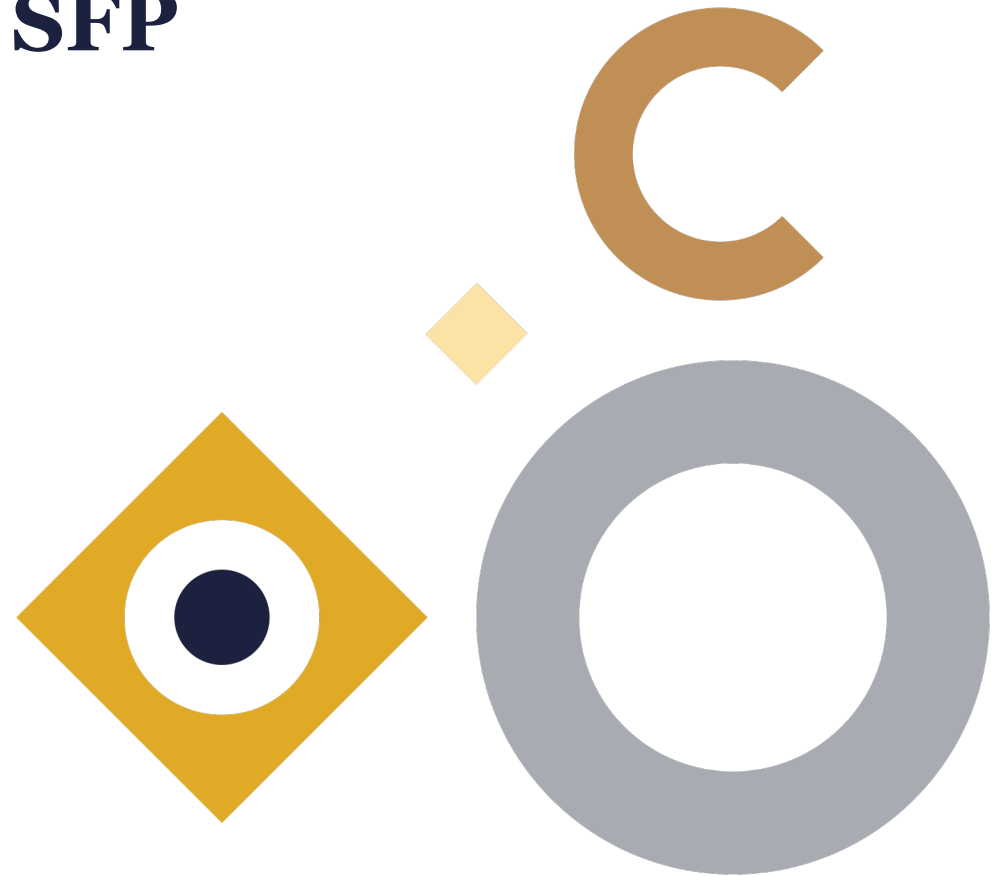
Transactions sometimes referred to as 'true mergers' or 'mergers of equals' are also **business combinations** as that term is used in **this IFRS**.



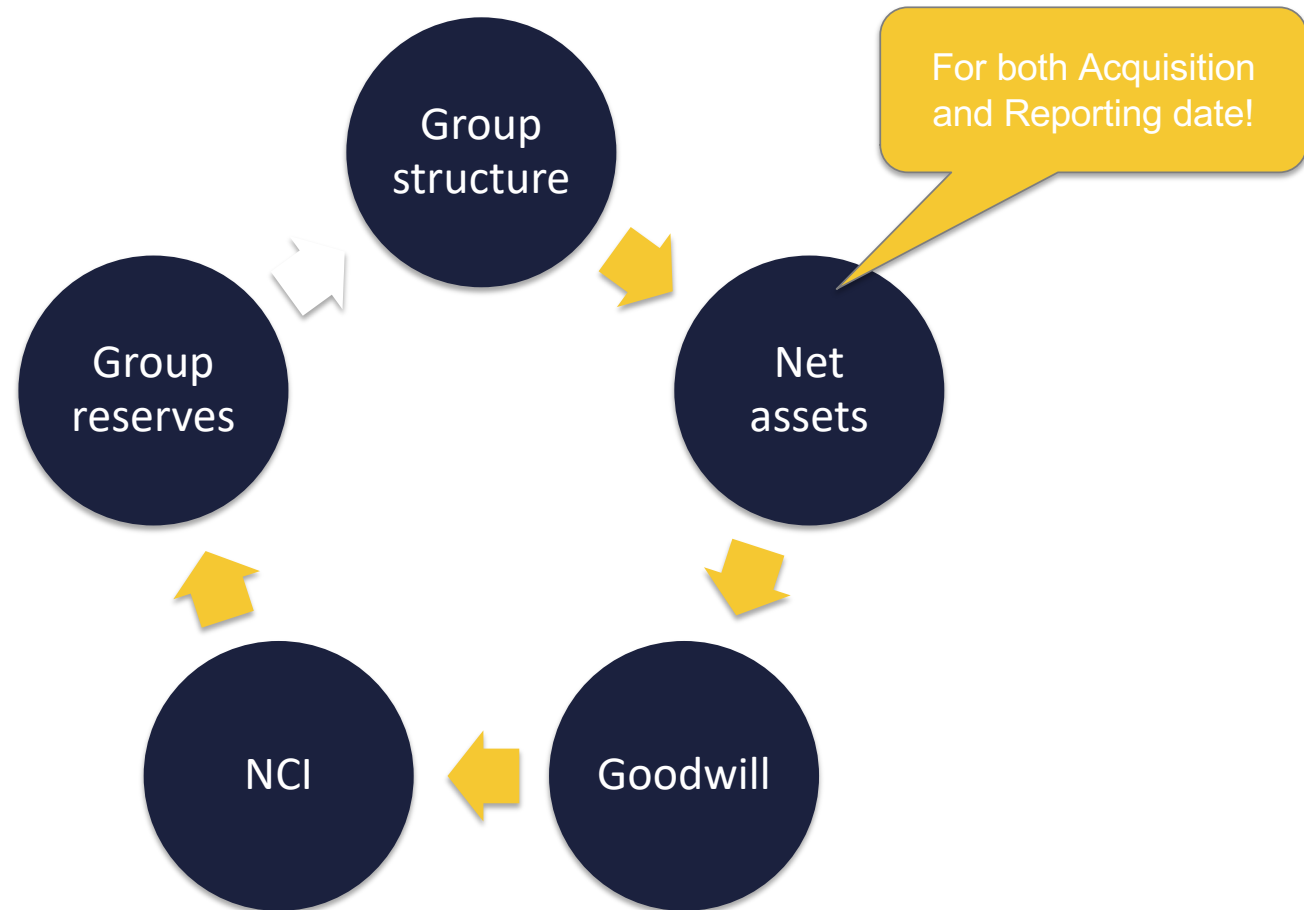
Okay, now I know what I have. What's next?



CONSOLIDATING THE SFP



So what I do at the end of every period?



W2: Net asset working

- ▶ This must be done for two dates:
 - ▶ acquisition date – this will never change
 - ▶ reporting date – this WILL change later
- ▶ For the first time ONLY! FRESH START ACCOUNTING MUST be used
- ▶ All the net assets of the acquired shall be revalued to fair value
- ▶ These new values will be included in the group statements (the previous numbers are disposed)
- ▶ The revaluation will effect the goodwill.

Yes, and what else? Those were just workings!

- ▶ Every entity has separate financial statement
- ▶ The separate statements are combined
- ▶ The intercompany balances are eliminated

- ▶ REMEMBER! Everything is mine, from the day I have acquired control
 - ▶ statement of financial position (include on reporting date)
 - ▶ statement of comprehensive income (include proportionally)
 - ▶ statement of changes in the equity (include proportionally)
 - ▶ statement of cash-flows (include proportionally)

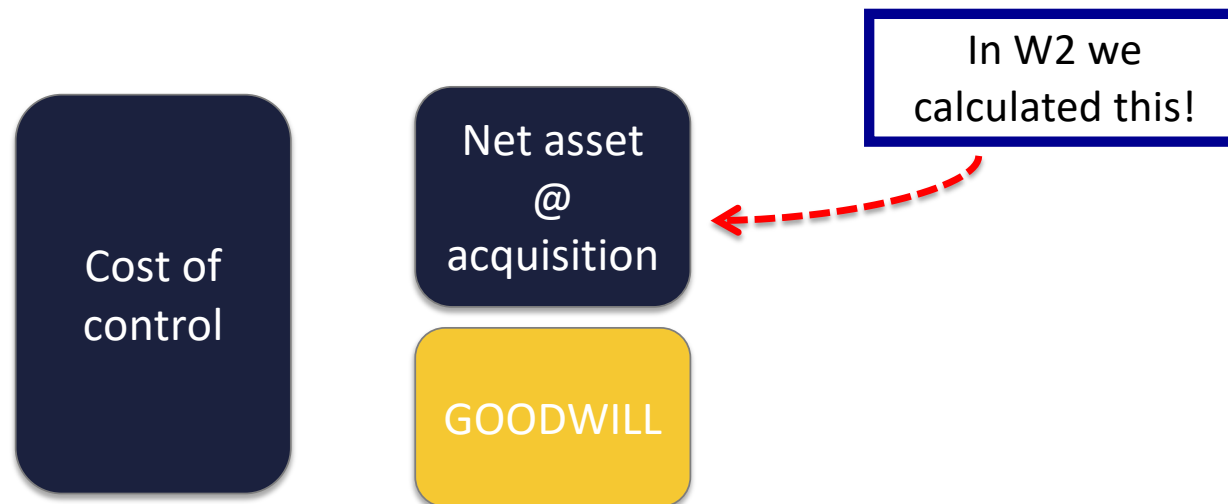
Group reserve working (W5)

NEVER EVER FORGET!
GROUP FINANCIAL STATEMENTS ARE DONE FOR THE
SHAREHOLDERS OF THE PARENT

- ▶ Issued capital (share capital): only the share capital of the parent [the same rule will apply to premium]
- ▶ Other reserves: they only belong to the group from the date of the acquisition (previous reserves are NOT group reserves)
- ▶ Investments of the consolidated companies may not be recognized in the group statement, since...

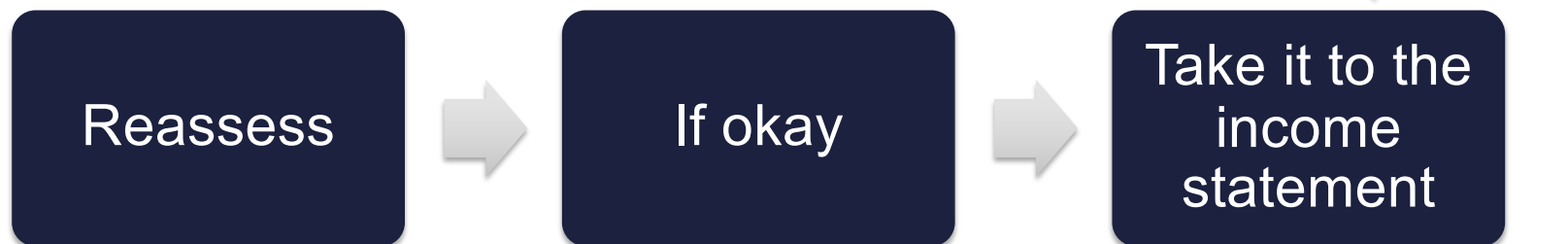
Calculation of the goodwill (W3)

- ▶ Goodwill: “An asset representing the future economic benefits arising from other assets acquired in a **business combination that are** not individually identified and separately recognised.” So you pay this in EXCESS – represents HOPE 😊
- ▶ Goodwill is then...



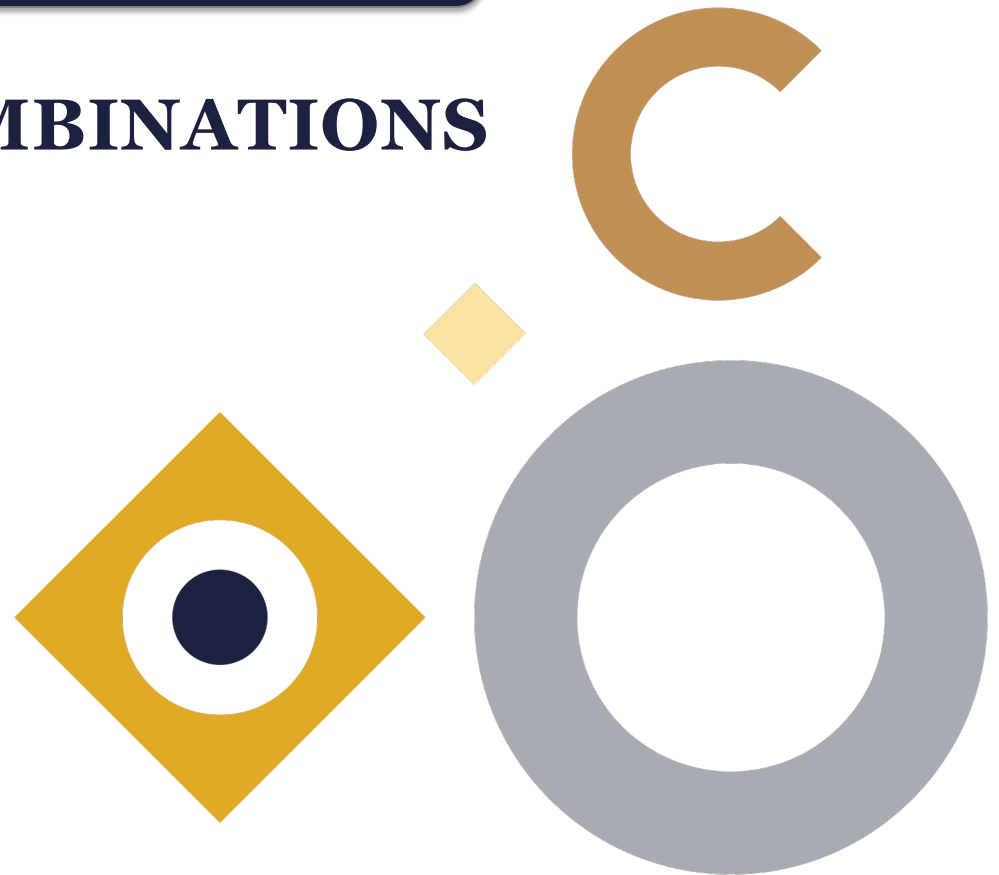
Goodwill (W3)

- ▶ What is included in cost of control?
 - ▶ cash payments
 - ▶ issued shares of the parent (be careful with pricing those shares)
 - ▶ other considerations (see later)
- ▶ May it be negative? Why not? What to do then?



A little detour...

FORMS OF BUSINESS COMBINATIONS



Legal Form of Combination

Merger

- ▶ Occurs when one corporation takes over all the operations of another business entity and that other entity is dissolved.

Consolidation

- ▶ Occurs when a new corporation is formed to take over the assets and operations of two or more separate business entities and dissolves the previously separate entities.

Mergers:

A + B = A

$$X + Y = X$$

Company A acquires the net assets of Company B for cash, other assets, or Company A debt/equity securities. Company B is dissolved; Company A survives with Company B's assets and liabilities.

Company X acquires the equity of Company Y from its shareholders for cash, other assets, or Company X debt/equity securities. Company Y is dissolved. Company X survives with Company Y's assets and liabilities.

Consolidations:

$$\mathbf{E + F = "D"}$$

$$\mathbf{K + L = "J"}$$

Company D is formed and acquires the net assets of companies E and F by issuing Company D stock. Companies E and F are dissolved. Company D survives with the assets and liabilities of both dissolved firms.

Company J is formed and acquires the shares of companies K and L from their respective shareholders by issuing Company J shares. Companies K and L are dissolved. Company J survives with the assets and liabilities of both firms.

Keeping the Terms Straight

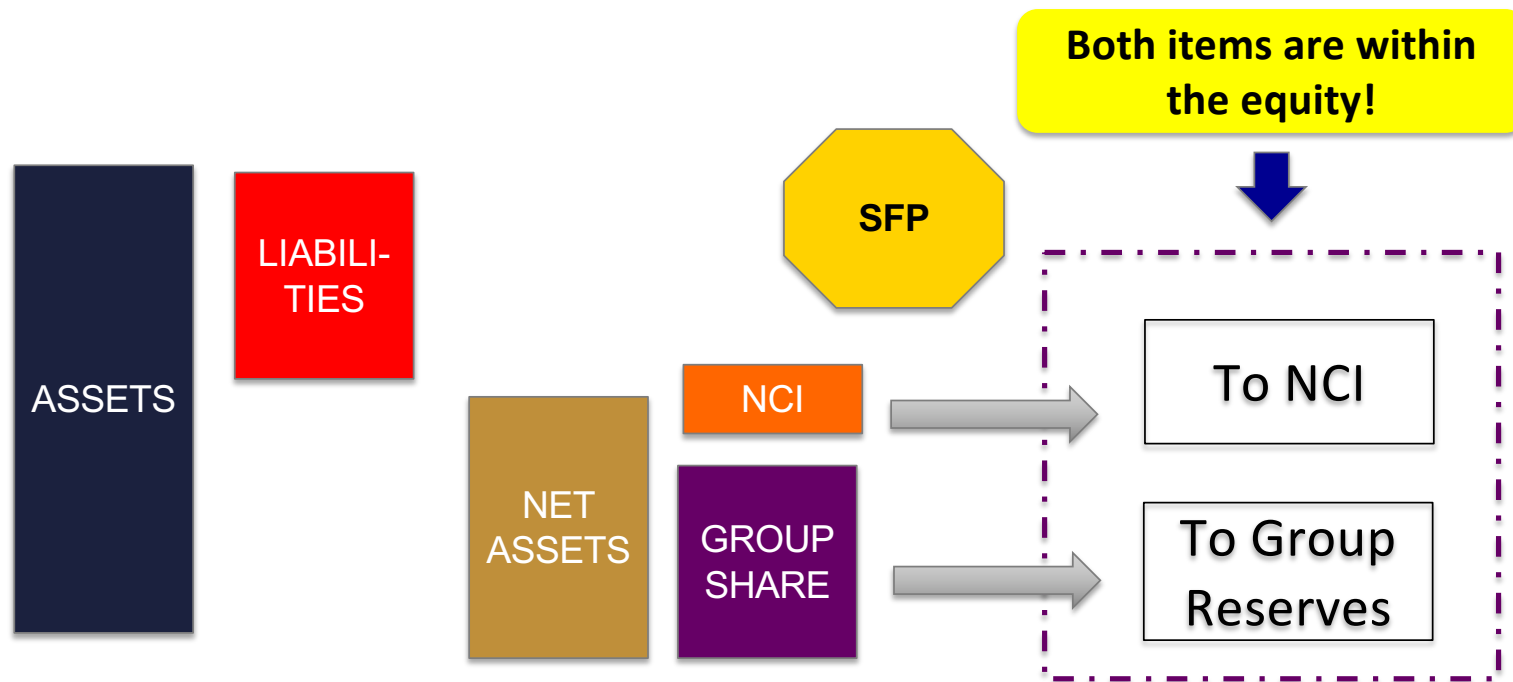
In the general business sense, mergers and consolidations are business combinations and may or may not involve the dissolution of the acquired firm(s).

Mergers and consolidations – at the very beginning – will involve only 100% acquisitions with the dissolution of the acquired firm(s). These assumptions will be relaxed in later.

“Consolidation” is also an accounting term used to describe the process of preparing consolidated financial statements for a parent and its subsidiaries.

Non controlling interest (NCI) (W4)

- ▶ The entity OWNs less than 100% but has the control (over all the net assets!)
- ▶ Remember! The SFP shows what assets and liabilities the group CONTROLS. The equity shows what the shareholders of the group OWN.



Intercompany trading – the rule

Intercompany items must be
eliminated IN FULL

BALANCES

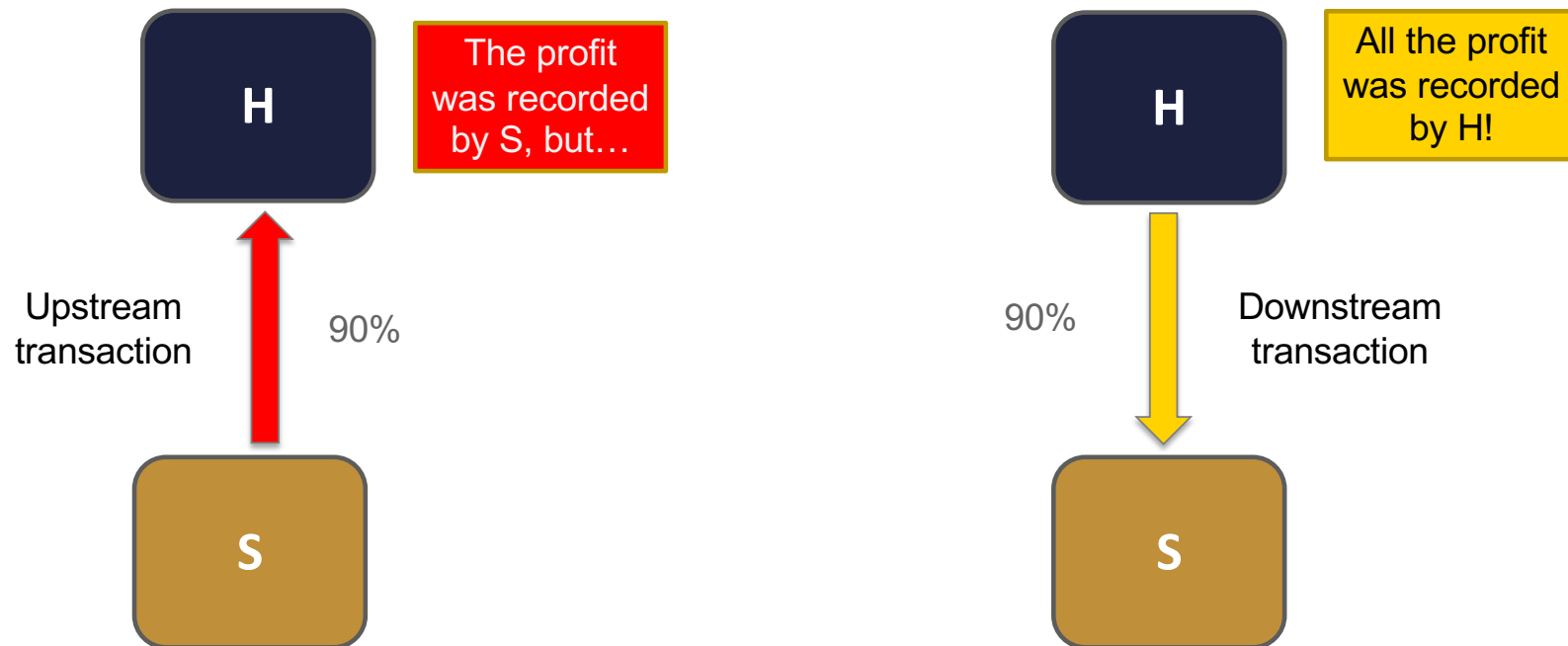
A/R, A/P, loans,
etc...

TRANSACTIONS

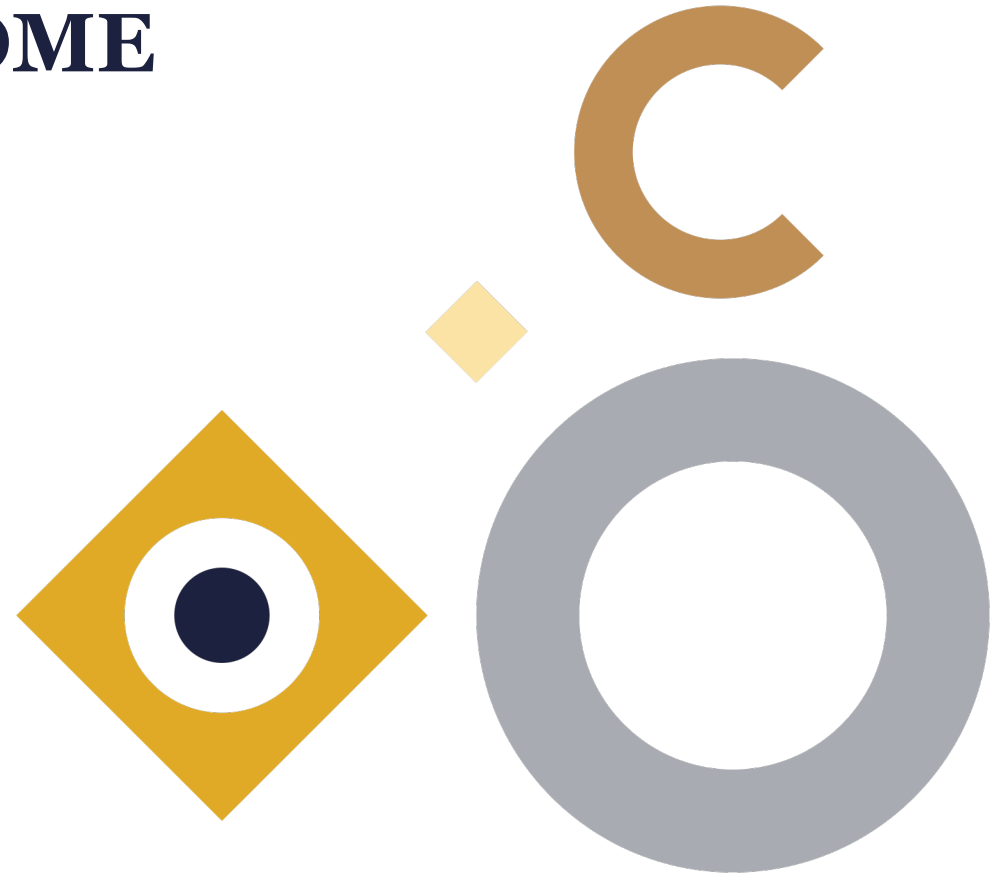
Profit, Items
“captured” in assets

Intercompany trading and the NCI

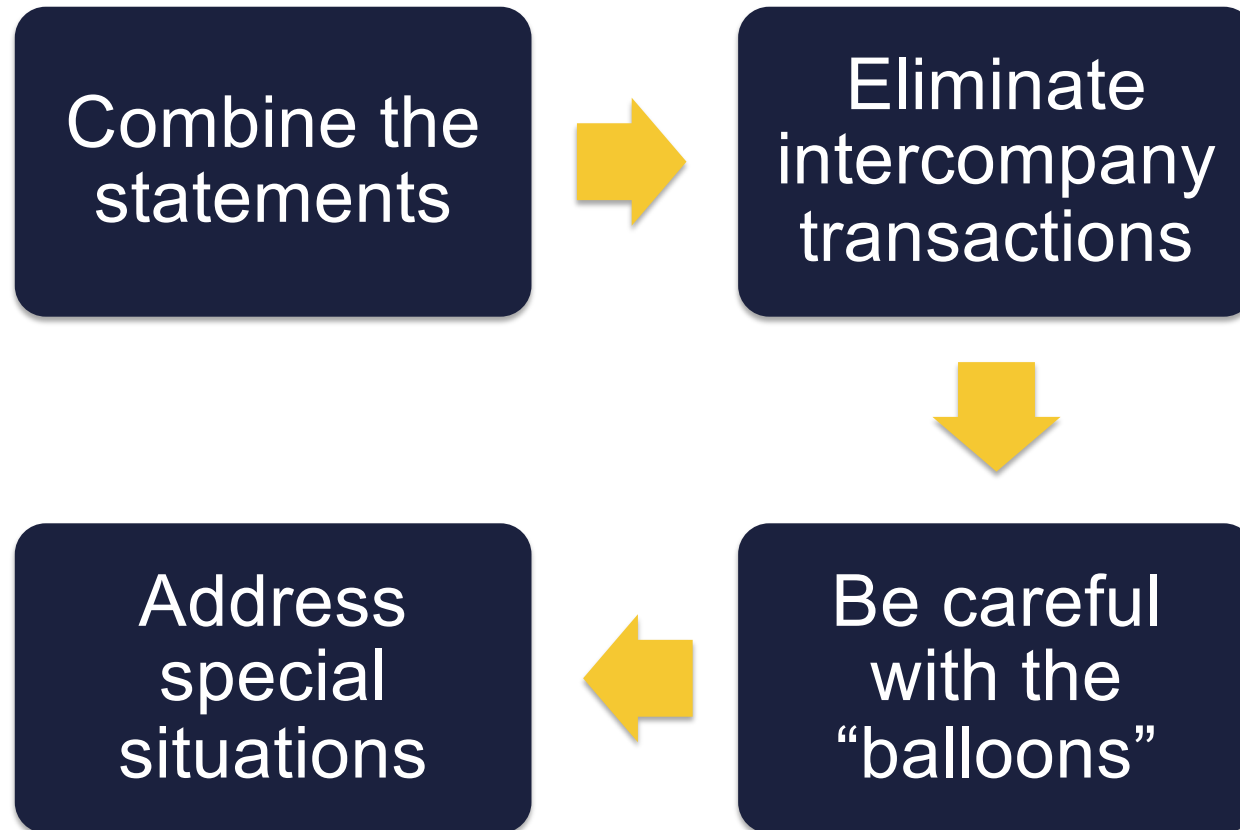
Assume that H has 90% of S. There was a transaction that led to 100 revenue which included 10 profit. Does it make a difference if the profit goes up or down?



CONSOLIDATING THE STATEMENT OF COMPREHENSIVE INCOME



Statement of incomes



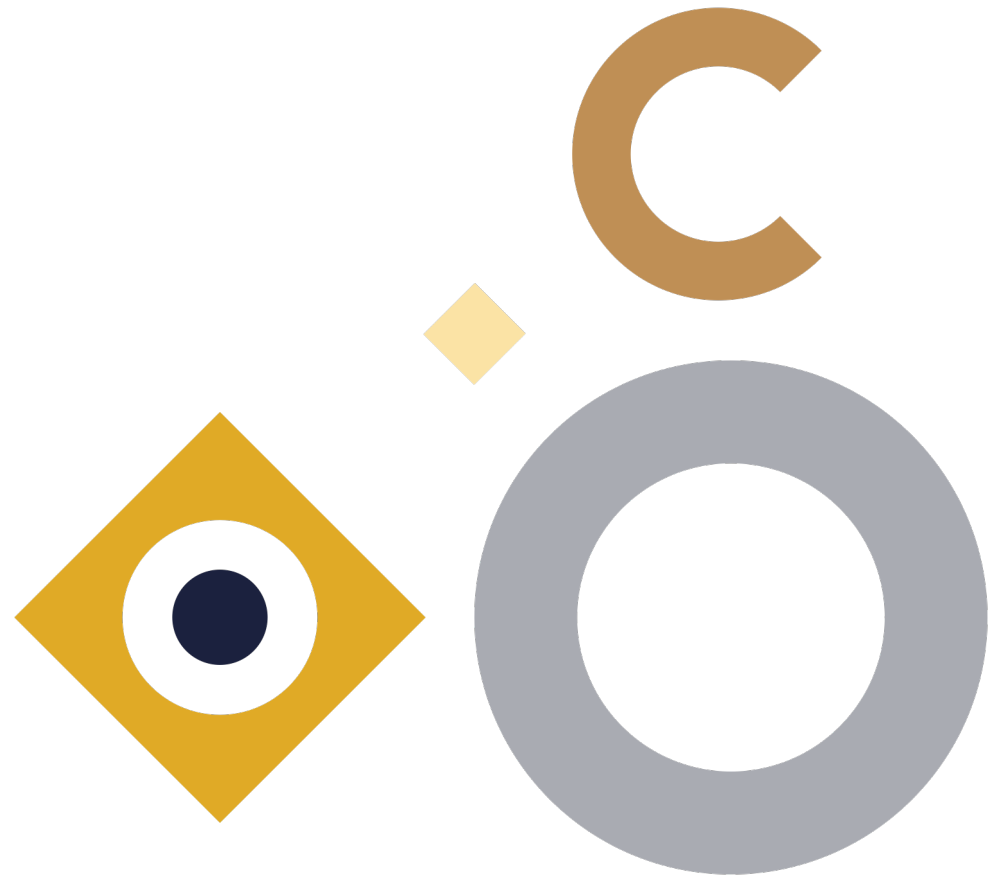
Special issues to address

▶ Dividends

- ▶ from the group's point of view dividend is ONLY the dividend of the Parent Company
- ▶ dividends from subsidiaries to the parent are eliminated (this is tricky!)
- ▶ dividends from S to NCI 😊 (leave me alone...)

▶ Midyear acquisitions

ADDITIONAL RULES





**Thank you for
your attention!**

Chapter 18

Capital Budgeting and Valuation with Leverage

18-1. Explain whether each of the following projects is likely to have risk similar to the average risk of the firm.

- a. **The Clorox Company considers launching a new version of Armor All designed to clean and protect notebook computers.**
 - b. **Google, Inc., plans to purchase real estate to expand its headquarters.**
 - c. **Target Corporation decides to expand the number of stores it has in the southeastern United States.**
 - d. **GE decides to open a new Universal Studios theme park in China.**
- a. While there may be some differences, the market risk of the cash flows from this new product is likely to be similar to Clorox's other household products. Therefore, it is reasonable to assume it has the same risk as the average risk of the firm.
- b. A real estate investment likely has very different market risk than Google's other investments in Internet search technology and advertising. It would not be appropriate to assume this investment as risk equal to the average risk of the firm.
- c. An expansion in the same line of business is likely to have risk equal to the average risk of the business.
- d. The theme park will likely be sensitive to the growth of the Chinese economy. Its market risk may be very different from GE's other division, and from the company as a whole. It would not be appropriate to assume this investment as risk equal to the average risk of the firm.

18-2. Suppose Caterpillar, Inc., has 665 million shares outstanding with a share price of \$74.77, and \$25 billion in debt. If in three years, Caterpillar has 700 million shares outstanding trading for \$83 per share, how much debt will Caterpillar have if it maintains a constant debt-equity ratio?

$$E = 665 \text{ million} \times \$74.77 = \$49.7 \text{ billion}, D = \$25 \text{ billion}, D/E = 25/49.722 = 0.503.$$

$$E = 700 \text{ million} \times \$83.00 = \$58.1 \text{ billion. Constant } D/E \text{ implies } D = 58.1 \times 0.503 = \$29.2 \text{ billion.}$$

18-3. In 2006, Intel Corporation had a market capitalization of \$112 billion, debt of \$2.2 billion, cash of \$9.1 billion, and EBIT of more than \$11 billion. If Intel were to increase its debt by \$1 billion and use the cash for a share repurchase, which market imperfections would be most relevant for understanding the consequence for Intel's value? Why?

Intel's debt is a tiny fraction of its total value. Indeed, Intel has more cash than debt, so its net debt is negative. Intel is also very profitable; at an interest rate of 6%, interest on Intel's debt is only \$132 million per year, which is less than 1.5% of its EBIT. Thus, the risk that Intel will default on its debt is extremely small. This risk will remain extremely small even if Intel borrows an additional \$1 billion.

Thus, adding debt will not really change the likelihood of financial distress for Intel (which is nearly zero), and thus will also not lead to agency conflicts. As a result, the most important financial friction for such a debt increase is the tax savings Intel would receive from the interest tax shield. A secondary issue may be the signaling impact of the transaction—borrowing to do a share repurchase is usually interpreted as a positive signal that management may view the shares to be underpriced.

- 18-4. Suppose Goodyear Tire and Rubber Company is considering divesting one of its manufacturing plants. The plant is expected to generate free cash flows of \$1.5 million per year, growing at a rate of 2.5% per year. Goodyear has an equity cost of capital of 8.5%, a debt cost of capital of 7%, a marginal corporate tax rate of 35%, and a debt-equity ratio of 2.6. If the plant has average risk and Goodyear plans to maintain a constant debt-equity ratio, what after-tax amount must it receive for the plant for the divestiture to be profitable?**

We can compute the levered value of the plant using the WACC method. Goodyear's WACC is

$$r_{\text{wacc}} = \frac{1}{1+2.6} 8.5\% + \frac{2.6}{1+2.6} 7\%(1-0.35) = 5.65\%.$$

$$\text{Therefore, } V^L = \frac{1.5}{0.0565 - 0.025} = \$47.6 \text{ million}$$

A divestiture would be profitable if Goodyear received more than \$47.6 million after tax.

- 18-5. Suppose Lucent Technologies has an equity cost of capital of 10%, market capitalization of \$10.8 billion, and an enterprise value of \$14.4 billion. Suppose Lucent's debt cost of capital is 6.1% and its marginal tax rate is 35%.**

a. What is Lucent's WACC?

b. If Lucent maintains a constant debt-equity ratio, what is the value of a project with average risk and the following expected free cash flows?

Year	0	1	2	3
FCF	-100	50	100	70

c. If Lucent maintains its debt-equity ratio, what is the debt capacity of the project in part (b)?

a. $r_{\text{wacc}} = \frac{10.8}{14.4} 10\% + \frac{14.4 - 10.8}{14.4} 6.1\%(1 - 0.35) = 8.49\%$

b. Using the WACC method, the levered value of the project at date 0 is

$$V^L = \frac{50}{1.0849} + \frac{100}{1.0849^2} + \frac{70}{1.0849^3} = 185.86.$$

Given a cost of 100 to initiate, the project's NPV is $185.86 - 100 = 85.86$.

c. Lucent's debt-to-value ratio is $d = (14.4 - 10.8) / 14.4 = 0.25$. The project's debt capacity is equal to d times the levered value of its remaining cash flows at each date.

Year	0	1	2	3
FCF	-100	50	100	70
VL	185.86	151.64	64.52	0
D = d*VL	46.47	37.91	16.13	0.00

- 18-6. Acort Industries has 10 million shares outstanding and a current share price of \$40 per share. It also has long-term debt outstanding. This debt is risk free, is four years away from maturity, has annual coupons with a coupon rate of 10%, and has a \$100 million face value. The first of the remaining coupon payments will be due in exactly one year. The riskless interest rates for all**

maturities are constant at 6%. Acort has EBIT of \$106 million, which is expected to remain constant each year. New capital expenditures are expected to equal depreciation and equal \$13 million per year, while no changes to net working capital are expected in the future. The corporate tax rate is 40%, and Acort is expected to keep its debt-equity ratio constant in the future (by either issuing additional new debt or buying back some debt as time goes on).

a. Based on this information, estimate Acort's WACC.

b. What is Acort's equity cost of capital?

- a. We don't know Acort's equity cost of capital, so we cannot calculate WACC directly. However, we can compute it indirectly by estimating the discount rate that is consistent with Acort's market value. First, $E = 10 \times 40 = \$400$ million. The market value of Acort's debt is

$$D = 10 \times \frac{1}{0.06} \left(1 - \frac{1}{1.06^4} \right) + \frac{100}{1.06^4} = \$113.86 \text{ million.}$$

Therefore, Acort's enterprise value is $E + D = 400 + 113.86 = 513.86$.

Acort's $\text{FCF} = \text{EBIT} \times (1 - \tau_c) + \text{Dep} - \text{Capex} - \text{Inc in NWC}$

$$\text{FCF} = 106 \times (1 - 0.40) = 63.6$$

Because Acort is not expected to grow,

$$V^L = 513.86 = \frac{63.6}{r_{\text{wacc}}} \text{ and so } r_{\text{wacc}} = \frac{63.6}{513.86} = 12.38\%.$$

$$\text{b. Using } r_{\text{wacc}} = \frac{E}{E + D} r_E + \frac{D}{D + E} r_D (1 - \tau_c),$$

$$12.38\% = \frac{400}{513.86} r_E + \frac{113.86}{513.86} 6\% (1 - 0.40)$$

solving for r_E :

$$r_E = \frac{513.86}{400} \left[12.38\% - \frac{113.86}{513.86} 6\% (1 - 0.40) \right] = 14.88\%.$$

18-7. Suppose Goodyear Tire and Rubber Company has an equity cost of capital of 8.5%, a debt cost of capital of 7%, a marginal corporate tax rate of 35%, and a debt-equity ratio of 2.6. Suppose Goodyear maintains a constant debt-equity ratio.

a. What is Goodyear's WACC?

b. What is Goodyear's unlevered cost of capital?

c. Explain, intuitively, why Goodyear's unlevered cost of capital is less than its equity cost of capital and higher than its WACC.

$$\text{a. } r_{\text{wacc}} = \frac{1}{1 + 2.6} 8.5\% + \frac{2.6}{1 + 2.6} 7\% (1 - 0.35) = 5.65\%$$

b. Because Goodyear maintains a target leverage ratio, we can use Eq. 18.6:

$$r_U = \frac{1}{1 + 2.6} 8.5\% + \frac{2.6}{1 + 2.6} 7\% = 7.42\%.$$

c. Goodyear's equity cost of capital exceeds its unlevered cost of capital because leverage makes equity riskier than the overall firm. Goodyear's WACC is less than its unlevered cost of capital because the WACC includes the benefit of the interest tax shield.

18-8. You are a consultant who was hired to evaluate a new product line for Markum Enterprises. The upfront investment required to launch the product line is \$10 million. The product will generate free cash flow of \$750,000 the first year, and this free cash flow is expected to grow at a rate of 4% per year. Markum has an equity cost of capital of 11.3%, a debt cost of capital of 5%, and a tax rate of 35%. Markum maintains a debt-equity ratio of 0.40.

- What is the NPV of the new product line (including any tax shields from leverage)?
- How much debt will Markum initially take on as a result of launching this product line?
- How much of the product line's value is attributable to the present value of interest tax shields?

a. $WACC = (1 / 1.4)(11.3\%) + (.4 / 1.4)(5\%)(1 - .35) = 9\%$

$$V^L = 0.75 / (9\% - 4\%) = \$15 \text{ million}$$

$$NPV = -10 + 15 = \$5 \text{ million}$$

b. Debt-to-Value ratio is $(0.4) / (1.4) = 28.57\%$.

$$\text{Therefore Debt is } 28.57\% \times \$15 \text{ million} = \$4.29 \text{ million.}$$

c. Discounting at r_u gives unlevered value. $r_u = (1 / 1.4)11.3\% + (.4 / 1.4)5\% = 9.5\%$

$$V^u = 0.75 / (9.5\% - 4\%) = \$13.64 \text{ million}$$

$$\text{Tax shield value is therefore } 15 - 13.64 = 1.36 \text{ million.}$$

Alternatively, initial debt is \$4.29 million, for a tax shield in the first year of $4.29 \times 5\% \times 0.35 = 0.075$ million. Then $PV(ITS) = 0.075 / (9.5\% - 4\%) = 1.36$ million.

Alternatively, initial debt is \$4.29 million, for a tax shield in the first year of $4.29 \times 5\% \times 0.35 = 0.075$ million. Then $PV(ITS) = 0.075 / (9.5\% - 4\%) = 1.36$ million.

18-9. Consider Lucent's project in Problem 5.

- What is Lucent's unlevered cost of capital?
- What is the unlevered value of the project?
- What are the interest tax shields from the project? What is their present value?
- Show that the APV of Lucent's project matches the value computed using the WACC method.

a. $r_U = \frac{10.8}{14.4}10\% + \frac{14.4 - 10.8}{14.4}6.1\% = 9.025\%$

b. $V^U = \frac{50}{1.09025} + \frac{100}{1.09025^2} + \frac{70}{1.09025^3} = 184.01$

- c. Using the results from problem 5(c):

Year	0	1	2	3
FCF	-100	50	100	70
VL	185.86	151.64	64.52	0
D = d*VL	46.47	37.91	16.13	0.00
Interest		2.83	2.31	0.98
Tax Shield		0.99	0.81	0.34

The present value of the interest tax shield is

$$PV(ITS) = \frac{0.99}{1.09025} + \frac{0.81}{1.09025^2} + \frac{0.34}{1.09025^3} = 1.85$$

d. $V^L = APV = 184.01 + 1.85 = 185.86$

This matches the answer in problem 5.

18-10. Consider Lucent's project in Problem 5.

- a. What is the free cash flow to equity for this project?
 - b. What is its NPV computed using the FTE method? How does it compare with the NPV based on the WACC method?
- a. Using the debt capacity calculated in problem 5, we can compute FCFE by adjusting FCF for after-tax interest expense ($D \times r_D \times (1 - t_c)$) and net increases in debt ($D_t - D_{t-1}$).

Year	0	1	2	3
D	46.47	37.91	16.13	0.00
FCF	-\$100.00	\$50.00	\$100.00	\$70.00
After-tax Interest Exp.	\$0.00	-\$1.84	-\$1.50	-\$0.64
Inc. in Debt	\$46.47	-\$8.55	-\$21.78	-\$16.13
FCFE	-\$53.53	\$39.60	\$76.72	\$53.23

b. $NPV = -53.53 + \frac{39.60}{1.10} + \frac{76.72}{1.10^2} + \frac{53.23}{1.10^3} = \85.86

18-11. In year 1, AMC will earn \$2000 before interest and taxes. The market expects these earnings to grow at a rate of 3% per year. The firm will make no net investments (i.e., capital expenditures will equal depreciation) or changes to net working capital. Assume that the corporate tax rate equals 40%. Right now, the firm has \$5000 in risk-free debt. It plans to keep a constant ratio of debt to equity every year, so that on average the debt will also grow by 3% per year. Suppose the risk-free rate equals 5%, and the expected return on the market equals 11%. The asset beta for this industry is 1.11.

- a. If AMC were an all-equity (unlevered) firm, what would its market value be?
 - b. Assuming the debt is fairly priced, what is the amount of interest AMC will pay next year? If AMC's debt is expected to grow by 3% per year, at what rate are its interest payments expected to grow?
 - c. Even though AMC's debt is *riskless* (the firm will not default), the future growth of AMC's debt is uncertain, so the exact amount of the future interest payments is risky. Assuming the future interest payments have the same beta as AMC's assets, what is the present value of AMC's interest tax shield?
 - d. Using the APV method, what is AMC's total market value, V^L ? What is the market value of AMC's equity?
 - e. What is AMC's WACC? (*Hint: Work backward from the FCF and V^L .*)
 - f. Using the WACC method, what is the expected return for AMC equity?
 - g. Show that the following holds for AMC: .[SHERYL: there's an equation that should be set here, but I can't get it out of the PDF in correct for. It's on page 631].
 - h. Assuming that the proceeds from any increases in debt are paid out to equity holders, what cash flows do the equity holders expect to receive in one year? At what rate are those cash flows expected to grow? Use that information plus your answer to part (f) to derive the market value of equity using the FTE method. How does that compare to your answer in part (d)?
- a. AMC has unlevered FCF of $\$2,000 \times 0.6 = \$1,200$.

From the CAPM, AMC's unlevered cost of capital is $5\% + 1.11 \times (11\% - 5\%) = 11.66\%$.

Discounting the FCF as a growing perpetuity tells us that the value of the firm, assuming growth of 3%, is:

$$V(\text{All Equity}) = \frac{\$1,200}{0.1166 - 0.03} = \$13,857.$$

- b. Since the debt is risk-free, the interest rate paid on it must equal the risk-free rate of 5% (or else there would be an arbitrage opportunity). The firm has \$5,000 of debt next year. The interest payment will be 5% of that, or \$250. If the debt grows by 3% per year, so will the interest payments.
- c. The expected value of next year's tax shield will be $\$250 \times 40\% = \100 , and it will grow (with the growth of the debt) at a rate of 3%. But the exact amount of the tax shield is uncertain, since AMC may add new debt or repay some debt during the year, depending on their cash flows. This makes the actual amount of the tax shield risky (even though the debt itself is not). Since the beta of the tax shield due to debt is 1.11, the appropriate discount rate is $5\% + 1.11(11\% - 5\%) = 11.67\%$. We can now use the growing perpetuity formula and conclude that

$$PV(\text{Interest Tax Shields}) = \frac{\$100}{0.1166 - 0.03} = \$1,155.$$

- d. The APV tells us that the value of a firm with debt equals the sum of the value of an all equity firm and the tax shield. From previous work (parts (a) and (c)), we get:

$$V(\text{AMC}) = \$13,857 + \$1,155 = \$15,012.$$

The market value of the equity is therefore $V - D = \$15,012 - \$5,000 = \$10,012$.

- e. Next year's FCF is $\$2,000 \times 0.6 = \$1,200$. It is expected to grow at 3%, so the WACC must satisfy:

$$V(\text{AMC}) = \frac{\$1,200}{r_{\text{wacc}} - 0.03} = \$15,000.$$

Solving for the WACC, we get $\text{WACC} = 11\%$.

- f. By definition, $r_{\text{wacc}} = \frac{E}{V} \times r_E + \frac{D}{V} \times r_D \times (1 - \tau_c)$.

The return on the debt is 5%; the value of the debt is \$5,000, the value of the firm is \$15,000 and therefore the value of the equity is $\$15,000 - \$5,000 = \$10,000$. Plugging into the above expression, we get:

$$11\% = \frac{\$10,000}{\$15,000} \times r_E + \frac{\$5,000}{\$15,000} \times 5\% \times (1 - 0.4) \Rightarrow r_E = 15\%.$$

- g. From the CAPM, β_E must satisfy $15\% = 5\% + \beta_E(11\% - 5\%)$, so we conclude $\beta_E = 1.66$.

The relationship holds since $(\$10,000/\$15,000) \times 1.66 = 1.11$, and the beta of the debt equals 0.

- h. The debt is expected to increase to $\$5,000 \times (1 + 0.03) = \$5,150$, so the equity holders will get \$150 due to the increase in debt. These proceeds will increase by 3% annually. (The second-year debt will be $\$5,000 \times (1 + 0.03)^2 = \$5,304.5$, with an increase in debt of \$154.5, 3% higher than the \$150 proceeds of year 1.) The expected FCF to equity at the end of the first year is therefore $\text{EBIT} - \text{Interest} - \text{Taxes} + \text{Debt proceeds}$, or $\text{FCFE} = (2000 - 250) \times (1 - .40) + 150 = \1200 .

This cash flow is expected to grow at 3% per year. Thus, another way to compute the value of equity is to discount these cash flows directly at the MCR for the equity of 15% (from (f)):

$$E = \frac{FCFE}{r_E - g} = \frac{1200}{15\% - 3\%} = 10,000.$$

This is the same value we computed in (d), using the APV.

- 18-12. Prokter and Gramble (PG)** has historically maintained a debt-equity ratio of approximately 0.20. Its current stock price is \$50 per share, with 2.5 billion shares outstanding. The firm enjoys very stable demand for its products, and consequently it has a low equity beta of 0.50 and can borrow at 4.20%, just 20 basis points over the risk-free rate of 4%. The expected return of the market is 10%, and PG's tax rate is 35%.

- a. This year, PG is expected to have free cash flows of \$6.0 billion. What constant expected growth rate of free cash flow is consistent with its current stock price?
- b. PG believes it can increase debt without any serious risk of distress or other costs. With a higher debt-equity ratio of 0.50, it believes its borrowing costs will rise only slightly to 4.50%. If PG announces that it will raise its debt-equity ratio to 0.5 through a leveraged recap, determine the increase in the stock price that would result from the anticipated tax savings.

a. $E = \$50 \times 2.5 \text{ B} = \125 B

$$D = 0.20 \times 125 \text{ B} = \$25 \text{ B}$$

$$V^L = E + D = \$150 \text{ B}$$

$$\text{From CAPM: Equity Cost of Capital} = 4\% + 0.5(10\% - 4\%) = 7\%$$

$$\text{WACC} = (125 / 150) 7\% + (25 / 150) 4.2\% (1 - 35\%) = 6.29\%$$

$$V^L = FCF / (r_{\text{wacc}} - g) \Rightarrow g = r_{\text{wacc}} - FCF / V = 6.29\% - 6 / 150 = 2.29\%$$

b. Initial Unlevered cost of capital (Eq. 18.6) $= (125 / 150) 7\% + (25 / 150) 4.2\% = 6.53\%$

$$\text{New Equity cost of capital (Eq. 18.10)} = 6.53\% + (.5)(6.53\% - 4.5\%) = 7.55\%$$

$$\text{New WACC} = (1 / 1.5) 7.55\% + (.5 / 1.5) 4.5\% (1 - 35\%) = 6.01\%$$

$$V^L = FCF / (r_{\text{wacc}} - g) = 6.0 / (6.01\% - 2.29\%) = 161.29$$

This is a gain of $161.29 - 150 = \$11.29 \text{ B}$ or $11.29 / 2.5 = \$4.52$ per share.

Thus, share price rises to \$54.52/share.

- 18-13. Amarindo, Inc. (AMR)**, is a newly public firm with 10 million shares outstanding. You are doing a valuation analysis of AMR. You estimate its free cash flow in the coming year to be \$15 million, and you expect the firm's free cash flows to grow by 4% per year in subsequent years. Because the firm has only been listed on the stock exchange for a short time, you do not have an accurate assessment of AMR's equity beta. However, you do have beta data for UAL, another firm in the same industry:

	Equity Beta	Debt Beta	Debt-Equity Ratio
UAL	1.5	0.30	1

AMR has a much lower debt-equity ratio of 0.30, which is expected to remain stable, and its debt is risk free. AMR's corporate tax rate is 40%, the risk-free rate is 5%, and the expected return on the market portfolio is 11%.

- a. Estimate AMR's equity cost of capital.
- b. Estimate AMR's share price.
- a. From Eq. 14.9, UAL Asset beta $= (1/2) 1.5 + (1/2) 0.3 = 0.90$

We can use this for AMR's asset beta.

To derive the equity beta, since AMR's debt is risk free we have (Eq. 14.10):

$$\text{Equity Beta} = \text{Asset Beta} \times (1 + D/E) = 0.9 \times 1.30 = 1.17.$$

From the SML

$$r_e = 5\% + 1.17(11\% - 5\%) = 12.02\%.$$

Alternatively, given an asset or unlevered beta of 0.90 for AMR, we have (from SML):

$$r_u = 5\% + 0.90(11\% - 5\%) = 10.4\%.$$

Then we can solve for r_e using Eq. 18.10:

$$r_e = 10.4\% + 0.30(10.4\% - 5\%) = 12.02\%.$$

- b. Since D/E ratio is stable, we can value AMR using the WACC approach.

$$\text{WACC} = (1/1.3) 12.02\% + (.3/1.3) 5\% (1 - 40\%) = 9.94\%$$

Levered value of AMR (as a constant growth perpetuity):

$$D + E = V^L = \text{FCF} / (r_{\text{wacc}} - g) = 15 / (9.94\% - 4\%) = \$252.52 \text{ million}$$

$$E = (E / (D + E)) \times V^L = 252.52 / 1.3 = \$194.25 \text{ million}$$

$$\text{Share price} = 194.25 / 10 = \$19.43$$

- 18-14. Remex (RMX) currently has no debt in its capital structure. The beta of its equity is 1.50. For each year into the indefinite future, Remex's free cash flow is expected to equal \$25 million. Remex is considering changing its capital structure by issuing debt and using the proceeds to buy back stock. It will do so in such a way that it will have a 30% debt-equity ratio after the change, and it will maintain this debt-equity ratio forever. Assume that Remex's debt cost of capital will be 6.5%. Remex faces a corporate tax rate of 35%. Except for the corporate tax rate of 35%, there are no market imperfections. Assume that the CAPM holds, the risk-free rate of interest is 5%, and the expected return on the market is 11%.**



- a. Using the information provided, complete the following table:

	Debt-Equity Ratio	Debt Cost of Capital	Equity Cost of Capital	Weighted Average Cost of Capital
Before change in capital structure	0	N/A		
After change in capital structure	0.30	6.5%		

- b. Using the information provided and your calculations in part (a), determine the value of the tax shield acquired by Remex if it changes its capital structure in the way it is considering.

- a. Before Change: From the SML, $r_E = 5\% + 1.50 \times 6\% = 14\%$

$$\text{Since the firm has no leverage, } r_{\text{wacc}} = r_U = r_E = 14\%.$$

After the change, from Eq. 18.10:

$$r_E = 14\% + 0.30(14\% - 6.5\%) = 16.25\%.$$

Since the firm has D/E of 0.30, the WACC formula is

$$\begin{aligned}
 r_{wacc} &= \frac{E}{D+E} R_E + \frac{D}{D+E} R_D (1 - T_C) \\
 &= \frac{1}{1.3} 16.25 + \frac{.3}{1.3} 6.5 (1 - .35) \\
 &= 13.475\%.
 \end{aligned}$$

We can also use Eq. 18.11: $r_{wacc} = 14\% - (.3 / 1.3)(.35)(6.5\%) = 13.475\%$.

- b. We can compare Remex's value with and without leverage. Without leverage (and no expected growth),

$$V^U = \frac{FCF}{r_U} = \frac{25}{14\%} = \$178.57 \text{ million.}$$

With leverage (and no expected growth):

$$V^L = \frac{FCF}{r_{wacc}} = \frac{25}{13.475\%} = \$185.53 \text{ million}$$

Therefore, $PV(ITS) = V^L - V^U = 185.53 - 178.57 = \6.96 million.

-  **18-15.** You are evaluating a project that requires an investment of \$90 today and provides a single cash flow of \$115 for sure one year from now. You decide to use 100% debt financing, that is, you will borrow \$90. The risk-free rate is 5% and the tax rate is 40%. Assume that the investment is fully depreciated at the end of the year, so without leverage you would owe taxes on the difference between the project cash flow and the investment, that is, \$15.

- Calculate the NPV of this investment opportunity using the APV method.
- Using your answer to part (a), calculate the WACC of the project.
- Verify that you get the same answer using the WACC method to calculate NPV.
- Finally, show that flow-to-equity also correctly gives the NPV of this investment opportunity.

- a. $FCF \text{ at year end (after tax)} = 115 - .40 \times 25 = 105$

$$V_U = 105 / 1.05 = 100$$

$$PV(its) = 40\% \times 5\% \times 90 / 1.05 = 1.71$$

$$V_L = 100 + 1.71 = 101.71$$

$$NPV = 101.71 - 90 = 11.71$$

- b. $r_u = r_d = 5\%$, $d = 90 / 101.71$

$$t_c = 40\%,$$

$$WACC = 5\% - (90 / 101.71)(40\%)(5\%) = 3.23\%$$

NOTE: if $r_u = r_d$, must use techniques in section 18.8 to calculate WACC.

- c. $V_L = 105 / 1.0323 = 101.71$

$$NPV = 101.71 - 90 = 11.71$$

- d. $FCFE_0 = 0$

$$FCFE_1 = 105 - 5\%(90)(1 - 40\%) - 90 = 12.3$$

$$R_e = 5\% \text{ (since no risk)}$$

$$\text{Value to equity} = 12.3 / 1.05 = 11.71$$

18-16. Tybo Corporation adjusts its debt so that its interest expenses are 20% of its free cash flow. Tybo is considering an expansion that will generate free cash flows of \$2.5 million this year and is expected to grow at a rate of 4% per year from then on. Suppose Tybo's marginal corporate tax rate is 40%.

- If the unlevered cost of capital for this expansion is 10%, what is its unlevered value?
- What is the levered value of the expansion?
- If Tybo pays 5% interest on its debt, what amount of debt will it take on initially for the expansion?
- What is the debt-to-value ratio for this expansion? What is its WACC?
- What is the levered value of the expansion using the WACC method?

a. Unlevered value $V^U = \text{FCF} / (r_U - g) = 2.5 / (10\% - 4\%) = \41.67 million

b. From Eq. 18.14, $V^L = (1 + \tau_c k) V^U = (1 + 0.40 \times 0.20) 41.67 = \45 million

c. Interest = 20%(FCF) = 20%(2.5) = \$0.5 million = $r_D D = 0.05 D$

Therefore, $D = 0.5 / 0.05 = \$10$ million

d. Debt-to-value $d = D / V^L = 10 / 45 = 0.2222$.

From Eq. 18.11, $r_{\text{wacc}} = 10\% - (0.2222)(0.40)5\% = 9.556\%$

e. Using the WACC method, $V^L = 2.5 / (9.556\% - 4\%) = \45 million

18-17. You are on your way to an important budget meeting. In the elevator, you review the project valuation analysis you had your summer associate prepare for one of the projects to be discussed:



	0	1	2	3	4
EBIT		10.0	10.0	10.0	10.0
Interest (5%)		-4.0	-4.0	-3.0	-2.0
Earnings Before Taxes		6.0	6.0	7.0	8.0
Taxes		-2.4	-2.4	-2.8	-3.2
Depreciation		25.0	25.0	25.0	25.0
Cap Ex	-100.0				
Additions to NWC	-20.0				20.0
Net New Debt	80.0	0.0	-20.0	-20.0	-40.0
FCFE	-40.0	28.6	8.6	9.2	9.8
NPV at 11% Equity Cost of Capital	5.9				

Looking over the spreadsheet, you realize that while all of the cash flow estimates are correct, your associate used the flow-to-equity valuation method and discounted the cash flows using the *company's* equity cost of capital of 11%. However, the project's incremental leverage is very different from the company's historical debt-equity ratio of 0.20: For this project, the company will instead borrow \$80 million upfront and repay \$20 million in year 2, \$20 million in year 3, and \$40 million in year 4. Thus, the *project's* equity cost of capital is likely to be higher than the firm's, not constant over time—invalidating your associate's calculation.

Clearly, the FTE approach is not the best way to analyze this project. Fortunately, you have your calculator with you, and with any luck you can use a better method before the meeting starts.

- What is the present value of the interest tax shield associated with this project?
- What are the free cash flows of the project?
- What is the best estimate of the project's value from the information given?

a. First,

Interest Payment = Interest Rate (5%) × Prior period debt

From the tax calculation in the spreadsheet, we can see that the tax rate is $2.4/6 = 40\%$. Therefore,

Interest Tax shield = Interest Payment × Tax Rate (40%)

Because the tax shields are predetermined, we can discount them using the 5% debt cost of capital.

$$\begin{aligned} PV(ITS) &= \frac{0.40(0.05)(80)}{1.05} + \frac{0.40(0.05)(80)}{1.05^2} + \frac{0.40(0.05)(60)}{1.05^3} + \frac{0.40(0.05)(40)}{1.05^4} \\ &= \$4.67 \text{ million} \end{aligned}$$

	Year 0	Year 1	Year 2	Year 3	Year 4
Debt	80	80	60	40	0
Interest at 5.0%		4	4	3	2
Tax shield 40.0%		1.6	1.6	1.2	0.8
PV 5.0%	\$4.67				

b. We can use Eq. 7.5:

	0	1	2	3	4
EBIT		10	10	10	10
Taxes		-4	-4	-4	-4
Unlevered Net Income		6	6	6	6
Depreciation			25	25	25
Cap Ex		-100			
Additions to NWC		-20			20
FCF	-120	31	31	31	51

$$FCF = EBIT \times (1 - T_c) + \text{Depreciation} - \text{CapEx} - \Delta NWC$$

Alternatively, we can use Eq. 18.9:

$$FCF = FCFE + \text{Int} \times (1 - T_c) - \text{Net New Debt}$$

	Year 0	Year 1	Year 2	Year 3	Year 4
FCFE	-40	28.6	8.6	9.2	9.8
+ After-tax Interest		2.4	2.4	1.8	1.2
- Net New Debt	-80	0	20	20	40
FCF	-120	31	31	31	51

c. With predetermined debt levels, the APV method is easiest.

Step 1: Determine r_U . Assuming the company has maintained a historical D/E ratio of 0.20, we can approximate its unlevered cost of capital using Eq. 18.6:

$$r_U = (1 / 1.2) 11\% + (.2 / 1.2) 5\% = 10\%$$

Step 2: Compute NPV of FCF without leverage

$$NPV = -120 + \frac{31}{1.10} + \frac{31}{1.10^2} + \frac{31}{1.10^3} + \frac{51}{1.10^4} = -8.1$$

Step 3: Compute APV

$$APV = NPV + PV(ITS) = -8.1 + 4.7 = -3.4$$

So the project actually has negative value.

18-18. Your firm is considering building a \$600 million plant to manufacture HDTV circuitry. You expect operating profits (EBITDA) of \$145 million per year for the next 10 years. The plant will be depreciated on a straight-line basis over 10 years (assuming no salvage value for tax purposes). After 10 years, the plant will have a salvage value of \$300 million (which, since it will be fully depreciated, is then taxable). The project requires \$50 million in working capital at the start, which will be recovered in year 10 when the project shuts down. The corporate tax rate is 35%. All cash flows occur at the end of the year.

- a. If the risk-free rate is 5%, the expected return of the market is 11%, and the asset beta for the consumer electronics industry is 1.67, what is the NPV of the project?
- b. Suppose that you can finance \$400 million of the cost of the plant using 10-year, 9% coupon bonds sold at par. This amount is incremental new debt associated specifically with this project and will not alter other aspects of the firm's capital structure. What is the value of the project, including the tax shield of the debt?

- a. First we compute the FCF:

$$FCF_0 = -600 \text{ (Capex)} - 50 \text{ (Inc in NWC)} = -650$$

Using Eq. 7.6:

$$FCF_{1-9} = 145 \times (1 - 0.35) + 0.35 \times 60 = 115.25$$

$$\text{After-tax Salvage Value} = 300 \times (1 - 0.35) = 195$$

$$FCF_{10} = 145 \times (1 - 0.35) + 0.35 \times 60 + 50 \text{ (Inc in NWC)} + 195 \text{ (salvage)} \\ = 360.25$$

$$\text{From the CAPM, } r_U = 5\% + 1.67(11\% - 5\%) = 15\%$$

Therefore,

$$NPV = -650 + 115.25 \times \frac{1}{.15} \left(1 - \frac{1}{1.15^9} \right) + \frac{360.25}{1.15^{10}} = -11.0$$

Without leverage, project NPV is -\$11 million.

- b. Because the debt level is predetermined, we can use the APV approach. Because the bonds initially trade at par, the interest payments are the 9% coupon payments of the bond. Assuming annual coupons:

$$PV(\text{ITS}) = 400 \times 0.09 \times 0.35 \times \frac{1}{.09} \left(1 - \frac{1}{1.09^{10}} \right) = \$80.9 \text{ million.}$$

Therefore,

$$APV = NPV + PV(\text{ITS}) = -11 + 81 = \$70 \text{ million.}$$

Note that this project is only profitable as a result of the tax benefits of leverage.

18-19. DFS Corporation is currently an all-equity firm, with assets with a market value of \$100 million and 4 million shares outstanding. DFS is considering a leveraged recapitalization to boost its share price. The firm plans to raise a fixed amount of permanent debt (i.e., the outstanding principal will remain constant) and use the proceeds to repurchase shares. DFS pays a 35% corporate tax rate, so one motivation for taking on the debt is to reduce the firm's tax liability. However, the upfront investment banking fees associated with the recapitalization will be 5% of the amount of debt raised. Adding leverage will also create the possibility of future financial distress or agency costs; shown below are DFS's estimates for different levels of debt:



Debt amount (\$ million):	0	10	20	30	40	50
Present value of expected distress and agency costs (\$ million):	0.0	−0.3	−1.8	−4.3	−7.5	−11.3

a. Based on this information, which level of debt is the best choice for DFS?

b. Estimate the stock price once this transaction is announced.

- a. Because the debt is permanent, the value of the tax shield is $35\% \times D$. From that we must deduct the 5% issuance cost, and the PV of distress and agency costs to determine the net benefit of leverage.

Debt Amount (\$M):	0	10	20	30	40	50
PV of Expected Distress and Agency Costs (\$M):	0	−	−	−	−	−
	0	0.	1.	4.	7.	11
	0	3	8	3	5	.3
	0	+	+	+	+	+
	0	3.	6.	9.	12	15
Tax Benefit less Issuance Cost (30%):	0	0	0	0	.0	.0
	0	+	+	+	+	+
	0	2.	4.	4.	4.	3.
Net Benefit:	0	7	2	7	5	7

Based on this information, the greatest net benefit occurs for debt = \$30 million.

- b. Value of assets goes up from \$100M to \$104.7M. Thus, the share price should rise to \$26.175.

18-20. Your firm is considering a \$150 million investment to launch a new product line. The project is expected to generate a free cash flow of \$20 million per year, and its unlevered cost of capital is 10%. To fund the investment, your firm will take on \$100 million in permanent debt.

- a. Suppose the marginal corporate tax rate is 35%. Ignoring issuance costs, what is the NPV of the investment?
- b. Suppose your firm will pay a 2% underwriting fee when issuing the debt. It will raise the remaining \$50 million by issuing equity. In addition to the 5% underwriting fee for the equity issue, you believe that your firm's current share price of \$40 is \$5 per share less than its true value. What is the NPV of the investment including any tax benefits of leverage? (Assume all fees are on an after-tax basis.)

- a. With permanent debt the APV method is simplest. $\text{NPV}(\text{unlevered}) = -150 + 20 / 0.10 = \50 million. $\text{PV}(\text{ITS}) = \tau_c \times D = 35\% \times 100 = \35 million. Thus, the NPV with leverage is $\text{APV} = \text{NPV} + \text{PV}(\text{ITS}) = 50 + 35 = \85 million.

- b. Financing costs = $2\% \times 100 + 5\% \times 50 = \4.5 million. (We assume these amounts are after-tax.)

Underpricing cost = $(5 / 40) \times 50 = \$6.25$ million

$\text{APV} = 85 - 4.5 - 6.25 = 74.25$ million

18-21. Consider Avco's RFX project from Section 18.3. Suppose that Avco is receiving government loan guarantees that allow it to borrow at the 6% rate. Without these guarantees, Avco would pay 6.5% on its debt.

- a. What is Avco's unlevered cost of capital given its true debt cost of capital of 6.5%?

- b. What is the unlevered value of the RFX project in this case? What is the present value of the interest tax shield?
- c. What is the NPV of the loan guarantees? (*Hint* : Because the actual loan amounts will fluctuate with the value of the project, discount the expected interest savings at the unlevered cost of capital.)
- d. What is the levered value of the RFX project, including the interest tax shield and the NPV of the loan guarantees?
- a. We use Eq. 18.6 with the true debt cost:

$$r_u = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D = 0.50 \times 10\% + 0.50 \times 6.50\% = 8.25\%$$

- b. The unlevered value is the PV of the FCF discounted at r_u :

$$V^U = 18 \times \frac{1}{.0825} \left(1 - \frac{1}{1.0825^4} \right) = \$59.29 \text{ million}$$

The amount of the interest tax shield each period is that same as computed in Table 18.5 in the text, but now we discount at $r_u = 8.25\%$:

$$PV(ITS) = \frac{0.73}{1.0825} + \frac{0.57}{1.0825^2} + \frac{0.39}{1.0825^3} + \frac{0.20}{1.0825^4} = \$1.62 \text{ million}$$

- c. The loan guarantee reduces the interest paid from 6.5% to 6% each year. Thus, the savings in year t is $0.5\% \times D_{t-1}$. The value of the loan guarantee is the present value of this savings. Because the debt amount D will vary with the value of the project over time, we discount the savings at rate r_u .

$$NPV(\text{Loan}) = \frac{.005 \times 30.62}{1.0825} + \frac{.005 \times 23.71}{1.0825^2} + \frac{.005 \times 16.32}{1.0825^3} + \frac{.005 \times 8.43}{1.0825^4} = \$0.34 \text{ million}$$

- d. $APV = V^U + PV(ITS) + NPV(\text{Loan}) = 59.29 + 1.62 + 0.34 = \61.25 million

Note that this is the same value we originally computed using the WACC method, where we used the firm's actual borrowing cost rather than the true rate it would have received.

18-22. Arden Corporation is considering an investment in a new project with an unlevered cost of capital of 9%. Arden's marginal corporate tax rate is 40%, and its debt cost of capital is 5%.

- a. Suppose Arden adjusts its debt continuously to maintain a constant debt-equity ratio of 50%. What is the appropriate WACC for the new project?
- b. Suppose Arden adjusts its debt once per year to maintain a constant debt-equity ratio of 50%. What is the appropriate WACC for the new project now?
- c. Suppose the project has free cash flows of \$10 million per year, which are expected to decline by 2% per year. What is the value of the project in parts (a) and (b) now?

a. $r_{wacc} = r_u - d\tau_c(r_D) = 9\% - (.5/1.5)(0.40)5\% = 8.333\%$

b. $r_{wacc} = r_u - d\tau_c(r_D + \phi(r_u - r_D))$

$$= 9\% - (.5/1.5)(0.40) \left(5\% + \frac{.05}{1.05}(9\% - 5\%) \right) = 8.308\%$$

Alternatively, from Eq. 18.17:

$$\begin{aligned} r_{\text{wacc}} &= r_u - d\tau_c r_D \frac{1+r_u}{1+r_D} \\ &= 9\% - (.5/1.5)(0.40)5\% \frac{1.09}{1.05} = 8.308\% \end{aligned}$$

- c. In case (a), $V^L = 10/((.08333 + .02) = \96.78 million.

In case (b), $V^L = 10/((.08308 + .02) = \97.01 million.

Note the minor difference in the two cases. Case (b) is higher because the tax shields are less risky when debt is fixed over the year.

18-23. XL Sports is expected to generate free cash flows of \$10.9 million per year. XL has permanent debt of \$40 million, a tax rate of 40%, and an unlevered cost of capital of 10%.

- What is the value of XL's equity using the APV method?
- What is XL's WACC? What is XL's equity value using the WACC method?
- If XL's debt cost of capital is 5%, what is XL's equity cost of capital?
- What is XL's equity value using the FTE method?

- a. $V^U = 10.9 / 10\% = \$109$ million. $PV(\text{ITS}) = 0.40 \times \$40 \text{ million} = \$16$ million.

$V^L = \text{APV} = 109 + 16 = \125 million, so $E = 125 - 40 = \$85$ million.

- b. $r_{\text{wacc}} = r_u - d\tau_c(r_D + \phi(r_u - r_D)) = r_u - d\tau_c(r_D + r_u - r_D)$
 $= r_u - d\tau_c r_u$
 $= 10\% - (40/125)(0.40)10\% = 8.72\%$

Using the WACC method, $V^L = 10.9 / 8.72\% = \$125$ million, so $E = 125 - 40 = \$85$ million.

- c. If XL's debt cost of capital is 5%, what is XL's equity cost of capital?

From Eq. 18.20:

$$\begin{aligned} r_E &= r_u + \frac{D^S}{E}(r_u - r_D) \\ &= 10\% + \frac{40-16}{125-40}(10\% - 5\%) = 11.412\% \end{aligned}$$

- d. $\text{FCFE} = \text{FCF} - \text{After-tax Interest} + \text{Net new debt} = 10.9 - 5\%(1 - 0.40)40 = 9.7$

$E = 9.7 / 0.11412 = \$85$ million.

18-24. Propel Corporation plans to make a \$50 million investment, initially funded completely with debt. The free cash flows of the investment and Propel's incremental debt from the project follow:



Year	0	1	2	3
Free cash flows	-50	40	20	25
Debt	50	30	15	0

Propel's incremental debt for the project will be paid off according to the predetermined schedule shown. Propel's debt cost of capital is 8%, and its tax rate is 40%. Propel also estimates an unlevered cost of capital for the project of 12%.

- Use the APV method to determine the levered value of the project at each date and its initial NPV.
 - Calculate the WACC for this project at each date. How does the WACC change over time? Why?
 - Compute the project's NPV using the WACC method.
 - Compute the equity cost of capital for this project at each date. How does the equity cost of capital change over time? Why?
 - Compute the project's equity value using the FTE method. How does the initial equity value compare with the NPV calculated in parts (a) and (c)?
- a. Note that this answer actually uses the APV method instead of the WACC method.

We compute V^U at each date by discounting the project's future FCF at rate $r_U = 12\%$.

($V_t^U = \text{NPV}(r_U, \text{FCF}_{t+1} : \text{FCF}_T)$):

Year	0	1	2	3
FCF	-50	40	20	25
V^u	\$69.45	\$37.79	\$22.32	

Then we compute the value of the future interest tax shields at each date by discounting at rate $r_D = 8\%$:

Year	0	1	2	3
D	50	30	15	0
interest at 8%		4	2.4	1.2
tax shield at 40%		1.6	0.96	0.48
PV(ITS)	\$2.69	\$1.30	\$0.44	

Finally, we compute $V^L = \text{APV} = V^U + \text{PV}(\text{ITS})$:

Year	0	1	2	3
V^u	\$69.45	\$37.79	\$22.32	
PV(ITS)	\$2.69	\$1.30	\$0.44	
V^L	\$72.14	\$39.09	\$22.77	

Given the initial investment of \$50, the project's NPV is $72.14 - 50 = \$22.14$.

- b. We can compute the WACC at each date using Eq. 18.21. The debt-to-value ratio, d , is given by D/V^L . The debt persistence ϕ is given by $T^s/(\tau_c D)$, where $T^s = \text{PV}(\text{ITS})$ (since all tax shields are predetermined):

Year	0	1	2	3
D	50	30	15	0
V^L	\$72.14	\$39.09	\$22.77	
$d = D/V^L$	69%	77%	66%	
$T^s = \text{PV}(\text{ITS})$	\$2.69	\$1.30	\$0.44	
$T^s/\tau_c D$	13.4%	10.8%	7.4%	
r_{wacc}	9.63%	9.41%	9.81%	

Note that the WACC changes over time, decreasing from date 0 to 1, and increasing from date 1 to 2. The WACC fluctuates because the leverage ratio of the project changes over time (as does the persistence of the debt).

- c. We can compute the levered value of the project by discounting the FCF using the WACC at each date:

$$V_2^L = \frac{FCF_3}{1 + r_{wacc}(2)} = \frac{25}{1.0981} = \$22.77$$

$$V_1^L = \frac{FCF_2 + V_2^L}{1 + r_{wacc}(1)} = \frac{20 + 22.77}{1.0941} = \$39.09$$

$$V_0^L = \frac{FCF_1 + V_1^L}{1 + r_{wacc}(0)} = \frac{40 + 39.09}{1.0963} = \$72.14.$$

Note that these results coincide with part (a).

- d. We can compute the project's equity cost of capital using Eq. 18.20. Note that $D^s = D - T^s = D - PV(ITS)$:

Year	<u>0</u>	<u>1</u>	<u>2</u>	<u>3</u>
$D^s = D - T^s$	\$47.31	\$28.70	\$14.56	
$E = V^L - D$	\$22.14	\$9.09	\$7.77	
D^s/E	2.14	3.16	1.87	
r_E	20.55%	24.63%	19.50%	

Note the equity cost of capital rises and then falls with the project's effective debt-equity ratio, D^s/E .

- e. We first compute FCFE at each date by deducting the after-tax interest expenses (equivalently, deducting interest and adding back the tax shield) and adding net increases in debt:

Year	<u>0</u>	<u>1</u>	<u>2</u>	<u>3</u>
FCF	-50	40	20	25
- Interest		-4	-2.4	-1.2
+ Tax shield		1.6	0.96	0.48
+ Inc. in Debt	50	-20	-15	-15
FCFE	0	17.6	3.56	9.28
E	22.14	9.09	7.77	

Then, we compute the equity value of the project by discounting FCFE using r_E at each date:

$$E_2 = \frac{FCFE_3}{1 + r_E(2)} = \frac{9.28}{1.1950} = \$7.77$$

$$E_1 = \frac{FCFE_2 + E_2}{1 + r_E(1)} = \frac{3.56 + 7.77}{1.2463} = \$9.09$$

$$E_0 = \frac{FCFE_1 + E_1}{1 + r_E(0)} = \frac{17.60 + 9.09}{1.2055} = \$22.14/$$

These values for equity match those computed earlier, and match the project's initial NPV.

Note that to use the WACC or FTE methods here, we relied on V^L computed in the APV method. We could also solve for the value using the WACC or FTE methods directly using the techniques in appendix 18A.3.

18-25. Gartner Systems has no debt and an equity cost of capital of 10%. Gartner's current market capitalization is \$100 million, and its free cash flows are expected to grow at 3% per year. Gartner's corporate tax rate is 35%. Investors pay tax rates of 40% on interest income and 20% on equity income.

- a. Suppose Gartner adds \$50 million in permanent debt and uses the proceeds to repurchase shares. What will Gartner's levered value be in this case?
- b. Suppose instead Gartner decides to maintain a 50% debt-to-value ratio going forward. If Gartner's debt cost of capital is 6.67%, what will Gartner's levered value be in this case?

- a. From Eq. 18.25, $\tau^* = 1 - (1 - 0.35)(1 - 0.20) / (1 - 0.40) = 13.333\%$.

Using the APV method, $V^L = V^U + \tau_c D = 100 + 0.1333 \times 50 = \106.67 million

- b. With a constant debt-to-value ratio, the WACC approach is easiest. We need to determine Gartner's WACC with this new leverage policy. To compute the WACC, we need to determine the new equity cost of capital using Eq. 18.24:

$$r_U = \frac{E}{E + D^s} r_E + \frac{D^s}{E + D^s} r_D^*$$

Because Gartner initially has no leverage, $r_U = r_E = 10\%$. Next, $r_D^* = r_D (1 - \tau_i) / (1 - \tau_e) = 6.67\%(1 - 0.40) / (1 - 0.20) = 5.00\%$. With a constant debt-to-value ratio, $T^s = 0$ and $D^s / (E + D^s) = D / (E + D) = 50\%$. Thus,

$$10\% = 0.50r_E + 0.50(5\%)$$

implying that r_E rises to 15%. Therefore, Gartner's WACC is

$$r_{wacc} = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - \tau_c) = 0.50(15\%) + 0.50(6.67\%)(1 - 0.35) = 9.67\%.$$

We also need to estimate Gartner's FCF. Based on its current market cap,

$$100 = FCF / (10\% - 3\%) \text{ implies } FCF = \$7 \text{ million.}$$

Therefore, with the new leverage,

$$V^L = 7 / (9.67\% - 3\%) = \$104.95 \text{ million.}$$

18-26. Revtek, Inc., has an equity cost of capital of 12% and a debt cost of capital of 6%. Revtek maintains a constant debt-equity ratio of 0.5, and its tax rate is 35%.



- a. What is Revtek's WACC given its current debt-equity ratio?
- b. Assuming no personal taxes, how will Revtek's WACC change if it increases its debt-equity ratio to 2 and its debt cost of capital remains at 6%?
- c. Now suppose investors pay tax rates of 40% on interest income and 15% on income from equity. How will Revtek's WACC change if it increases its debt-equity ratio to 2 in this case?
- d. Provide an intuitive explanation for the difference in your answers to parts (b) and (c).

$$a. \quad r_{wacc} = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - \tau_c) = (1/1.5)(12\%) + (.5/1.5)(6\%)(1 - 0.35) = 9.3\%$$

- b. From Eq. 18.6:

$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D = (1/1.5)(12\%) + (.5/1.5)(6\%) = 10\%$$

From Eq. 18.11:

$$r_{wacc} = r_U - d\tau_c r_D = 10\% - (2/3)(.35)6\% = 8.6\%$$

- c. Given their initial capital structure, we would estimate Revtek's unlevered cost of capital as (using Eq. 18.24):

$$r_U = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D^* = (1/1.5)(12\%) + (.5/1.5)(4.235\%) = 9.41\%.$$

We can also use Eq. 18.24 to calculate r_E with higher leverage:

$$9.41\% = (1/3)r_E + (2/3)4.235\% \text{ so that } r_E = 19.76\%.$$

Then,

$$r_{wacc} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D(1 - \tau_c) = (1/3)(19.76\%) + (2/3)(6\%)(1 - 0.35) = 9.19\%.$$

- d. When investors pay higher taxes on interest income than equity income, the tax benefit of leverage is reduced. Thus, for the same increase in leverage, the decline in the WACC is smaller in the presence of investor taxes.

Chapter 19

Valuation and Financial Modeling: A Case Study

- 19-1. You would like to compare Ideko's profitability to its competitors' profitability using the EBITDA/sales multiple. Given Ideko's current sales of \$75 million, use the information in Table 19.2 to compute a range of EBITDA for Ideko assuming it is run as profitably as its competitors.

Ideko's 2005 sales are \$75 million.

Find the highest and lowest EBITDA values across all three firms and the industry as a whole:

	EBITDA/Sales (%)	EBITDA (\$ mil)
Oakley	17.0	12.75
Luxcottica	18.5	13.875
Nike	15.9	11.925
Industry	12.1	9.075

This implies an EBITDA range of \$9.075 to \$13.875 million.

- 19-2. Assume that Ideko's market share will increase by 0.5% per year rather than the 1% used in the chapter. What production capacity will Ideko require each year? When will an expansion become necessary (when production volume will exceed the current level by 50%)?

First compute the projected annual market share:

			2005	2006	2007	2008	2009	2010
Sales Data		Growth/Yr						
1	Market Size (000 units)	5.0%	10,000	10,500	11,025	11,576	12,155	12,763
2	Market Share	0.5%	10.0%	10.5%	11.0%	11.5%	12.0%	12.5%
3	Ave. Sales Price (\$/unit)	2.0%	75.00	76.50	78.03	79.59	81.18	82.81

Using these projections, calculate the projected annual production volume:

			2005	2006	2007	2008	2009	2010
Production Volume (000 units)								
1	Market Size		10,000	10,500	11,025	11,576	12,155	12,763
2	Market Share		10.0%	10.5%	11.0%	11.5%	12.0%	12.5%
3	Production Volume (1x2)		1,000	1,103	1,213	1,331	1,459	1,595

Based on these estimates, it will be 2010 before current capacity is exceeded and an expansion becomes necessary.

- 19-3. Under the assumption that Ideko market share will increase by 0.5% per year, you determine that the plant will require an expansion in 2010. The cost of this expansion will be \$15 million. Assuming the financing of the expansion will be delayed accordingly, calculate the projected interest payments and the amount of the projected interest tax shields (assuming that the interest rates on the term loans remain the same as in the chapter) through 2010.

			2005	2006	2007	2008	2009	2010
Debt & Interest Table (\$000s)								
1	Outstanding Debt		100,000	100,000	100,000	100,000	100,000	115,000
2	Interest on Term Loan	6.80%		(6,800)	(6,800)	(6,800)	(6,800)	(6,800)
3	Interest Tax Shield			2,380	2,380	2,380	2,380	2,380

- 19-4. Under the assumption that Ideko's market share will increase by 0.5% per year (and the investment and financing will be adjusted as described in Problem 3), you project the following depreciation:



	Year	2005	2006	2007	2008	2009	2010
Fixed Assets and Capital Investment (\$ 000)							
2	New Investment	5,000	5,000	5,000	5,000	5,000	20,000
3	Depreciation	(5,500)	(5,450)	(5,405)	(5,365)	(5,328)	(6,795)

Using this information, project net income through 2010 (that is, reproduce Table 19.7 under the new assumptions).

	Year	2005	2006	2007	2008	2009	2010
INCOME STATEMENT (\$000s)							
1	Sales	75,000	84,341	94,631	105,956	118,413	132,105
2	Cost of Goods Sold						
3	Raw Materials	(16,000)	(17,816)	(19,794)	(21,946)	(24,285)	(26,828)
4	Direct Labor Costs	(18,000)	(20,639)	(23,611)	(26,955)	(30,715)	(34,938)
5	Gross Profit	41,000	45,886	51,226	57,056	63,413	70,339
6	Sales & Marketing	(11,250)	(13,916)	(17,034)	(20,662)	(23,683)	(26,421)
7	Administration	(13,500)	(12,651)	(14,195)	(14,834)	(15,394)	(17,174)
8	EBITDA	16,250	19,319	19,998	21,560	24,337	26,745
9	Depreciation	(5,500)	(5,450)	(5,405)	(5,365)	(5,328)	(6,795)
10	EBIT	10,750	13,869	14,593	16,196	19,009	19,949
11	Interest Expense (net)	(75)	(6,800)	(6,800)	(6,800)	(6,800)	(6,800)
12	Pretax Income	10,675	7,069	7,793	9,396	12,209	13,149
13	Income Tax	(3,736)	(2,474)	(2,728)	(3,289)	(4,273)	(4,602)
14	Net Income	6,939	4,595	5,065	6,107	7,936	8,547

- 19-5.** Under the assumptions that Ideko's market share will increase by 0.5% per year (implying that the investment, financing, and depreciation will be adjusted as described in Problems 3 and 4) and that the forecasts in Table 19.8 remain the same, calculate Ideko's working capital requirements though 2010 (that is, reproduce Table 19.9 under the new assumptions).



	Year	2005	2006	2007	2008	2009	2010
Working Capital (\$000s)							
Assets							
1	Accounts Receivable	18,493	13,864	15,556	17,418	19,465	21,716
2	Raw Materials	1,973	1,464	1,627	1,804	1,996	2,205
3	Finished Goods	4,192	4,741	5,351	6,029	6,781	7,615
4	Minimum Cash Balance	6,164	6,932	7,778	8,709	9,733	10,858
5	Total Current Assets	30,822	27,002	30,312	33,959	37,975	42,394
Liabilities							
6	Wages Payable	1,295	1,368	1,554	1,717	1,895	2,142
7	Other Accounts Payable	3,360	3,912	4,540	5,253	5,914	6,565
8	Total Current Liabilities	4,654	5,280	6,094	6,970	7,809	8,706
Net Working Capital		26,168	21,722	24,218	26,989	30,166	33,687
9	Increase in Net Working Capital		(4,446)	2,496	2,771	3,177	3,521

- 19-6.** Under the assumptions that Ideko's market share will increase by 0.5% per year (implying that the investment, financing, and depreciation will be adjusted as described in Problems 3 and 4) but that the projected improvements in net working capital do not transpire (so the numbers in Table 19.8 remain at their 2005 levels through 2010), calculate Ideko's working capital requirements though 2010 (that is, reproduce Table 19.9 under these assumptions).



	Year	2005	2006	2007	2008	2009	2010
Working Capital (\$000s)							
Assets							
1	Accounts Receivable	18,493	20,796	23,334	26,126	29,198	32,574
2	Raw Materials	1,973	2,197	2,440	2,706	2,994	3,308
3	Finished Goods	4,192	4,741	5,351	6,029	6,781	7,615
4	Minimum Cash Balance	6,164	6,932	7,778	8,709	9,733	10,858
5	Total Current Assets	30,822	34,666	38,903	43,569	48,705	54,354
Liabilities							
6	Wages Payable	1,295	1,368	1,554	1,717	1,895	2,142
7	Other Accounts Payable	3,360	3,912	4,540	5,253	5,914	6,565
8	Total Current Liabilities	4,654	5,280	6,094	6,970	7,809	8,706
Net Working Capital		26,168	29,386	32,809	36,599	40,897	45,648
9	Increase in Net Working Capital		3,218	3,423	3,790	4,297	4,751

- 19-7.  Forecast Ideko's free cash flow (reproduce Table 19.10), assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (that is, under the assumptions in Problem 5).

	Year	2005	2006	2007	2008	2009	2010
Free Cash Flow (\$000s)							
1	Net Income		4,595	5,065	6,107	7,936	8,547
2	Plus: After-Tax Interest Expense		4,420	4,420	4,420	4,420	4,420
3	Unlevered Net Income		9,015	9,485	10,527	12,356	12,967
4	Plus: Depreciation		5,450	5,405	5,365	5,328	6,795
5	Less: Increases in NWC		4,446	(2,496)	(2,771)	(3,177)	(3,521)
6	Less: Capital Expenditures		(5,000)	(5,000)	(5,000)	(5,000)	(20,000)
7	Free Cash Flow of Firm		13,911	7,394	8,121	9,507	(3,759)
8	Plus: Net Borrowing		-	-	-	-	15,000
9	Less: After-Tax Interest Expense		(4,420)	(4,420)	(4,420)	(4,420)	(4,420)
10	Free Cash Flow to Equity		9,491	2,974	3,701	5,087	6,821

- 19-8.  Forecast Ideko's free cash flow (reproduce Table 19.10), assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital do *not* occur (that is, under the assumptions in Problem 6).

	Year	2005	2006	2007	2008	2009	2010
Free Cash Flow (\$000s)							
1	Net Income		4,595	5,065	6,107	7,936	8,547
2	Plus: After-Tax Interest Expense		4,420	4,420	4,420	4,420	4,420
3	Unlevered Net Income		9,015	9,485	10,527	12,356	12,967
4	Plus: Depreciation		5,450	5,405	5,365	5,328	6,795
5	Less: Increases in NWC		(3,218)	(3,423)	(3,790)	(4,297)	(4,751)
6	Less: Capital Expenditures		(5,000)	(5,000)	(5,000)	(5,000)	(20,000)
7	Free Cash Flow of Firm		6,246	6,467	7,102	8,387	(4,989)
8	Plus: Net Borrowing		-	-	-	-	15,000
9	Less: After-Tax Interest Expense		(4,420)	(4,420)	(4,420)	(4,420)	(4,420)
10	Free Cash Flow to Equity		1,826	2,047	2,682	3,967	5,591

- 19-9.  Reproduce Ideko's balance sheet and statement of cash flows, assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (that is, under the assumptions in Problem 5).

	Year	2005	2006	2007	2008	2009	2010
BALANCE SHEET (\$000s)							
Assets							
1	Cash & Cash Equivalents	6,164	6,932	7,778	8,709	9,733	10,858
2	Accounts Receivable	18,493	13,864	15,556	17,418	19,465	21,716
3	Inventories	6,164	6,205	6,978	7,833	8,777	9,820
4	Total Current Assets	30,822	27,002	30,312	33,959	37,975	42,394
5	Property, Plant and Equipment	49,500	49,050	48,645	48,281	47,952	61,157
6	Goodwill	72,332	72,332	72,332	72,332	72,332	72,332
7	Total Assets	152,654	148,384	151,289	154,572	158,259	175,883
Liabilities							
8	Accounts Payable	4,654	5,280	6,094	6,970	7,809	8,706
9	Debt	100,000	100,000	100,000	100,000	100,000	115,000
10	Total Liabilities	104,654	105,280	106,094	106,970	107,809	123,706
Stockholders' Equity							
11	Starting Stockholders' Equity		48,000	43,104	45,195	47,601	50,451
12	Net Income		4,595	5,065	6,107	7,936	8,547
13	Dividends	(2,000)	(9,491)	(2,974)	(3,701)	(5,087)	(6,821)
14	Capital Contributions	50,000	-	-	-	-	-
15	Stockholders' Equity	48,000	43,104	45,195	47,601	50,451	52,177
16	Total Liabilities & Equity	152,654	148,384	151,289	154,572	158,259	175,883

	Year	2005	2006	2007	2008	2009	2010
STATEMENT OF CASH FLOWS (\$000s)							
1	Net Income		4,595	5,065	6,107	7,936	8,547
2	Depreciation		5,450	5,405	5,365	5,328	6,795
3	Changes in Working Capital						
4	Accounts Receivable		4,629	(1,691)	(1,862)	(2,048)	(2,251)
5	Inventory		(41)	(773)	(854)	(944)	(1,043)
6	Accounts Payable		626	814	876	838	898
7	Cash from Operating Activities		15,259	8,820	9,632	11,110	12,946
8	Capital Expenditures		(5,000)	(5,000)	(5,000)	(5,000)	(20,000)
9	Other Investment		-	-	-	-	-
10	Cash from Investing Activities		(5,000)	(5,000)	(5,000)	(5,000)	(20,000)
11	Net Borrowing		-	-	-	-	15,000
12	Dividends		(9,491)	(2,974)	(3,701)	(5,087)	(6,821)
13	Capital Contributions		-	-	-	-	-
14	Cash from Financing Activities		(9,491)	(2,974)	(3,701)	(5,087)	8,179
15	Change in Cash & Cash Equivalents		768	846	931	1,024	1,125

- 19-10. Reproduce Ideko's balance sheet and statement of cash flows, assuming Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital do *not* occur (that is, under the assumptions in Problem 6).



	Year	2005	2006	2007	2008	2009	2010
BALANCE SHEET (\$000s)							
Assets							
1	Cash & Cash Equivalents	6,164	6,932	7,778	8,709	9,733	10,858
2	Accounts Receivable	18,493	20,796	23,334	26,126	29,198	32,574
3	Inventories	6,164	6,938	7,792	8,734	9,775	10,922
4	Total Current Assets	30,822	34,666	38,903	43,569	48,705	54,354
5	Property, Plant and Equipment	49,500	49,050	48,645	48,281	47,952	61,157
6	Goodwill	72,332	72,332	72,332	72,332	72,332	72,332
7	Total Assets	152,654	156,048	159,880	164,182	168,990	187,844
Liabilities							
8	Accounts Payable	4,654	5,280	6,094	6,970	7,809	8,706
9	Debt	100,000	100,000	100,000	100,000	100,000	115,000
10	Total Liabilities	104,654	105,280	106,094	106,970	107,809	123,706
Stockholders' Equity							
11	Starting Stockholders' Equity		48,000	50,768	53,786	57,212	61,181
12	Net Income		4,595	5,065	6,107	7,936	8,547
13	Dividends	(2,000)	(1,826)	(2,047)	(2,682)	(3,967)	(5,591)
14	Capital Contributions	50,000	-	-	-	-	-
15	Stockholders' Equity	48,000	50,768	53,786	57,212	61,181	64,137
16	Total Liabilities & Equity	152,654	156,048	159,880	164,182	168,990	187,844

	Year	2005	2006	2007	2008	2009	2010
STATEMENT OF CASH FLOWS (\$000s)							
1	Net Income		4,595	5,065	6,107	7,936	8,547
2	Depreciation		5,450	5,405	5,365	5,328	6,795
3	Changes in Working Capital						
4	Accounts Receivable		(2,303)	(2,537)	(2,793)	(3,072)	(3,376)
5	Inventory		(773)	(854)	(943)	(1,040)	(1,148)
6	Accounts Payable		626	814	876	838	898
7	Cash from Operating Activities		7,594	7,893	8,613	9,990	11,717
8	Capital Expenditures		(5,000)	(5,000)	(5,000)	(5,000)	(20,000)
9	Other Investment		-	-	-	-	-
10	Cash from Investing Activities		(5,000)	(5,000)	(5,000)	(5,000)	(20,000)
11	Net Borrowing		-	-	-	-	15,000
12	Dividends		(1,826)	(2,047)	(2,682)	(3,967)	(5,591)
13	Capital Contributions		-	-	-	-	-
14	Cash from Financing Activities		(1,826)	(2,047)	(2,682)	(3,967)	9,409
15	Change in Cash & Cash Equivalents		768	846	931	1,024	1,125

- 19-11. Calculate Ideko's unlevered cost of capital when Ideko's unlevered beta is 1.1 rather than 1.2, and all other required estimates are the same as in the chapter.

$$r_u = r_f + \beta_u (E[R_{\text{mkt}}] - r_f) = 4\% + 1.1(5\%) = 9.5\%$$

- 19-12. Calculate Ideko's unlevered cost of capital when the market risk premium is 6% rather than 5%, the risk-free rate is 5% rather than 4%, and all other required estimates are the same as in the chapter.

$$r_u = r_f + \beta_u (E[R_{\text{mkt}}] - r_f) = 5\% + 1.2(6\%) = 12.2\%$$

- 19-13. Using the information produced in the income statement in Problem 4, use EBITDA as a multiple to estimate the continuation value in 2010, assuming the current value remains unchanged (reproduce Table 19.15). Infer the EV/sales and the unlevered and levered P/E ratios implied by the continuation value you calculated.

Continuation Value: Multiples Approach (\$000s)				
1	EBITDA in 2010	26,745	Common Multiples	
2	EBITDA multiple	9.1x		
3	Cont. Enterprise Value	243,377		
4	Debt	(115,000)		
5	Cont. Equity Value	128,377		
			EV/Sales	1.8x
			P/E (levered)	15.0x
			P/E (unlevered)	18.8x

- 19-14. How does the assumption on future improvements in working capital affect your answer to Problem 13?

It does not affect the answer because the working capital savings do not affect EBITDA or debt levels.

Continuation Value: Multiples Approach (\$000s)				
1	EBITDA in 2010	26,745	Common Multiples	
2	EBITDA multiple	9.1x		
3	Cont. Enterprise Value	243,377		
4	Debt	(115,000)		
5	Cont. Equity Value	128,377		
			EV/Sales	1.8x
			P/E (levered)	15.0x
			P/E (unlevered)	18.8x

- 19-15. Approximately what expected future long-run growth rate would provide the same EBITDA multiple in 2010 as Ideko has today (i.e., 9.1)? Assume that the future debt-to-value ratio is held constant at 40%; the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year until 2010; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (i.e., the assumptions in Problem 5).

Approximately 5.6%.

Continuation Value: DCF and EBITDA Multiple (\$000s)				
1	Long-term growth rate	5.60%	Target D/(E+D)	40.0%
2			Projected WACC	9.05%
Free Cash Flow in 2011				
3	Unlevered Net Income	13,693	Cont. Enterprise Value	243,098
4	Less: Inc. in NWC	(1,886)		
5	Less: Inc. in Fixed Assets	(3,425)	Implied EBITDA Multiple	9.1x
6	Free Cash Flow	8,382		

- 19-16. Approximately what expected future long-run growth rate would provide the same EBITDA multiple in 2010 as Ideko has today (i.e., 9.1). Assume that the future debt-to-value ratio is held constant at 40%; the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital do *not* occur (i.e., the assumptions in Problem 6).

Approximately 6.05%.

Continuation Value: DCF and EBITDA Multiple (\$000s)			
1	Long-term growth rate	6.05%	Target D/(E+D) 40.0%
2			Projected WACC 9.05%
Free Cash Flow in 2011			
3	Unlevered Net Income	13,752	Cont. Enterprise Value 243,161
4	Less: Inc. in NWC	(2,762)	
5	Less: Inc. in Fixed Assets	(3,700)	Implied EBITDA Multiple 9.1x
6	Free Cash Flow	7,290	

- 19-17. Using the APV method, estimate the value of Ideko and the NPV of the deal using the continuation value you calculated in Problem 13 and the unlevered cost of capital estimate in Section 19.4. Assume that the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year until 2010; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital occur (i.e., the assumptions in Problem 5).

The equity value is \$90 million so the NPV of the deal is $90 - 53 = \$37$ million.

	Year	2005	2006	2007	2008	2009	2010
APV Method (\$ millions)							
1	Free Cash Flow		13,911	7,394	8,121	9,507	(3,759)
2	Unlevered Value V^u	180,136	184,238	195,268	206,674	217,835	243,377
3	Interest Tax Shield		2,380	2,380	2,380	2,380	2,380
4	Tax Shield Value T^s	9,811	8,098	6,269	4,315	2,228	-
5	APV: $V^L = V^u + T^s$	189,946	192,336	201,537	210,989	220,063	243,377
6	Debt	(100,000)	(100,000)	(100,000)	(100,000)	(100,000)	(115,000)
7	Equity Value	89,946	92,336	101,537	110,989	120,063	128,377

- 19-18. Using the APV method, estimate the value of Ideko and the NPV of the deal using the continuation value you calculated in Problem 13 and the unlevered cost of capital estimate in Section 19.4. Assume that the debt cost of capital is 6.8%; Ideko's market share will increase by 0.5% per year; investment, financing, and depreciation will be adjusted accordingly; and the projected improvements in working capital do *not* occur (i.e., the assumptions in Problem 6).

The equity value is \$80 million so the NPV of the deal is $90 - 53 = \$27$ million.

	Year	2005	2006	2007	2008	2009	2010
APV Method (\$ millions)							
1	Free Cash Flow		6,246	6,467	7,102	8,387	(4,989)
2	Unlevered Value V^u	170,107	180,872	192,492	204,639	216,717	243,377
3	Interest Tax Shield		2,380	2,380	2,380	2,380	2,380
4	Tax Shield Value T^s	9,811	8,098	6,269	4,315	2,228	-
5	APV: $V^L = V^u + T^s$	179,918	188,970	198,760	208,954	218,945	243,377
6	Debt	(100,000)	(100,000)	(100,000)	(100,000)	(100,000)	(115,000)
7	Equity Value	79,918	88,970	98,760	108,954	118,945	128,377

- 19-19.** Use your answers from Problems 17 and 18 to infer the value today of the projected improvements in working capital under the assumptions that Ideko's market share will increase by 0.5% per year and that investment, financing, and depreciation will be adjusted accordingly.

The value of the savings in working capital management is the difference between the value with and without the savings—approximately \$10 million.

Chapter 20

Financial Options

20-1. Explain the meanings of the following financial terms:

- a. Option
- b. Expiration date
- c. Strike price
- d. Call
- e. Put

- a. Option: An option is a contract that gives one party the right, but not the obligation, to buy or sell an asset at some point in the future.
- b. Expiration date: The last date on which the holder still has the right to exercise the option. If the option is American, the right can be exercised until the exercise date; if it is European, the option can be exercised only on the exercise date.
- c. Strike price: the price at which the holder of the option has the right to buy or sell the asset.
- d. Call: An option that gives its holder the right to buy an asset.
- e. Put: An option that gives its holder the right to sell an asset.

20-2. What is the difference between a European option and an American option? Are European options available exclusively in Europe and American options available exclusively in America?

European options can be exercised *only* on the exercise date, while American options can be exercised on any date *prior* to the exercise date. Both types of options are traded in both Europe and America.

20-3. Below is an option quote on IBM from the CBOE Web site.

- a. Which option contract had the most trades today?
- b. Which option contract is being held the most overall?
- c. Suppose you purchase one option with symbol IBM GA-E. How much will you need to pay your broker for the option (ignoring commissions)?
- d. Explain why the last sale price is not always between the bid and ask prices.
- e. Suppose you sell one option with symbol IBM GA-E. How much will you receive for the option (ignoring commissions)?
- f. The calls with which strike prices are currently in-the-money? Which puts are in-the-money?
- g. What is the difference between the option with symbol IBM GS-E and the option with symbol IBM HS-E?

IBM

Jul 13 2009 @ 13:26 ET

Bid 102.2 Ask 102.22 Size 6 × 6

102.22 + 1.39

Vol 5683797

Calls	Last Sale	Net	Bid	Ask	Vol	Open Int	Puts	Last Sale	Net	Bid	Ask	Vol	Open Int
09 Jul 95.00 (IBM GS-E)	7.50	0.95	7.40	7.60	26	8159	09 Jul 95.00 (IBM SS-E)	0.31	-0.24	0.25	0.35	2039	11452
09 Jul 100.00 (IBM GT-E)	3.50	0.72	3.40	3.50	1764	14436	09 Jul 100.00 (IBM ST-E)	1.25	-0.65	1.20	1.25	2262	19401
09 Jul 105.00 (IBM GA-E)	0.91	0.26	0.90	1.00	1945	23210	09 Jul 105.00 (IBM SA-E)	3.79	-1.56	3.60	3.80	379	8000
09 Jul 110.00 (IBM GB-E)	0.15	0.07	0.10	0.15	632	20808	09 Jul 110.00 (IBM SB-E)	7.57	-1.53	7.80	8.00	35	6536
09 Aug 95.00 (IBM HS-E)	8.75	1.35	8.40	8.60	32	1532	09 Aug 95.00 (IBM TS-E)	1.51	-0.49	1.50	1.60	1076	2766
09 Aug 100.00 (IBM HT-E)	5.11	0.91	4.80	5.00	122	2754	09 Aug 100.00 (IBM TT-E)	2.90	-0.86	3.00	3.20	513	5322
09 Aug 105.00 (IBM HA-E)	2.40	0.44	2.35	2.40	456	6091	09 Aug 105.00 (IBM TA-E)	5.99	-0.81	5.50	5.70	52	1586
09 Aug 110.00 (IBM HB-E)	0.95	0.25	0.90	0.95	207	3429	09 Aug 110.00 (IBM TB-E)	10.60	-0.40	9.10	9.30	10	751

Source: Chicago Board Options Exchange at www.cboe.com

- 09 Jul 100 Put
- 09 Jul 105 call
- $\$1.00 \times 100 = \100
- Last sale may have happened earlier in the day, whereas bid/ask are current quotes.
- $\$0.90 \times 100 = \90
- Calls : 95, 100
Puts : 105, 110

Identical except that the second expires one month later than the first.

20-4. Explain the difference between a long position in a put and a short position in a call.

When a party has a long position in a put, it has the right to sell the underlying asset at the strike price; when it has a short position in a call, it has the obligation to sell the underlying asset at the strike price if exercised. These are clearly different positions.

20-5. Which of the following positions benefit if the stock price increases?

- Long position in a call
- Short position in a call
- Long position in a put
- Short position in a put

Long call & short put

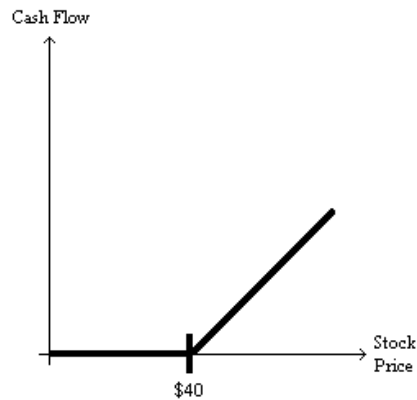
20-6. You own a call option on Intuit stock with a strike price of \$40. The option will expire in exactly three months' time.

- If the stock is trading at \$55 in three months, what will be the payoff of the call?
- If the stock is trading at \$35 in three months, what will be the payoff of the call?
- Draw a payoff diagram showing the value of the call at expiration as a function of the stock price at expiration.

Long call option: value at expiration:

- \$15
- 0\$

c. Draw graph:



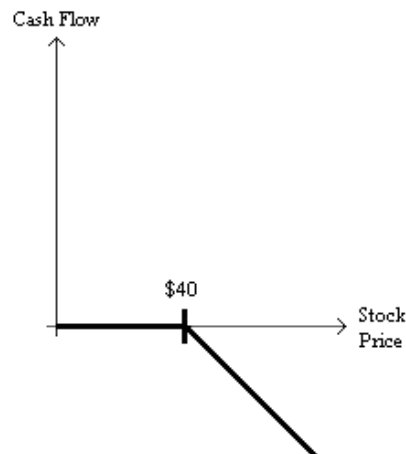
20-7. Assume that you have shorted the call option in Problem 6.



- If the stock is trading at \$55 in three months, what will you owe?
- If the stock is trading at \$35 in three months, what will you owe?
- Draw a payoff diagram showing the amount you owe at expiration as a function of the stock price at expiration.

Short call: value at expiration date:

- You owe \$15.
- You owe nothing.
- Draw the payoff diagram:



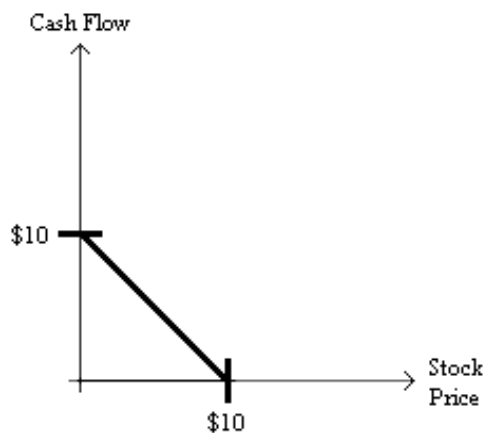
20-8. You own a put option on Ford stock with a strike price of \$10. The option will expire in exactly six months' time.



- If the stock is trading at \$8 in six months, what will be the payoff of the put?
- If the stock is trading at \$23 in six months, what will be the payoff of the put?
- Draw a payoff diagram showing the value of the put at expiration as a function of the stock price at expiration.

Long put value at expiration:

- \$2
- \$0
- Draw payoff diagram:



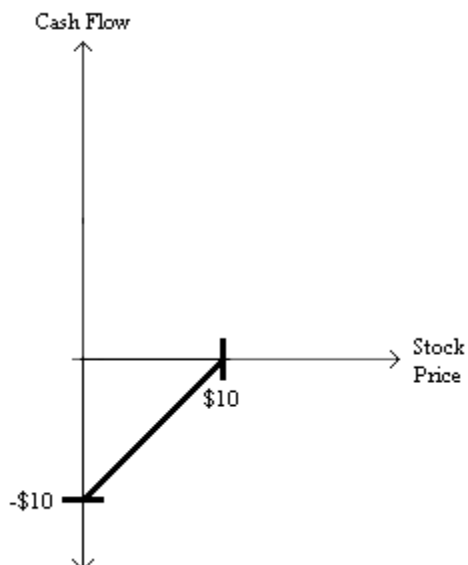
20-9. Assume that you have shorted the put option in Problem 8.



- If the stock is trading at \$8 in three months, what will you owe?
- If the stock is trading at \$23 in three months, what will you owe?
- Draw a payoff diagram showing the amount you owe at expiration as a function of the stock price at expiration.

Short put: value at expiration:

- You owe \$2.
- You owe nothing.
- Draw payoff diagram:



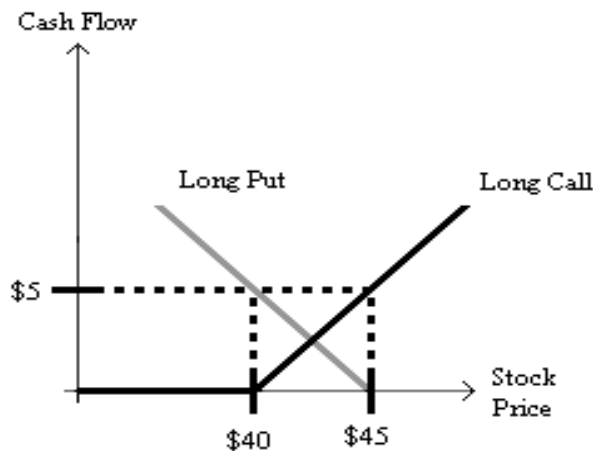
- 20-10. What position has more downside exposure: a short position in a call or a short position in a put? That is, in the worst case, in which of these two positions would your losses be greater?**

Downside exposure is larger with a short call (the downside is unlimited) than with a short put (the downside cannot be larger than the strike price).

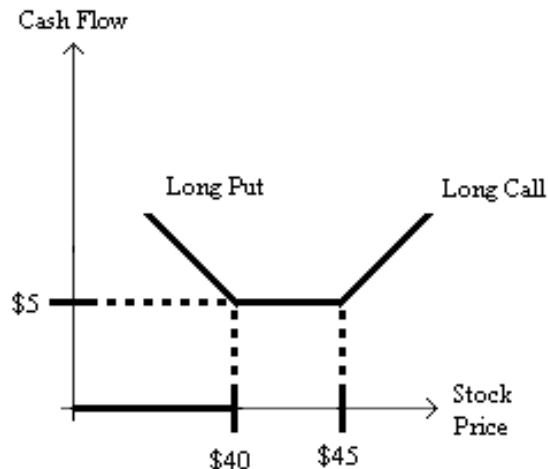
- 20-11. Consider the July 2009 IBM call and put options in Problem 3. Ignoring any interest you might earn over the remaining few days' life of the options:**

- Compute the break-even IBM stock price for each option (i.e., the stock price at which your total profit from buying and then exercising the option would be zero).
 - Which call option is most likely to have a return of -100% ?
 - If IBM's stock price is \$111 on the expiration day, which option will have the highest return?
- For calls, strike + ask. For puts, strike – ask.
 - 110 call option is worthless if IBM is below 110.
 - 110 call option has return of $1/.15 - 1 = 567\%$.

- 20-12. You are long both a call and a put on the same share of stock with the same exercise date. The exercise price of the call is \$40 and the exercise price of the put is \$45. Plot the value of this combination as a function of the stock price on the exercise date.**

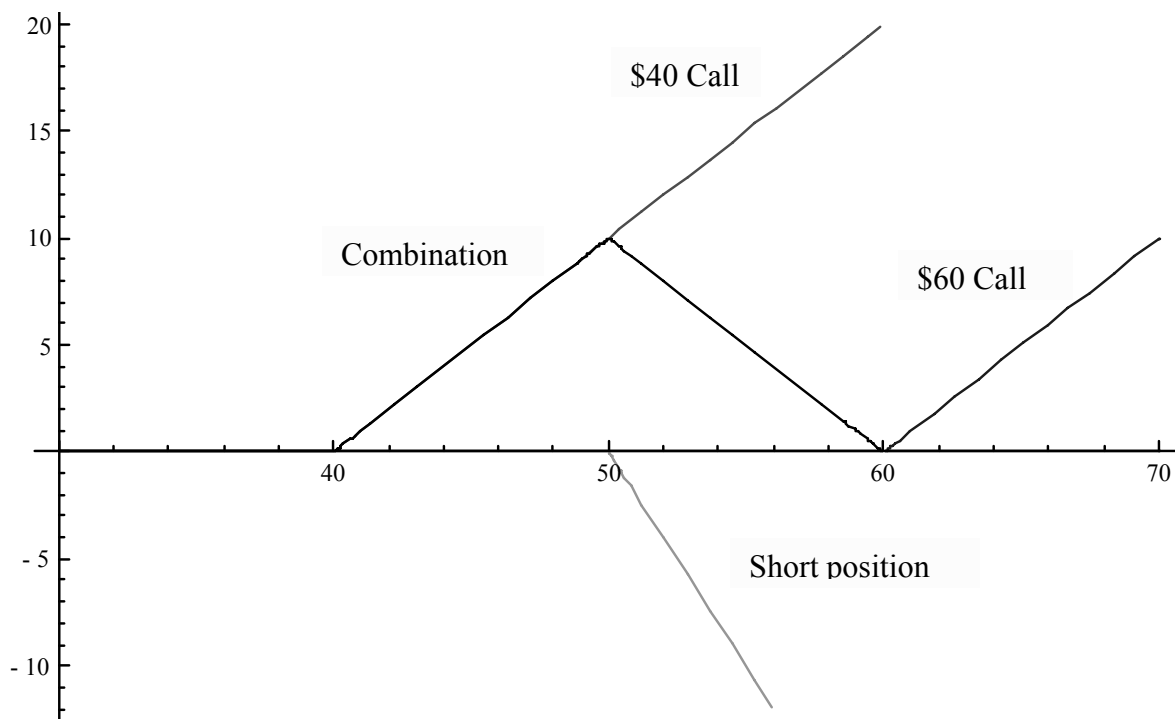


⇓



- 20-13.** You are long two calls on the same share of stock with the same exercise date. The exercise price of the first call is \$40 and the exercise price of the second call is \$60. In addition, you are short two otherwise identical calls, both with an exercise price of \$50. Plot the value of this combination as a function of the stock price on the exercise date. What is the name of this combination of options?

The top curve is the \$40 Call, the middle curve is the \$60 Call, the bottom curve is the short position in two \$50 calls and the up then down curve is the combination.



This is called a Butterfly Spread.

- 20-14.** A forward contract is a contract to purchase an asset at a fixed price on a particular date in the future. Both parties are obligated to fulfill the contract. Explain how to construct a forward contract on a share of stock from a position in options.

A forward with price p can be constructed longing a call and shorting a put with strike p .

- 20-15.** You own a share of Costco stock. You are worried that its price will fall and would like to insure yourself against this possibility. How can you purchase insurance against this possibility?

To protect against a fall in the price of Costco, you can buy a put with Costco as the underlying asset. By doing this, over the life of the option you are guaranteed to get at least the strike price from selling the stock you already have.

- 20-16.** It is July 13, 2009, and you own IBM stock. You would like to insure that the value of your holdings will not fall significantly. Using the data in Problem 3, and expressing your answer in terms of a percentage of the current value of your portfolio:

- a. What will it cost to insure that the value of your holdings will not fall below \$95 per share between now and the third Friday in July?
 - b. What will it cost to insure that the value of your holdings will not fall below \$95 per share between now and the third Friday in August?
 - c. What will it cost to insure that the value of your holdings will not fall below \$100 per share between now and the third Friday in August?
- a. To ensure that the value of your IBM does not fall significantly, you would purchase a protective put. The current ask price for a protective put with a strike price of \$95 that expires the third Friday of July is \$0.35 per share. As a percentage of your portfolio this cost to insure is $\$0.35/\$102.22 = .0034$ or 0.34% of the value of your portfolio.
 - b. The ask price of a protective put with a strike price of \$95 that expires on the third Friday of August is \$1.60. The cost to insure the value of your holdings will not fall is $\$1.60/\$102.22 = 0.0157$ or 1.57%.
 - c. The ask price of a protective put with a strike price of \$100 that expires on the third Friday of August is \$3.20. The cost to insure the value of your holdings will not fall is $\$3.20/\$102.22 = 0.0313$ or 3.13%.

- 20-17. Dynamic Energy Systems stock is currently trading for \$33 per share. The stock pays no dividends. A one-year European put option on Dynamic with a strike price of \$35 is currently trading for \$2.10. If the risk-free interest rate is 10% per year, what is the price of a one-year European call option on Dynamic with a strike price of \$35?**

Put-call parity:

$$C = P + S - \frac{K}{1+r} = 2.10 + 33 - \frac{35}{1.1} = 3.282$$

- 20-18. You happen to be checking the newspaper and notice an arbitrage opportunity. The current stock price of Intrawest is \$20 per share and the one-year risk-free interest rate is 8%. A one year put on Intrawest with a strike price of \$18 sells for \$3.33, while the identical call sells for \$7. Explain what you must do to exploit this arbitrage opportunity.**

The arbitrage opportunity exists because:

$$\$7 > \$3.33 + \$20 - \frac{\$18}{(1+0.08)} = \$6.66.$$

So the call is overpriced compared to the portfolio of a put, the stock, and risk-free borrowing.

As a result, the strategy would be to sell the call option, buy the put, buy the stock, and borrow \$16.67 (the present value of \$18).

The net amount left after doing this is \$.34, with no cash flows when the options expire.

- 20-19. Consider the July 2009 IBM call and put options in Problem 3. Ignoring the negligible interest you might earn on TBills over the remaining few days' life of the options, show that there is no arbitrage opportunity using put-call parity for the options with a \$100 strike price. Specifically:**

- a. What is your profit/loss if you buy a call and TBills, and sell IBM stock and a put option?
 - b. What is your profit/loss if you buy IBM stock and a put option, and sell a call and TBills?
 - c. Explain why your answers to (a) and (b) are not both zero.
- a. Consider : Buy call & TBills, sell stock & put $-3.50 - 100 + 102.2 + 1.20 = -0.10$
 - b. Sell call & tBills, buy stock & put $+3.40 + 100 - 102.22 - 1.25 = -.07$

- c. Both negative due to transactions costs: call spread (0.10) + put spread (0.05) + stock spread (0.02) = 0.17 in total loss in (a) & (b)

20-20. Suppose Amazon stock is trading for \$70 per share, and Amazon pays no dividends.

- a. What is the maximum possible price of a call option on Amazon?
- b. What is the maximum possible price of a put option on Amazon with a strike price of \$100?
- c. What is the minimum possible value of a call option on Amazon stock with a strike price of \$50?
- d. What is the minimum possible value of an American put option on Amazon stock with a strike price of \$100?
 - a. \$70 (Stock price)
 - b. \$100 (strike price)
 - c. Intrinsic value = \$20
 - d. Intrinsic value = \$30

20-21. Consider the data for IBM options in Problem 3. Suppose a new American-style put option on IBM is issued with a strike price of \$110 and an expiration date of August 1st.

- a. What is the maximum possible price for this option?
- b. What is the minimum possible price for this option?
 - a. No one will pay more than the price of the Aug option (which expires later), so \$9.30.
 - b. No one would sell for less than the sale price of the July option: \$7.80.

20-22. You are watching the option quotes for your favorite stock, when suddenly there is a news announcement. Explain what type of news would lead to the following effects:

- a. Call prices increase, and put prices fall.
- b. Call prices fall, and put prices increase.
- c. Both call and put prices increase.
 - a. Good news about the stock, which raises its stock price
 - b. Bad news, which lowers the stock's price
 - c. News that increases the volatility of the stock

20-23. Explain why an American call option on a non-dividend-paying stock always has the same price as its European counterpart.

Because the option to exercise early is worthless, the American option provides no more benefits than its European counterpart.

20-24. Consider an American put option on XAL stock with a strike price of \$55 and one year to expiration. Assume XAL pays no dividends, XAL is currently trading for \$10 per share, and the one-year interest rate is 10%. If it is optimal to exercise this option early:

- a. What is the price of a one-year American put option on XAL stock with a strike price of \$60 per share?
- b. What is the maximum price of a one-year American call option on XAL stock with a strike price of \$55 per share?
 - a. It is optimal to exercise early puts with higher strikes, so value = intrinsic value of $60 - 10 = \$50$.

b. Because put has no time value, call value must be less than $dis(55) = 55 - 55/1.10 = \5 .

- 20-25. The stock of Harford Inc. is about to pay a \$0.30 dividend. It will pay no more dividends for the next month. Consider call options that expire in one month. If the interest rate is 6% APR (monthly compounding), what is the maximum strike price where it could be possible that early exercise of the call option is optimal? (Round to the nearest dollar.)**

$$0 > \underbrace{dis(K) + P - PV(Div)}_{\text{Time value}} > dis(K) - PV(Div)$$

so

$$PV(Div) > dis(K)$$

i.e., the discount on strike must be smaller than dividend. Given $6\%/12 = 0.5\%$ interest over the month,

$$K - K/1.005 < 0.30$$

so

$$K < 0.30 \times 1.005/.005 = \$60.30.$$

So, strikes at or below \$60 per share could be exercised early.

- 20-26. Suppose the S&P 500 is at 900, and a one-year European call option with a strike price of \$400 has a negative time value. If the interest rate is 5%, what can you conclude about the dividend yield of the S&P 500? (Assume all dividends are paid at the end of the year.)**

Call has negative time value implies

$$dis(K) + P - PV(Div) < 0$$

which means that

$$dis(K) - PV(Div) < 0$$

or $PV(div) > dis(K)$, or $FV(divs) > \text{interest on } K = 5\% \times 400 = 20$. So dividend yield must be at least $20/900 = 2.22\%$.

- 20-27. Suppose the S&P 500 is at 900, and it will pay a dividend of \$30 at the end of the year. Suppose the interest rate is 2%. If a one-year European put option has a negative time value, what is the lowest possible strike price it could have?**

$$-dis(K) + C + PV(div) < 0$$

implies that

$$-dis(K) + PV(div) < 0$$

or

$$dis(K) > PV(div)$$

$$\frac{Kr}{1+r} > \frac{div}{1+r}$$

$$Kr > div$$

so interest on strike must exceed the dividend: $(.02 \times K) > \$30$, so $K > 1500$.

20-28. Wesley Corp. stock is trading for \$25/share. Wesley has 20 million shares outstanding and a market debt-equity ratio of 0.5. Wesley's debt is zero coupon debt with a 5-year maturity and a yield to maturity of 10%.

a. Describe Wesley's equity as a call option. What is the maturity of the call option? What is the market value of the asset underlying this call option? What is the strike price of this call option?

b. Describe Wesley's debt using a call option.

c. Describe Wesley's debt using a put option.

a. Maturity = 5 years

$$\text{Assets} = E + D = \$25 \times 20 + .5(\$25 \times 20) = \$500 + 250 = \$750 \text{ million}$$

$$\text{Strike price} = D = \$250 \text{ million}$$

b. Long the firm's assets and short the equity call option above

c. Long risk-free debt and short a put option on Wesley's assets with a 5-yr maturity and \$250 million face value

20-29. Express the position of an equity holder in terms of put options.

An equity holder is long a put on a share of the value of the firm assets with the per share value of debt as the strike price, long a share on the assets of the firm and short a loan worth the value per share of the debt.

20-30. Use the option data in Figure 20.10 to determine the rate Google would pay if it issued \$128 billion in zero-coupon debt due in January 2011. Suppose Google currently has 320 million shares outstanding, implying a market value of \$135.1 billion. (Assume perfect capital markets.)

Issuing \$128 billion in debt is equivalent to a claim on \$400 of Google's assets per share. From Table 20.5, the average of the Bid-Ask spread of the 11 Jan 400 Call is \$86.05. This gives a market value of the remaining equity of $86.05 \times 320 = \$27.54$ billion. Subtracting from the total value of \$135.1 billion gives the estimated value of the debt: $135.1 - 27.54 = 107.56$ billion. The yield to maturity is

$$\text{then } \left(\frac{128}{107.56} \right)^{12/18} - 1 = 12.3\% . \text{ The credit spread is therefore } 12.3\% - 1\% = 11.1\% .$$

20-31. Suppose Google were to issue \$96 billion in zero-coupon senior debt, and another \$32 billion in zero-coupon junior debt, both due in January 2011. Use the option data in Figure 20.10 to determine the rate Google would pay on the junior debt issue. (Assume perfect capital markets.)

We can compute the rate on the senior debt as in Example 20.10. The debt value is \$87.4 billion, and the rate is 6.46%.

Next, we can determine the value of equity. Because the firm has $\$96 + \$32 = \$128$ billion in total debt, or $128\text{b}/320\text{m} = \400 per share, from Figure 20.10, the average of the Bid-Ask spread of the 11 Jan 400 Call is \$86.05. This gives a market value of the remaining equity of $86.05 \times 320 = \$27.54$ bil.

Because Senior Debt + Junior Debt + Equity must equal the total value of \$135.1 billion,

Junior debt has a value of $135.1 - 87.4 - 27.54 = \$20.16$ billion.

Therefore, the yield on the junior debt is

$$\left(\frac{32}{20.16} \right)^{12/18} - 1 = 36.1\% .$$

Chapter 21

Option Valuation

- 21-1.** The current price of Estelle Corporation stock is \$25. In each of the next two years, this stock price will either go up by 20% or go down by 20%. The stock pays no dividends. The one-year risk-free interest rate is 6% and will remain constant. Using the Binomial Model, calculate the price of a one-year call option on Estelle stock with a strike price of \$25.

In this case, the stock price either rises to $S_u = 25 \times 1.20 = 30$ or falls to $S_d = 25 \times 0.80 = 20$. The option payoff is therefore either $C_u = 5$ or $C_d = 0$. The replicating portfolio is $\Delta = (5 - 0)/(30 - 20) = 0.5$ and

$$B = (0 - 20 \times 0.5)/1.06 = -9.43.$$

Therefore, $C = 0.5 \times 25 - 9.43 = \3.07 .

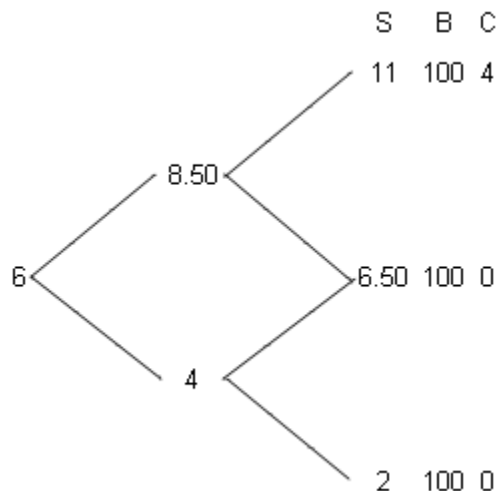
- 21-2.** Using the information in Problem 1, use the Binomial Model to calculate the price of a one year put option on Estelle stock with a strike price of \$25.

The parameters are the same as in 21-1, but the payoff of the put is 0 if the stock goes up and 5 if the stock goes down. Therefore, the replicating portfolio is

$$\Delta = (0 - 5)/(30 - 20) = -0.5 \text{ and } B = (5 - 20 \times (-0.5))/1.06 = 14.15.$$

Therefore, $P = -0.5 \times 25 + 14.15 = \1.65 .

- 21-3.** The current price of Natasha Corporation stock is \$6. In each of the next two years, this stock price can either go up by \$2.50 or go down by \$2. The stock pays no dividends. The one-year risk-free interest rate is 3% and will remain constant. Using the Binomial Model, calculate the price of a two-year call option on Natasha stock with a strike price of \$7.



Up state at time 1: $\Delta = (4 - 0)/(11 - 6.50) = 0.889$, $B = (0 - 6.50 \times 0.889)/1.03 = -5.61$, therefore $C_u = 0.889 \times 8.50 - 5.61 = \1.95 .

In the down state at time 1 the option is worth nothing. The call option at time 0 is therefore equivalent to the replicating portfolio $\Delta = (1.95 - 0)/(8.50 - 4) = 0.433$, $B = (0 - 4 \times 0.433)/1.03 = -1.68$ and so, by the Law of One Price, the initial option price is $0.433 \times 6 - 1.68 = \0.92 .

21-4. Using the information in Problem 3, use the Binomial Model to calculate the price of a two-year European put option on Natasha stock with a strike price of \$7.

If the stock goes up twice, the put is worth zero. If the stock ends up at \$6.50, the put is worth \$0.50; if the stock goes down twice, the put is worth \$5. Given these final values, we can calculate the value of the put at earlier dates using the binomial model.

Up state at time 1: $\Delta = (0 - 0.50)/(11 - 6.50) = -0.111$ and $B = (0.50 - 6.50 \times (-0.111))/1.03 = 1.19$.

Therefore, $P_u = -0.111 \times 8.50 + 1.19 = \0.25 .

Down state at time 1: $\Delta = (0.50 - 5)/(6.50 - 2) = -1$ and $B = (5 - 2 \times (-1))/1.03 = 6.80$

Therefore, $P_d = -1 \times 4 + 6.80 = \2.80 .

Time 0: $\Delta = (0.25 - 2.80)/(8.50 - 4) = -0.567$ and $B = (2.80 - 4 \times (-0.567))/1.03 = 4.92$

Therefore, $P_d = -0.567 \times 6 + 4.92 = \1.52 .

21-5. Suppose the option in Example 21.1 actually sold in the market for \$8. Describe a trading strategy that yields arbitrage profits.

In Example 21.1, the theoretical put price is \$3.30. If it actually sells for a higher price, it is overvalued, and we can sell it and buy the replicating portfolio to earn an arbitrage profit. This means that at $t = 0$, you will sell the put, short 0.3333 shares of stock, and invest \$23.30 in Treasury Bills.

Following this strategy, you will earn the put price less $(60 \times (-0.3333) + 23.30) = \3.30 upfront.

Then, if the stock goes up, our portfolio of the put, shares, and borrowing is worth $\$0 - 0.3333 \times 72 + 23.30 \times 1.03 = \0 . If it goes down, our portfolio is worth $-\$6 - 0.3333 \times \$54 + 23.30 \times 1.03 = \0 , so that at maturity, the payoff of the option and the value of the replicating portfolio cancel out.

(Note that this is not the only arbitrage strategy one can follow—e.g., we could invest more in TBills initially—but it is the one that generates the most cash upfront without any risk of loss in the future.)

21-6. Suppose the option in Example 21.2 actually sold today for \$5. You do not know what the option will trade for next period. Describe a trading strategy that will yield arbitrage profits.

In Example 21.2, the theoretical put price is \$8.68. If it actually sells for \$5, it is underpriced, which means we buy it and sell the replicating portfolio to earn an arbitrage profit. This means that at $t = 0$, you will buy the \$5 put, buy 0.6633 shares of stock, and borrow \$41.84.

Following this strategy, you will end up with $\$41.84 - 0.6633 \times 50 - 5 = \3.68 upfront.

At $t = 1$, you rebalance that portfolio according to the new Δ and B :

If the stock goes up at date 1: we reduce our stock holdings to 0.3333 shares, using the proceeds to reduce our debt to $41.84 \times 1.03 - (0.6633 - 0.3333) \times \$60 = \$23.30$. Then, if the stock goes up again, our portfolio of the put, shares, and borrowing is worth $\$0 + 0.3333 \times 72 - 23.30 \times 1.03 = \0 . If it goes down, our portfolio is worth $\$6 + 0.3333 \times \$54 - 23.30 \times 1.03 = \0 .

If the stock goes down at date 1: we increase our stock holdings to 1 share, and increase our debt to $41.84 \times 1.03 + (1 - 0.6633) \times \$45 = \$58.25$. Then, if the stock goes up, our portfolio of the put, shares, and borrowing is worth $\$6 + 1 \times 54 - 58.25 \times 1.03 = \0 . If it goes down, our portfolio is worth $\$19.50 + 1 \times 40.50 - 58.25 \times 1.03 = \0 .

Thus, we have made a profit of \$3.68 upfront, with a zero payoff no matter what happens in the future.

- 21-7. Eagletron's current stock price is \$10. Suppose that over the current year, the stock price will either increase by 100% or decrease by 50%. Also, the risk-free rate is 25% (EAR).**
- What is the value today of a one-year at-the-money European put option on Eagletron stock?**
 - What is the value today of a one-year European put option on Eagletron stock with a strike price of \$20?**
 - Suppose the put options in parts (a) and (b) could either be exercised immediately, or in one year. What would their values be in this case?**
 - $P_u = 0$, $P_d = 5$, $\text{delta} = (0 - 5)/(20 - 5) = -1/3$, $B = (5 - 5(-1/3))/1.25 = 5.33$
 $P = -1/3(10) + 5.33 = \$2.00$
 - The payoffs are $P_u = 0$, $P_d = 15$. Because these payoffs are three times the payoffs in (a), the put must be worth $3 \times \$2 = \6 .
 - We can exercise the put immediately and get its intrinsic value. In (a), intrinsic value = 0, so not relevant, value is still \$2
 In (b) intrinsic value = $20 - 10 = \$10$, so it is better to exercise now $\Rightarrow$ value is \$10 (not \$6).
- 21-8. What is the highest possible value for the delta of a call option? What is the lowest possible value? (Hint: See Figure 21.1.)**
 $\text{Delta} \leq 1$, $\text{Delta} \geq 0$
- 21-9. Hema Corp. is an all equity firm with a current market value of \$1000 million (i.e., \$1 billion), and will be worth \$900 million or \$1400 million in one year. The risk-free interest rate is 5%. Suppose Hema Corp. issues zero-coupon, one-year debt with a face value of \$1050 million, and uses the proceeds to pay a special dividend to shareholders. Assuming perfect capital markets, use the binomial model to answer the following:**
- What are the payoffs of the firm's debt in one year?**
 - What is the value today of the debt today?**
 - What is the yield on the debt?**
 - Using Modigliani-Miller, what is the value of Hema's equity before the dividend is paid? What is the value of equity just after the dividend is paid?**
 - Show that the ex-dividend value of Hema's equity is consistent with the binomial model. What is the Δ of the equity, when viewed as a call option on the firm's assets?**
 - Either 1050 (if the firm does well) or 900 (if the firm does poorly and defaults).
 - We can use the binomial model: $\text{delta} = (1050 - 900)/(1400 - 900) = 0.3$, $B = (900 - 900(.3))/(1.05) = 600$, Debt value = $0.3(1000) + 600 = 900$.
 - $1050/900 = 16.67\%$
 - MM: initial value should not change = \$1000. Subtract dividend of \$900 (debt value) to determine ex-div value = \$100.
 - Equity payoffs are $1400 - 1050 = 350$ or 0.
 $\text{Delta} = (350 - 0)/(1400 - 900) = 0.70$
 $B = (0 - 900(.7))/1.05 = -600$
 Equity Value = $.7(1000) - 600 = 100$

21-10. Consider the setting of Problem 9. Suppose that in the event Hema Corp. defaults, \$90 million of its value will be lost to bankruptcy costs. Assume there are no other market imperfections.

- a. What is the present value of these bankruptcy costs, and what is their delta with respect to the firm's assets?
- b. In this case, what is the value and yield of Hema's debt?
- c. In this case, what is the value of Hema's equity before the dividend is paid? What is the value of equity just after the dividend is paid?

a. Bankruptcy costs are 0 or 90.

$$\Delta = (0 - 90)/(1400 - 900) = -0.18$$

$$B = (90 - 900(-.18))/1.05 = 240$$

$$BC \text{ value} = -0.18(1000) + 240 = \$60 \text{ million}$$

b. Debt value = \$900 less BC = 900 - 60 = \$840

$$\text{Yield} = 1050/840 - 1 = 25\%$$

c. On announcement, equity value declines by BC = \$1000 - 60 = \$940. Subtract dividend of \$840 (debt value) to determine ex-div value = \$100 (Note this ex-div value is the same as in Problem 9 because equity holders have the same final payoffs.)

21-11. Roslin Robotics stock has a volatility of 30% and a current stock price of \$60 per share. Roslin pays no dividends. The risk-free interest is 5%. Determine the Black-Scholes value of a one-year, at-the-money call option on Roslin stock.

$$BS \text{ value} = \$8.50$$

21-12. Rebecca is interested in purchasing a European call on a hot new stock, Up, Inc. The call has a strike price of \$100 and expires in 90 days. The current price of Up stock is \$120, and the stock has a standard deviation of 40% per year. The risk-free interest rate is 6.18% per year.

- a. Using the Black-Scholes formula, compute the price of the call.
- b. Use put-call parity to compute the price of the put with the same strike and expiration date.

a. Using the Black-Scholes formula:

$$PV(K) = 100 / (1.0638)^{\frac{90}{365}} = 98.487, \quad d_1 = \frac{\ln\left(\frac{120}{98.487}\right)}{0.4\sqrt{90/365}} + \frac{0.4\sqrt{90/365}}{2} = 1.094$$

$$= 1.09167$$

$$d_2 = 1.094 - 0.4\sqrt{90/365} = 0.895$$

$$C = S \times N(d_1) - PV(K)N(d_2) = 120 \times 0.863 - 98.487 \times 0.815 = \$23.29$$

b. Using put-call parity:

$$P = C + PV(K) - S = 23.29 + 98.487 - 120 = \$1.78$$

21-13. Using the data in Table 21.1, compare the price on July 24, 2009, of the following options on JetBlue stock to the price predicted by the Black-Scholes formula. Assume that the standard deviation of JetBlue stock is 65% per year and that the short-term risk-free rate of interest is 1% per year.

- a. December 2009 call option with a \$5 strike price
- b. December 2009 put option with a \$6 strike price

c. March 2010 put option with a \$7 strike price

The January contract expires on the third Friday of January (20th); there are 45 days left until expiration.

$$PV(K) = 11 / (1 + 0.0438)^{45/365} = 10.942,$$

$$d_1 = \frac{\ln\left(\frac{S_t}{PV(K)}\right) + \frac{\sigma\sqrt{T}}{2}}{\frac{0.25\sqrt{45/365}}{2}} = \frac{\ln\left(\frac{12.585}{10.942}\right) + \frac{0.25\sqrt{45/365}}{2}}{0.25\sqrt{45/365}} = 1.638$$

$$d_2 = d_1 - \sigma\sqrt{T} = 1.638 - 0.25\sqrt{45/365} = 1.550$$

Substituting d_1 and d_2 into the Black-Scholes formula gives:

$$\begin{aligned} C_t(S_t, K, T, \sigma, r) &= S_t N(d_1) - PV(K)N(d_2) \\ &= 12.585 \times 0.949 - 10.942 \times 0.939 = 1.67. \end{aligned}$$

This value is between the bid and ask prices of \$1.65 and \$1.75.

a. BS price = \$0.85

b. BS price = \$1.45

c. BS price = \$2.42

21-14. Using the market data in Figure 20.10 and a risk-free rate of 1% per annum, calculate the implied volatility of Google stock in July 2009, using the 320 January 2011 call option.

The bid and ask prices of the call are \$133.90 and 135.90, the time to expiration is approximately 1.53 years (July 13, 2009 to Jan 21, 2011 = 18 + 153 + 365 + 21 = 557 days), and the price of the stock is \$422.27.

$$\text{With } PV(K) = 320 / (1 + 0.01)^{1.53} = 315.18,$$

$$d_1 = \frac{\ln\left(\frac{422.27}{315.18}\right) + \frac{\sigma\sqrt{1.53}}{2}}{\sigma\sqrt{1.53}}, \quad d_2 = d_1 - \sigma\sqrt{1.53}.$$

The Black-Scholes formula, Eq. 21.7, implies: $C = 422.27 \times N(d_1) - 315.18 \times N(d_2)$.

You can verify that C is between the bid and ask prices for the call option when $\sigma = 38\%$. (You can find the σ by guessing or using a calculator or spreadsheet program.) Thus, the implied volatility for Google derived from this call option is about 38.3%.

21-15. Using the implied volatility you calculated in Problem 14, and the information in that problem, use the Black-Scholes option pricing formula to calculate the value of the 340 January 2011 call option.

Implied volatility is 38.5% from Problem 21.9. The strike price = 340, current stock price is 405.85, and the risk-free rate is 4.5%, with a time to maturity of 2.12 years.

The Black-Scholes formula gives:

$$d_1 = \frac{\ln\left(\frac{405.85}{309.71}\right) + \frac{0.385\sqrt{2.12}}{2}}{0.385\sqrt{2.12}} = 0.763, \quad d_2 = 0.763 - 0.385\sqrt{2.12} = 0.202.$$

Therefore,

$$C = S \times N(d_1) - PV(K)N(d_2) = 405.85 \times 0.777 - 309.71 \times 0.580 = \$135.71$$

$$P = C + PV(K) - S = 135.71 + 309.71 - 405.85 = \$39.57.$$

Note that the call price is within the range of the quotes provided in Table 20.5.

BS value using 38.5% vol = \$122.53

- 21-16. Plot the value of a two-year European put option with a strike price of \$20 on World Wide Plants as a function of the stock price. Recall that World Wide Plants has a constant dividend yield of 5% per year and that its volatility is 20% per year. The two-year risk-free rate of interest is 4%. Explain why there is a region where the option trades for less than its intrinsic value.**

The price of the put is given by the standard Black-Scholes equation for puts, but with the stock price replaced everywhere with $S / (1+q)^T = S / (1+0.05)^2 = S / 1.1025$. Given $PV(K) = 20/1.04^2 = 18.491$. Therefore,

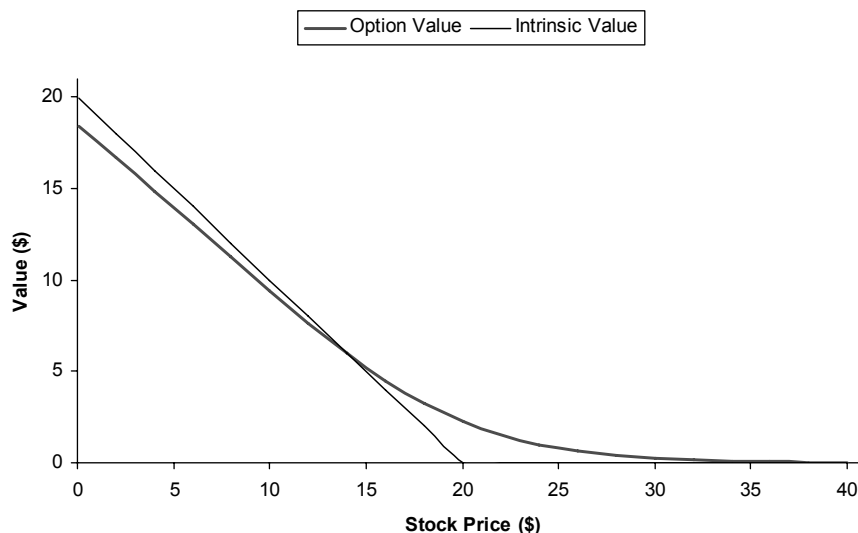
$$d_1 = \frac{\ln\left(\frac{S^x}{PV(K)}\right)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2} = \frac{\ln\left(\frac{S/1.1025}{18.491}\right)}{0.2\sqrt{2}} + 0.1\sqrt{2}$$

$$d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln\left(\frac{S/1.1025}{18.491}\right)}{0.2\sqrt{2}} - 0.1\sqrt{2}$$

so, the value of the put option given current stock price S is

$$\begin{aligned} P(S) &= PV(K) (1 - N(d_2)) - (S^x)(1 - N(d_1)) \\ &= 18.491 \times \left(1 - N\left(\frac{\ln\left(\frac{S/1.1025}{18.491}\right)}{0.2\sqrt{2}} - 0.1\sqrt{2} \right) \right) - S/1.1025 \times \left(1 - N\left(\frac{\ln\left(\frac{S/1.1025}{18.491}\right)}{0.2\sqrt{2}} + 0.1\sqrt{2} \right) \right) \end{aligned}$$

Plotting this function (the curved line below) gives:



Notice that when the put is deep in the money it is worth less than its intrinsic value. In this case the time value is negative because the size of the discount on a two year zero-coupon bond is larger than the value of the dividends and the call option, implying a negative time value for the option.

- 21-17. Consider the at-the-money call option on Roslin Robotics evaluated in Problem 11. Suppose the call option is not available for trade in the market. You would like to replicate a long position in 1000 call options.**
- What portfolio should you hold today?**
 - Suppose you purchase the portfolio in part (a). If Roslin stock goes up in value to \$62 per share today, what is the value of this portfolio now? If the call option were available for trade, what would be the difference in value between the call option and the portfolio (expressed as percent of the value of the call)?**
 - After the stock price change in part (b), how should you adjust your portfolio to continue to replicate the options?**
- a. $\Delta = N(d_1) = N(0.312634) = 0.6227$, so purchase 623 shares.
 $B = -28.859$, so borrow \$28,859.
 $\text{Cost} = 622.27 \times 60 - 28859 = \$8,504$
- b. $\text{Portfolio} = 622.27 \times 62 - 28859 = 9749$
 $\text{Call} = 9790$
 $\text{Difference} = (9749 - 9790)/9790 = -0.42\%$
- c. $\Delta = .663$
 Increase shares to 663.
 Borrow additional $(663 - 623) \times 62 = \2526 .
- 21-18. Consider again the at-the-money call option on Roslin Robotics evaluated in Problem 11. What is the impact on the value of this call option of each of the following changes (evaluated separately)?**
- The stock price increases by \$1 to \$61.**
 - The volatility of the stock goes up by 1% to 31%.**
 - Interest rates go up by 1% to 6%.**
 - One month elapses, with no other change.**
 - The firm announces a \$1 dividend, paid immediately.**
- Original BS price = \$8.50
- BS price = \$9.14
 - BS price = \$8.73
 - BS price = \$8.78
 - BS price = \$8.09
 - Ex-div stock price = \$59, BS price = \$7.89
- 21-19. Harbin Manufacturing has 10 million shares outstanding with a current share price of \$20 per share. In one year, the share price is equally likely to be \$30 or \$18. The risk-free interest rate is 5%.**
- What is the expected return on Harbin stock?**
 - What is the risk neutral probability that Harbin's stock price will increase?**
- a. $.5(30+18)/20 - 1 = 20\%$

b. 25%

21-20. Using the information on Harbin Manufacturing in Problem 19, answer the following:

- Using the risk neutral probabilities, what is the value of a one-year call option on Harbin stock with a strike price of \$25?
- What is the expected return of the call option?
- Using the risk neutral probabilities, what is the value of a one-year put option on Harbin stock with a strike price of \$25?
- What is the expected return of the put option?

a. $25\% \times (30-25)/1.05 = \1.19

b. $50\% \times (5)/1.19 - 1 = 110\%$

c. $75\% \times (25 - 18)/1.05 = \5

d. $50\% \times (7)/5 - 1 = -30\%$

21-21. Using the information in Problem 1, calculate the risk-neutral probabilities. Then use them to price the option.

The risk neutral probabilities can be calculated using:

$$\rho = \frac{(1+r_f)S - S_d}{S_u - S_d} = \frac{(1.06)25 - 20}{30 - 20} = 0.65.$$

Using these probabilities the price of the option is

$$\frac{5(0.65) + 0(0.35)}{1.06} = \$3.066.$$

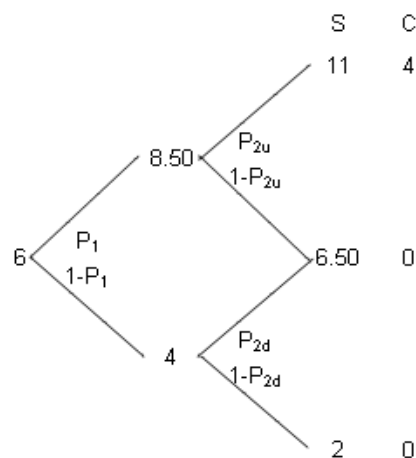
21-22. Using the information in Problem 3, calculate the risk-neutral probabilities. Then use them to price the option.

The risk neutral probabilities are

$$p_1 = \frac{(1.03)6 - 4}{8.5 - 4} = 48.44\%$$

$$p_{2u} = \frac{(1.03)8.5 - 6.5}{11 - 6.5} = 50.11\%$$

$$p_{2d} = \frac{(1.03)4 - 2}{6.5 - 2} = 47.11\%.$$



The value of the call is therefore:

$$\begin{aligned} & \frac{1}{1.03^2} (4p_1p_{2u} + 0(p_1(1-p_{2u}) + (1-p_1)(p_{2d})) + 0(1-p_1)(1-p_{2d})) \\ &= \frac{4(0.4844)(0.5011)}{1.03^2} = \$0.9153. \end{aligned}$$

- 21-23. Explain the difference between the risk-neutral and actual probabilities. In which states is one higher than the other? Why?**

Actual probabilities are the probabilities with which an event will happen. Risk-neutral probabilities are the probabilities of an event happening *in a world where investors are risk-neutral*. Thus, risk-neutral probabilities are a construction and do not reflect reality. Assuming that investors are risk-averse, risk-neutral probabilities are lower in good states and larger in bad states. This has to be the case because if investors were risk-neutral, then the expected returns are the risk-free rate, whereas risk-averse demand higher returns. And given the same payoffs, a higher expected return implies good states are more likely.

- 21-24. Explain why risk-neutral probabilities can be used to price derivative securities in a world where investors are risk averse.**

Risk neutral probabilities can be used to price derivative securities because the pricing of derivatives only depends on the characteristics of the underlying asset. By construction, the value of the underlying asset can be calculated using risk-neutral probabilities and therefore the value of the call will depend on these probabilities. Risk-neutral probabilities are the easiest probabilities to work with, given that they simplify the calculations, and that is why we use them.

- 21-25. Calculate the beta of the January 2010 \$9 call option on JetBlue listed in Table 21.1. Assume that the volatility of JetBlue is 65% per year and its beta is 0.85. The short-term risk-free rate of interest is 1% per year. What is the option's leverage ratio?**

$$\beta_{call} = \frac{\Delta S}{\Delta S + B} \beta_S = \frac{N(d_1)S}{C} \beta_S$$

Call option: 175 days to maturity, $S = 5.035$, $K = 9$, $\sigma = 65\%$,
 $r_f = 1\% \Rightarrow N(d_1) = 0.146$, $C = \$0.141$.

$$\beta_{call} = \frac{0.146 \times 5.035}{0.141} \times 0.85 = 4.4, \text{ and } \text{Leverage ratio} = \frac{N(d_1)S}{C} = 5.2$$

- 21-26. Consider the March 2010 \$5 put option on JetBlue listed in Table 21.1. Assume that the volatility of JetBlue is 65% per year and its beta is 0.85. The short-term risk-free rate of interest is 1% per year.**

- What is the put option's leverage ratio?
- What is the beta of the put option?
- If the expected risk premium of the market is 6%, what is the expected return of the put option based on the CAPM?
- Given its expected return, why would an investor buy a put option?

$$\beta_{put} = \frac{\Delta S}{\Delta S + B} \beta_S = \frac{-(1 - N(d_1))S}{P} \beta_S$$

Put option: 238 days to maturity, $S = 5.035$, $K = 5$, $\sigma = 65\%$, $r_f = 1\% \Rightarrow N(d_1) = 0.613$, $P = \$1.002$

$$\text{a. } \text{Leverage ratio} = \frac{-(1 - N(d_1))S}{P} = -1.9$$

$$\text{b. } \beta_{put} = \frac{-(1 - 0.613) \times 5.035}{1.002} \times 0.85 = -1.65$$

$$\text{c. } E(R) = r_f + \beta_{put} (E(R_{Mkt}) - r_f) = 0.01 + (-1.65)(0.06) = -0.089$$

- d. An investor would buy a put option given a negative expected return to act as a hedge against losses. The negative beta implies the return will move inverse to the market. Giving good returns when times are bad (when positive returns are the most valuable).

21-27. Return to Example 20.10, in which Google was contemplating issuing zero-coupon debt due in 18 months with a face value of \$96 billion, and using the proceeds to pay a special dividend. Google currently has a market value of \$135.1 billion and the risk-free rate is 1%. Using the market data in Figure 20.10, answer the following:

- a. If Google's current equity beta is 1.45, estimate Google's equity beta after the debt is issued.
b. Estimate the beta of the new debt.

$$\beta_E = \Delta \frac{A}{E} \beta_U = \Delta \left(1 + \frac{D}{E}\right) \beta_U$$

- a. We want the β_E following the debt issuance:

$$\beta_E = \left(\Delta \left(1 + \frac{D}{E}\right) \right) \beta_U = \left(N(d_1) \left(1 + \frac{D}{E}\right) \right) \beta_U$$

We have: $S = 135.1$, $K = 96$, $r = 1\%$, $T = 1.5$ years, $\sigma_{\text{implied}} = 38.6\%$ from problem 21.14

$$\Rightarrow N(d_1) = 0.839, C = 47.37.$$

So Google's equity beta would be $0.839 \times \left(1 + \frac{135.1 - 47.37}{47.37}\right) \times 1.45 = 3.47$.

b.
$$\beta_D = (1 - \Delta) \frac{A}{D} \beta_U = (1 - \Delta) \left(1 + \frac{E}{D}\right) \beta_U$$

$$\beta_D = \left((1 - \Delta) \left(1 + \frac{D}{E}\right) \right) \beta_U = \left((1 - N(d_1)) \left(1 + \frac{D}{E}\right) \right) \beta_U$$

So Google's debt beta would be $(1 - 0.839) \times \left(1 + \frac{135.1 - 47.37}{47.37}\right) \times 1.45 = 0.67$.

21-28. You would like to know the unlevered beta of Schwartz Industries (SI). SI's value of outstanding equity is \$400 million, and you have estimated its beta to be 1.2. SI has four-year zero-coupon debt outstanding with a face value of \$100 million that currently trades for \$75 million. SI pays no dividends and reinvests all of its earnings. The four-year risk-free rate of interest is currently 5.13%. Use the Black-Scholes formula to estimate the unlevered beta of the firm.

The equity can be interpreted as a four-year call option on the firm's assets with a strike price of \$100 million. The current market value of assets is $400 + 75 = \$475$ million.

Using the Black-Scholes formula, we can get the implied volatility of assets: Call option value = 400, $K = 100$, $S = 475$, $T = 4$, $r = 0.0513 \Rightarrow \sigma = 60.6\%$.

Using the volatility:

$$\Delta = N(d_1) = N \left[\frac{\ln\left(\frac{475}{100 / (1 + 0.0513)^4}\right)}{0.606\sqrt{4}} + \frac{0.606\sqrt{4}}{2} \right] = 0.980$$

$$\beta_U = \frac{\beta_E}{\Delta(1 + \frac{D}{E})} = \frac{1.2}{0.980 \times (1 + \frac{75}{400})} = 1.03$$

a: $D = 900/1.09^5 = 585$. $S = D + E = 585 + 20 \times 25 = 1085$, $r_f = 5\%$, $K = 900$, $T = 5$

σ Vol = 35%

b: $B_d D/B_e E = (1 - \Delta)/\Delta = (1 - .827)/.827 = 0.209$

c: Asset value could fall to 1008, and with volatility of 45%, equity call option is worth 500. Thus, NPV could be $1008 - 1085 = -77$ million, and equity holders still gain!

21-29. The J. Miles Corp. has 25 million shares outstanding with a share price of \$20 per share. Miles also has outstanding zero-coupon debt with a 5-year maturity, a face value of \$900 million, and a yield to maturity of 9%. The risk-free interest rate is 5%.

- a. What is the implied volatility of Miles' assets?
 - b. What is the minimum profitability index required for equity holders to gain by funding a new investment that does not change the volatility of the Miles' assets?
 - c. Suppose Miles is considering investing cash on hand in a new investment that will increase the volatility of its assets by 10%. What is the minimum NPV such that this investment will increase the value of Miles' shares?
- a: Miles' equity can be viewed as a call option on Miles' assets with a strike price of 900, five years to maturity, and a market value of $\$20/\text{share} \times 25 \text{ million shares} = \500 million. The market value of Miles' debt is $D = 900/1.09^5 = 585$. Using Black-Scholes with parameters $S = D + E = 585 + 500 = 1085$, $r_f = 5\%$, $K = 900$, $T = 5$, we find an implied volatility of Miles' assets of 35%.
- b: We can use Black-Scholes to calculate the delta of the equity call option in (a) to be $\Delta = 0.827$. Then, the profitability index must exceed (see Example 21.11 and also Eq. 16.2) $\beta_d D/(\beta_e E) = (1 - \Delta)/\Delta = (1 - .827)/.827 = 0.209$. (To understand this, note that a \$1 investment by equity holders with and NPV of \$0.209 will increase the value of the assets by \$1.209, and so increase the value of equity by approximately $0.827 \times 1.209 = \$1$.)
- c: Asset value could fall to 1009, and with volatility of 45%, equity call option has value just over 500.

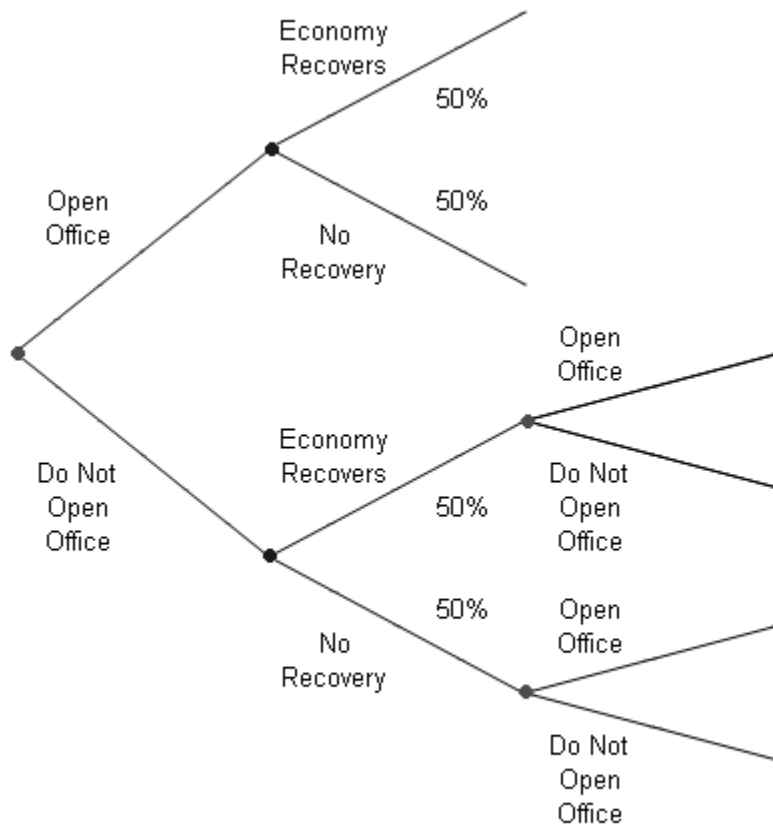
Thus, NPV could be $1009 - 1085 = -76$ million, and equity holders still gain!

Chapter 22

Real Options

- 22-1. Your company is planning on opening an office in Japan. Profits depend on how fast the economy in Japan recovers from its current recession. There is a 50% chance of recovery this year. You are trying to decide whether to open the office now or in a year. Construct the decision tree that shows the choices you have to open the office either today or one year from now.

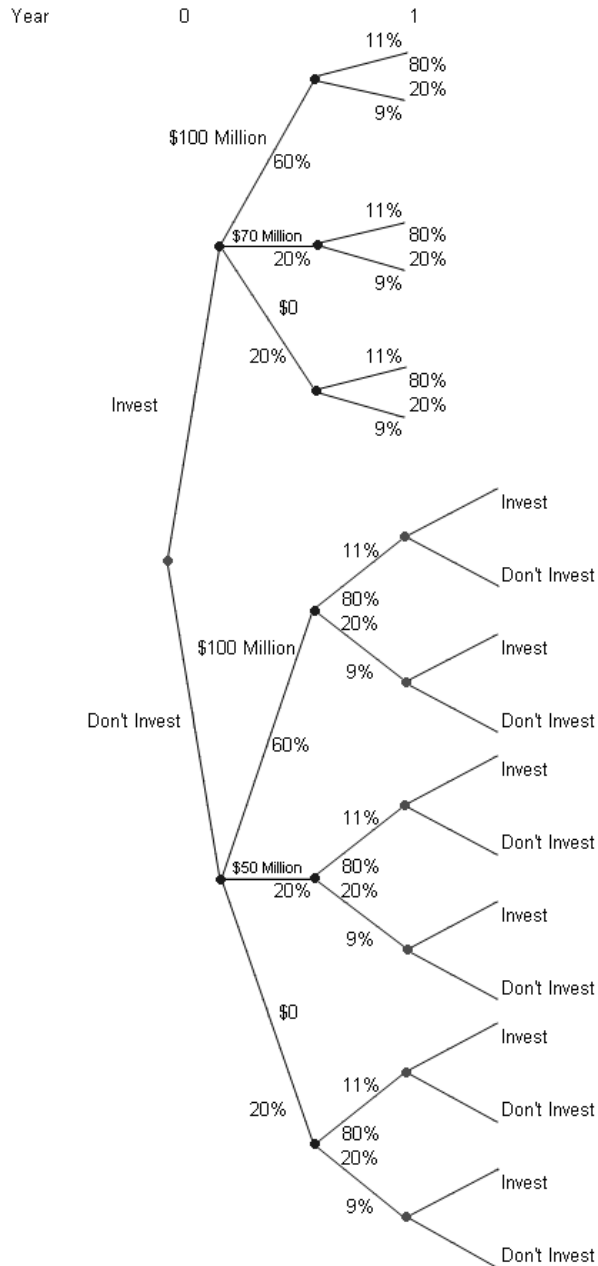
Decision Tree



- 22-2. You are trying to decide whether to make an investment of \$500 million in a new technology to produce Everlasting Gobstoppers. There is a 60% chance that the market for these candies will produce profits of \$100 million annually, a 20% chance the market will produce profits of \$50 million, and a 20% chance that there will be no profits. The size of the market will become clear one year from now. Currently, the cost of capital of the project is 11% per year. There is a 20% chance that the cost of capital will drop to 9% in a year and stay at that level forever, and an 80% chance that it will stay at 11% forever. Movements in the cost of capital are unrelated to the size of the candy market. Construct the decision tree that shows the choices you have to make the investment either today or one year from now.

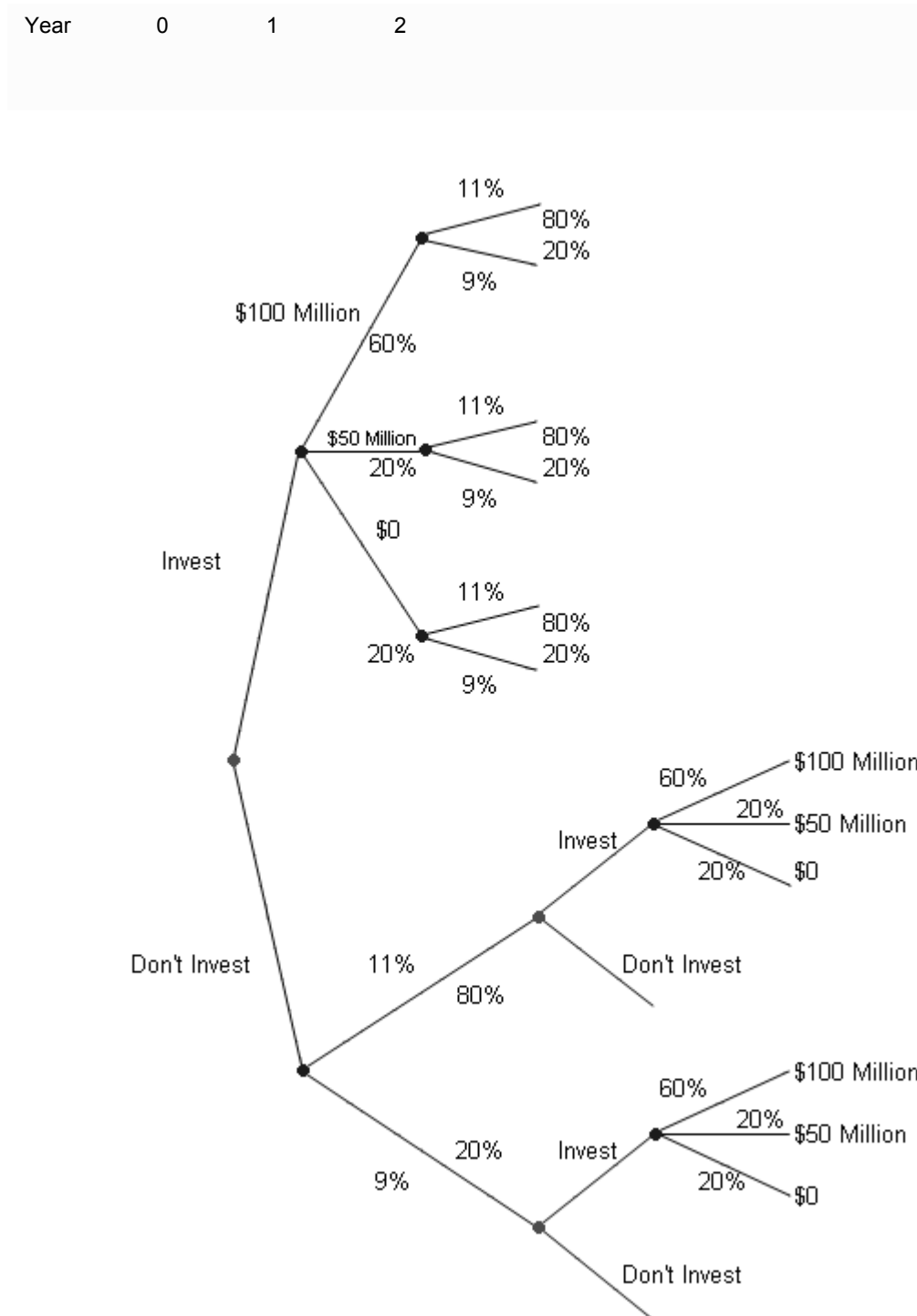


Decision Tree



- 22-3. Using the information in Problem 2, rework the problem assuming you find out the size of the Everlasting Gobstopper market one year *after* you make the investment. That is, if you do not make the investment, you do not find out the size of the market. Construct the decision tree that shows the choices you have under these circumstances.

Decision Tree



22-4. Describe the benefits and costs of delaying an investment opportunity.

By delaying, you delay the benefits of taking on the project and your competitors might take advantage of this delay. However, by delaying, uncertainty can be resolved, so you can become better informed and make better decisions.

22-5. You are a financial analyst at Global Conglomerate and are considering entering the shoe business. You believe that you have a very narrow window for entering this market. Because of Christmas demand, the time is right today and you believe that exactly a year from now would also be a good opportunity. Other than these two windows, you do not think another opportunity will exist to break into this business. It will cost you \$35 million to enter the market. Because other shoe manufacturers exist and are public companies, you can construct a perfectly comparable company. Hence, you have decided to use the Black-Scholes formula to decide when and if you should enter the shoe business. Your analysis implies that the current value of an operating shoe company is \$40 million. However, the flow of customers is uncertain, so the value of the company is volatile—your analysis indicates that the volatility is 25% per year. Fifteen percent of the value of the company is attributable to the value of the free cash flows (cash available to you to spend how you wish) expected in the first year. If the one-year risk-free rate of interest is 4%:

- a. Should Global enter this business and, if so, when?
- b. How will the decision change if the current value of a shoe company is \$36 million instead of \$40 million?
- c. Plot the value of your investment opportunity as a function of the current value of a shoe company.

a. $S^x = S - PV(\text{Div}) = 40(1 - 0.15) = 40(0.85)$

$$PV(K) = 35 / (1.04) = 33.6538$$

$$T = 1$$

$$\sigma = 0.25$$

$$d_1 = \frac{\ln(S^x / PV(K)) + \frac{\sigma\sqrt{T}}{2}}{\sigma\sqrt{T}} = 0.1659$$

$$d_2 = -0.0841$$

$$C = S^x N(d_1) - PV(K)N(d_2) = \$3.54$$

So the value of waiting is \$3.54 million. The value of investing today is $\$40 - \$35 = \$5$ million.

So they should enter the business now.

b. $S^x = S - PV(\text{Div}) = 36(1 - 0.15) = 36(0.85)$

$$PV(K) = 35 / (1.04) = 33.6538$$

$$T = 1$$

$$\sigma = 0.25$$

$$d_1 = \frac{\ln(S^x / PV(K)) + \frac{\sigma\sqrt{T}}{2}}{\sigma\sqrt{T}} = 0.2555$$

$$d_2 = -0.5055$$

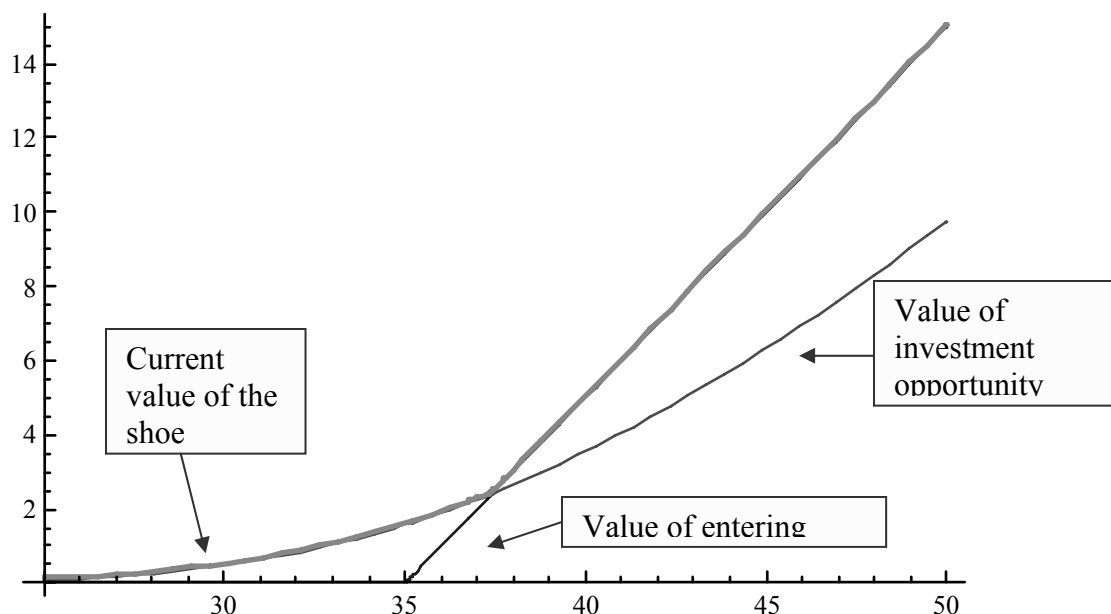
$$C = S^x N(d_1) - PV(K)N(d_2) = \$1.90$$

So the value of waiting is \$1.90 million. The value of investing today is $\$36 - \$35 = \$1$ million.

So they should not enter the business now.

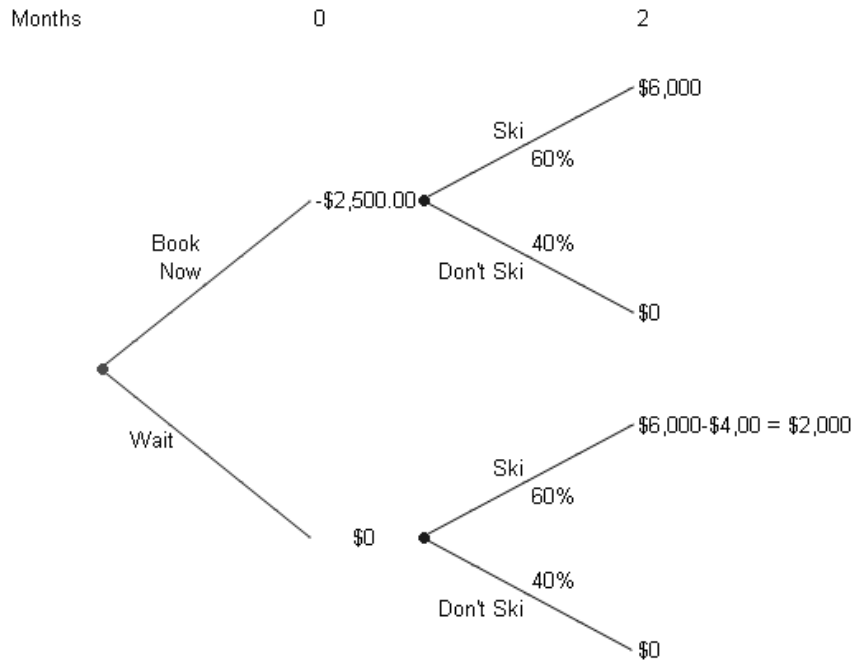
c.

Decision Tree



- 22-6. It is the beginning of September and you have been offered the following deal to go heli-skiing. If you pick the first week in January and pay for your vacation now, you can get a week of heli-skiing for \$2500. However, if you cannot ski because the helicopters cannot fly due to bad weather, there is no snow, or you get sick, you do not get a refund. There is a 40% probability that you will not be able to ski. If you wait until the last minute and go only if you know that the conditions are perfect and you are healthy, the vacation will cost \$4000. You estimate that the pleasure you get from heli-skiing is worth \$6000 per week to you (if you had to pay any more than that, you would choose not to go). If your cost of capital is 8% per year, should you book ahead or wait?

Decision Tree



If you book now, your expected benefit from skiing in 4 months is:

$$6,000(0.60) + 0(0.40) = \$3,600$$

The NPV of booking today is therefore:

$$NPV = \frac{3,600}{1.08^{12}} - 2,500 = \$1,008.82$$

If you wait to book the expected benefit in 4 months is:

$$2,000(0.60) + 0(0.40) = \$1,200$$

The PV of this today is

$$\frac{\$1,200}{1.08^{12}} = \$1,169.61$$

So you should wait.

22-7.



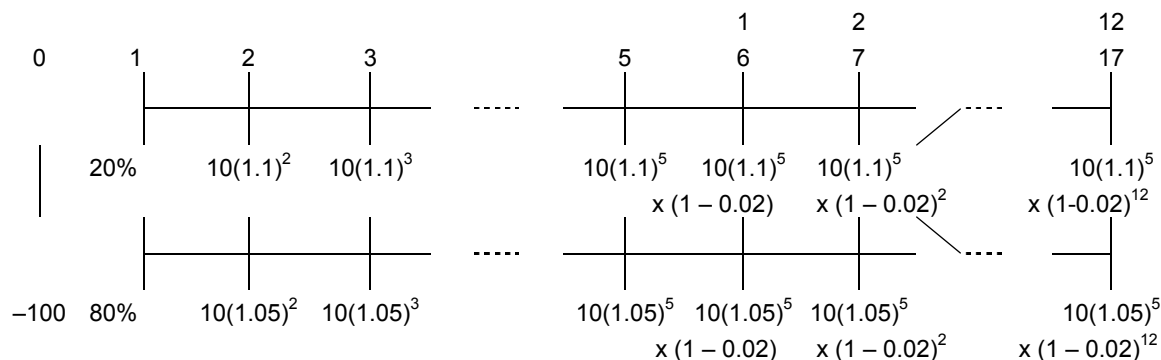
A professor in the Computer Science department at United States Institute of Technology has just patented a new search engine technology and would like to sell it to you, an interested venture capitalist. The patent has a 17-year life. The technology will take a year to implement (there are no cash flows in the first year) and has an up-front cost of \$100 million. You believe this technology will be able to capture 1% of the Internet search market, and currently this market generates profits of \$1 billion per year. Over the next five years, the risk-neutral probability that profits will grow at 10% per year is 20% and the risk-neutral probability that profits will grow at 5% per year is 80%. This growth rate will become clear one year from now (after the first year of growth). After five years, profits are expected to decline 2% annually. No profits are expected after the patent runs out. Assume that all risk-free interest rates are constant (regardless of the term) at 10% per year.

a. Calculate the NPV of undertaking the investment today.

b. Calculate the NPV of waiting a year to make the investment decision.

c. What is your optimal investment strategy?

a. Decision Tree



If the high growth rate state occurs, then the NPV is:

$$NPV_{high} = 4(10) + \frac{1}{(1.1)^5} \left(\frac{10(1.1)^5 (1-0.02)}{0.1+0.02} \right) \left(1 - \left(\frac{0.98}{1.1} \right)^{12} \right) - 100$$

$$= \$1,247,147.$$

Note: Since the first four cash flows grow out of the same rate as the discount rate, their present value is just the sum of the cash flows.

If the low growth rate state occurs, then the NPV is:

$$NPV_{low} = \frac{1}{1.1} \left(\frac{10(1.05)^2}{0.1-0.05} \left(1 - \left(\frac{1.05}{1.1} \right)^4 \right) \right) + \frac{1}{(1.1)^5} \left(\frac{10(1.05)^5 (1-0.02)}{0.1+0.02} \right) \left(1 - \left(\frac{0.98}{1.1} \right)^{12} \right) - 100$$

$$= -17.428 \text{ million.}$$

So the expected value is:

$$NPV = 1.25(0.20) + -17.428(0.80)$$

$$= -13.6927 \text{ million.}$$

b. Decision Tree

If the high growth rate state occurs, then the NPV at time 1 is:

$$NPV_{high} = 3(10)(1.1) + \frac{1}{(1.1)^4} \left(\frac{10(1.1)^5 (1-0.02)}{0.1+0.02} \right) \left(1 - \left(\frac{0.98}{1.1} \right)^{12} \right) - 100$$

$$= 0.372 \text{ million.}$$

Note: Since the first three cash flows grow at the same rate as the discount rate, their present value is just the sum of the cash flows, so the value at time 1 is just the PV compounded.

If the low growth rate state occurs, then the NPV at time 1 is:

$$NPV_{low} = \frac{1}{1.1} \left(\frac{10(1.05)^3}{0.1-0.05} \left(1 - \left(\frac{1.05}{1.1} \right)^3 \right) \right) + \frac{1}{(1.1)^4} \left(\frac{10(1.05)^5 (1-0.02)}{0.1+0.02} \right) \left(1 - \left(\frac{0.98}{1.1} \right)^{12} \right) - 100$$

$$= -19.19 \text{ million.}$$

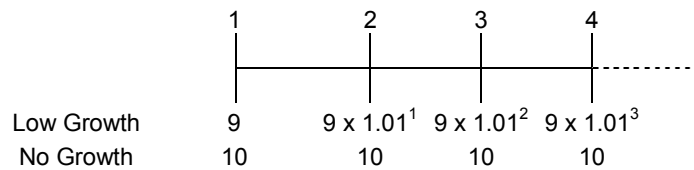
So investment will only occur in the high growth state. The value today of this is:

$$NPV_0 = (0.2) \left(\frac{0.372}{1.1} \right) = \$67.6 \text{ million}$$

- c. Since the NPV of investing today is negative, you should wait and only invest if the high state occurs.

22-8. The management of Southern Express Corporation is considering investing 10% of all future earnings in growth. The company has a single growth opportunity that it can take either now or in one period. Although the managers do not know the return on investment with certainty, they know it is equally likely to be either 10% or 14% per year. In one period, they will find out which state will occur. Currently the firm pays out all earnings as a dividend of \$10 million; if it does not make the investment, dividends are expected to remain at this level forever. If Southern Express undertakes the investment, the new dividend will reflect the realized return on investment and will grow at the realized rate forever. Assuming the opportunity cost of capital is 10.1%, what is the value of the company just before the current dividend is paid (the cum-dividend value)?

Notice that if the firm makes the investment, the current dividend will be $10(1 - 0.1) = \$9$ million. It will have two possible growth rates in one period. If the return on investment turns out to be 14%, then the growth rate will be $g = \text{retention ratio} \times \text{return on new investment} = 0.1 \times 0.14 = 1.4\%$. If the return on new investment is 10%, then the growth rate will be $g = 0.1 \times 0.1 = 1\%$. Assume the firm decides to wait to find out what the return on investment will be before making a decision. In this case, we need to decide whether to make the investment decision at time 1. If we make the investment and the return on investment is 10%, so the growth rate is 1% (the *low growth* state), the timeline will be as shown.



From the timeline we can see that the value of the firm at time 1 (before the dividend is paid) if the firm decides to invest and grow is:

$$P_{invest}^l = 9 + \frac{9(1.01)}{.101 - 0.01} = 108.89.$$

When the firm chooses not to invest, the value of the firm is:

$$P_{noinvest} = 10 + \frac{10}{.101} = 109.01.$$

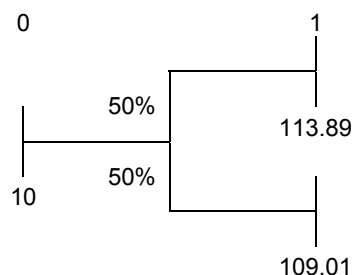
Clearly, the firm is better off not investing, so its value in this state is \$109.01 million. (One way to see this immediately is to note that in this state the return on investment is less than the opportunity cost of capital.)

The return on investment is 14%, so the growth rate is 1.4% (the *high growth* state); the same logic shows that if the investment is undertaken, the value of the firm is:

$$P_{invest}^h = 9 + \frac{9(1.014)}{.101 - 0.014} = 113.897.$$

This value exceeds \$109.01, the value if the investment is not undertaken, so the firm should undertake the investment. (Here the return on investment exceeds the opportunity cost of capital.)

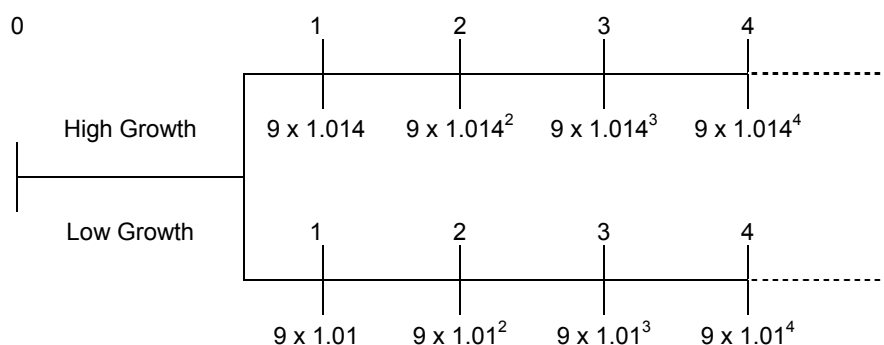
What is the value of this firm today? To solve this problem, first write down the decision tree:



Since each state is equally likely, the expected value of the firm in one period is $113.897 (0.5) + 109.01 (0.5)$. Computing the present value and adding today's dividend (remember, we are computing the value before the dividend is paid) gives the firm value today of:

$$P = \frac{\frac{1}{2}(109.01 + 113.897)}{1.101} + 10 = 111.229.$$

Consider what would have happened if instead of waiting one period to make the investment decision, Southern's managers decided to make the decision today, i.e., at time 0 on the time line. In this case, the decision tree looks like this:



We can derive the value of the firm using the same logic as above. The present value today (time 0) of the future dividends if the growth rate turns out to be high is:

$$P_{invest}^h = \frac{9(1.014)}{0.101 - 0.014} = 104.897.$$

When the growth rate is low, the present value of future dividends at time 0 is:

$$P_{invest}^l = \frac{9(1.01)}{0.101 - 0.01} = 99.89.$$

The value of the firm today is the present value of the expected future dividends plus the dividend today. Since both states are equally likely, we have:

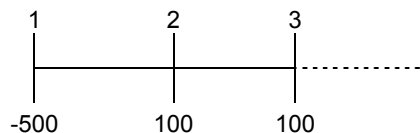
$$P_{invest} = 9 + \frac{1}{2}(99.89) + \frac{1}{2}(104.897) = 111.39.$$

Note that this value is *higher* than the value of the firm if management waits and makes the decision at time 1 when the return on investment becomes certain. So managers should give up the option to wait and invest today. Hence the value of the company is \$111.39 million.

First let's calculate the NPV of investing in 3 possible states:

1. \$100 Million State:

Timeline:



$$\text{NPV}(r) = \frac{100}{r} - 500$$

$$\text{NPV}(11\%) = \$409 \text{ million}$$

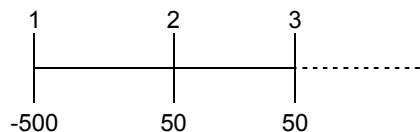
$$\text{NPV}(9\%) = 611.11 \text{ million}$$

So the expected value if this state occurs is

$$\begin{aligned} \text{EV}_{100} &= 409(0.8) + 611.11(0.2) \\ &= 450. \end{aligned}$$

2. \$50 Million State:

Timeline:



$$\text{NPV}(r) = \frac{100}{r} - 500$$

$$\text{NPV}(11\%) = -45.45$$

$$\text{NPV}(9\%) = 55.56$$

$$\begin{aligned} \text{EV}_{100} &= 55.56(0.2) \\ &= 11.11 \end{aligned}$$

3. NPV is zero in the worst state, since profits are zero so the project will not be undertaken.

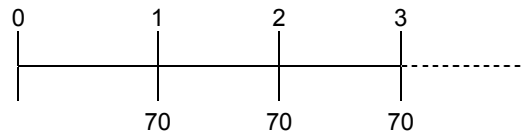
So the present value at time 0 of the expected value at time 1 is:

$$\frac{0.6 \times 450 + 0.2 \times 11.11}{1.1544} = \$235.54 \text{ million.}$$

If the investment is made at time 0, then the NPV is the PV of the expected cash flows minus the initial investment. The expected cash flows are:

$$0.6 \times 100 + 0.2 \times 50 = 70 \text{ per year.}$$

Timeline:



$$NPV = \frac{70}{0.11} - 500 = \$136.36 \text{ million}$$

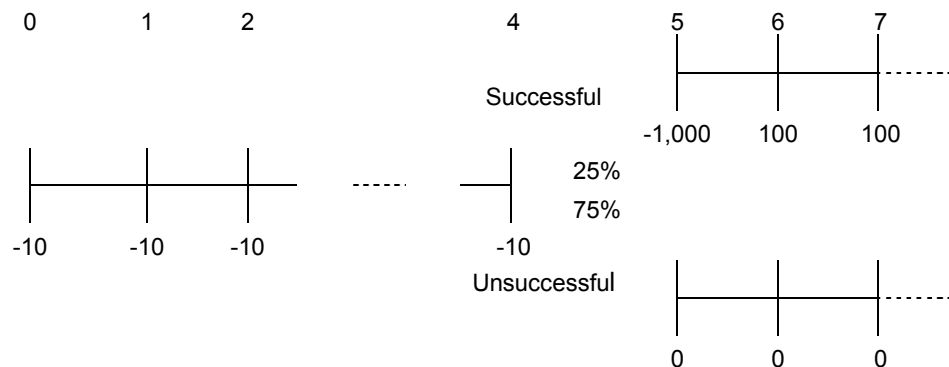
Alternatively, you can calculate the value of the project one year from now and discount that to get the same answer:

$$NPV = 0.2 \times \left(\frac{70}{0.09} \times \frac{1}{1.1544} + \frac{70}{1.1544} - 500 \right) + 0.8 \times \left(\frac{70}{1.1} \times \frac{1}{1.1544} + \frac{70}{1.1544} - 500 \right) = 136.39 \text{ million.}$$

So you are better off waiting.

- 22-10.** Your R&D division has just synthesized a material that will superconduct electricity at room temperature; you have given the go-ahead to try to produce this material commercially. It will take five years to find out whether the material is commercially viable, and you estimate that the probability of success is 25%. Development will cost \$10 million per year, paid at the beginning of each year. If development is successful and you decide to produce the material, the factory will be built immediately. It will cost \$1 billion to put in place, and will generate profits of \$100 million at the end of every year in perpetuity. Assume that the current five-year risk-free interest rate is 10% per year, and the yield on a perpetual risk-free bond will be 12%, 10%, 8%, or 5% in five years. Assume that the risk-neutral probability of each possible rate is the same. What is the value today of this project?

Decision Tree



If development is successful, then the NPV (at time 5) of producing the wire is:

$$NPV_5(r) = \frac{100}{r} - 1,000.$$

This is negative for $r > 10\%$, so the wire will only be produced if the rates are 8% or 5%:

$$NPV_5(0.08) = \frac{100}{0.08} - 1,000 = 250$$

$$NPV_5(0.05) = 1,000.$$

So the expected value of the growth opportunity at time 5 if development is successful is:

$$\begin{aligned} EV_S &= 250(0.25) + 1,000(0.25) \\ &= 312.5. \end{aligned}$$

There is a 25% chance of success so the expected value at time 5 of the investment opportunity is:

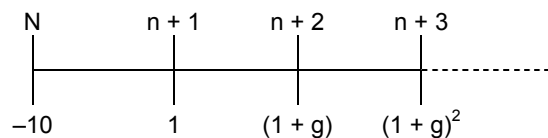
$$EV = 312.5 \times 0.25 = \$78.125.$$

The NPV of the development opportunity at time 0 is therefore

$$\begin{aligned} NPV &= -10 - \frac{10}{0.1} \left(1 - \left(\frac{1}{1.1} \right)^4 \right) + \frac{78.125}{(1.1)^5} \\ &= \$6.811 \text{ million.} \end{aligned}$$

- 22-11.** You are an analyst working for Goldman Sachs, and you are trying to value the growth potential of a large, established company, Big Industries. Big Industries has a thriving R&D division that has consistently turned out successful products. You estimate that, on average, the R&D division generates two new product proposals every three years, so that there is a 66% chance that a project will be proposed every year. Typically, the investment opportunities the R&D division produces require an initial investment of \$10 million and yield profits of \$1 million per year that grow at one of three possible growth rates in perpetuity: 3%, 0%, and -3%. All three growth rates are equally likely for any given project. These opportunities are always “take it or leave it” opportunities: If they are not undertaken immediately, they disappear forever. Assume that the cost of capital will always remain at 12% per year. What is the present value of all future growth opportunities Big Industries will produce?

Take a project that arrives in year n . The timeline is as follows.



The NPV of this investment opportunity is:

$$NPV_n(g) = \frac{1}{0.12 - g} - 10.$$

Now:

$$NPV_n(g = 3\%) = \frac{1}{0.12 - 0.03} - 10 = \$1.11 \text{ million}$$

$$NPV_n(g = 0\%) = \frac{1}{0.12 - 0} - 10 = -\$1.67 \text{ million}$$

$$NPV_n(g = -3\%) = \frac{1}{0.12 + 0.03} - 10 = -\$3.33 \text{ million}$$

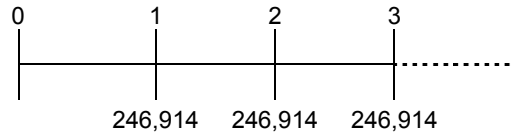
Therefore, only the projects with positive growth rates will be taken on. Thus, the expected value of any given investment opportunity is

$$EV_n = 1.11 \times \frac{1}{3} = \$370,370.$$

The probability that a project will arrive in any given year is $2/3$, and so the expected value of the growth opportunity that will arrive in year n is:

$$G_n = 370,370 \times \frac{2}{3} = \$246,914.$$

Putting this on a timeline:

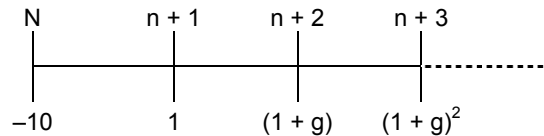


So the PV of all these opportunities today is:

$$PV = \frac{246,914}{0.12} = \$2.058 \text{ million.}$$

- 22-12. Repeat Problem 11, but this time assume that all the probabilities are risk-neutral probabilities, which means the cost of capital is always the risk-free rate and risk-free rates are as follows: The current interest rate for a risk-free perpetuity is 8%; in one year, there is a 64.375% chance that all risk-free interest rates will be 10% and stay there forever, and a 35.625% chance that they will be 6% and stay there forever. The current one-year risk-free rate is 7%.**

Take a project that arrives in year n . The timeline is as follows.



The NPV of this investment opportunity is:

$$NPV_n(g, r) = \frac{1}{r - g} - 10.$$

If the risk free rate is 10%:

$$NPV(3\%, 10\%) = \frac{1}{0.1 - 0.03} - 10 = \$4.286 \text{ million}$$

$$NPV(0\%, 10\%) = \frac{1}{0.1} - 10 = 0$$

$$NPV(-3\%, 10\%) = \frac{1}{0.1 + 0.03} - 10 = -\$2.308 \text{ million.}$$

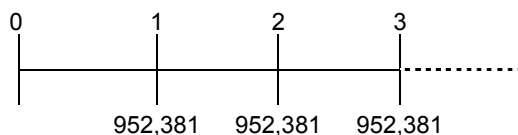
Therefore, only the projects with positive rates will be taken on. Thus, the expected value of any given investment opportunity is:

$$EV_n = 4.286 \times \frac{1}{3} = \$1.4286 \text{ million.}$$

The probability that a project will arrive in any given year is $2/3$, so the expected value of the growth opportunity that will arrive in year n is:

$$G_n = 1.4286 \times \frac{2}{3} = \$952,381.$$

Putting this on a timeline:



So the PV at time 1 of all these opportunities is:

$$PV = 952,381 + \frac{952,381}{0.1} = \$10.4762 \text{ million.}$$

Now, if the risk free rate is 6%:

$$NPV(3\%, 6\%) = \frac{1}{0.06 - 0.03} - 10 = 23.333$$

$$NPV(0\%, 6\%) = \frac{1}{0.06} - 10 = 6.667$$

$$NPV(-3\%, 6\%) = \frac{1}{0.06 + 0.03} - 10 = 1.111.$$

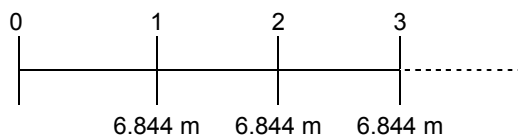
Therefore, regardless of the growth rate, all projects will be taken on. Thus, the expected value of any given investment opportunity is:

$$EV_n = 23.33 \times \frac{1}{3} + 6.67 \times \frac{1}{3} + 1.111 \times \frac{1}{3} = \$10.37 \text{ million.}$$

The probability that a project will arrive in any given year is $2/3$, and so the expected value of the growth opportunity that will arrive in year n is

$$G_n = 10.37 \times \frac{2}{3} = \$6.844 \text{ million.}$$

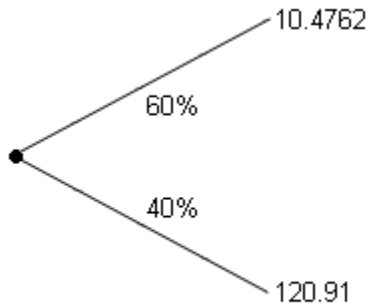
Putting this on a timeline:



So the PV at time 1 of all these opportunities is:

$$PV = 6.844 + \frac{6.844}{0.06} = \$120.91 \text{ million.}$$

There is a 64.375% chance of rates going to 10% and a 35.625% chance of rates going to 6%. Putting this on a decision tree



So the expected value is:

$$10.4762 \times 0.64375 + 120.91 \times 0.35625 = 49.82.$$

Computing the PV gives the answer:

$$\frac{49.82}{1.07} = \$46.56 \text{ million.}$$

- 22-13.** You own a small networking startup. You have just received an offer to buy your firm from a large, publicly traded firm, JCH Systems. Under the terms of the offer, you will receive 1 million shares of JCH. JCH stock currently trades for \$25 per share. You can sell the shares of JCH that you will receive in the market at any time. But as part of the offer, JCH also agrees that at the end of the next year, it will buy the shares back from you for \$25 per share if you desire. Suppose the current one-year risk-free rate is 6.18%, the volatility of JCH stock is 30%, and JCH does not pay dividends.

- a. Is this offer worth more than \$25 million? Explain.
 - b. What is the value of the offer?
- a. The offer is worth more than \$25 million because of the put option.
 - b. The value of the offer is the current value of the shares plus the value of the put option.

To calculate the value of the put:

$$S = 25$$

$$PV(K) = 25 / (1.0618)$$

$$T = 1$$

$$\sigma = 0.30$$

$$d_1 = \frac{\ln(S / PV(K)) + \frac{\sigma\sqrt{T}}{2}}{\sigma\sqrt{T}} = 0.3499$$

$$d_2 = 0.04989$$

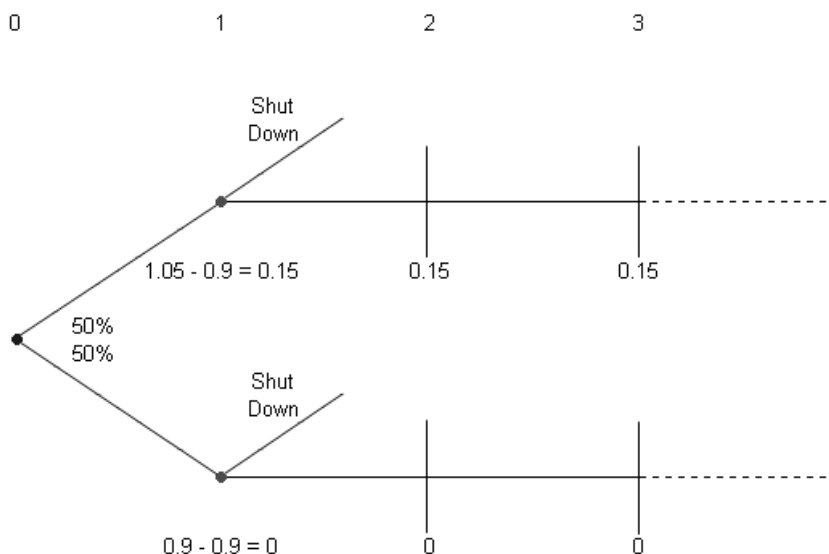
$$P = PV(K)(1 - N(d_2)) - S(1 - N(d_1)) = 2.224$$

So, the value of the offer is $25 + 2.224 = \$27.224$ million.

(Note that the actual value will be slightly higher because this uses the value of a European put.)

- 22-14. You own a wholesale plumbing supply store. The store currently generates revenues of \$1 million per year. Next year, revenues will either decrease by 10% or increase by 5%, with equal probability, and then stay at that level as long as you operate the store. You own the store outright. Other costs run \$900,000 per year. There are no costs for shutting down; in that case, you can always sell the store for \$500,000. What is the business worth today if the cost of capital is fixed at 10%?

Decision Tree



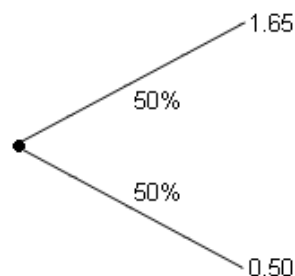
The value of the store is the maximum of the PV of all future profits and \$500,000, the value of selling the store.

If the revenues decrease, then future profits are zero so the store should be shut down. Hence the value in this state is \$500,000.

If the revenues increase, then the PV of future profits at time 1 are:

$$(1.05 - 0.9) + \frac{1.05 - 0.9}{0.1} = 0.15 + \frac{0.15}{0.1} = \$1.65 \text{ million.}$$

So it is optimal to keep the store running. Putting this on a decision tree:



So the PV of the time 1 expected value is:

$$\frac{1.65 \frac{1}{2} + 0.5 \frac{1}{2}}{1.1} = \$977,273.$$

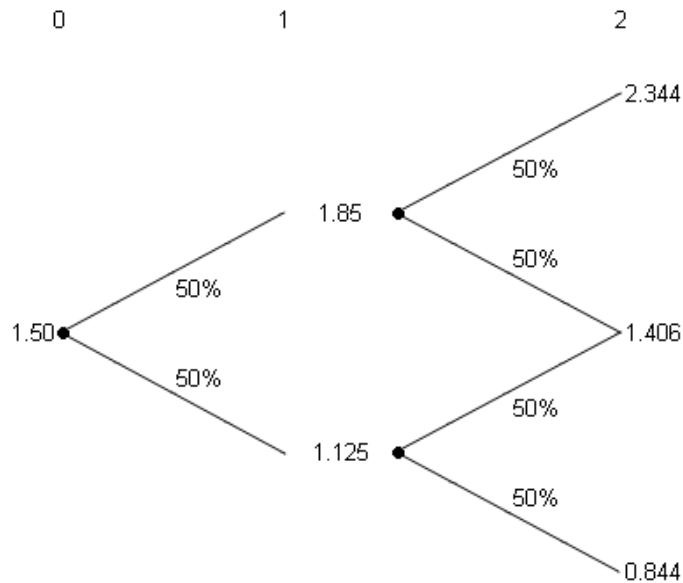
Since this is greater than \$500,000, the value of the store is \$977,273 with the option to abandon.

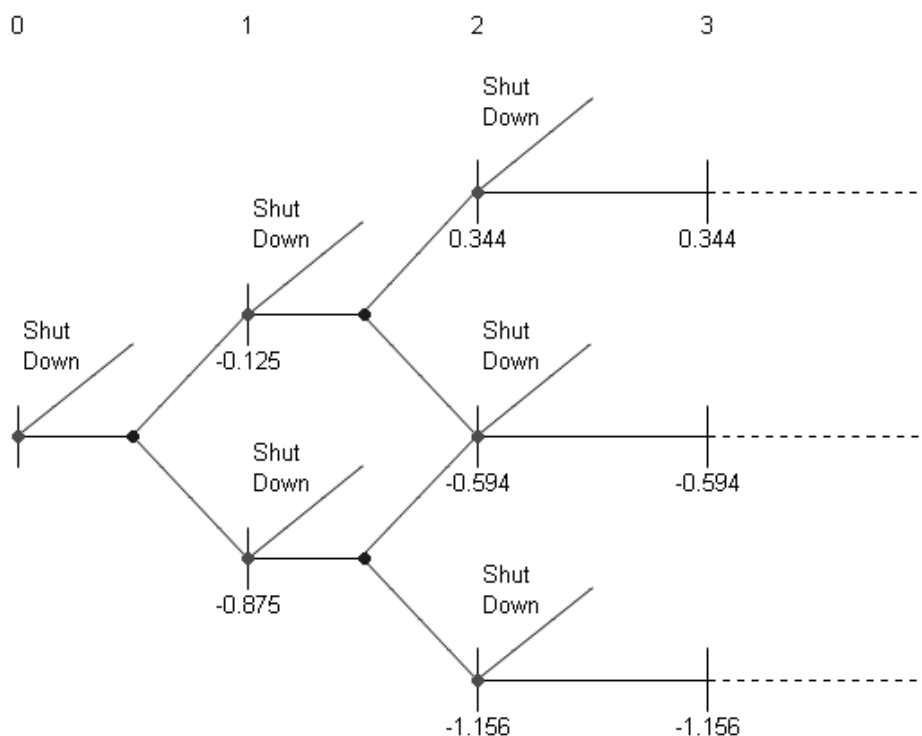
- 22-15. You own a copper mine. The price of copper is currently \$1.50 per pound. The mine produces 1 million pounds of copper per year and costs \$2 million per year to operate. It has enough copper to operate for 100 years. Shutting the mine down would entail bringing the land up to EPA standards and is expected to cost \$5 million. Reopening the mine once it is shut down would be an impossibility given current environmental standards. The price of copper has an equal (and independent) probability of going up or down by 25% each year for the next two years and then will stay at that level forever. Calculate the NPV of continuing to operate the mine if the cost of capital is fixed at 15%. Is it optimal to abandon the mine or keep it operating?



Decision Tree:

Copper Price:



Cash Flows

First, determine whether the mine is operating or shut down in each possible state at time 2.

When the copper price is \$2.344, the mine's profits are \$0.344 million/year for 98 years and then it will cost \$5 million to shut the mine down. The PV at time 2 of these cash flows is:

$$PV_{2,0.344} = \frac{0.344}{0.15} \left(1 - \frac{1}{(1.15)^{98}} \right) - \frac{5}{(1.15)^{98}} = \$2.293 \text{ million.}$$

Since it will cost \$5 million to shut the mine down, it is better to leave it operating.

Similarly, when the copper price is \$1.406,

$$PV_{2,-0.594} = \frac{-0.594}{0.15} \left(1 - \frac{1}{(1.15)^{98}} \right) - \frac{5}{(1.15)^{98}} = -\$3.96 \text{ million.}$$

Again, since the value is greater than -\$5 million, it is better to operate the mine.

Finally, when the copper price is \$0.844:

$$PV_{2,-1.156} = \frac{-1.156}{0.15} \left(1 - \frac{1}{(1.15)^{98}} \right) - \frac{5}{(1.15)^{98}} = -\$7.7 \text{ million.}$$

Since it is smaller than -\$5 million, the mine should be shut down.

Next, calculate the value of the mine if it is operating at time 1 when the copper price is \$1.85. Putting the value of the mine plus the cash flow at time 2 on a decision tree:



The present value at time 1 of the expected value at time 2 is

$$\frac{(2.293 + 0.344)\frac{1}{2} + (-3.96 - 0.594)\frac{1}{2}}{1.15} = -\$0.8337 \text{ million.}$$

So it is optimal to run the mine in this state.

Next, calculate the value of the mine if it is operating at time 1 when the copper price is \$1.125. Putting the value of the mine plus the cash flow at time 2 on a decision tree:



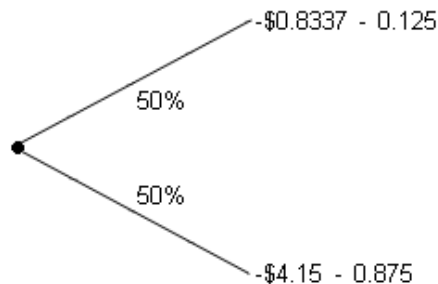
The present value at time 1 of the expected value at time 2 is

$$\frac{-5 \times \frac{1}{2} + (-3.96 - 1.594)\frac{1}{2}}{1.15} = -\$4.154 \text{ million.}$$

So it is optimal to run the mine in this state.

Finally, we calculate the value of the value of the mine at time 0 if it is operating.

Putting the value of the mine plus the cash flow at time 1 on a decision tree:



The present value at time 0 of the expected value at time 1 is:

$$\frac{(-0.8337 - 0.125)\frac{1}{2} + (-4.154 - 0.875)\frac{1}{2}}{1.15} = -\$2.603 \text{ million.}$$

So the mine is worth $-\$2.603$ million. Even though it is worth a negative amount because in most states it will lose money for the next 98 years, it is not optimal to shut down the mine at time 0 because the costs of shutting down are even greater.

- 22-16. An original silver dollar from the late eighteenth century consists of approximately 24 grams of silver. At a price of \$.019 per gram (\$6 per troy ounce), the silver content of the coin is currently worth about \$4.50. Assume that these coins are in plentiful supply and are not collector's items, so they have no numismatic value. If the current price of silver is \$0.19 per gram, will the price of the coin be greater than, less than, or equal to \$4.50? Justify your answer.**

The silver dollar is actually a real option because you always have the option to use it as a dollar coin. Although at the current price of silver this does not make sense, if the price of silver dropped below 4.1¢/gram (\$1.28/troy ounce) the value of the silver in the coin would drop below \$1. The coin can always be used as a dollar coin, so its price cannot fall below a dollar (by the Law of One Price). Since the price of 24 grams of silver can drop below \$1, the coin must be worth more than 24 grams of silver or \$4.50. So the coin must be worth \$4.50.

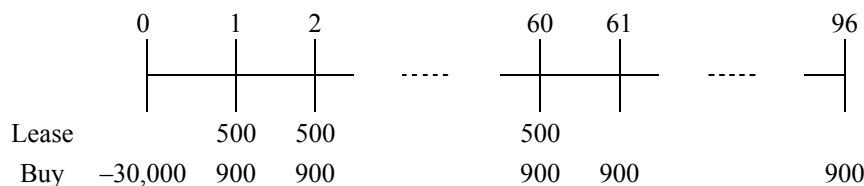
- 22-17. What implicit assumption is made when managers use the equivalent annual benefit method to decide between two projects with different lives that use the same resource?**

The equivalent annual cost method implicitly assumes that, at the end of the life of the shorter length project, you can replace the shorter length project on the original terms.

- 22-18. You own a cab company and are evaluating two options to replace your fleet. Either you can take out a five-year lease on the replacement cabs for \$500 per month per cab, or you can purchase the cabs outright for \$30,000, in which case the cabs will last eight years. You must return the cabs to the leasing company at the end of the lease. The leasing company is responsible for all maintenance costs, but if you purchase the cabs, you will buy a maintenance contract that will cost \$100 per month for the life of each cab. Each cab will generate revenues of \$1000 per month. Assume the cost of capital is fixed at 12%.**

- Calculate the NPV per cab of both possibilities: purchasing the cabs or leasing them.
- Calculate the equivalent monthly annual benefit of both opportunities.
- If you are leasing a cab, you have the opportunity to buy the used cab after five years. Assume that in five years a five-year-old cab will cost either \$10,000 or \$16,000 with equal likelihood, will have maintenance costs of \$500 per month, and will last three more years. Which option should you take?

- a. Timeline:



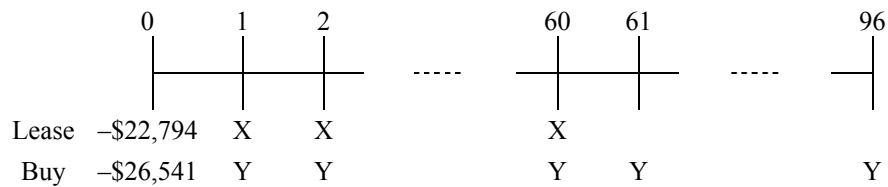
Converting the cost of capital to a monthly discount rate gives:

$(1.12)^{\frac{1}{12}} - 1 = .00949$, so the monthly discount rate is 0.949%.

$$\begin{aligned} \text{NPV}_{\text{Lease}} &= \frac{500}{0.00949} \left(1 - \frac{1}{(1.00949)^{60}} \right) \\ &= \$22,794 \end{aligned}$$

$$\begin{aligned} \text{NPV}_{\text{Buy}} &= -30,000 + \frac{900}{0.00949} \left(1 - \frac{1}{(1.00949)^{96}} \right) \\ &= \$26,541 \end{aligned}$$

b. Timeline:



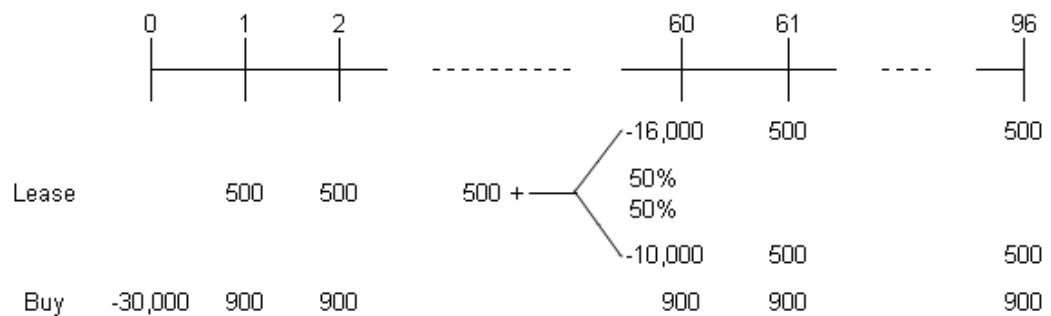
Solving for the EAB of leasing:

$$22,794 = \frac{X}{0.00949} \left(1 - \frac{1}{(1.00949)^{60}} \right) \Rightarrow X = \$500.$$

Solving for the EAB of buying:

$$26,541 = \frac{Y}{0.00949} \left(1 - \frac{1}{(1.00949)^{96}} \right) \Rightarrow Y = \$422.$$

c. Decision Tree



In month 60, the NPV of buying a cab is either

$$\text{NPV} = -10,000 + \frac{500}{0.00949} \left(1 - \frac{1}{(1.00949)^{36}} \right) = \$5,188.$$

Or

$$NPV = -16,000 + \frac{500}{0.00949} \left(1 - \frac{1}{(1.00949)^{36}} \right) = -\$813.$$

The expected value of replacing the cabs in year 5 is:

$$EV = 5188(0.5) = \$2,594.$$

The value today of this:

$$PV = \frac{2,594}{(1.00949)^{60}} = \$1,472.$$

Adding this to the NPV of leasing from part a gives:

$$NPV_{Lease} = 1,472 + 22,794 = \$24,266.$$

Since the NPV of buying as not changed

$$NPV_{Buy} = \$26,541.$$

So you should *buy* the cab.

22-19. You own a piece of raw land in an up-and-coming area in Gotham City. The costs to construct a building increase disproportionately with the size of the building. A building of q square feet costs $0.1 \times q^2$ to build. After you construct a building on the lot, it will last forever but you are committed to it: You cannot put another building on the lot. Buildings currently rent at \$100 per square foot per month. Rents in this area are expected to increase in five years. There is a 50% chance that they will rise to \$200 per square foot per month and stay there forever, and a 50% chance that they will stay at \$100 per square foot per month forever. The cost of capital is fixed at 12% per year.

- a. Should you construct a building on the lot right away? If so, how large should the building be?
 - b. If you choose to delay the decision, how large a building will you construct in each possible state in five years?
- a. Converting the cost of capital to a monthly discount rate gives:

$$(1.12)^{\frac{1}{12}} - 1 = .00949.$$

So 12% per year is equivalent to 0.949% per month. If rents rise then a building of size q will be worth:

$$\frac{200q}{0.00949}.$$

This building will cost $0.1q^2$ to build, so the NPV is:

$$\frac{200q}{0.00949} - 0.1q^2.$$

To find the maximum value of this function, take the derivative and set the results equal to zero:

$$\frac{200}{0.00949} - 0.2q = 0.$$

Solving for q gives:

$$q = 105,387 \text{ sq. ft.}$$

Doing the same thing in the state with \$100/sq. ft. rents gives:

$$q = 52,694.$$

- b. If you wait then the NPV of building in 200 sq. ft. state is

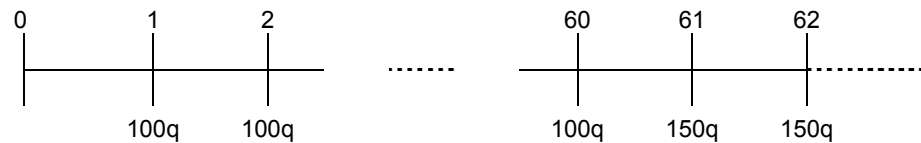
$$\frac{200q}{0.00949} - 0.1q^2 = \frac{200 \times 105,387}{0.00949} - 0.1(105,387)^2 = \$1.11 \text{ billion.}$$

Repeating for the \$100 sq ft state gives \$278 million.

So the PV of this today is:

$$PV = \frac{1,110(0.5) + 278(0.5)}{(1.00949)^{60}} = \$394 \text{ million.}$$

If we build today the expected cash flows are:



$$NPV = \frac{100q}{0.00949} \left(1 - \frac{1}{(1.00949)^{60}} \right) + \frac{150q}{0.00949} \left(\frac{1}{(1.00949)^{60}} \right) - 0.1q^2.$$

Taking the derivative, setting the result equal to zero and solving for q gives the optimal size to the building today:

$$\frac{100}{0.00949} \left(1 - \frac{1}{(1.00949)^{60}} \right) + \frac{150}{0.00949} \left(\frac{1}{(1.00949)^{60}} \right) - 0.2q = 0.$$

$$q = 67,644 \text{ sq. ft.}$$

So the NPV is:

$$NPV = \$458 \text{ million.}$$

So we should build a building of 67,644 sq. ft. today.

- 22-20. Genenco is developing a new drug that will slow the aging process. In order to succeed, two breakthroughs are needed, one to increase the potency of the drug, and the second to eliminate toxic side effects. Research to improve the drug's potency is expected to require an upfront investment of \$10 million and take 2 years; the drug has a 5% chance of success. Reducing the drug's toxicity will require a \$30 million up-front investment, take 4 years, and has a 20% chance of success. If both efforts are successful, Genenco can sell the patent for the drug to a major drug company for \$2 billion. All risk is idiosyncratic, and the risk-free rate is 6%.**

- a. What is the NPV of launching both research efforts simultaneously?

- b. What is the optimal order to stage the investments?

- c. What is the NPV with the optimal staging?

a. $-10 - 30 + .05 \times 20 \times 2000/1.06^4 = -24$ million

b. Potency : $(1 - .05/1.06^2)/10 = 0.096$

$$\text{Toxicity: } (1 - .20/1.06^4)/30 = 0.028$$

So, work on potency, then toxicity—higher risk and smaller investment.

- c. If potency works, then the NPV of continuing is

$$(-30 + .2 \cdot 2000/1.06^4)$$

So the NPV of starting is

$$-10 + 0.05 (\text{NPV of starting toxicity})/1.06^2$$

$$= -10 + 0.05 (-30 + .2 \cdot 2000/1.06^4)/1.06^2 = \$2.76 \text{ million.}$$

- 22-21. Your engineers are developing a new product to launch next year that will require both software and hardware innovations. The software team requests a budget of \$5 million and forecasts an 80% chance of success. The hardware team requests a \$10 million budget and forecasts a 50% chance of success. Both teams will need 6 months to work on the product, and the risk-free interest rate is 4% APR with semiannual compounding.**

- a. Which team should work on the project first?
- b. Suppose that before anyone has worked on the project, the hardware team comes back and revises their proposal, changing the estimated chance of success to 75% based on new information. Will this affect your decision in (a)?

- a. Software: $(1 - .80/1.02)/5 = 0.043$

$$\text{Hardware: } (1 - .50/1.02)/10 = 0.051$$

>> Hardware should go first

- b. Hardware: $(1 - .75/1.02)/10 = 0.026$

>> Yes, software should go first now.

- 22-22. Your firm is thinking of expanding. If you invest today, the expansion will generate \$10 million in FCF at the end of the year, and will have a continuation value of either \$150 million (if the economy improves) or \$50 million (if the economy does not improve). If you wait until next year to invest, you will lose the opportunity to make \$10 million in FCF, but you will know the continuation value of the investment in the following year (that is, in a year from now, you will know what the investment continuation value will be in the following year). Suppose the risk-free rate is 5%, and the risk-neutral probability that the economy improves is 45%. Assume the cost of expanding is the same this year or next year.**

- a. If the cost of expanding is \$80 million, should you do so today, or wait until next year to decide?
- b. At what cost of expanding would there be no difference between expanding now and waiting? To what profitability index does this correspond?

- a. $V_0 = (.45 \times (150) + .55 \times (50))/1.05 = 100$

$$\text{NPV}(\text{expand}) = 100 - 80 = 20$$

$$\text{NPV}(\text{wait}) = .45 \times (150 - 80)/1.05 = 30$$

So, optimal to wait.

- b. $\text{NPV}(\text{expand}) = 100 - C =$

$$\text{NPV}(\text{wait}) = .45 \times (150 - C)/1.05$$

$$\text{So, } C = 62.5$$

$$\text{Prof Index} = \text{NPV}/C = (100 - 62.5)/62.5 = 0.6$$

- 22-23. Assume that the project in Example 22.5 pays an annual cash flow of \$100,000 (instead of \$90,000).**
- What is the NPV of investing today?**
 - What is the NPV of waiting and investing tomorrow?**
 - Verify that the hurdle rate rule of thumb gives the correct time to invest in this case.**
 - $100k/.054 - 1m = 851.852k$
 - $NPV(up) = 0, NPV(down) = 100k/.05 - 1m = 1m$
 $PV = 0.9 \times 1m/1.08 = \$833,333k$
 - Hurdle rate rule: $NPV = 100k/.09 - 1m = 111.11k > 0$, so invest now, correct.
- 22-24. Assume that the project in Example 22.5 pays an annual cash flow of \$80,000 (instead of \$90,000).**
- What is the NPV of investing today?**
 - What is the NPV of waiting and investing tomorrow?**
 - Verify that the hurdle rate rule of thumb gives the correct time to invest in this case.**
 - $80k/.054 - 1m = 481.48k$
 - $Npv(down) = 80k/.05 - 1m = 600k$
 $PV = .9 \times 600/1.08 = 500k$
 - $80/.09 - 1m = -111k$, so wait, correct

Chapter 23

Raising Equity Capital

23-1. What are some of the alternative sources from which private companies can raise equity capital?

Private companies can raise equity capital from angel investors, venture capitalists, institutional investors, or corporate investors.

23-2. What are the advantages and the disadvantages to a private company of raising money from a corporate investor?

Advantages of raising money from a corporate investor are that the large corporate partner may provide benefits such as capital, expertise, or access to distribution channels. The corporate partner may become an important customer or supplier for the startup firm, and the willingness of an established company to invest may be an important endorsement of the new company.

The disadvantages are that not all corporate investments are successful. The corporate partner may gain access to proprietary technology, or eventually even become a competitor. Once a young firm has aligned itself with one corporate partner, the competitors of this partner may be unwilling to do business with the startup.

23-3. Starware Software was founded last year to develop software for gaming applications. Initially, the founder invested \$800,000 and received 8 million shares of stock. Starware now needs to raise a second round of capital, and it has identified an interested venture capitalist. This venture capitalist will invest \$1 million and wants to own 20% of the company after the investment is completed.

a. How many shares must the venture capitalist receive to end up with 20% of the company? What is the implied price per share of this funding round?

b. What will the value of the whole firm be after this investment (the post-money valuation)?

a. After the funding round, the founder's 8 million shares will represent 80% ownership of the firm. To solve for the new total number of shares (TOTAL):

$$8,000,000 = .80 \times \text{TOTAL}$$

So $\text{TOTAL} = 10,000,000$ shares. If the new total is 10 million shares, and the venture capitalist will end up with 20%, then the venture capitalist must buy 2 million shares. Given the investment of \$1 million for 2 million shares, the implied price per share is \$0.50.

b. After this investment, there will be 10 million shares outstanding, with a price of \$0.50 per share, so the post-money valuation is \$5 million.

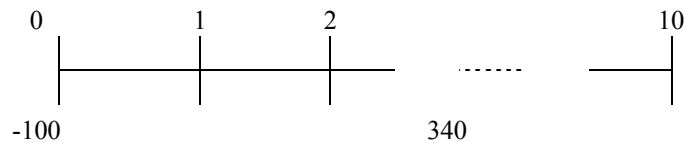
-  **23-4.** Suppose venture capital firm GSB partners raised \$100 of committed capital. Each year over the 10-year life of the fund, 2% of this committed capital will be used to pay GSB's management fee. As is typical in the venture capital industry, GSB will only invest \$80 million (committed capital less lifetime management fees). At the end of 10 years, the investments made by the fund are worth \$400 million. GSB also charges 20% carried interest on the profits of the fund (net of management fees).

- Assuming the \$80 million in invested capital is invested immediately and all proceeds were received at the end of 10 years, what is the IRR of the investments GSB partners made? That is, compute IRR ignoring all management fees.
- Of course, as an investor, or limited partner, you are more interested in your own IRR, that is the IRR including all fees paid. Assuming that investors gave GSB partners the full \$100 million up front, what is the IRR for GSB's limited partners (that is, the IRR net of *all* fees paid).

a. IRR solves $\text{NPV}(\text{Total invested}) = \frac{400}{(1+r)^{10}} - 80 = 0$

$$\text{So } r = \left(\frac{400}{80} \right)^{1/10} - 1 = 17.46\%$$

b.



\$100 million invested

Profit = 400 – 100 = 300

Carried interest = 20% × 300 = \$60 million

LP payoff = 400 – 60 = 340

$$\text{So IRR} = \left(\frac{340}{100} \right)^{1/10} - 1 = 13.02\% \text{ solves}$$

-  **23-5.** Three years ago, you founded your own company. You invested \$100,000 of your money and received 5 million shares of Series A preferred stock. Since then, your company has been through three additional rounds of financing.

Round	Price (\$)	Number of Shares
Series B	0.50	1,000,000
Series C	2.00	500,000
Series D	4.00	500,000

- What is the pre-money valuation for the Series D funding round?
- What is the post-money valuation for the Series D funding round?

c. Assuming that you own only the Series A preferred stock (and that each share of all series of preferred stock is convertible into one share of common stock), what percentage of the firm do you own after the last funding round?

a. Before the Series D funding round, there are $(5,000,000 + 1,000,000 + 500,000 = 6,500,000)$ shares outstanding. Given a Series D funding price of \$4.00 per share, the pre-money valuation is $(6,500,000) \times \$4.00/\text{share} = \26 million.

b. After the funding round, there will be $(6,500,000 + 500,000 = 7,000,000)$ shares outstanding, so the post-money valuation is $(7,000,000) \times \$4.00/\text{share} = \$28,000,000$.

c. You will own $5,000,000 / 7,000,000 = 71.4\%$ of the firm after the last funding round.

23-6. What are the main advantages and disadvantages of going public?

The two main advantages of going public are liquidity and access to capital. One of the major disadvantages of an IPO is that once a company becomes a public company, it must satisfy all of the requirements of being a public company such as SEC filings and listing requirements of the securities exchanges.

23-7. Do underwriters face the most risk from a best-efforts IPO, a firm commitment IPO, or an auction IPO? Why?

Underwriters face the most risk from a firm commitment IPO. With this method, they guarantee that they will sell all of the stock at the offer price. If the entire issue does not sell at the IPO price, the remaining shares must be sold at a lower price and the underwriter must take the loss.

With a best-efforts IPO, the underwriter does not guarantee that the stock will be sold, but instead tries to sell the stock for the best possible price. In an auction IPO, the underwriters let the market determine the price by auctioning off the company.

23-8. Roundtree Software is going public using an auction IPO. The firm has received the following bids:

Price (\$)	Number of Shares
14.00	100,000
13.80	200,000
13.60	500,000
13.40	1,000,000
13.20	1,200,000
13.00	800,000
12.80	400,000

Assuming Roundtree would like to sell 1.8 million shares in its IPO, what will the winning auction offer price be?

First, compute the cumulative total number of shares demanded at or above any given price:

Price	Cumulative Demand
14.00	100,000
13.80	300,000
13.60	800,000
13.40	1,800,000
13.20	3,000,000
13.00	3,800,000
12.80	4,200,000

The winning price should be \$13.40, because investors have placed orders for a total of 1.8 million shares at a price of \$13.40 or higher.

- 23-9.** Three years ago, you founded Outdoor Recreation, Inc., a retailer specializing in the sale of equipment and clothing for recreational activities such as camping, skiing, and hiking. So far, your company has gone through three funding rounds:



Round	Date	Investor	Shares	Share Price (\$)
Series A	Feb. 2002	You	500,000	1.00
Series B	Aug. 2003	Angels	1,000,000	2.00
Series C	Sept. 2004	Venture capital	2,000,000	3.50

Currently, it is 2007 and you need to raise additional capital to expand your business. You have decided to take your firm public through an IPO. You would like to issue an additional 6.5 million new shares through this IPO. Assuming that your firm successfully completes its IPO, you forecast that 2007 net income will be \$7.5 million.

- Your investment banker advises you that the prices of other recent IPOs have been set such that the P/E ratios based on 2007 forecasted earnings average 20.0. Assuming that your IPO is set at a price that implies a similar multiple, what will your IPO price per share be?
 - What percentage of the firm will you own after the IPO?
- With a P/E ratio of 20.0x, and 2005 earnings of \$7.5 million, the total value of the firm at the IPO should be:

$$\frac{P}{7.5} = 20.0x \Rightarrow P = \$150 \text{ million.}$$

There are currently $(500,000 + 1,000,000 + 2,000,000) = 3,500,000$ shares outstanding (before the IPO). At the IPO, the firm will issue an additional 6.5 million shares, so there will be 10 million shares outstanding immediately after the IPO. With a total market value of \$150 million, each share should be worth $\$150 / 10 = \15 per share.

- After the IPO, you will own 500,000 of the 10 million shares outstanding, or 5% of the firm.

- 23-10.** What is IPO underpricing? If you decide to try to buy shares in every IPO, will you necessarily make money from the underpricing?

Underpricing refers to the fact that, on average, underwriters pick the IPO issue price so that the average first-day return is positive. If you followed a strategy of placing an order for a fixed number of shares on every IPO, your order will be completely filled when the stock price goes down, but you will be rationed when it goes up. In effect you only get substantial amounts of stock when you do not want it. The winners' curse is substantial enough so that the strategy of investing in every IPO does not yield above market returns.

- 23-11.** Margoles Publishing recently completed its IPO. The stock was offered at a price of \$14 per share. On the first day of trading, the stock closed at \$19 per share. What was the initial return on Margoles? Who benefited from this underpricing? Who lost, and why?

The initial return on Margoles Publishing stock is $(\$19.00 - \$14.00) / (\$14.00) = 35.7\%$.

Who gains from the price increase? Investors who were able to buy at the IPO price of \$14/share see an immediate return of 35.7% on their investment. Owners of the other shares outstanding that were not sold as part of the IPO see the value of their shares increase. To the extent that the investors who were able to obtain shares in the IPO have other relationships with the investment banks, the investment banks may benefit indirectly from the deal through their future business with these customers.

Who loses from the price increase? The original shareholders lose, because they sold stock for \$14.00 per share when the market was willing to pay \$19.00 per share.

- 23-12. Chen Brothers, Inc., sold 4 million shares in its IPO, at a price of \$18.50 per share. Management negotiated a fee (the underwriting spread) of 7% on this transaction. What was the dollar cost of this fee?**

The total dollar value of the IPO was $(\$18.50) \times (4 \text{ million}) = \74 million . The spread equaled $(0.07) \times (\$74 \text{ million})$ or \$5.18 million.

- 23-13. Your firm has 10 million shares outstanding, and you are about to issue 5 million new shares in an IPO. The IPO price has been set at \$20 per share, and the underwriting spread is 7%. The IPO is a big success with investors, and the share price rises to \$50 the first day of trading.**

- How much did your firm raise from the IPO?**
- What is the market value of the firm after the IPO?**
- Assume that the post IPO value of your firm is its fair market value. Suppose your firm could have issued shares directly to investors at their fair market value, in a perfect market with no underwriting spread and no underpricing. What would the share price have been in this case, if you raise the same amount as in part (a)?**
- Comparing part (b) and part (c), what is the total cost to the firm's original investors due to market imperfections from the IPO?**

- $5\text{m} \times (20 - 7\% \times 20) = \93 million
- $15\text{m} \times 50 = \$750 \text{ million}$
- Market value of firm assets absent new cash raised = $750 - 93 = \$657 \text{ million}$.
 $\$657\text{m}/(10\text{m original shares}) = \65.70 per share
 Check: $93\text{m}/65.70 = 1.4155\text{m new shares}$,
 $\$750/11.4155 = \65.7
- $(65.7 - 50) \times 10\text{m} = \157 million

- 23-14. You have an arrangement with your broker to request 1000 shares of all available IPOs. Suppose that 10% of the time, the IPO is "very successful" and appreciates by 100% on the first day, 80% of the time it is "successful" and appreciates by 10%, and 10% of the time it "fails" and falls by 15%.**

- By what amount does the average IPO appreciate the first day; that is, what is the average IPO underpricing?**
 - Suppose you expect to receive 50 shares when the IPO is very successful, 200 shares when it is successful, and 1000 shares when it fails. Assume the average IPO price is \$15. What is your expected one-day return on your IPO investments?**
- $.10(100\%) + .80(10\%) + .10(-15\%) = 16.5\%$
 - Average investment = $.10(50 \times 15) + .80(200 \times 15) + .10(1000 \times 15) = \3975
 Ave gain = $.10(50 \times 15 \times 100\%) + .80(200 \times 15 \times 10\%) + .10(1000 \times 15 \times -15\%) = \90
 Return = $90/3975 = 2.3\%$

- 23-15. On January 20, Metropolitan, Inc., sold 8 million shares of stock in an SEO. The current market price of Metropolitan at the time was \$42.50 per share. Of the 8 million shares sold, 5 million shares were primary shares being sold by the company, and the remaining 3 million shares were being sold by the venture capital investors. Assume the underwriter charges 5% of the gross proceeds as an underwriting fee (which is shared proportionately between primary and secondary shares).**

a. How much money did Metropolitan raise?

b. How much money did the venture capitalists receive?

a. The company sold 5 million shares at \$42.50 per share, so it raised $(\$42.50) \times (5,000,000) = \212.5 million. After underwriting fees, it will keep $212.5 \times (1 - 0.05) = \201.875 million.

b. The venture capitalists raised $(\$42.50) \times (3,000,000) = \127.5 million. After underwriting fees, they will keep $127.5 \times (1 - 0.05) = \121.125 million.

So, in total, the SEO was worth $\$201.875 + \$121.125 = (\$42.50) \times (8,000,000) \times 0.95 = \323 million.

23-16. What are the advantages to a company of selling stock in an SEO using a cash offer? What are the advantages of a rights offer?

A cash offer is when a company offers the new shares to investors at large. A rights offer is when the new shares are offered only to existing shareholders. Rights offers protect existing shareholders from underpricing. However, with a rights offer, only existing shareholders are offered stock to purchase. Demand may be lower, because existing shareholders are only a subset of all possible investors, and because they may not want to increase the percentage weight of this stock in their portfolios. If demand is lower, firms may receive a lower price from rights offers.

23-17. MacKenzie Corporation currently has 10 million shares of stock outstanding at a price of \$40 per share. The company would like to raise money and has announced a rights issue. Every existing shareholder will be sent one right per share of stock that he or she owns. The company plans to require five rights to purchase one share at a price of \$40 per share.

a. Assuming the rights issue is successful, how much money will it raise?

b. What will the share price be after the rights issue? (Assume perfect capital markets.)

Suppose instead that the firm changes the plan so that *each* right gives the holder the right to purchase one share at \$8 per share.

c. How much money will the new plan raise?

d. What will the share price be after the rights issue?

e. Which plan is better for the firm's shareholders? Which is more likely to raise the full amount of capital?

a. $10\text{m shares}/5 \times 40 = \80 million

b. $12\text{m total shares, Value} = \$400 \text{ million} + 80 \text{ million in new capital} = \480

Share price = $480/12 = \$40$

c. $10\text{m} \times \$8 = \80 million

d. $\$480/20 = \24 per share

e. Shareholders are the same either way. In the first case, each share is worth \$40, and exercising the right has 0 npv, so the total value of a share is \$40.

In the second case, the share is worth \$24, but the right is worth $(24 - 8) = \$16$, so the total value from owning a share is $\$24 + \$16 = \$40$ per share.

However, the second plan is much more likely to be fully subscribed, because exercising the right is a good deal. In the first case, shareholders are indifferent between exercising and not exercising.

Chapter 24

Debt Financing

24-1. Explain some of the differences between a public debt offering and a private debt offering.

In a public debt offering, a prospectus is created with details of the offering and a formal contract between the bond issuer and the trust company is signed. The trust company makes sure the terms of the contract are enforced. In a private offering there is no need for a prospectus or a formal contract. Instead a promissory note can be enough. Moreover, the contract in a private placement does not have to be standard.

24-2. Why do bonds with lower seniority have higher yields than equivalent bonds with higher seniority?

Requiring coupon payments protects the bondholders from waiting a long time in case the debtor defaults. Without coupon payments default only happens when the bond matures, but by then the corporation might have depleted all of its assets. In contrast, with coupon payments the debtor would be in default the moment it misses one of the coupon payments, and the bondholders can then force the firm into bankruptcy. At this stage, they might be able to get a larger fraction of the value of the original debt than if they waited until maturity.

24-3. Explain the difference between a secured corporate and an unsecured corporate bond.

A secured corporate bond gives the bondholder the right over particular assets that serve as collateral in case of default. An unsecured corporate bond does not offer such protection to the bondholder. Thus, with an unsecured corporate bond the bondholders are residual claimants in the case of bankruptcy after the secured assets have been given to the corresponding bondholders.

24-4. What is the difference between a foreign bond and a Eurobond?

A foreign bond is a bond issued by a foreign company in a local market. Eurobonds, on the other hand, are bonds denominated in a different currency of the country in which they are issued.

24-5. Describe the kinds of securities the U.S. government uses to finance the federal debt.

The U.S. government use treasury bills, note, bonds, and TIPS. Treasury bills are pure discount bonds with maturities of one year or less. Treasury notes are coupon bonds with semi-annual coupon payments with maturities between 1 and 10 years. Treasury bonds are semi-annual coupon bonds with maturities longer than 10 years. Finally, TIPS are bonds with coupon payments that adjust with the rate of inflation. The final payment is protected against deflation since the value of the final payment is the maximum between the face value and the inflation adjusted face value.

24-6. On January 15, 2010, the U.S. Treasury issued a five-year inflation-indexed note with a coupon of 3%. On the date of issue, the consumer price index (CPI) was 250. By January 15, 2015, the CPI had increased to 300. What principal and coupon payment was made on January 15, 2015?

The CPI index appreciated by:

$$\frac{300}{250} = 1.2.$$

Consequently, the principal amount of the bond increased by this amount; that is, the original face value of \$1,000 increased to \$1,200.

Since the bond pays semi-annual coupons, the coupon payment is:

$$1.2 \times \left(\frac{0.03}{2} \right) \times \$1,000 = \$18.$$

- 24-7. On January 15, 2020, the U.S. Treasury issued a 10-year inflation-indexed note with a coupon of 6%. On the date of issue, the CPI was 400. By January 15, 2030, the CPI had decreased to 300. What principal and coupon payment was made on January 15, 2030?**

The CPI index depreciated by

$$\frac{300}{400} = 0.75.$$

Consequently, the principal amount of the bond decreased by this amount; that is, the original face value \$1,000 decreased to \$750.

Since the bond pays semi-annual coupons, the coupon payment is:

$$0.75 \times \left(\frac{0.06}{2} \right) \times \$1,000 = \$22.5.$$

However, the final payment of the maturity (i.e. the principal) is protected against deflation. So since \$750 is less than the original face value of \$1,000, the original amount is repaid, i.e. \$1,000.

- 24-8. Describe what prepayment risk in a GNMA is.**

Holders of the GNMA securities face payment risk because homeowners have the option to prepay their debt whenever they decide to do so. In particular, they will prepay if interest rates fall and they can obtain new debt at a lower interest rate. This is precisely when the holders of GNMA securities would like to *avoid* payments, since they can only reinvest at a lower interest rate.

- 24-9. What is the distinguishing feature of how municipal bonds are taxed?**

The distinguishing feature is that income from municipal bonds is not taxed at the federal level.

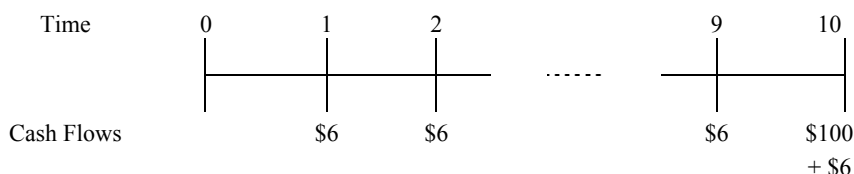
- 24-10. Explain why bond issuers might voluntarily choose to put restrictive covenants into a new bond issue.**

Bond issuers benefit from placing restricting covenants because by doing so they can obtain a lower interest rate.

- 24-11. General Electric has just issued a callable 10-year, 6% coupon bond with annual coupon payments. The bond can be called at par in one year or anytime thereafter on a coupon payment date. It has a price of \$102. What is the bond's yield to maturity and yield to call?**



Timeline:



The present value formula to be solved is:

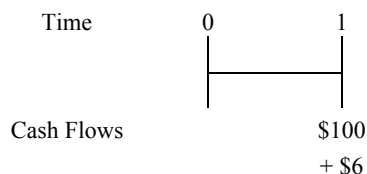
$$102 = \frac{6}{\text{YTM}} \left(1 - \frac{1}{(1 + \text{YTM})^{10}} \right) + \frac{100}{(1 + \text{YTM})^{10}}$$

Using the annuity calculator:

$$\text{YTM} = 5.73\%$$

YTC:

Timeline:



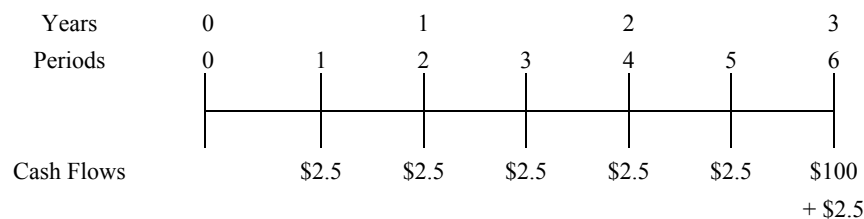
The present value formula to be solved is:

$$102 = \frac{106}{1 + \text{YTC}}$$

$$\Rightarrow \text{YTC} = \frac{106}{102} - 1 = 3.92\%.$$

- 24-12.**  **Boeing Corporation has just issued a callable (at par) three-year, 5% coupon bond with semiannual coupon payments. The bond can be called at par in two years or anytime thereafter on a coupon payment date. It has a price of \$99. What is the bond's yield to maturity and yield to call?**

Timeline:



The present value formula to be solved is:

$$99 = \frac{2.5}{i} \left(1 - \frac{1}{(1 + i)^6} \right) + \frac{100}{(1 + i)^6}.$$

Using the annuity calculator:

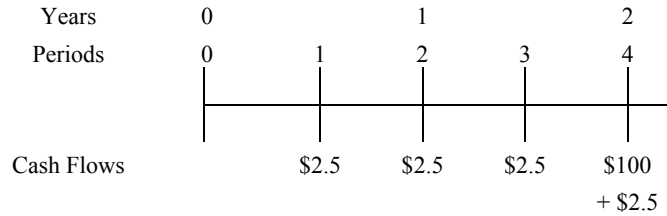
$$i = 2.68\%.$$

So since YTM are quoted as APR's:

$$\text{YTM} = i \times 2 = 2.68\% \times 2 = 5.36\%.$$

YTC:

Timeline:



The present value formula to be solved is:

$$99 = \frac{2.5}{i} \left(1 - \frac{1}{(1+i)^4} \right) + \frac{100}{(1+i)^4}.$$

Using the annuity calculator:

$$i = 2.77\%.$$

Since YTM (and therefore YTC) are quoted as APR's:

$$\begin{aligned} \text{YTC} &= i \times 2 \\ &= 5.54\%. \end{aligned}$$

- 24-13. Explain why the yield on a convertible bond is lower than the yield on an otherwise identical bond without a conversion feature.**

The option to convert the bond into stock is valuable, hence its price will be higher and its yield lower.

- 24-14. You own a bond with a face value of \$10,000 and a conversion ratio of 450. What is the conversion price?**



The conversion price is the face value of the bond divided by the conversion ratio. In this case:

$$P = \frac{\text{Face value}}{\text{Conversion ratio}} = \frac{\$10,000}{450}$$

$$P = \$22.22.$$

Chapter 25

Leasing

25-1. Suppose an H1200 supercomputer has a cost of \$200,000 and will have a residual market value of \$60,000 in five years. The risk-free interest rate is 5% APR with monthly compounding.

- a. What is the risk-free monthly lease rate for a five-year lease in a perfect market?
- b. What would be the monthly payment for a five-year \$200,000 risk-free loan to purchase the H1200?

- a. From Eq. 25.1, for a five-year (60 month) lease,

$$PV(\text{Lease payments}) = 200,000 - 60,000/(1 + .05/12)^{60} = \$153,248.$$

Because the first lease payment is paid upfront, and the remaining 59 payments are paid as an annuity:

$$153,248 = L \left(1 + \frac{1}{.05/12} \left(1 - \frac{1}{(1 + .05/12)^{59}} \right) \right)$$

Therefore, $L = \$2,880$.

- b. From Eq. 25.2, (see also Example 25.2)

$$200,000 = M \frac{1}{.05/12} \left(1 - \frac{1}{(1 + .05/12)^{60}} \right)$$

Therefore, $M = \$3,774$.

25-2. Suppose the risk-free interest rate is 5% APR with monthly compounding. If a \$2 million MRI machine can be leased for seven years for \$22,000 per month, what residual value must the lessor recover to break even in a perfect market with no risk?

From Eq. 25.1, $PV(\text{Residual Value}) = \text{Purchase Price} - PV(\text{Lease Payments})$

$$= \$2 \text{ million} - 22,000 \left(1 + \frac{1}{.05/12} \left(1 - \frac{1}{(1 + .05/12)^{83}} \right) \right)$$

$= \$436,974$.

The future residual value in 84 months is therefore:

$$\text{Residual Value} = \$436,974 \times (1 + .05/12)^{84} = \$619,645.$$

25-3. Consider a five-year lease for a \$400,000 bottling machine, with a residual market value of \$150,000 at the end of the five years. If the risk-free interest rate is 6% APR with monthly compounding, compute the monthly lease payment in a perfect market for the following leases:

- a. A fair market value lease

b. A \$1.00 out lease**c. A fixed price lease with an \$80,000 final price**

- a. From Eq. 25.1, for a five-year (60 month) lease with a monthly interest rate of $6\%/12 = 0.5\%$,

$$PV(\text{Lease payments}) = 400,000 - 150,000/(1.005)^{60} = \$288,794.$$

Because the first lease payment is paid upfront, and the remaining 59 payments are paid as an annuity:

$$288,794 = L \left(1 + \frac{1}{.005} \left(1 - \frac{1}{1.005^{59}} \right) \right).$$

Therefore, $L = \$5555$.

- b. In this case, the lessor will only receive \$1 at the conclusion of the lease. Therefore, the present value of the lease payments should be \$400,000:

$$400,000 = L \left(1 + \frac{1}{.005} \left(1 - \frac{1}{1.005^{59}} \right) \right).$$

Therefore, $L = \$7695$.

- c. In this case the lessor will receive \$80,000 at the conclusion of the lease. Thus,

$$PV(\text{Lease payments}) = 400,000 - 80,000/(1.005)^{60} = \$340,690.$$

Because the first lease payment is paid upfront, and the remaining 59 payments are paid as an annuity:

$$340,690 = L \left(1 + \frac{1}{.005} \left(1 - \frac{1}{1.005^{59}} \right) \right).$$

Therefore, $L = \$6554$.

25-4. Acme Distribution currently has the following items on its balance sheet:

Assets		Liabilities	
Cash	20	Debt	70
Property, Plant, and Equipment	175	Equity	125

How will Acme's balance sheet change if it enters into an \$80 million capital lease for new warehouses? What will its book debt-equity ratio be? How will Acme's balance sheet and debt-equity ratio change if the lease is an operating lease?

(See Example 25.4) Capital Lease: property added to balance sheet, lease added to debt –

Assets		Liabilities	
Cash	20	Debt	150
Prop., Plant, Equip.	255	Equity	125

Book D/E = $150 / 120 = 1.25$

Operating Lease: no change to balance sheet. Book D/E = $70/125 = 0.56$

25-5. Your firm is considering leasing a \$50,000 copier. The copier has an estimated economic life of eight years. Suppose the appropriate discount rate is 9% APR with monthly compounding.

Classify each lease below as a capital lease or operating lease, and explain why:

- a. A four-year fair market value lease with payments of \$1150 per month
- b. A six-year fair market value lease with payments of \$790 per month
- c. A five-year fair market value lease with payments of \$925 per month
- d. A five-year fair market value lease with payments of \$1000 per month and an option to cancel after three years with a \$9000 cancellation penalty

- a. A four-year fair market value lease with payments of \$1,150 per month.

$$PV(\text{Lease Payments}) = 1150 \times \left(1 + \frac{1}{.09/12} \left(1 - \frac{1}{(1 + .09/12)^{47}} \right) \right) = \$46,559.$$

This is $46,559/50,000 = 93\%$ of the purchase price. Because it exceeds 90% of the purchase price, this is a capital lease.

- b. A six-year fair market value lease with payments of \$790 per month.

The lease term is 75% or more of the economic life of the asset ($75\% \times 8 \text{ years} = 6 \text{ years}$), and so this is a capital lease.

- c. A five-year fair market value lease with payments of \$925 per month.

$$PV(\text{Lease Payments}) = 925 \times \left(1 + \frac{1}{.09/12} \left(1 - \frac{1}{(1 + .09/12)^{59}} \right) \right) = \$44,895.$$

This is $44,895/50,000 = 89.8\%$ of the purchase price. Because it is less than 90% of the purchase price, and the term is less than 6 years, and it is a fair market value lease, this is an operating lease.

- d. A five-year fair market value lease with payments of \$1000 per month and an option to cancel after three years with a \$9000 cancellation penalty.

Without the cancellation option, the PV of the lease payments would exceed 90% of the purchase price.

With the cancellation option,

$PV(\text{Min. Lease Pmts}) =$

$$1000 \times \left(1 + \frac{1}{.09/12} \left(1 - \frac{1}{(1 + .09/12)^{35}} \right) \right) + \frac{9000}{(1 + .09/12)^{36}} = \$38,560.$$

As this is less than 90% of the purchase price, the lease qualifies as an operating lease.

25-6. Craxton Engineering will either purchase or lease a new \$756,000 fabricator. If purchased, the fabricator will be depreciated on a straight-line basis over seven years. Craxton can lease the fabricator for \$130,000 per year for seven years. Craxton's tax rate is 35%. (Assume the fabricator has no residual value at the end of the seven years.)

- a. What are the free cash flow consequences of buying the fabricator if the lease is a true tax lease?
- b. What are the free cash flow consequences of leasing the fabricator if the lease is a true tax lease?
- c. What are the incremental free cash flows of leasing versus buying?

- a. $FCF_0 = \text{Capital Expenditure} = \$756,000$

$$FCF_{1-7} = \text{Depreciation tax shield} = 35\% \times 756,000/7 = \$37,800$$

- b. $FCF_{0-6} = \text{After-tax lease payment} = 130,000 \times (1 - 35\%) = \$84,500$
- c. $FCF_0 = -84,500 - (-756,000) = \$671,500$
 $FCF_{1-6} = -84,500 - (37,800) = \$122,300$
 $FCF_7 = 0 - (37,800) = \$37,800$

25-7. Riverton Mining plans to purchase or lease \$220,000 worth of excavation equipment. If purchased, the equipment will be depreciated on a straight-line basis over five years, after which it will be worthless. If leased, the annual lease payments will be \$55,000 per year for five years. Assume Riverton's borrowing cost is 8%, its tax rate is 35%, and the lease qualifies as a true tax lease.

- a. **If Riverton purchases the equipment, what is the amount of the lease-equivalent loan?**
- b. **Is Riverton better off leasing the equipment or financing the purchase using the lease equivalent loan?**
- c. **What is the effective after-tax lease borrowing rate? How does this compare to Riverton's actual after-tax borrowing rate?**
- a. If Riverton buys the equipment, it will pay \$220,000 upfront and have depreciation expenses of $220,000 / 5 = \$44,000$ per year, generating a depreciation tax shield of $35\% \times 44,000 = \$15,400$ per year for years 1–5.

If it leases, the after-tax lease payments are $\$55,000 \times (1 - .35) = \$35,750$. Thus, the FCF of leasing versus buying is $-35,750 - (-220,000) = 184,250$ in year 0, $-35,750 - (15,400) = -51,150$ in years 1–4, and $0 - (15,400) = -15,400$ in year 5.

The initial amount of the lease equivalent loan is the PV of the incremental free cash flows in years 1–5 at Riverton's after-tax borrowing rate of $8\%(1 - .35) = 5.2\%$:

$$\begin{aligned} \text{Loan Amt} &= \frac{-51,150}{1.052} + \frac{-51,150}{1.052^2} + \frac{-51,150}{1.052^3} + \frac{-51,150}{1.052^4} + \frac{-15,400}{1.052^5} \sqrt{2} \\ &= -192,488. \end{aligned}$$

That is, leasing leads to the same future cash flows as buying the equipment and borrowing \$192,488 initially.

- b. If Riverton leases, it pays \$35,750 after-tax as an initial lease payment. If it buys using the lease equivalent loan, it pays $220,000 - 192,488 = \$27,512$ upfront. Because the future liabilities are the same, buying with the lease equivalent loan is cheaper by $35,750 - 27,512 = \$8,238$ today. Thus, the lease is not attractive.
- c. We compute the effective after-tax lease borrowing rate as the IRR of the incremental FCF calculated in (a): 184,250; -51,150; -51,150; -51,150; -51,150; -15,400.

Using Excel, we find the IRR is 7.0%, which is higher than Riverton's actual after-tax borrowing rate of $8\% \times (1 - .35) = 5.2\%$. Thus, the lease is not attractive.

25-8. Suppose Clorox can lease a new computer data processing system for \$975,000 per year for five years. Alternatively, it can purchase the system for \$4.25 million. Assume Clorox has a borrowing cost of 7% and a tax rate of 35%, and the system will be obsolete at the end of five years.



- a. **If Clorox will depreciate the computer equipment on a straight-line basis over the next five years, and if the lease qualifies as a true tax lease, is it better to lease or finance the purchase of the equipment?**
- b. **Suppose that if Clorox buys the equipment, it will use accelerated depreciation for tax purposes. Specifically, suppose it can expense 20% of the purchase price immediately and**

can take depreciation deductions equal to 32%, 19.2%, 11.52%, 11.52%, and 5.76% of the purchase price over the next five years. Compare leasing with purchase in this case.

- a. If Clorox buys the equipment, it will pay \$4.25 million upfront and have depreciation expenses of $4.25 / 5 = \$850,000$ per year, generating a depreciation tax shield of $35\% \times \$850,000 = \$297,500$ per year for years 1–5.

If it leases, the after-tax lease payments are $\$975,000 \times (1 - .35) = \$633,750$. Thus, the FCF of leasing versus buying is $-633,750 - (-4,250,000) = 3,616,250$ in year 0, $-633,750 - (297,500) = -931,250$ in years 1–4, and $0 - (297,500) = -297,500$ in year 5. We can determine the gain from leasing by discounting the incremental cash flows at Clorox's after-tax borrowing rate of 7% ($1 - .35 = 4.55\%$):

$$\begin{aligned} \text{NPV(Lease-Buy)} &= 3,616,250 - \frac{931,250}{1.0455} - \frac{931,250}{1.0455^2} - \frac{931,250}{1.0455^3} - \frac{931,250}{1.0455^4} - \frac{297,500}{1.0455^5} \\ &= \$41,112. \end{aligned}$$

Under these assumptions, the lease is more attractive than financing a purchase of the computer.

- b. The depreciation tax shield if Clorox buys is now $35\% \times (\$4.25 \text{ million} \times 20\%) = \$297,500$ in year 0. Therefore, the incremental FCF from leasing is $633,750 - (-4,250,000) - 297,500 = \$3,318,750$ in year 0. In year 1, the depreciation tax shield is $35\% \times (\$4.25 \text{ million} \times 32\%) = \$476,000$, and the incremental cash flow is $-633,750 - (476,000) = -1,109,750$. We can continue in this way each year as shown in the spreadsheet below:

	A	B	C	D	E	F	G	H	I
41			Year	0	1	2	3	4	5
42	Buy:								
43	1	Capital Expenditures		(4,250,000)	-	-	-	-	-
46	2	Depreciation tax shield at 35%		297,500	476,000	285,600	171,360	171,360	85,680
47	3	Free Cash Flow (Buy)		(3,952,500)	476,000	285,600	171,360	171,360	85,680
48	Lease:								
49	4	Lease payments		(975,000)	(975,000)	(975,000)	(975,000)	(975,000)	-
50	5	Income tax savings at 35%		341,250	341,250	341,250	341,250	341,250	-
51	6	Free Cash Flow (Lease)		(633,750)	(633,750)	(633,750)	(633,750)	(633,750)	-
52	Lease vs. Buy:								
53	7	Lease - Buy		3,318,750	(1,109,750)	(919,350)	(805,110)	(805,110)	(85,680)
56		NPV L		(30,712)					

Therefore:

$$\begin{aligned} \text{NPV(Lease-Buy)} &= 3,318,750 - \frac{1,109,750}{1.0455} - \frac{919,350}{1.0455^2} - \frac{805,110}{1.0455^3} - \frac{805,110}{1.0455^4} - \frac{85,680}{1.0455^5} \\ &= -\$30,712 \end{aligned}$$

and so the lease is no longer attractive.

- 25-9. Suppose Procter and Gamble (P&G) is considering purchasing \$15 million in new manufacturing equipment. If it purchases the equipment, it will depreciate it on a straight-line basis over the five years, after which the equipment will be worthless. It will also be responsible for maintenance expenses of \$1 million per year. Alternatively, it can lease the equipment for \$4.2 million per year for the five years, in which case the lessor will provide necessary maintenance. Assume P&G's tax rate is 35% and its borrowing cost is 7%.

- a. What is the NPV associated with leasing the equipment versus financing it with the lease equivalent loan?
- b. What is the break-even lease rate—that is, what lease amount could P&G pay each year and be indifferent between leasing and financing a purchase?
- a. If P&G buys the equipment, it will pay \$15 million upfront, and have depreciation expenses of $15 / 5 = \$3$ million per year, generating a depreciation tax shield of $35\% \times 3 = \$1.05$ million per year

for years 1–5. It will also have after tax maintenance expenses of \$1 million $\times (1 - .35) = 0.65$ million. Thus, the FCF from buying is $1.05 - .65 = \$0.4$ million in years 1–5.

If it leases, the after-tax lease payments are $\$4.2 \text{ million} \times (1 - .35) = \2.73 million. Thus, the FCF of leasing versus buying is $-2.73 - (-15) = 12.27$ million in year 0, $-2.73 - (0.4) = -3.13$ million in years 1–4, and $0 - (0.4) = -0.4$ million in year 5. We can determine the gain from leasing by discounting the incremental cash flows at P&G's after-tax borrowing rate of $7\%(1 - .35) = 4.55\%$:

$$\begin{aligned} \text{NPV}(\text{Lease-Buy}) &= 12.27 - \frac{3.13}{1.0455} - \frac{3.13}{1.0455^2} - \frac{3.13}{1.0455^3} - \frac{3.13}{1.0455^4} - \frac{0.4}{1.0455^5} \\ &= \$733,955. \end{aligned}$$

Under these assumptions, the lease is more attractive than financing a purchase of the computer.

- b. We can increase the after-tax lease payments by an amount with present value equal to the NPV in (a). Thus,

$$733,955 = (\text{Increase in } L) \times (1 - 0.35) \times \left(1 + \frac{1}{.0455} \left(1 - \frac{1}{1.0455^4} \right) \right)$$

so that $\text{Increase in } L = 246,363$. Therefore, the break-even lease rate is $4.2 + 0.246363 = \$4.446363$ million per year, as verified in the following spreadsheet:

	Year	0	1	2	3	4	5
Buy							
1	Capital Expenditures	(15,000,000)	—	—	—	—	—
2	After-tax maintenance cost	—	(650,000)	(650,000)	(650,000)	(650,000)	(650,000)
3	Depreciation tax shield at 35%	—	1,050,000	1,050,000	1,050,000	1,050,000	1,050,000
4	Free Cash Flow (Buy)	(15,000,000)	400,000	400,000	400,000	400,000	400,000
Lease							
5	Lease Payments	(4,446,363)	(4,446,363)	(4,446,363)	(4,446,363)	(4,446,363)	—
6	Income tax savings at 35%	1,556,227	1,556,227	1,556,227	1,556,227	1,556,227	—
7	Free Cash Flow (Lease)	(2,890,136)	(2,890,136)	(2,890,136)	(2,890,136)	(2,890,136)	—
Lease vs. Buy							
8	Lease-Buy	12,109,864	(3,290,136)	(3,290,136)	(3,290,136)	(3,290,136)	(400,000)
	NPV(Lease-Buy)	(0)					

-  **25-10.** Suppose Netflix is considering the purchase of computer servers and network infrastructure to facilitate its move into video-on-demand services. In total, it will purchase \$48 million in new equipment. This equipment will qualify for accelerated depreciation: 20% can be expensed immediately, followed by 32%, 19.2%, 11.52%, 11.52%, and 5.76% over the next five years. However, because of the firm's substantial loss carryforwards, Netflix estimates its marginal tax rate to be 10% over the next five years, so it will get very little tax benefit from the depreciation expenses. Thus, Netflix considers leasing the equipment instead. Suppose Netflix and the lessor face the same 8% borrowing rate, but the lessor has a 35% tax rate. For the purpose of this question, assume the equipment is worthless after five years, the lease term is five years, and the lease qualifies as a true tax lease.

- What is the lease rate for which the lessor will break even?
- What is the gain to Netflix with this lease rate?
- What is the source of the gain in this transaction?

- a. The break-even lease rate for the lessor is 11,080,000 as shown in the spreadsheet:

LESSOR	tcb Year	35% 0	5.2% 1	2	3	4	5
Buy							
1	Capital Expenditures	(48,000)	—	—	—	—	—
2	Depreciation tax shield at 35%	3,360	5,376	3,226	1,935	1,935	968
3	Free Cash Flow (Buy)	(44,640)	5,376	3,226	1,935	1,935	968
Lease							
4	Lease Payments	11,080	11,080	11,080	11,080	11,080	—
5	Income tax at 35%	(3,878)	(3,878)	(3,878)	(3,878)	(3,878)	—
6	Free Cash Flow (Lease)	7,202	7,202	7,202	7,202	7,202	—
Lessor Free Cash Flow							
7	Buy & Lease	(37,438)	12,578	10,428	9,137	9,137	968
	NPV(Buy & Lease)	0					

To compute this amount, first we compute the FCF from buying the machine. The depreciation tax shield is $0.35 \times (\$48\text{m} \times 0.20) = \3.36 million in year 0, $0.35 \times (\$48\text{m} \times 0.32) = \5.376 million in year 1, etc., as shown in line 2. The NPV of the FCF from buying the machine (line 3) is:

$$\begin{aligned}\text{NPV}(\text{Buy}) &= -44.64 + \frac{5.376}{1.052} + \frac{3.226}{1.052^2} + \frac{1.935}{1.052^3} + \frac{1.935}{1.052^4} + \frac{0.968}{1.052^5} \\ &= -32.622.\end{aligned}$$

Therefore, to break-even, the PV of the after-tax lease payments must equal \$32.622 million:

$$32.622 = L \times (1 - 0.35) \times \left(1 + \frac{1}{.052} \left(1 - \frac{1}{1.052^4} \right) \right)$$

and so $L = 11.080$ million.

- b. At a lease rate of \$11.080 and a tax rate of 10%, Netflix has a gain of \$0.145 million.

	A	B	C	D	E	F	G	H	I
38			rD	8%					
39	LESSEE		tca	10%	7.200%				
40									
41			Year	0	1	2	3	4	5
42	Buy:								
43	1	Capital Expenditures		(48,000)	-	-	-	-	-
46	2	Depreciation tax shield at 10%		960	1,536	922	553	553	276
47	3	Free Cash Flow (Buy)		(47,040)	1,536	922	553	553	276
48	Lease:								
49	4	Lease payments		(11,080)	(11,080)	(11,080)	(11,080)	(11,080)	-
50	5	Income tax savings at 10%		1,108	1,108	1,108	1,108	1,108	-
51	6	Free Cash Flow (Lease)		(9,972)	(9,972)	(9,972)	(9,972)	(9,972)	-
52	Lease vs. Buy:								
53	7	Lease - Buy		37,068	(11,508)	(10,894)	(10,525)	(10,525)	(276)
56		NPV(Lease-Buy)		145					

- c. The source of the gain is the difference in tax rates between the two parties. Because the depreciation tax shield is more accelerated than the lease payments, there is a gain from shifting the depreciation tax shields to the party with the higher tax rate.

Chapter 26

Working Capital Management

26-1. Answer the following:

- a. **What is the difference between a firm's cash cycle and its operating cycle?**
 - b. **How will a firm's cash cycle be affected if a firm increases its inventory, all else being equal?**
 - c. **How will a firm's cash cycle be affected if a firm begins to take the discounts offered by its suppliers, all else being equal?**
- a. A firm's cash cycle is the average length of time from when a firm pays cash for its inventory to when it receives cash from the sale of that inventory (or the end product that the firm produced with the inventory). It is calculated as the average number of days between the purchase of the initial inventory and the sale of the end product plus the average number of days it takes the firm's customers to pay cash for the inventory they purchase minus the average number of days the firm takes to pay its suppliers for the inventory. A firm's operating cycle is the average length of time between when a firm purchases its inventory and when the firm receives cash from the sale of the inventory. It is calculated as the average number of days between the purchase of the initial inventory and the sale of the end product plus the average number of days it takes the firm's customers to pay for the inventory they purchase. If a firm were to pay cash for its inventory, rather than buying the inventory on credit, the cash cycle and the operating cycle of the firm would be identical. In most cases, however, firms buy their inventory on credit, so the cash cycle is shorter than the operating cycle of the firm.
- b. If a firm increases its inventory, its inventory days will increase, all else equal. This will, therefore, increase the cash cycle of the firm.
- c. If a firm begins to take discounts offered by its suppliers, its accounts payable days will decrease. All else equal, this will cause the cash cycle of the firm to increase.

26-2. Does an increase in a firm's cash cycle necessarily mean that a firm is managing its cash poorly?

No. An increase in a firm's cash cycle does not necessarily mean that the firm is managing its cash poorly. The increase may be due to a conscious management decision. For example, a firm may decide to increase its inventory in order if it has been experiencing excessive stock-outs. All else equal, this would result in an increase in its cash cycle. Or a firm may decide to loosen its credit policy in order to attract customers from its competitors. This would result in an increase in accounts receivable days, and all else equal, result in an increase in the firm's cash cycle. And if a firm chooses to take the discounts offered by its suppliers, the accounts payable days will decrease, leading to an increase in the firm's cash cycle.

26-3. Aberdeen Outboard Motors is contemplating building a new plant. The company anticipates that the plant will require an initial investment of \$2 million in net working capital today. The plant will last 10 years, at which point the full investment in net working capital will be recovered. Given an annual discount rate of 6%, what is the net present value of this working capital investment?

Ignoring revenues and other expenses associated with the new plant, the NPV of the \$2 million investment in net working capital is simply the present value of the \$2 million that the firm will recoup at the end of ten years minus the initial \$2 million investment.

$$\text{NPV} = -\$2,000,000 + \frac{\$2,000,000}{(1.06)^{10}} = -\$883,210$$

- 26-4.  **The Greek Connection had sales of \$32 million in 2009, and a cost of goods sold of \$20 million. A simplified balance sheet for the firm appears below:**

THE GREEK CONNECTION
Balance Sheet
As of December 31, 2009
(\$ thousands)

Assets		Liabilities and Equity	
Cash	\$ 2,000	Accounts payable	\$ 1,500
Accounts receivable	3,950	Notes payable	1,000
Inventory	1,300	Accruals	1,220
Total current assets	\$ 7,250	Total current liabilities	\$ 3,720
		Long-term debt	3,000
Net plant, property, and equipment	\$ 8,500	Total liabilities	\$ 6,720
Total assets	\$ 15,750	Common equity	9,030
		Total liabilities and equity	\$ 15,750

- a. Calculate The Greek Connection's net working capital in 2009.
 - b. Calculate the cash conversion cycle of The Greek Connection in 2009.
 - c. The industry average accounts receivable days is 30 days. What would the cash conversion cycle for The Greek Connection have been in 2009 had it matched the industry average for accounts receivable days?
- a. Net working capital is current assets minus current liabilities. Using this definition, The Greek Connection's net working capital is $\$7,250 - \$3,720 = \$3,530$. Some analysts calculate the net operating working capital instead, which is the non-interest earning current assets minus the non-interest bearing current liabilities. In this case, the notes payable would not be included in the calculation since they are assumed to be interest bearing. Net operating working capital for The Greek Connection is $\$7,250 - (\$1,500 + \$1,220) = \$4,530$.
 - b. The cash conversion cycle (CCC) is equal to the inventory days plus the accounts receivable days minus the accounts payable days. The Greek Connection's cash conversion cycle for 2004 was 41.4 days.

$$\text{CCC} = \frac{\text{inventory}}{\text{average daily COGS}} + \frac{\text{accounts receivable}}{\text{average daily sales}} - \frac{\text{accounts payable}}{\text{average daily COGS}}$$

$$\text{CCC}_{2004} = \frac{\$1,300}{\frac{\$20,000}{365}} + \frac{\$3,950}{\frac{\$32,000}{365}} - \frac{\$1,500}{\frac{\$20,000}{365}}$$

$$= 23.7 \text{ days} + 45.1 \text{ days} - 27.4 \text{ days} = 41.4 \text{ days}$$

- c. If The Greek Connection accounts receivable days had been 30 days, its cash conversion cycle would have been only 26.3 days:

$$\text{CCC} = 23.7 \text{ days} + 30 \text{ days} - 27.4 \text{ days} = 26.3 \text{ days}$$

- 26-5. Assume the credit terms offered to your firm by your suppliers are 3/5, Net 30. Calculate the cost of the trade credit if your firm does not take the discount and pays on day 30.**

In this instance, the customer will have the use of \$97 for an additional 25 days ($30 - 5$) if he chooses not to take the discount. It will cost him \$3 to do so since he must pay \$100 for the goods if he pays after the 5-day discount period. Thus, the interest rate per period is:

$$\frac{\$3}{\$97} = 0.0309 = 3.09\%.$$

The number of 25-day periods in a year is $365/25 = 14.6$ periods. So the effective annual cost of the trade credit is:

$$\text{EAR} = (1.0309)^{14.6} - 1 = 55.94\%.$$

- 26-6. Your supplier offers terms of 1/10, Net 45. What is the effective annual cost of trade credit if you choose to forgo the discount and pay on day 45?**

If you were to pay within the 10-day discount period, you would pay \$99 for \$100 worth of goods. If you wait until day 45, you will owe \$100. Thus, you are paying \$1 in interest for a 35-day ($45 - 10$) loan. The interest rate per period is:

$$\frac{\$1}{\$99} = 0.0101 = 1.01\%.$$

The number of 35-day periods in a year is $365 / 35 = 10.43$ periods. So the effective annual cost of the trade credit is:

$$\text{EAR} = (1.0101)^{10.43} - 1 = 11.05\%.$$

- 26-7. The Fast Reader Company supplies bulletin board services to numerous hotel chains nationwide. The owner of the firm is investigating the benefit of employing a billing firm to do her billing and collections. Because the billing firm specializes in these services, collection float will be reduced by 20 days. Average daily collections are \$1200, and the owner can earn 8% annually (expressed as an APR with monthly compounding) on her investments. If the billing firm charges \$250 per month, should the owner employ the billing firm?**

The benefit of outsourcing the billing and collection to the other firm is equal to what Fast Reader can earn on the funds that are freed up. Since average daily collections are \$1,200 and float will be reduced by 20 days, Fast Reader will have an additional \$24,000 ($\$1,200 \times 20$). (Think about this as follows. Immediately after hiring the billing firm, its collection float drops by 20 days, so all collections due within the next 20 days are immediately available.) The billing firm charges \$250 per month. At an 8% annual rate, the monthly discount rate is $1.081 / 12 - 1 = 6.43\%$, so the present value of these charges in perpetuity is $250/0.0643 = \$38,856$. Thus, the costs (\$38,856) exceed the benefits (\$24,000), and Fast Reader should not employ the billing firm.

- 26-8. The Saban Corporation is trying to decide whether to switch to a bank that will accommodate electronic funds transfers from Saban's customers. Saban's financial manager believes the new system would decrease its collection float by as much as five days. The new bank would require a compensating balance of \$30,000, whereas its present bank has no compensating balance requirement. Saban's average daily collections are \$10,000, and it can earn 8% on its short-term investments. Should Saban make the switch? (Assume the compensating balance at the new bank will be deposited in a non-interest-earning account.)**

The electronic funds transfer system will free up \$50,000 ($= 5 \times \$10,000$) (Think about this as follows. Immediately after switching banks its collections due within the next five days are immediately available.) On the other hand, Saban will have to pay a cost because it has to hold \$30,000 in a non-interest earning account, which means it has essentially given up these funds. Because the benefits (\$50,000) are larger than the costs (\$30,000) it should switch banks.

26-9. What are the three steps involved in establishing a credit policy?

The three steps involved in establishing a credit policy are:

- (1) Establish credit standards. In this step, the firm must decide how much credit risk it is willing to accept.
- (2) Establish credit terms. Here, the firm decides on the length of time before payment must be made and whether or not it will offer a discount. If a discount is to be offered, the amount of the discount and the length of the discount period must also be established.
- (3) Establish a collection policy. In this step, the firm must decide how it will handle late payers.

26-10. The Manana Corporation had sales of \$60 million this year. Its accounts receivable balance averaged \$2 million. How long, on average, does it take the firm to collect on its sales?

If we assume all the sales were made on credit, the average length of time it takes Manana to collect on its sales is 12.2 days:

$$\text{Accounts receivable days} = \frac{\text{Accounts receivable}}{\text{Average daily sales}} = \frac{\$2,000,000}{\frac{\$60,000,000}{365}} = 12.2 \text{ days.}$$

26-11. The Mighty Power Tool Company has the following accounts on its books:

Customer	Amount Owed (\$)	Age (days)
ABC	50,000	35
DEF	35,000	5
GHI	15,000	10
KLM	75,000	22
NOP	42,000	40
QRS	18,000	12
TUV	82,000	53
WXY	36,000	90

The firm extends credit on terms of 1/15, Net 30. Develop an aging schedule using 15-day increments through 60 days, and then indicate any accounts that have been outstanding for more than 60 days.

Mighty Power Tool Company Aging Schedule

Percent of

Days Outstanding	Amount Owed	Accounts Receivable
0-15	\$ 68,000	19.3%
16-30	\$ 75,000	21.2%
31-45	\$ 92,000	26.1%
46-60	\$ 82,000	23.2%
over 60	\$ 36,000	10.2%
	\$353,000	100.0%

26-12. What is meant by “stretching the accounts payable”?

“Stretching the accounts payable” refers to a customer’s not paying by the payment date specified.

 26-13. Simple Simon’s Bakery purchases supplies on terms of 1/10, Net 25. If Simple Simon’s chooses to take the discount offered, it must obtain a bank loan to meet its short-term financing needs. A local bank has quoted Simple Simon’s owner an interest rate of 12% on borrowed funds. Should Simple Simon’s enter the loan agreement with the bank and begin taking the discount?

If Simple Simon’s takes the discount, it must pay \$99 in 10 days for every \$100 of purchases. If it elects not to take the discount, it will owe the full \$100 in 25 days. The interest rate on the loan is:

$$\frac{\$1}{\$99} = 1.01\%.$$

The loan period is 15 days (= 25 – 10). The effective annual cost of the trade credit is:

$$\text{EAR} = (1.0101)^{365/15} - 1 = 27.7\%.$$

Since the bank loan is only 12%, Simple Simon should borrow the funds from the bank in order to take advantage of the discount.

26-14. Your firm purchases goods from its supplier on terms of 3/15, Net 40.

- a. What is the effective annual cost to your firm if it chooses not to take the discount and makes its payment on day 40?
- b. What is the effective annual cost to your firm if it chooses not to take the discount and makes its payment on day 50?
- a. Your firm is paying \$3 to borrow \$97 for 25 days (= 40 – 15). The interest rate per period is:

$$\frac{\$3}{\$97} = 0.0309 = 3.09\%.$$

The effective annual rate is $(1.0309)^{365/25} - 1 = 55.9\%$.

- b. In this case, your firm is stretching its accounts payable. You are still paying \$3 to borrow \$97, so the interest rate per period is 3.09%. However, the loan period is now 35 days (= 50 – 15). The effective annual rate is reduced to 37.4% because your firm has use of the money for a longer period of time:

$$\text{EAR} = (1.0309)^{365/35} - 1 = 37.4\%.$$

 26-15. Use the financial statements supplied below for International Motor Corporation (IMC) to answer the following questions.

- a. Calculate the cash conversion cycle for IMC for both 2009 and 2010. What change has occurred, if any? All else being equal, how does this change affect IMC’s need for cash?
- b. IMC’s suppliers offer terms of Net 30. Does it appear that IMC is doing a good job of managing its accounts payable?

INTERNATIONAL MOTOR CORPORATION

Income Statement (in millions)
for the Years Ending December 31

	2009	2010
Sales	\$ 60,000	\$ 75,000
Cost of goods sold	52,000	61,000
Gross profit	\$ 8,000	\$ 14,000
Selling and general and administrative expenses	6,000	8,000
Operating profit	\$ 2,000	\$ 6,000
Interest expense	1,400	1,300
Earnings before tax	\$ 600	\$ 4,700
Taxes	300	2,350
Earnings after tax	\$ 300	\$ 2,350

INTERNATIONAL MOTOR CORPORATION

Balance Sheet (in millions)
as of December 31

	2009	2010		2009	2010
Assets			Liabilities		
Cash	\$ 3,080	\$ 6,100	Accounts payable	\$ 3,600	\$ 4,600
Accounts receivable	2,800	6,900	Notes payable	1,180	1,250
Inventory	6,200	6,600	Accruals	5,600	6,211
Total current assets	\$ 12,080	\$ 19,600	Total current liabilities	\$ 10,380	\$ 12,061
Net plant, property, and equipment	\$ 23,087	\$ 20,098	Long-term debt	\$ 6,500	\$ 7,000
Total assets	\$ 35,167	\$ 39,698	Total liabilities	\$ 16,880	\$ 19,061
			Equity		
			Common stock	\$ 2,735	\$ 2,735
			Retained earnings	\$ 15,552	\$ 17,902
			Total equity	\$ 18,287	\$ 20,637
			Total liabilities and equity	\$ 35,167	\$ 39,698

- a. The cash conversion cycle (CCC) is equal to the inventory days plus the accounts receivable days minus the accounts payable days. IMC's cash conversion cycle for 2003 was 35.2 days, and for 2004, it was 45.6 days.

$$CCC = \frac{\text{inventory}}{\text{average daily COGS}} + \frac{\text{accounts receivable}}{\text{average daily sales}} - \frac{\text{accounts payable}}{\text{average daily COGS}}$$

$$CCC_{2003} = \frac{\$6,200}{\frac{\$52,000}{365}} + \frac{\$2,800}{\frac{\$60,000}{365}} - \frac{\$3,600}{\frac{\$52,000}{365}}$$

$$= 43.5 \text{ days} + 17.0 \text{ days} - 25.3 \text{ days} = 35.2 \text{ days}$$

$$CCC_{2004} = \frac{\$6,600}{\frac{\$61,000}{365}} + \frac{\$6,900}{\frac{\$75,000}{365}} - \frac{\$4,600}{\frac{\$61,000}{365}}$$

$$= 39.5 \text{ days} + 33.6 \text{ days} - 27.5 \text{ days} = 45.6 \text{ days}$$

IMC's cash conversion cycle has lengthened in 2004, due to an increase in its accounts receivable days. The number of days goods are held in inventory has decreased, and IMC is taking longer to pay its suppliers, both of which would decrease the cash conversion cycle all else equal. These changes were not enough to offset the increase in the amount of time it is taking IMC's customers

to pay for purchases made on credit. The lengthening of the cash conversion cycle means that IMC will require more cash.

- b. If IMC's suppliers are offering terms of net 30, IMC should consider waiting longer to pay for its purchases. In 2003, it paid nearly five days earlier than necessary, and in 2004, it paid 2.5 days earlier. IMC could, therefore, have kept the money working for it longer because there was no discount offered for early payment. The early payment may give IMC a preferred position with its suppliers, however, which may have benefits that are not presented here. IMC's decision on whether to extend its accounts payable days would have to take these benefits into consideration.

26-16. Ohio Valley Homecare Suppliers, Inc. (OVHS), had \$20 million in sales in 2009. Its cost of goods sold was \$8 million, and its average inventory balance was \$2,000,000.

- a. Calculate the average number of inventory days outstanding for OVHS.
- b. The average days of inventory in the industry is 73 days. By how much would OVHS reduce its investment in inventory if it could improve its inventory days to meet the industry average?

- a. The inventory days ratio is equal to the inventory divided by average daily cost of goods sold. This was 91.25 days for Ohio Valley Homecare Suppliers in 2004:

$$\text{inventory days} = \frac{\$2,000,000}{\frac{\$8,000,000}{365}} = 91.25 \text{ days.}$$

The inventory turnover ratio for Ohio Valley Homecare Supplies was four times in 2004:

$$\text{inventory turnover} = \frac{\text{cost of goods sold}}{\text{inventory}} = \frac{\$8,000,000}{\$2,000,000} = 4X.$$

- b. OVHS could increase its inventory turnover to five times by reducing its inventory to \$1,600,000:

$$5 = \frac{\$8,000,000}{\text{inventory}} \Rightarrow \text{inventory} = \$1,600,000.$$

This means OVHS could reduce its investment in inventory by \$400,000 (= \$2,000,000 – \$1,600,000).

26-17. Which of the following short-term securities would you expect to offer the highest before-tax return: Treasury bills, certificates of deposit, short-term tax exempts, or commercial paper? Why?

Commercial paper. Treasury bills and certificates of deposit are considered to be free of default risk, so they offer lower returns than a security that has default risk associated with it. Short-term tax exempts have default risk, but the interest on these instruments is free from federal taxation; thus, they offer a lower yield than a similarly risky, fully taxable security. Commercial paper exposes the investor to default risk and the interest earned is fully taxable. It would, therefore, have to offer the investor a higher before-tax return than the other instruments.

Chapter 27

Short-Term Financial Planning

27-1. Which of the following companies are likely to have high short-term financing needs? Why?

- a. A clothing retailer
- b. A professional sports team
- c. An electric utility
- d. A company that operates toll roads
- e. A restaurant chain

The professional sports team is likely to have very high short-term financing needs because of the seasonality of its revenue stream. Several months before the season actually begins for the team, it may have large cash flow requirements—equipment, tickets, advertising, etc.—that it will need to finance since the bulk of its cash inflows will occur during the season.

The clothing retailer may have high short-term financing needs because of the seasonality of its business. Sales are often highest in the fall and spring, so the retailer may need short-term financing in order to purchase inventory prior to the high seasons. It may need even more short-term financing due to a negative cash flow shock. For example, if the store's buyer misjudged the fashion trends and overbought a particular style that the store was unable to sell, the store might experience a large net cash outflow for that season. In that case, it may require more short-term financing than normal in order to purchase inventory for the upcoming season.

27-2. Sailboats Etc. is a retail company specializing in sailboats and other sailing-related equipment. The following table contains financial forecasts as well as current (month 0) working capital levels. During which months are the firm's seasonal working capital needs the greatest? When does it have surplus cash?

	Month						
(\$000)	0	1	2	3	4	5	6
Net Income		\$10	\$12	\$15	\$25	\$30	\$18
Depreciation		2	3	3	4	5	4
Capital Expenditures		1	0	0	1	0	0
Levels of Working Capital							
Accounts Receivable	\$2	3	4	5	7	10	6
Inventory	3	2	4	5	5	4	2
Accounts Payable	2	2	2	2	2	2	2

To determine Sailboats seasonal working capital needs, we calculate the changes in net working capital for the firm.

Changes in working capital	Month					
	1	2	3	4	5	6
Accounts receivable	\$1	\$1	\$1	\$2	\$3	-\$4
Inventory	-\$1	\$2	\$1	0	-\$1	-\$2
Accounts payable	0	0	0	0	0	0
Change in net working capital	\$0	\$3	\$2	\$2	\$2	-\$6

From the table it can be seen that Sailboat's working capital needs are highest in Month 2 because its investments in accounts receivable and in inventory increased the most in that month.

Sailboats, Etc. has a surplus cash position in every month as shown below:

	Month					
	1	2	3	4	5	6
(\$000)						
Net income	\$10	\$12	\$15	\$25	\$30	\$18
<i>plus</i> depreciation	2	3	3	4	5	4
<i>minus</i> changes in net working capital	0	3	2	2	2	-6
Cash flow from operations	\$12	\$12	\$16	\$27	\$33	\$28
<i>minus</i> capital expenditures	1	0	0	1	0	0
Change in cash	\$11	\$12	\$16	\$26	\$33	\$28

27-3. What is the difference between permanent working capital and temporary working capital?

Permanent working capital is the amount that a firm has to keep invested in its short-term assets in order to support its operations. For example, a toy retailer may need to keep a minimum amount invested in inventory at all times. Temporary working capital, on the other hand, is the result of seasonal fluctuations and/or unanticipated cash flow shocks. The toy retailer, for example, may need to have more invested in inventory just prior to its peak season. Temporary working capital is the difference between the actual level of investment in short-term assets and the permanent working capital investment.

27-4. Quarterly working capital levels for your firm for the next year are included in the following table. What are the permanent working capital needs of your company? What are the temporary needs?



(\$000)	Quarter			
	1	2	3	4
Cash	\$100	\$100	\$100	\$100
Accounts Receivable	200	100	100	600
Inventory	200	500	900	50
Accounts Payable	100	100	100	100

The net working capital for each quarter is calculated below:

(\$000)	Quarter			
	1	2	3	4
Cash	\$100	\$100	\$ 100	\$100
Accounts receivable	\$200	\$100	\$ 100	\$600
Inventory	\$200	\$500	\$ 900	\$ 50
Accounts payable	\$100	\$100	\$ 100	\$100
Net working capital	\$400	\$600	\$1,000	\$650

The minimum level of net working capital—\$400,000 in Quarter 1—represents the firm's permanent working capital. The difference between the higher net working capital levels in each quarter and the permanent working capital needs represents the firm's temporary working capital needs. Thus the firm has temporary working capital needs of \$200,000 in Quarter 2, \$600,000 in Quarter 3, and \$250,000 in Quarter 4.

27-5. Why might a company choose to finance permanent working capital with short-term debt?

Financing permanent working capital with short-term debt is an aggressive financing policy and is considered risky. Although under a normal term structure, the interest rates on short-term debt are lower than those on long-term debt, the firm faces the risk that it will have to pay more when it needs to refinance the debt in the future. This increased risk will be reflected in a higher cost of equity for the firm. Nevertheless, a firm may decide to use short-term debt to finance permanent working capital if it believes that one or more market imperfections exist. For example, short-term debt is less sensitive to a firm's credit quality than is long-term debt and so will be less affected by management's actions or information. Thus, short-term debt may have lower agency and lemons costs than long-term debt. Management may also believe it has superior knowledge regarding the future cash flows for the firm, which is not yet reflected in the firm's credit rating. If management believes its ability to produce future cash flows will have a positive impact on its credit rating in the future, management may elect to use lower-cost, short-term debt to finance its permanent working capital for the time being with the expectation that it will refinance it with long-term debt in the near future when the market has recognized the firm's improved future prospects and rewarded it with a higher credit rating, which can result in a lower cost of long-term debt to the firm.

27-6. The Hand-to-Mouth Company needs a \$10,000 loan for the next 30 days. It is trying to decide which of three alternatives to use:



Alternative A: Forgo the discount on its trade credit agreement that offers terms of 2/10, Net 30.

Alternative B: Borrow the money from Bank A, which has offered to lend the firm \$10,000 for 30 days at an APR of 12%. The bank will require a (no-interest) compensating balance of 5% of the face value of the loan and will charge a \$100 loan origination fee, which means Hand-to-Mouth must borrow even more than the \$10,000.

Alternative C: Borrow the money from Bank B, which has offered to lend the firm \$10,000 for 30 days at an APR of 15%. The loan has a 1% loan origination fee.

Which alternative is the cheapest source of financing for Hand-to-Mouth?

Alternative A: The effective annual cost of the trade credit is 44.6%, calculated as follows:

$$\text{Interest rate per period} = \frac{\$2}{\$98} = 2.041\%.$$

The loan period is 20 (= 30 – 10) periods, and there are 18.25 periods in a year (365 / 20 = 18.25).

$$\text{EAR} = 1.02041^{18.25} - 1 = 44.6\%$$

Alternative B: Hand-to-Mouth will need to borrow \$10,100 just to cover its loan origination fee. Beyond that, it needs to have enough to meet the compensating balance requirement. So the total amount that Hand-to-Mouth must borrow is \$10,632:

$$\text{Amount needed} = \frac{\$10,100}{1 - 0.05} = \$10,632.$$

At a 12% APR, the interest expense for the 30-day loan will be $0.01(\$10,632) = \106.32 .

Since the loan origination fee is simply additional interest, the total interest on the 30-day loan is $\$106.32 + \$100 = \$206.32$. The firm's usable proceeds from the loan is \$10,000. So the interest rate per period is $2.063\% \left(= \frac{\$206.32}{\$10,000} \right)$.

The effective annual rate of alternative B is $(1.02063)^{365/30} - 1 = 28.2\%$.

Alternative C: Hand-to-Mouth will need to borrow \$10,100 in order to cover the loan origination fee.

An APR of 15% translates to an interest rate of $1.25\% \left(= \frac{15\%}{12} \right)$ for 30 days. The interest expense for one month is $0.0125 \times \$10,100 = \126.25 . This with the loan origination fee makes the total interest charge \$226.25 for the 30-day loan. This amounts to 2.263% for 30 days: $\frac{\$226.25}{\$10,000}$.

The effective annual rate is $(1.02263)^{365/30} - 1 = 31.3\%$.

Thus in Alternative A, the cost of trade credit is the most expensive at an effective annual rate of 44.6%.

- 27-7. Consider two loans with a 1-year maturity and identical face values: an 8% loan with a 1% loan origination fee and an 8% loan with a 5% (no-interest) compensating balance requirement. Which loan would have the higher effective annual rate? Why?**

The loan with the 1% loan origination fee would cost the most since the loan origination fee is just another form of interest, so on a \$1,000 loan, the borrower is paying \$90 in interest and will have the use of only \$990 for the period, making the effective annual cost of the loan over 9% ($\$90/\$990 = 9.1\%$). The compensating balance requirement of 5% on a \$1,000 loan reduces the usable proceeds of the firm by 5% to \$950, but the interest rate is still 8%, so the effective annual cost of that arrangement is $8.4\% \left(= \frac{\$80}{\$950} \right)$. The effective annual rate is not increased by a full percentage.

- 27-8. What is the difference between evergreen credit and a revolving line of credit?**

The major difference between a revolving line of credit and evergreen credit is the commitment period. A revolving line of credit is a committed line of credit that involves a solid commitment from the bank for a period that is longer than the typical one-year term of a regular line of credit. The revolving line of credit is usually good for two or three years before it must be renegotiated. Evergreen credit is a revolving line of credit that has *no* fixed maturity.

- 27-9. Which of the following one-year \$1000 bank loans offers the lowest effective annual rate?**

- A loan with an APR of 6%, compounded monthly
- A loan with an APR of 6%, compounded annually, that also has a compensating balance requirement of 10% (on which no interest is paid)
- A loan with an APR of 6%, compounded annually, that has a 1% loan origination fee

The effective annual rates of each of the alternatives are calculated as follows.

- a. Since the APR is 6%, the monthly rate is $6\%/12 = 0.5\%$. This translates to an effective annual rate of $(1.005)^{12} - 1 = 6.2\%$.
- b. The compensating balance is $\$1,000 \times 0.10 = \100 . Therefore, the borrower will have use of only \$900 of the \$1,000. The interest is $0.06 \times \$1,000 = \60 . The interest rate per period is $\$60 / \$900 = 6.7\%$. Since this alternative assumes annual compounding, the effective annual rate is 6.7% as well.
- c. The interest expense is $0.06 \times \$1,000$, and the loan origination fee is $0.01 \times \$1,000 = \10 . The loan origination fee reduces the usable proceeds of the loan to \$990 because it is paid at the beginning of the loan. The interest rate per period is $\$70 / \$990 = 7.1\%$. Since the loan is compounded annually in this case, 7.1% is the effective annual rate.

Thus, alternative (a) offers the lowest effective annual cost.

- 27-10. The Needy Corporation borrowed \$10,000 from Bank Ease. According to the terms of the loan, Needy must pay the bank \$400 in interest every three months for the three-year life of the loan, with the principal to be repaid at the maturity of the loan. What effective annual rate is Needy paying?**

In this problem, Needy must pay \$400 every three months to have the use of \$10,000. Thus, the interest rate per period is $\$400 / \$10,000 = 4\%$. Since there are four three-month periods in a year, the effective annual rate is $(1.04)^4 - 1 \approx 17\%$.

- 27-11. The Treadwater Bank wants to raise \$1 million using three-month commercial paper. The net proceeds to the bank will be \$985,000. What is the effective annual rate of this financing for Treadwater?**

Treadwater is paying \$15,000 ($= \$1,000,000 - \$985,000$) to use \$985,000 for three months, so the three-month interest rate is $\$15,000 / \$985,000 = 1.523\%$. There are four three-month periods in a year, making the effective annual rate $(1.01523)^4 - 1 = 6.2\%$.

- 27-12. Magna Corporation has an issue of commercial paper with a face value of \$1,000,000 and a maturity of six months. Magna received net proceeds of \$973,710 when it sold the paper. What is the effective annual rate of the paper to Magna?**

Magna is paying \$26,290 ($= \$1,000,000 - \$973,710$) to use \$973,710 for six months. The six-month interest rate is $\$26,290 / \$973,710 = 2.7\%$. There are two six-month periods in one year, so the effective annual rate is $(1.027)^2 - 1 = 5.5\%$.

- 27-13. What is the difference between direct paper and dealer paper?**

Direct paper is a method by which a firm sells its commercial paper directly to investors. Dealer paper refers to the sale of commercial paper through dealers. The dealers get a fee for their services, thus reducing the proceeds that the issuing firm receives (and increasing the effective cost to the firm).

- 27-14. The Signet Corporation has issued four-month commercial paper with a \$6 million face value. The firm netted \$5,870,850 on the sale. What effective annual rate is Signet paying for these funds?**

Signet's interest expense on this loan is \$129,150 ($= \$6,000,000 - \$5,870,850$), and the usable proceeds are \$5,870,850. The interest rate for the four-month period is $\$129,150 / \$5,870,850 = 2.2\%$. The effective annual rate is $(1.022)^3 - 1 = 6.7\%$.

27-15. What is the difference between pledging accounts receivable to secure a loan and factoring accounts receivable?

When accounts receivable are pledged, the borrowing firm is simply using its accounts receivable as collateral for a loan. The lender reviews the invoices for the credit sales of the borrower and determines which accounts are acceptable collateral. The lender will then lend some percentage of the dollar amount of the accepted invoices. If one or more of the borrowing firm's customers fail to pay, the firm is still responsible to the lender for the money it has borrowed. In a factoring arrangement, the accounts receivable are sold to the lender (i.e., factor), and the firm's customers typically make their payments directly to the lender. The lender agrees to pay the borrowing firm the amount due from the firm's customers at the end of the firm's payment period, but the firm may borrow a certain percentage of the face value of its receivables in order to receive the money in advance. The factoring arrangement may be "with recourse," which means that if any of the borrowing firm's customers defaults on its bills, the factor can require the borrowing firm to make the payment. It may also be "without recourse," in which case the lender bears the risk that one or more customers will default on their bills. The borrowing firm is not responsible for the payments.

27-16. The Ohio Valley Steel Corporation has borrowed \$5 million for one month at a stated annual rate of 9%, using inventory stored in a field warehouse as collateral. The warehouse charges a \$5000 fee, payable at the end of the month. What is the effective annual rate of this loan?

The monthly interest rate is $9\% / 12 = 0.75\%$, so Ohio Valley Steel must pay $0.0075 \times \$5,000,000 = \$37,500$ in interest on the loan. Combining this with the \$5,000 warehouse fee makes the monthly cost of the loan \$42,500. Since the fee is paid at the end of the month, Ohio Valley Steel has use of the full \$5,000,000 for the month. The interest rate per period is $\$42,500 / \$5,000,000 = 0.85\%$. There are 12 months in a year, so the effective annual rate is $(1.0085)^{12} - 1 = 10.7\%$.

27-17. Discuss the three different arrangements under which a firm may use inventory to secure a loan.

The three different methods under which inventory is used as collateral for a loan are floating liens, trust receipts, and warehouse arrangements.

With a floating lien (also called a "general lien" or a "blanket lien"), all of the borrower's inventory serves as collateral for a loan. The value of the collateral declines as the firm sells its inventory. In times of financial distress, a firm may decide to sell its inventory without making payments on the loan and may not have enough money to replenish the inventory it has sold. The loan is then under-collateralized. This is the riskiest arrangement from the lender's standpoint, and the loan will carry a higher interest rate than if one of the other two methods is used. Additionally, the lender will lend a much smaller percentage of the inventory value under this arrangement.

In a trust receipt (or floor planning) arrangement, specific inventory items are identified as collateral for the loan. As the specified inventory is sold, the firm uses the cash received to repay the loan. The lender will send someone to the borrower's premises periodically to ensure that none of the specified inventory has been sold without a repayment made.

In a warehouse arrangement, the inventory serving as collateral is stored in a warehouse. One type of warehouse is a public warehouse, which is a business that exists for the sole purpose of tracking the flow of the inventory. The inventory is delivered to the public warehouse by the borrowing firm, and the lender extends a loan based on the value of that inventory. When the borrowing firm needs the inventory to sell, it must return to the warehouse to retrieve it after receiving permission from the lender to do so. This arrangement is the least risky from the standpoint of the lender since it allows the lender to maintain the tightest control over the inventory, but it is only feasible for some types of inventory. This method would not be usable for inventory that is subject to spoilage or that is bulky and difficult to transport. In this latter case, a field warehouse might be a good alternative. A field warehouse is established on the borrower's premises, but it is separated from the borrower's main plant. It is operated by a third party. This type of arrangement is more convenient for the borrower, but it still gives the lender the added security of having the inventory that serves as collateral tracked by a third party.

- 27-18. The Rasputin Brewery is considering using a public warehouse loan as part of its short-term financing. The firm will require a loan of \$500,000. Interest on the loan will be 10% (APR, annual compounding) to be paid at the end of the year. The warehouse charges 1% of the face value of the loan, payable at the beginning of the year. What is the effective annual rate of this warehousing arrangement?**

Rasputin's interest expense is $0.10(\$500,000) = \$50,000$. The warehouse fee is $0.01(\$500,000) = \$5,000$. Because the warehouse fee must be paid at the beginning of the year, Rasputin's usable proceeds from the loan are only $\$495,000 (= \$500,000 - \$5,000)$. The effective annual rate is $\$55,000 / \$495,000 = 11.1\%$.

Chapter 28

Mergers and Acquisitions

28-1. What are the two primary mechanisms under which ownership and control of a public corporation can change?

Either another corporation or group of individuals can acquire the target firm, or the target firm can merge with another firm.

28-2. Why do you think mergers cluster in time, causing merger waves?

There are many competing theories as to why this is so. They generally fall into two camps: either stock market valuations drive merger activity or industry shocks accompanying economic expansions drive merger activity. It is clear that merger activity is much greater during economic expansions than during contractions and that merger activity strongly correlates with bull markets. Thus, there must be something about economic expansions in general and higher stock market valuations in particular that grease the wheels of the merger process. However, unless you are willing to believe that the majority of managers simply buy other companies because they can, without regard to economic reasoning, this can't be the whole story. There must be real economic impetus to the activity. Many of the same technological and economic conditions that lead to bull markets also motivate managers to reshuffle assets through merger and acquisitions. Thus, it takes a combination of forces usually only present during strong economic expansions to drive peaks in merger activity.

28-3. What are some reasons why a horizontal merger might create value for shareholders?

Horizontal mergers are more likely to create value for acquiring shareholders. Horizontal mergers combine two firms in the same industry. This provides for greater potential synergies in eliminating redundant functions within the two firms and potentially increased pricing power with both vendors and customers.

28-4. Why do you think shareholders from target companies enjoy an average gain when acquired, while acquiring shareholders on average often do not gain anything?

The acquiring firm has to compete against other firms, thus reducing the gains it can obtain from the transaction. Target shareholders benefit from this competition, as they obtain higher bids for the company.

28-5. If you are planning an acquisition that is motivated by trying to acquire expertise, you are basically seeking to gain intellectual capital. What concerns would you have in structuring the deal and the post-merger integration that would be different from the concerns you would have when buying physical capital?

In cases where you are buying a lot of intangible assets, especially human capital, you have to be particularly worried about how you are going to create incentives for the target's employees to stay-on. Retention bonuses are common for key employees in these types of acquisitions. It is also hard to be successful with a hostile acquisition when retention of target employees is critical. Keeping uncertainty low and moving quickly during the integration phase are both critical to acquisitions of expertise.

28-6. Do you agree that the European Union should be able to block mergers between two U.S.-based firms? Why or why not?

The argument can go either way on this. Some of the critical factors to consider are: What is the social good created by antitrust regulation? Do European regulators have a right to regulate firms doing business in Europe, regardless of where those firms are headquartered? What would be the alternatives?

28-7. How do the carryforward and carryback provisions of the U.S. tax code affect the benefits of merging to capture operating losses?

Carryforward and carryback provisions generally reduce the attractiveness of tax losses as a motivation to merger. Since the tax loss motivation is based on the ability of a larger firm to capture the tax deduction from the losses of the target, it requires that the target not be able to capture the value of that deduction itself. Carryforward and carryback provisions give the target more opportunities to capture the deduction either through recapture of previously paid taxes or by applying the deduction in the future when the company returns to profitability.

28-8. Diversification is good for shareholders. So why shouldn't managers acquire firms in different industries to diversify a company?

Yes, diversification is good for shareholders, and they can do it efficiently themselves by purchasing shares in different companies. There is no need for managers to do this for them by creating a conglomerate through purchasing other companies at a premium over market prices. Given the premium paid in an acquisition and the differing preferences of shareholders, it cannot be efficient for managers to diversify the company rather than leaving it to shareholders to diversify their portfolio.

28-9. Your company has earnings per share of \$4. It has 1 million shares outstanding, each of which has a price of \$40. You are thinking of buying TargetCo, which has earnings per share of \$2, 1 million shares outstanding, and a price per share of \$25. You will pay for TargetCo by issuing new shares. There are no expected synergies from the transaction.

- a. If you pay no premium to buy TargetCo, what will your earnings per share be after the merger?
 - b. Suppose you offer an exchange ratio such that, at current pre-announcement share prices for both firms, the offer represents a 20% premium to buy TargetCo, what will your earnings per share be after the merger?
 - c. What explains the change in earnings per share in part (a)? Are your shareholders any better or worse off?
 - d. What will your price-earnings ratio be after the merger (if you pay no premium)? How does this compare to your P/E ratio before the merger? How does this compare to TargetCo's premerger P/E ratio?
- a. TargetCo's shares are worth \$25, and your shares are worth \$40. You will have to issue $25/40 (= 5/8)$ shares per share of TargetCo to buy it. That means that in aggregate, you have to issue $(5/8) \times 1 \text{ million} = 625,000$ new shares. After the merger, you will have a total of 1,625,000 shares outstanding (the original 1 million plus the 625,000 new shares). Your total earnings will be \$6 million. This comes from the \$4 per share $\times$ 1 million shares = \$4 million you were earning before the merger and the \$2 per share $\times$ 1 million shares = \$2 million that TargetCo was earning. Thus, your new EPS will be \$6 million/1.625 million shares = \$3.69.
 - b. A 20% premium means that you will have to pay \$30 per share to buy TargetCo ($= \$25 \times 1.20$). Thus, you will have to issue $\$30/\$40 = 0.75$ of your shares per share of TargetCo, or a total of 750,000 new shares. With total earnings of \$6 million and total shares outstanding after the merger of 1,750,000, you will have EPS of \$6 million/1.75 million shares = \$3.43.
 - c. In part (a), the change in the EPS simply came from combining the two companies, one of which was earning \$4 per share and the other was earning \$2 per share. However, you will notice that

even though TargetCo has half your EPS, it is trading for more than half your value. That is possible if TargetCo's earnings are less risky or if they are expected to grow more in the future. Thus, although your shareholders end-up with lower EPS after the transaction, they have paid a fair price, exchanging their \$4 per share before the transaction for either lower, but safer EPS after the transaction, or lower EPS that are expected to grow more in the future. Either way, focusing on EPS alone cannot tell you whether shareholders are better or worse off.

- d. If you simply combine the two companies without any indicated synergies, then the total value of the company will be \$40 million + \$25 million = \$65 million. You will have earnings totaling \$6 million, so your P/E ratio is $\$65 / \$6 = 10.83$. Your P/E ratio before the merger was $\$40 / \$4 = 10$, and TargetCo's was $\$25 / \$2 = 12.5$. You can see that by buying TargetCo for its market price and creating no synergies, the transaction simply ends-up with a company whose P/E ratio is between the P/E ratios of the two companies going into the transaction. Again, simply focusing on metrics like P/E does not tell you whether you are better or worse off. (Your P/E went-up from 10 to 10.83, but your shareholders are no better or worse off.)

28-10. If companies in the same industry as TargetCo (from Problem 9) are trading at multiples of 14 times earnings, what would be one estimate of an appropriate premium for TargetCo?



TargetCo has \$2 in earnings, so if other companies in its industry are trading at 14 times earnings, then a starting point for a valuation of TargetCo in this transaction might be \$28 per share, implying a 12% premium ($\$28 / \25).

28-11. You are invested in GreenFrame, Inc. The CEO owns 3% of GreenFrame and is considering an acquisition. If the acquisition destroys \$50 million of GreenFrame's value, but the present value of the CEO's compensation increases by \$5 million, will he be better or worse off?

The CEO will be better off. His portion of the \$50 million loss in firm value is 3%, or \$1.5 million. If his compensation increases by \$5 million, even for only one year, he will be better off by \$3.5 million.

28-12. Loki, Inc., and Thor, Inc., have entered into a stock swap merger agreement whereby Loki will pay a 40% premium over Thor's premerger price. If Thor's premerger price per share was \$40 and Loki's was \$50, what exchange ratio will Loki need to offer?

The premium is 40%, so the compensation to Thor shareholders must be 1.4(40) or \$56. Loki's shares are worth \$50, so it will need to offer $\$56 / \$50 = 1.12$ shares of Loki for every share of Thor.

28-13. The NFF Corporation has announced plans to acquire LE Corporation. NFF is trading for \$35 per share and LE is trading for \$25 per share, implying a premerger value of LE of approximately \$4 billion. If the projected synergies are \$1 billion, what is the maximum exchange ratio NFF could offer in a stock swap and still generate a positive NPV?

First, calculate the number of shares of LE:

$$\text{Number of shares} = \frac{\$4,000,000,000}{\$25} = 1,600,000,000.$$

Including synergies, LE will be worth \$4 billion + \$1 billion = \$5 billion, or \$31.25 per share (= \$5 billion / 1.6 billion).

Hence the maximum exchange ratio that NFF can offer is:

$$\text{Exchange ratio} = \frac{\$31.25}{\$35} = 0.893.$$

Thus, NFF can offer a maximum exchange ratio of 0.893 of its share in exchange of each share of LE.

- 28-14.** Let's reconsider part (b) of Problem 9. The actual premium that your company will pay for TargetCo will not be 20%, because on the announcement the target price will go up and your price will go down to reflect the fact that you are willing to pay a premium for TargetCo. Assume that the takeover will occur with certainty and all market participants know this on the announcement of the takeover.



- a. What is the price per share of the combined corporation immediately after the merger is completed?
 - b. What is the price of your company immediately after the announcement?
 - c. What is the price of TargetCo immediately after the announcement?
 - d. What is the actual premium your company will pay?
- a. Since 0.75 million new shares will be issued, the share price will be $(40 + 25)/1.75 = \$37.143$
 - b. Same as the price after the merger, \$37.143.
 - c. Since TargetCo shareholders will receive $0.75 \times 37.143 = 27.86$ million, and there are 1 million shareholders the share price will be \$27.86.
 - d. $27.86/25 - 1 = 11.43\%$

- 28-15.** ABC has 1 million shares outstanding, each of which has a price of \$20. It has made a takeover offer of XYZ Corporation which has 1 million shares outstanding, and a price per share of \$2.50. Assume that the takeover will occur with certainty and all market participants know this. Furthermore, there are no synergies to merging the two firms.

- a. Assume ABC made a cash offer to purchase XYZ for \$3 million. What happens to the price of ABC and XYZ on the announcement? What premium over the current market price does this offer represent?
 - b. Assume ABC makes a stock offer with an exchange ratio of 0.15. What happens to the price of ABC and XYZ this time? What premium over the current market price does this offer represent?
 - c. At current market prices, both offers are offers to purchase XYZ for \$3 million. Does that mean that your answers to parts (a) and (b) must be identical? Explain.
- a. Price of XYZ = \$3. Premium = 20%
Price of ABC = $20 - \text{premium} \times 2.5 = \19.50
 - b. ABC price = price of combine entity = $22.50/1.15 = \$19.5652$.
XYZ price = amount shareholders will receive = $0.15 \times \$19.5652 = \2.9345
Premium = $2.9345/2.5 = 17.4\%$ premium
 - c. No, the premium in the stock offer is lower because market prices change to reflect the fact that ABC shareholders are giving XYZ shareholders money because they are paying a premium. The part (b) announcement means XYZ stock goes up and ABC stock goes down, which lowers the premium relative to the cash offer.

-  **28-16.** BAD Company's stock price is \$20, and the firm has 2 million shares outstanding. You believe you can increase the company's value if you buy it and replace the management. Assume that BAD has a poison pill with a 20% trigger. If it is triggered, all BAD's shareholders—other than the acquirer—will be able to buy one new share in BAD for each share they own at a 50% discount. Assume that the price remains at \$20 while you are acquiring your shares. If BAD's management decides to resist your buyout attempt, and you cross the 20% threshold of ownership:

- a. How many new shares will be issued and at what price?
 - b. What will happen to your percentage ownership of BAD?
 - c. What will happen to the price of your shares of BAD?
 - d. Do you lose or gain from triggering the poison pill? If you lose, where does the loss go (who benefits)? If you gain, from where does the gain come (who loses)?
- a. If you trigger the poison pill, then you own 20% of the company, or 400,000 shares ($= 20\% \times 2,000,000$ shares). When you trigger the poison pill, every other shareholder will buy a new share for every share they hold, so 1,600,000 shares ($= 2,000,000 - 400,000$) will be issued. These shares will be issued at \$10, which is 50% of the price immediately before triggering the poison pill (which we assume stays constant at \$20).
 - b. After the new 1,600,000 shares are issued, there will be a total of 3,600,000 shares ($= 2,000,000 + 1,600,000$). You will own 400,000 of them, so your participation will be 11.11% ($= 400,000 / 3,600,000$).
 - c. When the poison pill is triggered, the market value of the firm will increase to \$56 million [$= (\$20 \times 2,000,000) + (\$10 \times 1,600,000)$]. The new stock price will be \$15.56 ($= \$56 \text{ million} / 3,600,000$).
 - d. You lose from triggering the poison pill (you bought shares at \$20 that are now worth \$15.56). Every other shareholder in the target firm gains (they end with \$31.12 ($= \15.56×2) worth of shares for which they only paid \$30 ($= \$20 + \10)).

28-17. How does a toehold help overcome the free rider problem?

Since the acquirer gains the full amount of the value improvement on the shares acquired as a toehold, a toehold provides an incentive to undertake the acquisition, even if the acquirer must pay a price equal to the with-improvement value for the rest of the shares.

- 28-18.** You work for a leveraged buyout firm and are evaluating a potential buyout of UnderWater Company. UnderWater's stock price is \$20, and it has 2 million shares outstanding. You believe that if you buy the company and replace its management, its value will increase by 40%. You are planning on doing a leveraged buyout of UnderWater, and will offer \$25 per share for control of the company.

- a. Assuming you get 50% control, what will happen to the price of non-tendered shares?
 - b. Given the answer in part (a), will shareholders tender their shares, not tender their shares, or be indifferent?
 - c. What will your gain from the transaction be?
- a. The value should reflect the expected improvement that you will make by replacing the management, so the value of the company will be \$40 million plus 40% = \$56 million. If you buy 50% of the shares for \$25 apiece, you will buy 1 million shares, paying \$25 million. However, you will borrow this money, pledging the shares as collateral and then assign the loan to the company once you have control. This means that the new value of the equity will be \$56 million – \$25 million in debt = \$31 million. With 2 million shares outstanding, the price of the equity will drop to \$15.50.

- b. Since the price of the shares will drop from \$20 to \$15.50 after the tender offer, everyone will want to tender their shares for \$25.
- c. Assuming that everyone tenders their shares and you buy them all at \$25 apiece, you will pay \$50 million to acquire the company and it will be worth \$56 million. You will own 100% of the equity, which will be \$56 million – \$50 million loan to buy the shares = \$6 million.

Chapter 29

Corporate Governance

29-1. What inherent characteristic of corporations creates the need for a system of checks on manager behavior?

The corporation allows for the separation of management and ownership. Thus, those who control the operations of the corporation and how its money is spent are not the same who have invested in the corporation. This creates a clear conflict of interest and this conflict between the investors and managers creates the need for investors to devise a system of checks on managers—the system of corporate governance.

29-2. What are some examples of agency problems?

Examples of agency problems are excessive perquisite consumption (more company jets/company jet travel than needed, nicer office than necessary, etc.). Others are value-destroying acquisitions that nonetheless increase the pecuniary or non-pecuniary benefits to the CEO on net.

29-3. What are the advantages and disadvantages of the corporate organizational structure?

The corporate organizational form allows those who have the capital to fund an enterprise to be different from those who have the expertise to manage the enterprise. This critical separation allows a wide class of investors to share the risk of the enterprise. However, as mentioned in the answer to question 1, this separation comes at a cost—the managers will act in their own best interests, not in the best interests of the shareholders who own the firm.

29-4. What is the role of the board of directors in corporate governance?

The board of directors is the primary internal control mechanism and the first line of defense to prevent fraud, agency conflicts, and mismanagement. The board is empowered to hire and fire managers, set compensation contracts, approve major investment decisions, etc.

29-5. How does a board become captured by a CEO?

Over time, a long-standing CEO can maneuver the nomination process so that his or her associates and friends are nominated to the board. Additionally, board members representing customers, suppliers, or others who have the potential for business relationships with the firm will sometimes compromise their fiduciary duty in order to keep the management of the firm happy. This desire to keep the CEO happy or a reluctance to challenge him or her interferes with the board's primary function of monitoring the management.

29-6. What role do security analysts play in monitoring?

By knowing a company and its industry as well as possible, they are in a position to uncover irregularities. They also participate in earnings calls with the CEO and CFO, asking difficult and probing questions.

29-7. How are lenders part of corporate governance?

Lenders are exposed to the firm as creditors and so are motivated to carefully monitor the firm. They often include covenants in their loans that require the company to maintain certain profitability and liquidity levels. Breaking these covenants can be a warning sign of deeper trouble.

29-8. What is a whistleblower?

Whistleblowers can be anyone but are typically employees who uncover outright wrongdoing and “blow the whistle,” on the fraud by reporting it to the authorities.

29-9. What are the advantages and disadvantages of increasing the options granted to CEOs?

The advantages are that, since options increase in value when the firm’s stock price increases, the CEO’s wealth and incentives will be more closely tied to the shareholders’ wealth. The disadvantage is that option grants can increase a CEO’s incentives to game the system by timing the release of information to fit the option granting schedule or to artificially smooth earnings.

29-10. Is it necessarily true that increasing managerial ownership stakes will improve firm performance?

No. There are two counter arguments here. First, as Demsetz and Lehn (1985) argue, there is no reason to expect a simple relation between ownership and performance. There are many dimensions to the corporate governance system and a one-size-fits-all approach is too simplistic; the correct ownership level for one firm may not be the correct level for another. Second, some studies have shown a non-linear relationship between firm valuation and ownership—specifically that increasing ownership is good at first, but that in a certain range, managers can use their ownership level to partially block efforts to constrain them, even though they still own a minority of the shares. In this “entrenching” range, increasing ownership could reduce performance.

29-11. How can proxy contests be used to overcome a captured board?

Proxy contests are simply contested elections for directors. In a proxy contest, two competing slates of directors rather than just one slate are proposed by the company. If a board has become captured or unresponsive to shareholder demands, shareholders can put their own slate of new directors up for election. If the dissident slate wins, then shareholders will have succeeded in placing new directors, presumably not beholden to the CEO, on the board.

29-12. What is a say-on-pay vote?

A say-on-pay vote is a non-binding vote whereby the shareholders indicate whether they approve of an executive’s pay package or not.

29-13. What are a board’s options when confronted with dissident shareholders?

When confronted with a dissident shareholder, a board can:

- Ignore the shareholder, which will result in either the shareholder going away or launching a proxy fight, in which case the board will need to expend resources in an attempt to convince shareholders not to side with the dissident; or
- Negotiate with the dissident shareholder to come to a solution on which the board and the shareholder can agree.

29-14. What is the essential trade-off faced by government in designing regulation of public firms?

The government should be trying to maximize societal welfare. Thus, in designing regulation, it must trade off the effects of direct and indirect enforcement, compliance and other costs associated with regulation against the aggregate benefits that accrue to shareholders and the economy as a whole.

29-15. Many of the provisions of the Sarbanes-Oxley Act of 2002 were aimed at auditors. How does this affect corporate governance?

Auditors are important to corporate governance. Auditors ensure that the financial picture of the firm presented to outside investors is clear and accurate. Part of the role of auditors is to detect financial fraud before it threatens the viability of the firm. Sarbanes-Oxley included measures designed to reduce conflicts of interest among auditors and to increase the penalties for fraud.

29-16. What are the costs and benefits of prohibiting insider trading?

Trading is how prices come to reflect all material information about a company's prospects. By restricting a set of investors from trading, we decrease the efficiency of the prices because it will take longer for the prices to reflect that private information. We rely on efficient prices to make sure that capital is allocated to its best use. While that is a cost of prohibiting insider trading, there is also a benefit. In order for a capital market to fulfill its function, uninformed investors must be willing to invest their money—providing liquidity and lowering the cost of capital. If investors thought that the stock market was just a fools' game where they lost to insiders, they would be unwilling to invest or would price their expected loss into their required return. This increases the cost of capital for companies and slows economic growth.

29-17. How do the laws on insider trading differ for merger- versus non-merger-related trading?

The laws are much stricter for merger-related trading. Anyone who has information about a pending merger is restricted from trading. Non-merger restrictions depend on the source of the material non-public information. If the source violated a fiduciary duty to the shareholders, then the trading is prohibited.

29-18. Are the rights of shareholders better protected in the United States or in France?

They are better protected in the United States. The U.S. legal system is based on British common law, which offers considerably more protection to minority shareholders than French civil law does.

29-19. How can a controlling family use a pyramidal control structure to benefit itself at the expense of other shareholders?

Because pyramidal structures allow a controlling family to control firms in which they have little actual cash flow rights, the family can use their control to move profits away from firms where they get a small percentage of the cash flows to firms in which they can claim a larger fraction of the cash flows. For example, they can have one firm sell to another at a reduced price.

Chapter 30

Risk Management

- 30-1.** The William Companies (WMB) owns and operates natural gas pipelines that deliver 12% of the natural gas consumed in the United States. WMB is concerned that a major hurricane could disrupt its Gulfstream pipeline, which runs 691 miles through the Gulf of Mexico. In the event of a disruption, the firm anticipates a loss of profits of \$65 million. Suppose the likelihood of a disruption is 3% per year, and the beta associated with such a loss is -0.25 . If the risk-free interest rate is 5% and the expected return of the market is 10%, what is the actuarially fair insurance premium?

From the SML, the required return for a beta of -0.25 is $r_L = 5\% - 0.25(10\% - 5\%) = 3.75\%$. From Eq. 30.1:

$$\text{Premium} = \frac{3\% \times \$65 \text{ million}}{1.0375} = \$1.88 \text{ million.}$$

- 30-2.** Genentech's main facility is located in South San Francisco. Suppose that Genentech would experience a direct loss of \$450 million in the event of a major earthquake that disrupted its operations. The chance of such an earthquake is 2% per year, with a beta of -0.5 .

- If the risk-free interest rate is 5% and the expected return of the market is 10%, what is the actuarially fair insurance premium required to cover Genentech's loss?
- Suppose the insurance company raises the premium by an additional 15% over the amount calculated in part (a) to cover its administrative and overhead costs. What amount of financial distress or issuance costs would Genentech have to suffer if it were not insured to justify purchasing the insurance?

- From the SML, the required return for a beta of -0.5 is $r_L = 5\% - 0.5(10\% - 5\%) = 2.5\%$. From Eq. 30.1:

$$\text{Premium} = \frac{2\% \times \$450 \text{ million}}{1.025} = \$8.78 \text{ million.}$$

- With 15% overhead costs, the insurance premium will be $\$8.78 \times (1.15) = \10.098 million. Buying insurance is positive NPV for Genentech if it experiences distress or issuance costs equal to 15% of the amount of the loss. That is, it must experience distress or issuance costs of $15\% \times 450 = \$67.5$ million in the event of a loss. In that case:

$$\text{NPV}(\text{buy insurance}) = -10.098 + \frac{2\% \times \$ (450 + 67.5) \text{ million}}{1.025} = 0.$$

- 30-3.** Your firm imports manufactured goods from China. You are worried that U.S.–China trade negotiations could break down next year, leading to a moratorium on imports. In the event of a moratorium, your firm expects its operating profits to decline substantially and its marginal tax rate to fall from its current level of 40% to 10%.

An insurance firm has agreed to write a trade insurance policy that will pay \$500,000 in the event of an import moratorium. The chance of a moratorium is estimated to be 10%, with a beta of -1.5 . Suppose the risk-free interest rate is 5% and the expected return of the market is 10%.

- a. What is the actuarially fair premium for this insurance?
- b. What is the NPV of purchasing this insurance for your firm? What is the source of this gain?

- a. From the SML, the required return for a beta of -1.5 is $r_L = 5\% - 1.5(10\% - 5\%) = -2.5\%$. From Eq. 30.1:

$$\text{Premium} = \frac{10\% \times \$500,000}{1 - 0.025} = \$51,282.$$

- b. If we consider after-tax cash flows:

$$\text{NPV} = -\$51,282 \times (1 - 0.40) + \frac{10\% \times \$500,000 \times (1 - 0.10)}{1 - 0.025} = \$15,385.$$

The gain arises because the firm pays for the insurance when its tax rate is high, but receives the insurance payment when its tax rate is low.

- 30-4.** Your firm faces a 9% chance of a potential loss of \$10 million next year. If your firm implements new policies, it can reduce the chance of this loss to 4%, but these new policies have an upfront cost of \$100,000. Suppose the beta of the loss is 0, and the risk-free interest rate is 5%.

- a. If the firm is uninsured, what is the NPV of implementing the new policies?
- b. If the firm is fully insured, what is the NPV of implementing the new policies?
- c. Given your answer to part (b), what is the actuarially fair cost of full insurance?
- d. What is the minimum-size deductible that would leave your firm with an incentive to implement the new policies?
- e. What is the actuarially fair price of an insurance policy with the deductible in part (d)?

- a. New policies reduce the chance of loss by $9\% - 4\% = 5\%$, for an expected savings of $5\% \times \$10 \text{ million} = \$500,000$. Therefore, the NPV is

$$\text{NPV} = -\$100,000 + \$500,000/1.05 = \$376,190.$$

- b. If the firm is fully insured, then it will not experience a loss. Thus, there is no benefit to the firm from the new policies. Therefore, $\text{NPV} = -\$100,000$.
- c. If the firm insures fully, it will not have an incentive to implement the new safety policies. Therefore, the insurance company will expect a 9% chance of loss. Therefore, the actuarially fair premium would be

$$\text{Premium} = 9\% \times \$10 \text{ million}/1.05 = \$857,143.$$

- d. If the insurance policy has a deductible, then the firm will benefit from the new policies because it will avoid a loss, and therefore avoid paying the deductible, 5% of the time. Let D be the amount of the deductible. Then the NPV of the new policies is $\text{NPV} = -\$100,000 + 5\%(D)/1.05$.

Setting the NPV to 0 and solving for D we get $D = \$2.1 \text{ million}$.

- e. With a deductible of 2.1 million, the insurance company can expect the firm to implement the new policies. Therefore, it can expect a 4% chance of loss. In the event of a loss, the insurance will pay $(10 - 2.1) = \$7.9 \text{ million}$. Therefore:

$$\text{Premium} = 4\% \times \$7.9 \text{ million}/1.05 = \$300,952.$$

Aside: With this policy, the firm will pay \$300,952 for insurance, \$100,000 to implement the new policies, and $4\% \times \$2.1 \text{ million} = \$84,000$ in expected deductibles. Thus, the firm will pay

$\$300,952 + 100,000 + 84,000 = \$484,952$ in total, which is much less than the amount it would pay for full insurance in (c).

30-5. BHP Billiton is the world's largest mining firm. BHP expects to produce 2 billion pounds of copper next year, with a production cost of \$0.90 per pound.

- What will be BHP's operating profit from copper next year if the price of copper is \$1.25, \$1.50, or \$1.75 per pound, and the firm plans to sell all of its copper next year at the going price?
- What will be BHP's operating profit from copper next year if the firm enters into a contract to supply copper to end users at an average price of \$1.45 per pound?
- What will be BHP's operating profit from copper next year if copper prices are described as in part (a), and the firm enters into supply contracts as in part (b) for only 50% of its total output?
- Describe situations for which each of the strategies in parts (a), (b), and (c) might be optimal.

- Operating profit = 2 billion pounds $\times$ (Price per pound – \$0.90/lb). Thus:

Price (\$/lb)	1.25	1.50	1.75
Operating Profit (\$ billion)	0.70	1.20	1.70

- In this case, they will sell for the contract price of \$1.45/lb, no matter what the spot price of copper is next year:

Contract price (\$/lb)	1.45
Operating Profit (\$ billion)	1.10

That is, Oper Profit = $2 \times (1.45 - 0.90) = \1.10 billion.

- In this case, Operating Profit = $1 \times (1.45 - 0.90) + 1 \times (\text{Price} - 0.90)$. Therefore:

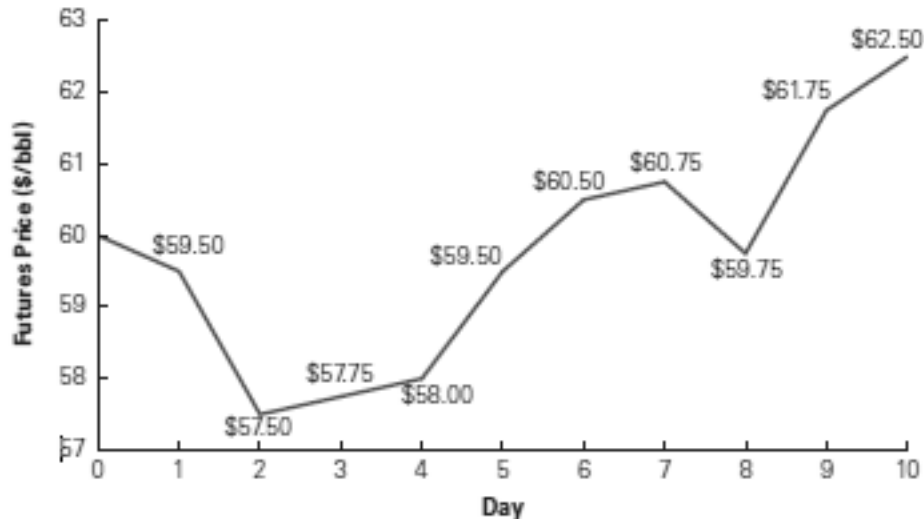
Contract price (\$/lb)	1.45		
Contract Amount	1.00	billion pounds	
Spot Price (\$/lb)	1.25	1.50	1.75
Operating Profit (\$ billion)	0.90	1.15	1.40

- Strategy (a) could be optimal if the firm is sufficiently profitable that it will not be distressed even if the copper price next year is low. Equity holders will in this case bear the risk of copper price fluctuations, and there is no gain from hedging the risk. It could also be optimal if the firm is currently in or near financial distress. Then by not hedging, the firm increases its risk. Equity holders can benefit if the price of copper is high, but debt holders suffer if the price is low. (Recall the discussion in Chapter 16 regarding equity holders incentive to increase risk when the firm is in or near financial distress.)

Strategy (b) could be optimal if the firm is not in distress now, but would be if the price of copper next year is low and it does not hedge. Then, by locking in the price it will receive at \$1.45/lb, the firm can avoid financial distress costs next year.

Strategy (c) could be optimal if the firm would risk distress with operating profits of \$0.7 billion from copper but would not with operating profits of \$0.9 billion. In that case, the firm can partially hedge and avoid any risk of financial distress.

- 30-6.  Your utility company will need to buy 100,000 barrels of oil in 10 days time, and it is worried about fuel costs. Suppose you go long 100 oil futures contracts, each for 1000 barrels of oil, at the current futures price of \$60 per barrel. Suppose futures prices change each day as follows:



- What is the mark-to-market profit or loss (in dollars) that you will have on each date?
 - What is your total profit or loss after 10 days? Have you been protected against a rise in oil prices?
 - What is the largest cumulative loss you will experience over the 10-day period? In what case might this be a problem?
- a. You have gone long $100 \times 1000 = 100,000$ barrels of oil. Therefore, the mark-to-market profit or loss will equal 100,000 times the change in the futures price each day.

Day	Price	Price Change	Profit/Loss
0	\$ 60.00		
1	\$ 59.50	(\$0.50)	(\$50,000)
2	\$ 57.50	(\$2.00)	(\$200,000)
3	\$ 57.75	\$0.25	\$25,000
4	\$ 58.00	\$0.25	\$25,000
5	\$ 59.50	\$1.50	\$150,000
6	\$ 60.50	\$1.00	\$100,000
7	\$ 60.75	\$0.25	\$25,000
8	\$ 59.75	(\$1.00)	(\$100,000)
9	\$ 61.75	\$2.00	\$200,000
10	\$ 62.50	\$0.75	\$75,000

- Summing the daily profit/loss amounts, the total is a gain of \$250,000. This gain offsets your increase in cost from the overall \$2.50 increase in oil prices over the 10 days, which increases your total cost of oil by $100,000 \times \$2.50 = \$250,000$.
- After the second day, you have lost a total of \$250,000. This loss could be a problem if you do not have sufficient resources to cover the loss. In that case, your position would have been liquidated on day 2, and you would have been stuck with the loss and had to pay the higher cost of oil on day 10.

- 30-7.** Suppose Starbucks consumes 100 million pounds of coffee beans per year. As the price of coffee rises, Starbucks expects to pass along 60% of the cost to its customers through higher prices per cup of coffee. To hedge its profits from fluctuations in coffee prices, Starbucks should lock in the price of how many pounds of coffee beans using supply contracts?

If the price of coffee goes up by \$0.01 per pound, Starbucks' cost of coffee will go up by $\$0.01 \times 100$ million = \$1 million. But because it can charge higher prices, its revenues will go up by $60\% \times \$1$ million = \$0.6 million. To hedge this risk, Starbucks should lock in the price for 40 million pounds of coffee, so that it will only suffer an increase in cost for the remaining 60 million pounds of coffee.

-  **30-8.** Your start-up company has negotiated a contract to provide a database installation for a manufacturing company in Poland. That firm has agreed to pay you \$100,000 in three months time when the installation will occur. However, it insists on paying in Polish zloty (PLN). You don't want to lose the deal (the company is your first client!), but are worried about the exchange rate risk. In particular, you are worried the zloty could depreciate relative to the dollar. You contact Fortis Bank in Poland to see if you can lock in an exchange rate for the zloty in advance.

- a. You find the following table posted on the bank's Web site, showing zloty per dollar, per euro, and per British pound:

	1 week	2 weeks	1 month	2 months	3 months
USD					
purchase	3.1433	3.1429	3.1419	3.1390	3.1361
sale	3.1764	3.1761	3.1755	3.1735	3.1712
EUR					
purchase	3.7804	3.7814	3.7836	3.7871	3.7906
sale	3.8214	3.8226	3.8254	3.8298	3.8342
GBP					
purchase	5.5131	5.5131	5.5112	5.5078	5.5048
sale	5.5750	5.5750	5.5735	5.5705	5.5681

What exchange rate could you lock in for the zloty in three months? How many zloty should you demand in the contract to receive \$100,000?

- b. Given the bank forward rates in part (a), were short-term interest rates higher or lower in Poland than in the United States at the time? How did Polish rates compare to euro or pound rates? Explain.
- a. Check out the Web site for Fortis Bank (www.fortisbank.com.pl). In the upper left of the page you can choose "English" from the menu, and then "currency exch." There you will be able to find exchange rates for currency forward contracts. Find the rates that applied on Mar 3, 2006 at 4:15pm. What exchange rate could you lock-in for zloty in three months? How many zloty should you demand in the contract in order to receive \$100,000?

Here is the table from the Web site, showing zloty per \$, per euro, and per British pound:

	1 week	2 weeks	1 month	2 months	3 months
USD 					
purchase	3.1433	3.1429	3.1419	3.1390	3.1361
sale	3.1764	3.1761	3.1755	3.1735	3.1712
EUR 					
purchase	3.7804	3.7814	3.7836	3.7871	3.7906
sale	3.8214	3.8226	3.8254	3.8298	3.8342
GBP 					
purchase	5.5131	5.5131	5.5112	5.5078	5.5048
sale	5.5750	5.5750	5.5735	5.5705	5.5681

Thus, you could lock in an exchange rate of 3.1712 zloty per U.S. dollar in three months time through a forward contract with the bank. (Note that when converting zloty to \$, you pay the higher rate.) In order to receive \$100,000, you would therefore need to write the contract for $100,000 \times 3.1712 = 317,120$ zloty.

- b. The forward rates show that fewer zloty per \$ are needed for longer maturities. From Eq 30.3, in terms of zloty per \$,

$$F_T = S \times \frac{(1 + r_z)^T}{(1 + r_s)^T}.$$

Thus, the zloty interest rate is below the \$ interest rate.

In general, from Eq. 30.2, we can tell which rate is higher by seeing if the forward rate is above or below the spot rate. From the table, the forward rates appear to be lower for the British pound, so the pound interest rate was higher at the time of these quotes (March 2006). The euro forward rates are higher than the spot rates, however, suggesting that Polish interest rates were higher than those for the euro.

30-9. You are a broker for frozen seafood products for Choyce Products. You just signed a deal with a Belgian distributor. Under the terms of the contract, in one year you will deliver 4000 kilograms of frozen king crab for 100,000 euros. Your cost for obtaining the king crab is \$110,000. All cash flows occur in exactly one year.



- Plot your profits in one year from the contract as a function of the exchange rate in one year, for exchange rates from \$0.75/€ to \$1.50/€. Label this line “Unhedged Profits.”
- Suppose the one-year forward exchange rate is \$1.25/€. Suppose you enter into a forward contract to sell the euros you will receive at this rate. In the figure from part (a), plot your combined profits from the crab contract and the forward contract as a function of the exchange rate in one year. Label this line “Forward Hedge.”
- Suppose that instead of using a forward contract, you consider using options. A one-year call option to buy euros at a strike price of \$1.25/€ is trading for \$0.10/€. Similarly a one year put option to sell euros at a strike price of \$1.25/€ is trading for \$0.10/€. To hedge the risk of your profits, should you buy or sell the call or the put?

- d. In the figure from parts (a) and (b), plot your “all in” profits using the option hedge (combined profits of crab contract, option contract, and option price) as a function of the exchange rate in one year. Label this line “Option Hedge.” (Note : You can ignore the effect of interest on the option price.)
- e. Suppose that by the end of the year, a trade war erupts, leading to a European embargo on U.S. food products. As a result, your deal is cancelled, and you don’t receive the euros or incur the costs of procuring the crab. However, you still have the profits (or losses) associated with your forward or options contract. In a new figure, plot the profits associated with the forward hedge and the options hedge (labeling each line). When there is a risk of cancellation, which type of hedge has the least downside risk? Explain briefly.
- a. Unhedged profit = (100,000 euros) $\times$ (S_1 \$/euro) – 110,000. See figure below.
- b. Forward Hedged profit = (100,000 euros) $\times$ (1.25 \$/euro) – 110,000 = \$15,000. See figure below.
- c. You want to sell euros in exchange for dollars. Thus, buying put options will protect the price at which you can sell euros.
- d. Buying put options for 100,000 euros costs $100,000 \times \$0.10 = \$10,000$.

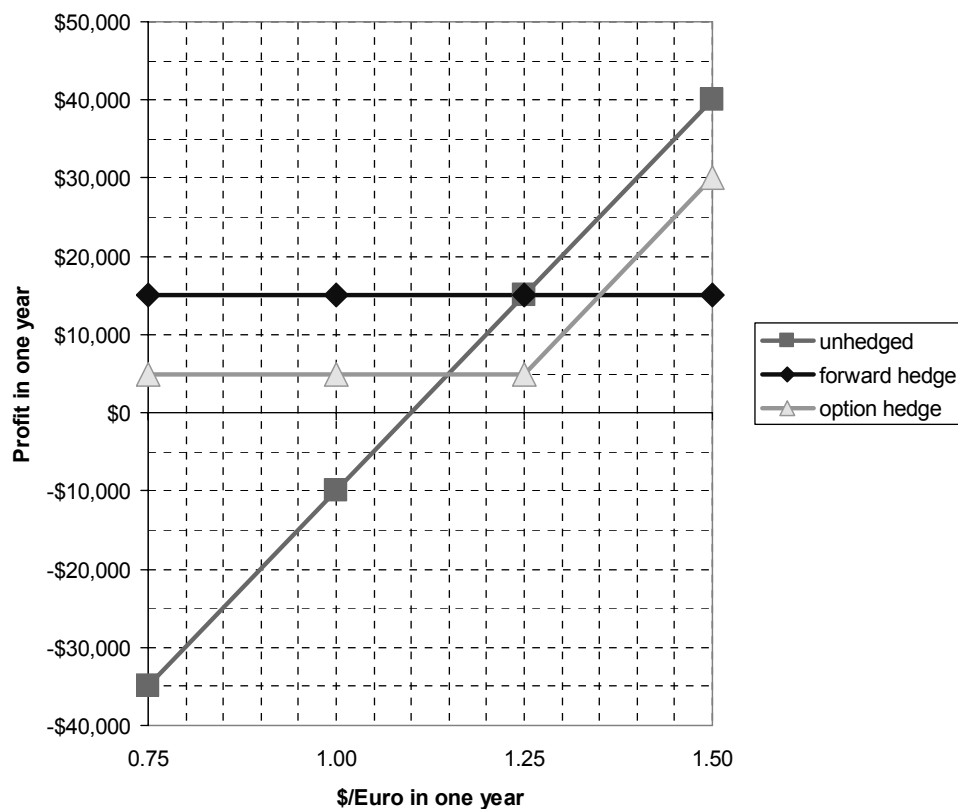
The put allows you to sell the euros for a minimum of \$1.25/euro. Therefore,

All-in Option hedged profit

$$= (100,000 \text{ euros}) \times (\max[1.25, S_1] \text{ $/euro}) - 110,000 - 10,000$$

$$= \max[\$5,000, (100,000 \text{ euros}) \times (S_1 \text{ $/euro}) - 120,000].$$

See figure below.



- e. For the forward hedge, you receive a payoff from the forward if the value of the euro declines. If the euro appreciates, you have a loss on your forward position. The profit is:

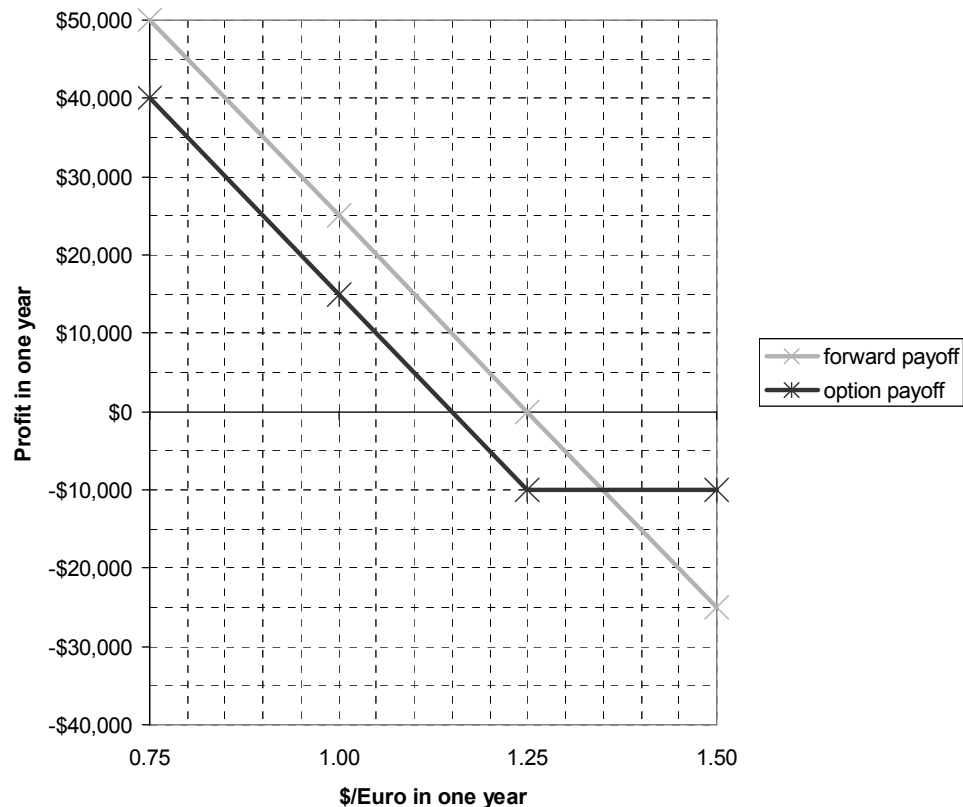
$$(100,000 \text{ euros}) \times (1.25 - S_1 \text{ \$/euro}).$$

If the euro appreciates significantly, the loss from the forward hedge can be very large. (See figure below.)

For the option hedge, you receive a payoff from the put if the value of the euro declines. If the euro appreciates, the put is worthless (and are out the original purchase price of the puts). The profit is:

$$\max[0, (100,000 \text{ euros}) \times (1.25 - S_1 \text{ \$/euro})] - 10,000.$$

With the puts, the maximum loss is their initial cost of \$10,000. (See figure below.)



From the picture, an advantage of hedging with options is the limited downside risk in the event of cancellation.

- 30-10.** Suppose the current exchange rate is \$1.80/£, the interest rate in the United States is 5.25%, the interest rate in the United Kingdom is 4%, and the volatility of the \$/£ exchange rate is 10%. Use the Black-Scholes formula to determine the price of a six-month European call option on the British pound with a strike price of \$1.80/£.

The inputs are S = spot exchange rate = 1.80, K = strike price = 1.80, T = 0.5, r_s = 5.25%, r_f = 4.0%, σ = volatility = 10%. From Eq. 30.3,

$$F_T = S(1 + r_s)^T / (1 + r_f)^T = 1.80(1.0525)^{0.5} / (1.04)^{0.5} = \$1.8108/\text{£}.$$

Therefore, from Eq. 30.5

$$d_1 = \frac{\ln(1.8108/1.80)}{10\%\sqrt{0.5}} + \frac{10\%\sqrt{0.5}}{2} = 0.120, \text{ and } d_2 = d_1 - 10\%\sqrt{0.5} = 0.049$$

and so $N(d_1) = 0.548$ and $N(d_2) = 0.520$.

From Eq. 30.4, $C = (1.80 / (1.04)^{0.5}) \times (0.548) - (1.80 / (1.0525)^{0.5}) \times (0.520) = \0.0549

Thus, the call option price is \$0.0549/£.

- 30-11. Assume each of the following securities has the same yield-to-maturity: a five-year, zero-coupon bond; a nine-year, zero-coupon bond; a five-year annuity; and a nine-year annuity. Rank these securities from lowest to highest duration.**

The duration of a security is equal to the weighted-average maturity of its cash flows (Eq. 30.6). Thus, the duration of a five-year zero coupon bond is five years, and the duration of the nine-year zero-coupon bond is nine years (see Ex 30.11).

We cannot determine the durations of the annuities exactly without knowing the current interest rate. But because the cash flows of an annuity are equal and at regular intervals, the duration of an annuity must be less than its average maturity (because weighting by present values will put less weight on later cash flows). Thus, the five-year annuity has a duration of less than $(1 + 2 + 3 + 4 + 5) / 5 = 3$ years, and the nine-year annuity has a duration of less than $(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9) / 9 = 5$ years. The ranking is therefore:

five-year annuity, nine-year annuity, five-year zero, nine-year zero.

- 30-12. You have been hired as a risk manager for Acorn Savings and Loan. Currently, Acorn's balance sheet is as follows (in millions of dollars):**



Assets		Liabilities	
Cash Reserves	50	Checking and Savings	80
Auto Loans	100	Certificates of Deposit	100
Mortgages	150	Long-Term Financing	100
Total Assets	300	Total Liabilities	280
		Owner's Equity	20
		Total Liabilities and Equity	300

When you analyze the duration of loans, find that the duration of the auto loans is two years, while the mortgages have a duration of seven years. Both the cash reserves and the checking and savings accounts have a zero duration. The CDs have a duration of two years and the long-term financing has a 10-year duration.

- What is the duration of Acorn's equity?
- Suppose Acorn experiences a rash of mortgage prepayments, reducing the size of the mortgage portfolio from \$150 million to \$100 million, and increasing cash reserves to \$100 million. What is the duration of Acorn's equity now? If interest rates are currently 4% but fall to 3%, estimate the approximate change in the value of Acorn's equity.
- Suppose that after the prepayments in part (b), but before a change in interest rates, Acorn considers managing its risk by selling mortgages and/or buying 10-year Treasury STRIPS (zero-coupon bonds). How many should the firm buy or sell to eliminate its current interest rate risk?

- a. From Eq. 30.8,

$$\text{Asset Duration} = \frac{50}{300}(0\text{yrs}) + \frac{100}{300}(2\text{yrs}) + \frac{150}{300}(7\text{yrs}) = 4.17\text{yrs}$$

$$\text{Liability Duration} = \frac{80}{280}(0\text{yrs}) + \frac{100}{280}(2\text{yrs}) + \frac{100}{280}(10\text{yrs}) = 4.29\text{yrs}$$

From Eq. 30.9,

$$\text{Equity Duration} = \frac{300}{20}(4.17\text{yrs}) - \frac{280}{20}(4.29\text{yrs}) = 2.49\text{yrs}$$

- b. $\text{Asset Duration} = \frac{100}{300}(0\text{yrs}) + \frac{100}{300}(2\text{yrs}) + \frac{100}{300}(7\text{yrs}) = 3.0\text{yrs}$

$$\text{Equity Duration} = \frac{300}{20}(3.0\text{yrs}) - \frac{280}{20}(4.29\text{yrs}) = -15.0\text{yrs}$$

Therefore, if interest rates drop by 1%, we would expect the value of Acorn's equity to drop by about 15% (or more precisely from Eq. 30.7, $15\%/1.04$, which equals 14.4%).

- c. Acorn would like to increase the duration of its assets, so it should use cash to buy long-term bonds. Because 10-year STRIPS (zero-coupon bonds) have a 10-year duration, we can use Eq. 30.10:

$$\text{Amount} = \frac{\overbrace{\text{change in equity duration}}^{(15\text{yrs})} \times \overbrace{\text{equity value}}^{20}}{\underbrace{10\text{yrs}}_{\text{duration of STRIPS (vs. cash)}}} = 30$$

That is, we should buy \$30 million worth of 10-year STRIPS.

- 30-13.** The Citrix Fund has invested in a portfolio of government bonds that has a current market value of \$44.8 million. The duration of this portfolio of bonds is 13.5 years. The fund has borrowed to purchase these bonds, and the current value of its liabilities (i.e., the current value of the bonds it has issued) is \$39.2 million. The duration of these liabilities is four years. The equity in the Citrix Fund (or its net worth) is obviously \$5.6 million. The market-value balance sheet below summarizes this information:



Assets		Liabilities (Debt) and Equity	
Portfolio of Government Bonds (duration = 13.5)	\$44,800,000	Short- and Long-Term Debt (duration = 4.0)	\$39,200,000
		Equity	\$5,600,000
Total	\$44,800,000	Total	\$44,800,000

Assume that the current yield curve is flat at 5.5%. You have been hired by the board of directors to evaluate the risk of this fund.

- Consider the effect of a surprise increase in interest rates, such that the yields rise by 50 basis points (i.e., the yield curve is now flat at 6%). What would happen to the value of the assets in the Citrix Fund? What would happen to the value of the liabilities? What can you conclude about the change in the value of the equity under these conditions?
- What is the initial duration of the Citrix Fund (i.e., the duration of the equity)?
- As a result of your analysis, the board of directors fires the current manager of the fund. You are hired and given the objective of minimizing the fund's exposure to interest rate

fluctuations. You are instructed to do so by liquidating a portion of the fund's assets and reinvesting the proceeds in short-term Treasury bills and notes with an average duration of two years. How many dollars do you need to liquidate and reinvest to minimize the fund's interest rate sensitivity?

- d. Rather than immunizing the fund using the strategy in part (c), you consider using a swap contract. If the duration of a 10-year, fixed-coupon bond is seven years, what is the notational amount of the swap you should enter into? Should you receive or pay the fixed rate portion of the swap?

- a. The duration of the assets is 13.5 years. To estimate the effect of a parallel interest-rate of 0.5%, we use the duration formula:

$$\% \text{change} = -\text{Duration} \times \frac{\varepsilon}{1+r} = -13.5 \times \frac{0.50\%}{1.055} = -6.40\%,$$

or a drop in value of $6.40\% \times \$44.8 \text{ million} = \2.87 million .

Similarly, for the liabilities:

$$\% \text{change} = -\text{Duration} \times \frac{\varepsilon}{1+r} = -4.0 \times \frac{0.50\%}{1.055} = -1.90\%,$$

or a drop in value of $1.90\% \times \$39.2 \text{ million} = \0.74 million .

As a result, the value of equity will decline by about $2.87 - 0.74 = \$2.13 \text{ million}$ (a loss of 38% of its value!).

- b. From Eq. 30.9:

$$\text{Equity Duration} = \frac{44.8}{5.6}(13.5\text{yrs}) - \frac{39.2}{5.6}(4.0\text{yrs}) = 80\text{yrs}$$

This explains the extreme sensitivity of the equity value to changes in interest rates.

- c. Liquidating a portion of the assets and investing in T-bills and notes will reduce the duration of these assets by $13.5 - 2 = 11.5$ years. From Eq. 30.10:

$$\text{Amount} = \frac{\overbrace{\text{change in equity duration}}^{(80\text{yrs})} \times \overbrace{\text{equity value}}^{5.6}}{\underbrace{\text{change in asset duration}}_{11.5\text{yrs}}} = \$38.96 \text{ million.}$$

That is, we should liquidate \$38.96 million of the fund's assets.

- d. We can also reduce the duration of the fund by entering into a swap contract in which Citrix will receive a floating rate and pay a fixed rate. This swap will increase in value when interest rates rise, offsetting the decline in the value of the rest of the fund. To determine the size of the swap, we proceed as in Ex. 30.14:

$$\text{Amount} = \frac{\overbrace{\text{change in equity duration}}^{(80\text{yrs})} \times \overbrace{\text{equity value}}^{5.6}}{\underbrace{\text{difference in duration of fixed and floating rate 10-yr bond}}_{(7.0 - 0.5)\text{yrs}}} = \$68.92 \text{ million.}$$

That is, we should enter a swap with a notional value of \$68.92 million.

- 30-14.** Your firm needs to raise \$100 million in funds. You can borrow short term at a spread of 1% over LIBOR. Alternatively, you can issue 10-year, fixed-rate bonds at a spread of 2.50% over 10-year Treasuries, which currently yield 7.60%. Current 10-year interest rate swaps are quoted at LIBOR versus the 8% fixed rate.

Management believes that the firm is currently “underrated” and that its credit rating is likely to improve in the next year or two. Nevertheless, the managers are not comfortable with the interest rate risk associated with using short-term debt.

- a. Suggest a strategy for borrowing the \$100 million. What is your effective borrowing rate?
- b. Suppose the firm’s credit rating does improve three years later. It can now borrow at a spread of 0.50% over Treasuries, which now yield 9.10% for a seven-year maturity. Also, seven-year interest rate swaps are quoted at LIBOR versus 9.50%. How would you lock in your new credit quality for the next seven years? What is your effective borrowing rate now?
- a. Borrow \$100m short term and paying LIBOR + 1.0%. Then enter a \$100m notional swap to receive LIBOR and pay 8.0% fixed. Effective borrowing rate is $(\text{LIBOR} + 1.0\%) - \text{LIBOR} + 8.0\% = 9.0\%$.

(Note: borrowing long-term would have cost $7.6\% + 2.5\% = 10.1\%$.)

- b. Refinance \$100m short-term loan with long-term loan at $9.10\% + 0.50\% = 9.60\%$. Unwind swap by entering new swap to pay LIBOR and receive 9.50%. Effective borrowing cost now:

$$9.60\% + (-\text{LIBOR} + 8.0\%) + (\text{LIBOR} - 9.50\%) = 8.10\%.$$

(Note: This rate is equal to the original long-term rate, less the 2% decline in the firm’s credit spread. The firm gets the benefit of its improved credit quality without being exposed to the increase in interest rates that occurred.)

Chapter 31

International Corporate Finance

31-1. You are a U.S. investor who is trying to calculate the present value of a €5 million cash inflow that will occur one year in the future. The spot exchange rate is $S = \$1.25/€$ and the forward rate is $F_1 = \$1.215/€$. You estimate that the appropriate dollar discount rate for this cash flow is 4% and the appropriate euro discount rate is 7%.

- a. What is the present value of the €5 million cash inflow computed by first discounting the euro and then converting it into dollars?
- b. What is the present value of the €5 million cash inflow computed by first converting the cash flow into dollars and then discounting?
- c. What can you conclude about whether these markets are internationally integrated, based on your answers to parts (a) and (b)?

a. $€5 / (1.07) \times 1.25 = \5.84112 million

b. $(€5 \times 1.214953) / 1.04 = \5.84112 million

- c. Yes, the markets are internationally integrated because the answers to (a) and (b) are identical.

31-2. Mia Caruso Enterprises, a U.S. manufacturer of children's toys, has made a sale in Cyprus and is expecting a C£4 million cash inflow in one year. (The currency of Cyprus is the Cypriot pound, C£. Cyprus is a member of the European Union, but has not yet adopted the euro.) The current spot rate is $S = \$1.80/C£$ and the one-year forward rate is $F_1 = \$1.8857/C£$.

- a. What is the present value of Mia Caruso's C£4 million inflow computed by first discounting the cash flow at the appropriate Cypriot pound discount rate of 5%, and then converting the result into dollars?
- b. What is the present value of Mia Caruso's C£4 million inflow computed by first converting the cash flow into dollars, and then discounting at the appropriate dollar discount rate of 10%?
- c. What can you conclude about whether these markets are internationally integrated, based on your answers to parts (a) and (b)?

a. $C£4 / (1.05) \times 1.80 = \6.8571 million

b. $C£4 / (1.10) \times 1.885 = \6.8545 million

- c. No, the markets are not internationally integrated because the answers to (a) and (b) are not the same.

Chapter 26

Working Capital Management

26-1. Answer the following:

- a. What is the difference between a firm's cash cycle and its operating cycle?**
 - b. How will a firm's cash cycle be affected if a firm increases its inventory, all else being equal?**
 - c. How will a firm's cash cycle be affected if a firm begins to take the discounts offered by its suppliers, all else being equal?**
- a. A firm's cash cycle is the average length of time from when a firm pays cash for its inventory to when it receives cash from the sale of that inventory (or the end product that the firm produced with the inventory). It is calculated as the average number of days between the purchase of the initial inventory and the sale of the end product plus the average number of days it takes the firm's customers to pay cash for the inventory they purchase minus the average number of days the firm takes to pay its suppliers for the inventory. A firm's operating cycle is the average length of time between when a firm purchases its inventory and when the firm receives cash from the sale of the inventory. It is calculated as the average number of days between the purchase of the initial inventory and the sale of the end product plus the average number of days it takes the firm's customers to pay for the inventory they purchase. If a firm were to pay cash for its inventory, rather than buying the inventory on credit, the cash cycle and the operating cycle of the firm would be identical. In most cases, however, firms buy their inventory on credit, so the cash cycle is shorter than the operating cycle of the firm.
- b. If a firm increases its inventory, its inventory days will increase, all else equal. This will, therefore, increase the cash cycle of the firm.
- c. If a firm begins to take discounts offered by its suppliers, its accounts payable days will decrease. All else equal, this will cause the cash cycle of the firm to increase.

26-2. Does an increase in a firm's cash cycle necessarily mean that a firm is managing its cash poorly?

No. An increase in a firm's cash cycle does not necessarily mean that the firm is managing its cash poorly. The increase may be due to a conscious management decision. For example, a firm may decide to increase its inventory in order if it has been experiencing excessive stock-outs. All else equal, this would result in an increase in its cash cycle. Or a firm may decide to loosen its credit policy in order to attract customers from its competitors. This would result in an increase in accounts receivable days, and all else equal, result in an increase in the firm's cash cycle. And if a firm chooses to take the discounts offered by its suppliers, the accounts payable days will decrease, leading to an increase in the firm's cash cycle.

26-3. Aberdeen Outboard Motors is contemplating building a new plant. The company anticipates that the plant will require an initial investment of \$2 million in net working capital today. The plant will last 10 years, at which point the full investment in net working capital will be recovered. Given an annual discount rate of 6%, what is the net present value of this working capital investment?

Ignoring revenues and other expenses associated with the new plant, the NPV of the \$2 million investment in net working capital is simply the present value of the \$2 million that the firm will recoup at the end of ten years minus the initial \$2 million investment.

$$\text{NPV} = -\$2,000,000 + \frac{\$2,000,000}{(1.06)^{10}} = -\$883,210$$

- 26-4.  **The Greek Connection had sales of \$32 million in 2009, and a cost of goods sold of \$20 million. A simplified balance sheet for the firm appears below:**

THE GREEK CONNECTION
Balance Sheet
As of December 31, 2009
(\$ thousands)

Assets		Liabilities and Equity	
Cash	\$ 2,000	Accounts payable	\$ 1,500
Accounts receivable	3,950	Notes payable	1,000
Inventory	1,300	Accruals	1,220
Total current assets	\$ 7,250	Total current liabilities	\$ 3,720
		Long-term debt	3,000
Net plant, property, and equipment	\$ 8,500	Total liabilities	\$ 6,720
Total assets	\$ 15,750	Common equity	9,030
		Total liabilities and equity	\$ 15,750

- Calculate The Greek Connection's net working capital in 2009.
 - Calculate the cash conversion cycle of The Greek Connection in 2009.
 - The industry average accounts receivable days is 30 days. What would the cash conversion cycle for The Greek Connection have been in 2009 had it matched the industry average for accounts receivable days?
- Net working capital is current assets minus current liabilities. Using this definition, The Greek Connection's net working capital is $\$7,250 - \$3,720 = \$3,530$. Some analysts calculate the net operating working capital instead, which is the non-interest earning current assets minus the non-interest bearing current liabilities. In this case, the notes payable would not be included in the calculation since they are assumed to be interest bearing. Net operating working capital for The Greek Connection is $\$7,250 - (\$1,500 + \$1,220) = \$4,530$.
 - The cash conversion cycle (CCC) is equal to the inventory days plus the accounts receivable days minus the accounts payable days. The Greek Connection's cash conversion cycle for 2004 was 41.4 days.

$$\text{CCC} = \frac{\text{inventory}}{\text{average daily COGS}} + \frac{\text{accounts receivable}}{\text{average daily sales}} - \frac{\text{accounts payable}}{\text{average daily COGS}}$$

$$\text{CCC}_{2004} = \frac{\$1,300}{\frac{\$20,000}{365}} + \frac{\$3,950}{\frac{\$32,000}{365}} - \frac{\$1,500}{\frac{\$20,000}{365}}$$

$$= 23.7 \text{ days} + 45.1 \text{ days} - 27.4 \text{ days} = 41.4 \text{ days}$$

- c. If The Greek Connection accounts receivable days had been 30 days, its cash conversion cycle would have been only 26.3 days:

$$\text{CCC} = 23.7 \text{ days} + 30 \text{ days} - 27.4 \text{ days} = 26.3 \text{ days}$$

- 26-5. Assume the credit terms offered to your firm by your suppliers are 3/5, Net 30. Calculate the cost of the trade credit if your firm does not take the discount and pays on day 30.**

In this instance, the customer will have the use of \$97 for an additional 25 days ($30 - 5$) if he chooses not to take the discount. It will cost him \$3 to do so since he must pay \$100 for the goods if he pays after the 5-day discount period. Thus, the interest rate per period is:

$$\frac{\$3}{\$97} = 0.0309 = 3.09\%.$$

The number of 25-day periods in a year is $365/25 = 14.6$ periods. So the effective annual cost of the trade credit is:

$$\text{EAR} = (1.0309)^{14.6} - 1 = 55.94\%.$$

- 26-6. Your supplier offers terms of 1/10, Net 45. What is the effective annual cost of trade credit if you choose to forgo the discount and pay on day 45?**

If you were to pay within the 10-day discount period, you would pay \$99 for \$100 worth of goods. If you wait until day 45, you will owe \$100. Thus, you are paying \$1 in interest for a 35-day ($45 - 10$) loan. The interest rate per period is:

$$\frac{\$1}{\$99} = 0.0101 = 1.01\%.$$

The number of 35-day periods in a year is $365 / 35 = 10.43$ periods. So the effective annual cost of the trade credit is:

$$\text{EAR} = (1.0101)^{10.43} - 1 = 11.05\%.$$

- 26-7. The Fast Reader Company supplies bulletin board services to numerous hotel chains nationwide. The owner of the firm is investigating the benefit of employing a billing firm to do her billing and collections. Because the billing firm specializes in these services, collection float will be reduced by 20 days. Average daily collections are \$1200, and the owner can earn 8% annually (expressed as an APR with monthly compounding) on her investments. If the billing firm charges \$250 per month, should the owner employ the billing firm?**

The benefit of outsourcing the billing and collection to the other firm is equal to what Fast Reader can earn on the funds that are freed up. Since average daily collections are \$1,200 and float will be reduced by 20 days, Fast Reader will have an additional \$24,000 ($\$1,200 \times 20$). (Think about this as follows. Immediately after hiring the billing firm, its collection float drops by 20 days, so all collections due within the next 20 days are immediately available.) The billing firm charges \$250 per month. At an 8% annual rate, the monthly discount rate is $1.081 / 12 - 1 = 6.43\%$, so the present value of these charges in perpetuity is $250/0.0643 = \$38,856$. Thus, the costs (\$38,856) exceed the benefits (\$24,000), and Fast Reader should not employ the billing firm.

- 26-8. The Saban Corporation is trying to decide whether to switch to a bank that will accommodate electronic funds transfers from Saban's customers. Saban's financial manager believes the new system would decrease its collection float by as much as five days. The new bank would require a compensating balance of \$30,000, whereas its present bank has no compensating balance requirement. Saban's average daily collections are \$10,000, and it can earn 8% on its short-term investments. Should Saban make the switch? (Assume the compensating balance at the new bank will be deposited in a non-interest-earning account.)**

The electronic funds transfer system will free up \$50,000 ($= 5 \times \$10,000$) (Think about this as follows. Immediately after switching banks its collections due within the next five days are immediately available.) On the other hand, Saban will have to pay a cost because it has to hold \$30,000 in a non-interest earning account, which means it has essentially given up these funds. Because the benefits (\$50,000) are larger than the costs (\$30,000) it should switch banks.

26-9. What are the three steps involved in establishing a credit policy?

The three steps involved in establishing a credit policy are:

- (1) Establish credit standards. In this step, the firm must decide how much credit risk it is willing to accept.
- (2) Establish credit terms. Here, the firm decides on the length of time before payment must be made and whether or not it will offer a discount. If a discount is to be offered, the amount of the discount and the length of the discount period must also be established.
- (3) Establish a collection policy. In this step, the firm must decide how it will handle late payers.

26-10. The Manana Corporation had sales of \$60 million this year. Its accounts receivable balance averaged \$2 million. How long, on average, does it take the firm to collect on its sales?

If we assume all the sales were made on credit, the average length of time it takes Manana to collect on its sales is 12.2 days:

$$\text{Accounts receivable days} = \frac{\text{Accounts receivable}}{\text{Average daily sales}} = \frac{\$2,000,000}{\frac{\$60,000,000}{365}} = 12.2 \text{ days.}$$

26-11. The Mighty Power Tool Company has the following accounts on its books:

Customer	Amount Owed (\$)	Age (days)
ABC	50,000	35
DEF	35,000	5
GHI	15,000	10
KLM	75,000	22
NOP	42,000	40
QRS	18,000	12
TUV	82,000	53
WXY	36,000	90

The firm extends credit on terms of 1/15, Net 30. Develop an aging schedule using 15-day increments through 60 days, and then indicate any accounts that have been outstanding for more than 60 days.

Mighty Power Tool Company Aging Schedule

Percent of

Days Outstanding	Amount Owed	Accounts Receivable
0-15	\$ 68,000	19.3%
16-30	\$ 75,000	21.2%
31-45	\$ 92,000	26.1%
46-60	\$ 82,000	23.2%
over 60	\$ 36,000	10.2%
	\$353,000	100.0%

26-12. What is meant by “stretching the accounts payable”?

“Stretching the accounts payable” refers to a customer’s not paying by the payment date specified.

 26-13. Simple Simon’s Bakery purchases supplies on terms of 1/10, Net 25. If Simple Simon’s chooses to take the discount offered, it must obtain a bank loan to meet its short-term financing needs. A local bank has quoted Simple Simon’s owner an interest rate of 12% on borrowed funds. Should Simple Simon’s enter the loan agreement with the bank and begin taking the discount?

If Simple Simon’s takes the discount, it must pay \$99 in 10 days for every \$100 of purchases. If it elects not to take the discount, it will owe the full \$100 in 25 days. The interest rate on the loan is:

$$\frac{\$1}{\$99} = 1.01\%.$$

The loan period is 15 days (= 25 – 10). The effective annual cost of the trade credit is:

$$\text{EAR} = (1.0101)^{365/15} - 1 = 27.7\%.$$

Since the bank loan is only 12%, Simple Simon should borrow the funds from the bank in order to take advantage of the discount.

26-14. Your firm purchases goods from its supplier on terms of 3/15, Net 40.

- a. What is the effective annual cost to your firm if it chooses not to take the discount and makes its payment on day 40?
- b. What is the effective annual cost to your firm if it chooses not to take the discount and makes its payment on day 50?
- a. Your firm is paying \$3 to borrow \$97 for 25 days (= 40 – 15). The interest rate per period is:

$$\frac{\$3}{\$97} = 0.0309 = 3.09\%.$$

The effective annual rate is $(1.0309)^{365/25} - 1 = 55.9\%$.

- b. In this case, your firm is stretching its accounts payable. You are still paying \$3 to borrow \$97, so the interest rate per period is 3.09%. However, the loan period is now 35 days (= 50 – 15). The effective annual rate is reduced to 37.4% because your firm has use of the money for a longer period of time:

$$\text{EAR} = (1.0309)^{365/35} - 1 = 37.4\%.$$

26-15. Use the financial statements supplied below for International Motor Corporation (IMC) to answer the following questions.

- a. Calculate the cash conversion cycle for IMC for both 2009 and 2010. What change has occurred, if any? All else being equal, how does this change affect IMC’s need for cash?
- b. IMC’s suppliers offer terms of Net 30. Does it appear that IMC is doing a good job of managing its accounts payable?

INTERNATIONAL MOTOR CORPORATION

Income Statement (in millions)
for the Years Ending December 31

	2009	2010
Sales	\$ 60,000	\$ 75,000
Cost of goods sold	52,000	61,000
Gross profit	\$ 8,000	\$ 14,000
Selling and general and administrative expenses	6,000	8,000
Operating profit	\$ 2,000	\$ 6,000
Interest expense	1,400	1,300
Earnings before tax	\$ 600	\$ 4,700
Taxes	300	2,350
Earnings after tax	\$ 300	\$ 2,350

INTERNATIONAL MOTOR CORPORATION

Balance Sheet (in millions)
as of December 31

	2009	2010		2009	2010
Assets			Liabilities		
Cash	\$ 3,080	\$ 6,100	Accounts payable	\$ 3,600	\$ 4,600
Accounts receivable	2,800	6,900	Notes payable	1,180	1,250
Inventory	6,200	6,600	Accruals	5,600	6,211
Total current assets	\$ 12,080	\$ 19,600	Total current liabilities	\$ 10,380	\$ 12,061
Net plant, property, and equipment	\$ 23,087	\$ 20,098	Long-term debt	\$ 6,500	\$ 7,000
Total assets	\$ 35,167	\$ 39,698	Total liabilities	\$ 16,880	\$ 19,061
			Equity		
			Common stock	\$ 2,735	\$ 2,735
			Retained earnings	\$ 15,552	\$ 17,902
			Total equity	\$ 18,287	\$ 20,637
			Total liabilities and equity	\$ 35,167	\$ 39,698

- a. The cash conversion cycle (CCC) is equal to the inventory days plus the accounts receivable days minus the accounts payable days. IMC's cash conversion cycle for 2003 was 35.2 days, and for 2004, it was 45.6 days.

$$CCC = \frac{\text{inventory}}{\text{average daily COGS}} + \frac{\text{accounts receivable}}{\text{average daily sales}} - \frac{\text{accounts payable}}{\text{average daily COGS}}$$

$$CCC_{2003} = \frac{\$6,200}{\frac{\$52,000}{365}} + \frac{\$2,800}{\frac{\$60,000}{365}} - \frac{\$3,600}{\frac{\$52,000}{365}}$$

$$= 43.5 \text{ days} + 17.0 \text{ days} - 25.3 \text{ days} = 35.2 \text{ days}$$

$$CCC_{2004} = \frac{\$6,600}{\frac{\$61,000}{365}} + \frac{\$6,900}{\frac{\$75,000}{365}} - \frac{\$4,600}{\frac{\$61,000}{365}}$$

$$= 39.5 \text{ days} + 33.6 \text{ days} - 27.5 \text{ days} = 45.6 \text{ days}$$

IMC's cash conversion cycle has lengthened in 2004, due to an increase in its accounts receivable days. The number of days goods are held in inventory has decreased, and IMC is taking longer to pay its suppliers, both of which would decrease the cash conversion cycle all else equal. These changes were not enough to offset the increase in the amount of time it is taking IMC's customers

to pay for purchases made on credit. The lengthening of the cash conversion cycle means that IMC will require more cash.

- b. If IMC's suppliers are offering terms of net 30, IMC should consider waiting longer to pay for its purchases. In 2003, it paid nearly five days earlier than necessary, and in 2004, it paid 2.5 days earlier. IMC could, therefore, have kept the money working for it longer because there was no discount offered for early payment. The early payment may give IMC a preferred position with its suppliers, however, which may have benefits that are not presented here. IMC's decision on whether to extend its accounts payable days would have to take these benefits into consideration.

26-16. Ohio Valley Homecare Suppliers, Inc. (OVHS), had \$20 million in sales in 2009. Its cost of goods sold was \$8 million, and its average inventory balance was \$2,000,000.

- a. Calculate the average number of inventory days outstanding for OVHS.
- b. The average days of inventory in the industry is 73 days. By how much would OVHS reduce its investment in inventory if it could improve its inventory days to meet the industry average?

- a. The inventory days ratio is equal to the inventory divided by average daily cost of goods sold. This was 91.25 days for Ohio Valley Homecare Suppliers in 2004:

$$\text{inventory days} = \frac{\$2,000,000}{\frac{\$8,000,000}{365}} = 91.25 \text{ days.}$$

The inventory turnover ratio for Ohio Valley Homecare Supplies was four times in 2004:

$$\text{inventory turnover} = \frac{\text{cost of goods sold}}{\text{inventory}} = \frac{\$8,000,000}{\$2,000,000} = 4X.$$

- b. OVHS could increase its inventory turnover to five times by reducing its inventory to \$1,600,000:

$$5 = \frac{\$8,000,000}{\text{inventory}} \Rightarrow \text{inventory} = \$1,600,000.$$

This means OVHS could reduce its investment in inventory by \$400,000 (= \$2,000,000 – \$1,600,000).

26-17. Which of the following short-term securities would you expect to offer the highest before-tax return: Treasury bills, certificates of deposit, short-term tax exempts, or commercial paper? Why?

Commercial paper. Treasury bills and certificates of deposit are considered to be free of default risk, so they offer lower returns than a security that has default risk associated with it. Short-term tax exempts have default risk, but the interest on these instruments is free from federal taxation; thus, they offer a lower yield than a similarly risky, fully taxable security. Commercial paper exposes the investor to default risk and the interest earned is fully taxable. It would, therefore, have to offer the investor a higher before-tax return than the other instruments.

Chapter 27

Short-Term Financial Planning

27-1. Which of the following companies are likely to have high short-term financing needs? Why?

- a. A clothing retailer
- b. A professional sports team
- c. An electric utility
- d. A company that operates toll roads
- e. A restaurant chain

The professional sports team is likely to have very high short-term financing needs because of the seasonality of its revenue stream. Several months before the season actually begins for the team, it may have large cash flow requirements—equipment, tickets, advertising, etc.—that it will need to finance since the bulk of its cash inflows will occur during the season.

The clothing retailer may have high short-term financing needs because of the seasonality of its business. Sales are often highest in the fall and spring, so the retailer may need short-term financing in order to purchase inventory prior to the high seasons. It may need even more short-term financing due to a negative cash flow shock. For example, if the store's buyer misjudged the fashion trends and overbought a particular style that the store was unable to sell, the store might experience a large net cash outflow for that season. In that case, it may require more short-term financing than normal in order to purchase inventory for the upcoming season.

27-2. Sailboats Etc. is a retail company specializing in sailboats and other sailing-related equipment. The following table contains financial forecasts as well as current (month 0) working capital levels. During which months are the firm's seasonal working capital needs the greatest? When does it have surplus cash?

	Month						
(\$000)	0	1	2	3	4	5	6
Net Income		\$10	\$12	\$15	\$25	\$30	\$18
Depreciation		2	3	3	4	5	4
Capital Expenditures		1	0	0	1	0	0
Levels of Working Capital							
Accounts Receivable	\$2	3	4	5	7	10	6
Inventory	3	2	4	5	5	4	2
Accounts Payable	2	2	2	2	2	2	2

To determine Sailboats seasonal working capital needs, we calculate the changes in net working capital for the firm.

Changes in working capital	Month					
	1	2	3	4	5	6
Accounts receivable	\$1	\$1	\$1	\$2	\$3	-\$4
Inventory	-\$1	\$2	\$1	0	-\$1	-\$2
Accounts payable	0	0	0	0	0	0
Change in net working capital	\$0	\$3	\$2	\$2	\$2	-\$6

From the table it can be seen that Sailboat's working capital needs are highest in Month 2 because its investments in accounts receivable and in inventory increased the most in that month.

Sailboats, Etc. has a surplus cash position in every month as shown below:

	Month					
	1	2	3	4	5	6
(\$000)						
Net income	\$10	\$12	\$15	\$25	\$30	\$18
<i>plus</i> depreciation	2	3	3	4	5	4
<i>minus</i> changes in net working capital	0	3	2	2	2	-6
Cash flow from operations	\$12	\$12	\$16	\$27	\$33	\$28
<i>minus</i> capital expenditures	1	0	0	1	0	0
Change in cash	\$11	\$12	\$16	\$26	\$33	\$28

27-3. What is the difference between permanent working capital and temporary working capital?

Permanent working capital is the amount that a firm has to keep invested in its short-term assets in order to support its operations. For example, a toy retailer may need to keep a minimum amount invested in inventory at all times. Temporary working capital, on the other hand, is the result of seasonal fluctuations and/or unanticipated cash flow shocks. The toy retailer, for example, may need to have more invested in inventory just prior to its peak season. Temporary working capital is the difference between the actual level of investment in short-term assets and the permanent working capital investment.

27-4. Quarterly working capital levels for your firm for the next year are included in the following table. What are the permanent working capital needs of your company? What are the temporary needs?



(\$000)	Quarter			
	1	2	3	4
Cash	\$100	\$100	\$100	\$100
Accounts Receivable	200	100	100	600
Inventory	200	500	900	50
Accounts Payable	100	100	100	100

The net working capital for each quarter is calculated below:

(\$000)	Quarter			
	1	2	3	4
Cash	\$100	\$100	\$ 100	\$100
Accounts receivable	\$200	\$100	\$ 100	\$600
Inventory	\$200	\$500	\$ 900	\$ 50
Accounts payable	\$100	\$100	\$ 100	\$100
Net working capital	\$400	\$600	\$1,000	\$650

The minimum level of net working capital—\$400,000 in Quarter 1—represents the firm's permanent working capital. The difference between the higher net working capital levels in each quarter and the permanent working capital needs represents the firm's temporary working capital needs. Thus the firm has temporary working capital needs of \$200,000 in Quarter 2, \$600,000 in Quarter 3, and \$250,000 in Quarter 4.

27-5. Why might a company choose to finance permanent working capital with short-term debt?

Financing permanent working capital with short-term debt is an aggressive financing policy and is considered risky. Although under a normal term structure, the interest rates on short-term debt are lower than those on long-term debt, the firm faces the risk that it will have to pay more when it needs to refinance the debt in the future. This increased risk will be reflected in a higher cost of equity for the firm. Nevertheless, a firm may decide to use short-term debt to finance permanent working capital if it believes that one or more market imperfections exist. For example, short-term debt is less sensitive to a firm's credit quality than is long-term debt and so will be less affected by management's actions or information. Thus, short-term debt may have lower agency and lemons costs than long-term debt. Management may also believe it has superior knowledge regarding the future cash flows for the firm, which is not yet reflected in the firm's credit rating. If management believes its ability to produce future cash flows will have a positive impact on its credit rating in the future, management may elect to use lower-cost, short-term debt to finance its permanent working capital for the time being with the expectation that it will refinance it with long-term debt in the near future when the market has recognized the firm's improved future prospects and rewarded it with a higher credit rating, which can result in a lower cost of long-term debt to the firm.

27-6. The Hand-to-Mouth Company needs a \$10,000 loan for the next 30 days. It is trying to decide which of three alternatives to use:



Alternative A: Forgo the discount on its trade credit agreement that offers terms of 2/10, Net 30.

Alternative B: Borrow the money from Bank A, which has offered to lend the firm \$10,000 for 30 days at an APR of 12%. The bank will require a (no-interest) compensating balance of 5% of the face value of the loan and will charge a \$100 loan origination fee, which means Hand-to-Mouth must borrow even more than the \$10,000.

Alternative C: Borrow the money from Bank B, which has offered to lend the firm \$10,000 for 30 days at an APR of 15%. The loan has a 1% loan origination fee.

Which alternative is the cheapest source of financing for Hand-to-Mouth?

Alternative A: The effective annual cost of the trade credit is 44.6%, calculated as follows:

$$\text{Interest rate per period} = \frac{\$2}{\$98} = 2.041\%.$$

The loan period is 20 (= 30 – 10) periods, and there are 18.25 periods in a year (365 / 20 = 18.25).

$$\text{EAR} = 1.02041^{18.25} - 1 = 44.6\%$$

Alternative B: Hand-to-Mouth will need to borrow \$10,100 just to cover its loan origination fee. Beyond that, it needs to have enough to meet the compensating balance requirement. So the total amount that Hand-to-Mouth must borrow is \$10,632:

$$\text{Amount needed} = \frac{\$10,100}{1 - 0.05} = \$10,632.$$

At a 12% APR, the interest expense for the 30-day loan will be $0.01(\$10,632) = \106.32 .

Since the loan origination fee is simply additional interest, the total interest on the 30-day loan is $\$106.32 + \$100 = \$206.32$. The firm's usable proceeds from the loan is \$10,000. So the interest rate per period is $2.063\% \left(= \frac{\$206.32}{\$10,000} \right)$.

The effective annual rate of alternative B is $(1.02063)^{365/30} - 1 = 28.2\%$.

Alternative C: Hand-to-Mouth will need to borrow \$10,100 in order to cover the loan origination fee.

An APR of 15% translates to an interest rate of $1.25\% \left(= \frac{15\%}{12} \right)$ for 30 days. The interest expense for one month is $0.0125 \times \$10,100 = \126.25 . This with the loan origination fee makes the total interest charge \$226.25 for the 30-day loan. This amounts to 2.263% for 30 days: $\frac{\$226.25}{\$10,000}$.

The effective annual rate is $(1.02263)^{365/30} - 1 = 31.3\%$.

Thus in Alternative A, the cost of trade credit is the most expensive at an effective annual rate of 44.6%.

- 27-7. Consider two loans with a 1-year maturity and identical face values: an 8% loan with a 1% loan origination fee and an 8% loan with a 5% (no-interest) compensating balance requirement. Which loan would have the higher effective annual rate? Why?**

The loan with the 1% loan origination fee would cost the most since the loan origination fee is just another form of interest, so on a \$1,000 loan, the borrower is paying \$90 in interest and will have the use of only \$990 for the period, making the effective annual cost of the loan over 9% ($\$90/\$990 = 9.1\%$). The compensating balance requirement of 5% on a \$1,000 loan reduces the usable proceeds of the firm by 5% to \$950, but the interest rate is still 8%, so the effective annual cost of that arrangement is $8.4\% \left(= \frac{\$80}{\$950} \right)$. The effective annual rate is not increased by a full percentage.

- 27-8. What is the difference between evergreen credit and a revolving line of credit?**

The major difference between a revolving line of credit and evergreen credit is the commitment period. A revolving line of credit is a committed line of credit that involves a solid commitment from the bank for a period that is longer than the typical one-year term of a regular line of credit. The revolving line of credit is usually good for two or three years before it must be renegotiated. Evergreen credit is a revolving line of credit that has *no* fixed maturity.

- 27-9. Which of the following one-year \$1000 bank loans offers the lowest effective annual rate?**

- A loan with an APR of 6%, compounded monthly
- A loan with an APR of 6%, compounded annually, that also has a compensating balance requirement of 10% (on which no interest is paid)
- A loan with an APR of 6%, compounded annually, that has a 1% loan origination fee

The effective annual rates of each of the alternatives are calculated as follows.

- a. Since the APR is 6%, the monthly rate is $6\%/12 = 0.5\%$. This translates to an effective annual rate of $(1.005)^{12} - 1 = 6.2\%$.
- b. The compensating balance is $\$1,000 \times 0.10 = \100 . Therefore, the borrower will have use of only \$900 of the \$1,000. The interest is $0.06 \times \$1,000 = \60 . The interest rate per period is $\$60 / \$900 = 6.7\%$. Since this alternative assumes annual compounding, the effective annual rate is 6.7% as well.
- c. The interest expense is $0.06 \times \$1,000$, and the loan origination fee is $0.01 \times \$1,000 = \10 . The loan origination fee reduces the usable proceeds of the loan to \$990 because it is paid at the beginning of the loan. The interest rate per period is $\$70 / \$990 = 7.1\%$. Since the loan is compounded annually in this case, 7.1% is the effective annual rate.

Thus, alternative (a) offers the lowest effective annual cost.

- 27-10. The Needy Corporation borrowed \$10,000 from Bank Ease. According to the terms of the loan, Needy must pay the bank \$400 in interest every three months for the three-year life of the loan, with the principal to be repaid at the maturity of the loan. What effective annual rate is Needy paying?**

In this problem, Needy must pay \$400 every three months to have the use of \$10,000. Thus, the interest rate per period is $\$400 / \$10,000 = 4\%$. Since there are four three-month periods in a year, the effective annual rate is $(1.04)^4 - 1 \approx 17\%$.

- 27-11. The Treadwater Bank wants to raise \$1 million using three-month commercial paper. The net proceeds to the bank will be \$985,000. What is the effective annual rate of this financing for Treadwater?**

Treadwater is paying \$15,000 ($= \$1,000,000 - \$985,000$) to use \$985,000 for three months, so the three-month interest rate is $\$15,000 / \$985,000 = 1.523\%$. There are four three-month periods in a year, making the effective annual rate $(1.01523)^4 - 1 = 6.2\%$.

- 27-12. Magna Corporation has an issue of commercial paper with a face value of \$1,000,000 and a maturity of six months. Magna received net proceeds of \$973,710 when it sold the paper. What is the effective annual rate of the paper to Magna?**

Magna is paying \$26,290 ($= \$1,000,000 - \$973,710$) to use \$973,710 for six months. The six-month interest rate is $\$26,290 / \$973,710 = 2.7\%$. There are two six-month periods in one year, so the effective annual rate is $(1.027)^2 - 1 = 5.5\%$.

- 27-13. What is the difference between direct paper and dealer paper?**

Direct paper is a method by which a firm sells its commercial paper directly to investors. Dealer paper refers to the sale of commercial paper through dealers. The dealers get a fee for their services, thus reducing the proceeds that the issuing firm receives (and increasing the effective cost to the firm).

- 27-14. The Signet Corporation has issued four-month commercial paper with a \$6 million face value. The firm netted \$5,870,850 on the sale. What effective annual rate is Signet paying for these funds?**

Signet's interest expense on this loan is \$129,150 ($= \$6,000,000 - \$5,870,850$), and the usable proceeds are \$5,870,850. The interest rate for the four-month period is $\$129,150 / \$5,870,850 = 2.2\%$. The effective annual rate is $(1.022)^3 - 1 = 6.7\%$.

27-15. What is the difference between pledging accounts receivable to secure a loan and factoring accounts receivable?

When accounts receivable are pledged, the borrowing firm is simply using its accounts receivable as collateral for a loan. The lender reviews the invoices for the credit sales of the borrower and determines which accounts are acceptable collateral. The lender will then lend some percentage of the dollar amount of the accepted invoices. If one or more of the borrowing firm's customers fail to pay, the firm is still responsible to the lender for the money it has borrowed. In a factoring arrangement, the accounts receivable are sold to the lender (i.e., factor), and the firm's customers typically make their payments directly to the lender. The lender agrees to pay the borrowing firm the amount due from the firm's customers at the end of the firm's payment period, but the firm may borrow a certain percentage of the face value of its receivables in order to receive the money in advance. The factoring arrangement may be "with recourse," which means that if any of the borrowing firm's customers defaults on its bills, the factor can require the borrowing firm to make the payment. It may also be "without recourse," in which case the lender bears the risk that one or more customers will default on their bills. The borrowing firm is not responsible for the payments.

27-16. The Ohio Valley Steel Corporation has borrowed \$5 million for one month at a stated annual rate of 9%, using inventory stored in a field warehouse as collateral. The warehouse charges a \$5000 fee, payable at the end of the month. What is the effective annual rate of this loan?

The monthly interest rate is $9\% / 12 = 0.75\%$, so Ohio Valley Steel must pay $0.0075 \times \$5,000,000 = \$37,500$ in interest on the loan. Combining this with the \$5,000 warehouse fee makes the monthly cost of the loan \$42,500. Since the fee is paid at the end of the month, Ohio Valley Steel has use of the full \$5,000,000 for the month. The interest rate per period is $\$42,500 / \$5,000,000 = 0.85\%$. There are 12 months in a year, so the effective annual rate is $(1.0085)^{12} - 1 = 10.7\%$.

27-17. Discuss the three different arrangements under which a firm may use inventory to secure a loan.

The three different methods under which inventory is used as collateral for a loan are floating liens, trust receipts, and warehouse arrangements.

With a floating lien (also called a "general lien" or a "blanket lien"), all of the borrower's inventory serves as collateral for a loan. The value of the collateral declines as the firm sells its inventory. In times of financial distress, a firm may decide to sell its inventory without making payments on the loan and may not have enough money to replenish the inventory it has sold. The loan is then under-collateralized. This is the riskiest arrangement from the lender's standpoint, and the loan will carry a higher interest rate than if one of the other two methods is used. Additionally, the lender will lend a much smaller percentage of the inventory value under this arrangement.

In a trust receipt (or floor planning) arrangement, specific inventory items are identified as collateral for the loan. As the specified inventory is sold, the firm uses the cash received to repay the loan. The lender will send someone to the borrower's premises periodically to ensure that none of the specified inventory has been sold without a repayment made.

In a warehouse arrangement, the inventory serving as collateral is stored in a warehouse. One type of warehouse is a public warehouse, which is a business that exists for the sole purpose of tracking the flow of the inventory. The inventory is delivered to the public warehouse by the borrowing firm, and the lender extends a loan based on the value of that inventory. When the borrowing firm needs the inventory to sell, it must return to the warehouse to retrieve it after receiving permission from the lender to do so. This arrangement is the least risky from the standpoint of the lender since it allows the lender to maintain the tightest control over the inventory, but it is only feasible for some types of inventory. This method would not be usable for inventory that is subject to spoilage or that is bulky and difficult to transport. In this latter case, a field warehouse might be a good alternative. A field warehouse is established on the borrower's premises, but it is separated from the borrower's main plant. It is operated by a third party. This type of arrangement is more convenient for the borrower, but it still gives the lender the added security of having the inventory that serves as collateral tracked by a third party.

- 27-18. The Rasputin Brewery is considering using a public warehouse loan as part of its short-term financing. The firm will require a loan of \$500,000. Interest on the loan will be 10% (APR, annual compounding) to be paid at the end of the year. The warehouse charges 1% of the face value of the loan, payable at the beginning of the year. What is the effective annual rate of this warehousing arrangement?**

Rasputin's interest expense is $0.10(\$500,000) = \$50,000$. The warehouse fee is $0.01(\$500,000) = \$5,000$. Because the warehouse fee must be paid at the beginning of the year, Rasputin's usable proceeds from the loan are only $\$495,000 (= \$500,000 - \$5,000)$. The effective annual rate is $\$55,000 / \$495,000 = 11.1\%$.

1.

Hobart acquired 100% of the shares of Sydney on 1 January 20X0 for \$5,000. At that date the statements of financial position were as follows:

	Hobart	Sydney
	\$	\$
Investment in Sydney	5,000	--
Other assets	<u>12,000</u>	<u>6,000</u>
Total assets	<u>17,000</u>	<u>6,000</u>
Issued capital	8,000	5,000
Retained earnings	<u>7,000</u>	--
Total equity	15,000	5,000
Liabilities	<u>2,000</u>	<u>1,000</u>
Total equity and liabilities	<u>17,000</u>	<u>6,000</u>

Required

Prepare the consolidated statement of financial position of the Hobart group as at 1 January 20X0.

GROUP STRUCTURE

(v1) H
| 100%
S

(w2) NET ASSET OF SUBS.

@A @R

IC	5	5
RE	-	-
	5	5

(w3,4) ???

(W3,7) !!!

(W5) Group **RESERVES**

IC

H → 8

PE

H

PE:

7

S

PAP:

0

7

B/S of HOBART GROUP

1/SAN/XO

INV. IN SYDNEY

OTHER ASSETS [12+6]

IC

PE(W5)

LIAB [2+1]

X

18

18

8

7

15

3

18



2.

@R

One year later, 31 December 20X0, the statements of financial position of Hobart and Sydney are as follows:

	Hobart	Sydney
	\$	\$
Investment in Sydney	5,000	--
Other assets	<u>12,000</u>	<u>9,000</u>
Total assets	<u>17,000</u>	<u>9,000</u>
Issued capital	<u>8,000</u>	5,000
Retained earnings	<u>7,000</u>	<u>3,000</u>
Total equity	15,000	8,000
Liabilities	<u>2,000</u>	<u>1,000</u>
Total equity and liabilities	<u>17,000</u>	<u>9,000</u>

Required

Prepare the consolidated statement of financial position of the Hobart group as at 31 December 20X0.

@R

(w1) H
100%
S

(w5)

(w2) @A @R

IC	5	5
RE	-	3
	5	8
	(3)	→

H RE: 7
S RE: 3
10

INV. IN SYDNEY
OTHER ASSETS [12+9] 21
IC 8
10

IC
RE (WT)

LIAB [2+1]

$$\begin{array}{r} 8 \\ 10 \\ \hline 18 \\ 3 \\ \hline 21 \\ \hline \end{array}$$



CONS

EQUITY $\begin{cases} \nearrow \text{INTRODUCED [IC; SP]} \\ \searrow \text{GENERATED [RE; REV]} \end{cases}$

H

H+S PAP

3.

Hercules acquired the entire Issued capital of Samson for \$6,000 on 1 April 20X0 when the statements of financial position of the two companies were as follows:

	Hercules	Samson
	\$	\$
Investment in Samson	6,000	--
Other assets	<u>11,500</u>	<u>7,000</u>
Total assets	<u>17,500</u>	<u>7,000</u>
Issued capital	5,000	4,000
Retained earnings	<u>11,000</u>	<u>2,000</u>
Total equity	16,000	6,000
Liabilities	<u>1,500</u>	<u>1,000</u>
Total equity and liabilities	<u>17,500</u>	<u>7,000</u>

Required

Prepare the consolidated statement of financial position of the Hercules group as at 1 April 20X0.

(w1) H
1 100%
S

(w2) @A = @R

IC	4	4
RE	2	2
	<u>6</u>	<u>6</u>

(w3,4) ???

INVESTMENT IN SAMSON
OTHER ASSETS [11.5 + 7]

12

(w5)

H RE: 11
S P+P: ~~11~~
11

X
18.5
18.5
5

IC
PE (WS)

LIAB[1.5+1]

$$\begin{array}{r} \hline 5 \\ 11 \\ \hline 16 \\ 2.5 \\ \hline 18.5 \\ \hline \hline \end{array}$$



4.

Homer acquired 100% of Simpson for \$7,000 when the statement of financial position of Simpson was as follows:

\$		
Total assets		<u>11.000</u>
Issued capital		2.000
Retained earnings		5.000
Total equity		<u>7.000</u>
Liabilities		<u>4.000</u>
Total equity and liabilities		<u>11.000</u>

@A

On 31 December 20X0 the statements of financial position of the two companies are as follows:

	Homer	Simpson
	\$	\$
Investment in Simpson	7,000	--
Other assets	<u>16,000</u>	<u>17,000</u>
Total assets	<u>23,000</u>	<u>17,000</u>
Issued capital	5,000	2,000
Retained earnings	<u>14,000</u>	<u>11,000</u>
Total equity	19,000	13,000
Liabilities	<u>4,000</u>	<u>4,000</u>
Total equity and liabilities	<u>23,000</u>	<u>17,000</u>

@R

Required

Prepare the consolidated statement of financial position of the Homer group as at 31 December 20X0.

@R

(w1) H
| 100%
S

(w5) IC → H → 5

(w2) @A @R
IC 2 2
NE 5 11

H RE: 14
→ S PAP: 6
20

12	5	11
12E		
	7	13
	(6)	5

INVESTMENT IN SIMPSON
OTHER ASSETS [16+17]

1C
12E (W5)

LIAB [4+4]

$$\begin{array}{r}
 \text{X} \\
 33 \\
 \hline
 33 \\
 \hline
 5 \\
 20 \\
 \hline
 25 \\
 8 \\
 \hline
 33 \\
 \hline
 \end{array}$$



5.

Hop acquired the entire Issued capital of Skip for \$8,000 on 1 February 20X0 when the statements of financial position of the two companies were as follows:

	Hop	Skip
	\$	\$
Investment in Skip	8,000	--
Other assets	9,500	6,500
Total assets	17,500	6,500
Issued capital	9,000	3,000
Retained earnings	6,000	2,000
Total equity	15,000	5,000
Liabilities	2,500	1,500
Total equity and liabilities	17,500	6,500

Required

Prepare the consolidated statement of financial position of the Hop group as at 1 February 20X0.

(W1) H
| 100%
S

(W2) @A @R
IC 3 3
RE 2 2

5 5

(W5) H RE:
S PAP

(W3) NEW!
COST OF CONTROL
(PAID)

NA @A
GOODWILL

(W4)???

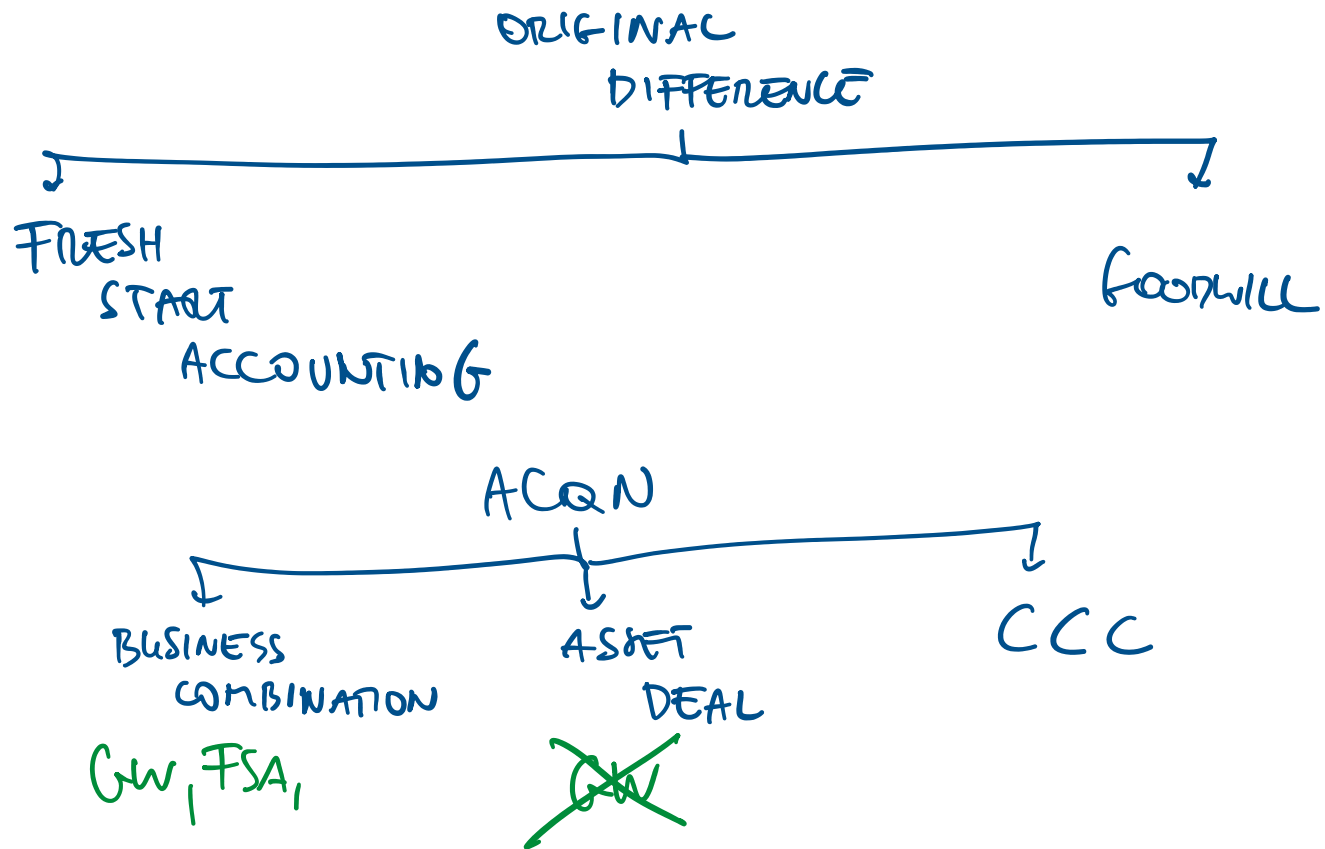
(5)
3

6
0

6

GOODWILL (W3)	3
INV IN SKIP	X
OTHER ASSET [9.5+6.5]	16
	19
IC	9
NE(W5)	6
	15
LIAB [2.5+1.5]	4
	19





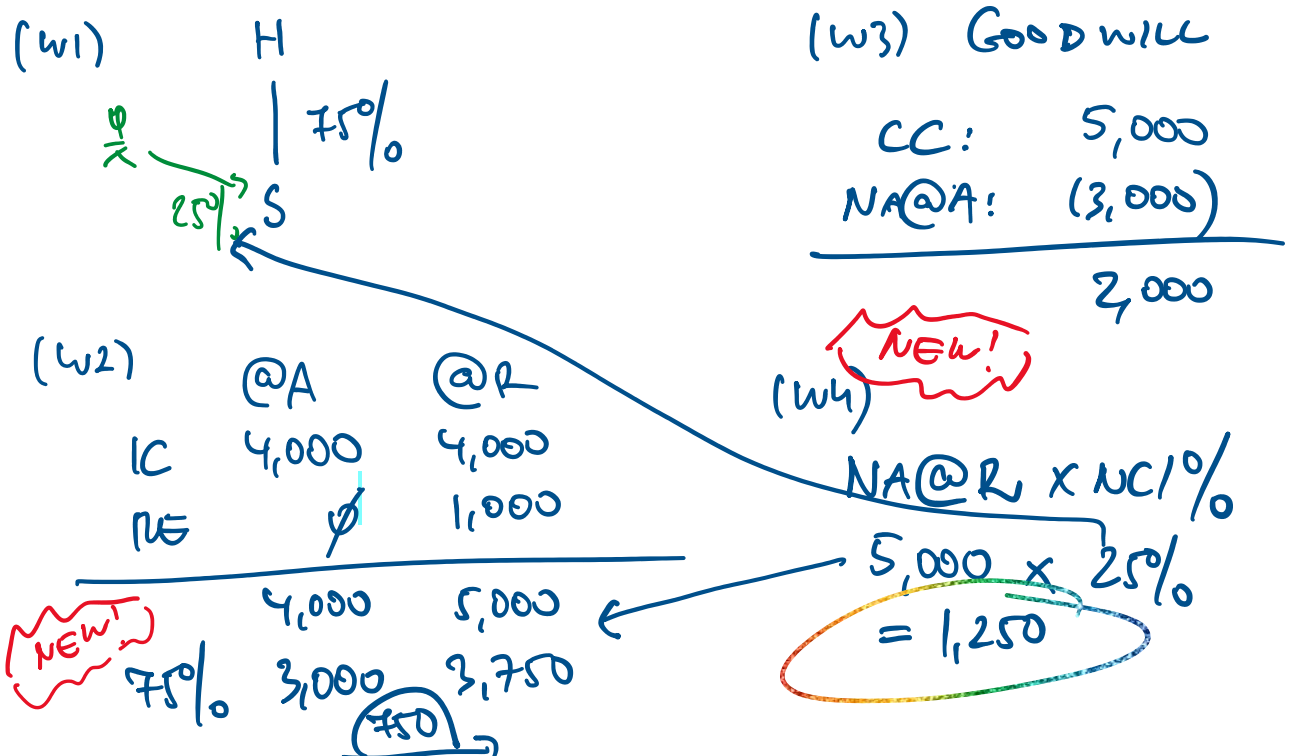
6.

Hide acquires 75% of the issued capital of Seek at the date of the latter company's incorporation on 1 January 20X5. One year later the two companies have the following statements of financial position.

	Hide	Seek
	\$	\$
Goodwill		
Investment in Seek	5,000	--
Other assets	<u>12,500</u>	<u>8,000</u>
Total assets	<u>17,500</u>	<u>8,000</u>
Issued capital	12,000	4,000
Retained earnings	<u>1,500</u>	<u>1,000</u>
Total equity	13,500	5,000
Non-controlling interests		
Total equity		
Liabilities	<u>4,000</u>	<u>3,000</u>
Total equity and liabilities	<u>17,500</u>	<u>8,000</u>

Required

Produce the consolidated statement of financial position of Hide and its subsidiary company at this date.



(W5)

H RE: 1,500

S PAF: 750

2,250

GOODWILL (W3)

OTHER ASSETS [12,500 + 8,000 ~~(750/6)~~]

2,000

20,500

22,500

IC

RE (W5)

FA SHP

NCL (W4)

EQUITY:

LIAB [4,000 + 3,000 ~~(750/6)~~]

12,000

2,250

14,250

1,250

15,500

7,000

22,500

← OWN

← CONTROL



8.

Hit acquired 80% of the issued capital of Score on 30 September 20X7.

At the year end 31 December 20X7 the two companies have the following statements of financial position:

	@A		@R	
	\$	\$	\$	\$
Investment in Score		2,000		
Property, plant and equipment		1,000		
Other assets		9,500		6,000
Total assets		12,500		6,000
Issued capital		4,000		1,000
Share premium		-		500
Retained earnings				
1 Jan 20X7	4,000		1,500	
Profit for 20X7	2,000		1,000	
		6,000		2,500
		10,000		4,000
Non-controlling interest				
Total equity				
Liabilities		2,500		2,000
Total equity and liabilities		12,500		6,000

The non-controlling interests of Score are to be valued at the proportionate share of identifiable net assets.

Required

Prepare the consolidated statement of financial position of Hit and its subsidiary company as at 31 December 20X7. Goodwill/negative goodwill arising is to be treated in accordance with IFRS 3.

(w1) H

80% 3MTHS
S

(w2)

@A

@R

IC

1,000

1,000

SP

500

500

NE

- 2,250 *

2,500

3,750

4,000

80%

3,000

3,200

$$* 1,500 + 1,000 \times 9/12 = 2,250$$

(w4)

11.1

80% 3,000 3,200
 (200) →

(W4)

NCI

4,000 × 20% = 800

(W3)

CC:	2,000
NA@A:	(3,000)
	(1,000)
REDUCTION	1,000
	0

Dr. GW

(W5)

H RE	6,000
S PAP:	200
NEGATIVE:	1,000
GW	

Cr. P/L → RE

7,200

GW (W3)

PPE

OTHER A [9,500 + 6,000]

1,000

15,500

16,500

IC

4,000

SP

7,200

RE (W5)

11,200

NCI (W4)

800

12,000

NCI (W4)

LIAB [2,500 + 2,000]

800
12,000
4,500
<u>16,500</u>



9.

Hard acquired 60% of the Issued capital of Soft on its incorporation. The statement of financial position of the two companies as at 31 December 20X8 are as follows:

	Hard \$'000	Soft \$'000
Non-current assets		
Property, plant and equipment	200	50
Investment in Soft	<u>6</u>	<u>—</u>
	206	50
Current assets		
Inventory	22	18
Receivables - Soft	<u>40</u>	<u>—</u>
- Other	<u>56</u>	<u>39</u>
Cash	<u>4</u>	<u>15</u>
	122	72
Total assets	<u>328</u>	<u>122</u>
Equity and liabilities		
Issued capital	100	10
Retained earnings	<u>147</u>	<u>42</u>
	247	52
Non-controlling interest		
Total equity		
Current liabilities		
Trade payables	81	50
Hard	<u>—</u>	<u>20</u>
	81	70
Total equity and liabilities	<u>328</u>	<u>122</u>

Notes:

- There was cash in transit of \$4,000 from Soft to Hard at the year-end.
- Goods dispatched by Hard to Soft before the year end with the related invoices to the value of \$16,000 had not been received by Soft until 4 January 20X9. The original cost of the goods was \$16,000.

Required

Prepare a consolidated statement of financial position as at 31 December 20X8.

(w1) H
| 60%.

(w3)
CC: 6
NA&A: (6)

| 60%
S

LC:
NA@A: 6
(6)

(W2)	@A	@R
IC	10	10
NE	8	42
	10	52
60%	6	31.2

$\xrightarrow{25.2}$

(W4)
 $52 \times 40\% = 20.8$

(W5)
H: 147
S RAP: $\frac{25.2}{172.2}$

TB.	H	S
BIT	40	(20)
11T	(4)	(15)
	36	(36)

(W6)	INV.	REC.	CASH	TP.
H	22	96	4	81
S	18	39	15	70
CIT		(4)	4	
11T	16			16
ELIM		(36)		(36)

CONS	56	95	23	131
------	----	----	----	-----

(W7)
PFE
H 200
S 50

H	...
S	50
CONS	250

GOODWILL	(W3)
PPE	(W7)
INV.	↑
REC.	(W6)
CASH	↓

-
250
56
95
23
424

IC	
RE	(W5)

100
172.2
272.2
20.8
293

NCI	(W ⁴ ₈)
-----	--------------------------------

TP	(WG)
----	------

131
424



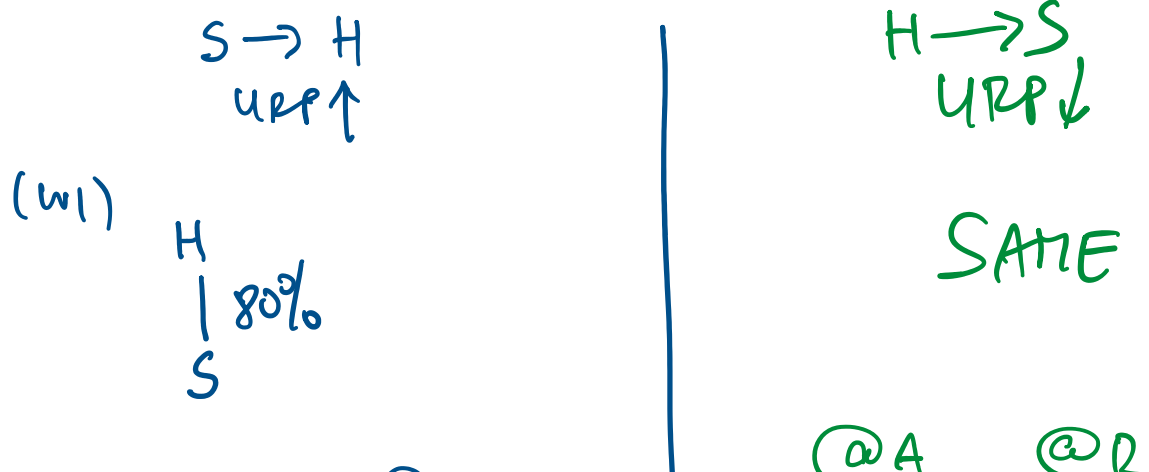
10.

Head acquired 80% of the Issued capital of Shoulder on its incorporation. The statements of financial position of the two companies as at 31 December 20X8 are as follows:

	Head \$'000	Shoulder \$'000
Non-current assets		
→ Property, plant and equipment	200	80
Investment in Shoulder	<u>8</u>	<u>—</u>
	208	80
Current assets		
→ Inventory	50	25
→ Others	<u>100</u>	<u>55</u>
	150	80
Total assets	<u>358</u>	<u>160</u>
	-	-
Equity and liabilities		
Issued capital	100	10
Retained earnings	<u>183</u>	<u>90</u>
	283	100
Non-controlling interest		
Total equity		
→ Current liabilities	<u>75</u>	<u>60</u>
Total equity and liabilities	<u>358</u>	<u>160</u>

Notes:

- There were no intercompany balances at the year-end.
- During December 20X8 Shoulder sold goods to Head for \$25,000. Shoulder sells goods at a mark-up of 25%. Head had not sold any of these goods at the year-end.

Required**Prepare a consolidated statement of financial position as at 31 December 20X8.**

(w2)	@A	@R
IC	10	10
PE	0	90
URP ↑	X	(5) ●
	10	95
80%	8	76
	(68)	

(w3) GW

CE:
NA@A

$$\frac{8}{(8)}$$

(w4) NC/
 $95 \times 20\% = 19$

(w5)

H PE : 183

.....

S PAP: $\frac{68}{251}$

(w6,7)

	PPE	INV.	OA.	CL.
H	200	50	100	75
S	80	25	55	60

URP ↑

(5) ●

180 70 155 135

@A	@R
10	10
0	90
URP ↑	X
	100
80%	8
	(72)

SAME

$100 \times 20\% = 20$

H PE:
URP ↓

S PAP: $\frac{72}{250}$

SAME ●

4.1.1

CONS 280 ✓ 70 ✓ 155 ✓ 135 ✓

GOODWILL (W3)

PPE (W7)

INV (W6)

O.A.

IC

RE (W5)

NCI (W4)

CL (W6)

S → H

-

280

70

155

505

100

251

351

19

370

135

505



H → S

S

A

M

E

-

505

100

250

350

20

SAME

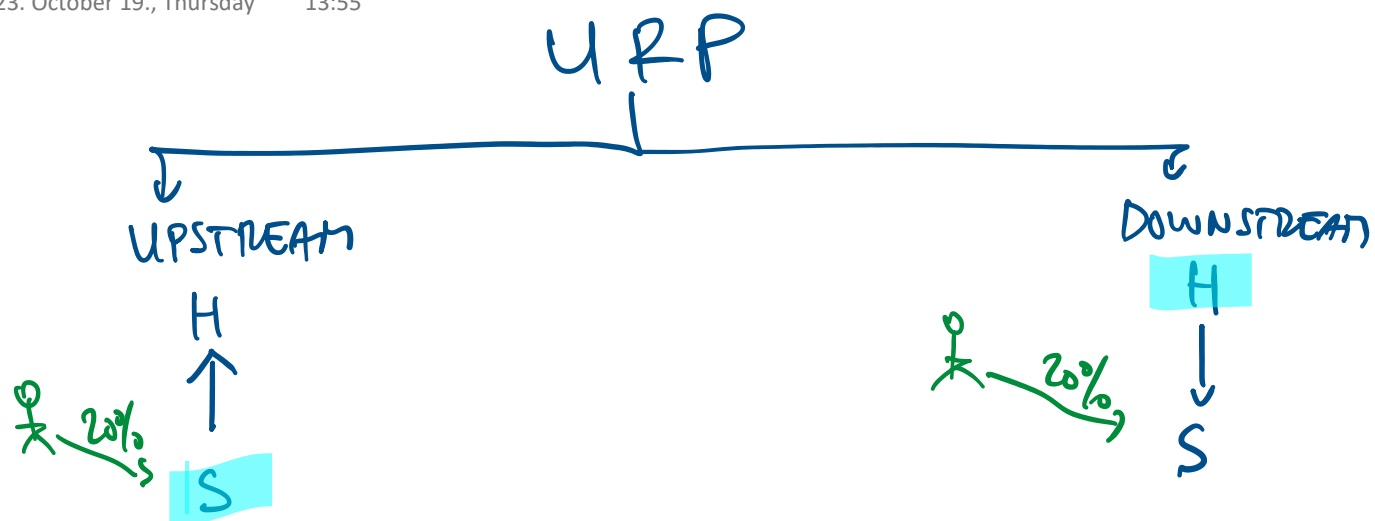
SAME

505



71





14. Hutch

Hutch acquired 90% of the voting Issued capital of Starsky on 1st January 2009 when Starsky's retained earnings stood at 15 million. @A RE

The statements of financial position of the two companies are as follows for 31st December 2009:

	Hutch	Starsky
	MFt	MFt
Investment in Starsky	100	-
Land and building	20	30
Plants	25	15
Inventory	30	40
Receivables	45	-
Cash and equivalents	5	5
Assets	225	90
Issued capital (1 each)	30	20
Share premium	10	10
Retained earnings	65	40
Equity	105	70
Loans	100	
Payables	20	20
Liabilities	120	20
Equity and liabilities	225	90

1. When Starsky was purchased the following differences were identified in the fair value:
- Starsky was purchased partly because of its contracts. The value of this was valued at 10 MFt at the date of acquisition. It is assumed that the contracts will lead to net cash inflows for five years.
 - The fair value of Starsky's plants has a fair value excess of 5 million (as opposed to book value). The useful life of these assets are deemed to be five years at the date of acquisition.
 - Starsky's inventories worth 1 million more on the date of acquisition. Those inventories were all sold before the current yearend.
 - The liabilities of Starsky did not include a possible liability worth 10 million, since IAS 37 does not allow the recognition of those items. However, on the 31st December 2009 it became obvious that no payment will be done due to this obligation.
2. The goodwill suffered a 40% impairment. (Only the parent's goodwill is shown.)
3. The following IC transactions and balances existed at the yearend:
- Starsky sold Hutch a land worth 4 million for 5 million. The price was paid during 2009.
 - Hutch sold inventory to Starsky in December. Hutch sent the goods with the invoice to Starsky. The price of the inventory was 3 million and there were no profit on the transaction.

Prepare the consolidated statement of financial position of the Hutch group!

(w1) H

(w1) H | 90%

S

(w2)

	@A	@R
IC	20	20
SP	10	10
DE	15	40
CONTRACT	10	8 ①
PLANT	5	4 ②
INVENTORY	1	- ③
CONT. LIAB	(10)	- ④
URP ↑	X	(1) ⑤
	51	81
90%	45.9	72.9

FAIR VALUE

BOOK VALUE OF NA.

FSA

27 →

(w4) NC/

$$81 \times 10\% = 8.1$$

(w3)

CC:	100
NA@A:	(45.9)
GW@A:	54.1
IMPAIRMENT	(21.64) ⑥
GW@R	32.46

(w5)

H RE:	65
S PAP:	27
GW IMP	(21.64) ⑥ ^{6.}
URP ↓	(10) ⑦ ^{7.}
CONS:	70.36

(w6)

	INV.	REC	CASH	PAYABLES	CONT. L.
H	30	45	5	20	X
S	40	-	5	20	X
FSA	0 ③ ^{3.}				0 ④ ^{4.}
IIT	3 ③ ^{3.}			3 ③ ^{3.}	
ELIM		(3) ⑩ ^{10.}		(3) ⑩ ^{10.}	
URP	(10) ⑦ ^{7.}				
CONS	73 ✓	42 ✓	10 ✓	40 ✓	- ✓

(w7)

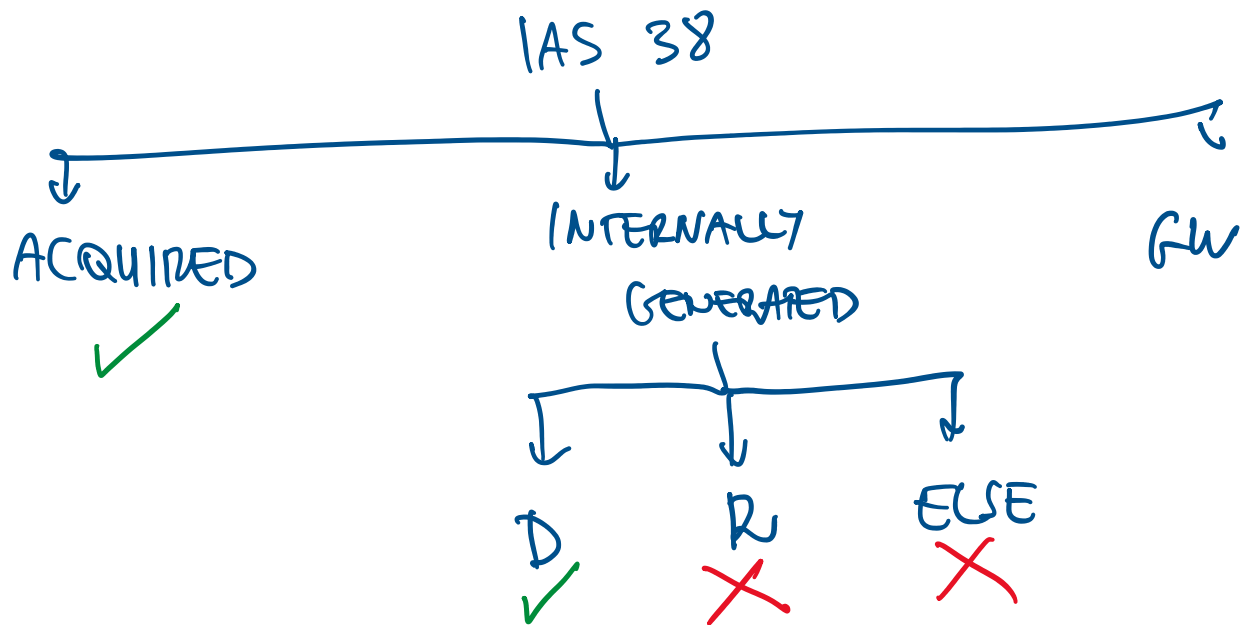
	LAND	PLANT	CONTRACT	LOAN
H	20	25	X	100
S	30	15	X	
FSA		4 ② ^{2.}	8 ① ^{1.}	
URP ↑	(1) ⑤ ^{5.}			
	49 ✓	44 ✓	8 ✓	100 ✓

HUTCH GROUP B/S 2009/12/31.

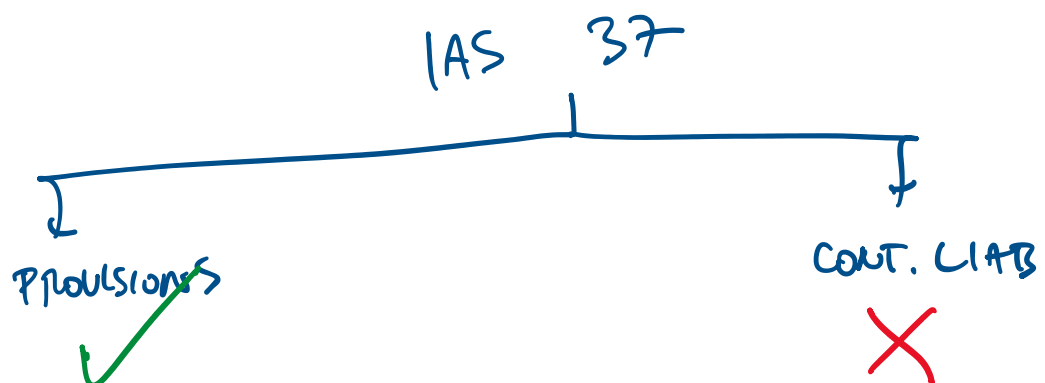
GOODWILL (W3)		32.46	IC	30
LAND	↑	49	SP.	10
PLANT	(W7)	44	RE (W5)	70.36
CONTRACT	↓	8		<u>110.36</u>
INV.	↑	73	NCI (W4)	8.1
REC.	(W6)	42		<u>118.46</u>
CASH	↓	10	LOAN (W7)	100
			PAYABLE	40
			CONT. L.	<u>—</u>
		<u>258.46</u>		<u>258.46</u>



I.



II.



III. INTANGIBLE ASSET